FORMOSA-FX TRAINING MATERIAL

FORMOSA-FX 教育訓練講義

FPC EG 台塑電子組

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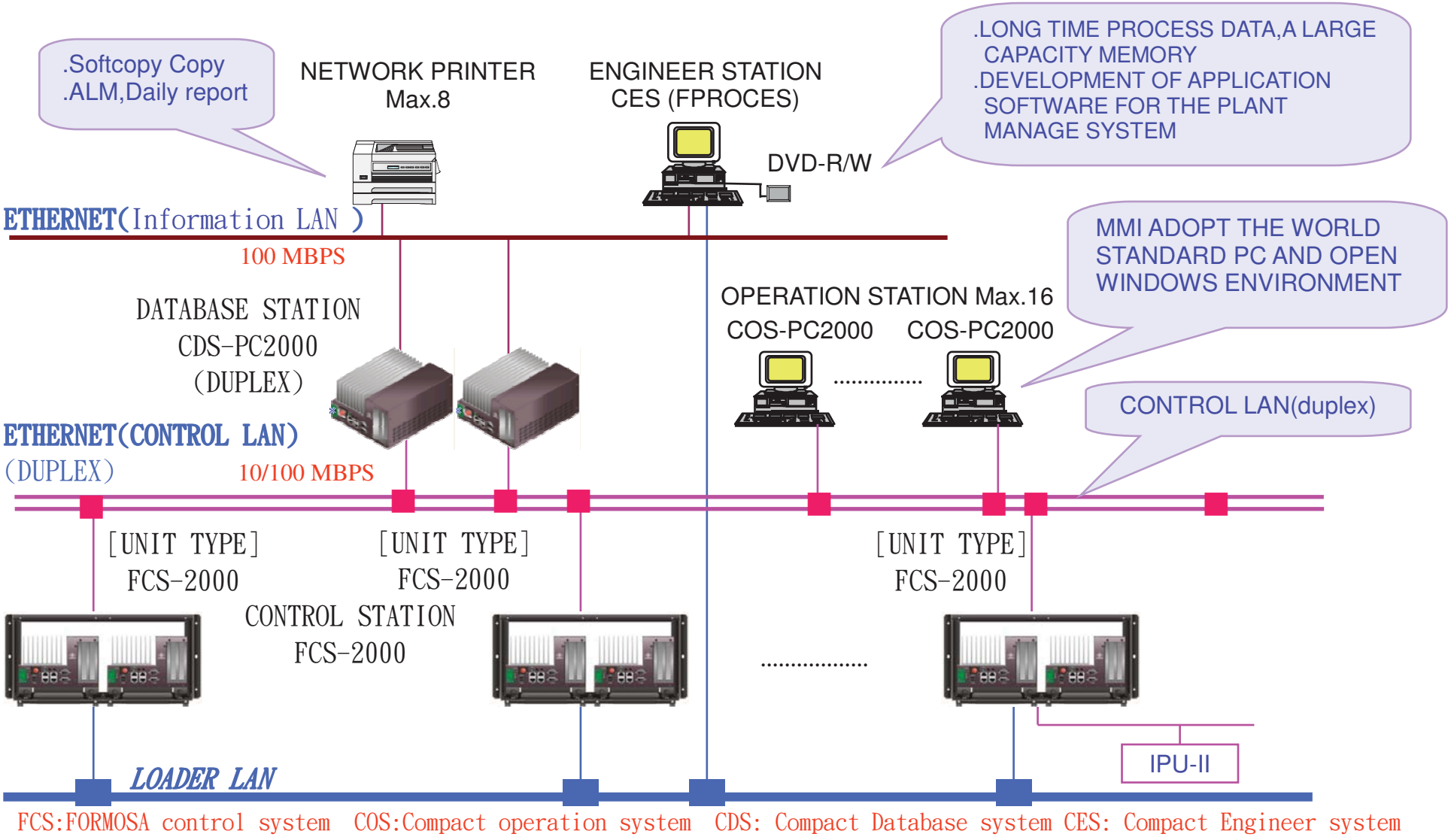
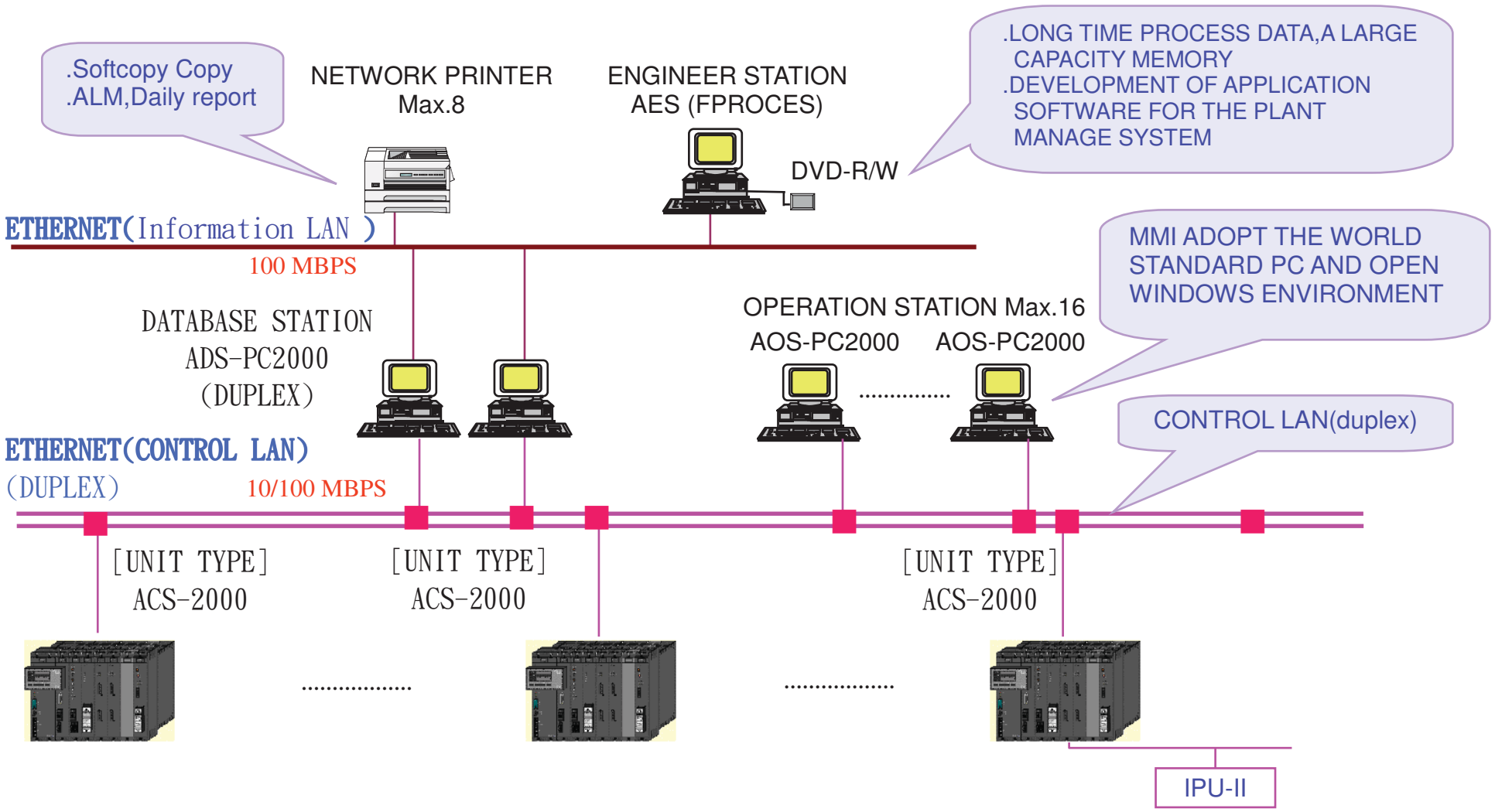
★FORMOSA-CX SYSTEM STRUCTURE AND CHARACTER



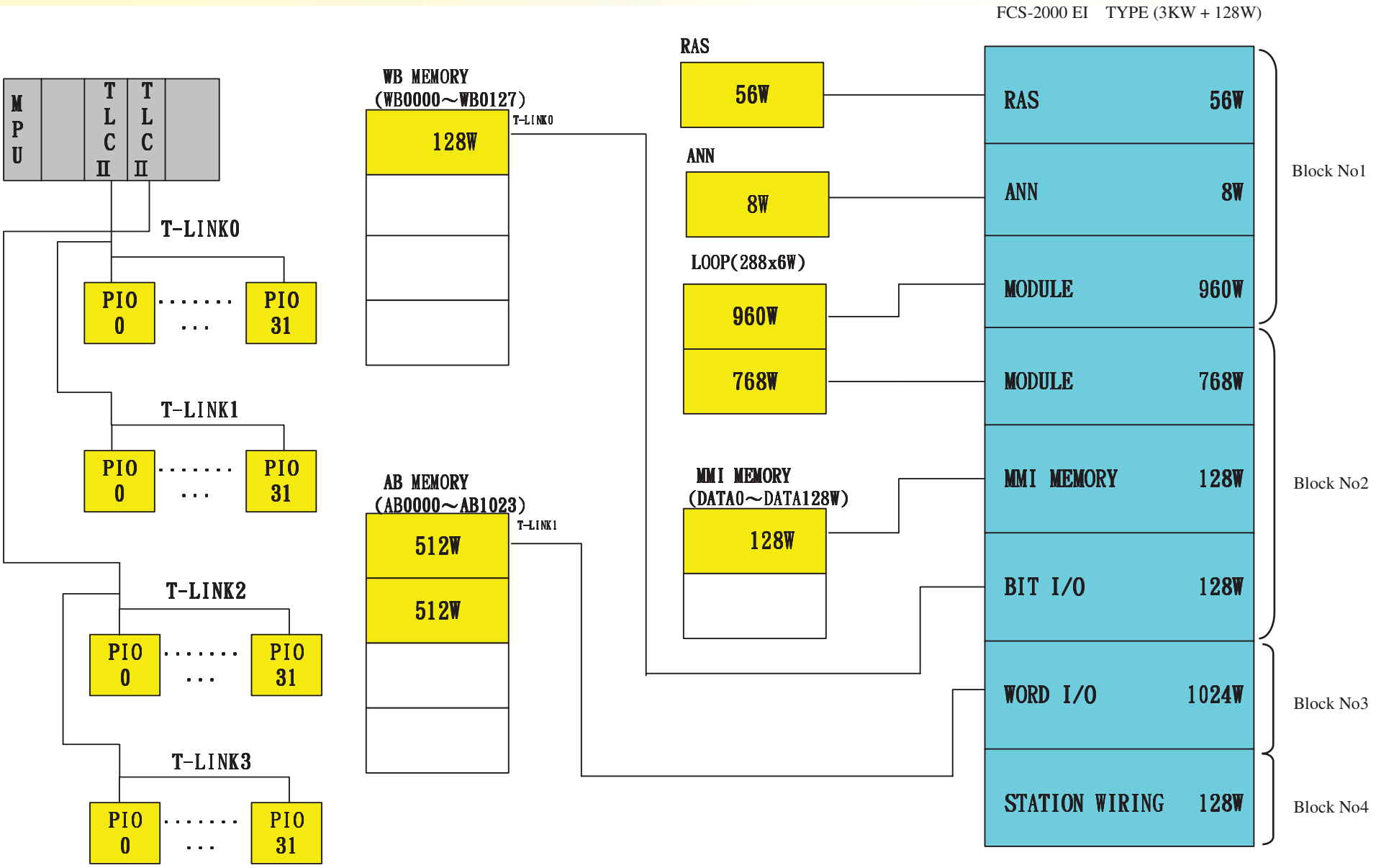
FCS:FORMOSA control system COS:Compact operation system CDS: Compact Database system CES: Compact Engineer system

★FORMOSA-AX SYSTEM STRUCTURE AND CHARACTER

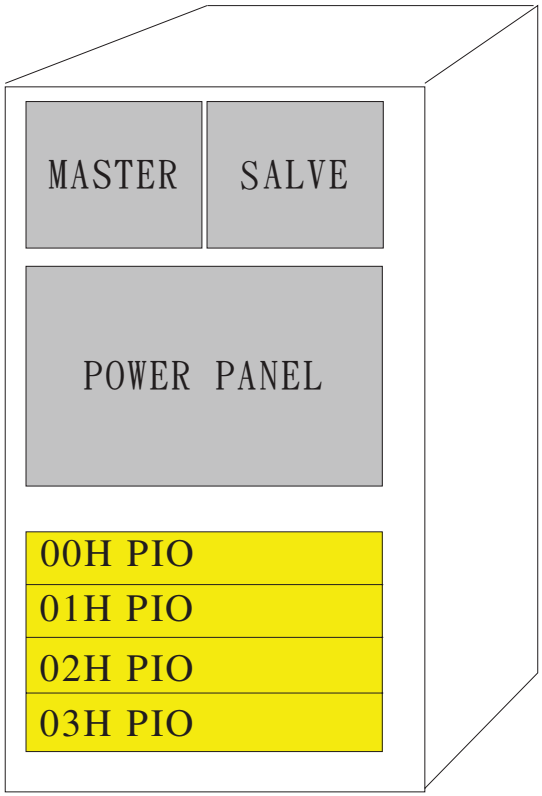


ACS:Advanced control system AOS:Advanced operation system ADS: Advanced Database system AES: Advanced Engineer system

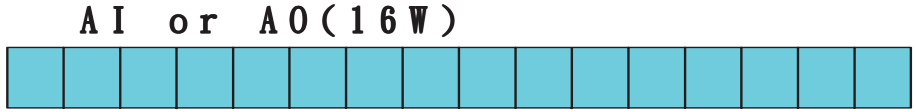
★T-LINK I/O Module



★T-LINK I/O ASSIGNMENT (IPU-1)



AI/AO

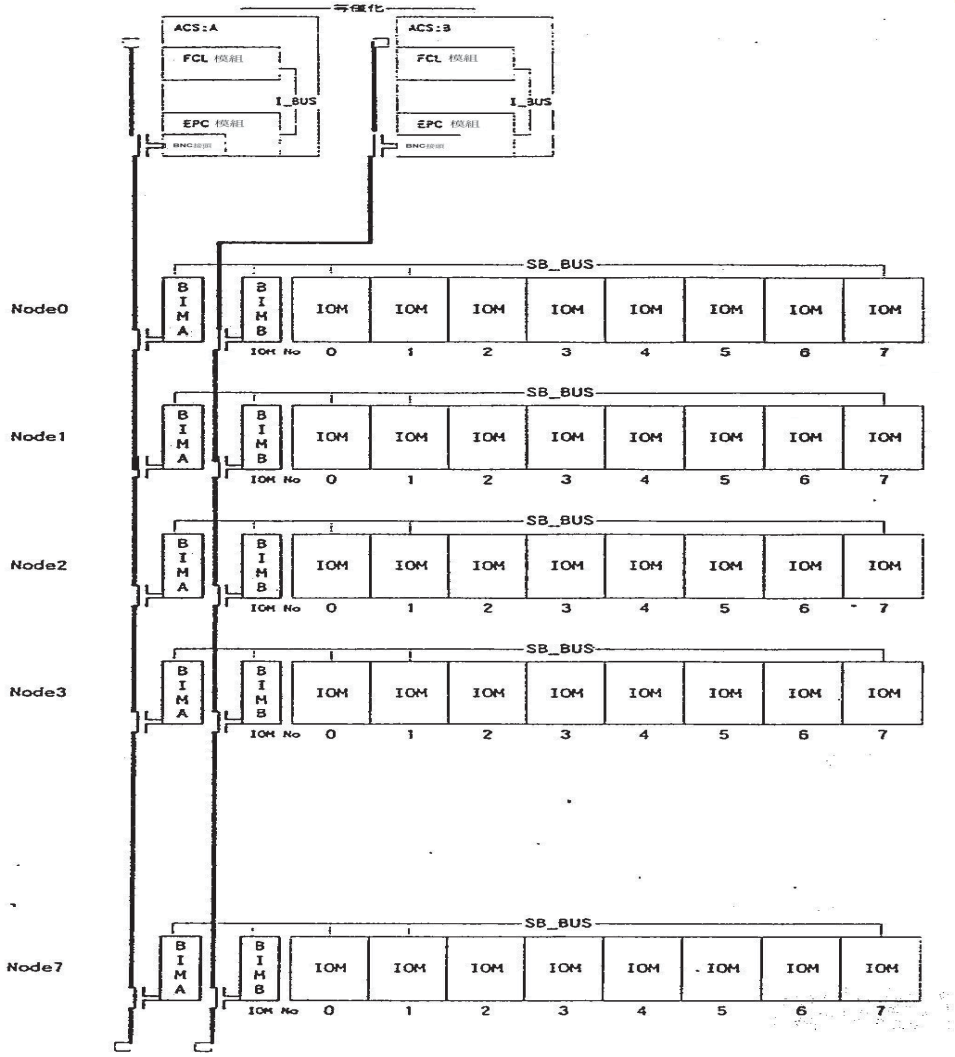


Every PIO can mount MAX :16 AI or AO module,memory used WORD address.
The N Module of PIO ,Start address(WORD), N*16.
If put AI module to PIO ,it will express by AIXXXX.
If put AO module to PIO ,it will express by AOXXXX.
XXXX: Express address serial number AB0000 - AB1023
PI module the same as AI module, PO module occupies the address of 2 AO module ,AO module need assign after AI module.

DI/DO

Every PIO can mount MAX : 2 DI or DO module,every PIO MAX: can put 64 port digital I/O , memory used BIT address ,
The N Module of PIO ,Start address(WORD), N*4.
If put DI module to PIO, it will express by WIXXXX.
If put DO module to PIO, it will express by WOXXXX.
DI/DO express with Bit and DI module to express by I.XXXX, DO module to express by O.XXXX
XXXX: Express address serial number WB0000 - WB0127

★EPC I/O MODULE (IPU-2)



EPC

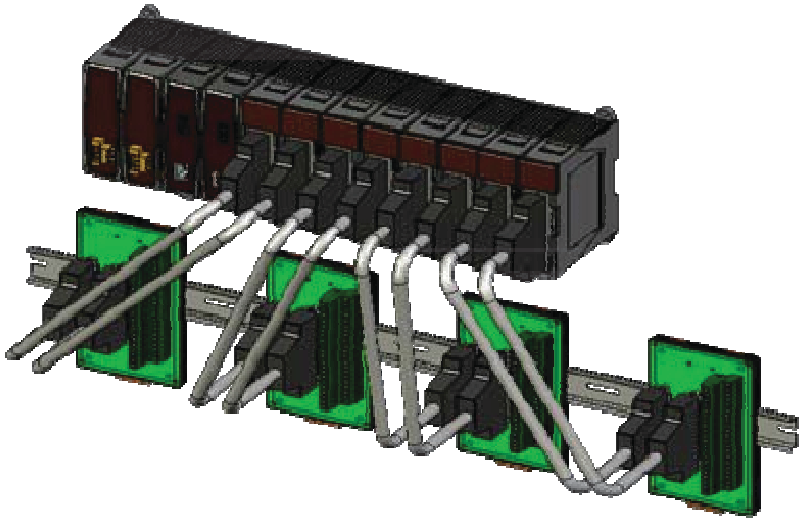
MPU have 2 EPC , EPC0 and EPC1,

Each EPC has 8 NODEs, NODE0 - NODE7

Each NODE has 8 IOMs, IOM0 - IOM7

Each IOM can connect different Module I/O,

★I/O MODULE (IPU-3)



MBA

MPU have 2 MBA, RIO0 and RIO1,

Every MBA has 8 Nodes, NODE0 - NODE7

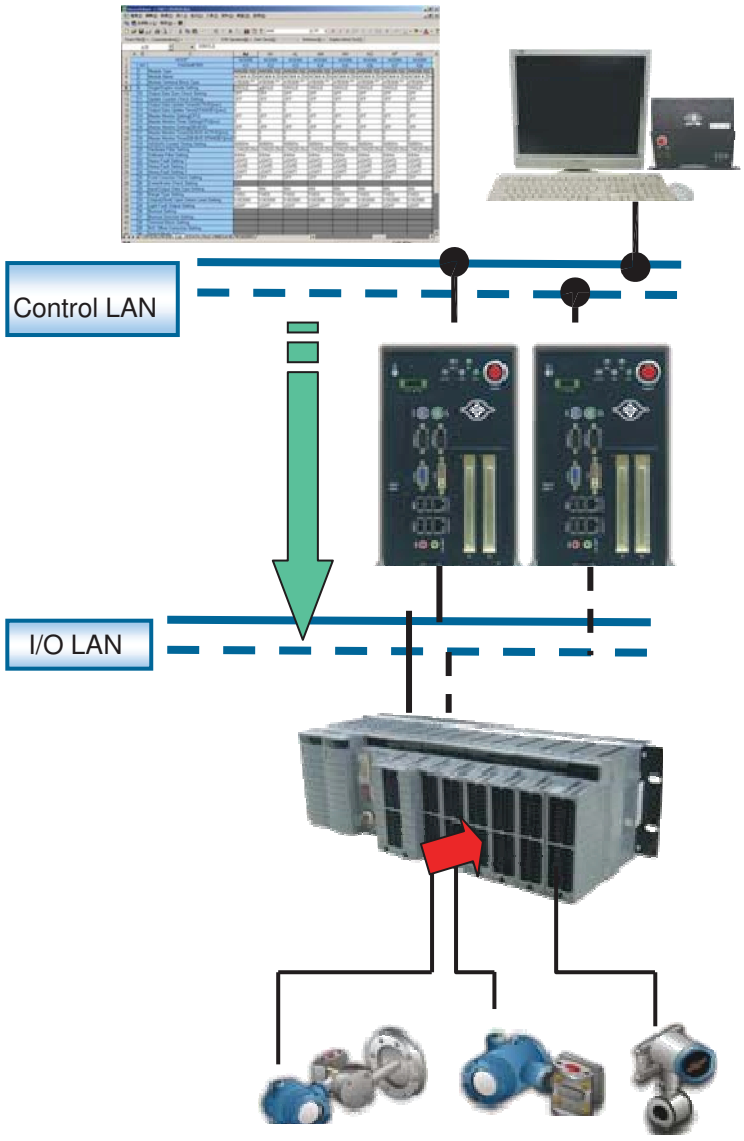
Every NODE has 8 IOMs, IOM0 - IOM7

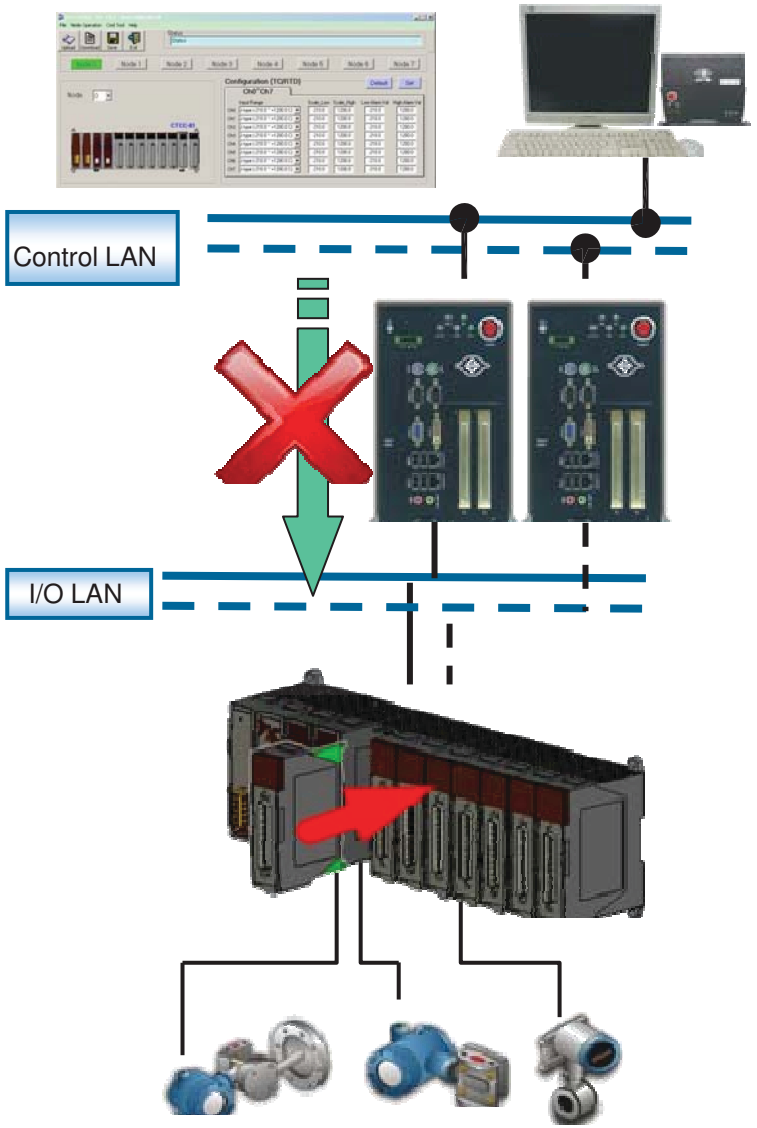
Every IOM can connect different Module I/O,

On-line hat swapping, change IOM do not need setup software.

Terminal board and I/O module connected to pre-wired cable, we may reduce the wiring work.

★I/O MODULE (IPU-3)





T-LINK0 (IPU-1)

I/O IPU	PIO NO DEC	PIO NO HEX	D I/O ADDR(WB***)	A I/O ADDR(AB***)
1	0	0	000 , 001, 002 , 003	0000 – 0007 , 0008 – 0015
	1	1	004 , 005, 006 , 007	0016 – 0023 , 0024 – 0031
	2	2	008 , 009, 010 , 011	0032 – 0039 , 0040 – 0047
	3	3	012 , 013, 014 , 015	0048 – 0055 , 0056 – 0063
2	4	4	016 , 017, 018 , 019	0064 – 0071 , 0072 – 0079
	5	5	020 , 021, 022 , 023	0080 – 0087 , 0088 – 0095
	6	6	024 , 025, 026 , 027	0096 – 0103 , 0104 – 0111
	7	7	028 , 029, 030 , 031	0112 – 0119 , 0120 – 0127
3	8	8	032 , 033, 034 , 035	0128 – 0135 , 0136 – 0143
	9	9	036 , 037, 038 , 039	0144 – 0151 , 0152 – 0159
	10	A	040 , 041, 042 , 043	0160 – 0167 , 0168 – 0175
	11	B	044 , 045, 046 , 047	0176 – 0183 , 0184 – 0191
4	12	C	048 , 049, 050 , 051	0192 – 0199 , 0200 – 0207
	13	D	052 , 053, 054 , 055	0208 – 0215 , 0216 – 0223
	14	E	056 , 057, 058 , 059	0224 – 0231 , 0232 – 0239
	15	F	060 , 061, 062 , 063	0240 – 0247 , 0248 – 0255
5	16	10	064 , 065, 066 , 067	0256 – 0263 , 0264 – 0271
	17	11	068 , 069, 070 , 071	0272 – 0279 , 0280 – 0287
	18	12	072 , 073, 074 , 075	0288 – 0295 , 0296 – 0303
	19	13	076 , 077, 078 , 079	0304 – 0311 , 0312 – 0319
6	20	14	080 , 081, 082, 083	0320 – 0327 , 0328 – 0335
	21	15	084 , 085, 086 , 087	0336 – 0343 , 0344 – 0351
	22	16	088 , 089, 090 , 091	0352 – 0359 , 0360 – 0367
	23	17	092 , 093, 094 , 095	0368 – 0375 , 0376 – 0383
7	24	18	096 , 097, 098 , 099	0384 – 0391 , 0392 – 0399
	25	19	100 , 101, 102 , 103	0400 – 0407 , 0408 – 0415
	26	1A	104 , 105, 106 , 107	0416 – 0423 , 0424 – 0431
	27	1B	108 , 109, 110 , 111	0432 – 0439 , 0440 – 0447
8	28	1C	112 , 113, 114 , 115	0448 – 0455 , 0456 – 0463
	29	1D	116 , 117, 118 , 119	0464 – 0471 , 0472 – 0479
	30	1E	120 , 121, 122 , 123	0480 – 0487 , 0488 – 0495
	31	1F	124 , 125, 126 , 127	0496 – 0503 , 0504 – 0511

T-LINK1 (IPU-1)

I/O IPU	PIO NO DEC	PIO NO HEX	D I/O ADDR(WB***)	A I/O ADDR(AB***)
1	32	20	128 , 129, 130 , 131	0512 – 0519 , 0520 – 0527
	33	21	132 , 133, 134 , 135	0528 – 0535 , 0536 – 0543
	34	22	136 , 137, 138 , 139	0544 – 0551 , 0552 – 0559
	35	23	140 , 141, 142 , 143	0560 – 0567 , 0568 – 0575
2	36	24	144 , 145, 146 , 147	0576 – 0583 , 0584 – 0591
	37	25	148 , 149, 150 , 151	0592 – 0599 , 0600 – 0607
	38	26	152 , 153, 154 , 155	0608 – 0615 , 0616 – 0623
	39	27	156 , 157, 158 , 159	0624 – 0631 , 0632 – 0639
3	40	28	160 , 161, 162 , 163	0640 – 0647 , 0648 – 0655
	41	29	164 , 165, 166 , 167	0656 – 0663 , 0664 – 0671
	42	2A	168 , 169, 170 , 171	0672 – 0679 , 0680 – 0687
	43	2B	172 , 173, 174 , 175	0688 – 0695 , 0696 – 0703
4	44	2C	176 , 177, 178 , 179	0704 – 0711 , 0712 – 0719
	45	2D	180 , 181, 182 , 183	0720 – 0727 , 0728 – 0735
	46	2E	184 , 185, 186 , 187	0736 – 0743 , 0744 – 0751
	47	2F	188 , 189, 190 , 191	0752 – 0759 , 0760 – 0767
5	48	30	192 , 193, 194 , 195	0768 – 0775 , 0776 – 0783
	49	31	196 , 197, 198 , 199	0784 – 0791 , 0792 – 0799
	50	32	200 , 201, 202 , 203	0800 – 0807 , 0808 – 0815
	51	33	204 , 205, 206 , 207	0816 – 0823 , 0824 – 0831
6	52	34	208 , 209, 210 , 211	0832 – 0839 , 0840 – 0847
	53	35	212 , 213, 214 , 215	0848 – 0855 , 0856 – 0863
	54	36	216 , 217, 218 , 219	0864 – 0871 , 0872 – 0879
	55	37	220 , 221, 222 , 223	0880 – 0887 , 0888 – 0895
7	56	38	224 , 225, 226 , 227	0896 – 0903 , 0904 – 0911
	57	39	228 , 229, 230 , 231	0912 – 0919 , 0920 – 0927
	58	3A	232 , 233, 234 , 235	0928 – 0935 , 0936 – 0943
	59	3B	236 , 237, 238 , 239	0944 – 0951 , 0952 – 0959
8	60	3C	240 , 241, 242 , 243	0960 – 0967 , 0968 – 0975
	61	3D	244 , 245, 246 , 247	0976 – 0983 , 0984 – 0991
	62	3E	248 , 249, 250 , 251	0992 – 0999 , 1000 – 1007
	63	3F	252 , 253, 254 , 255	1008 – 1015 , 1016 – 1023

T-LINK2 (IPU-1)

I/O IPU	PIO NO DEC	PIO NO HEX	D I/O ADDR(WB***)	A I/O ADDR(AB***)
1	64	40	256 , 257, 258 , 259	1024 – 1031 , 1032 – 1039
	65	41	260 , 261, 262 , 263	1040 – 1047 , 1048 – 1055
	66	42	264 , 265, 266 , 267	1056 – 1063 , 1064 – 1071
	67	43	268 , 269, 270 , 271	1072 – 1079 , 1080 – 1087
2	68	44	272 , 273, 274 , 275	1088 – 1095 , 1096 – 1103
	69	45	276 , 277, 278 , 279	1104 – 1111 , 1112 – 1119
	70	46	280 , 281, 282 , 283	1120 – 1127 , 1128 – 1135
	71	47	284 , 285, 286 , 287	1136 – 1143 , 1144 – 1151
3	72	48	288 , 289, 290 , 291	1152 – 1159 , 1160 – 1167
	73	49	292 , 293, 294 , 295	1168 – 1175 , 1176 – 1183
	74	4A	296 , 297, 298 , 299	1184 – 1191 , 1192 – 1199
	75	4B	300 , 301, 302 , 303	1200 – 1207 , 1208 – 1215
4	76	4C	304 , 305, 306 , 307	1216 – 1223 , 1224 – 1231
	77	4D	308 , 309, 310 , 311	1232 – 1239 , 1240 – 1247
	78	4E	312 , 313, 314 , 315	1248 – 1255 , 1256 – 1263
	79	4F	316 , 317, 318 , 319	1264 – 1271 , 1272 – 1279
5	80	50	320 , 321, 322 , 323	1280 – 1287 , 1288 – 1295
	81	51	324 , 325, 326 , 327	1296 – 1303 , 1304 – 1311
	82	52	328 , 329, 330 , 331	1312 – 1319 , 1320 – 1327
	83	53	332 , 333, 334 , 335	1328 – 1335 , 1336 – 1343
6	84	54	336 , 337, 338 , 339	1344 – 1351 , 1352 – 1359
	85	55	340 , 341, 342 , 343	1360 – 1367 , 1368 – 1375
	86	56	344 , 345, 346 , 347	1376 – 1383 , 1384 – 1391
	87	57	348 , 349, 350 , 351	1392 – 1399 , 1400 – 1407
7	88	58	352 , 353, 354 , 355	1408 – 1415 , 1416 – 1423
	89	59	356 , 357, 358 , 359	1424 – 1431 , 1432 – 1439
	90	5A	360 , 361, 362 , 363	1440 – 1447 , 1448 – 1455
	91	5B	364 , 365, 366 , 367	1456 – 1463 , 1464 – 1471
8	92	5C	368 , 369, 370 , 371	1472 – 1479 , 1480 – 1487
	93	5D	372 , 373, 374 , 375	1488 – 1495 , 1496 – 1503
	94	5E	376 , 377, 378 , 379	1504 – 1511 , 1512 – 1519
	95	5F	380 , 381, 382 , 383	1520 – 1527 , 1528 – 1535

T-LINK3 (IPU-1)

I/O IPU	PIO NO DEC	PIO NO HEX	D I/O ADDR(WB***)	A I/O ADDR(AB***)
1	96	60	384 , 385, 386 , 387	1536 – 1543 , 1544 – 1551
	97	61	388 , 389, 390 , 391	1552 – 1559 , 1560 – 1567
	98	62	392 , 393, 394 , 395	1568 – 1575 , 1576 – 1583
	99	63	396 , 397, 398 , 399	1584 – 1591 , 1592 – 1599
2	100	64	400 , 401, 402 , 403	1600 – 1607 , 1608 – 1615
	101	65	404 , 405, 406 , 407	1616 – 1623 , 1624 – 1631
	102	66	408 , 409, 410 , 411	1632 – 1639 , 1640 – 1647
	103	67	412 , 413, 414 , 415	1648 – 1655 , 1656 – 1663
3	104	68	416 , 417, 418 , 419	1664 – 1671 , 1672 – 1679
	105	69	420 , 421, 422 , 423	1680 – 1687 , 1688 – 1695
	106	6A	424 , 425, 426 , 427	1696 – 1703 , 1704 – 1711
	107	6B	428 , 429, 430 , 431	1712 – 1719 , 1720 – 1727
4	108	6C	432 , 433, 434 , 435	1728 – 1735 , 1736 – 1743
	109	6D	436 , 437, 438 , 439	1744 – 1751 , 1752 – 1759
	110	6E	440 , 441, 442 , 443	1760 – 1767 , 1768 – 1775
	111	6F	444 , 445, 446 , 447	1776 – 1783 , 1784 – 1791
5	112	70	448 , 449, 450 , 451	1792 – 1799 , 1800 – 1807
	113	71	452 , 453, 454 , 455	1808 – 1815 , 1816 – 1823
	114	72	456 , 457, 458 , 459	1824 – 1831 , 1832 – 1839
	115	73	460 , 461, 462 , 463	1840 – 1847 , 1848 – 1855
6	116	74	464 , 465, 466 , 467	1856 – 1863 , 1864 – 1871
	117	75	468 , 469, 470 , 471	1872 – 1879 , 1880 – 1887
	118	76	472 , 473, 474 , 475	1888 – 1895 , 1896 – 1903
	119	77	476 , 477, 478 , 479	1904 – 1911 , 1912 – 1919
7	120	78	480 , 481, 482 , 483	1920 – 1927 , 1928 – 1935
	121	79	484 , 485, 486 , 487	1936 – 1943 , 1944 – 1951
	122	7A	488 , 489, 490 , 491	1952 – 1959 , 1960 – 1967
	123	7B	492 , 493, 494 , 495	1968 – 1975 , 1976 – 1983
8	124	7C	496 , 497, 498 , 499	1984 – 1991 , 1992 – 1999
	125	7D	500 , 501, 502 , 503	2000 – 2007 , 2008 – 2015
	126	7E	504 , 505, 506 , 507	2016 – 2023 , 2024 – 2031
	127	7F	508 , 509, 510 , 511	2032 – 2039 , 2040 – 2047

EPC0 (IPU-2)

NODE	IOM	PIO NO DEC	PIO NO HEX	D I/O ADDR(WB***)	A I/O ADDR(AB***)
0	0	0	0	000 , 001, 002 , 003	0000 – 0007 , 0008 – 0015
	1	1	1	004 , 005, 006 , 007	0016 – 0023 , 0024 – 0031
	2	2	2	008 , 009, 010 , 011	0032 – 0039 , 0040 – 0047
	3	3	3	012 , 013, 014 , 015	0048 – 0055 , 0056 – 0063
	4	4	4	016 , 017, 018 , 019	0064 – 0071 , 0072 – 0079
	5	5	5	020 , 021, 022 , 023	0080 – 0087 , 0088 – 0095
	6	6	6	024 , 025, 026 , 027	0096 – 0103 , 0104 – 0111
	7	7	7	028 , 029, 030 , 031	0112 – 0119 , 0120 – 0127
1	0	8	8	032 , 033, 034 , 035	0128 – 0135 , 0136 – 0143
	1	9	9	036 , 037, 038 , 039	0144 – 0151 , 0152 – 0159
	2	A	10	040 , 041, 042 , 043	0160 – 0167 , 0168 – 0175
	3	B	11	044 , 045, 046 , 047	0176 – 0183 , 0184 – 0191
	4	C	12	048 , 049, 050 , 051	0192 – 0199 , 0200 – 0207
	5	D	13	052 , 053, 054 , 055	0208 – 0215 , 0216 – 0223
	6	E	14	056 , 057, 058 , 059	0224 – 0231 , 0232 – 0239
	7	F	15	060 , 061, 062 , 063	0240 – 0247 , 0248 – 0255
2	0	10	16	064 , 065, 066 , 067	0256 – 0263 , 0264 – 0271
	1	11	17	068 , 069, 070 , 071	0272 – 0279 , 0280 – 0287
	2	12	18	072 , 073, 074 , 075	0288 – 0295 , 0296 – 0303
	3	13	19	076 , 077, 078 , 079	0304 – 0311 , 0312 – 0319
	4	14	20	080 , 081, 082, 083	0320 – 0327 , 0328 – 0335
	5	15	21	084 , 085, 086 , 087	0336 – 0343 , 0344 – 0351
	6	16	22	088 , 089, 090 , 091	0352 – 0359 , 0360 – 0367
	7	17	23	092 , 093, 094 , 095	0368 – 0375 , 0376 – 0383
3	0	18	24	096 , 097, 098 , 099	0384 – 0391 , 0392 – 0399
	1	19	25	100 , 101, 102 , 103	0400 – 0407 , 0408 – 0415
	2	1A	26	104 , 105, 106 , 107	0416 – 0423 , 0424 – 0431
	3	1B	27	108 , 109, 110 , 111	0432 – 0439 , 0440 – 0447
	4	1C	28	112 , 113, 114 , 115	0448 – 0455 , 0456 – 0463
	5	1D	29	116 , 117, 118 , 119	0464 – 0471 , 0472 – 0479
	6	1E	30	120 , 121, 122 , 123	0480 – 0487 , 0488 – 0495
	7	1F	31	124 , 125, 126 , 127	0496 – 0503 , 0504 – 0511

EPC0 (IPU-2)

NODE	IOM	PIO NO DEC	PIO NO HEX	D I/O ADDR(WB***)	A I/O ADDR(AB***)
4	0	20	32	128 , 129, 130 , 131	0512 – 0519 , 0520 – 0527
	1	21	33	132 , 133, 134 , 135	0528 – 0535 , 0536 – 0543
	2	22	34	136 , 137, 138 , 139	0544 – 0551 , 0552 – 0559
	3	23	35	140 , 141, 142 , 143	0560 – 0567 , 0568 – 0575
	4	24	36	144 , 145, 146 , 147	0576 – 0583 , 0584 – 0591
	5	25	37	148 , 149, 150 , 151	0592 – 0599 , 0600 – 0607
	6	26	38	152 , 153, 154 , 155	0608 – 0615 , 0616 – 0623
	7	27	39	156 , 157, 158 , 159	0624 – 0631 , 0632 – 0639
5	0	28	40	160 , 161, 162 , 163	0640 – 0647 , 0648 – 0655
	1	29	41	164 , 165, 166 , 167	0656 – 0663 , 0664 – 0671
	2	2A	42	168 , 169, 170 , 171	0672 – 0679 , 0680 – 0687
	3	2B	43	172 , 173, 174 , 175	0688 – 0695 , 0696 – 0703
	4	2C	44	176 , 177, 178 , 179	0704 – 0711 , 0712 – 0719
	5	2D	45	180 , 181, 182 , 183	0720 – 0727 , 0728 – 0735
	6	2E	46	184 , 185, 186 , 187	0736 – 0743 , 0744 – 0751
	7	2F	47	188 , 189, 190 , 191	0752 – 0759 , 0760 – 0767
6	0	30	48	192 , 193, 194 , 195	0768 – 0775 , 0776 – 0783
	1	31	49	196 , 197, 198 , 199	0784 – 0791 , 0792 – 0799
	2	32	50	200 , 201, 202 , 203	0800 – 0807 , 0808 – 0815
	3	33	51	204 , 205, 206 , 207	0816 – 0823 , 0824 – 0831
	4	34	52	208 , 209, 210 , 211	0832 – 0839 , 0840 – 0847
	5	35	53	212 , 213, 214 , 215	0848 – 0855 , 0856 – 0863
	6	36	54	216 , 217, 218 , 219	0864 – 0871 , 0872 – 0879
	7	37	55	220 , 221, 222 , 223	0880 – 0887 , 0888 – 0895
7	0	38	56	224 , 225, 226 , 227	0896 – 0903 , 0904 – 0911
	1	39	57	228 , 229, 230 , 231	0912 – 0919 , 0920 – 0927
	2	3A	58	232 , 233, 234 , 235	0928 – 0935 , 0936 – 0943
	3	3B	59	236 , 237, 238 , 239	0944 – 0951 , 0952 – 0959
	4	3C	60	240 , 241, 242 , 243	0960 – 0967 , 0968 – 0975
	5	3D	61	244 , 245, 246 , 247	0976 – 0983 , 0984 – 0991
	6	3E	62	248 , 249, 250 , 251	0992 – 0999 , 1000 – 1007
	7	3F	63	252 , 253, 254 , 255	1008 – 1015 , 1016 – 1023

EPC1 (IPU-2)

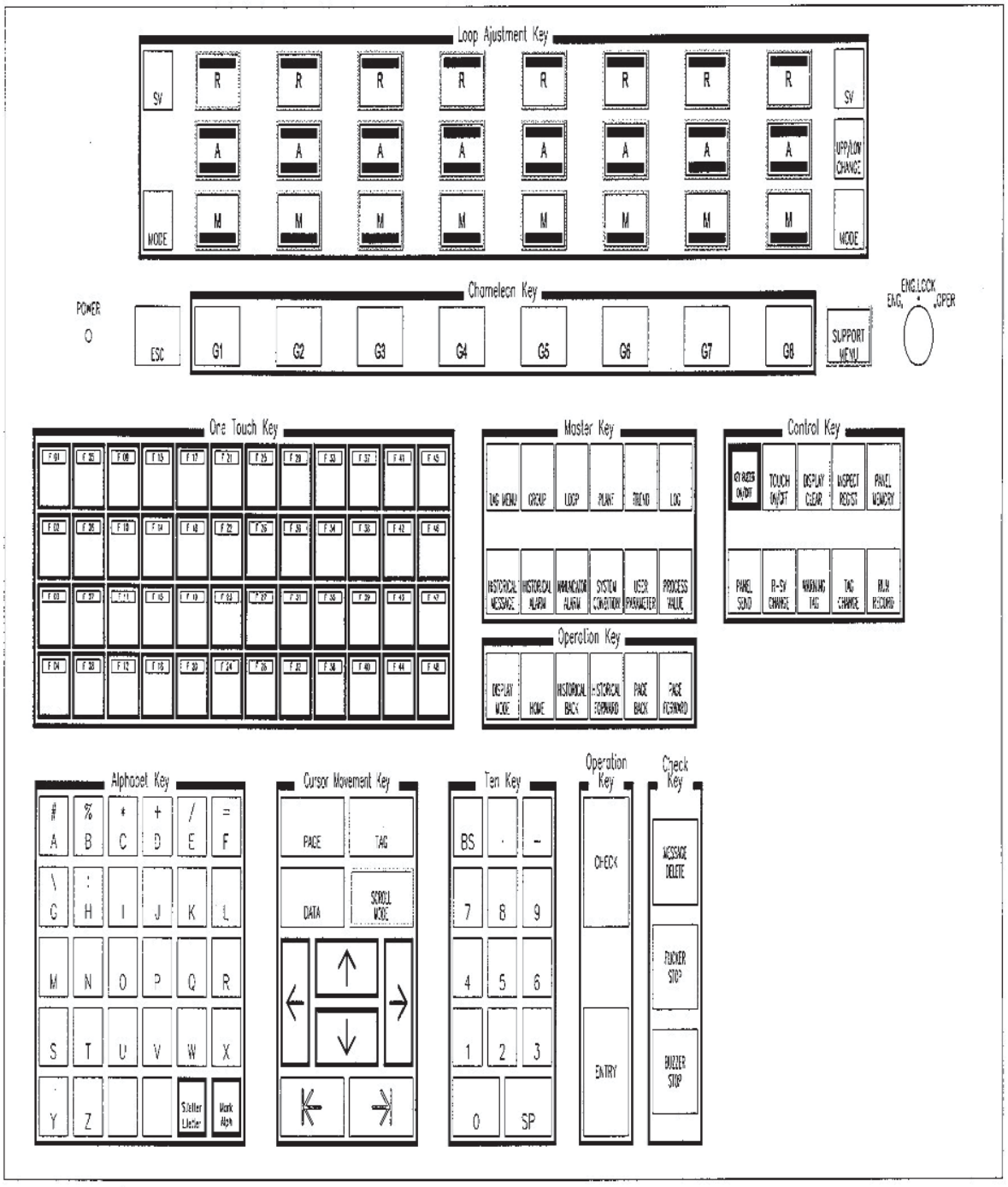
NODE	IOM	PIO NO DEC	PIO NO HEX	D I/O ADDR(WB***)	A I/O ADDR(AB***)
0	0	40	64	256 , 257, 258 , 259	1024 – 1031 , 1032 – 1039
	1	41	65	260 , 261, 262 , 263	1040 – 1047 , 1048 – 1055
	2	42	66	264 , 265, 266 , 267	1056 – 1063 , 1064 – 1071
	3	43	67	268 , 269, 270 , 271	1072 – 1079 , 1080 – 1087
	4	44	68	272 , 273, 274 , 275	1088 – 1095 , 1096 – 1103
	5	45	69	276 , 277, 278 , 279	1104 – 1111 , 1112 – 1119
	6	46	70	280 , 281, 282 , 283	1120 – 1127 , 1128 – 1135
	7	47	71	284 , 285, 286 , 287	1136 – 1143 , 1144 – 1151
1	0	48	72	288 , 289, 290 , 291	1152 – 1159 , 1160 – 1167
	1	49	73	292 , 293, 294 , 295	1168 – 1175 , 1176 – 1183
	2	4A	74	296 , 297, 298 , 299	1184 – 1191 , 1192 – 1199
	3	4B	75	300 , 301, 302 , 303	1200 – 1207 , 1208 – 1215
	4	4C	76	304 , 305, 306 , 307	1216 – 1223 , 1224 – 1231
	5	4D	77	308 , 309, 310 , 311	1232 – 1239 , 1240 – 1247
	6	4E	78	312 , 313, 314 , 315	1248 – 1255 , 1256 – 1263
	7	4F	79	316 , 317, 318 , 319	1264 – 1271 , 1272 – 1279
2	0	50	80	320 , 321, 322 , 323	1280 – 1287 , 1288 – 1295
	1	51	81	324 , 325, 326 , 327	1296 – 1303 , 1304 – 1311
	2	52	82	328 , 329, 330 , 331	1312 – 1319 , 1320 – 1327
	3	53	83	332 , 333, 334 , 335	1328 – 1335 , 1336 – 1343
	4	54	84	336 , 337, 338 , 339	1344 – 1351 , 1352 – 1359
	5	55	85	340 , 341, 342 , 343	1360 – 1367 , 1368 – 1375
	6	56	86	344 , 345, 346 , 347	1376 – 1383 , 1384 – 1391
	7	57	87	348 , 349, 350 , 351	1392 – 1399 , 1400 – 1407
3	0	58	88	352 , 353, 354 , 355	1408 – 1415 , 1416 – 1423
	1	59	89	356 , 357, 358 , 359	1424 – 1431 , 1432 – 1439
	2	5A	90	360 , 361, 362 , 363	1440 – 1447 , 1448 – 1455
	3	5B	91	364 , 365, 366 , 367	1456 – 1463 , 1464 – 1471
	4	5C	92	368 , 369, 370 , 371	1472 – 1479 , 1480 – 1487
	5	5D	93	372 , 373, 374 , 375	1488 – 1495 , 1496 – 1503
	6	5E	94	376 , 377, 378 , 379	1504 – 1511 , 1512 – 1519
	7	5F	95	380 , 381, 382 , 383	1520 – 1527 , 1528 – 1535

EPC1 (IPU-2)

NODE	IOM	PIO NO DEC	PIO NO HEX	D I/O ADDR(WB***)	A I/O ADDR(AB***)
4	0	60	96	384 , 385, 386 , 387	1536 – 1543 , 1544 – 1551
	1	61	97	388 , 389, 390 , 391	1552 – 1559 , 1560 – 1567
	2	62	98	392 , 393, 394 , 395	1568 – 1575 , 1576 – 1583
	3	63	99	396 , 397, 398 , 399	1584 – 1591 , 1592 – 1599
	4	64	100	400 , 401, 402 , 403	1600 – 1607 , 1608 – 1615
	5	65	101	404 , 405, 406 , 407	1616 – 1623 , 1624 – 1631
	6	66	102	408 , 409, 410 , 411	1632 – 1639 , 1640 – 1647
	7	67	103	412 , 413, 414 , 415	1648 – 1655 , 1656 – 1663
5	0	68	104	416 , 417, 418 , 419	1664 – 1671 , 1672 – 1679
	1	69	105	420 , 421, 422 , 423	1680 – 1687 , 1688 – 1695
	2	6A	106	424 , 425, 426 , 427	1696 – 1703 , 1704 – 1711
	3	6B	107	428 , 429, 430 , 431	1712 – 1719 , 1720 – 1727
	4	6C	108	432 , 433, 434 , 435	1728 – 1735 , 1736 – 1743
	5	6D	109	436 , 437, 438 , 439	1744 – 1751 , 1752 – 1759
	6	6E	110	440 , 441, 442 , 443	1760 – 1767 , 1768 – 1775
	7	6F	111	444 , 445, 446 , 447	1776 – 1783 , 1784 – 1791
6	0	70	112	448 , 449, 450 , 451	1792 – 1799 , 1800 – 1807
	1	71	113	452 , 453, 454 , 455	1808 – 1815 , 1816 – 1823
	2	72	114	456 , 457, 458 , 459	1824 – 1831 , 1832 – 1839
	3	73	115	460 , 461, 462 , 463	1840 – 1847 , 1848 – 1855
	4	74	116	464 , 465, 466 , 467	1856 – 1863 , 1864 – 1871
	5	75	117	468 , 469, 470 , 471	1872 – 1879 , 1880 – 1887
	6	76	118	472 , 473, 474 , 475	1888 – 1895 , 1896 – 1903
	7	77	119	476 , 477, 478 , 479	1904 – 1911 , 1912 – 1919
7	0	78	120	480 , 481, 482 , 483	1920 – 1927 , 1928 – 1935
	1	79	121	484 , 485, 486 , 487	1936 – 1943 , 1944 – 1951
	2	7A	122	488 , 489, 490 , 491	1952 – 1959 , 1960 – 1967
	3	7B	123	492 , 493, 494 , 495	1968 – 1975 , 1976 – 1983
	4	7C	124	496 , 497, 498 , 499	1984 – 1991 , 1992 – 1999
	5	7D	125	500 , 501, 502 , 503	2000 – 2007 , 2008 – 2015
	6	7E	126	504 , 505, 506 , 507	2016 – 2023 , 2024 – 2031
	7	7F	127	508 , 509, 510 , 511	2032 – 2039 , 2040 – 2047

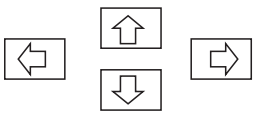
FCS/ACS MEMORY

Item	TAGNO	Type	Bit		Word		Remarks	
			Input	Out-put	Input	Out-put		
Input/output	I. xxx# Bit I/O address (0000 to 639F)		O				DI	
	O. xxx# Bit I/O address (0000 to 639F)		O	O			DO	
	B. xxx# Bit I/O address (0000 to 639F)		O	O			DI or DO	
	AI xxxx Word I/O address (0000 to 2559)				O		AI	
	AO xxxx Word I/O address (0000 to 2559)				O	O	AO	
	AB xxxx Word I/O address (0000 to 2559)				O	O	AI or AO	
	WI xxx Bit I/O address (000 to 639)				O		DI (for word access)	
	WO xxx Bit I/O address (000 to 639)				O	O	DO (for word access)	
	WB xxx Bit I/O address (000 to 639)				O	O	DI or DO (for word access)	
Memory	M. xxx# Relay NO. (0000 to 511F)		O	O			Auxiliary relay (for bit access)	
	WM xxx Relay NO. (000 to 511)				O	O	Auxiliary relay (for word access)	
	FLAG□□□□ Memory NO. (0001 to 1024)		O	O			Internal bit memory	
	DATA□□□□ Memory NO. (0001to 1024)				O	O	Internal word memory	
	MMI□□□ Memory NO. (001 to 128)				O	O	MMI terminal	
Keep relay	K. xxx# Relay NO.(0000 to 063F)		O	O			Keep relay (for bit access)	
	WK. xxx Relay NO.(0000 to 063)					O	Keep relay (for word access)	
Timer	TS xxxx (0000 to 1023)				O	O	Timer set value	
	TR xxxx (0000 to 1023)				O	O	Timer current value	
Counter	CS xxx (000 to 127)				O	O	Counter set value	
	CR xxx (0000 to 127)				O	O	Counter current value	
Analog calculation	AM xxx (0000 to 511)				O	O	Analog calculation memory	
SI memory	SI xxxx **Controller 9999** Memory NO.(0000 to 8191)				O	O	SI terminal	
DI memory	DI xxxx Memory NO. (0000 to 8191) **Controller 9999**				O	O	DI area (Low-order word)	Low-order word
		W01			O	O	DI area (High-order word)	High-order word
FP memory	FP xxxx Memory NO. (0000 to **4095**)				O	O	Floating decimal point	


Loop Ajustment Key
SV
R
A
M
MODE
UP/DOWN CHANGE
Chameleon Key
POWER
ESC
G1
G2
G3
G4
G5
G6
G7
G8
SUPPORT MENU
ENG.LOCK
ENG.
OPER
One Touch Key
Master Key
TAG MENU
GROUP
LOOP
PLANT
TREND
LOG
HISTORICAL MESSAGE
HISTORICAL ALARM
WARNINGTION ALARM
SYSTEM CONDITION
USER PARAMETER
PROCESS VALUE
Control Key
KEY BUZZER ON/OFF
TOUCH ON/OFF
DISPLAY CLEAR
INSPECT RESET
PANEL MEMORY
PANEL SEND
R-SV CHANGE
WARNING TAG
TAG CHANGE
MAN RECORD
Operation Key
DISPLAY MODE
HOME
HISTORICAL BACK
HISTORICAL FORWARD
PAGE BACK
PAGE FORWARD
Alphaset Key
Stelter Lletter
Mark Alph
Cursor Movement Key
PAGE
TAG
DATA
SCROLL MODE
Ten Key
BS
SP
Operation Key
CHECK
ENTRY
Check Key
MESSAGE DELETE
FLICKER STOP
BUZZER STOP

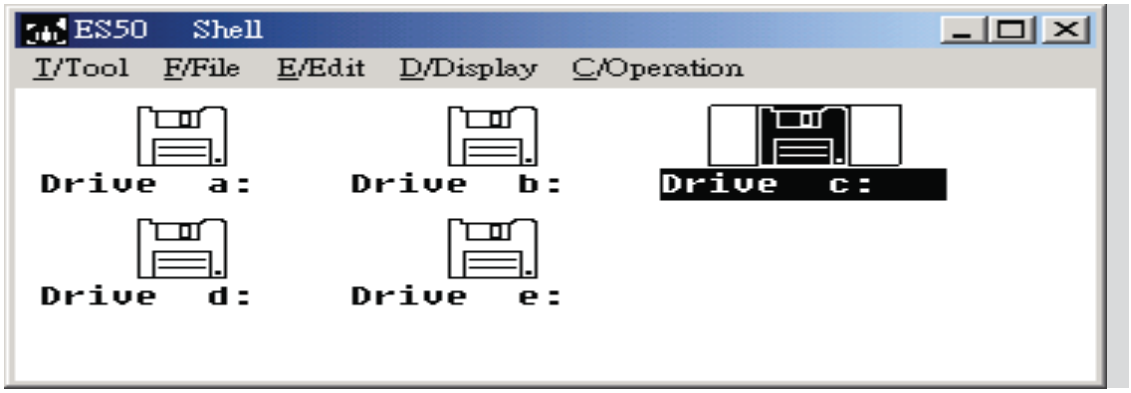
NO	KEY	DESCRIPTION
1	TAG MENU	LOOP TAG MENU
2	GROUP	LOOP GROUP
3	LOOP	LOOP PARAMETER SET
4	PLANT	PLANT PANEL DISPLAY
5	TREND	TREND PANEL DISPLAY
6	LOGING	LOG PANEL DISPLAY
7	HISTORICAL MESSAGE	Operated the panel of record and history message of record
8	HISTORICAL ALARM	ALL DCS HISTORY ALARM OF RECORD
9	ANNUNCIATOR ALARM	ANNUNCIATOR ALARN DISPLAY PANEL
10	SYSTEM CONDITION	DCS SYSTEM CONDITION DISPLAY
11	USER PARAMETER	USER CAN SETUP PARAMETER PANEL
12	DISPLAY MODE	DISPLAY MODE
13	HOME	RETURN TO MENU DISPLAY

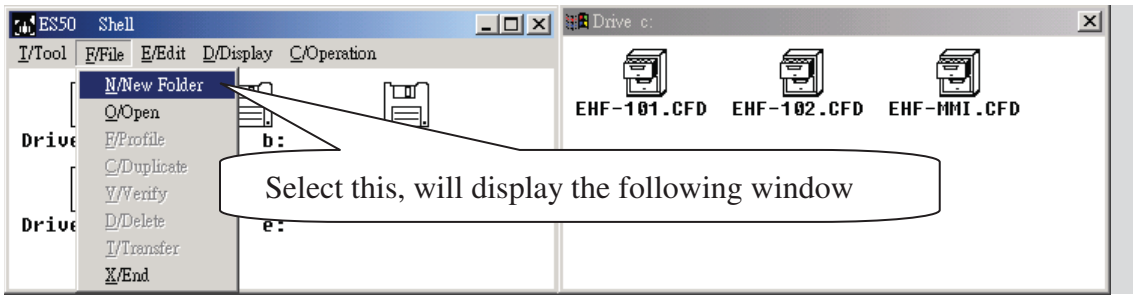
NO	KEY	DESCRIPTION
14	HISTORICAL BACK	RETURN TO HISTORY OPERATING PAGE
15	HISTORICAL FORWARD	HISTORY OPERATING PAGE FORWORD
16	PAGE BACK	RETURN TO OPERATING PAGE
17	PAGE FORWARD	OPERATING PAGE FORWARD
18	PANEL SEND	PANEL SEND TO OTHER OPERATION PANEL
19	R-SV CHANGE	R-SV DISPLAY/CLEAR
20	WARNING TAG	LOOP GROUP PANEL WARNNING TAG SETUP
21	KEY BUZZER ON/OFF	KEY BUZZER ON OR OFF
22	TOUCH ON/OFF	TOUCH SCREEN FUNCTION ON/OFF
23	CRT CHANGE	CHANGE CRT PANEL
24	PANEL MEMORY	PANEL MEMORY FUNCTION NEED PUSH ONE TOUCH KEY
25	SUPPORT MENU	SUPPORT FUNCTION MENU
26	R A M	LOOP OPARATING MODE SETUP R:REMOTE A:AUTO M:MANUAL

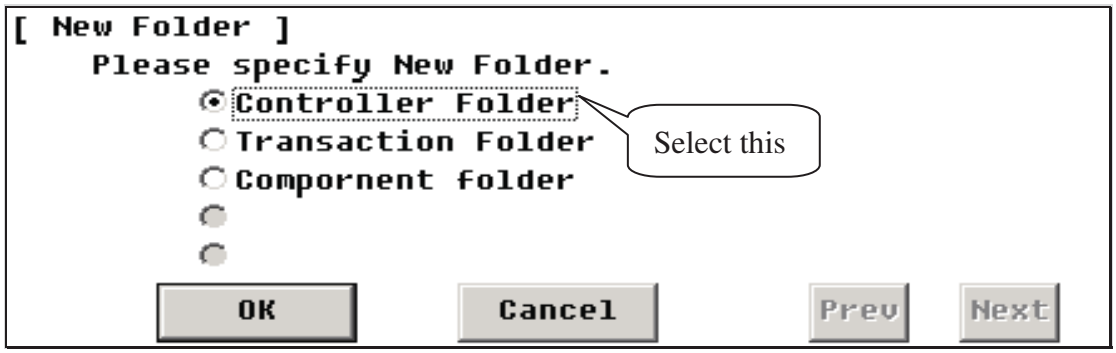
NO	KEY	DESCRIPTION
27	MODE	MODE CHANGE
28	UP/DOWN CHANGE	GROUP PANEL UP/DOWN CHANGE
29	ESC	ESCAPE KEY
30	MARK ALPHABET	SPECIAL SYMBOL / THE ENGLISH LETTER CHANGES THE KEY
31	S-letter M-letter	THE ENGLISH LETTER IS CAPITALIZED /SMALL LETTER CHANGES THE KEY
32	A,B,C..Z	THE ENGLISH LETTER
33	#,%,*,+,-...	SPECIAL SYMBOL
34	PAGE	PAGE INPUT
35	TAG	TAG INPUT
36	DATA	PARAMETER DATA SET
37	SCROLL MODE	SCROLL MODE
38		DIRECTION KEY
39		TAB KEY

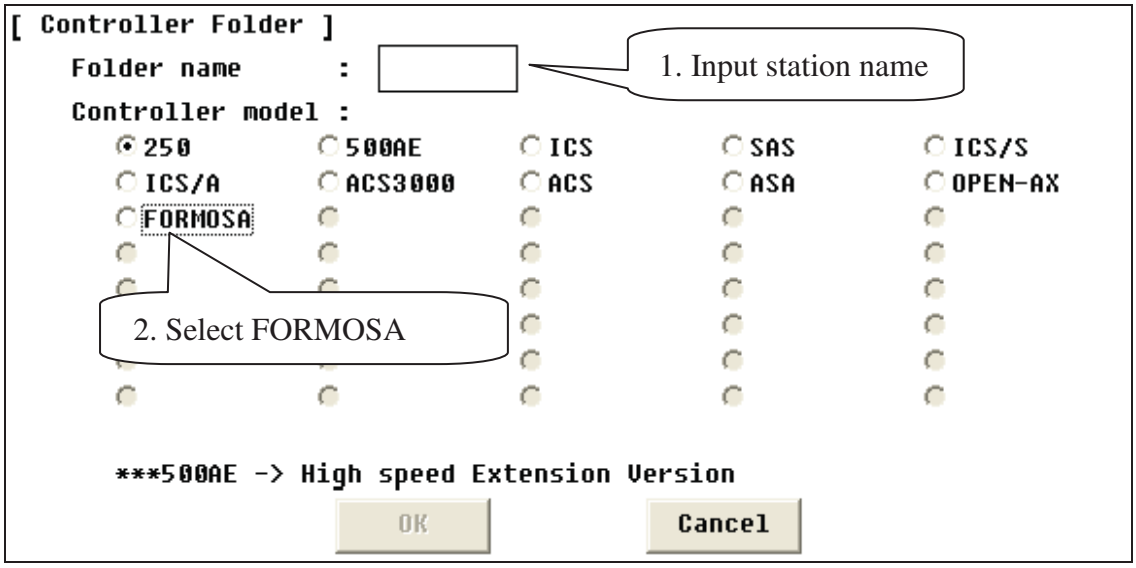
NO	KEY	DESCRIPTION
40	BS	BACKSPACE
41	SP	SPACE
42	0,1,2,3,4,5..9	NUMERIC
43	CHECK	INPUT DATA CHECK
44	ENTRY	ENTRY
45	MESSAGE DELETE	MESSAGE DELETE
46	FLICKER STOP	FLICK MESSAGE STOP
47	BUZZER STOP	BUZZER STOP KEY
48	TAG CHANGE	TAG NO.AND TAG COMMENT CHANGE
49		
50		
51		
52		

HOW TO OPEN A CFD FOLDER: CFD folder have two type
(1)Control station , (2)MMI
(1)How to open a Control station:First open the C drive(or D drive)

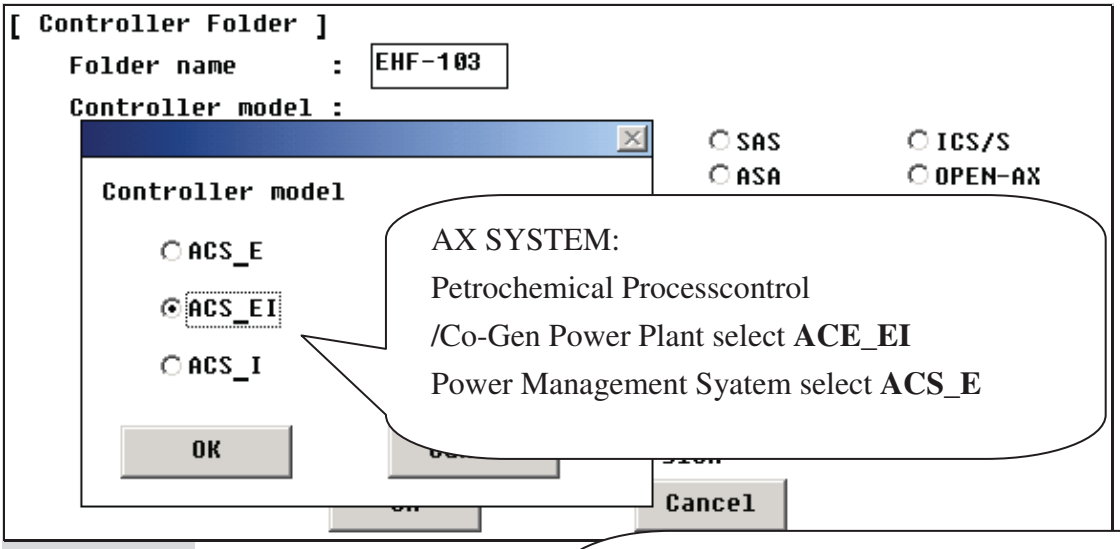




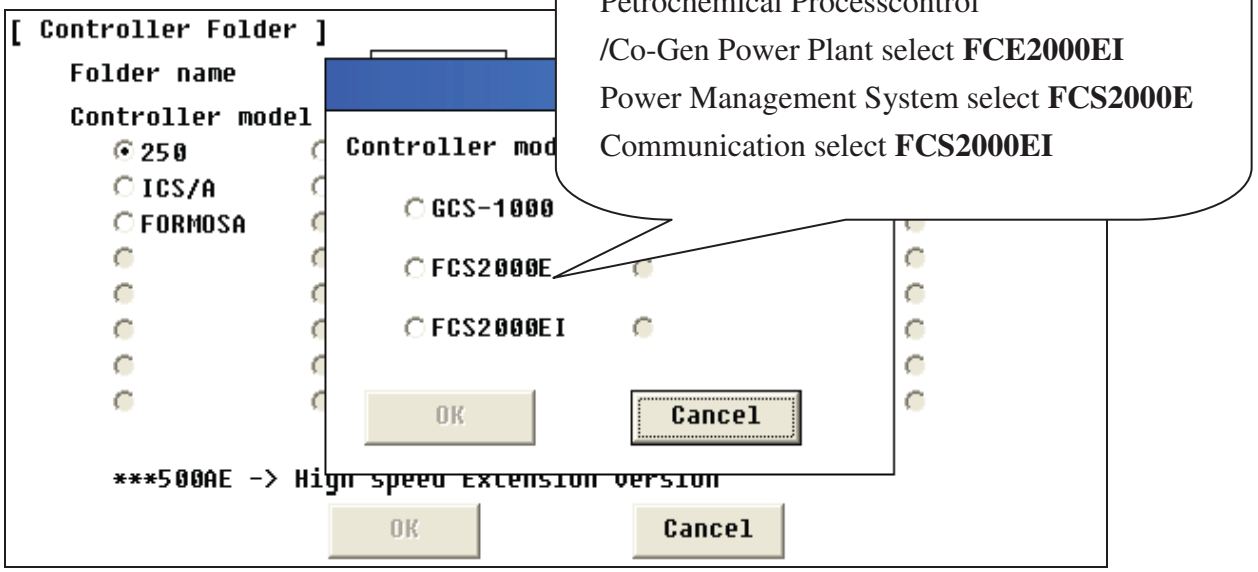


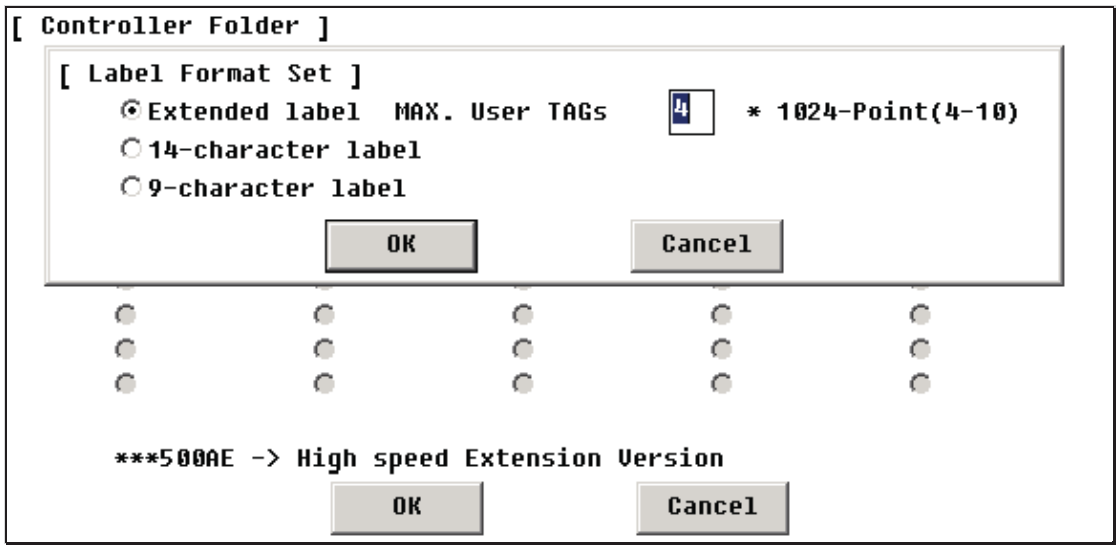


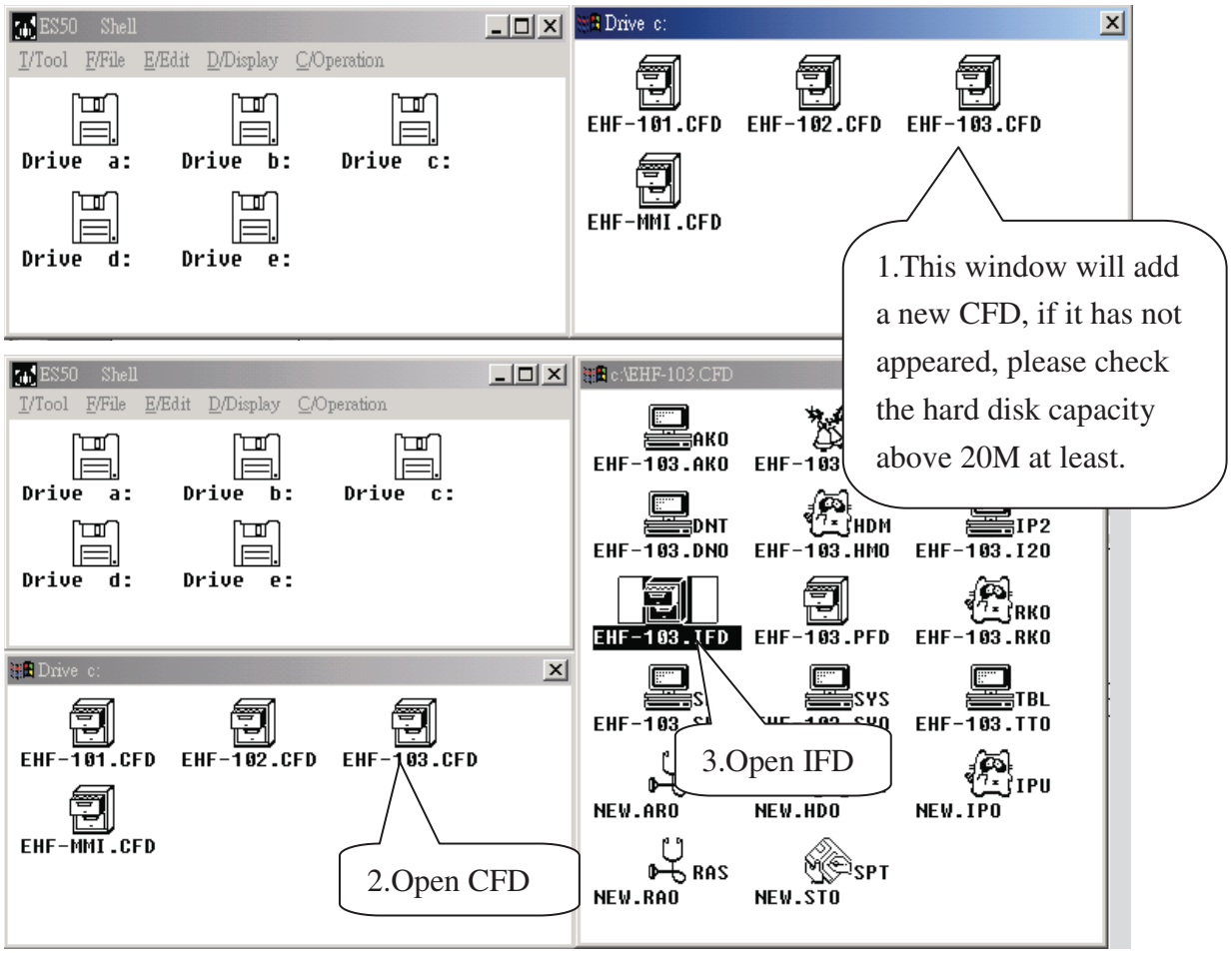
AX SYSTEM

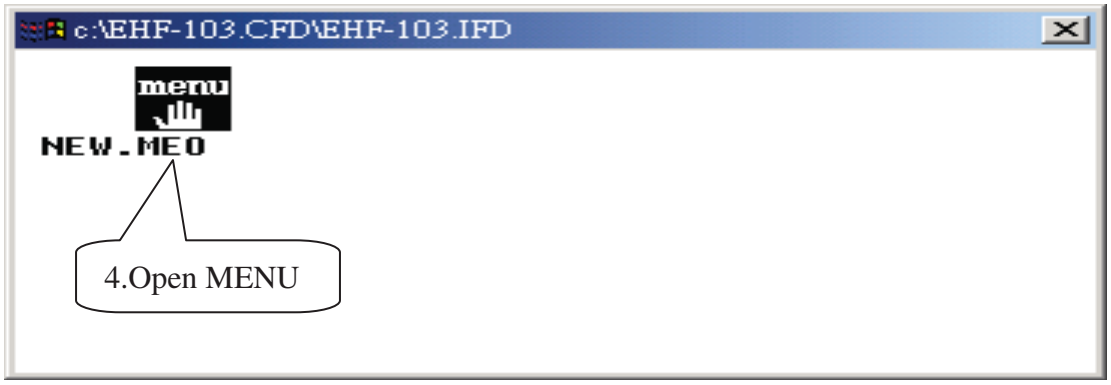


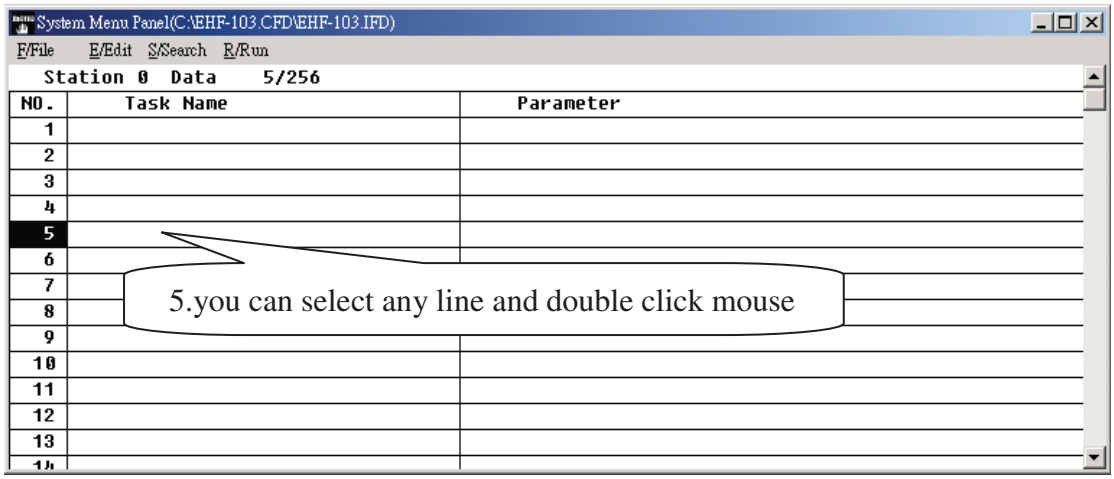
CX SYSTEM



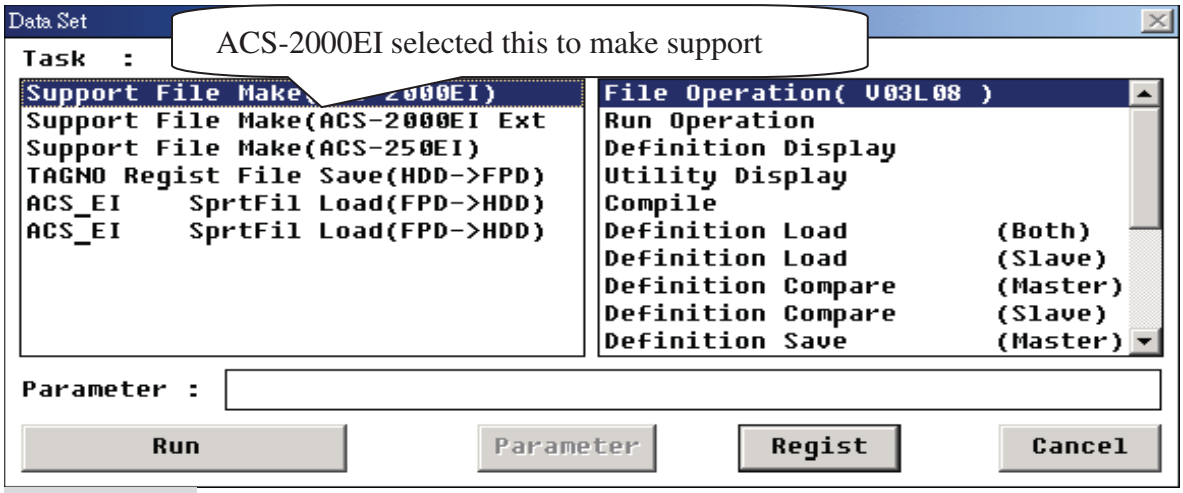



ES50 Shell
T/Tool F/File E/Edit D/Display C/Operation
Drive a:
Drive b:
Drive c:
Drive d:
Drive e:
Drive c:
EHF-101.CFD
EHF-102.CFD
EHF-103.CFD
EHF-MMI.CFD
1.This window will add a new CFD, if it has not appeared, please check the hard disk capacity above 20M at least.
c:\EHF-103.CFD
EHF-103.AKO
EHF-103.DNO
EHF-103.HMO
EHF-103.I20
EHF-103.IFD
EHF-103.PFD
EHF-103.RKO
EHF-103.TTO
EHF-103.IPO
NEW.ARO
NEW.HDO
NEW.RAO
NEW.STO
3.Open IFD
2.Open CFD

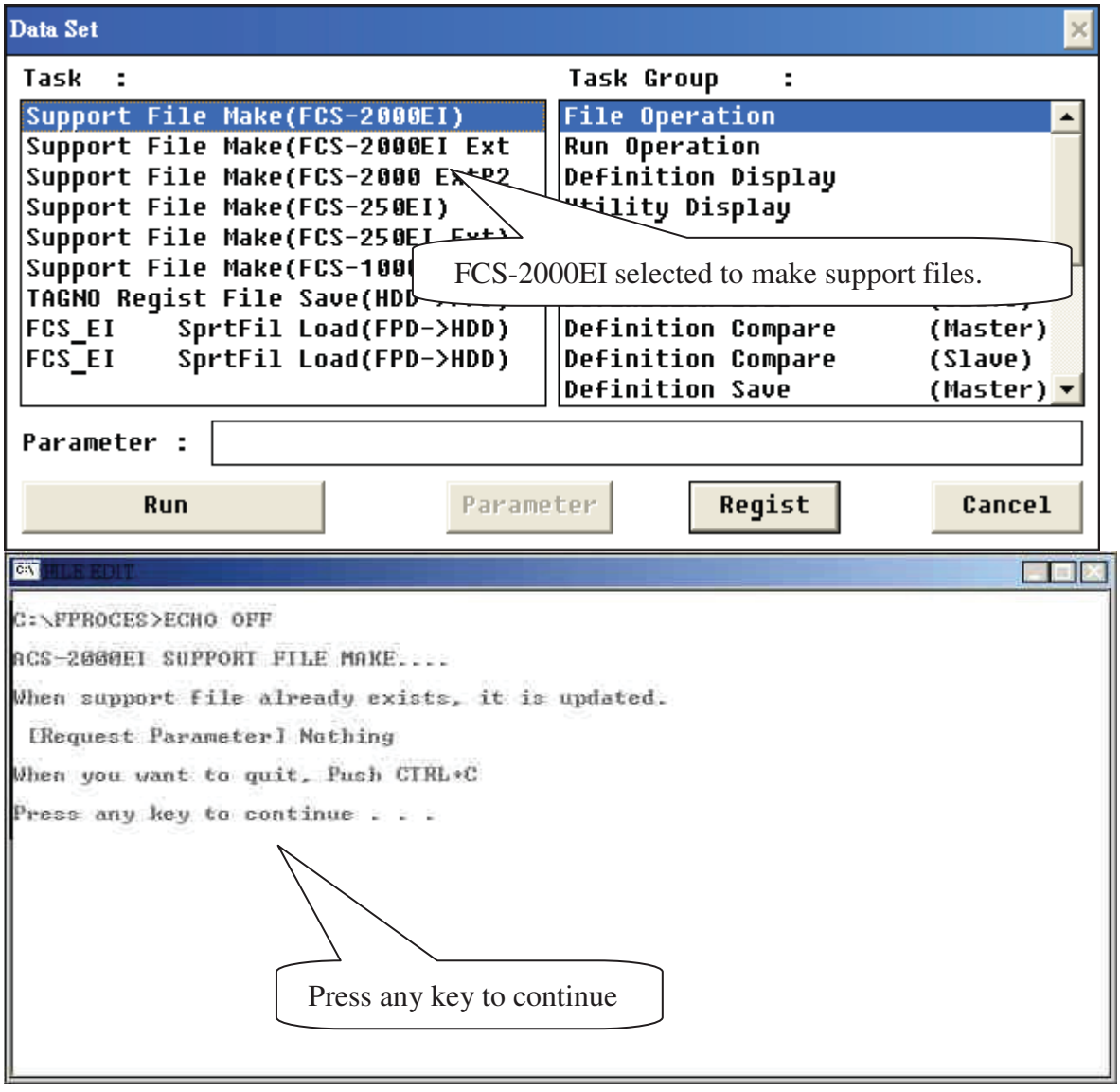

c:\EHF-103.CFD\EHF-103.IFD
menu
NEW.MEO
4.Open MENU

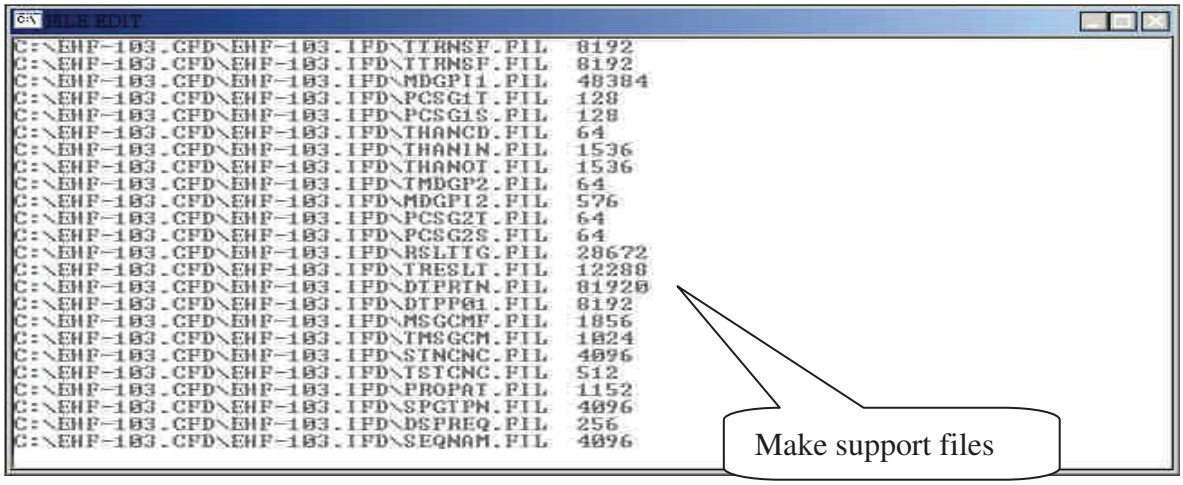

System Menu Panel(C:\EHF-103.CFD\EHF-103.IFD)
F/File E/Edit S/Search R/Run
Station 0 Data 5/256
NO.
Task Name
Parameter
5.you can select any line and double click mouse

AX SYSTEM

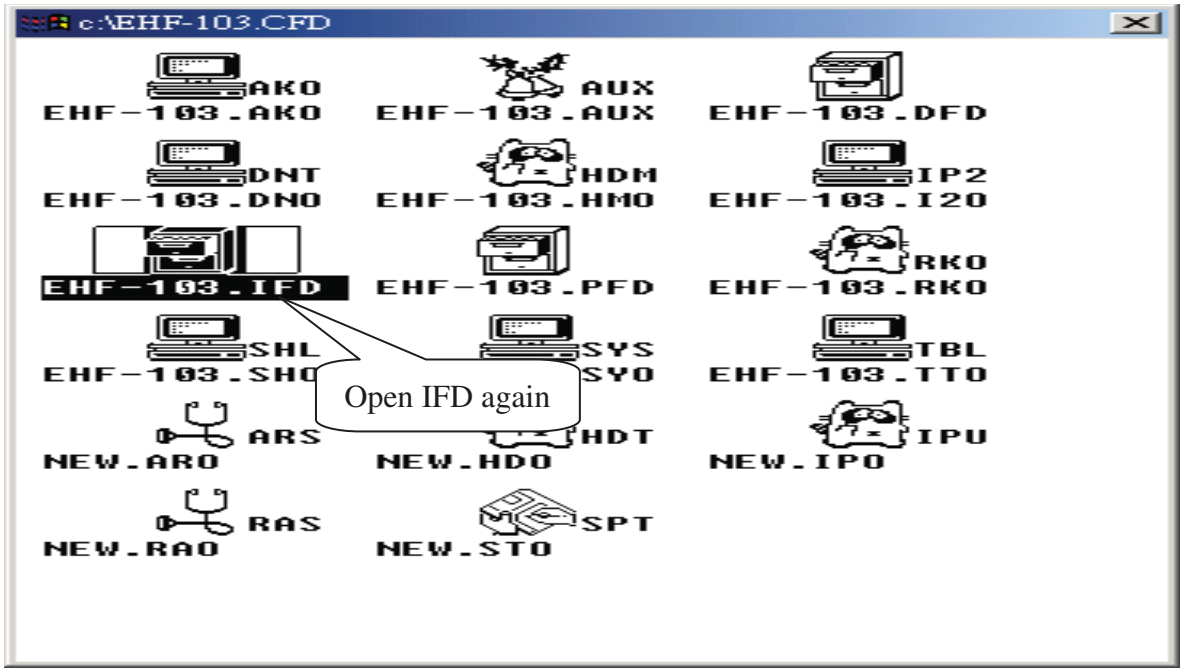


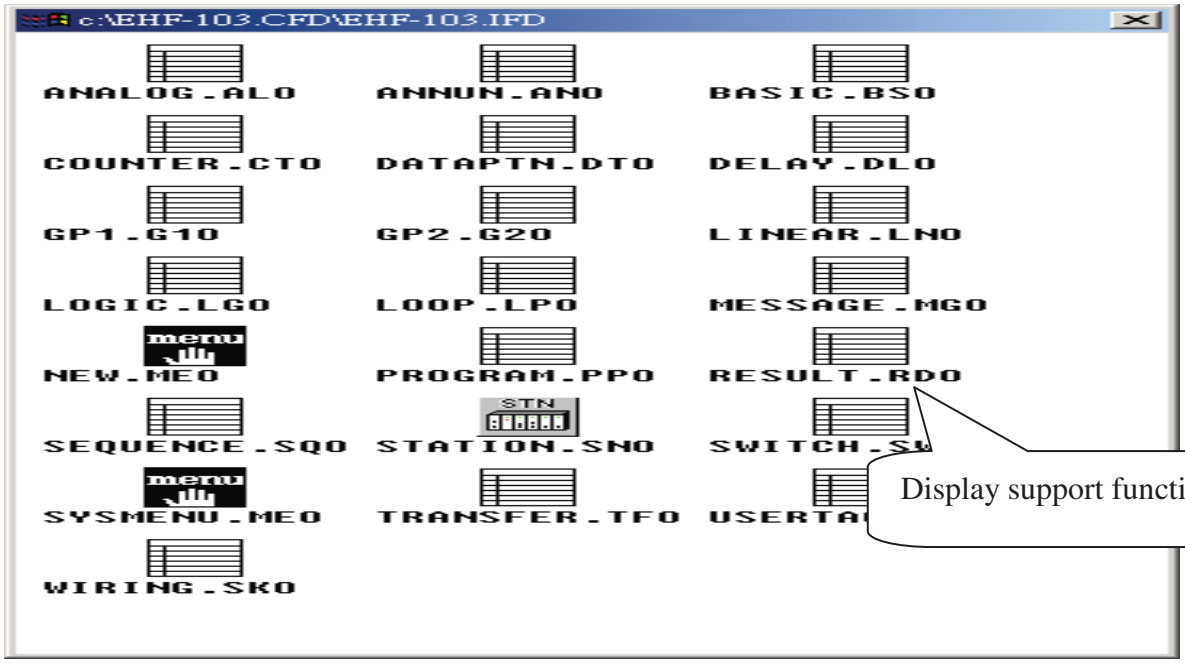
CX SYSTEM





Close the window after finishing above-mentioned steps.

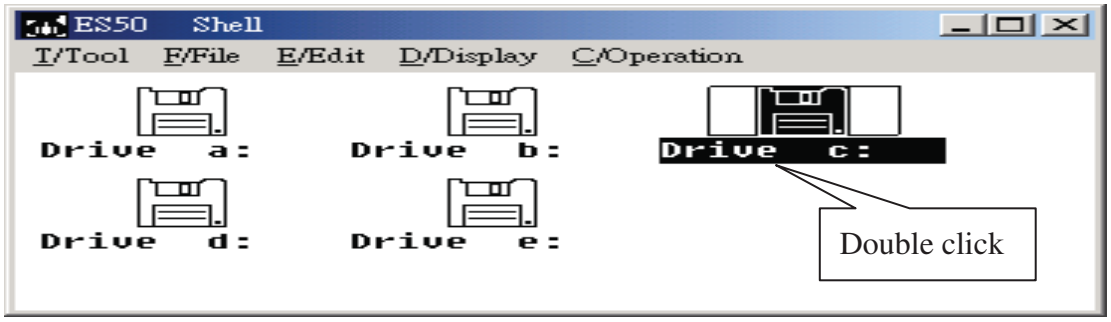




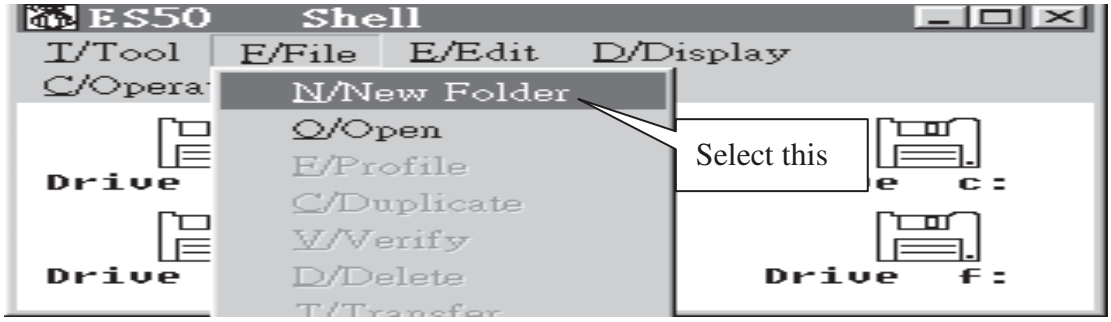
(2) How to open a MMI folder :

Drive C:\MMI.CFD\MMI.GFD

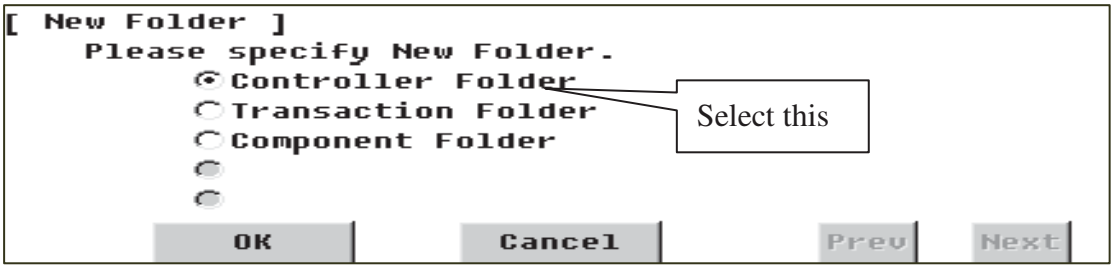
1.First open Drive c :



2.Sleect new folder

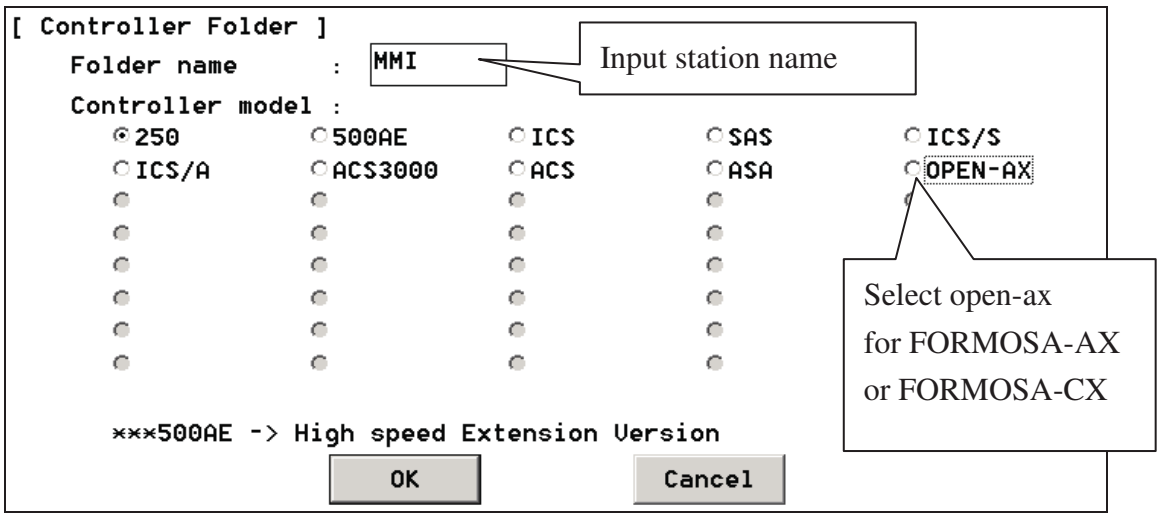


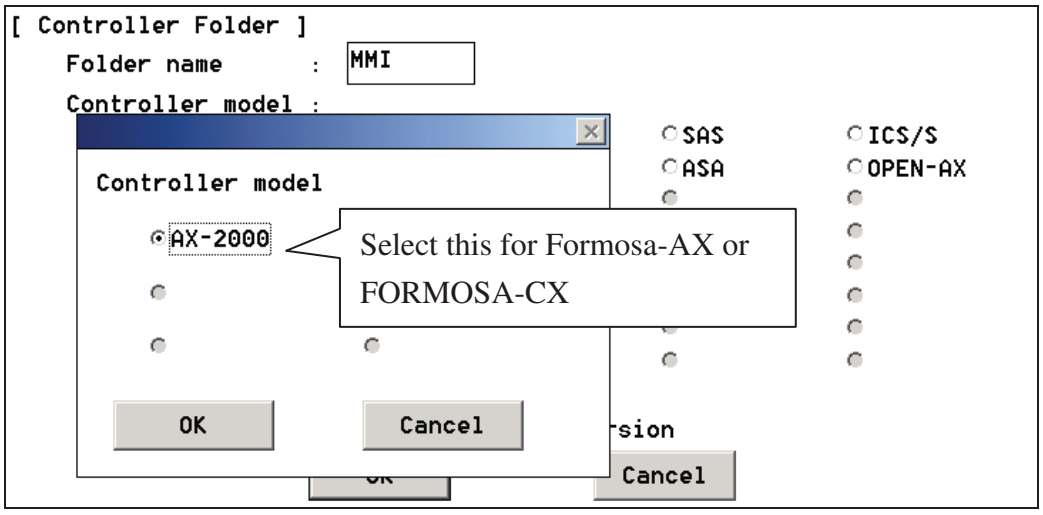
3.select controller folder



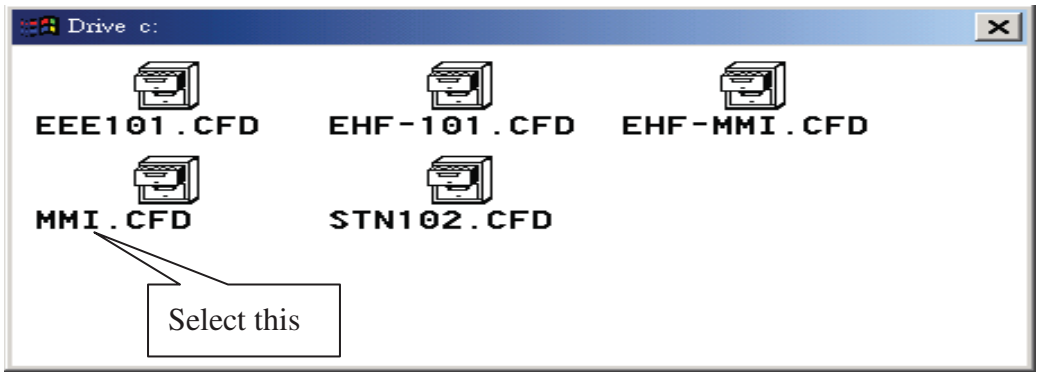
Please input XXMMI for MMI’s name,

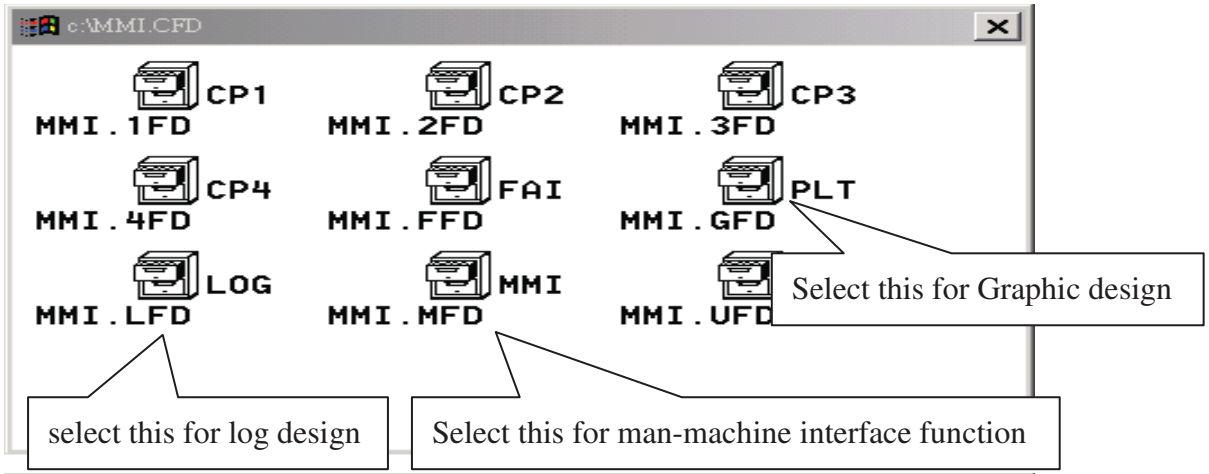
then select OPEN-AX .

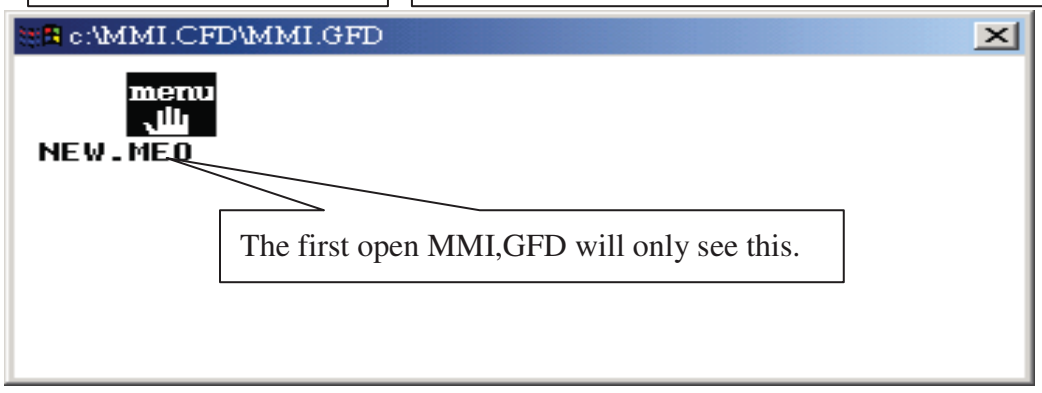




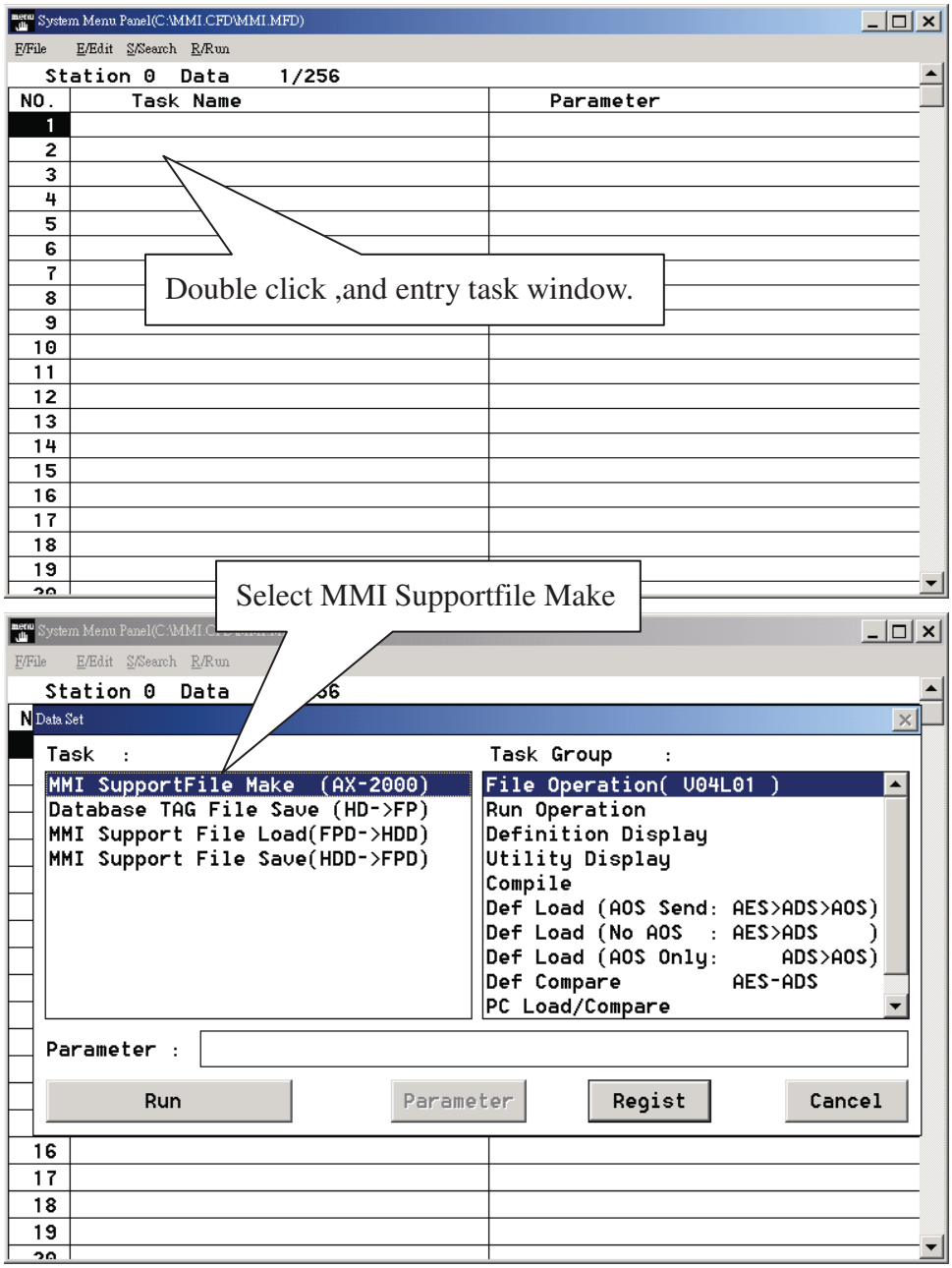
You can find increased MMI.CFD from drive C.







MMI.MFD (man-machine interface) and MMI.LFD (LOG) both are the same as MMI.GFD



MMI.MFD (man-machine interface) and MMI.LFD (LOG) both are the same as MMI.GFD do support file make.

FCS-2000/ACS-2000 FUNCTION

The following is FCS-2000/ACS-2000 Support function :

1.STATION.SNO : Set up the name of station and station number.

2.USERTAG.TGO : Set up the name of screen and operate DATA by making USER’ s TAG NO.

3.BASIC.BSO : Set up the base function and the common function PHASE /CYCLE.

4.LOOP.LPO : Define the inside instrument (LOOP type, Cycle, Phase,etc.) Signal link,address input, set up input press, define instrument TAG.

5.PROGRAM.PPO : Set up the procedure time PATTERN, support PGCS, PGSS, PGCM,PGSM,LOOP Type.

6.LINEAR.LNO : Set up linear curve function.

7. SEQUENCE.SQO : Set up sequence control.

8. ANALOG.ALO : The analog signal defines value upper limit and lower limit.

9. ANNUN.ANO : The logic operation of digit alarm signal, they can broadcast.

10. LOGIC.LGO : The logic operation of digit signal.

11. TRANSFER.TFO : The data transmit BIT (WORD) to BIT (WORD).

12. GP1.G10 : Call FM number, to run function module(FM), set up cycle/Phase ,and the address of input, and output data.

13. GP2.G20 : Call FM number, to run function module(FM), set up cycle/Phase.

14. DATAPTN.DTO : Set up constant DATA, and the value of constant

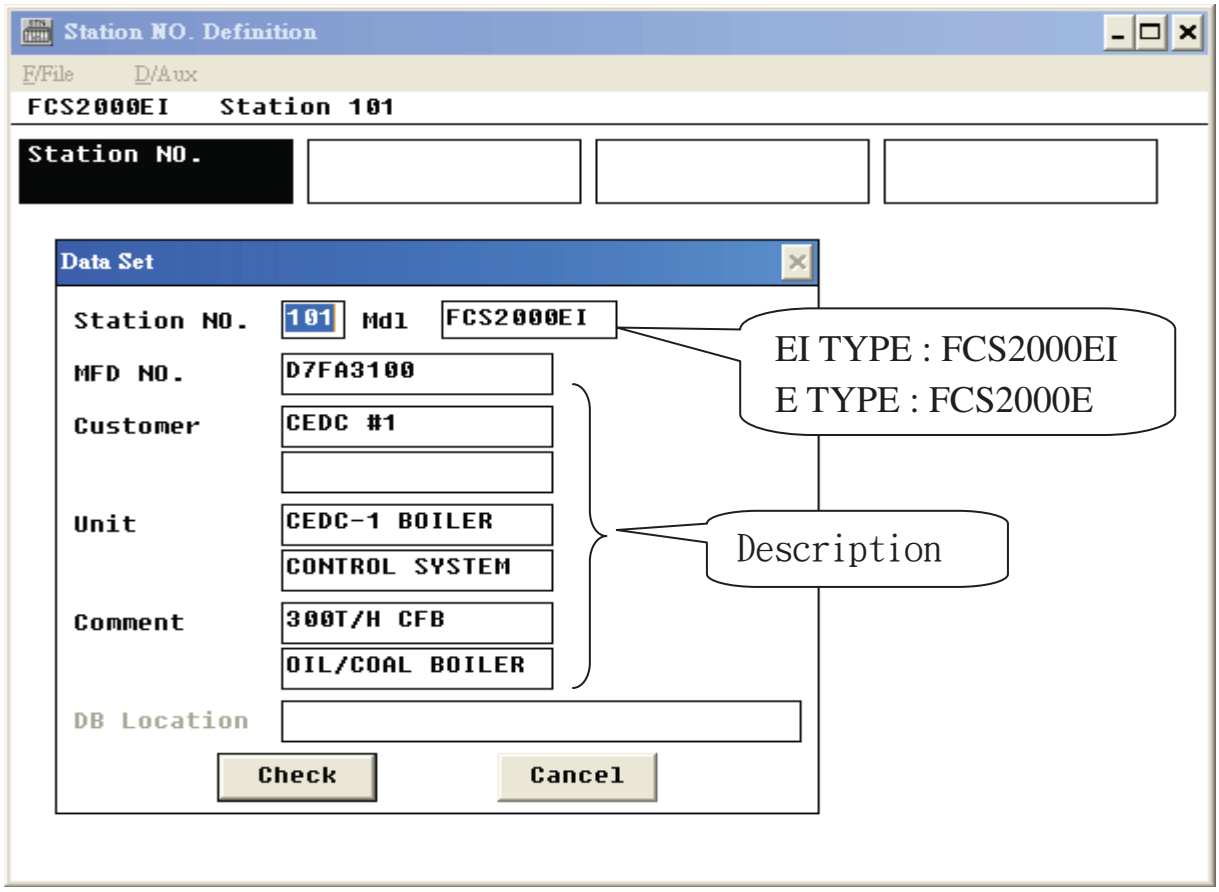
15. WRING.SKO : Define inside station wiring, it is adopted broadcast to communication with all other station.

16. MESSAGE.MGO : MESSAGE communication with other station , it need to Define station and the type of message communition.

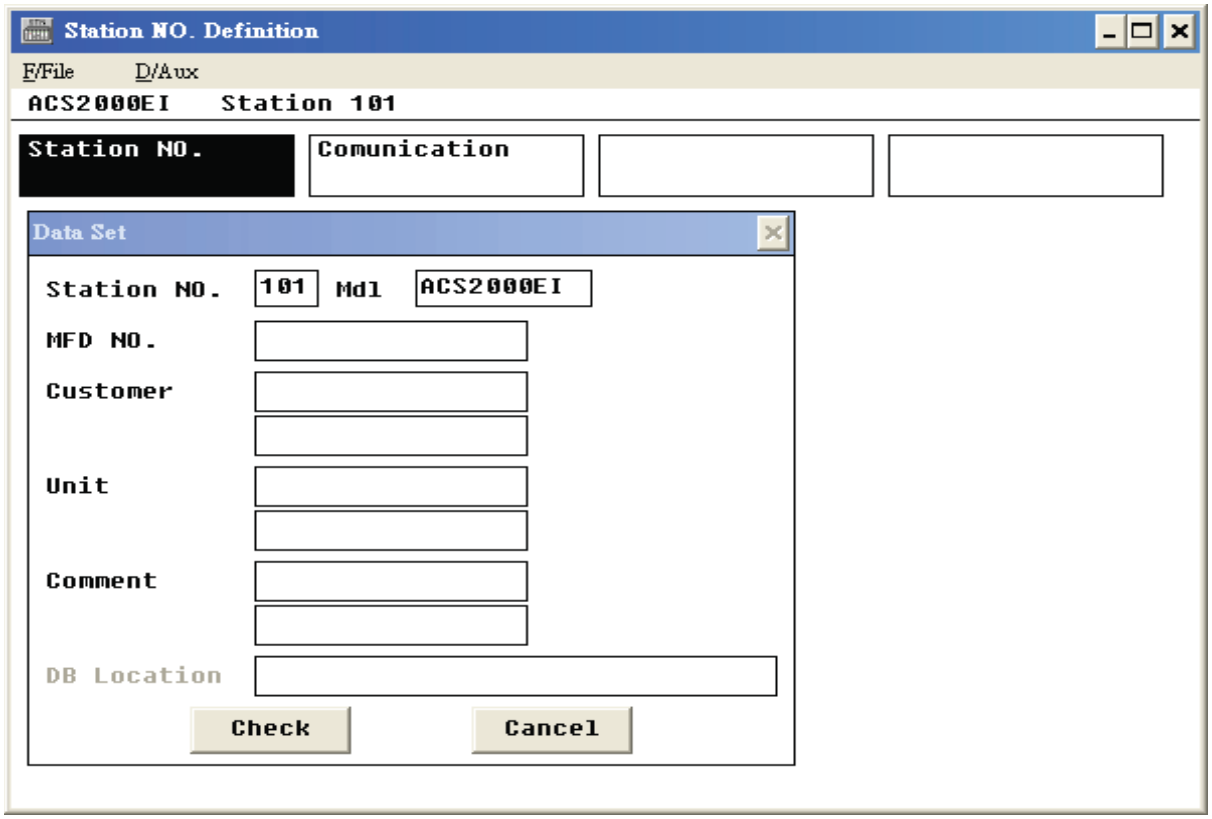
STATION.SNO

STATION.SNO

CX syetem:FCS2000EI/FCS2000E setup like the following.



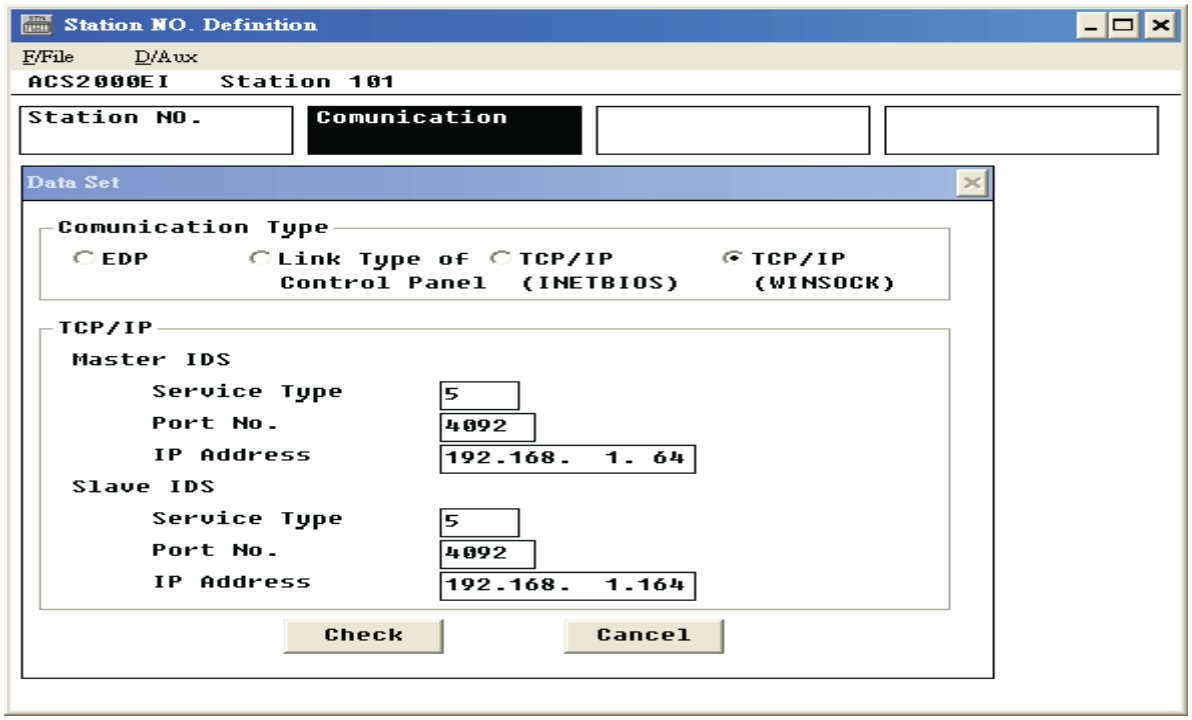
AX system:ACS2000EI/ACS2000E setup like the following.



Communication means AES Download to ACS communicate by the following 2 ways:

(1) TCP/IP:AES communicate through ADS to ACS, it may download to ACS Both master/slave.

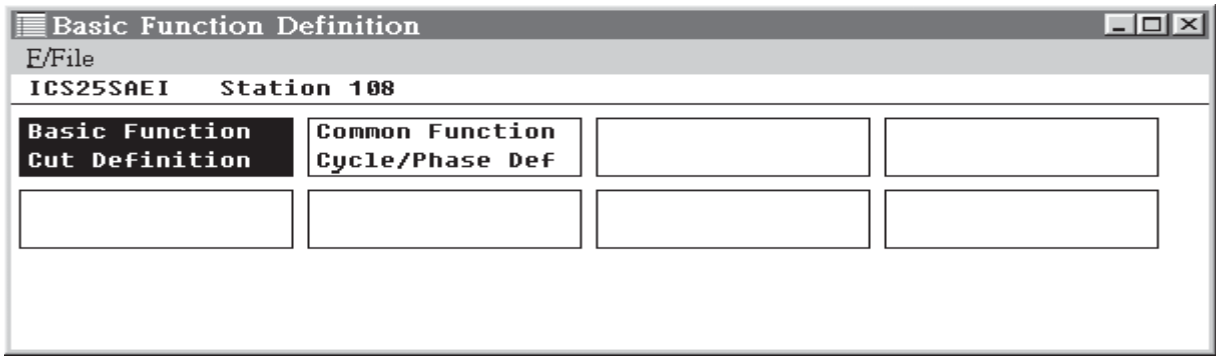
(2) Link Type of Control Panel: AES communicate use Control Panel Page 6 ，link by RS232C、TCP/IP(Ethernet)、FL-NET。



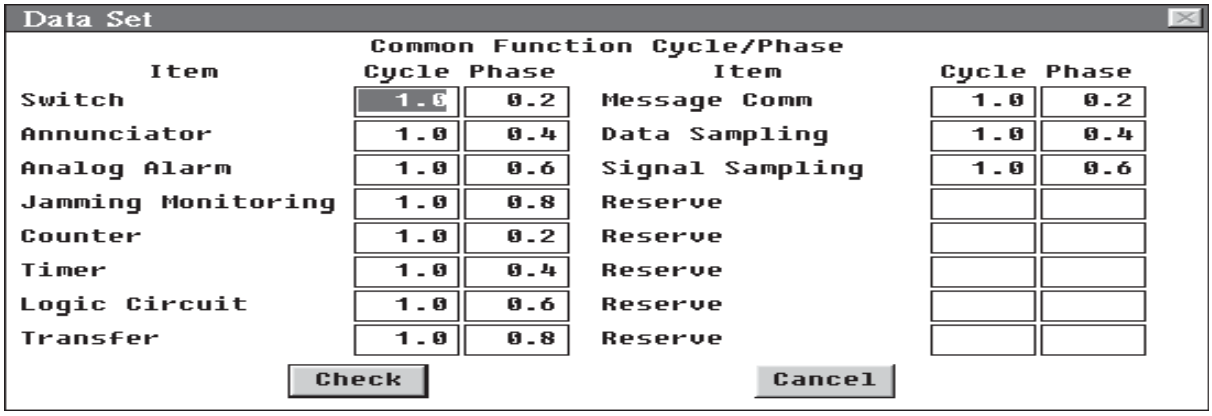
BASIC.BSO

Define: select the type of basic function, common function PHASE (delay time),CYCLE (cycle time)

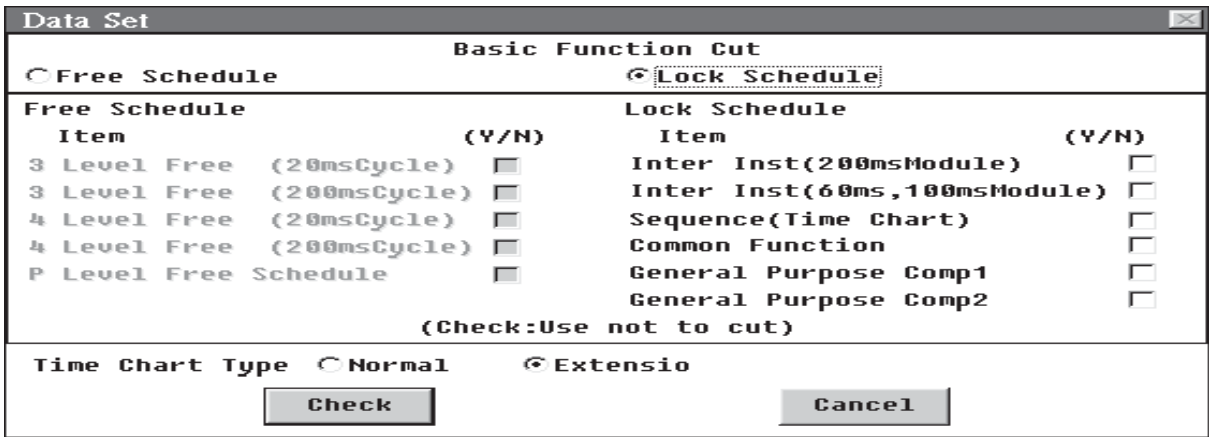
Drive C:\XXX101.CFD\XXX101.IFD\BASIC.BSO



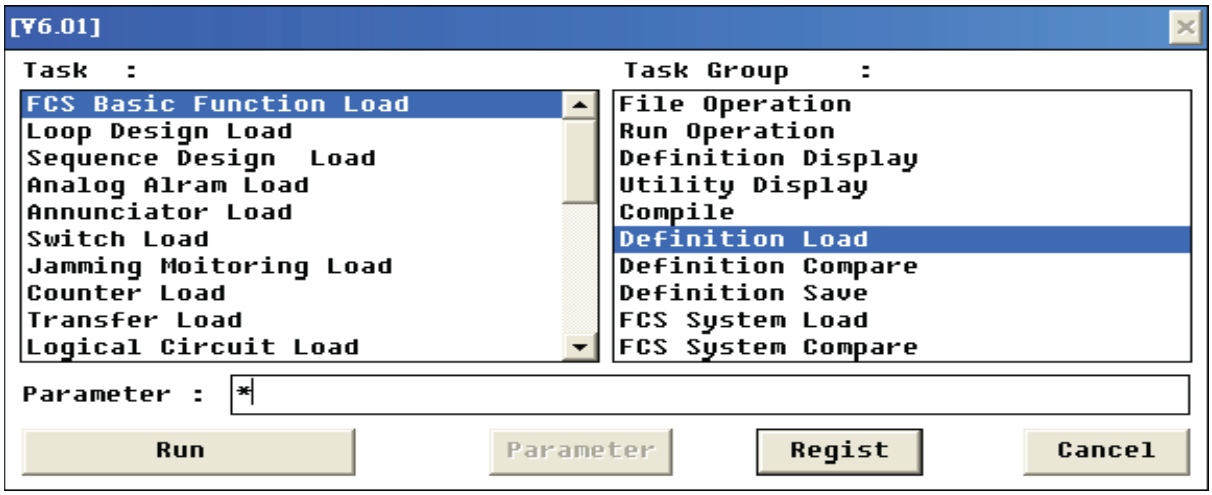
The following window is data set for common function cycle/phase.



If this station has sequence control, you need to select the Time Chart Type definition.
Time Chart Type:
select Normal :means this station sequence step max. 32 steps.
select extension :means this station sequence step max. 64 steps



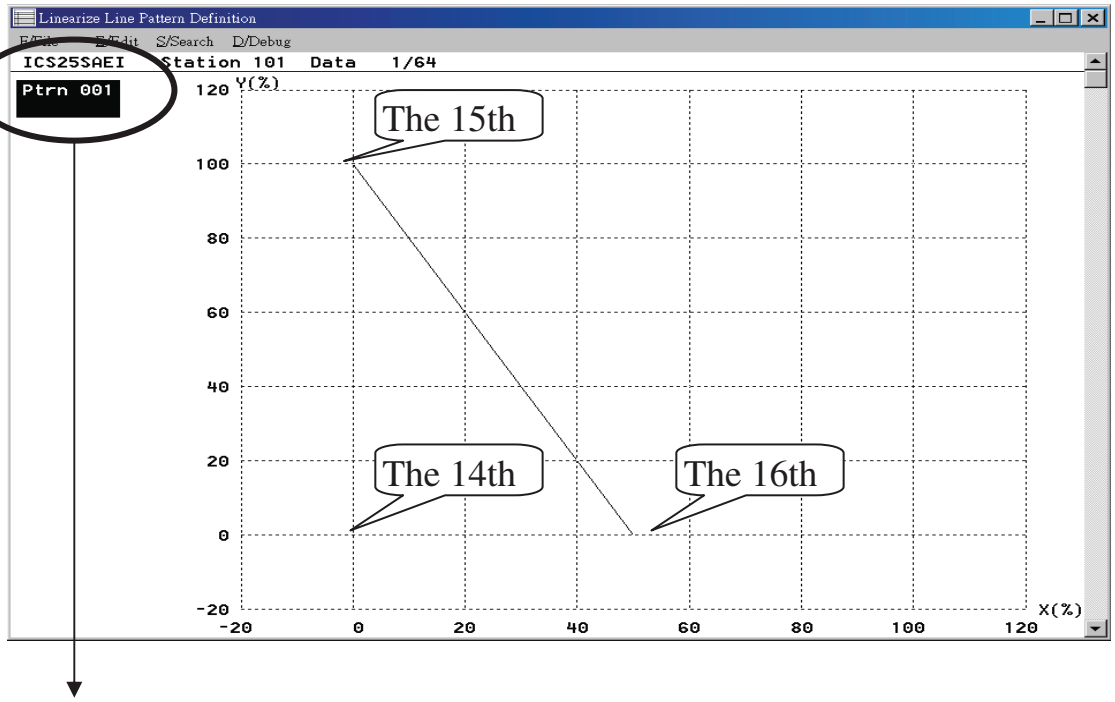
we will do basic function load after FCS initial，then restart FCS/ACS。



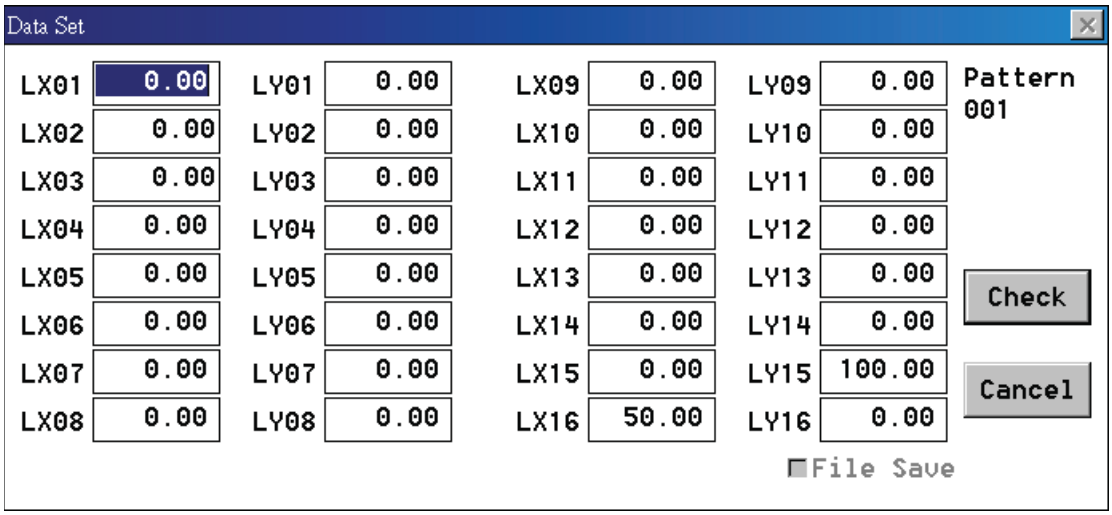
LINEAR.LNO

Function : Can define linear curve, enter input process (linearize1,

linearize2) support signal to control , use the Genius mouse to double click on Ptrn 001 , the curve Will display after input x,y data set,see the following figure.



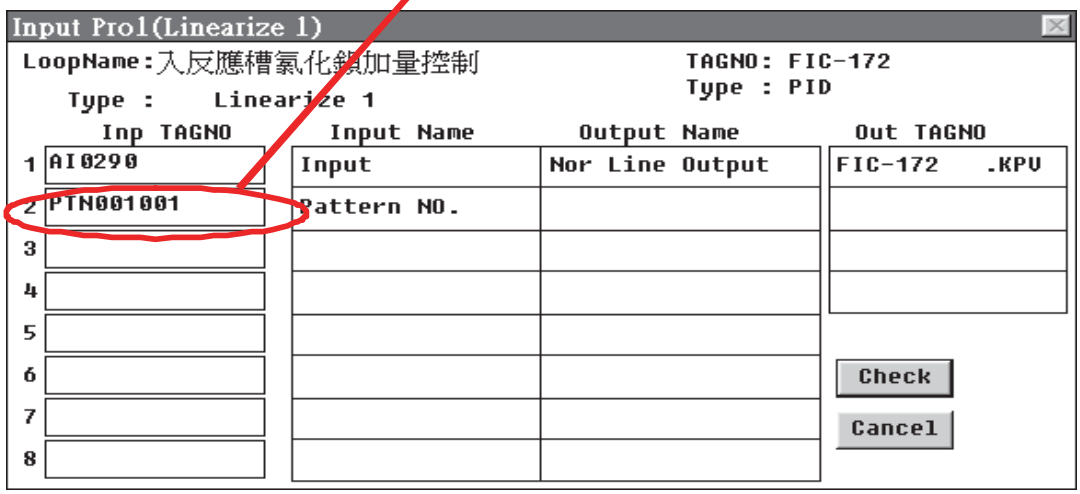
Double click will display the following figure, Please begin to input from line point of the most right,from the right side to the left side,input the coordinate of x, y and start to input from the 16th.



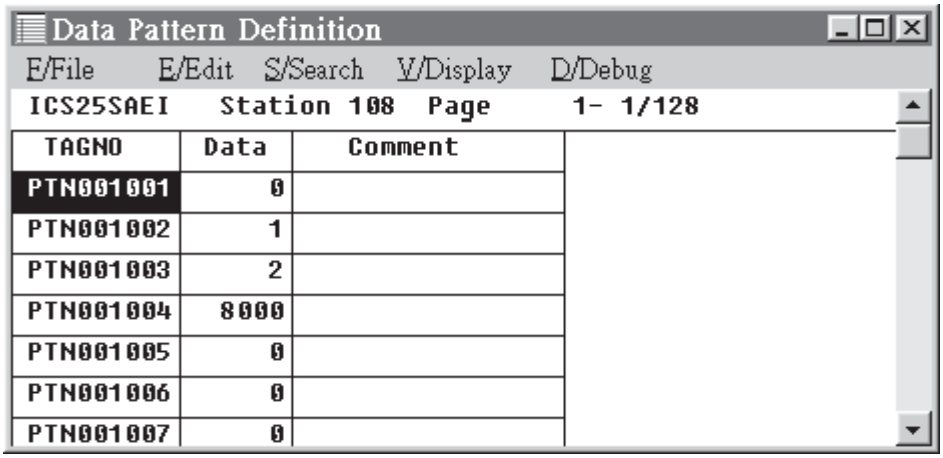
Ptrn001 → 0 Ptrn002 → 1 Ptrn003 → 2 Ptrn004 → 3

How to define input process :

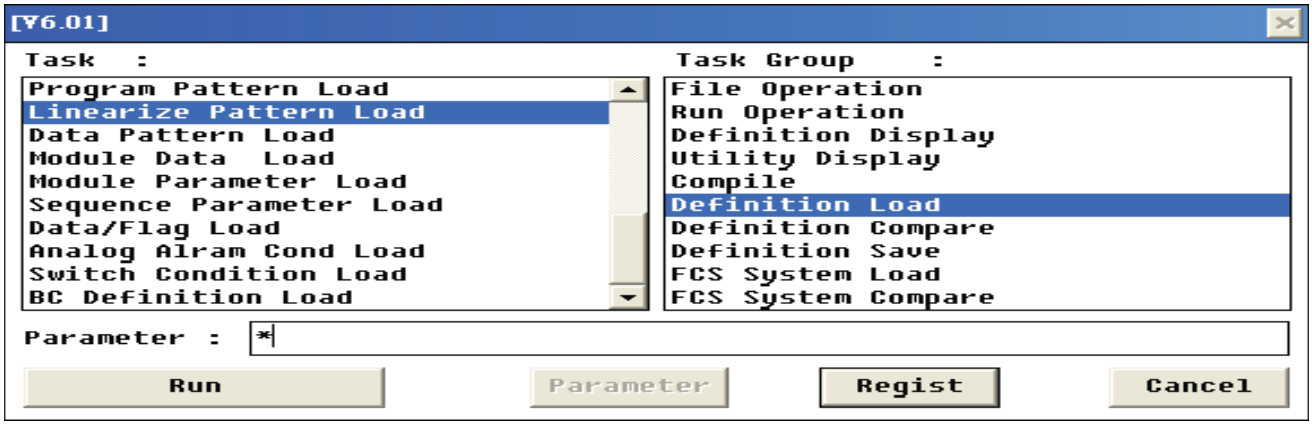
PTN001001 means the address of pattern NO. the value of this address is 0 ,see the following picture.



Pattern NO input address (only accept address, can use SI address), This address may put the value range 0～63 , corresponding LINEAR.LNO 1～64 figure.
The value of Addrress(PTN001001) is 0 corresponding the first Figure. See the following picture.

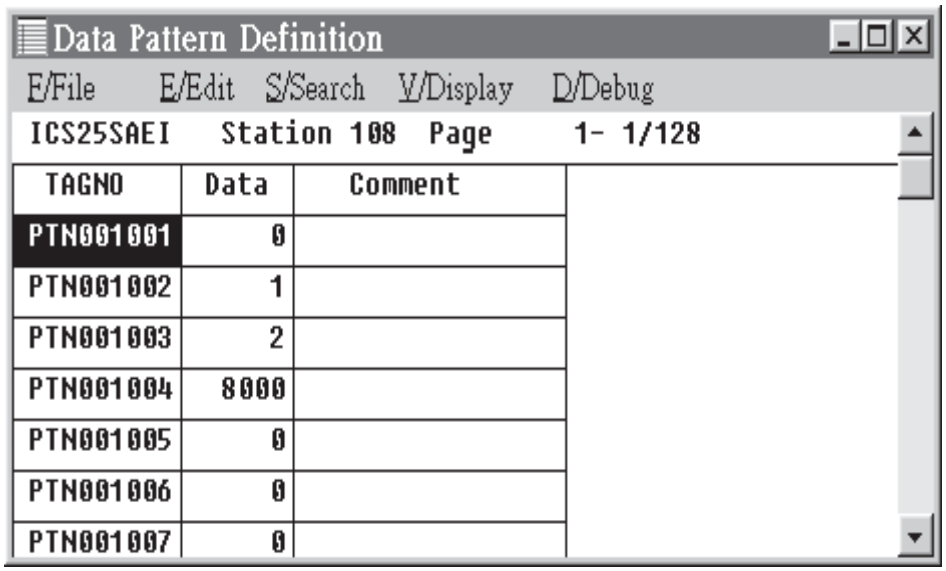


Parameter may use * or assign number.

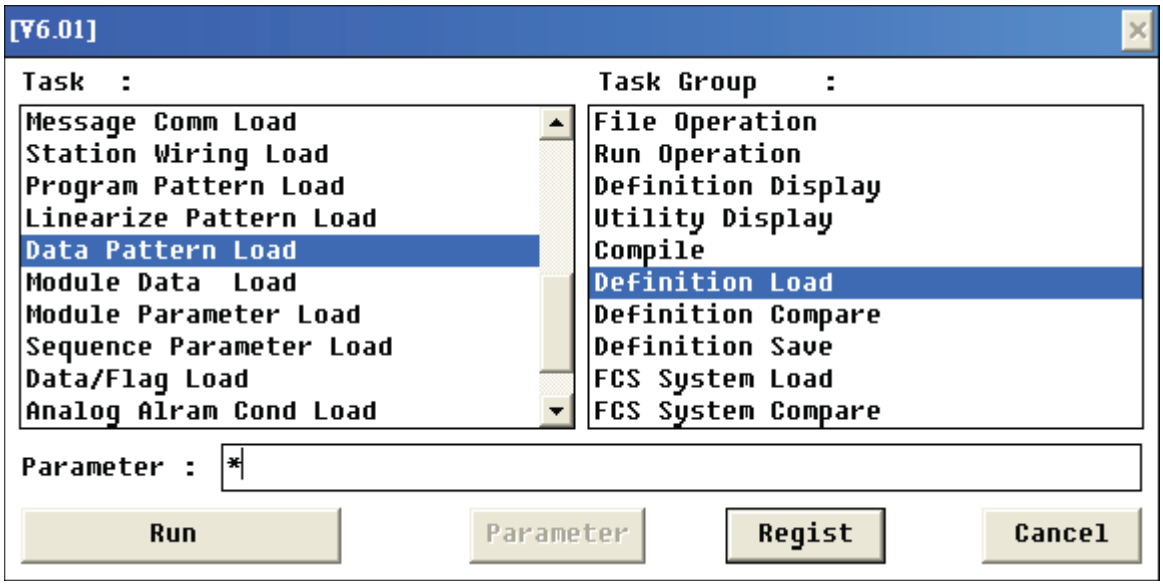


DATAPTN.DTO

Function: Setting up constant data, 128 PATTERNSs and each PATTERN have 32 DATAs, can set up, so can set up 128*32 DATAs.

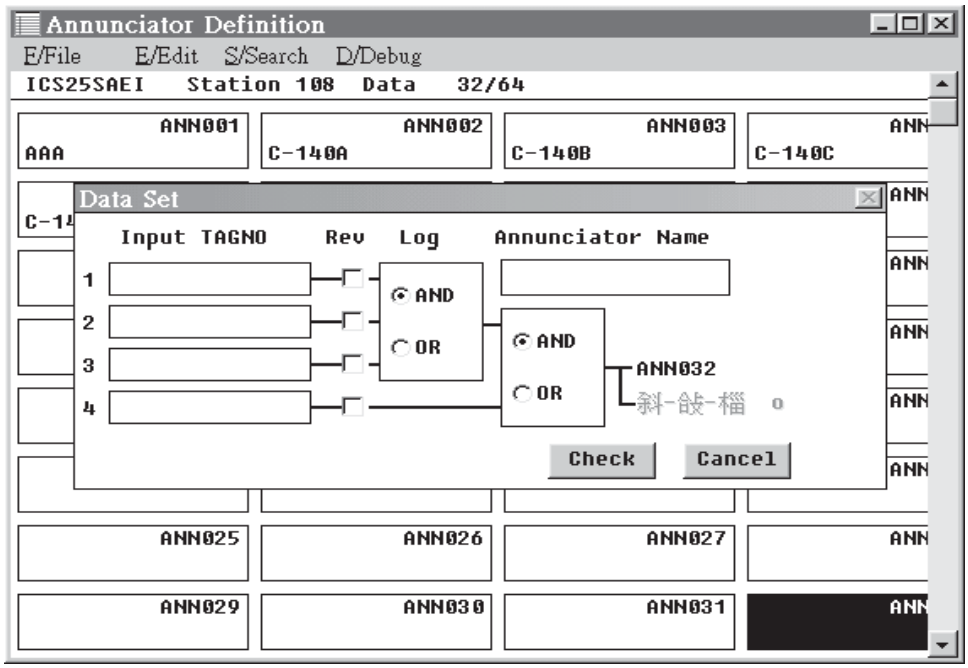


PTN001001 can be address.it may used in program ,also may used in graphic Or input address for COS/AOS。

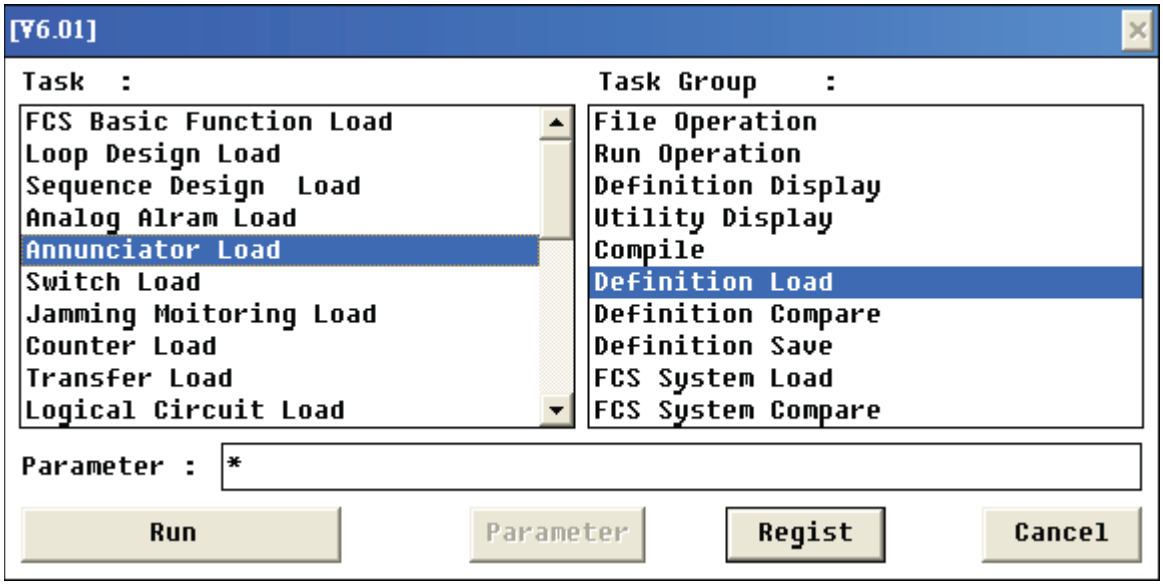


ANNUN.ANO

Function ：The logic operation of alarm signal, it can offer 64 output point.

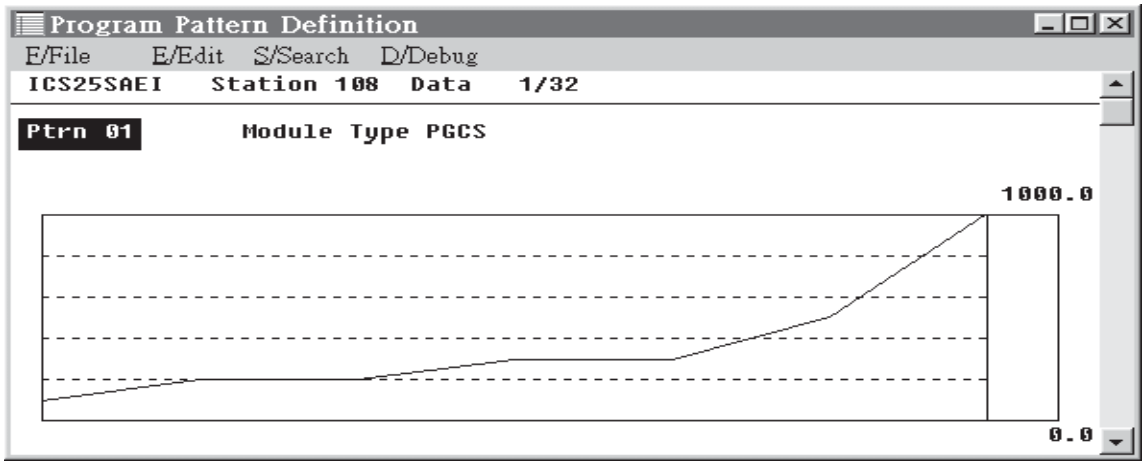


Parameter may use * or assign number.

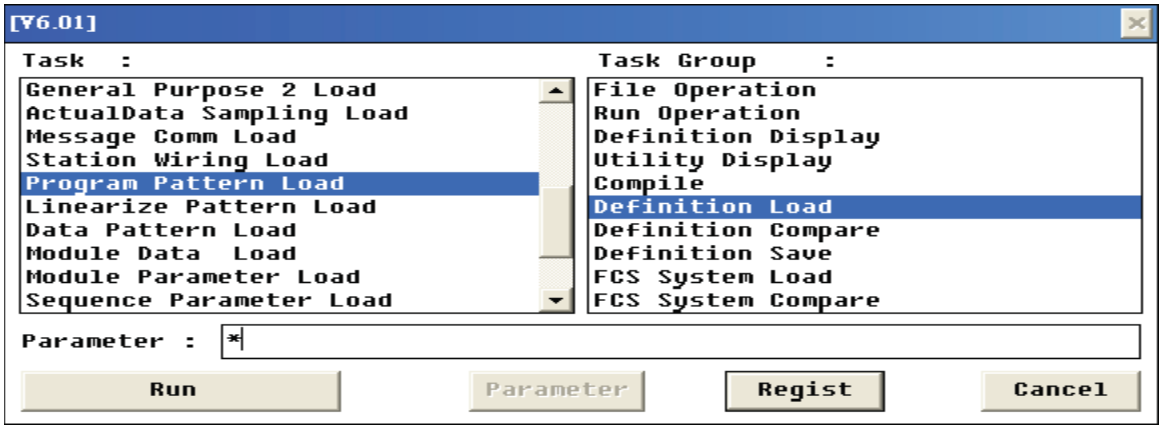


PROGRAM.PPO

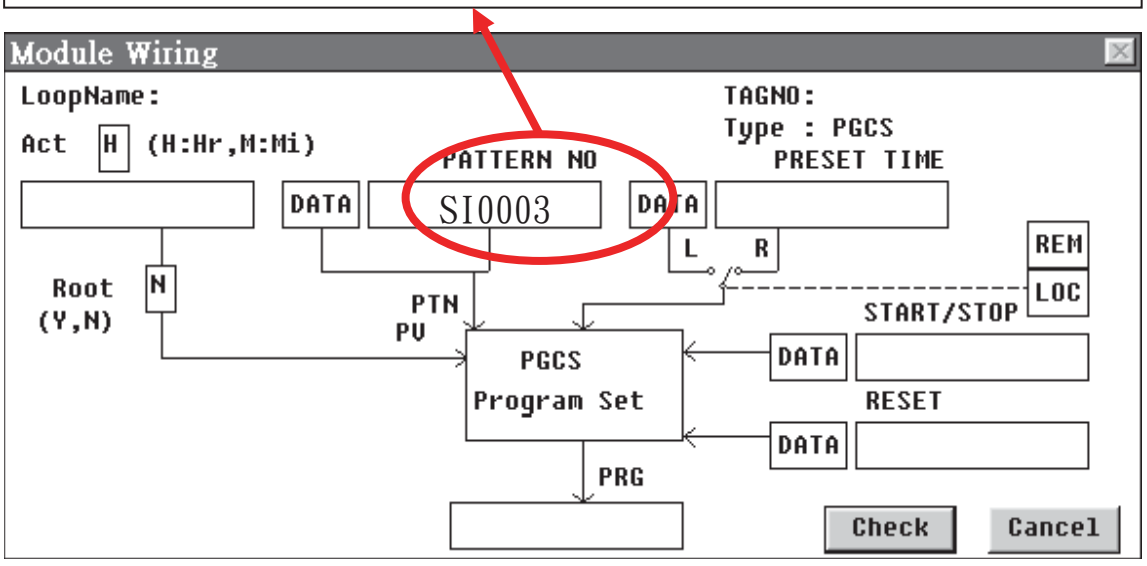
Function : Proived LOOP TYPE are PGCS , PGSS , PGCM , PGSM



Parameter may use * or assign number.



Input the System Tag (for example SI address) ,
(address's value add 1)the same as the PROGRAM NO.



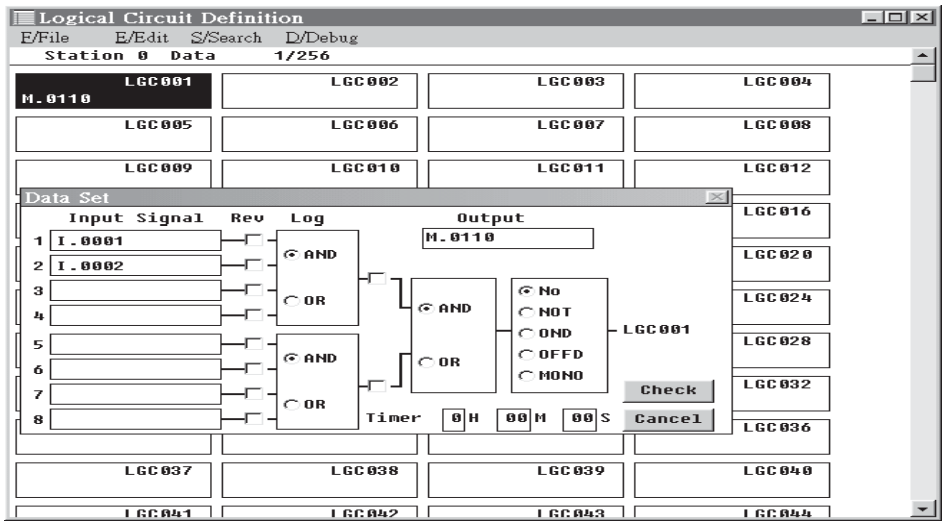
SI0003 : 0 (value)

LOGIC.LGO

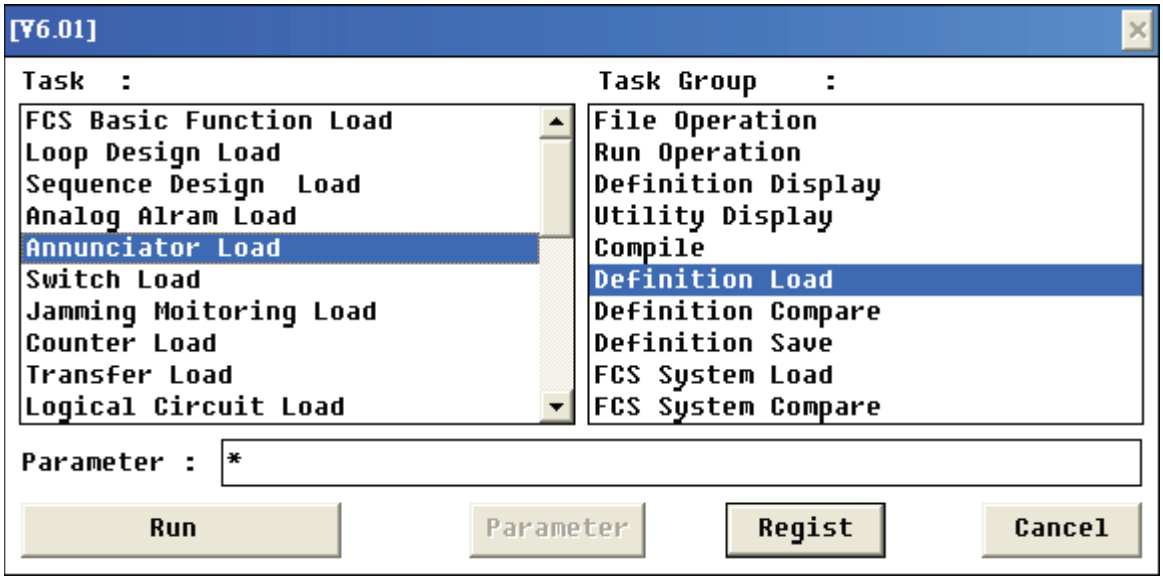
LOGIC.LGO can offer 256 logic operation, output LGCXXX.UP, can appoint one BIT to output,The following picture M.0110 (output)

Output type:

NO	Logic operation establishe and output ON signal.
NOT	Logic operation establishe and output OFF signal.
OND	Logic operation establishe and output ON signal after A period delay time ,delay time can set on TIMER.
OFFD	Logic operation establishe and output OFF signal after A period delay time ,delay time can set on TIMER.
MONO	Logic operation establishe and output pulse signal , ,this pulse spread can set on TIMER.

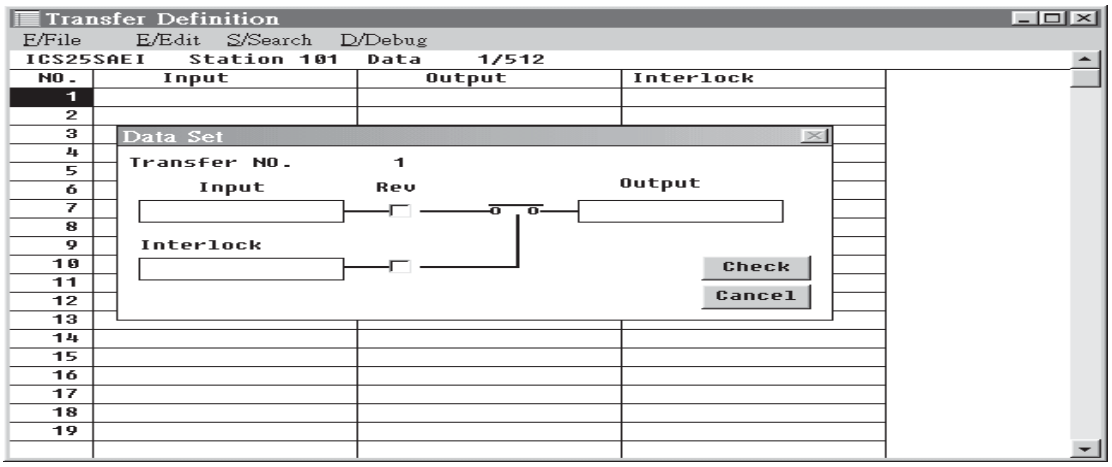


Parameter may use * or assign number.



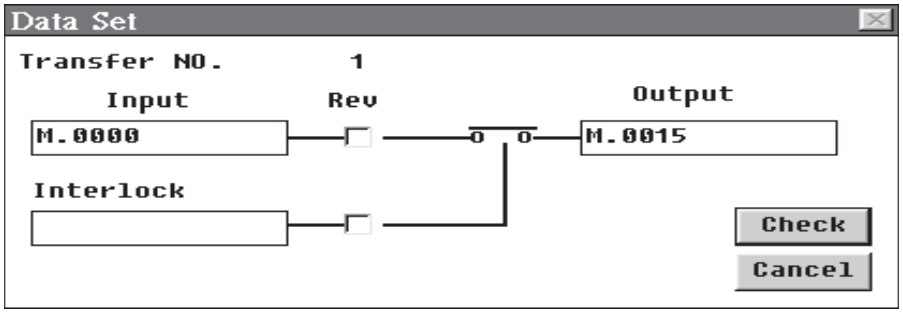
TRANSFER.TFO

TRANSFER.TFO provide 512 item may used to transfer,
BIT to BIT / WORD to WORD

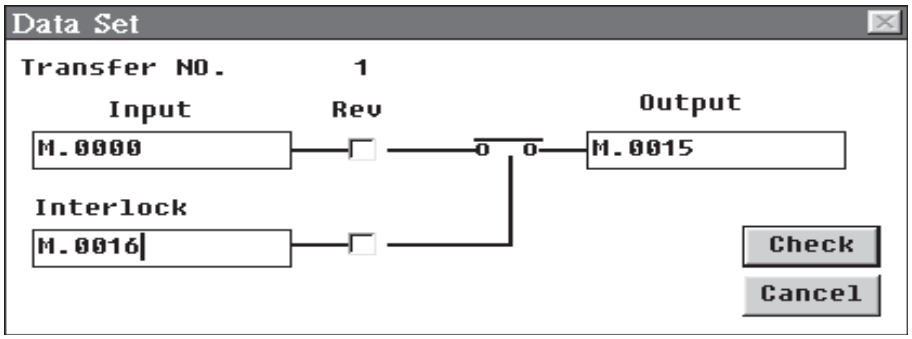


We usually use two ways:

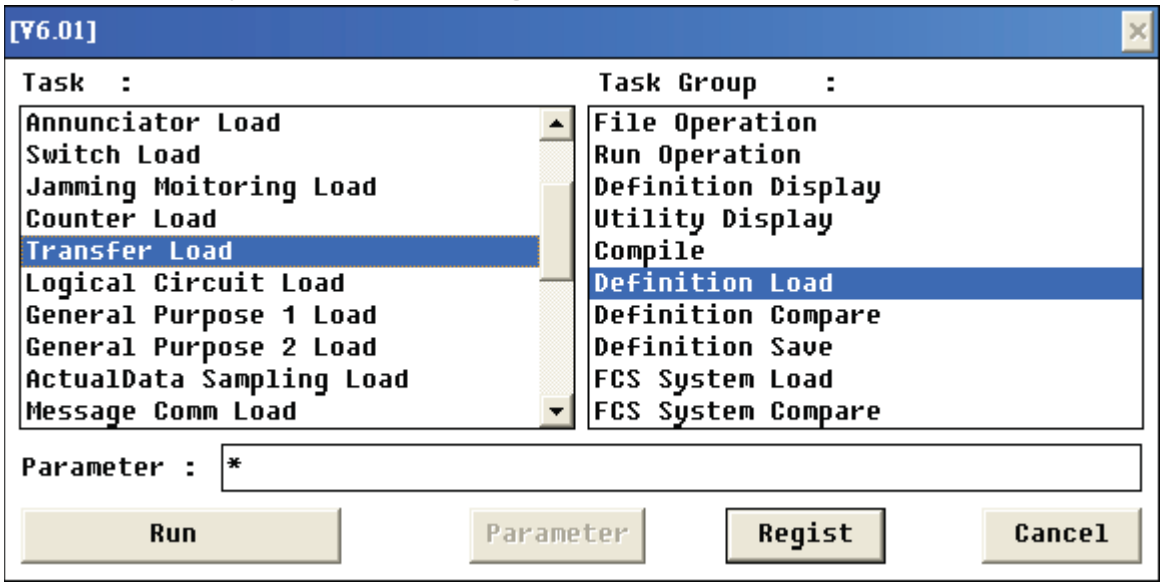
1. The following way means copy M.0000 to M.0015 every scan time of system.



2. The following way means when M.0016 ON, will copy M.0000 to M.0015.



Parameter may use * or assign number.



WIRING.SKO

Station wiring :every station provide MAX 128 word to boardcast word signal and every station can receive boardcast data of Station wiring , ,we usually use the following TAG to express word.

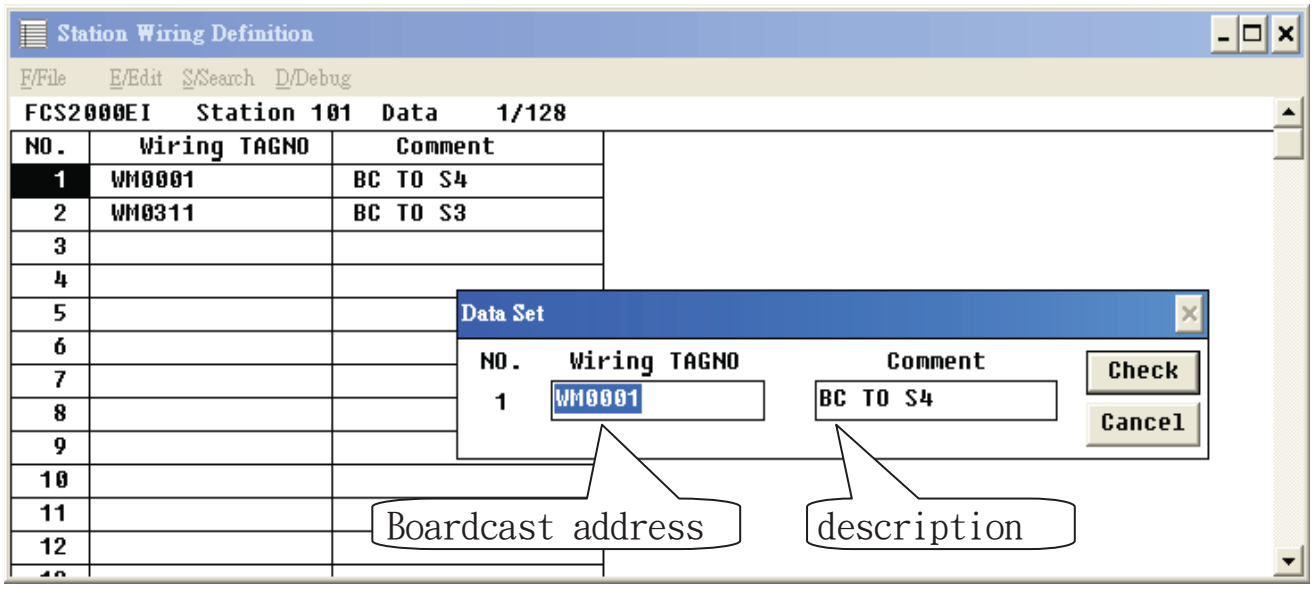
STN108 001

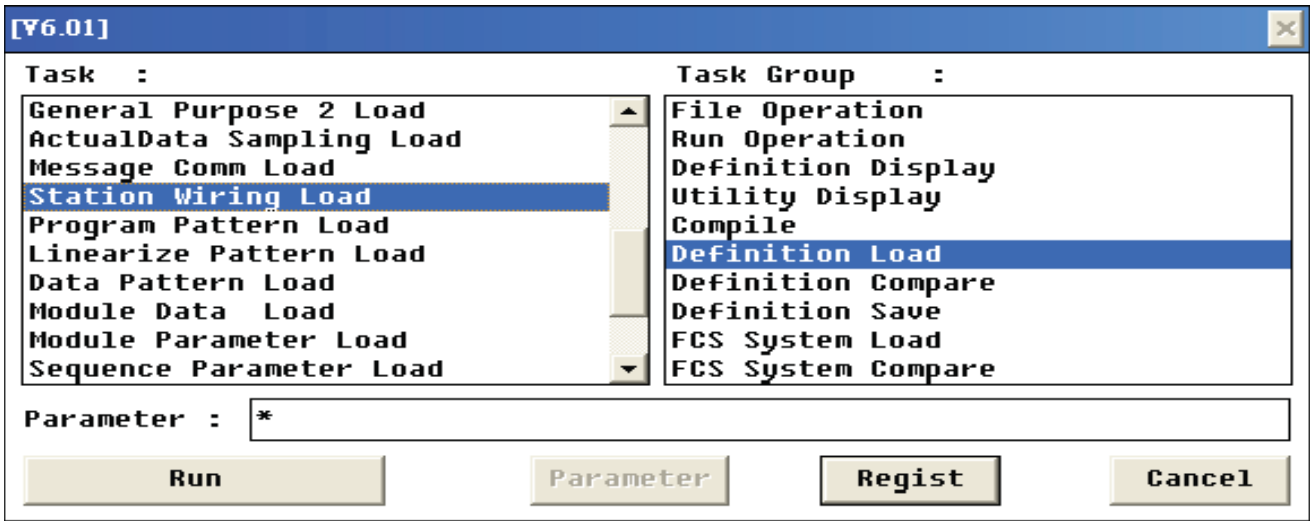
Station NO Station wiring

we usually use STN108001.B00～STN108001.B15 to express bits.

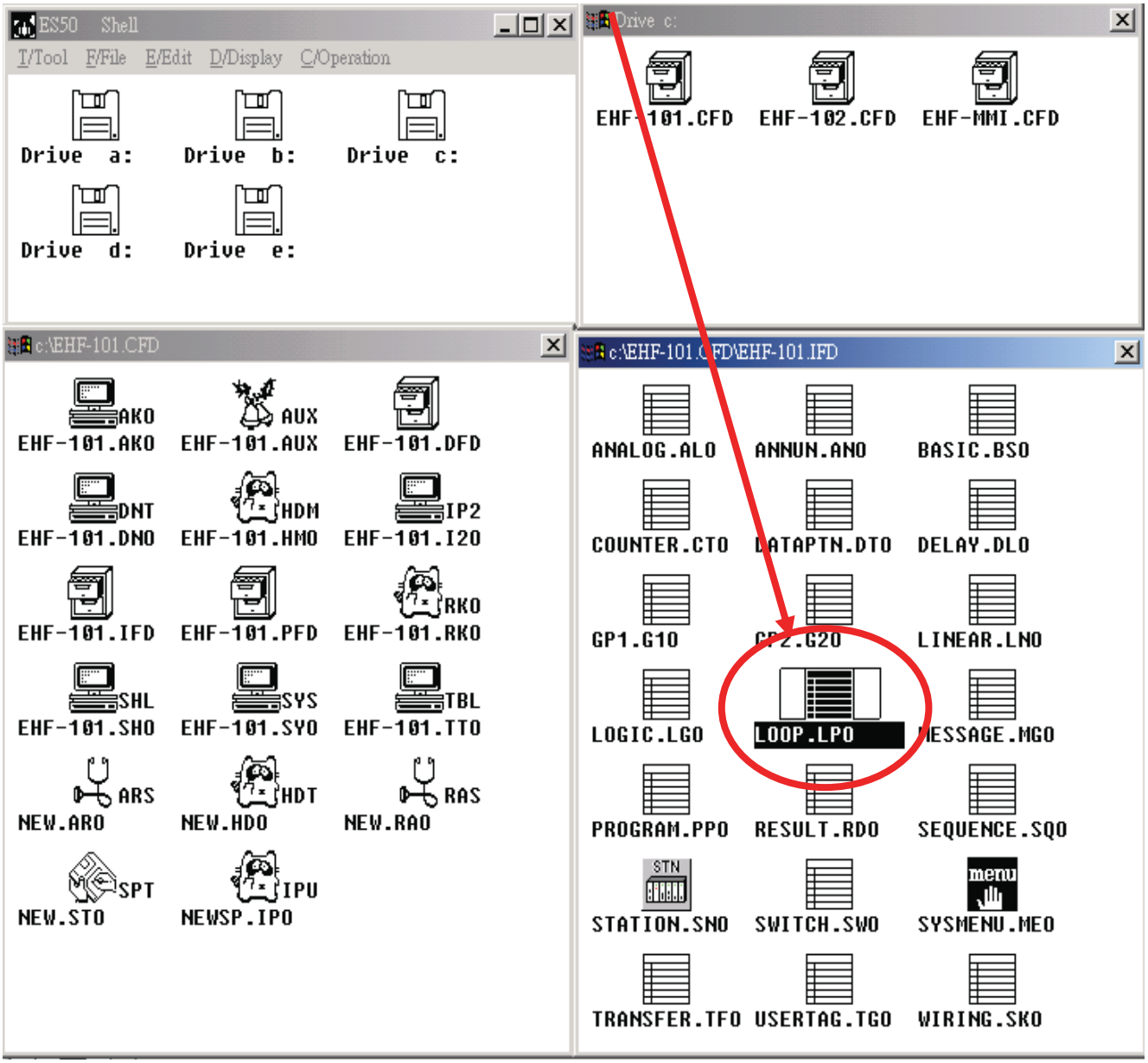
For example ： STN108001.B00 means first BIT of WM0001, just like M.0010.

STN108001 means the all word of WM0001.



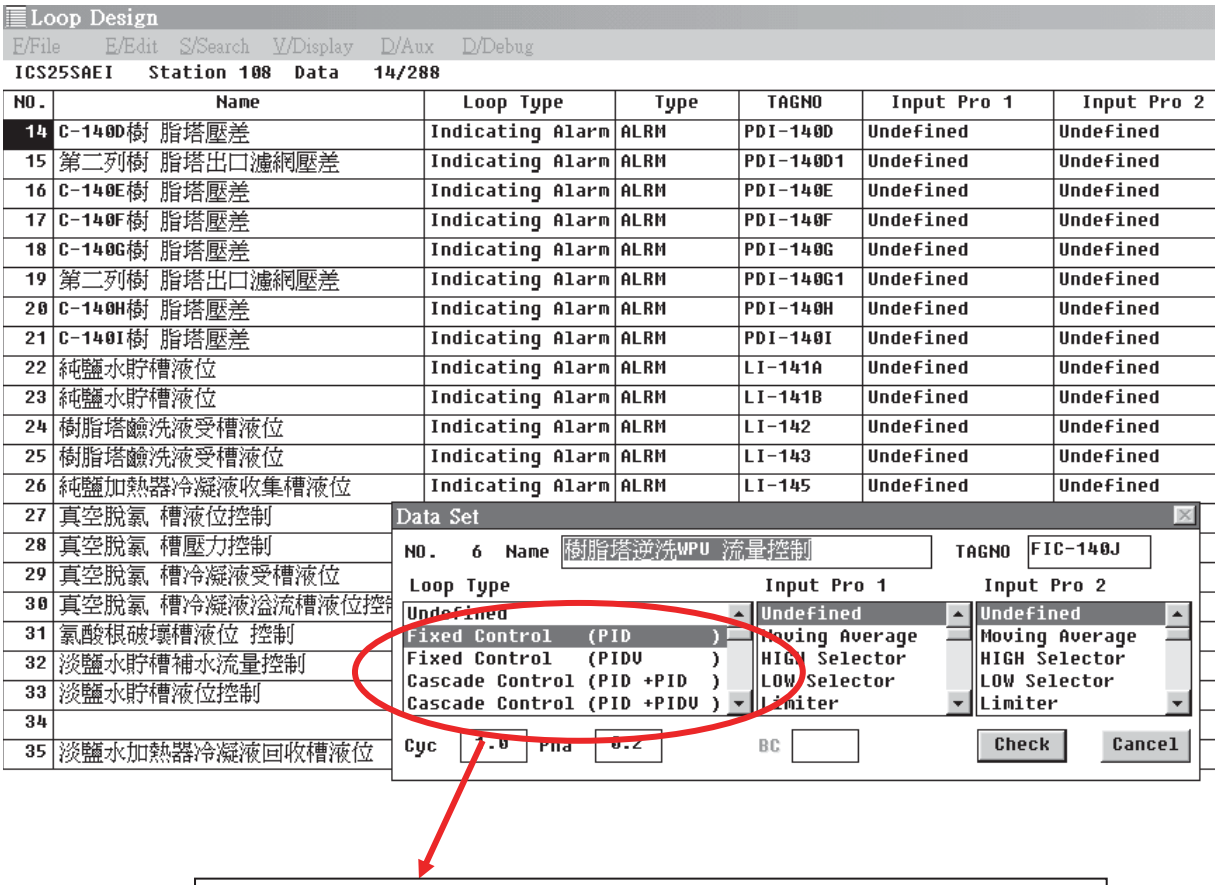


Enter LOOP from LOOP.LPO and increase/revise LOOP



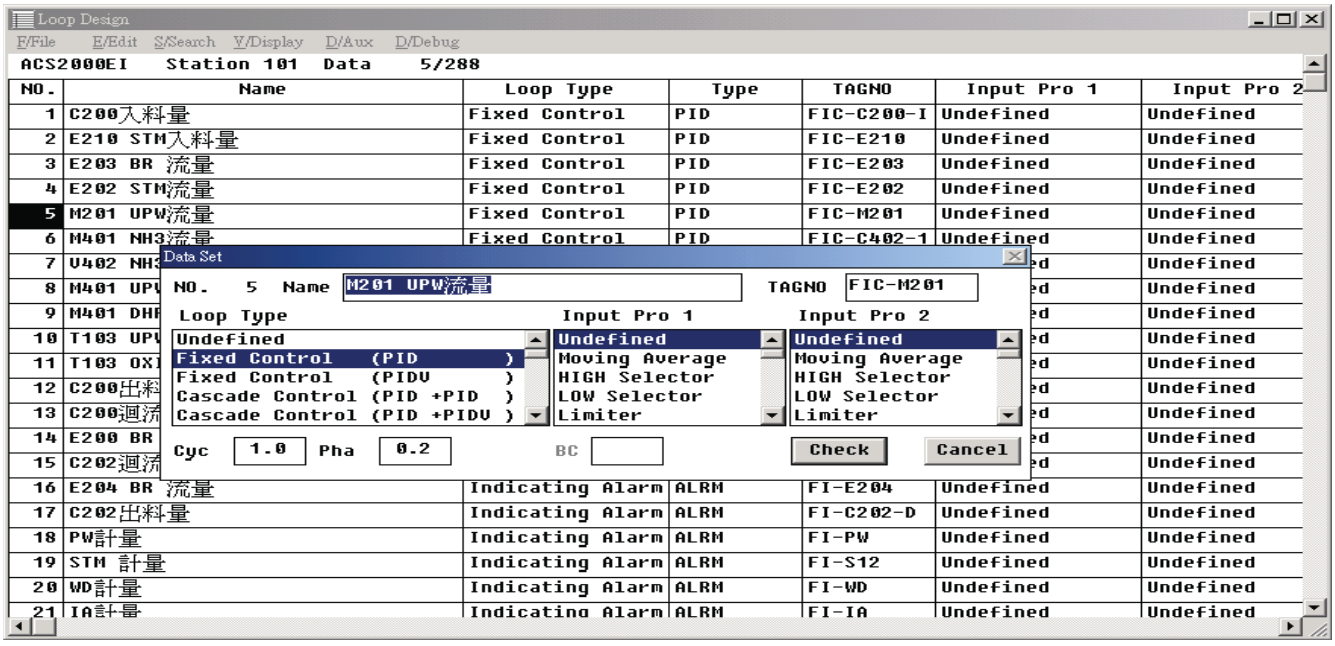
LOOP's PV input(AI/AO address) follow I/O assignment ,please reserve some spare I/O , it will be convenient to increase LOOP in the future.

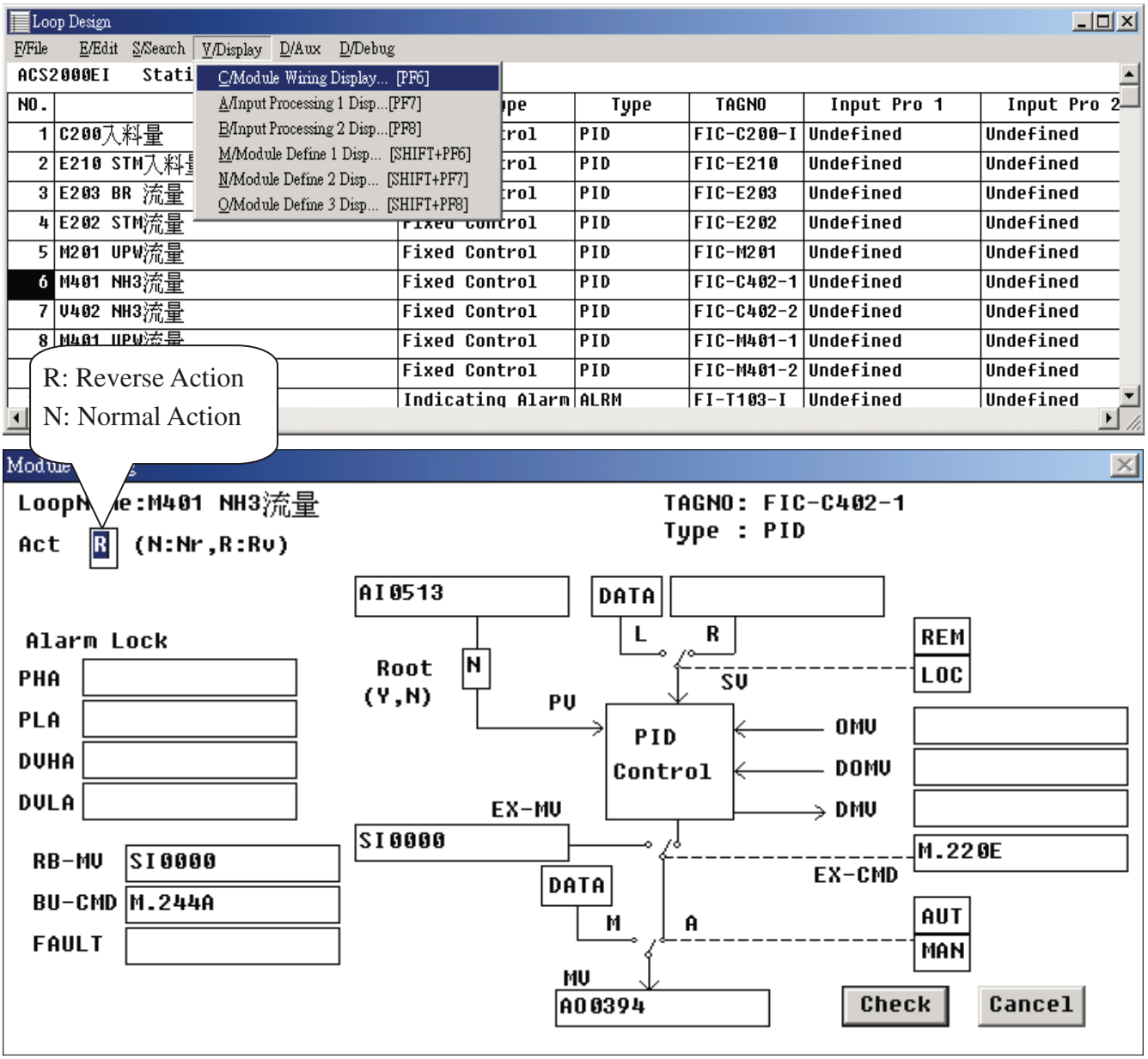
THE SELECTION OF LOOP TYPE:



User can choose loop type , they usually depend on equipment of the plant , for example : ALM, PID, MAN, SUMA, SUMP

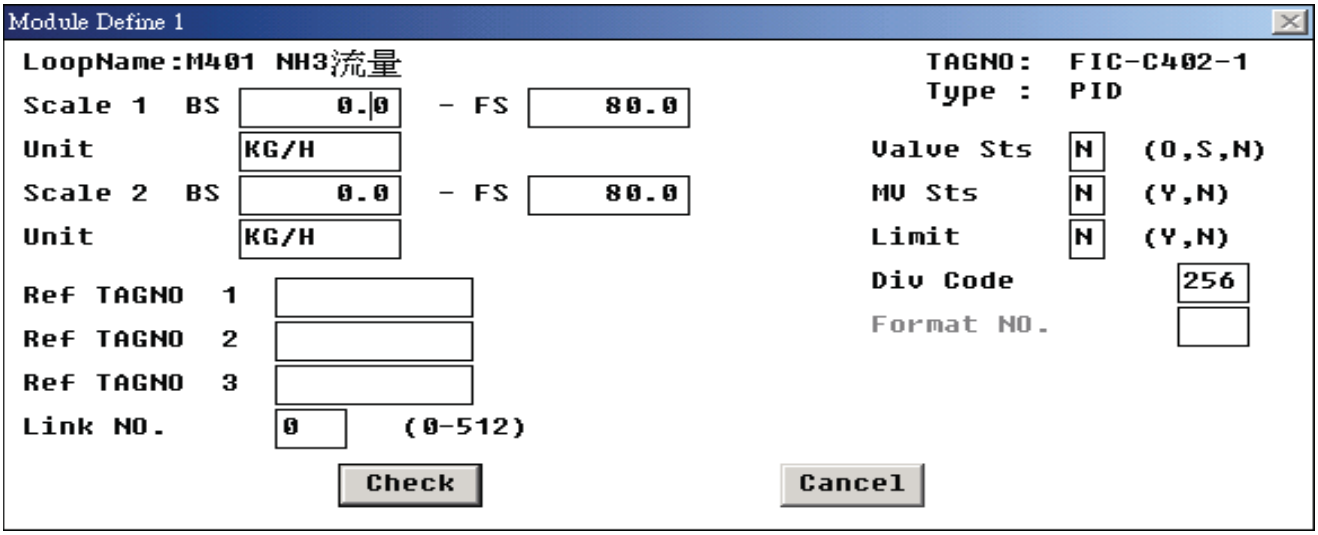
LOOP TYPE : PID

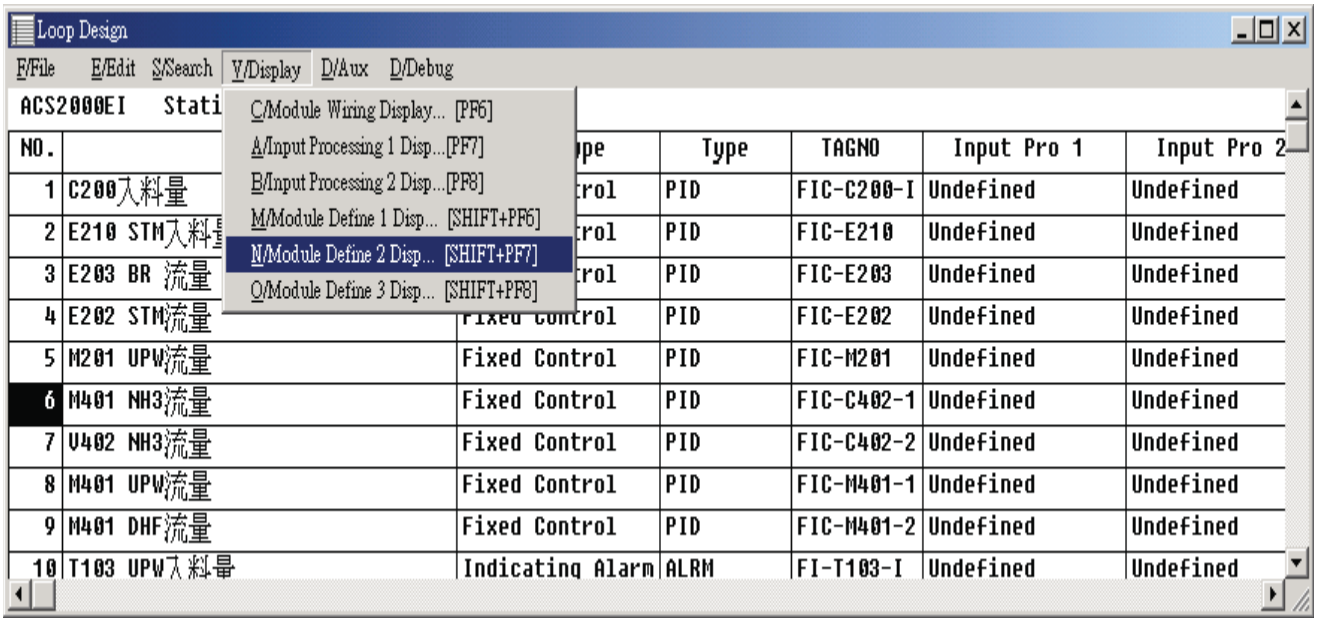




Loop Design
F/File E/Edit S/Search V/Display D/Aux D/Debug
C/Module Wiring Display... [PF6]
A/Input Processing 1 Disp...[PF7]
B/Input Processing 2 Disp...[PF8]
M/Module Define 1 Disp... [SHIFT+PF6]
N/Module Define 2 Disp... [SHIFT+PF7]
O/Module Define 3 Disp... [SHIFT+PF8]
ACS2000EI Stati
NO. Type TAGNO Input Pro 1 Input Pro 2
1 C200入料量 PID FIC-C200-I Undefined Undefined
2 E210 STM入料量 PID FIC-E210 Undefined Undefined
3 E203 BR 流量 PID FIC-E203 Undefined Undefined
4 E202 STM流量 PID FIC-E202 Undefined Undefined
5 M201 UPW流量 Fixed Control PID FIC-M201 Undefined Undefined
6 M401 NH3流量 Fixed Control PID FIC-C402-1 Undefined Undefined
7 V402 NH3流量 Fixed Control PID FIC-C402-2 Undefined Undefined
8 M401 UPW流量 Fixed Control PID FIC-M401-1 Undefined Undefined
9 M401 DHF流量 Fixed Control PID FIC-M401-2 Undefined Undefined
10 T103 UPW入料量 Indicating Alarm ALRM FI-T103-I Undefined Undefined

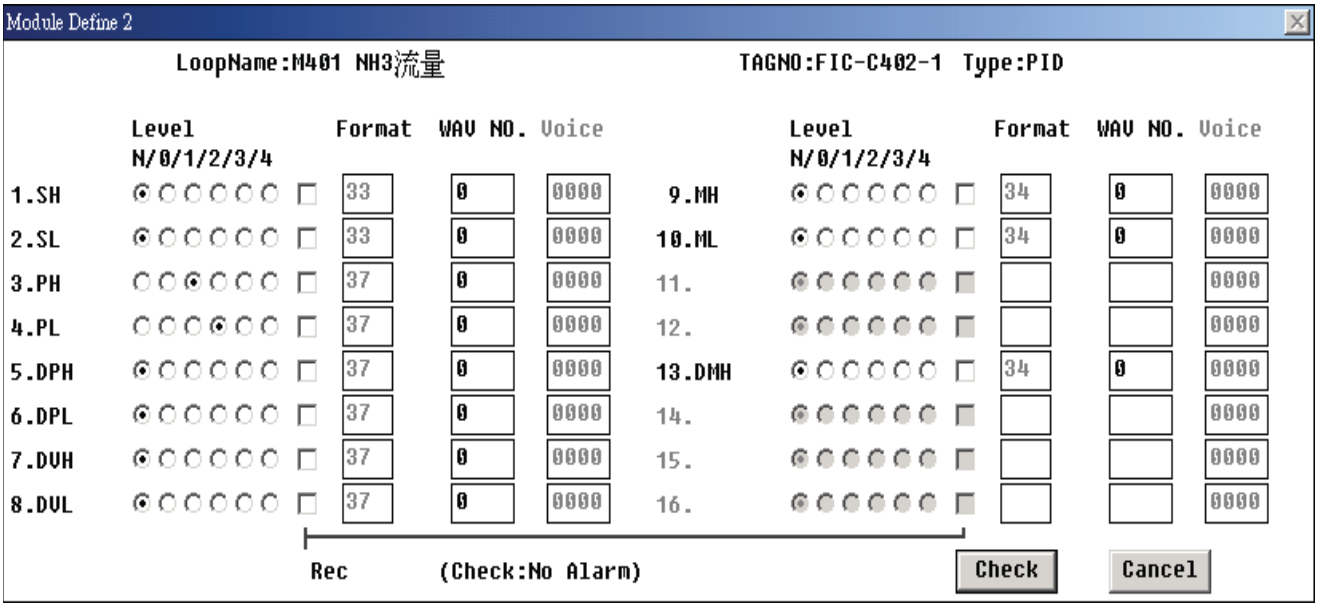
5---The selection of Loop type 3





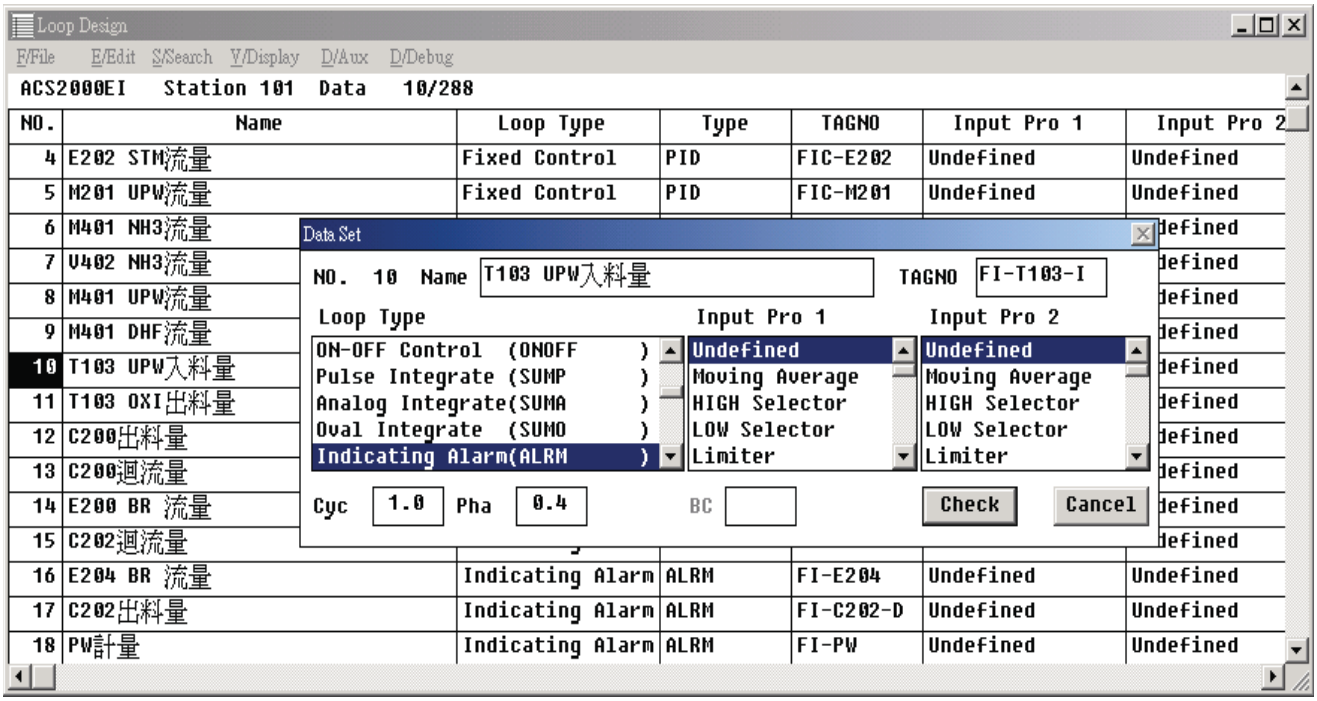
N:NO Alarm, 0:Level0(System alarm),1:Level1(Heavy alarm),2:level2(Middle alarm), 3:level3(Light alarm),4:level4 (signal occur)

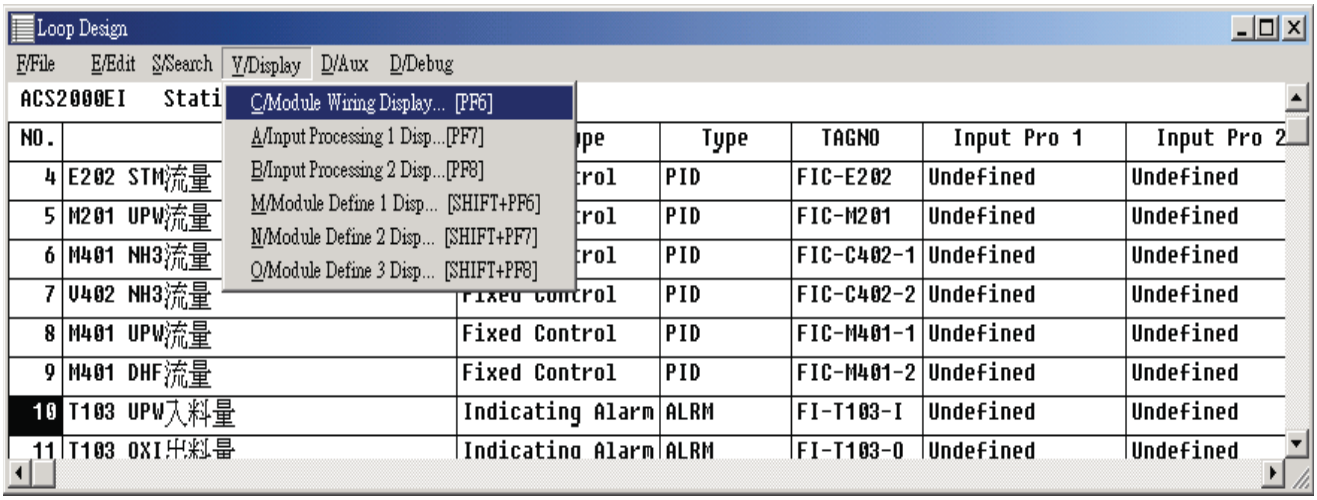
Priority: Level0> Level1> Level2> Level3> Level4

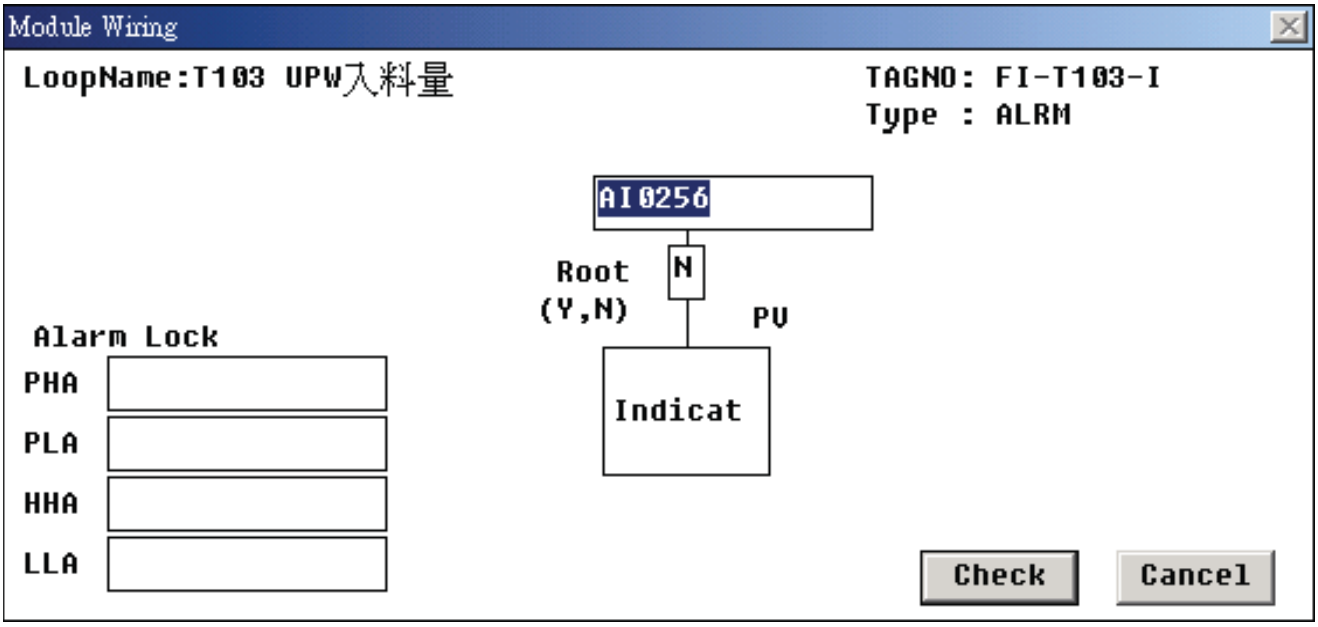


5---The selection of Loop type 4

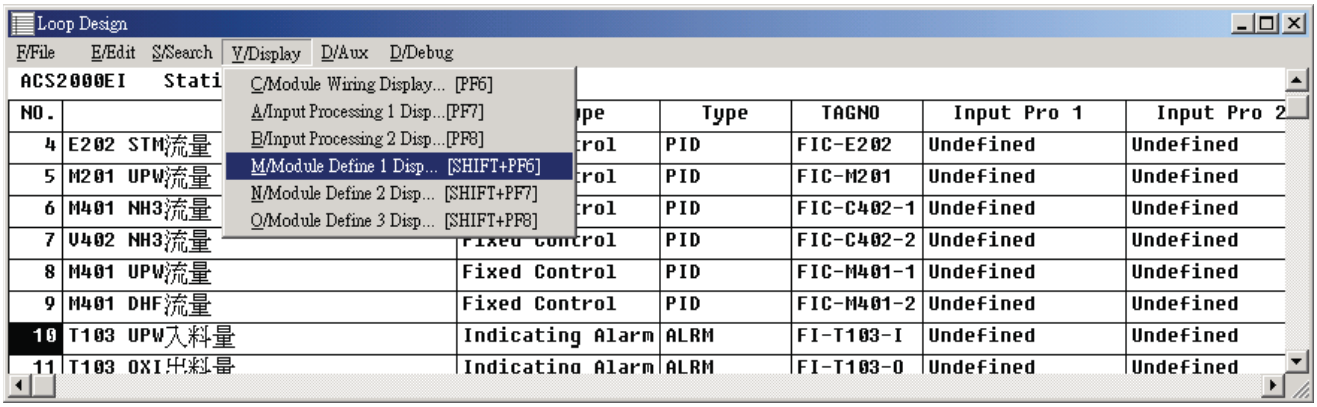
LOOP TYPE : ALRM

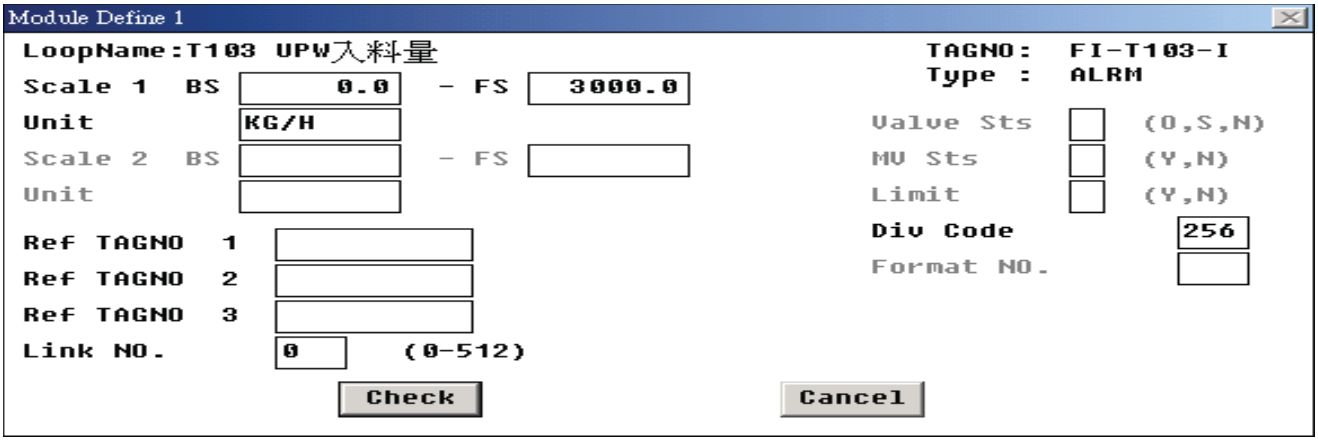


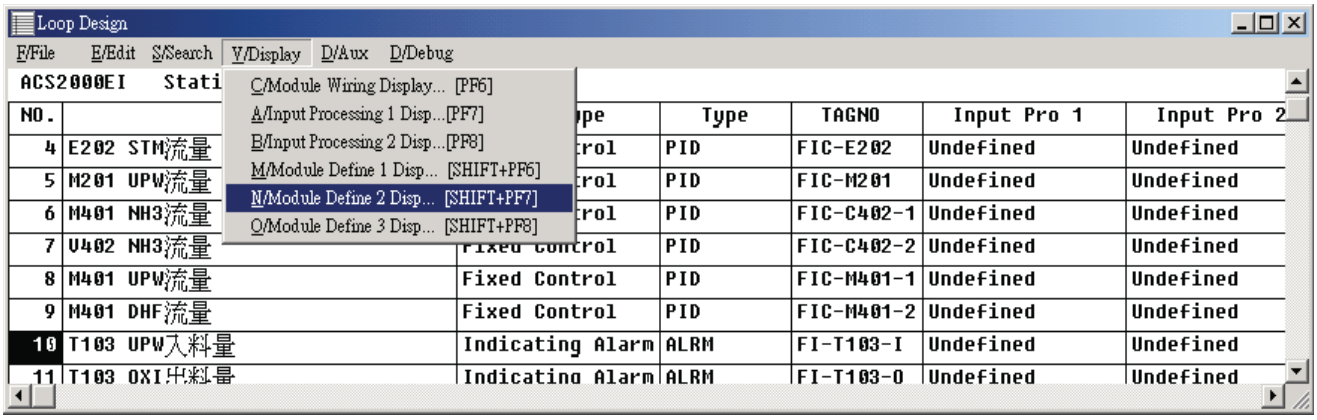




5---The selection of Loop type 5

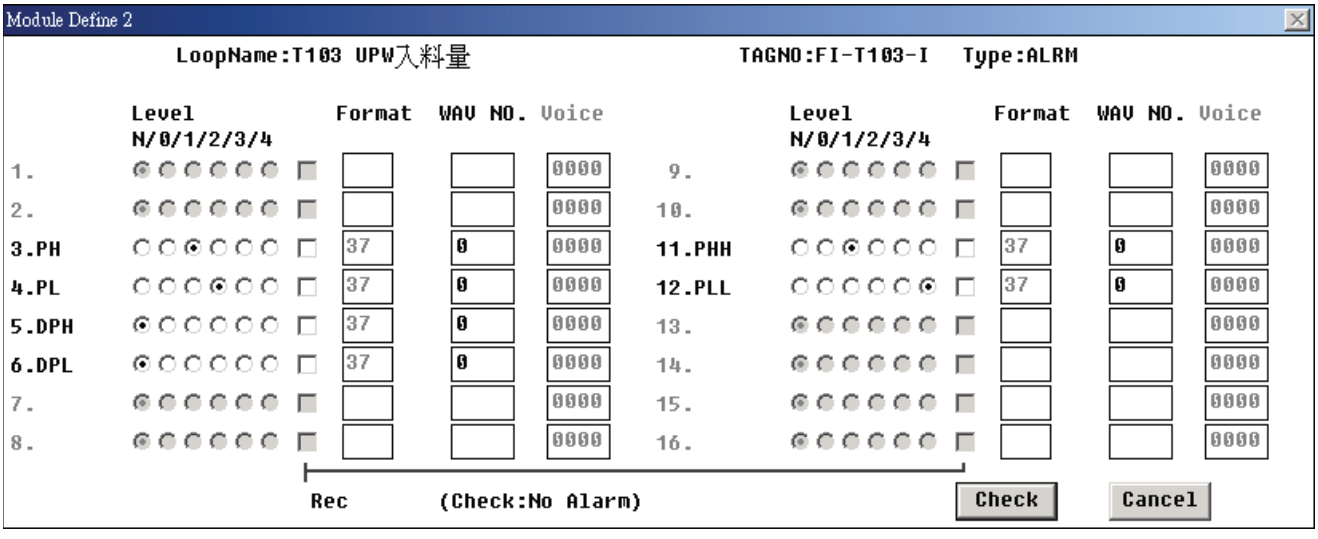






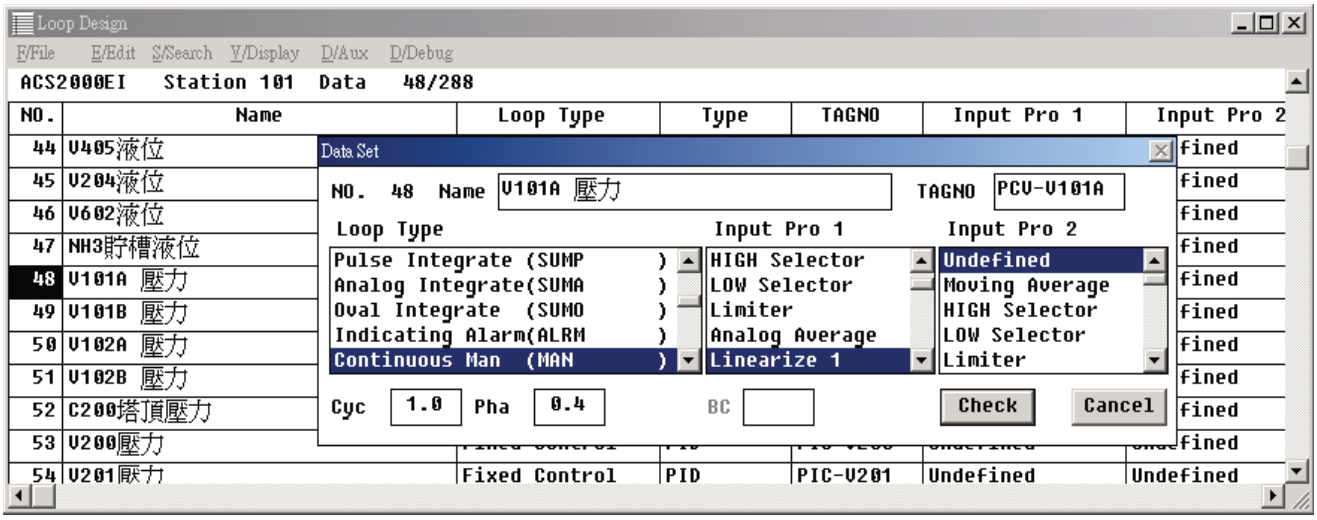
N:NO Alarm, 0:Level0(System alarm),1:Level1(Heavy alarm),2:level2(Middle alarm), 3:level3(Light alarm),4:level4 (signal occur)

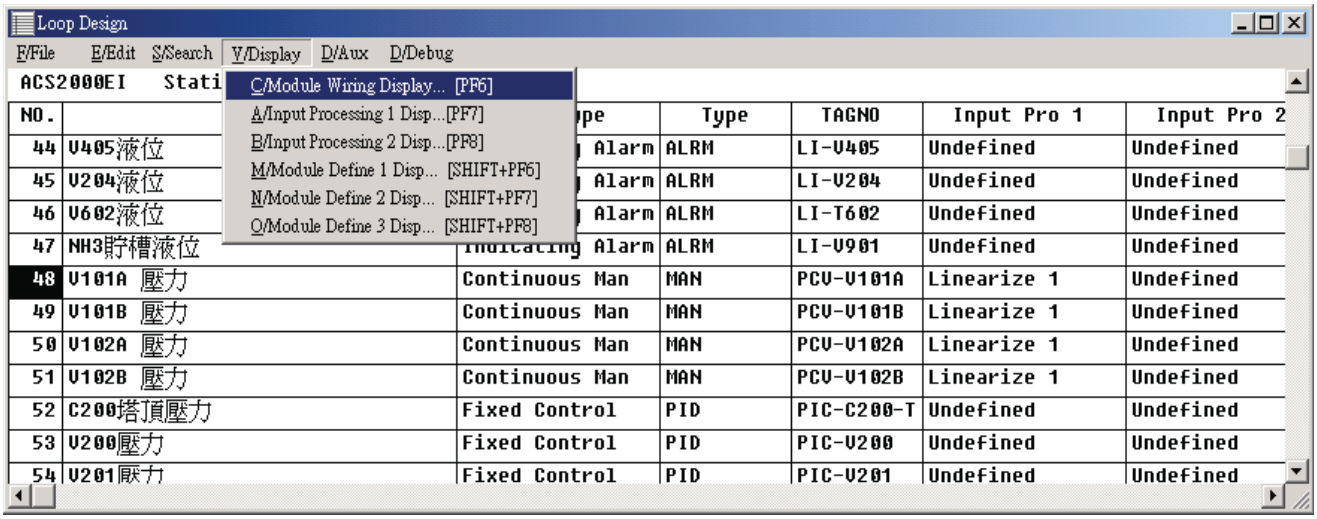
Priority: Level0> Level1> Level2> Level3> Level4

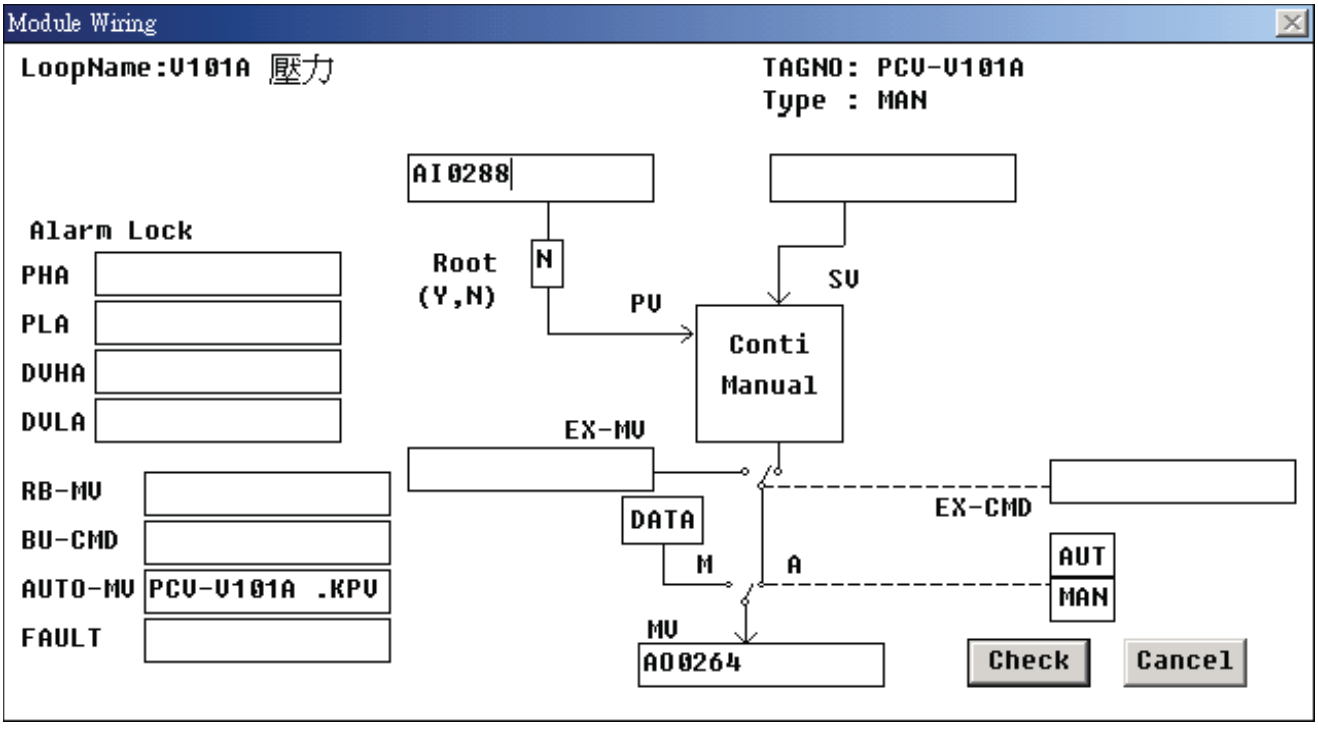


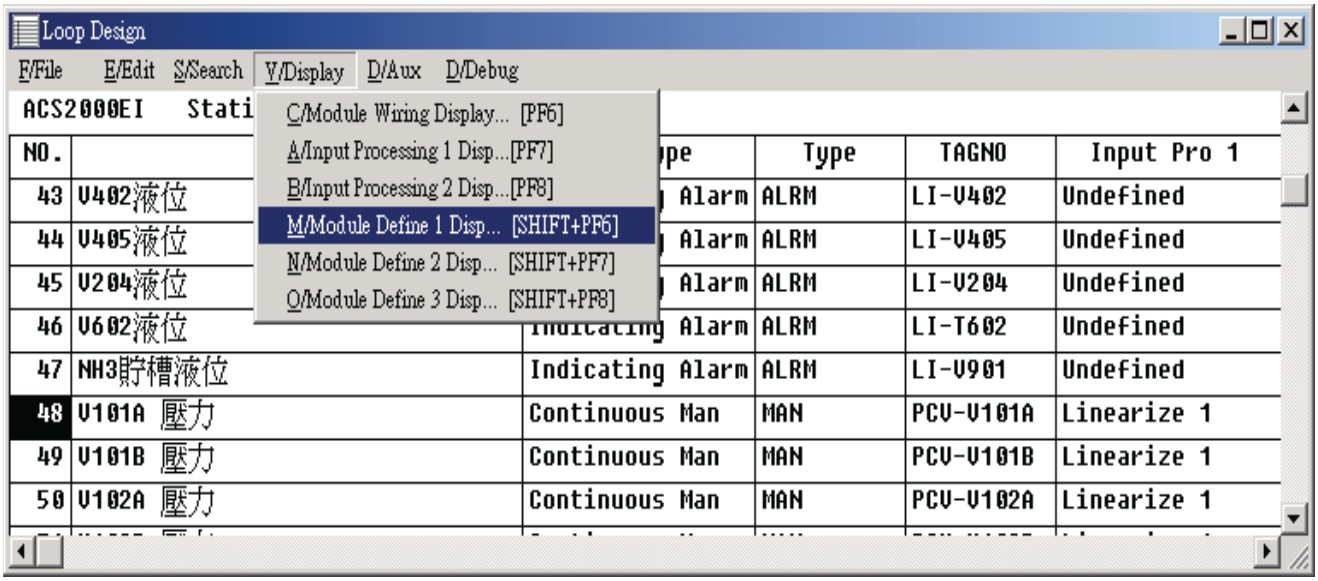
5---The selection of Loop type 6

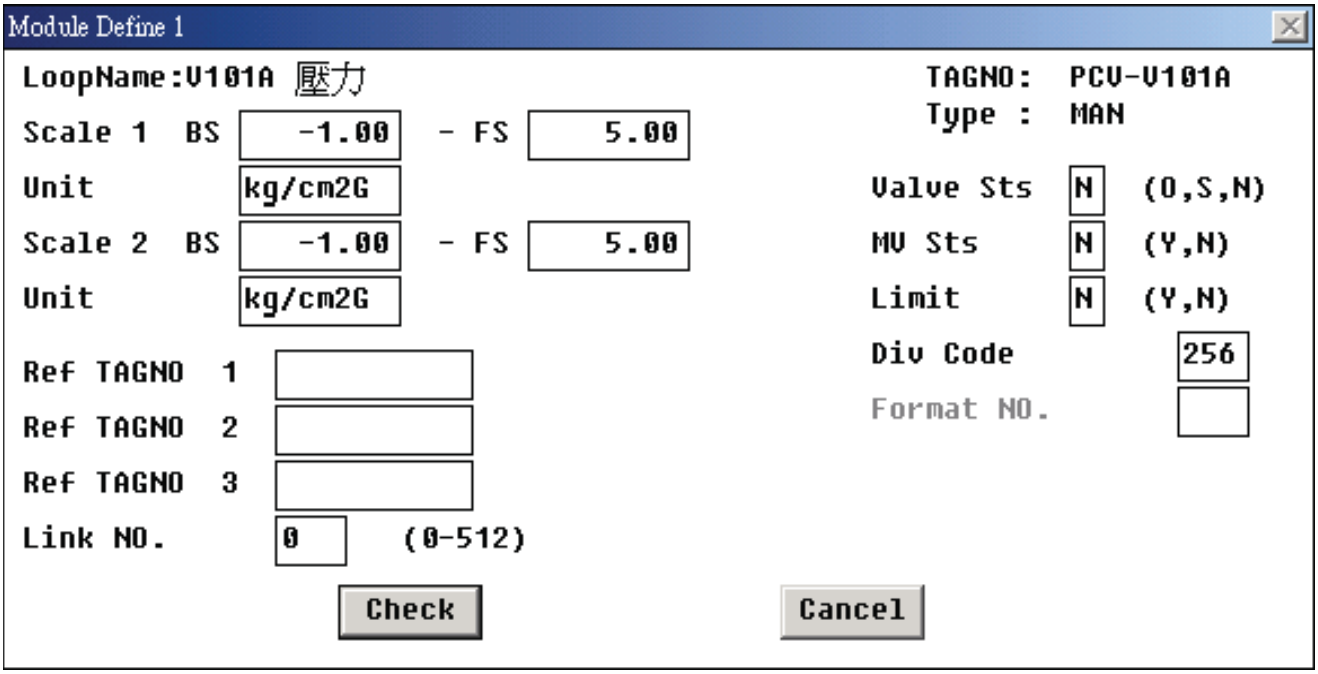
LOOP TYPE : MAN

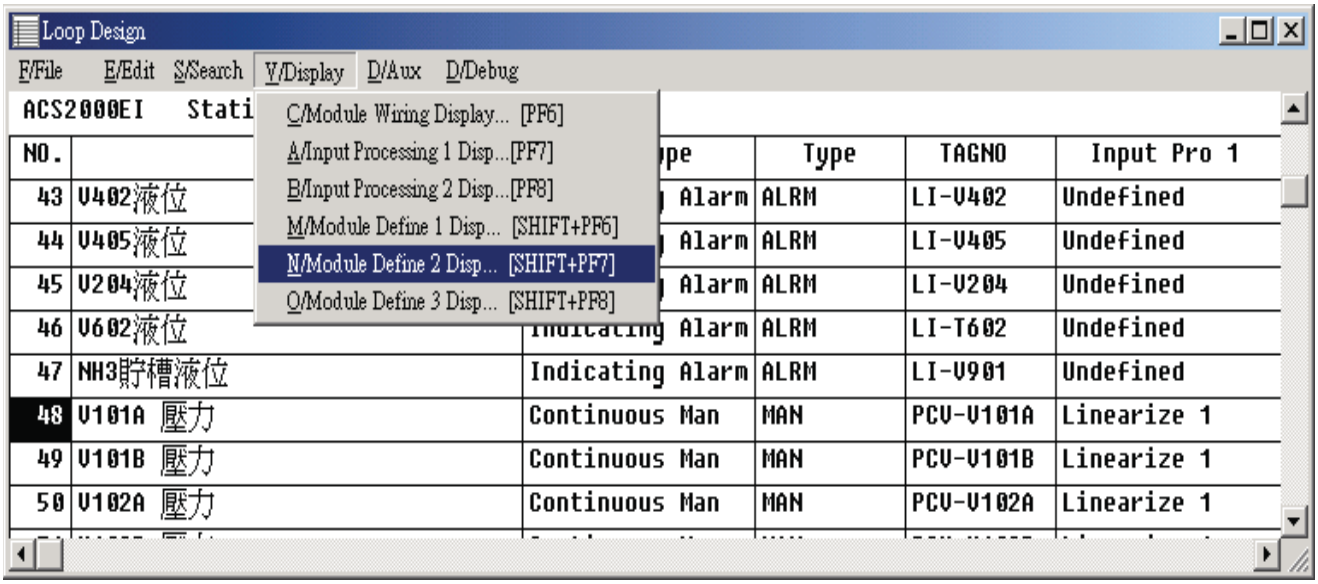








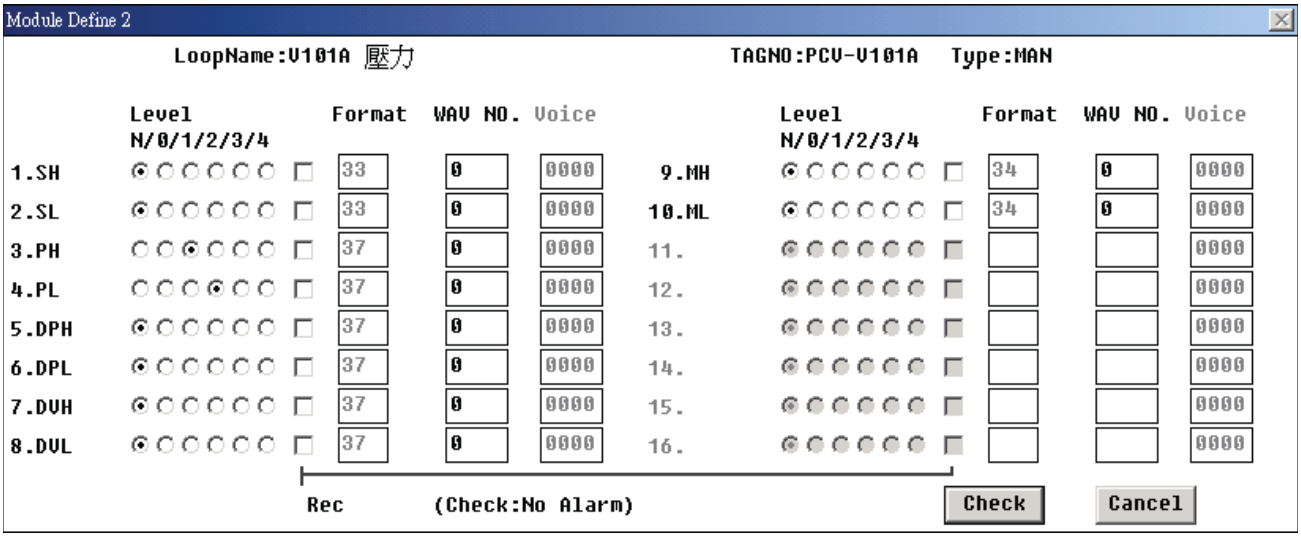




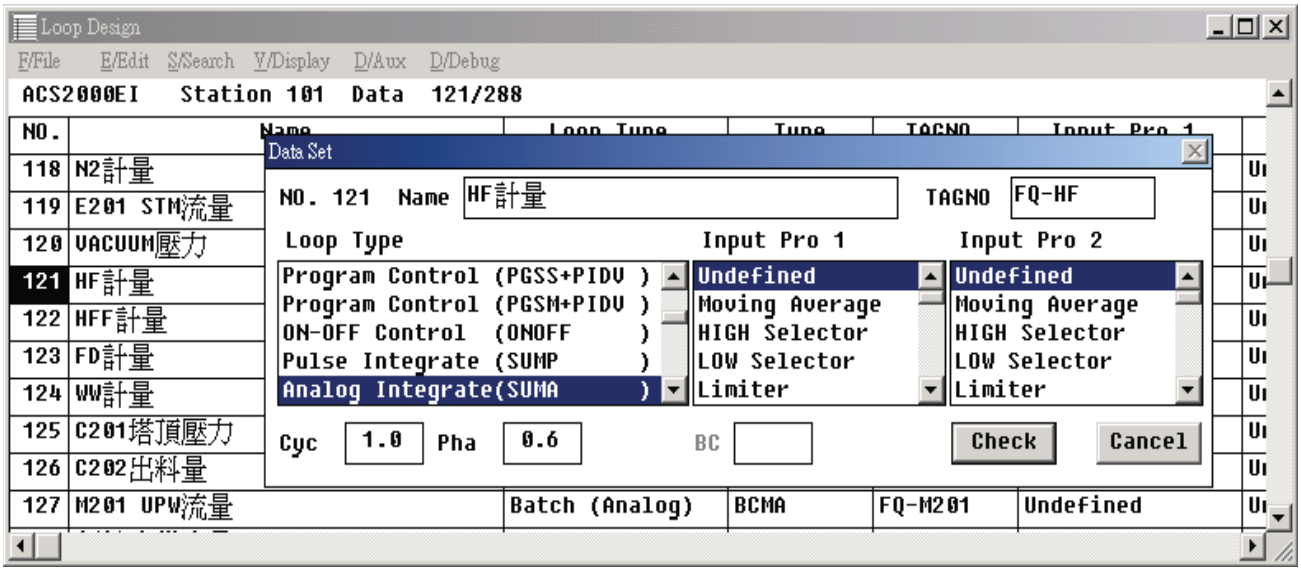
5---The selection of Loop type 8

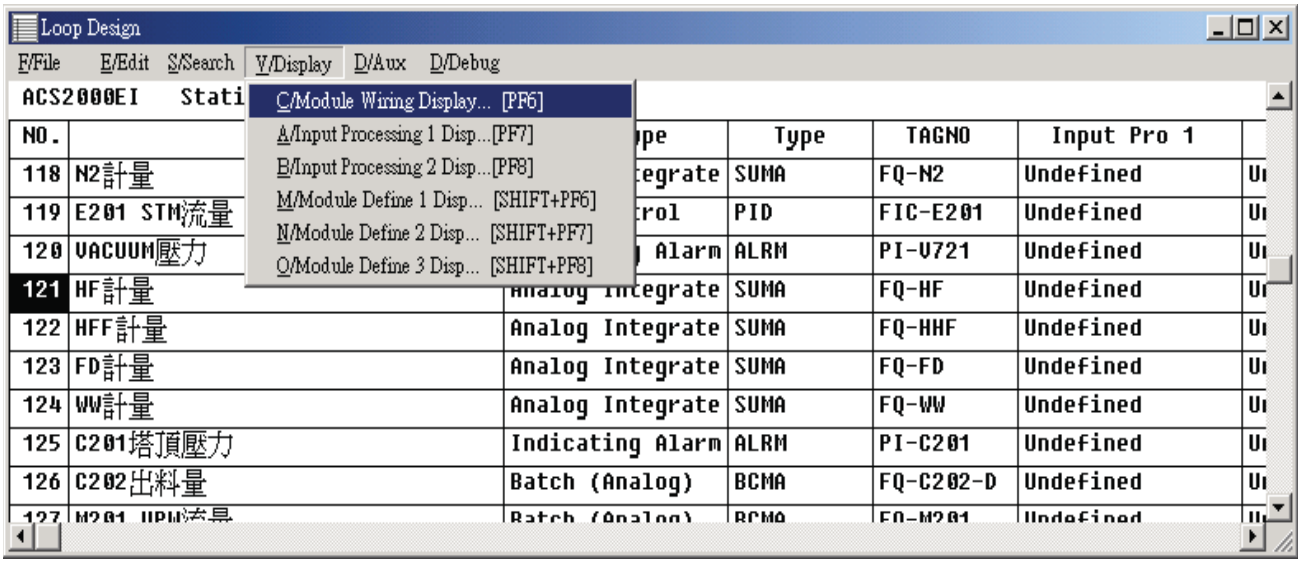
N:NO Alarm, 0:Level0(System alarm),1:Level1(Heavy alarm),2:level2(Middle alarm), 3:level3(Light alarm),4:level4 (signal occur)

Priority: Level0> Level1> Level2> Level3> Level4

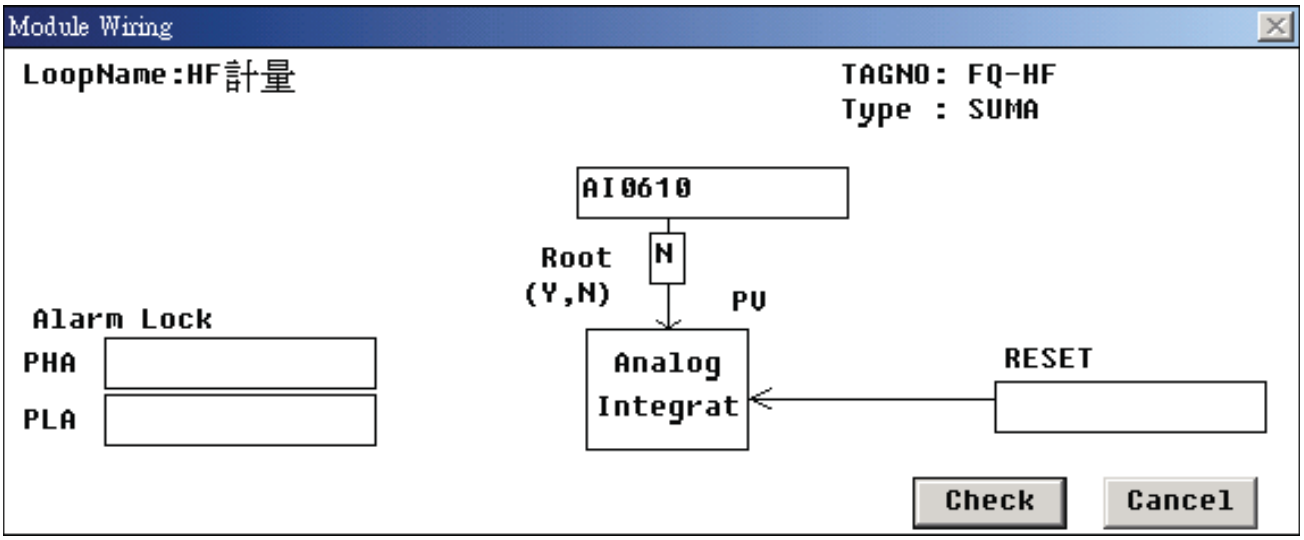


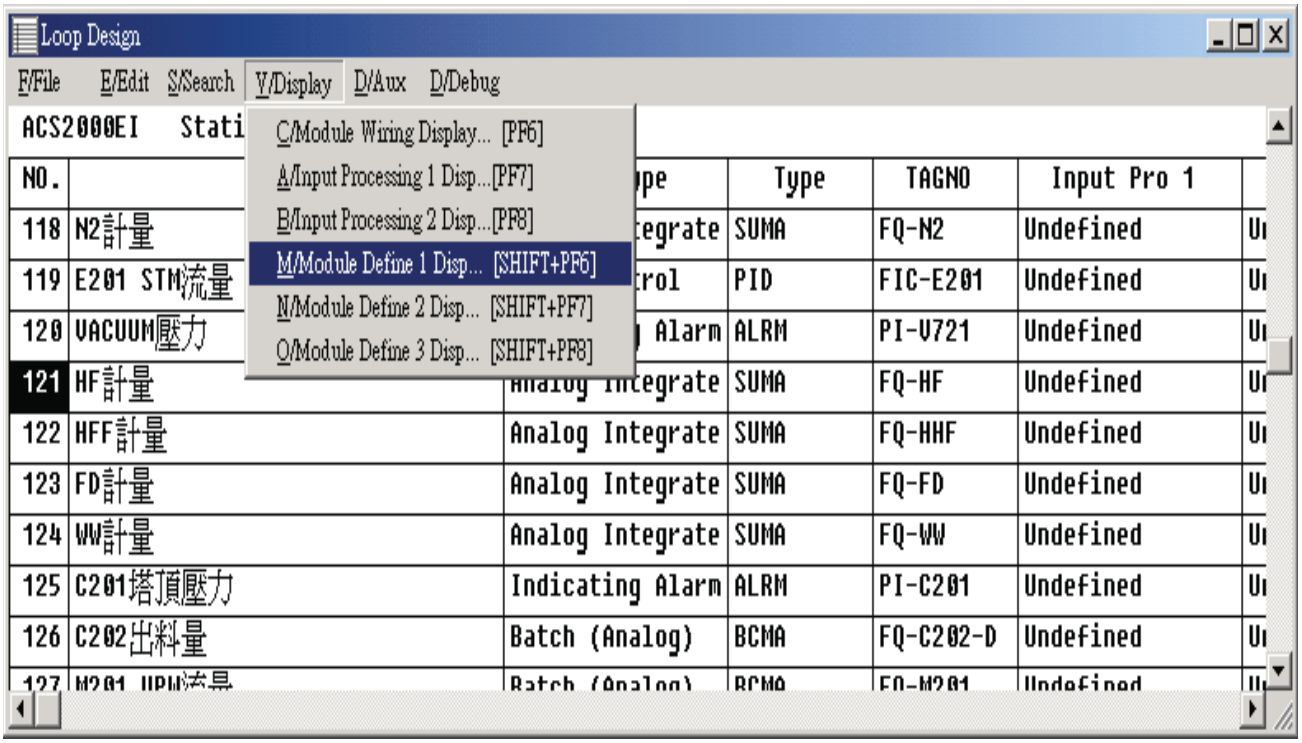
LOOP TYPE : SUMA

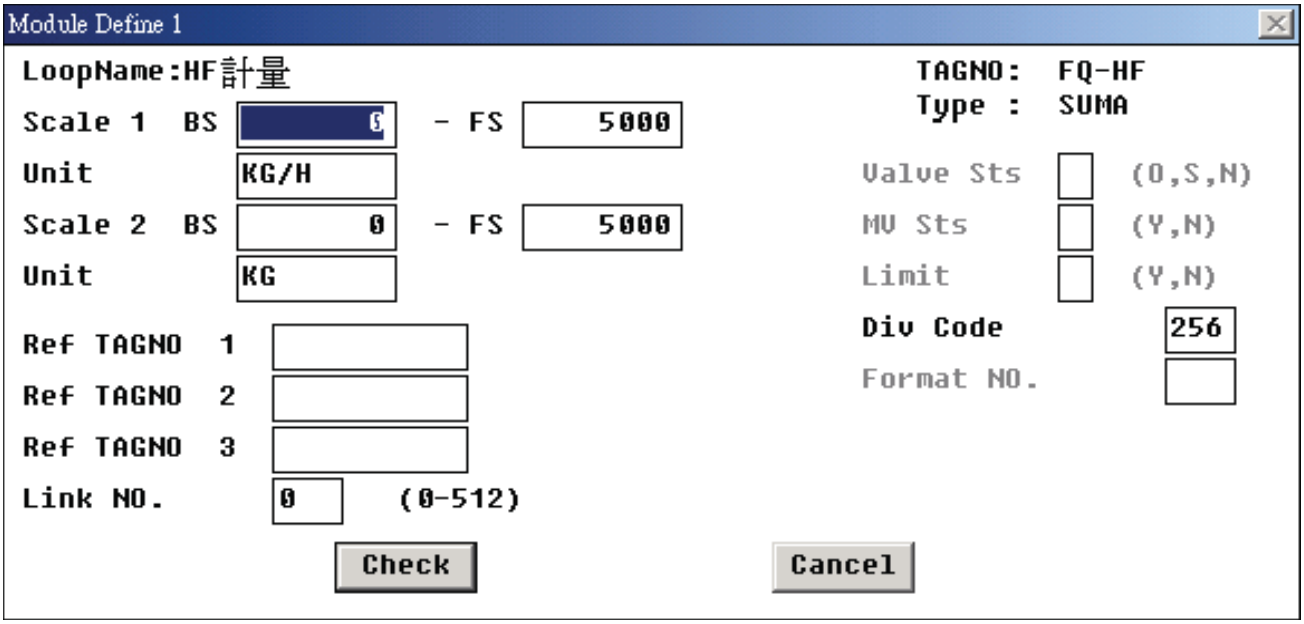




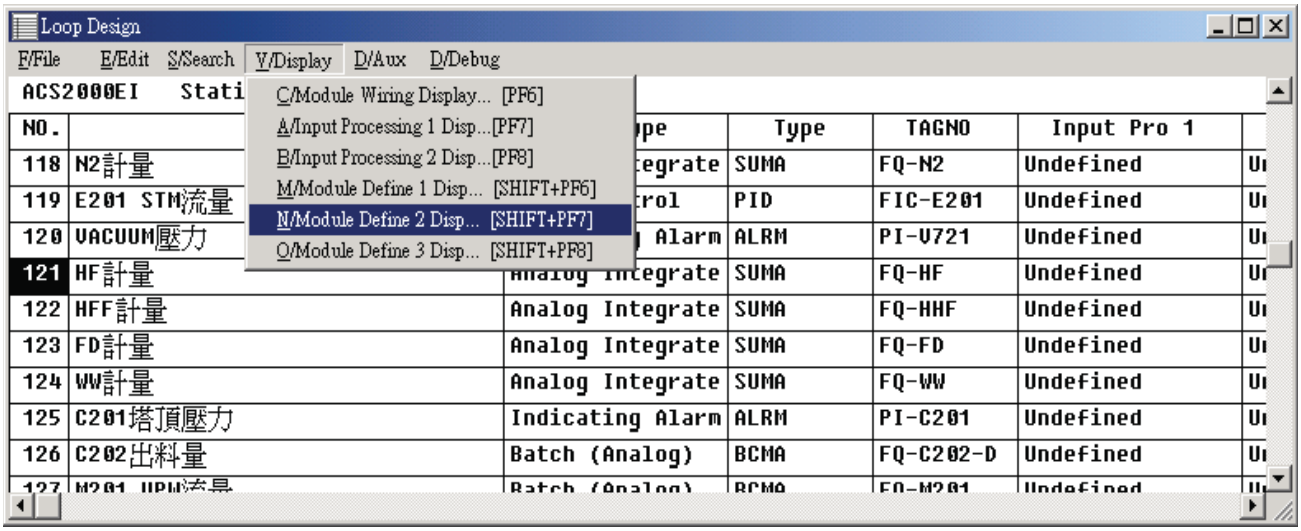
5---The selection of Loop type 9





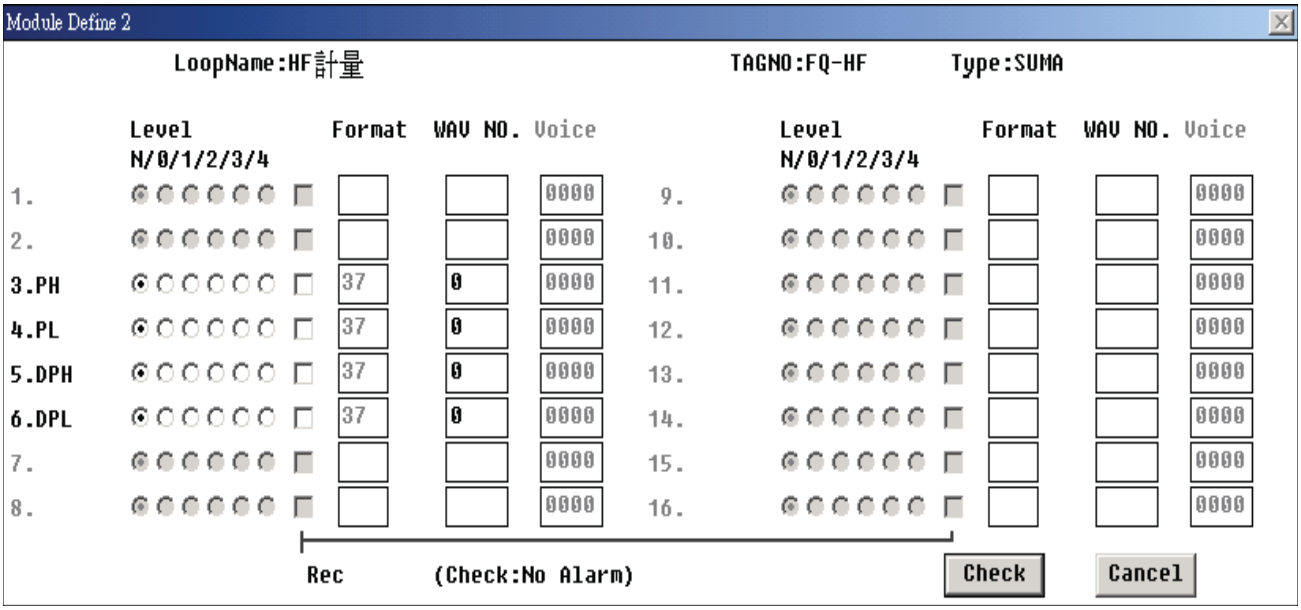


5---The selection of Loop type 10

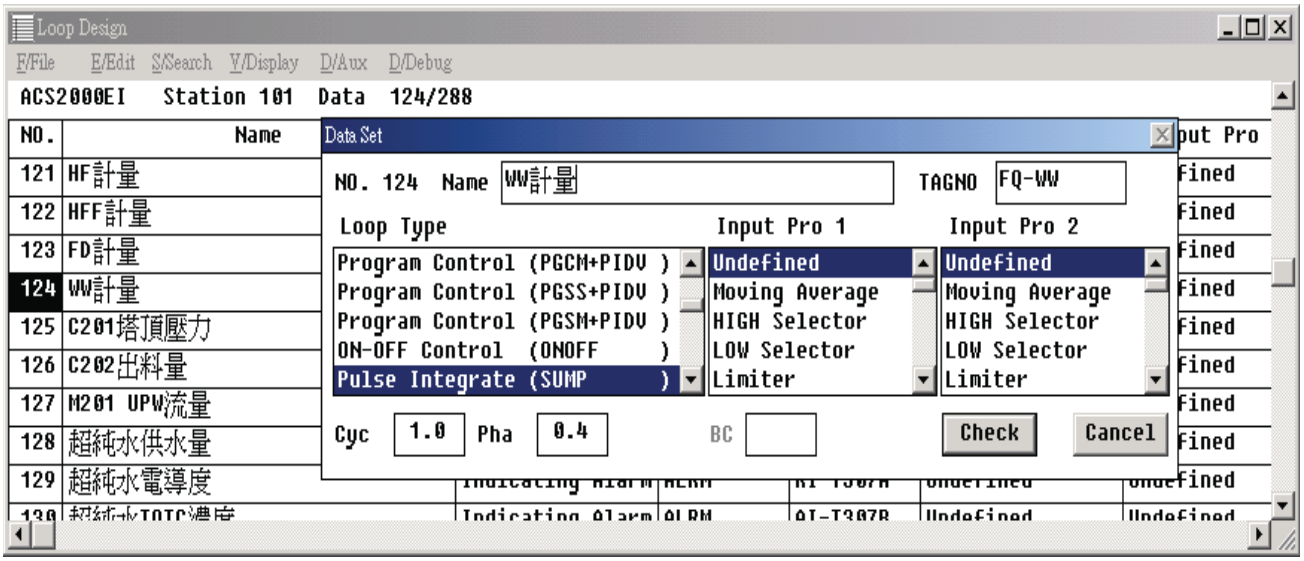


N:NO Alarm, 0:Level0(System alarm),1:Level1(Heavy alarm),2:level2(Middle alarm), 3:level3(Light alarm),4:level4 (signal occur)

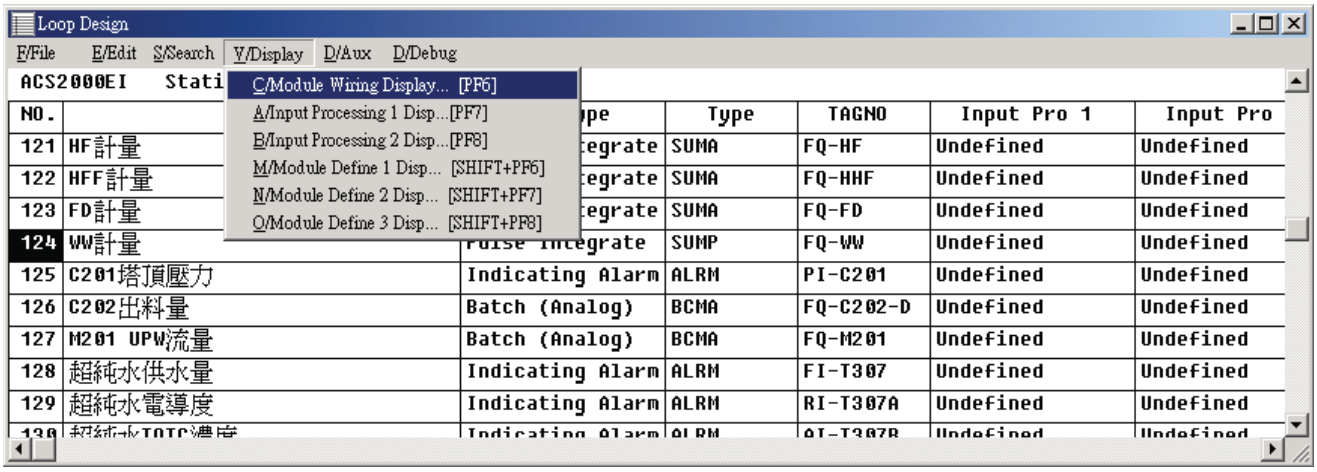
Priority: Level0> Level1> Level2> Level3> Level4

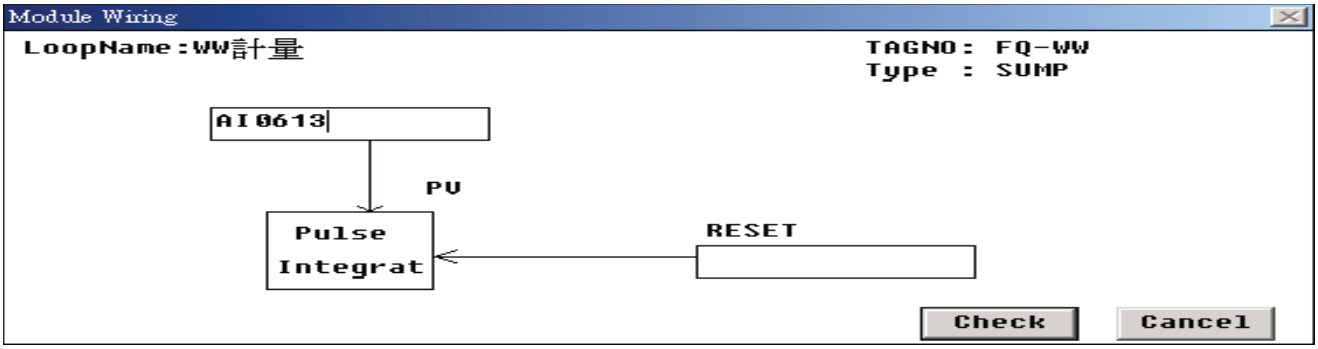


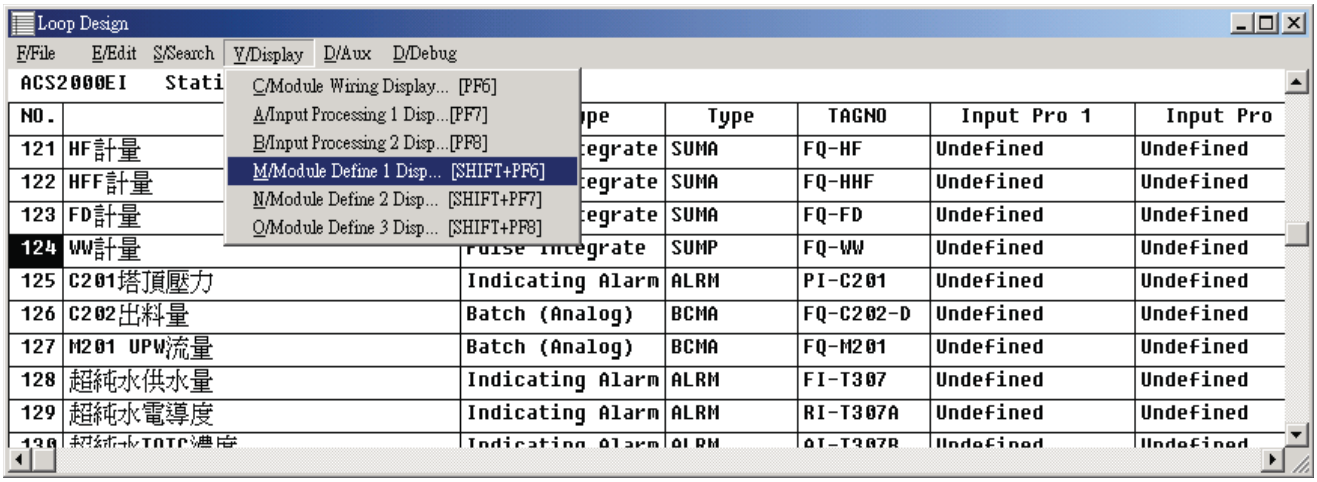
LOOP TYPE : SUMP

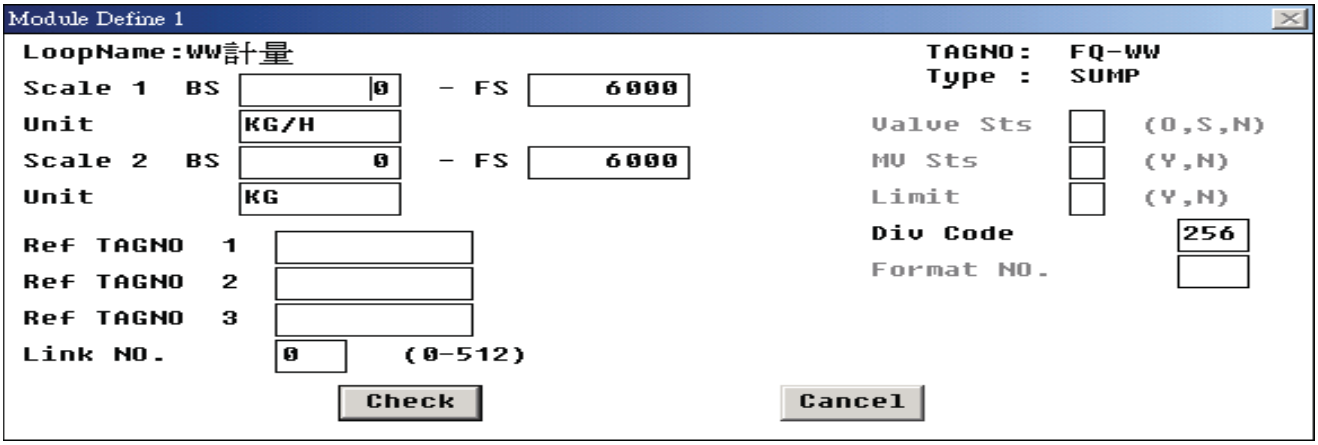


5---The selection of Loop type 11









Before you exit,please remember to save loop ,
how to download loop function ,please see Page 8~23 of loop function.

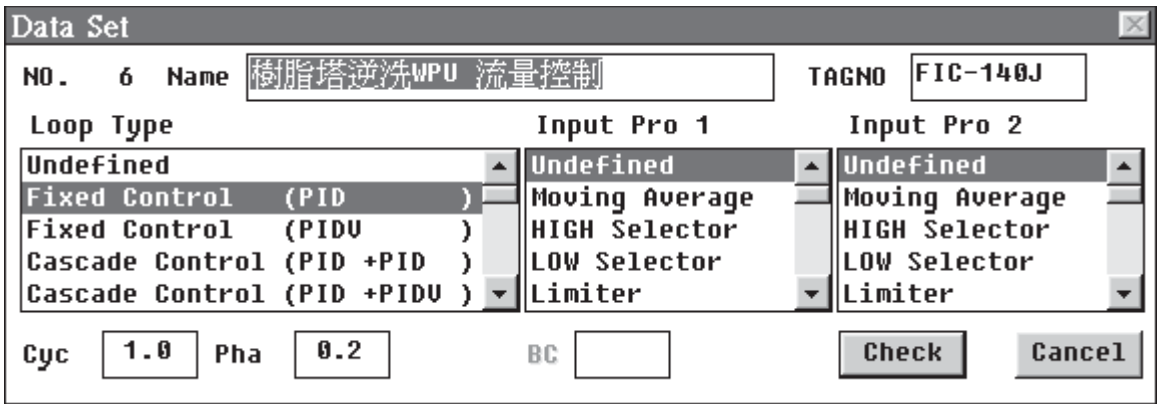
INPUT PROCESS : Every loop can provide 32 types input process for user.

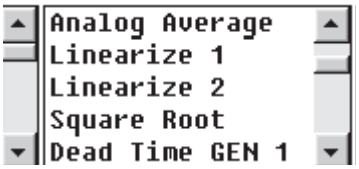
Every loop may define 2 input process, INPUT PRO1 and INPUT PRO2
Input process have 32 types, see the following picture.

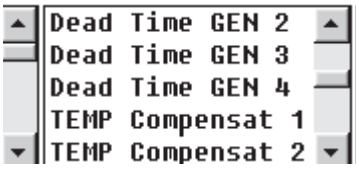
INPUT PRO 1's output tag use XXXX . KPV to express.

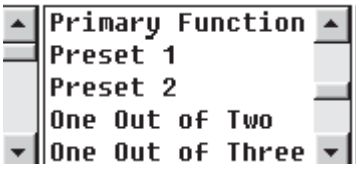
INPUT PRO 2's output tag use XXXX . KDT to express.

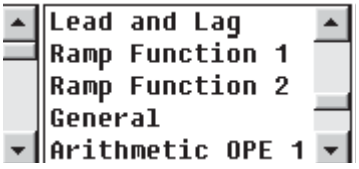
(PS: XXXX mean the TAGNO of loop)

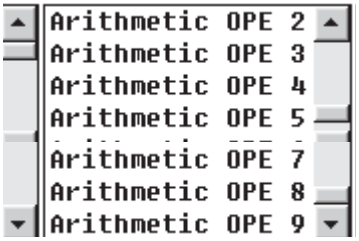








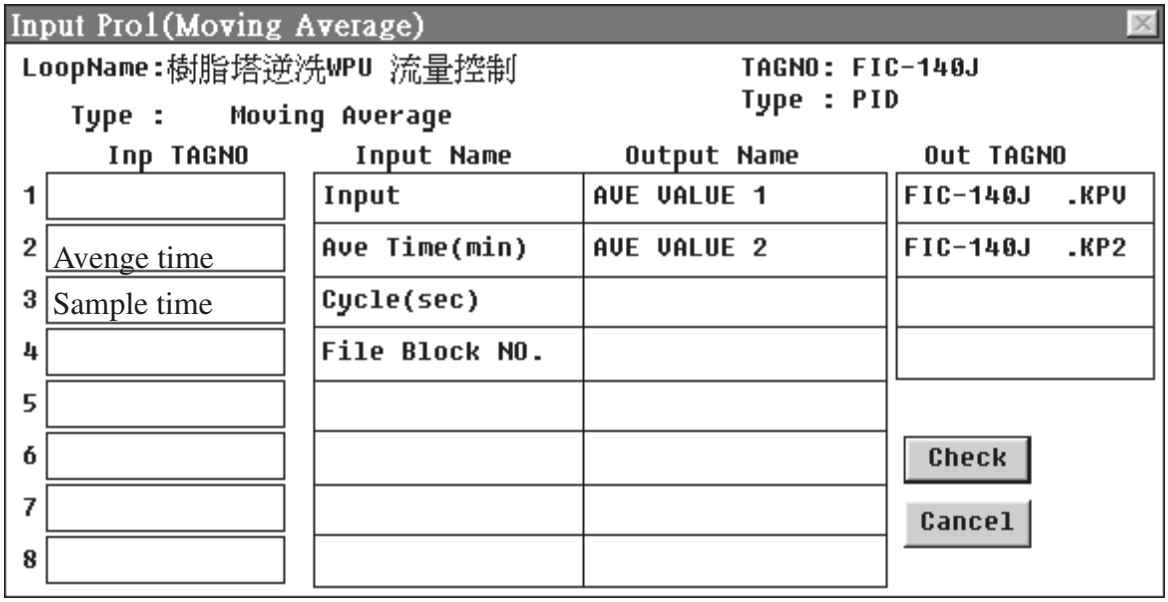




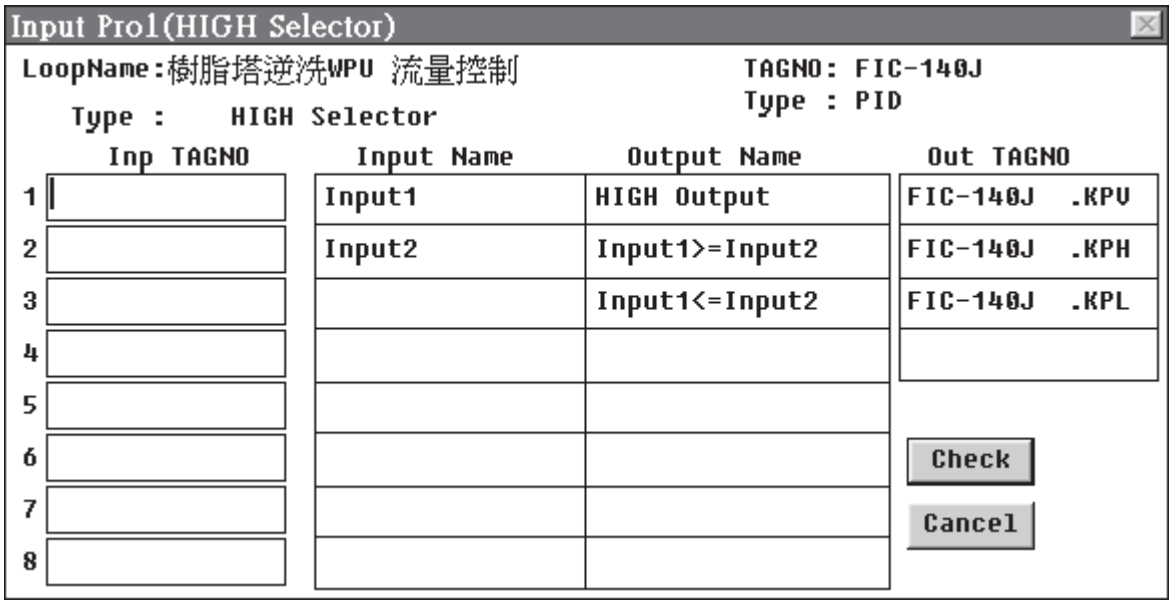
6---INPUT PROCESS OF LOOP 1

MOVING AVERAGE:

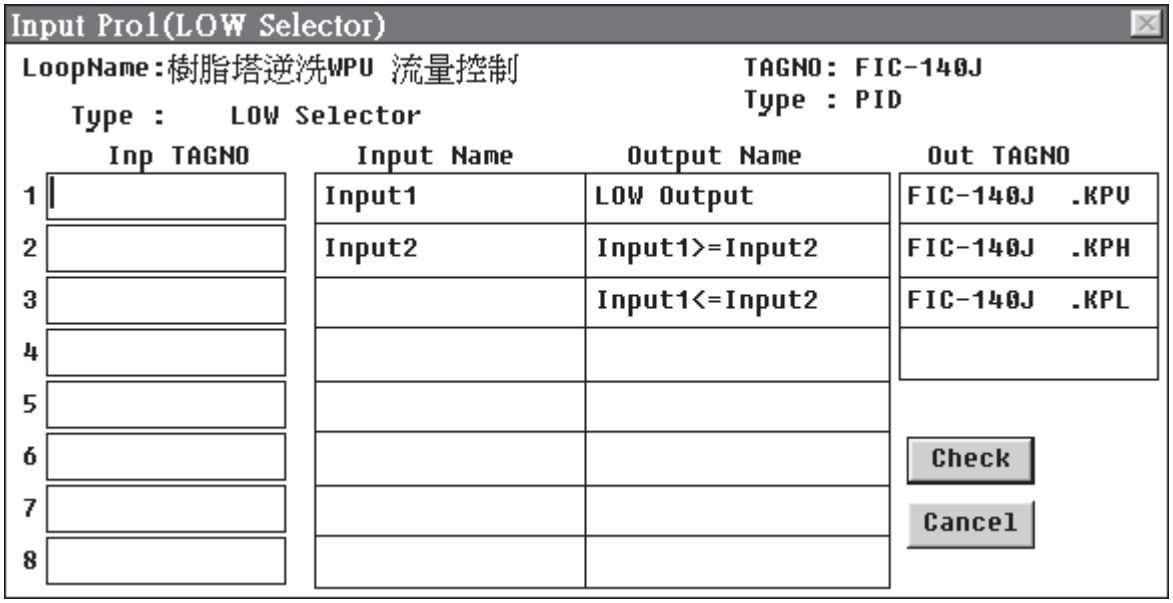
Field 1(Input) : Input the address of Moving avarage.
Field 4(File Block No) : Input the address of File Block No is 0～7 for one station.



HIGHT SELECTOR: mean comparing INPUT1 and INPUT2 , the heavier value export to FIC-140J.KPV

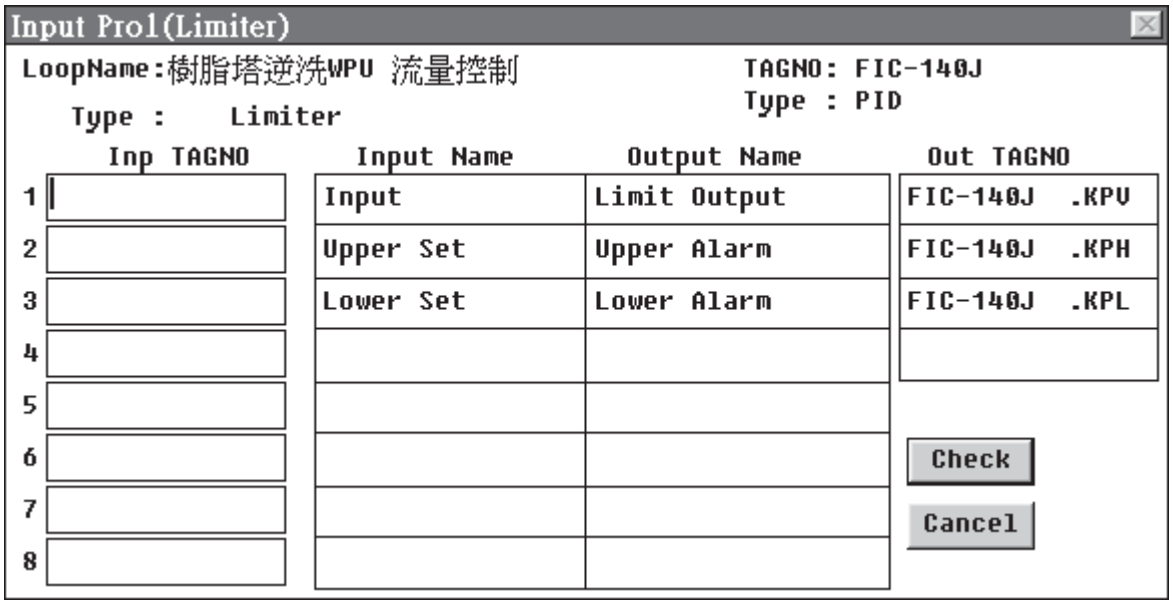


LOW SELECTOR: mean comparing INPUT1 and INPUT2 , the lighter value export to FIC-140J.KPV

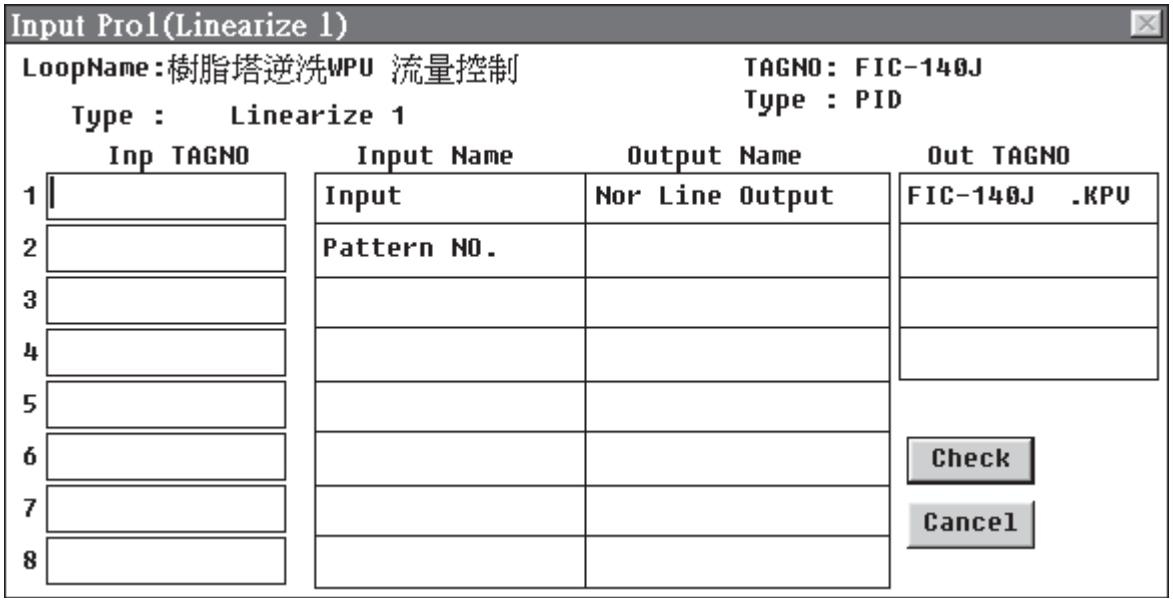


LIMITER: INPUT1 is compared with LOWER SET and UPPER SET,

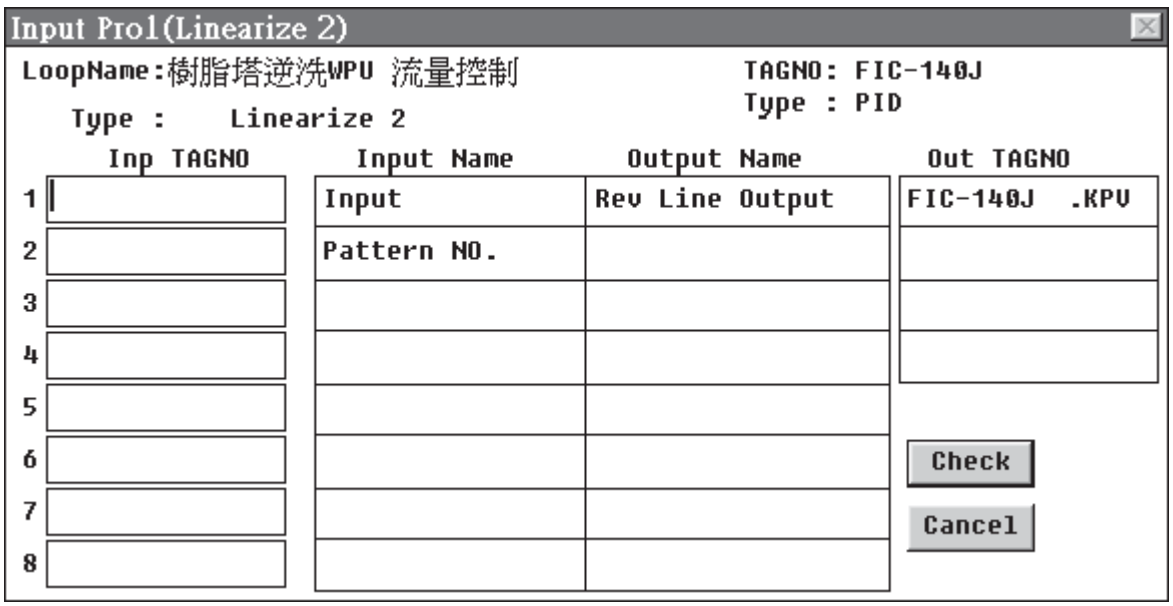
when UPPER SET > = INPUT1 > =LOWER SET input1's value is exported to FIC-140J.KPV



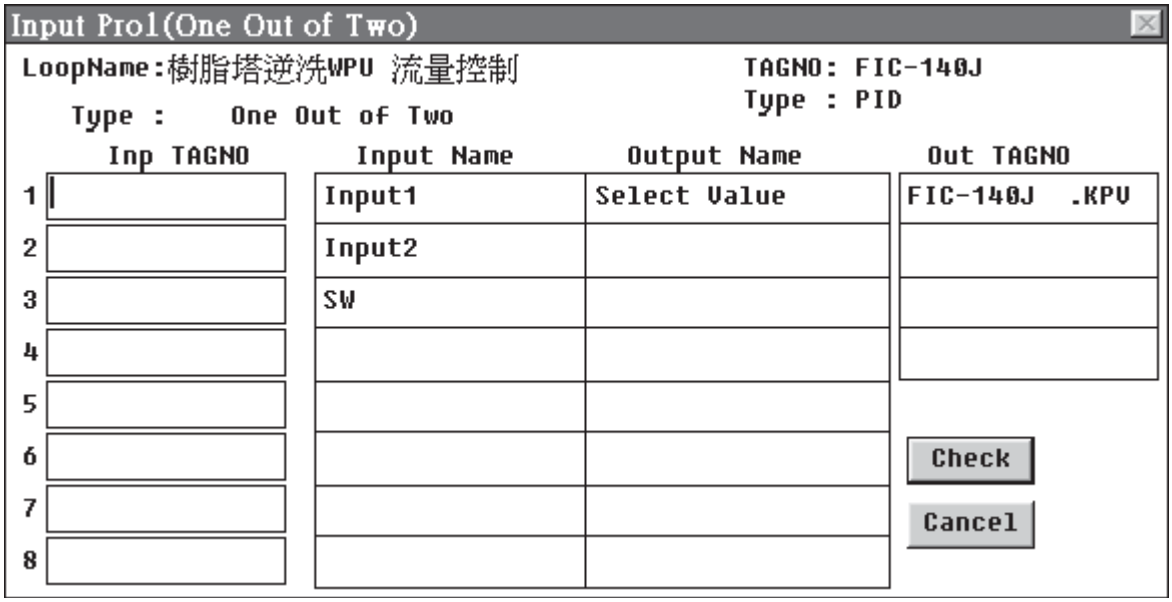
Linearize 1: Output is correspondent to pattern define, output TAG is FIC-140J.KPV, X axle express INPUT TAG , Y axle express OUTPUT TAG. It still need a linearize figure. The linearize figure define linear curve of figure by LINEAR.LNO of IFD function.



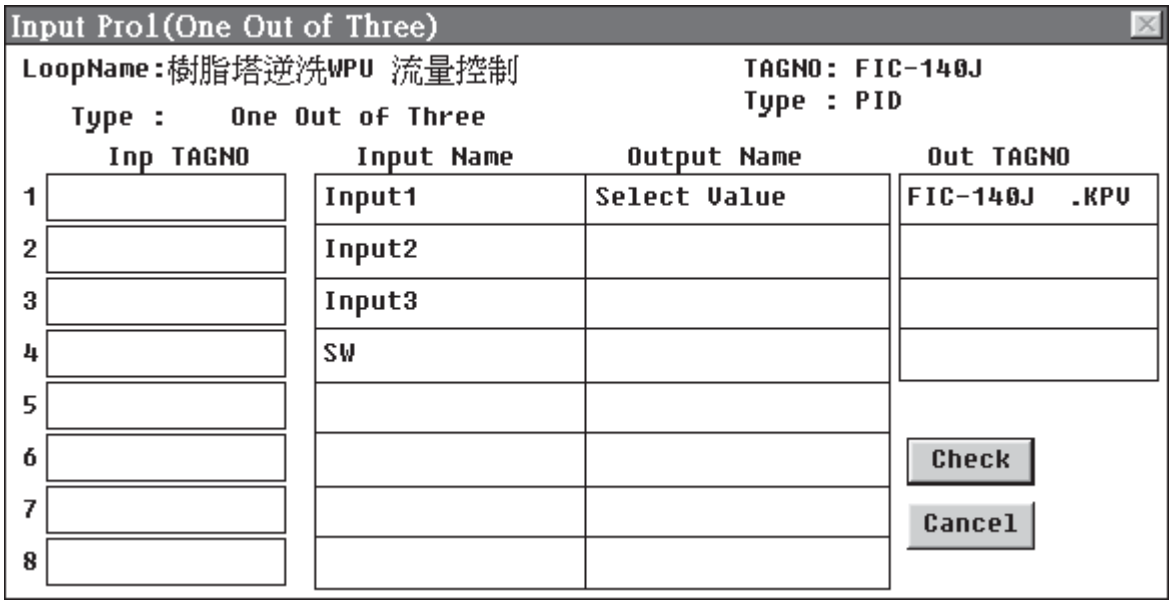
Linearize 2: Output is correspondent to pattern define, output TAG is FIC-140J.KPV, X axle express OUTPUT TAG , Y axle express INPUT TAG. It need a linearize figure. The linearize figure define linear curve of figure by LINEAR.LNO in IFD function.



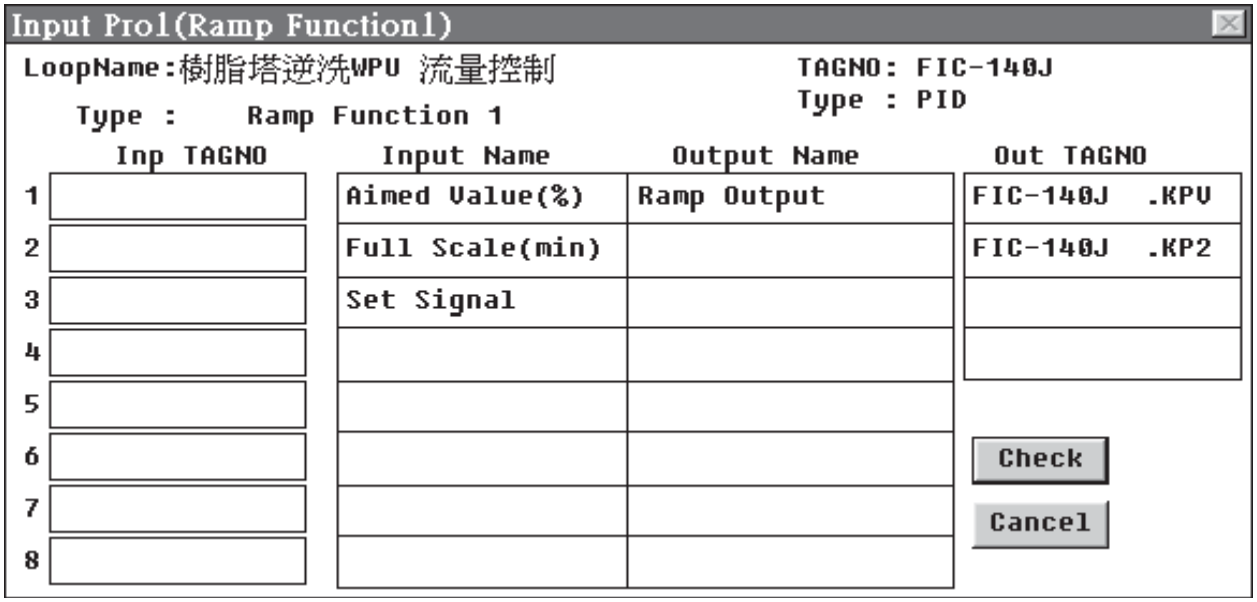
One out of two : Two Inputs (INPUT1, INPUT2) choose one to export, exports to FIC-140J.KPV , when SW = 0 use INPUT1 to export.
, when SW = 1 use INPUT2 to exports.
SW control use bit address.



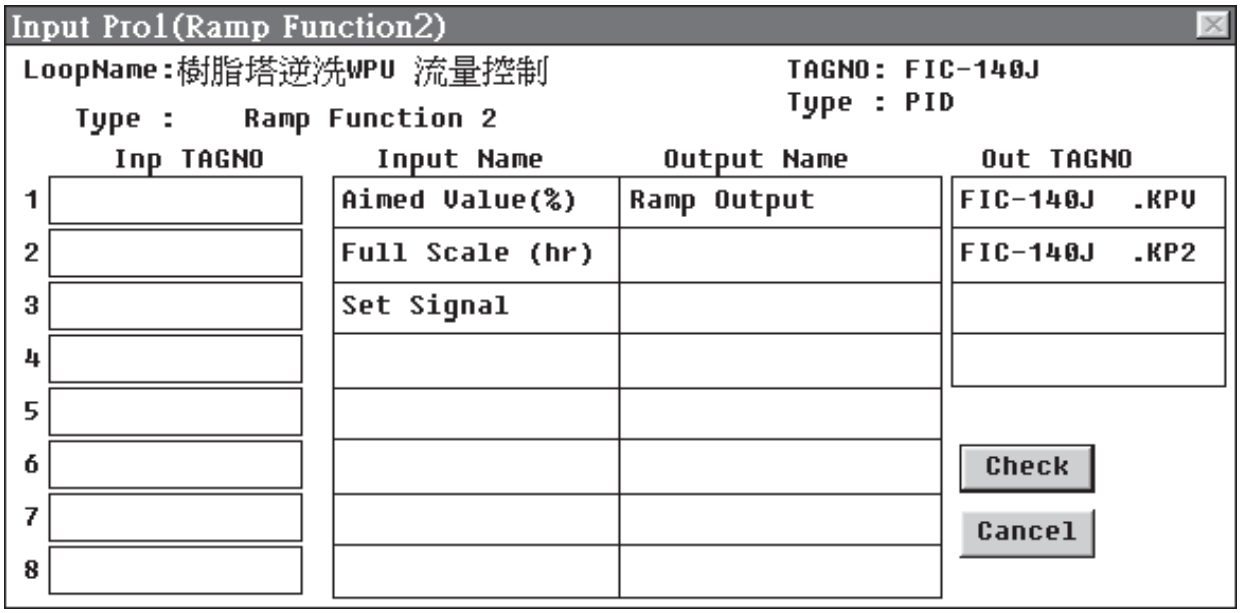
One out of three : Three Inputs (INPUT1, INPUT2, INPUT3) choose one to export, exports to FIC-140J.KPV, when SW = 0 exports to INPUT1
, when SW = 1 exports to INPUT2
, when SW = 2 exports to INPUT3
SW control use word address.



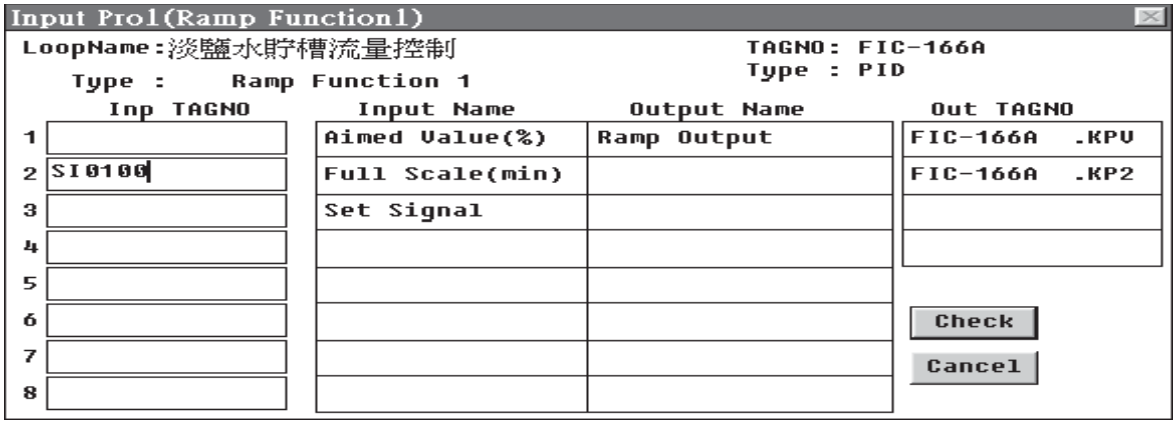
Ramp1 : When SET SIGNAL = 0 enable ramp function, the value of FIC-140J.KPV will rise to the setting up value within a specific time.
When SET SIGNAL = 1, FIC-140J.KPV = Aimed Value(%).
Regard minute as the unit.



Ramp2 : When SET SIGNAL = 0 enable ramp function, the value of FIC-140J.KPV will rise to the setting up value within a specific time.
When SET SIGNAL = 1, FIC-140J.KPV = Aimed Value(%).
Regard hour as the unit



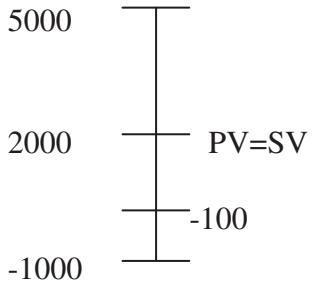
Ramp1: How to calculate the value of Full Scale Traveling Time by Ramp Time ?



Full Scale Traveling Time is Regarded minute as the unit, not second,
Full scale traveling time mean the period time from 0% rise to 100%.

$$\text{Full Scale Traveling Time} = \left[\left(\frac{\text{Input ramp time (sec)}}{\text{The resolution of (PV - Target value)}} \Big/ 60 \right) \right] \times 100$$

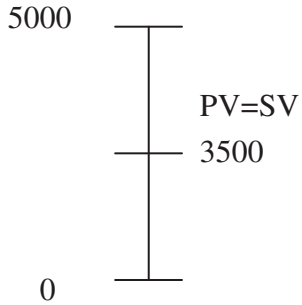
Ex: PIC-315X base scale = -1000, full scale = 5000 , now process value = 2000 ,
we hope it will reach goal value - 100 in 120 sec.



$$\text{Full Scale Traveling Time (SI0100)} = \left[\frac{120 \times 10000}{\left(\frac{2000-(-100)}{6000} \times 10000 \right)} \Big/ 60 \right] \times 100$$

$$= 571$$

Ex: PIC-400X base scale = 0 , full scale = 5000 , now process value = 3500 ,
we hope it will reach goal value 0 in 90 sec.



$$\text{Full Scale Traveling Time (SI0100)} = \left[\frac{90 \times 10000}{\left(\frac{3500-0}{5000} \times 10000 \right)} \Big/ 60 \right] \times 100$$

$$= 214$$

Arithmetic operation 1 :

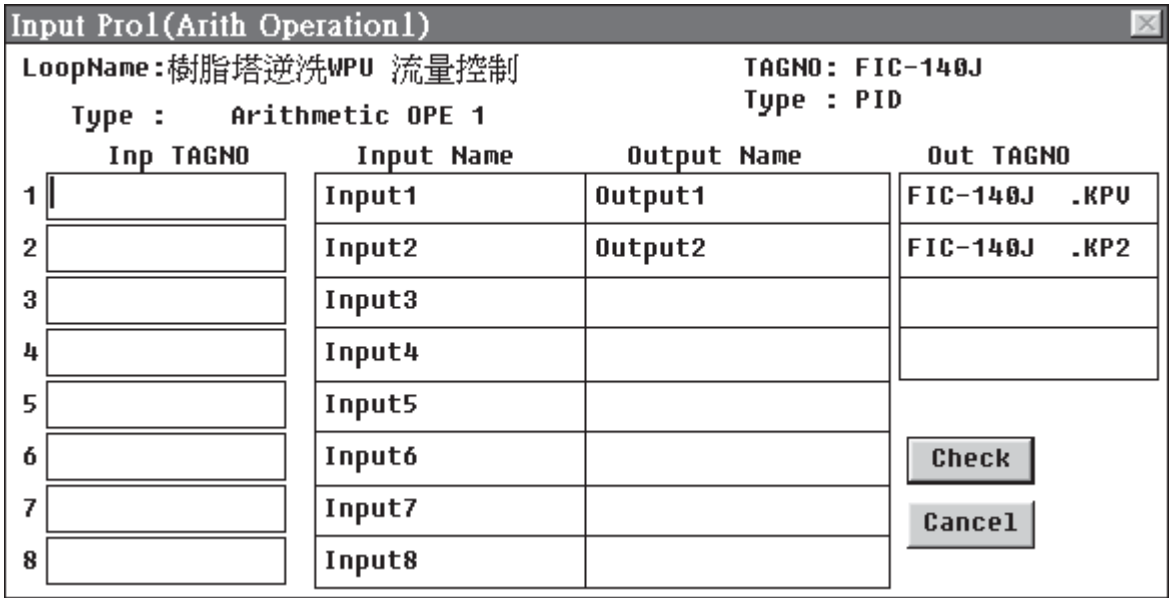
FUNCTION :

Y = A + B + C + D

Z = E + F + G + H

A:INPUT1 B:INPUT2 C:INPUT3 D:INPUT4 Y:OUTPUT1

E:INPUT5 F:INPUT6 G:INPUT7 H:INPUT8 Z:OUTPUT2



Arithmetic operation 2 :

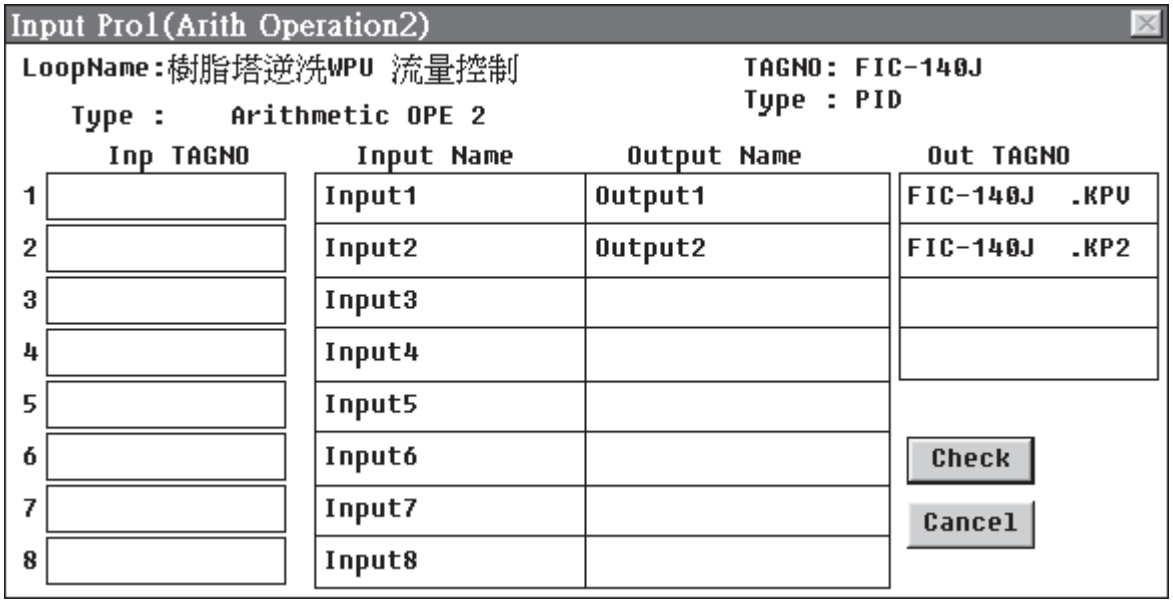
FUNCTION :

Y = A - B - C - D

Z = E - F - G - H

A:INPUT1 B:INPUT2 C:INPUT3 D:INPUT4 Y:OUTPUT1

E:INPUT5 F:INPUT6 G:INPUT7 H:INPUT8 Z:OUTPUT2

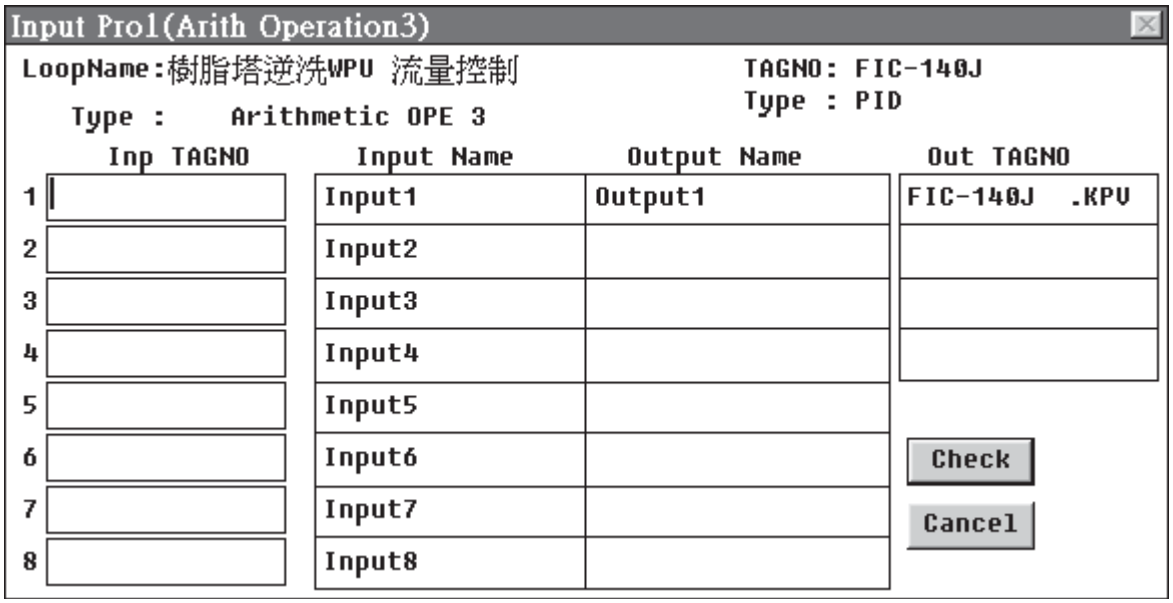


Arithmetic operation 3 :

FUNCTION :

$$Y = (A + B) \times C/ D + (E + F) \times G/ H$$

A:INPUT1 B:INPUT2 C:INPUT3 D:INPUT4 Y:OUTPUT1
E:INPUT5 F:INPUT6 G:INPUT7 H:INPUT8



If you want to understand the other function of input process , please refer to the instruction manul of ICS CONTROL FUNCTIONS EDITION.

The following is the parameter of flow meter in 100，700 area：

NO	TAGNO	PULSE RATE	MAX FLOW RATE	K(P/L)	KG	KPT	PMAX(P/S)	P16	STN	RMK
01	FQ136A	1L/P	65M^3/HR	0.01P/0.01L	100	4	18.06	0	101	OVAL
02	WQ136A	1L/P	65M^3/HR	0.01P/0.01L	100	4	18.06	0	101	OVAL
03	FS136A	1L/P	65M^3/HR	0.01P/0.01L	100	4	18.06	0	101	OVAL
04	FQ136B	0.01KG/P	72T/HR	1P/0.01KG	100	2	200	1	101	MASS
05	WQ136B	0.01KG/P	72T/HR	1P/0.01KG	100	2	200	1	101	MASS
06	FS136B	0.01KG/P	72T/HR	1P/0.01KG	100	2	200	1	101	MASS
07	WQ705A	10L/P	340 M^3/HR	0.1P/L	10	2	9.44	0	102	OVAL
08	WQ705B	0.01KG/P	340M^3/HR	36P/KG	360	1	34	2	102	MASS
09	FQ706A	10L/P	340M^3/HR	0.1P/L	10	2	9.44	0	102	OVAL
10	WQ706A	10L/P	340M^3/HR	0.1P/L	10	2	9.44	0	102	OVAL
11	WQ706B	0.01KG/P	340T/HR	36P/KG	360	1	34	2	102	MASS
12										
13										
14										

Notes :

1. $K=KG * 10^{-KPT}$

2.KG : 1 ～ 32767 , KPT : 0 ～ 4 , PMAX : 0.01 ～ 327.67

$K=1L/P=1\ P/L=0.01\ P/\ 0.01L=100 * 10^{-4}$

65 M^3/HR=(65 * 1000 / 60 * 60 L/SEC) * 1P/L =18.06 P/SEC

MODE INSTRUCTION

0	1	2	3	4	5	6	7	8	9	A	B	C	D	E	F
8	4	2	1	8	4	2	1	8	4	2	1	8	4	2	1

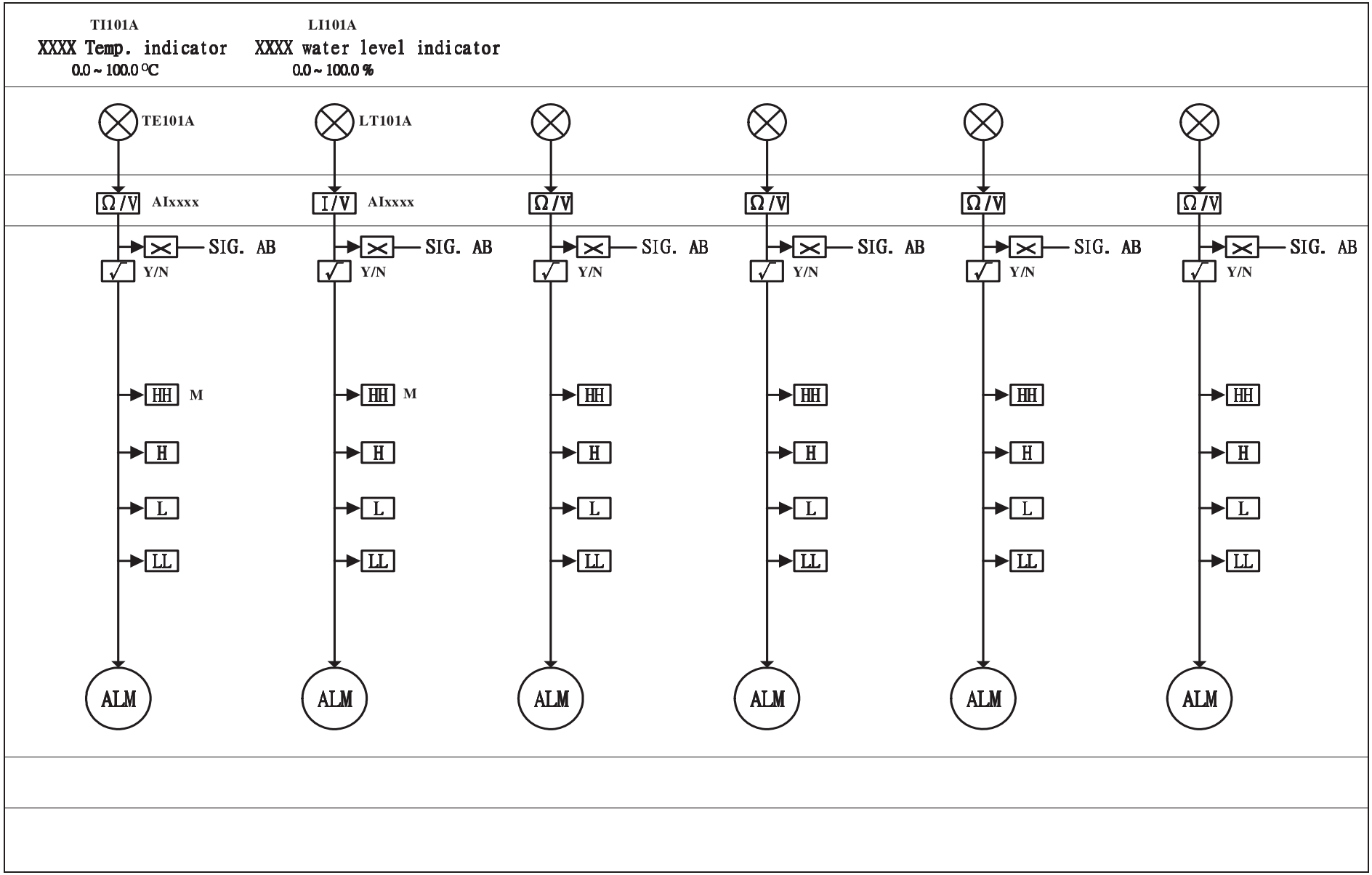
		PID	PIDV	RF	PGCS	PGCM	PGSS	PGSM	ON OFF	SUMA	SUMP	SUMO	ALRM	MAN	MPO	MOV	AICON
.BUP	0	BACK UP	BACK UP	—	OCS PRESET	OCS PRESET	OCS PRESET	OCS PRESET	—	OCS PRESET	OCS PRESET	OCS PRESET	—	BACK UP	BACK UP	BACK UP	—
.MAN	1	MANUAL	MANUAL	—	SEQ PRESET	SEQ PRESET	SEQ PRESET	SEQ PRESET	MANUAL	SEQ PRESET	SEQ PRESET	SEQ PRESET	—	MANUAL	MANUAL	MANUAL	—
.EMD	2	EXT-MV	EXT-MV	—	OCS STOP	OCS STOP	OCS STOP	OCS STOP	EXT-MV	—	—	—	—	EXT-MV	—	CENTER	—
	3	PAUSE	PAUSE	—	SEQ STOP	SEQ STOP	SEQ STOP	SEQ STOP	—	—	—	—	—	PAUSE	—	—	—
	4	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
.REM	5	REMOTE	REMOTE	REMOTE	REMOTE	REMOTE	REMOTE	REMOTE	REMOTE	—	—	—	—	—	—	—	—
.LOC	6	LOCAL	LOCAL	LOCAL	LOCAL	LOCAL	LOCAL	LOCAL	LOCAL	—	—	—	—	—	—	—	—
.PVT	7	PV-TRACK	PV-TRACK	—	—	—	—	—	—	—	—	—	—	—	—	—	—
	8	NORMAL	NORMAL	—	NORMAL	NORMAL	NORMAL	NORMAL	NORMAL	—	—	—	—	—	—	—	—
	9	R-INRBT	R-INRBT	R-INRBT	—	—	—	—	R-INHBT	—	—	—	—	—	—	—	—
	A	—	—	—	PRESRT	PRESRT	PRESRT	PRESRT	—	PRESRT	PRESRT	PRESRT	—	—	—	—	—
.LCK	B	LOCK	LOCK	LOCK(*)	RUN	RUN	RUN	RUN	LOCK	LOCK	—	—	LOCK	LOCK	LOCK	LOCK	LOCK
	C	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
	D	—	—	—	RUNNING	RUNNING	RUNNING	RUNNING	—	—	—	—	—	—	—	—	休止指令
	E	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	代替值指令
.FLT	F	FAULT	FAULT	FAULT	—	—	—	—	—	—	—	—	—	—	FAULT	—	FAULT

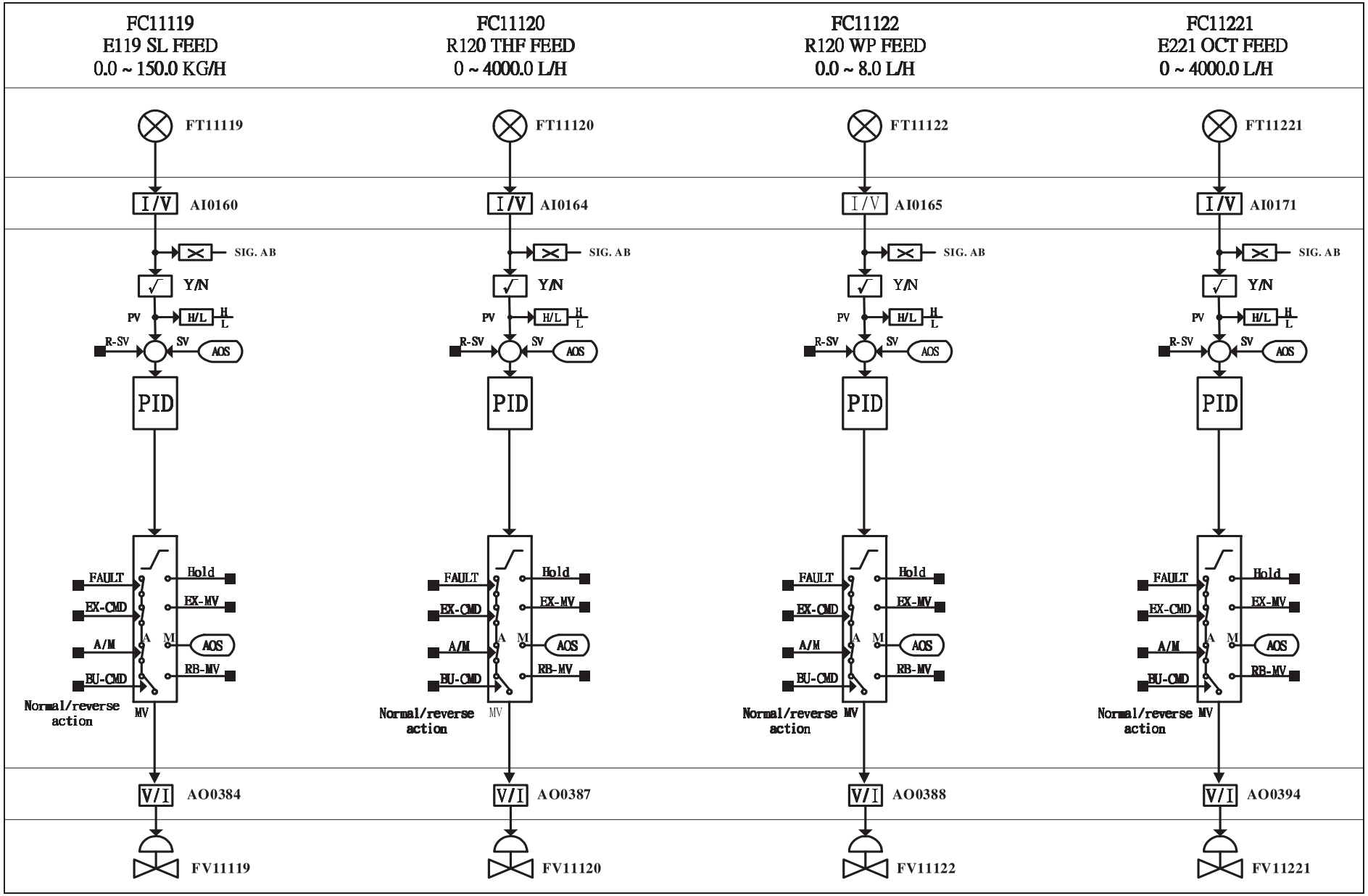
休止指令: 休止=1 ,運轉=0　　代替值指令: 代替值=1,運轉=0

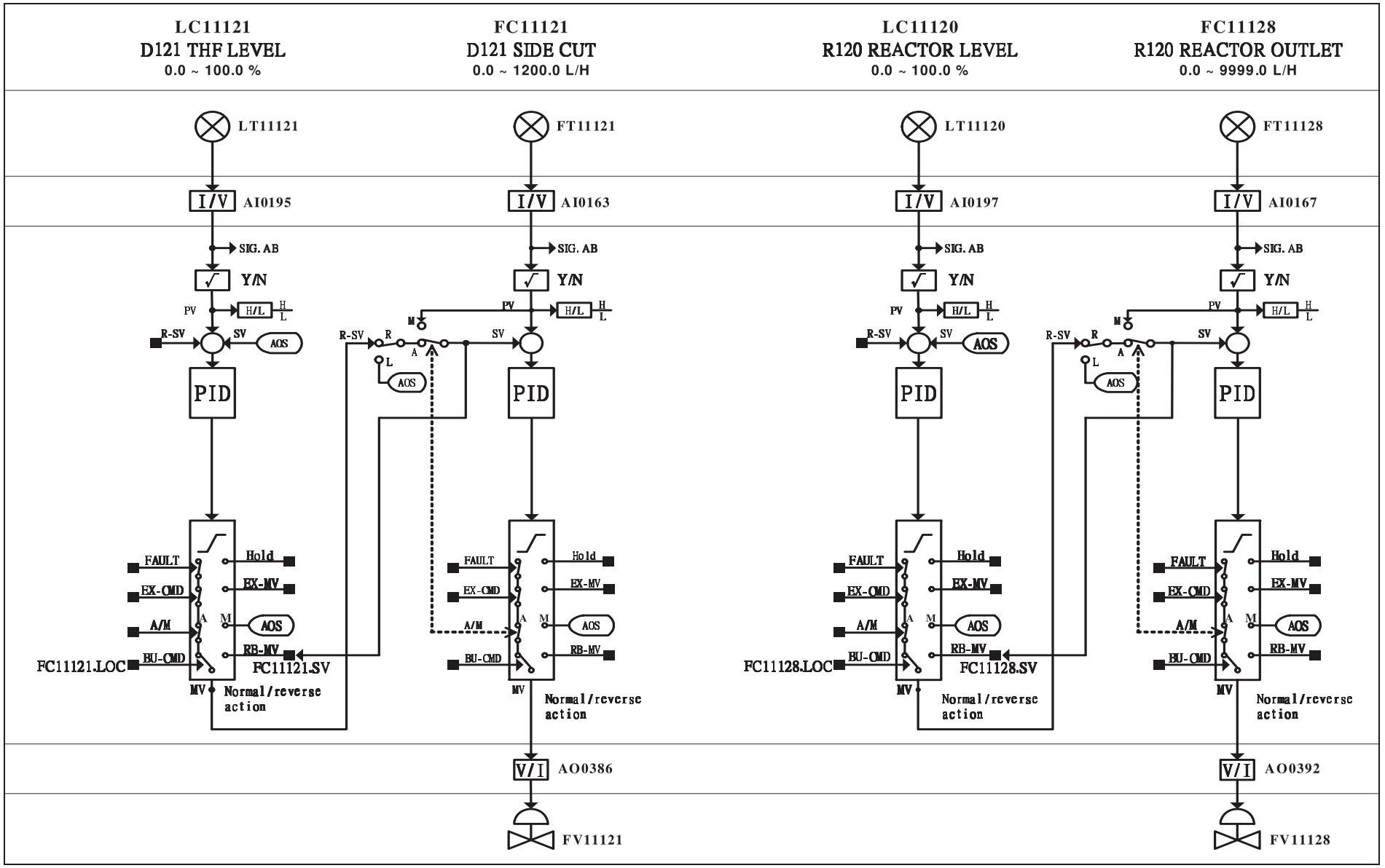
ALRM INSTRUCTION

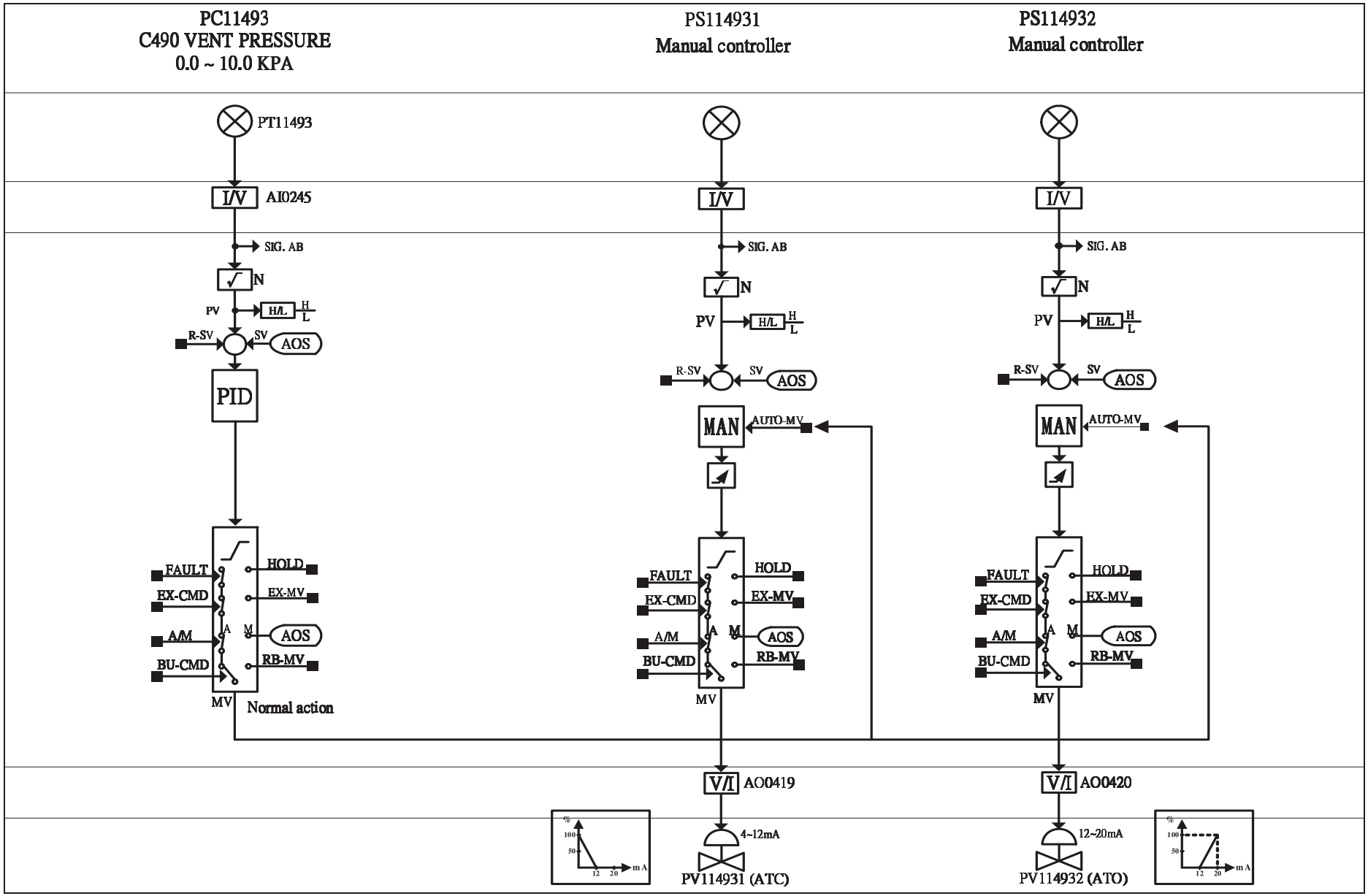
0	1	2	3	4	5	6	7	8	9	A	B	C	D	E	F
8	4	2	1	8	4	2	1	8	4	2	1	8	4	2	1

		PID	PIDV	RF	PGCS	PGCM	PGSS	PGSM	ON OFF	SUMA	SUMP	SUMO	ALRM	MAN	MPO	MOV	AICON
.SH	0	SH	SH	SH	—	—	—	—	SH	—	—	—	—	SH	—	—	—
.SL	1	SL	SL	SL	—	—	—	—	SL	—	—	—	—	SL	—	—	—
.PH	2	PH	PH	PH	—	—	—	—	PH	PH	—	—	PH	PH	PH	PH	AH
.PL	3	PL	PL	PL	—	—	—	—	PL	PL	—	—	PL	PL	PL	PL	AL
.DPH	4	△PVH	△PVH	△PVH	—	—	—	—	△PVH	△PVH	—	—	△PVH	△PVH	—	△PVH	—
.DPL	5	△PVL	△PVL	△PVL	—	—	—	—	△PVL	△PVL	—	—	△PVL	△PVL	—	△PVL	—
.DVH	6	DVH	DVH	—	—	—	—	—	DVH	—	—	—	—	DVH	—	—	INPUT ERROR
.DVL	7	DVL	DVL	—	—	—	—	—	DVL	—	—	—	—	DVH	—	—	INPUT ERROR
.MH	8	MH	MH	MH	—	—	—	—	—	—	—	—	—	MH	MH	—	INPUT ERROR
.ML	9	ML	ML	ML	—	—	—	—	—	—	—	—	—	ML	ML	—	—
.PHH	A	—	—	—	—	—	—	—	—	—	—	—	HH	—	—	—	AHH
.PLL	B	—	—	—	—	—	—	—	—	—	—	—	LL	—	—	—	ALL
.DMH	C	△MVH	△MVH	—	—	—	—	—	—	—	—	—	—	△MVH	△MVH	—	HALTING
	D	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	ALTERNATING
	E	—	—	—	—	—	—	—	—	—	—	—	—	—	—	DLY	POWER FAILURE
	F	—	—	—	—	—	—	—	—	—	—	—	—	—	—	ANN	—

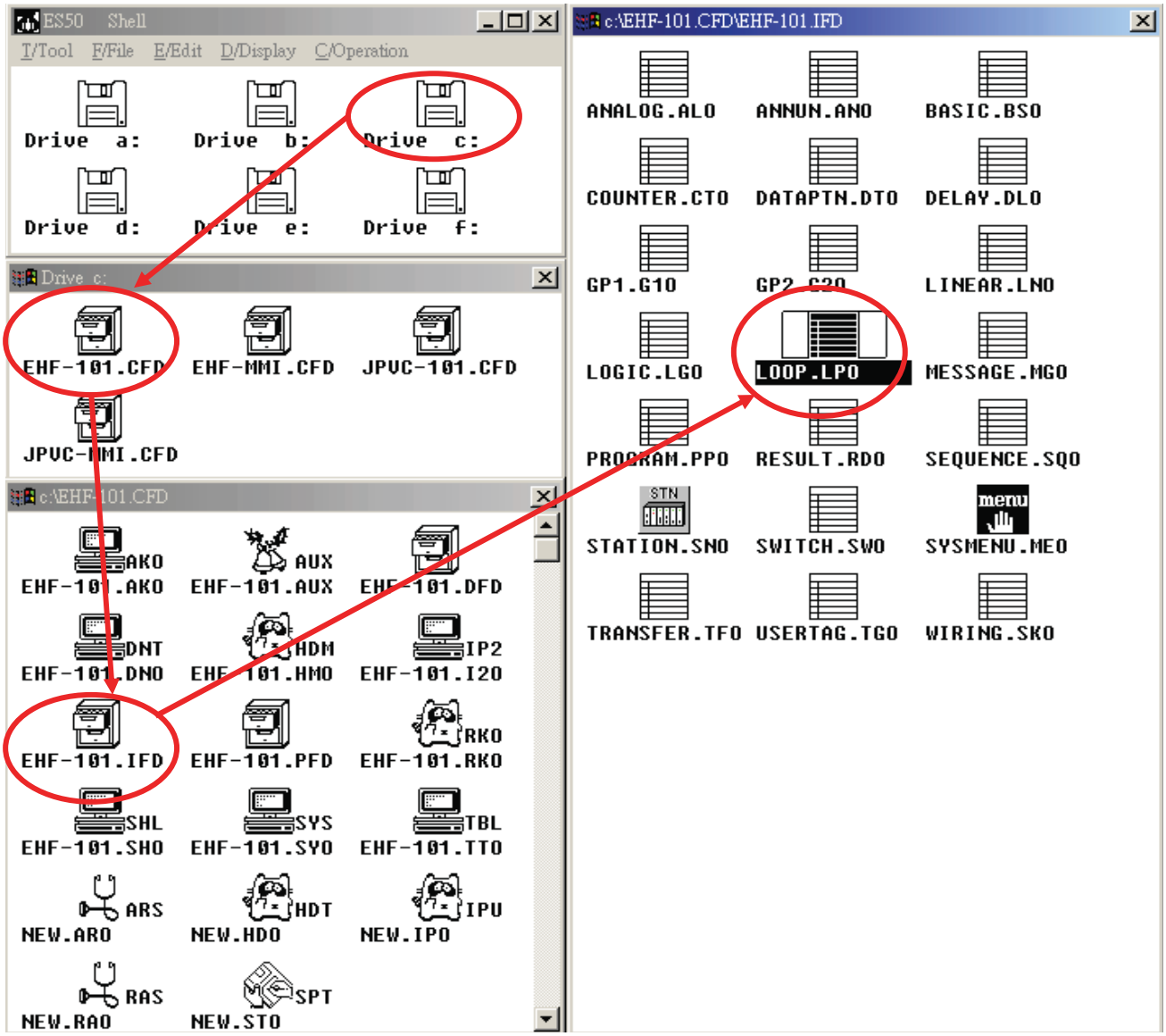

TI101A
XXXX Temp. indicator
0.0 ~ 100.0 ℃
LI101A
XXXX water level indicator
0.0 ~ 100.0 %
TE101A
LT101A
Ω/V
I/V
AIxxxx
SIG. AB
Y/N
HH
M
H
L
LL
ALM


FC11119
E119 SL FEED
0.0 ~ 150.0 KG/H
FC11120
R120 THF FEED
0 ~ 4000.0 L/H
FC11122
R120 WP FEED
0.0 ~ 8.0 L/H
FC11221
E221 OCT FEED
0 ~ 4000.0 L/H
FT11119
FT11120
FT11122
FT11221
I/V
AI0160
AI0164
AI0165
AI0171
SIG. AB
Y/N
PV
H/L
H
L
R-SV
SV
AOS
PID
FAULT
EX-CMD
A/M
BU-CMD
Hold
EX-MV
AOS
RB-MV
A
M
Normal/reverse action
MV
V/I
AO0384
AO0387
AO0388
AO0394
FV11119
FV11120
FV11122
FV11221


LC11121
D121 THF LEVEL
0.0 ~ 100.0 %
FC11121
D121 SIDE CUT
0.0 ~ 1200.0 L/H
LC11120
R120 REACTOR LEVEL
0.0 ~ 100.0 %
FC11128
R120 REACTOR OUTLET
0.0 ~ 9999.0 L/H
LT11121
FT11121
LT11120
FT11128
I/V
AI0195
AI0163
AI0197
AI0167
SIG. AB
Y/N
PV
H/L
H
L
R-SV
SV
AOS
R
A
M
PID
FAULT
EX-CMD
A/M
BU-CMD
Hold
EX-MV
RB-MV
MV
FC11121.LOC
FC11121.SV
FC11128.LOC
FC11128.SV
Normal/reverse action
V/I
AO0386
AO0392
FV11121
FV11128


PC11493
C490 VENT PRESSURE
0.0 ~ 10.0 KPA
PS114931
Manual controller
PS114932
Manual controller
PT11493
I/V
AI0245
SIG. AB
N
PV
H/L
H
L
R-SV
SV
AOS
PID
FAULT
EX-CMD
A/M
BU-CMD
HOLD
EX-MV
AOS
RB-MV
A
M
MV
Normal action
MAN
AUTO-MV
V/I
AO0419
AO0420
4~12mA
12~20mA
PV114931 (ATC)
PV114932 (ATO)
%
100
50
12
20
mA

We provide a example for creating new PID ,see Page8~Page23.

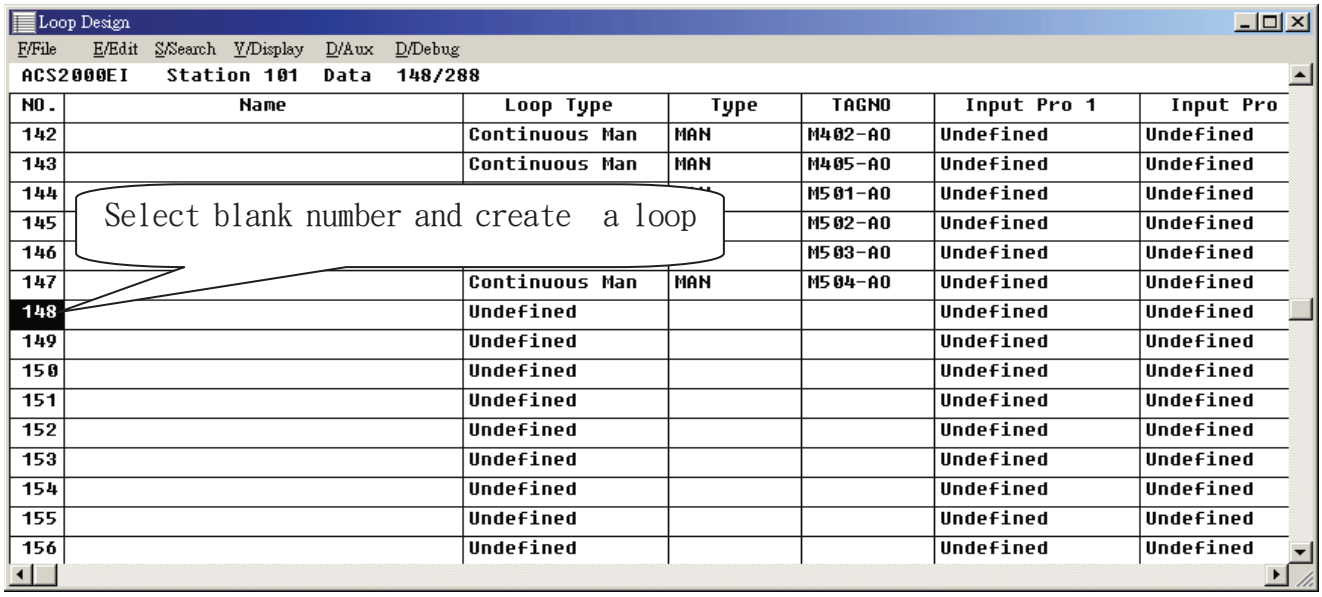


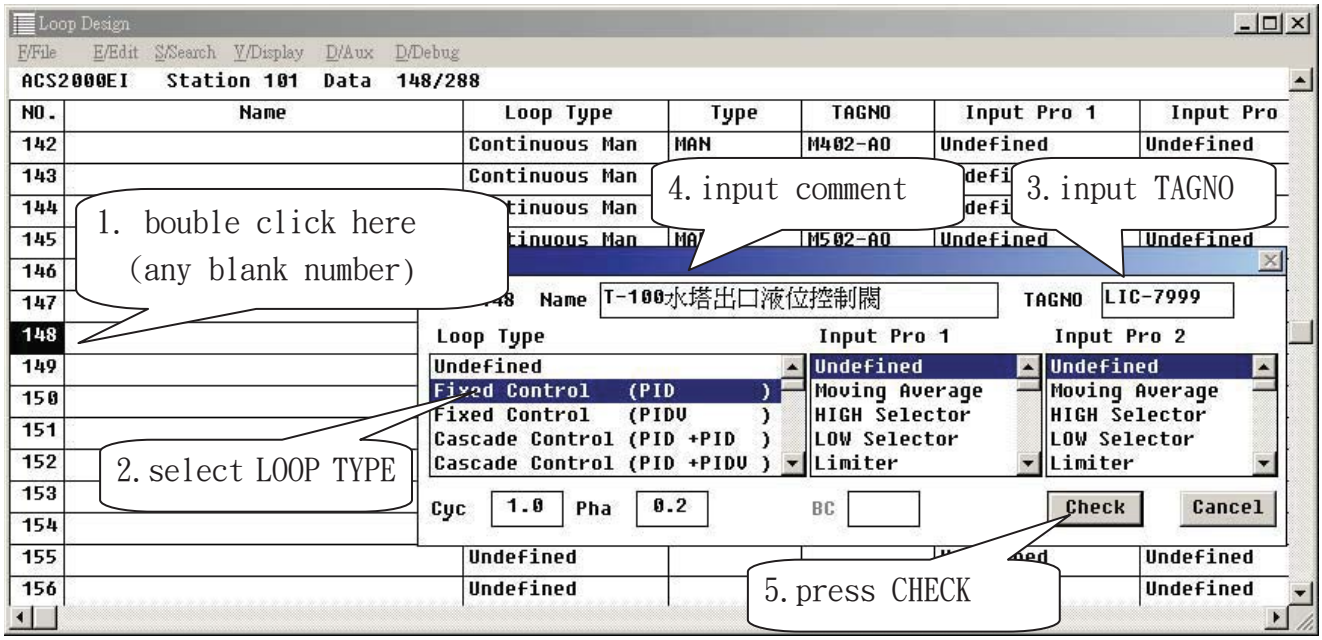
Loop Design

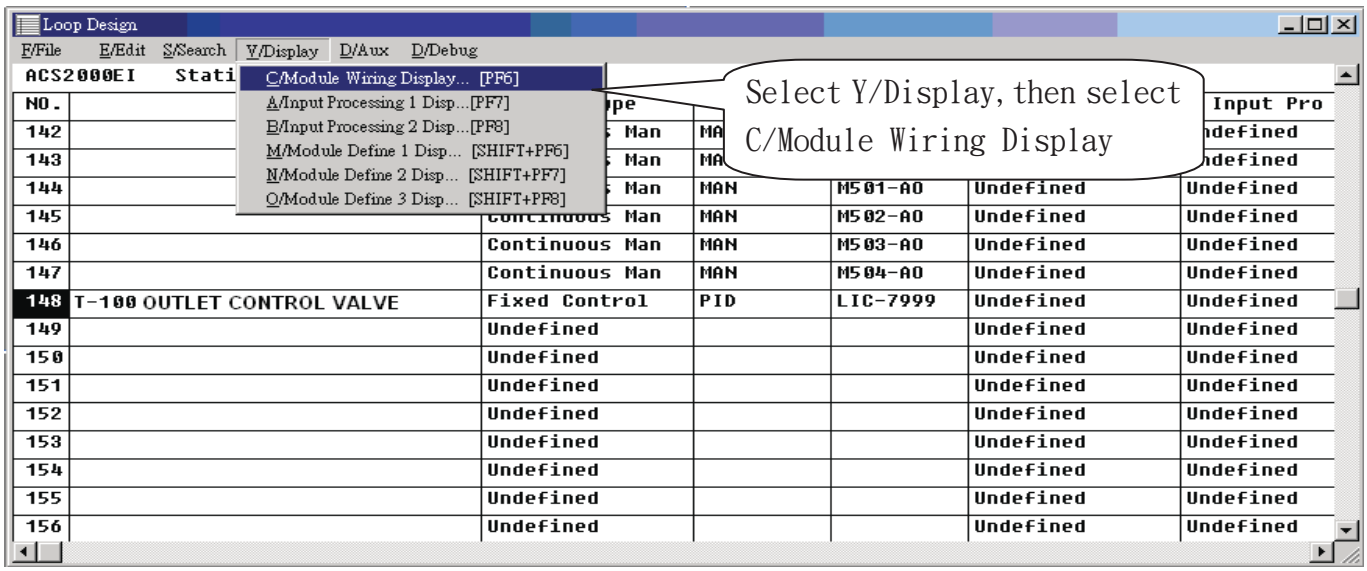
F/File E/Edit S/Search V/Display D/Aux D/Debug

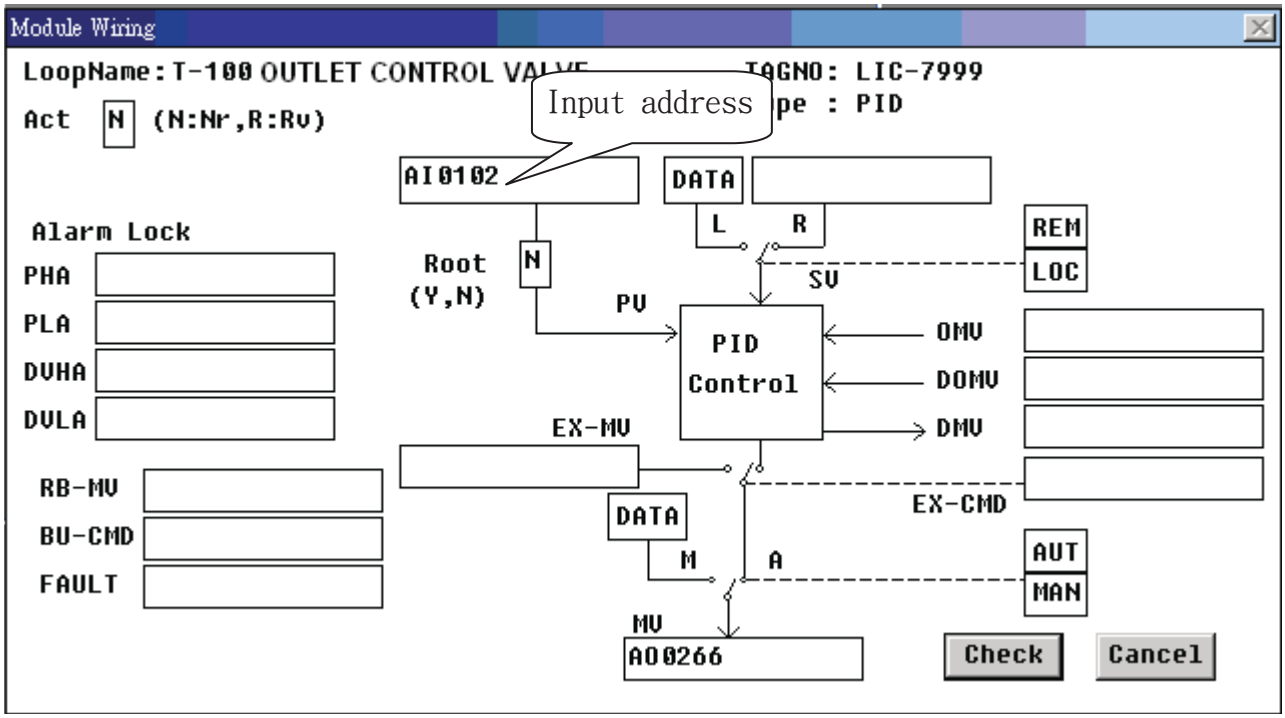
ACS2000EI Station 101 Data 1/288

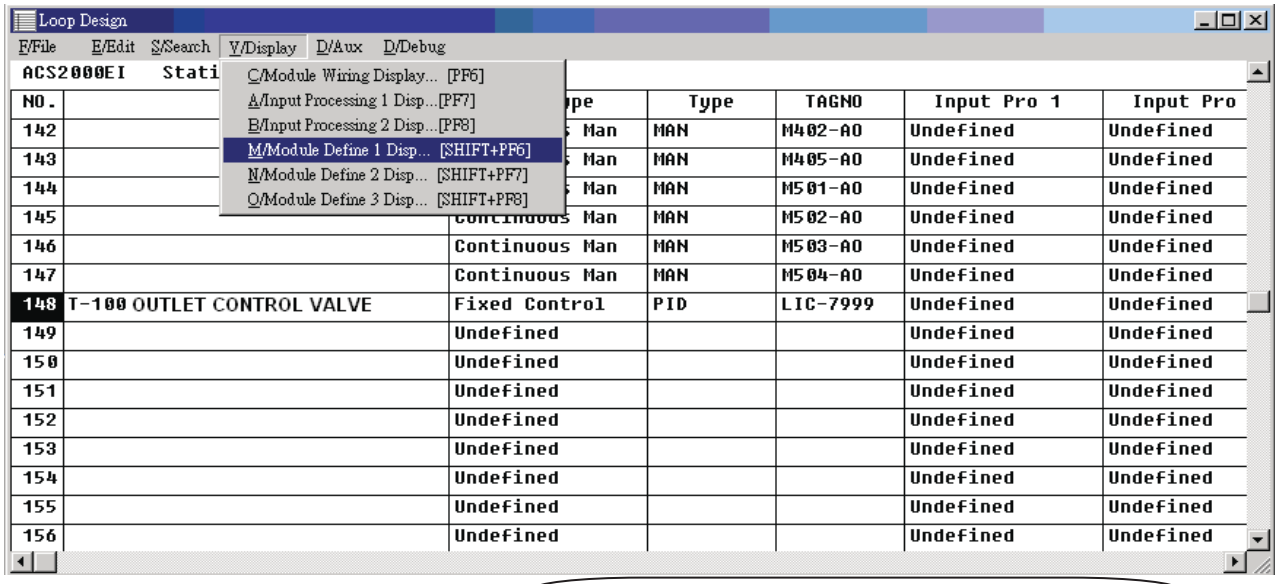
NO.	Name	Loop Type	Type	TAGNO	Input Pro 1	Input Pro
1	C200 INLET	Fixed Control	PID	FIC-C200-I	Undefined	Undefined
2	E210 STM	Fixed Control	PID	FIC-E210	Undefined	Undefined
3	E203 BR FLOW	Fixed Control	PID	FIC-E203	Undefined	Undefined
4	E202 STM FLOW	Fixed Control	PID	FIC-E202	Undefined	Undefined
5	M201 UPW FLOW	Fixed Control	PID	FIC-M201	Undefined	Undefined
6	M401 NH3 FLOW	Fixed Control	PID	FIC-C402-1	Undefined	Undefined
7	V402 NH3 FLOW	Fixed Control	PID	FIC-C402-2	Undefined	Undefined
8	M401 UPW FLOW	Fixed Control	PID	FIC-M401-1	Undefined	Undefined
9	M401 DHF FLOW	Fixed Control	PID	FIC-M401-2	Undefined	Undefined
10	T103 UPW INLET	Indicating Alarm	ALRM	FI-T103-I	Undefined	Undefined
11	T103 OXI OUTLET	Indicating Alarm	ALRM	FI-T103-O	Undefined	Undefined
12	C200 OUTLET	Indicating Alarm	ALRM	FI-C200-D	Undefined	Undefined
13	C200 FLOW	Indicating Alarm	ALRM	FI-C200-R	Undefined	Undefined
14	E200 BR FLOW	Indicating Alarm	ALRM	FI-E200	Undefined	Undefined
15	C202 FLOW	Indicating Alarm	ALRM	FI-C202-R	Undefined	Undefined

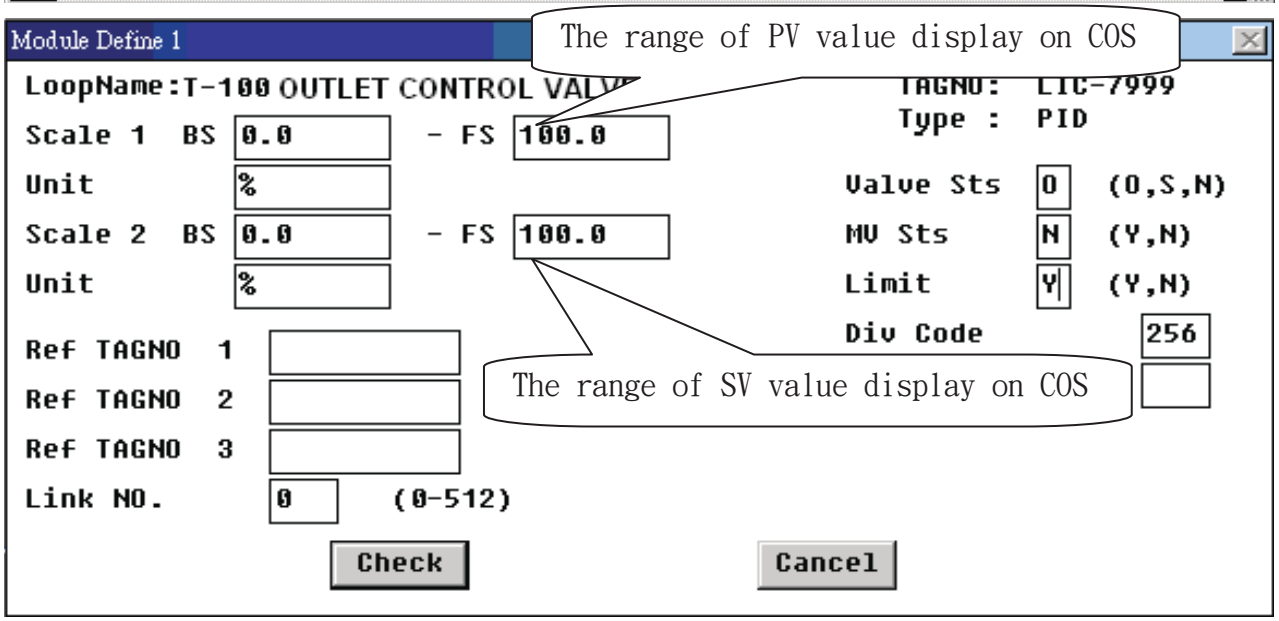

Loop Design
F/File E/Edit S/Search V/Display D/Aux D/Debug
ACS2000EI Station 101 Data 148/288
Select blank number and create a loop

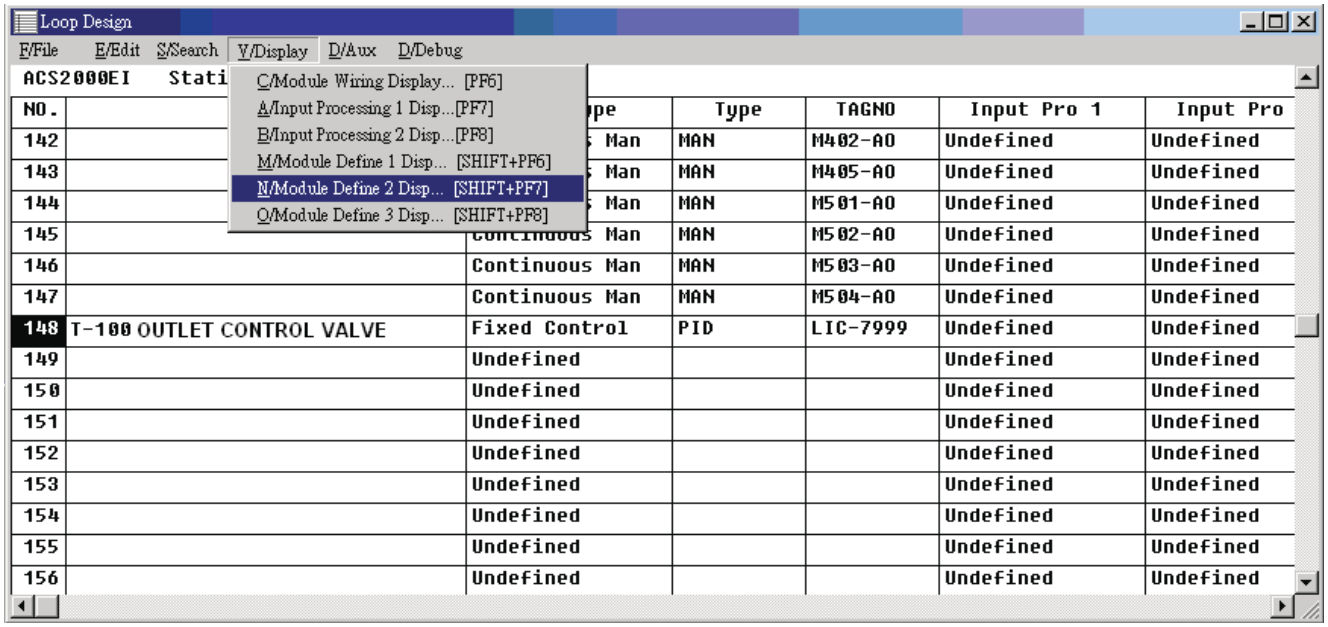

Loop Design
1. bouble click here
(any blank number)
4. input comment
3. input TAGNO
Name T-100水塔出口液位控制閥
TAGNO LIC-7999
Loop Type
Input Pro 1
Input Pro 2
Undefined
Fixed Control (PID)
Fixed Control (PIDV)
Cascade Control (PID +PID)
Cascade Control (PID +PIDV)
Moving Average
HIGH Selector
LOW Selector
Limiter
Cyc 1.0 Pha 0.2 BC
Check
Cancel
2. select LOOP TYPE
5. press CHECK


Loop Design
F/File E/Edit S/Search V/Display D/Aux D/Debug
C/Module Wiring Display... [PF6]
A/Input Processing 1 Disp...[PF7]
B/Input Processing 2 Disp...[PF8]
M/Module Define 1 Disp... [SHIFT+PF6]
N/Module Define 2 Disp... [SHIFT+PF7]
O/Module Define 3 Disp... [SHIFT+PF8]
Select Y/Display, then select
C/Module Wiring Display
148 T-100 OUTLET CONTROL VALVE Fixed Control PID LIC-7999

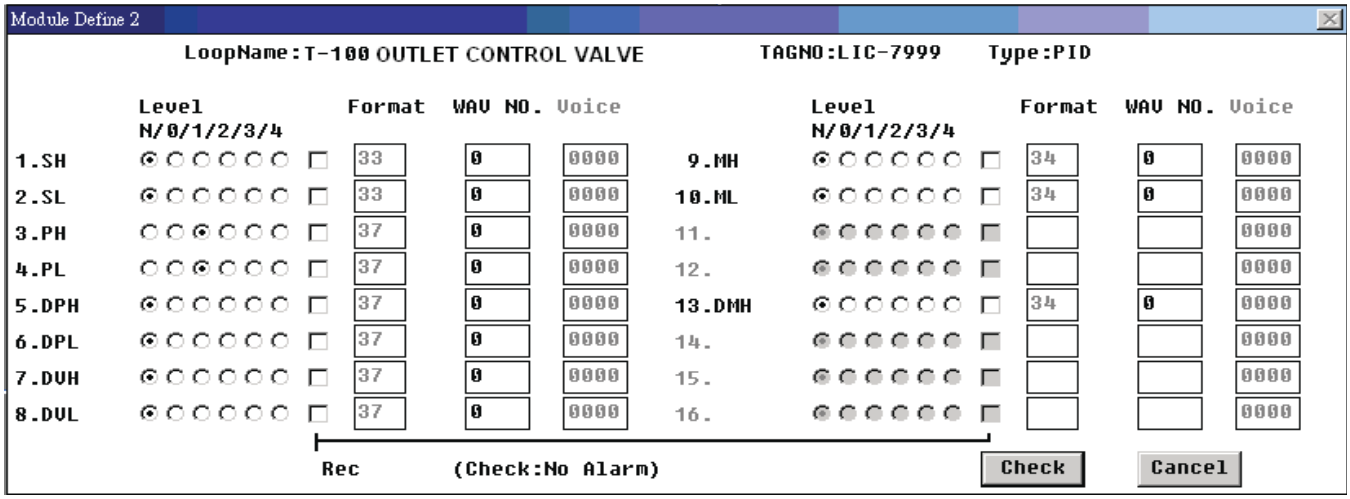

Module Wiring
LoopName:T-100 OUTLET CONTROL VALVE
TAGNO: LIC-7999
Act N (N:Nr,R:Rv)
Type : PID
Input address
AI0102
DATA
L
R
REM
LOC
SV
Alarm Lock
PHA
PLA
DVHA
DVLA
Root (Y,N)
N
PV
PID Control
OMV
DOMV
DMV
EX-MV
EX-CMD
RB-MV
BU-CMD
FAULT
DATA
M
A
AUT
MAN
MV
AO0266
Check
Cancel

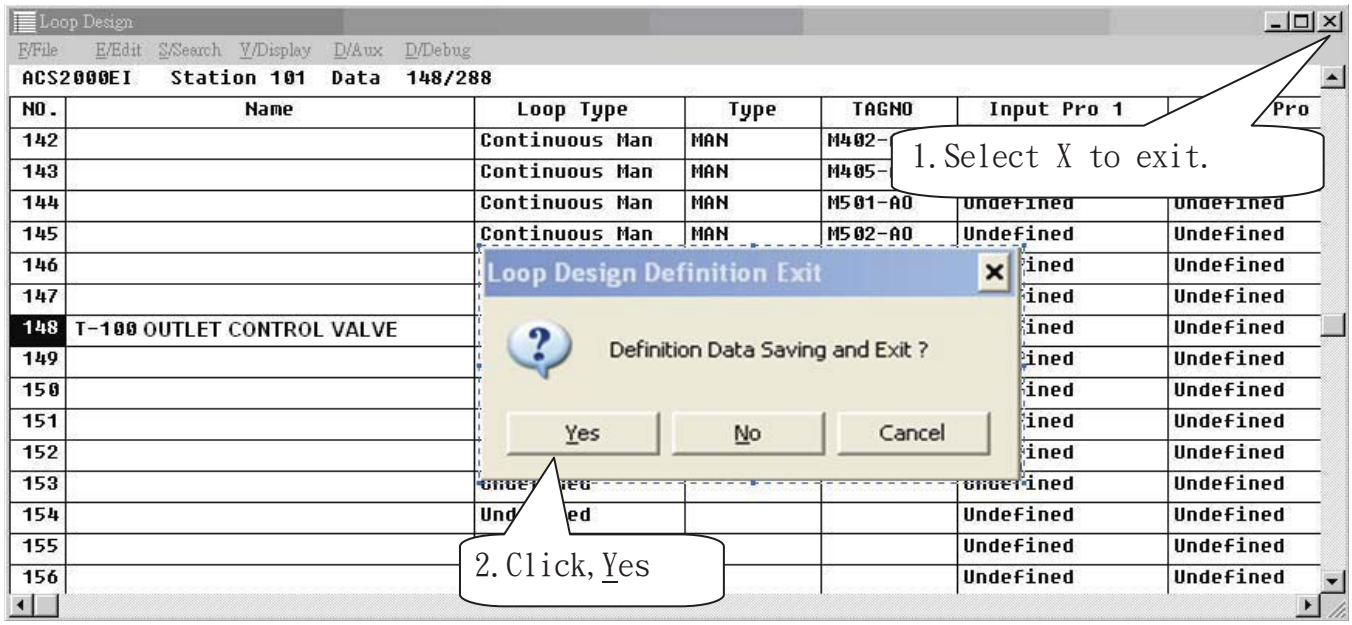

Loop Design
F/File E/Edit S/Search V/Display D/Aux D/Debug
ACS2000EI Stati
C/Module Wiring Display... [PF6]
A/Input Processing 1 Disp...[PF7]
B/Input Processing 2 Disp...[PF8]
M/Module Define 1 Disp... [SHIFT+PF6]
N/Module Define 2 Disp... [SHIFT+PF7]
O/Module Define 3 Disp... [SHIFT+PF8]
NO. Type TAGNO Input Pro 1 Input Pro
142 Man MAN M402-AO Undefined Undefined
143 Man MAN M405-AO Undefined Undefined
144 Man MAN M501-AO Undefined Undefined
145 Man MAN M502-AO Undefined Undefined
146 Continuous Man MAN M503-AO Undefined Undefined
147 Continuous Man MAN M504-AO Undefined Undefined
148 T-100 OUTLET CONTROL VALVE Fixed Control PID LIC-7999 Undefined Undefined
149 Undefined Undefined Undefined
150 Undefined Undefined Undefined
151 Undefined Undefined Undefined
152 Undefined Undefined Undefined
153 Undefined Undefined Undefined
154 Undefined Undefined Undefined
155 Undefined Undefined Undefined
156 Undefined Undefined Undefined

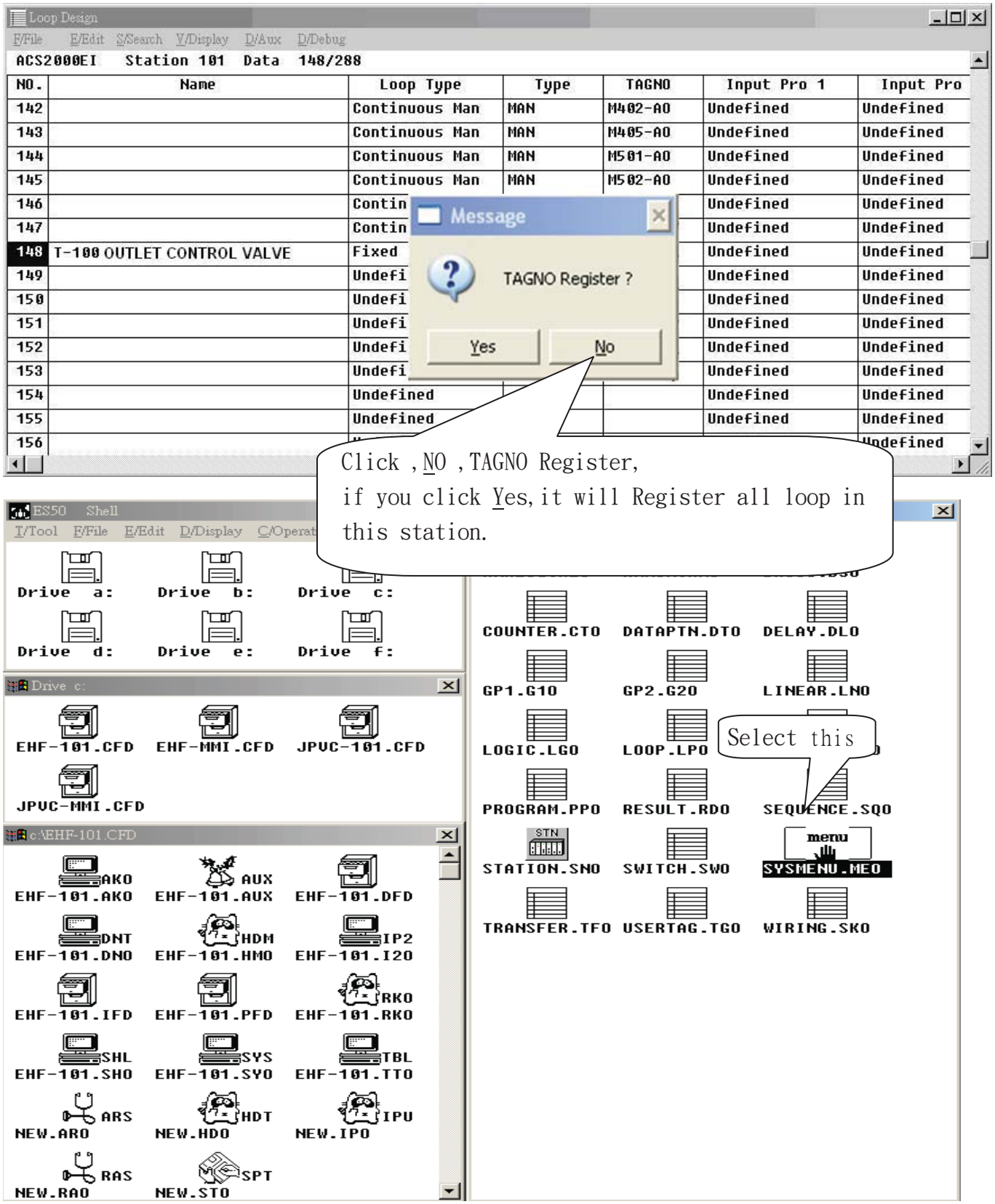

Module Define 1
The range of PV value display on COS
LoopName:T-100 OUTLET CONTROL VALVE
TAGNO: LIC-7999
Type : PID
Scale 1 BS 0.0 - FS 100.0
Unit %
Scale 2 BS 0.0 - FS 100.0
Unit %
Valve Sts O (O,S,N)
MV Sts N (Y,N)
Limit Y (Y,N)
Div Code 256
Ref TAGNO 1
Ref TAGNO 2
Ref TAGNO 3
Link NO. 0 (0-512)
The range of SV value display on COS
Check
Cancel

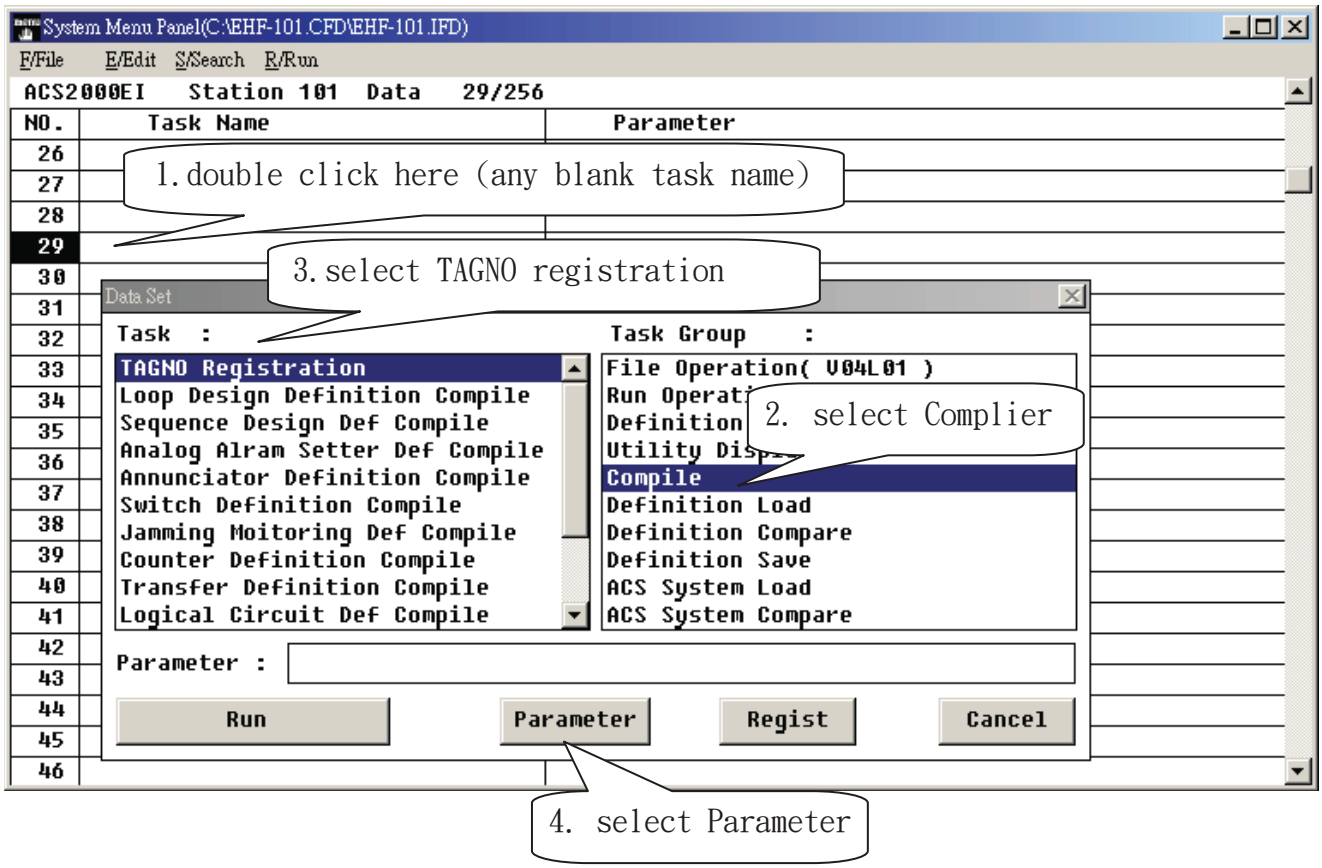


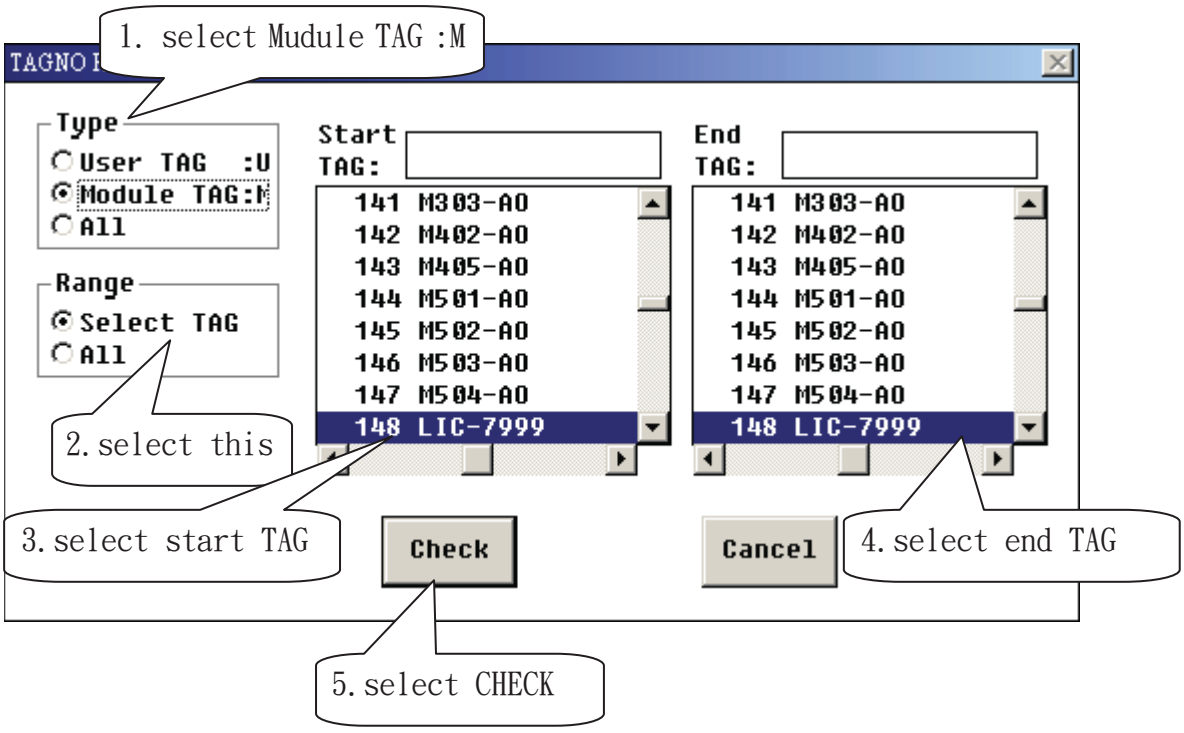
The alarm level setting , priority LEVEL0> LEVEL1> LEVEL2> LEVEL3> LEVEL4 LEVEL N means no alarm.

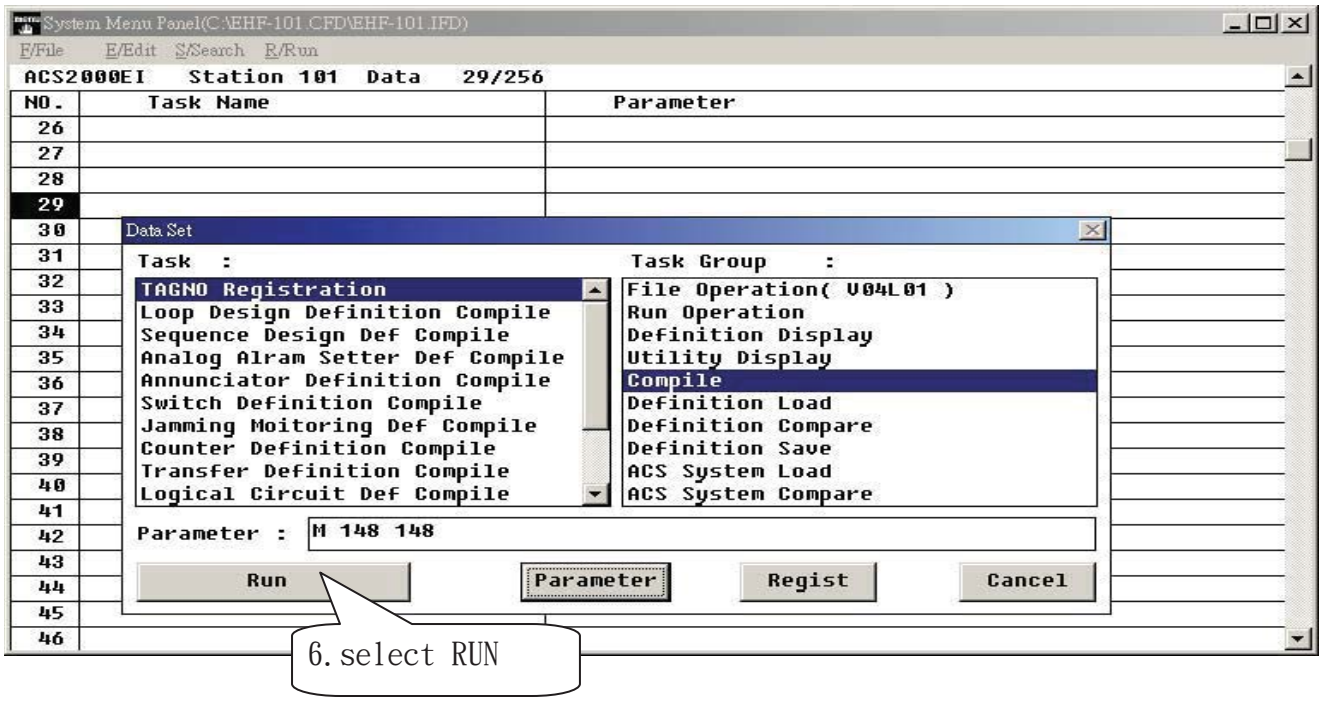


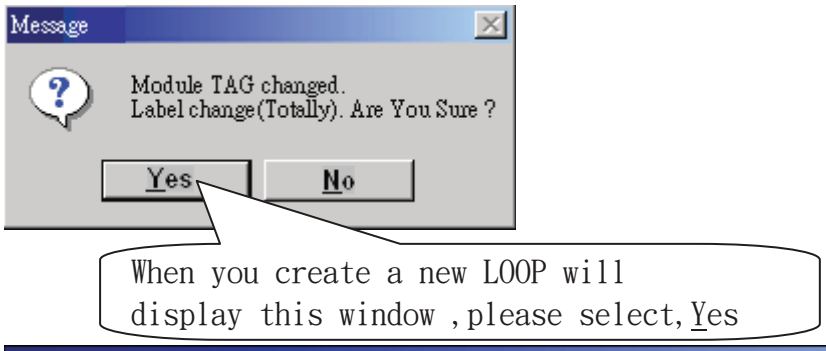


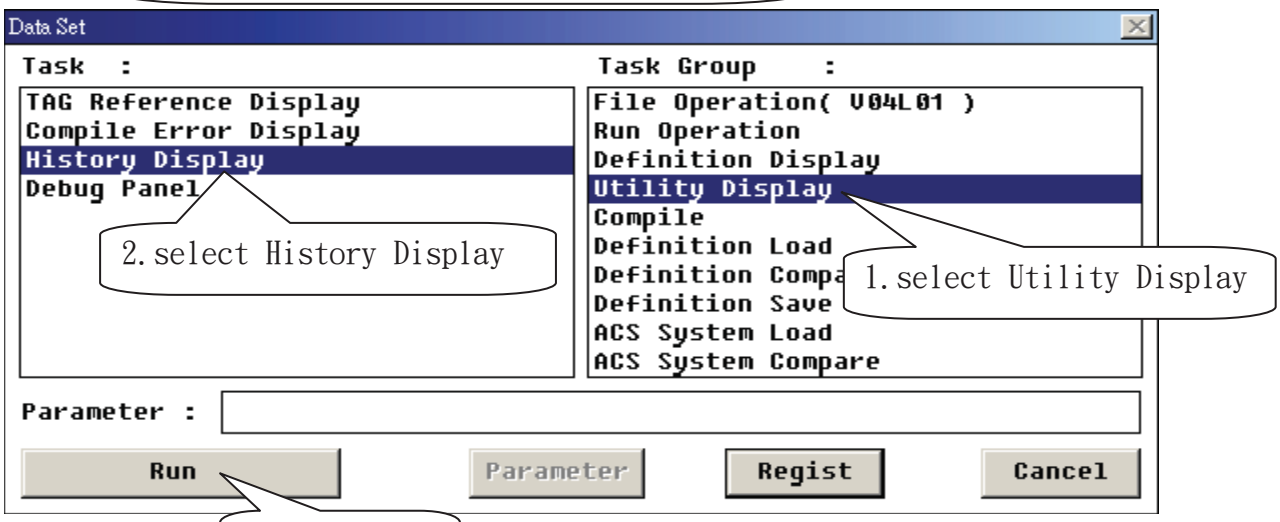

Loop Design
F/File E/Edit S/Search V/Display D/Aux D/Debug
ACS2000EI Station 101 Data 148/288
NO. Name Loop Type Type TAGNO Input Pro 1 Input Pro
142 Continuous Man MAN M402-AO Undefined Undefined
143 Continuous Man MAN M405-AO Undefined Undefined
144 Continuous Man MAN M501-AO Undefined Undefined
145 Continuous Man MAN M502-AO Undefined Undefined
146 Undefined Undefined
147 Undefined Undefined
148 T-100 OUTLET CONTROL VALVE Fixed Undefined Undefined
149 Undefined Undefined
150 Undefined Undefined
151 Undefined Undefined
152 Undefined Undefined
153 Undefined Undefined
154 Undefined Undefined
155 Undefined Undefined
156 Undefined
Message
TAGNO Register ?
Yes
No
Click ,NO ,TAGNO Register,
if you click Yes,it will Register all loop in this station.
ES50 Shell
T/Tool F/File E/Edit D/Display C/Operat
Drive a:
Drive b:
Drive c:
Drive d:
Drive e:
Drive f:
Drive c:
EHF-101.CFD
EHF-MMI.CFD
JPVC-101.CFD
JPVC-MMI.CFD
c:\EHF-101.CFD
AKO EHF-101.AKO
AUX EHF-101.AUX
EHF-101.DFD
DNT EHF-101.DNO
HDM EHF-101.HMO
IP2 EHF-101.I2O
EHF-101.IFD
EHF-101.PFD
RKO EHF-101.RKO
SHL EHF-101.SHO
SYS EHF-101.SYO
TBL EHF-101.TTO
ARS NEW.ARO
HDT NEW.HDO
IPU NEW.IPO
RAS NEW.RAO
SPT NEW.STO
COUNTER.CTO
DATAPTN.DTO
DELAY.DLO
GP1.G10
GP2.G20
LINEAR.LNO
LOGIC.LGO
LOOP.LPO
Select this
PROGRAM.PPO
RESULT.RDO
SEQUENCE.SQO
STN
STATION.SNO
SWITCH.SWO
menu
SYSMENU.MEO
TRANSFER.TFO
USERTAG.TGO
WIRING.SKO

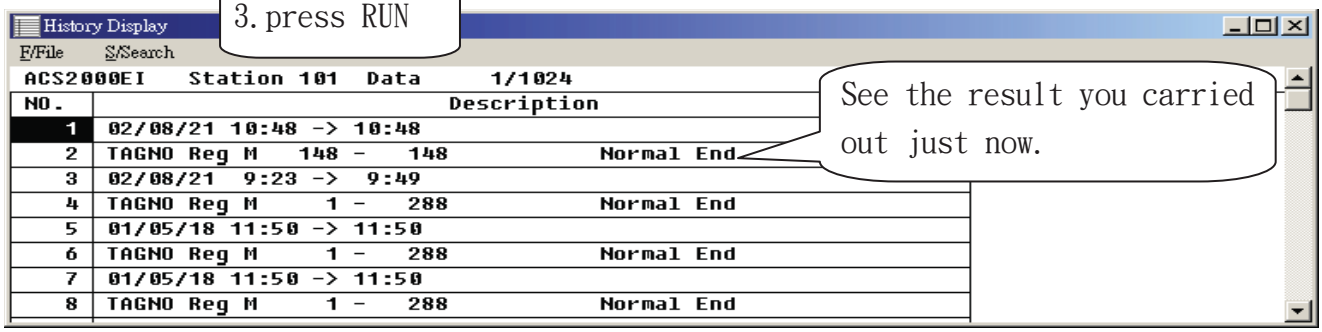

System Menu Panel(C:\EHF-101.CFD\EHF-101.IFD)
F/File E/Edit S/Search R/Run
ACS2000EI Station 101 Data 29/256
NO. Task Name Parameter
1.double click here (any blank task name)
3.select TAGNO registration
Data Set
Task :
Task Group :
TAGNO Registration
Loop Design Definition Compile
Sequence Design Def Compile
Analog Alram Setter Def Compile
Annunciator Definition Compile
Switch Definition Compile
Jamming Moitoring Def Compile
Counter Definition Compile
Transfer Definition Compile
Logical Circuit Def Compile
File Operation(V04L01)
Compile
Definition Load
Definition Compare
Definition Save
ACS System Load
ACS System Compare
2. select Complier
Parameter :
Run
Parameter
Regist
Cancel
4. select Parameter

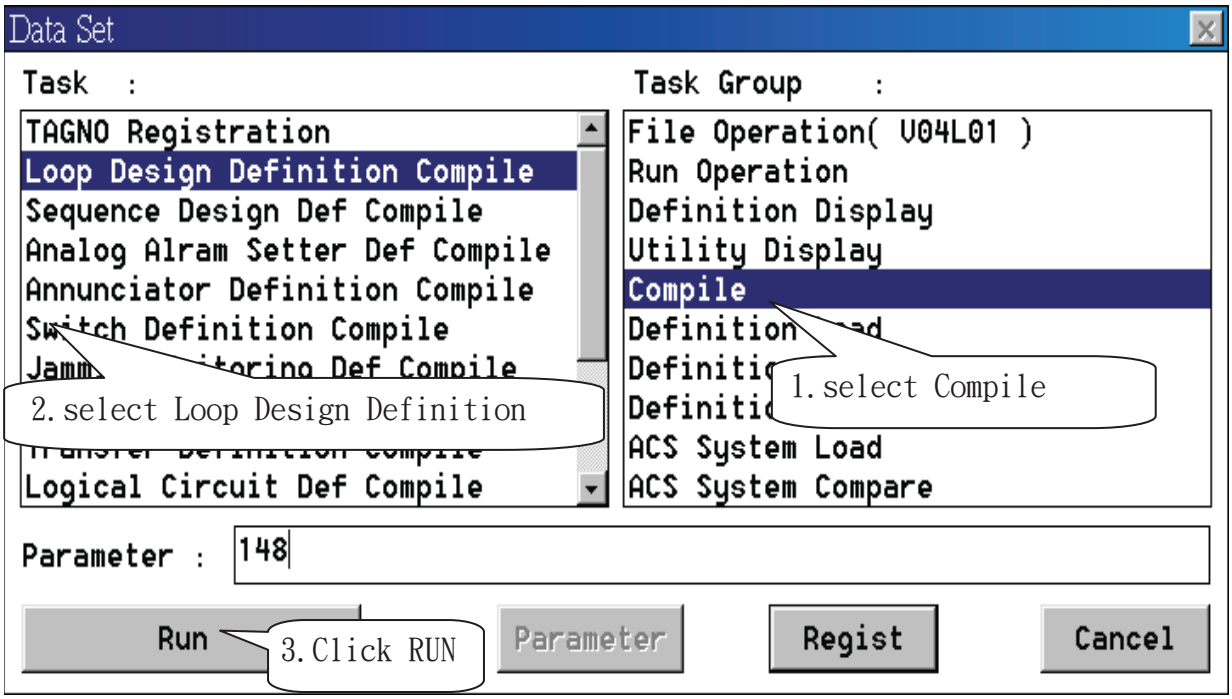

1. select Mudule TAG :M
Type
User TAG :U
Module TAG:M
All
Range
Select TAG
All
Start TAG:
End TAG:
141 M303-AO
142 M402-AO
143 M405-AO
144 M501-AO
145 M502-AO
146 M503-AO
147 M504-AO
148 LIC-7999
2.select this
3.select start TAG
Check
Cancel
4.select end TAG
5.select CHECK


System Menu Panel(C:\EHF-101.CFD\EHF-101.IFD)
F/File E/Edit S/Search R/Run
ACS2000EI Station 101 Data 29/256
NO. Task Name Parameter
Data Set
Task :
Task Group :
TAGNO Registration
Loop Design Definition Compile
Sequence Design Def Compile
Analog Alram Setter Def Compile
Annunciator Definition Compile
Switch Definition Compile
Jamming Moitoring Def Compile
Counter Definition Compile
Transfer Definition Compile
Logical Circuit Def Compile
File Operation(V04L01)
Run Operation
Definition Display
Utility Display
Compile
Definition Load
Definition Compare
Definition Save
ACS System Load
ACS System Compare
Parameter : M 148 148
Run
Parameter
Regist
Cancel
6. select RUN

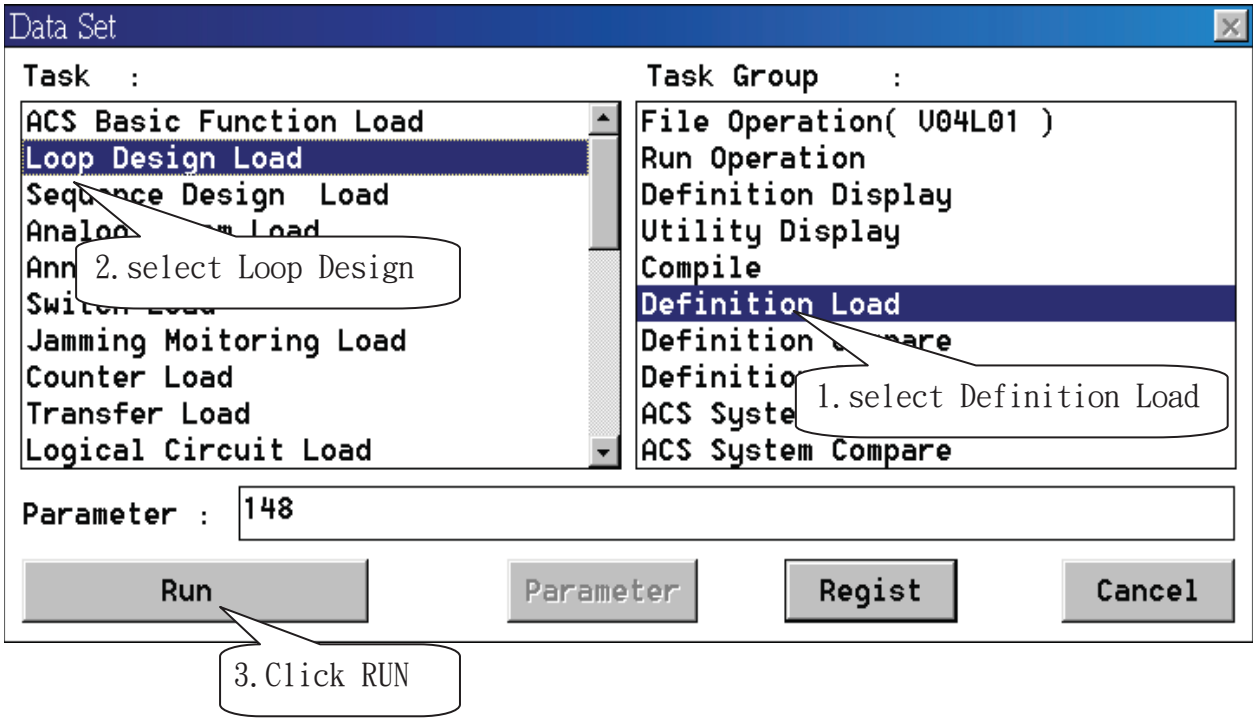

Message
Module TAG changed.
Label change(Totally). Are You Sure ?
Yes
No
When you create a new LOOP will display this window ,please select, Yes


Data Set
Task :
Task Group :
TAG Reference Display
Compile Error Display
History Display
Debug Panel
File Operation(V04L01)
Run Operation
Definition Display
Utility Display
Compile
Definition Load
Definition Compa
Definition Save
ACS System Load
ACS System Compare
2. select History Display
1. select Utility Display
Parameter :
Run
Parameter
Regist
Cancel
3. press RUN

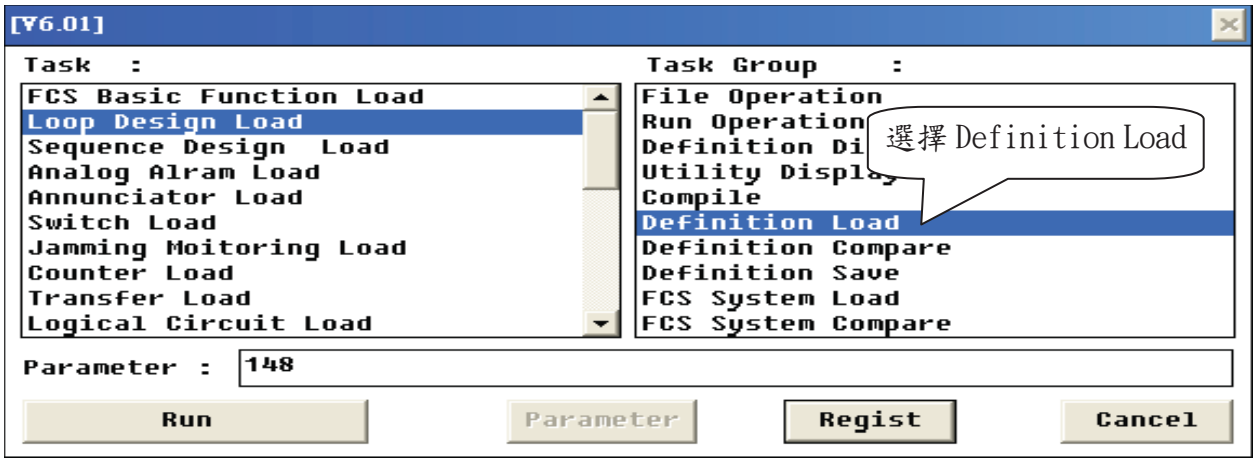

History Display
F/File S/Search
ACS2000EI Station 101 Data 1/1024
NO. Description
1 02/08/21 10:48 -> 10:48
2 TAGNO Reg M 148 - 148 Normal End
3 02/08/21 9:23 -> 9:49
4 TAGNO Reg M 1 - 288 Normal End
5 01/05/18 11:50 -> 11:50
6 TAGNO Reg M 1 - 288 Normal End
7 01/05/18 11:50 -> 11:50
8 TAGNO Reg M 1 - 288 Normal End
See the result you carried out just now.

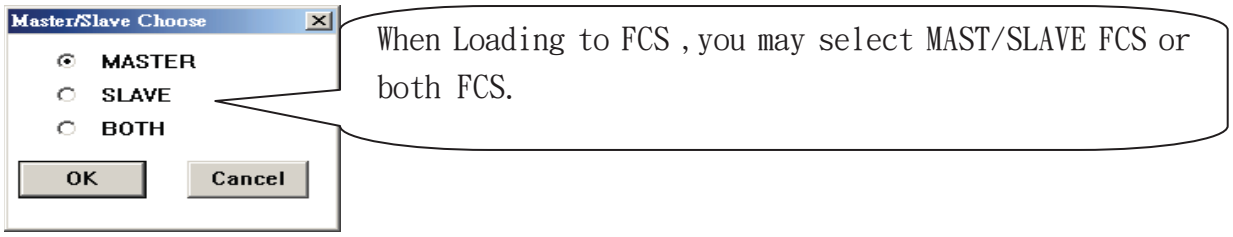


AX system: After click RUN, it will load to ACS, for AX system you should select MASTER/SLAVE ,BOTH in STATION.SNO ➔ Communication before your loading.

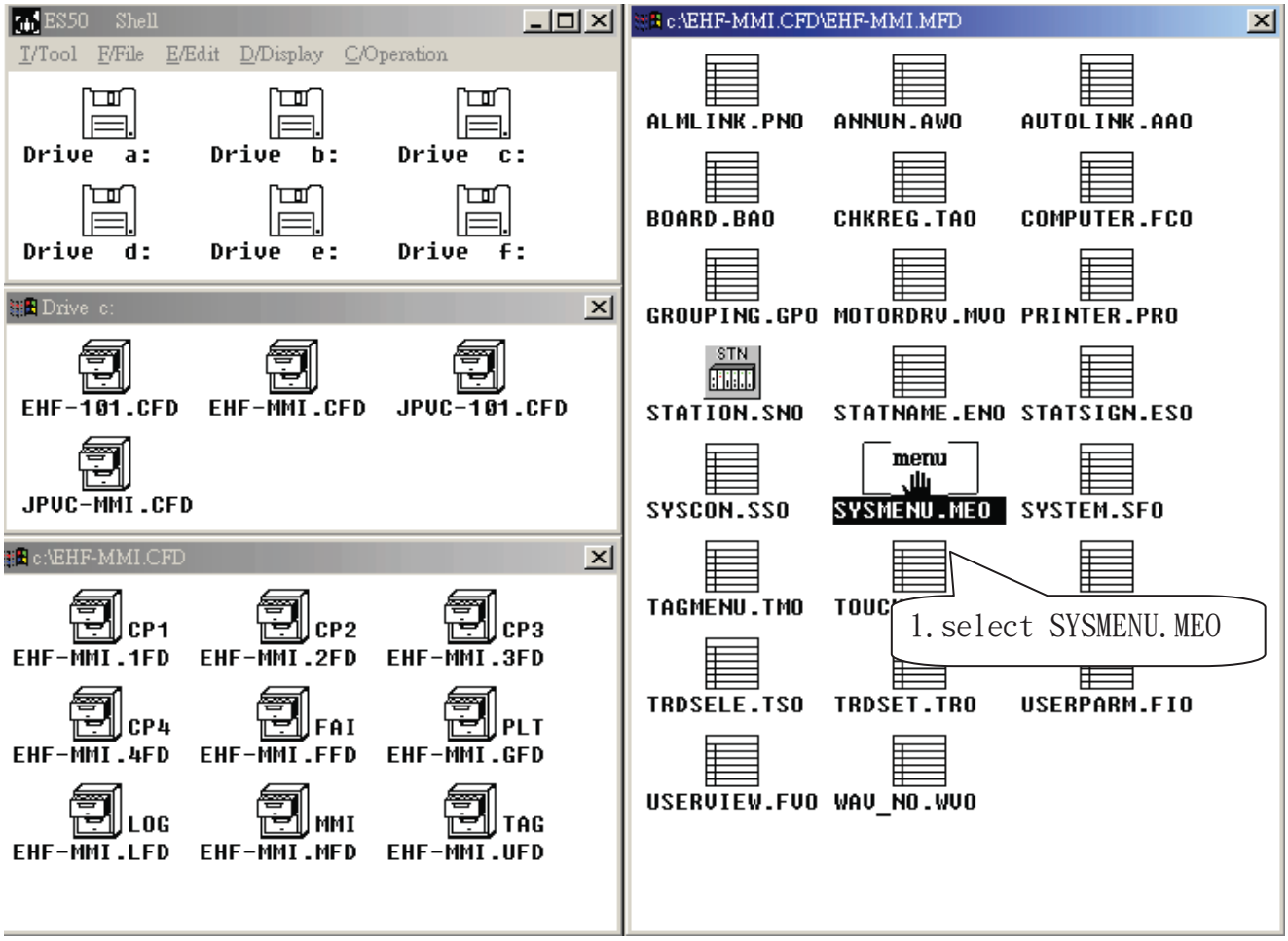


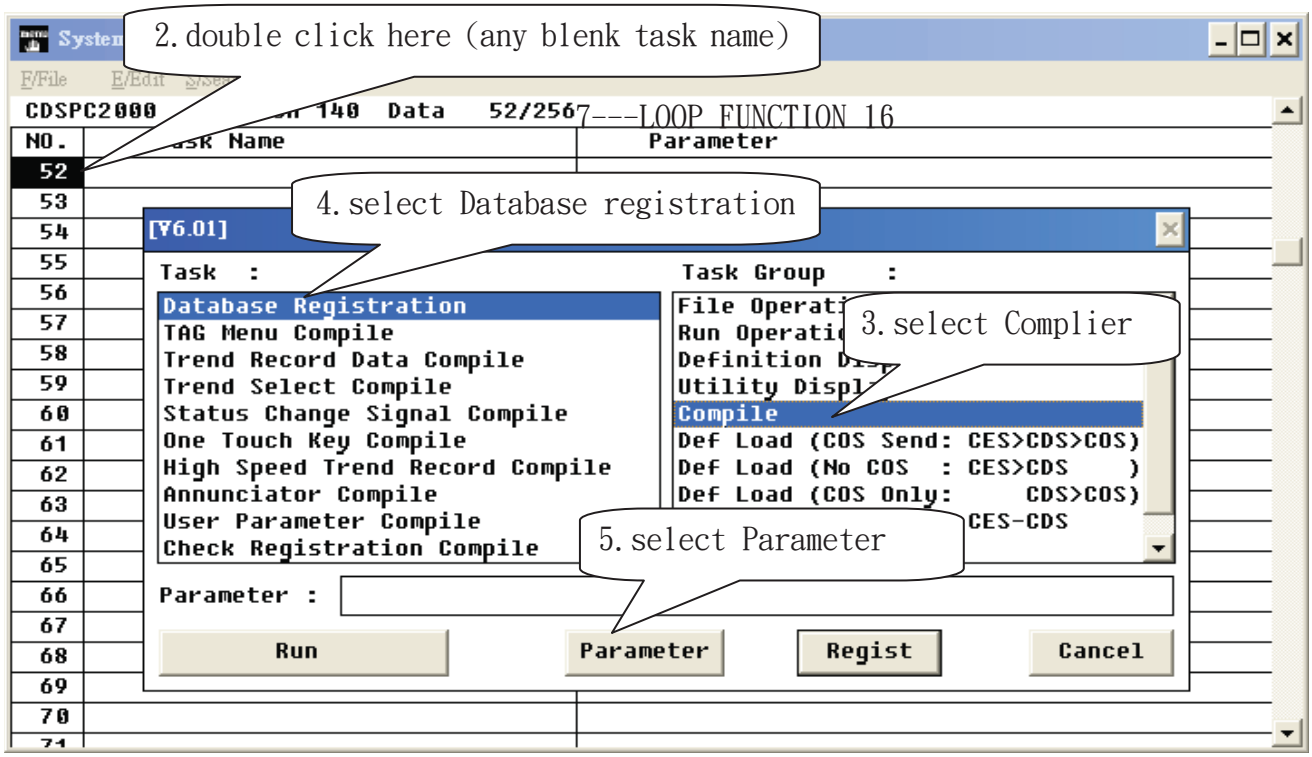
CX system: After your clicking on RUN, it will ask you load to FCS MASTER/SLAVE, BOTH. It automatically Communicate to FCS by MASTER/SLAVE ,BOTH.

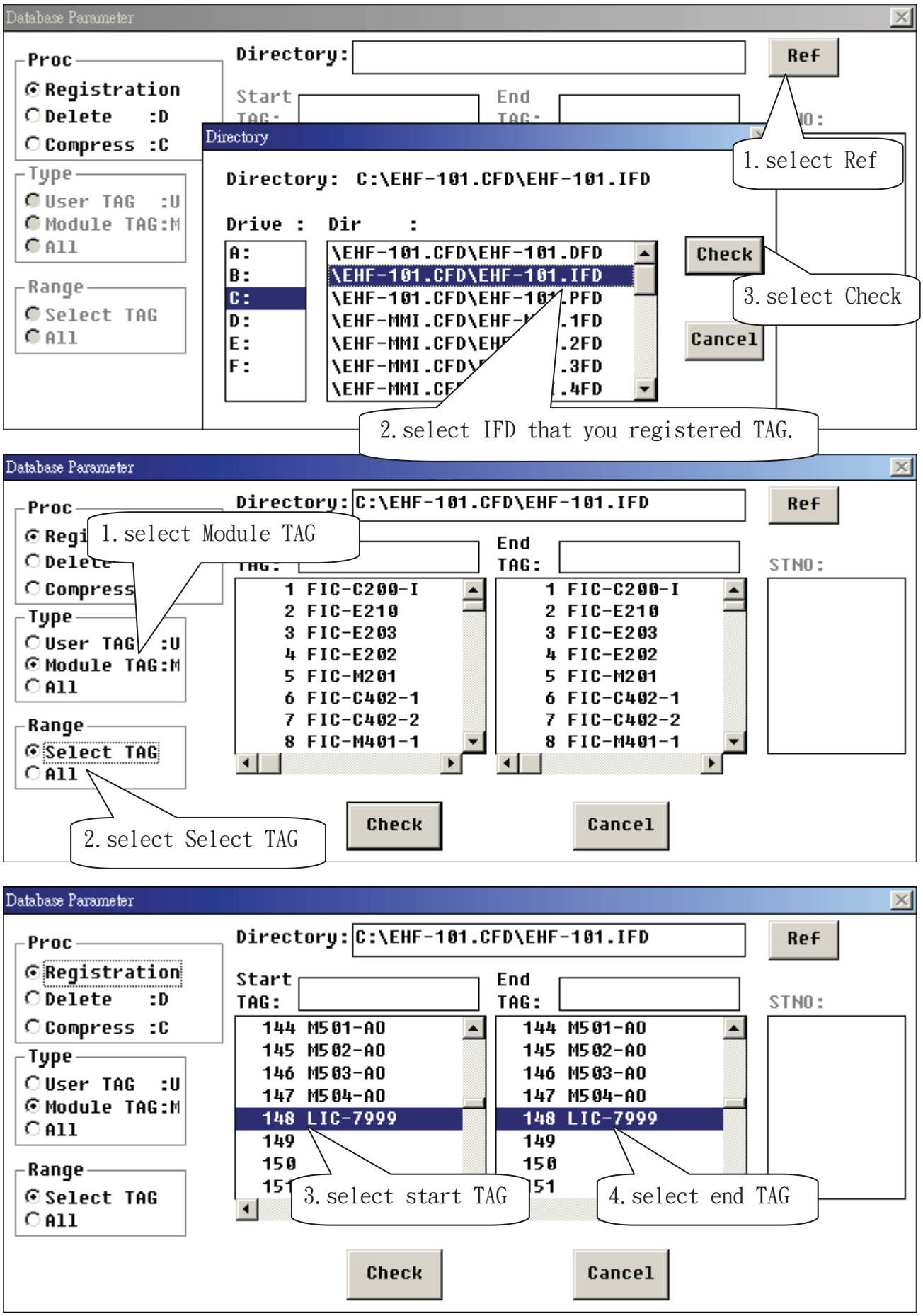


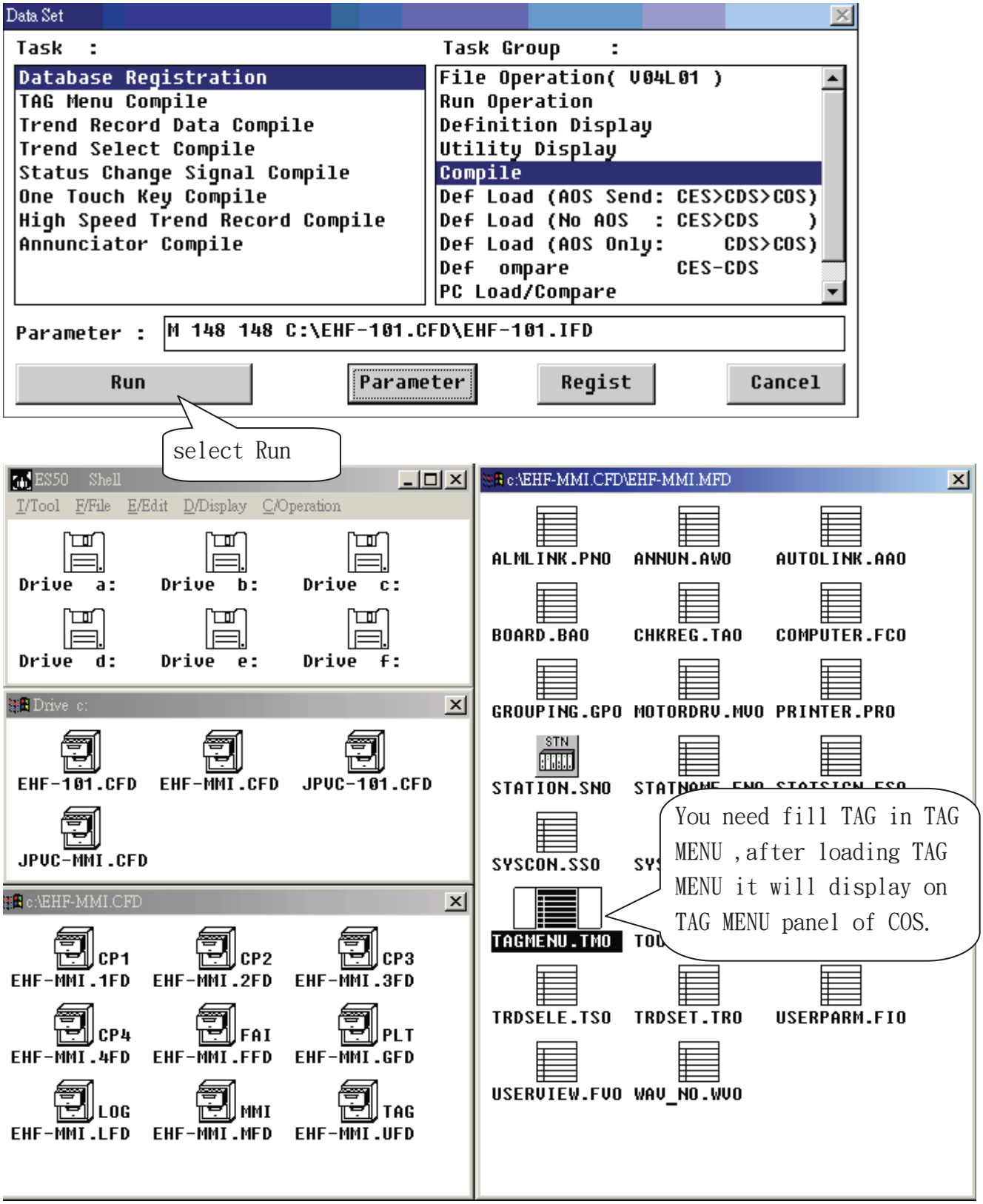


Move to MMI, Please select C:\XXXMMI.CFD\XXXMMI.MFD

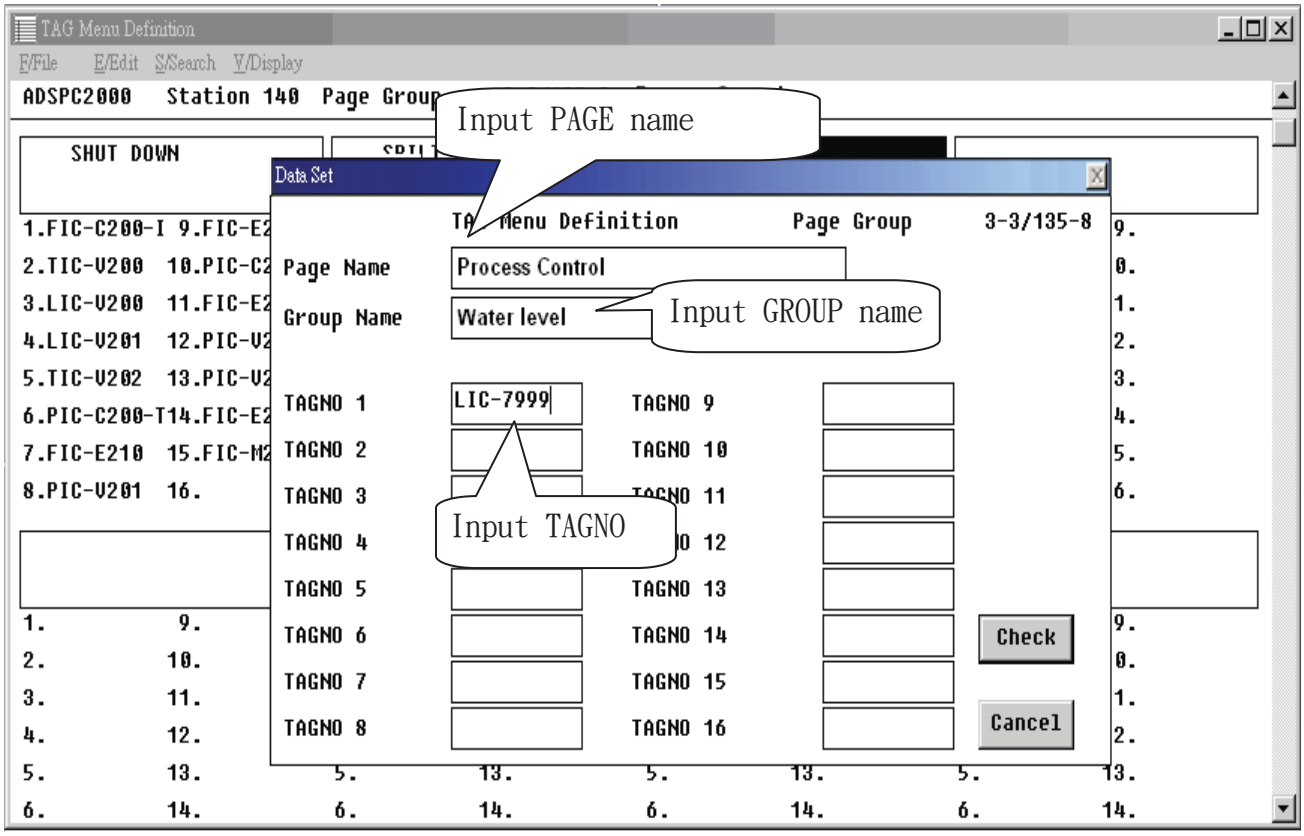




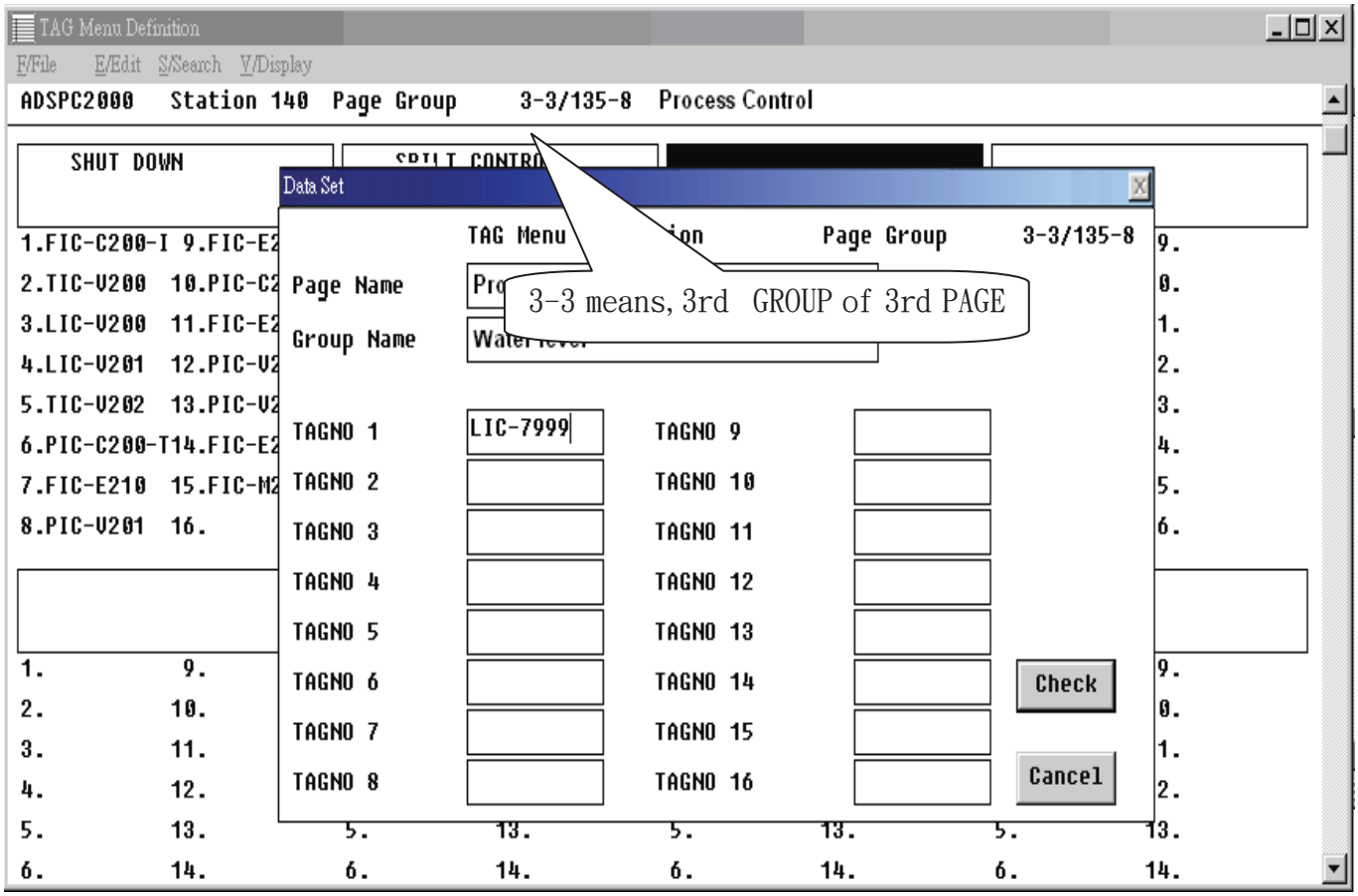

Database Parameter
Proc
Registration
Delete :D
Compress :C
Type
User TAG :U
Module TAG:M
All
Range
Select TAG
All
Directory:
Ref
Start TAG:
End TAG:
STNO:
Directory
Directory: C:\EHF-101.CFD\EHF-101.IFD
Drive :
Dir :
A:
B:
C:
D:
E:
F:
\EHF-101.CFD\EHF-101.DFD
\EHF-101.CFD\EHF-101.IFD
Check
Cancel
1. select Ref
2. select IFD that you registered TAG.
3. select Check
1. select Module TAG
2. select Select TAG
1 FIC-C200-I
2 FIC-E210
3 FIC-E203
4 FIC-E202
5 FIC-M201
6 FIC-C402-1
7 FIC-C402-2
8 FIC-M401-1
144 M501-AO
145 M502-AO
146 M503-AO
147 M504-AO
148 LIC-7999
149
150
3. select start TAG
4. select end TAG

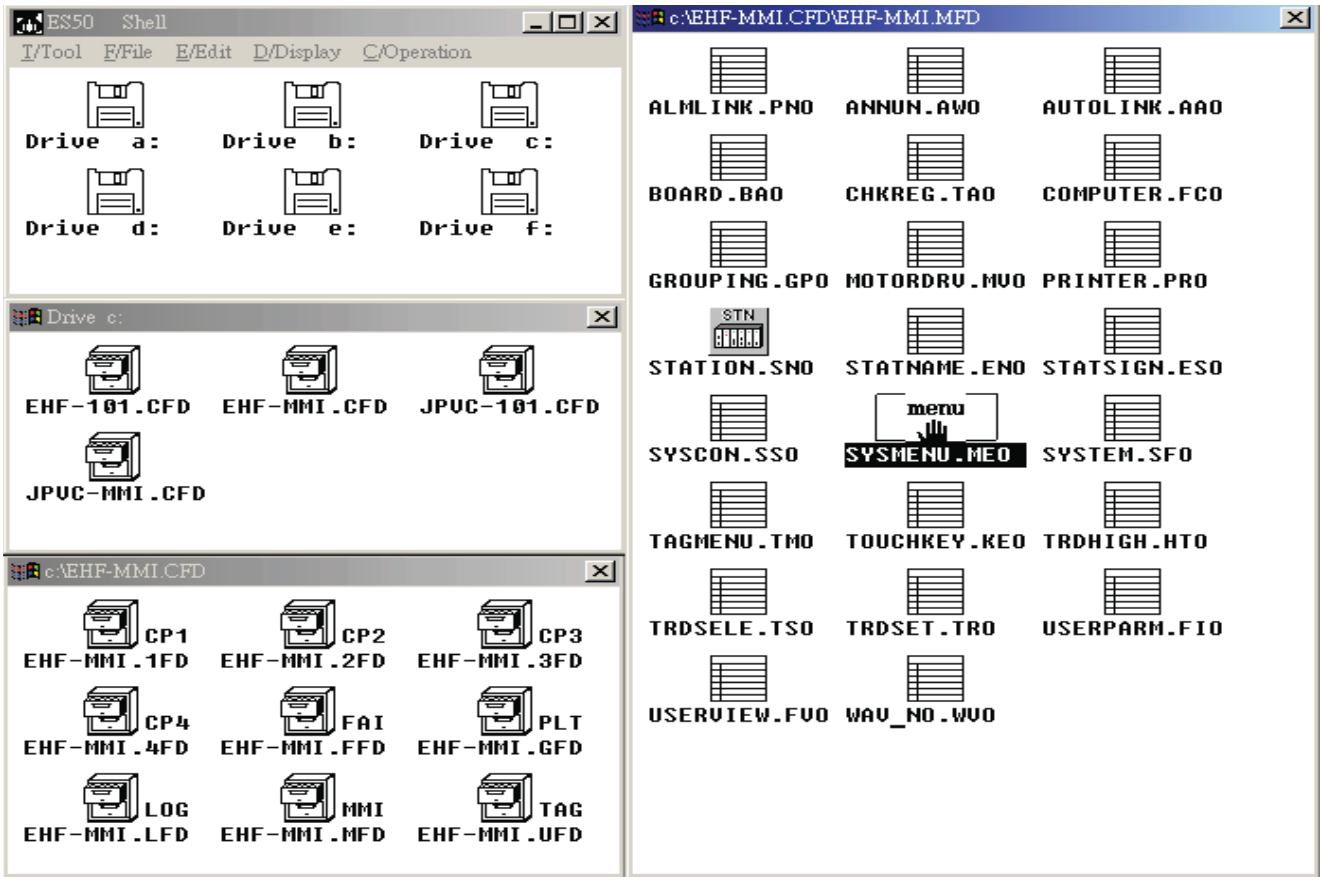


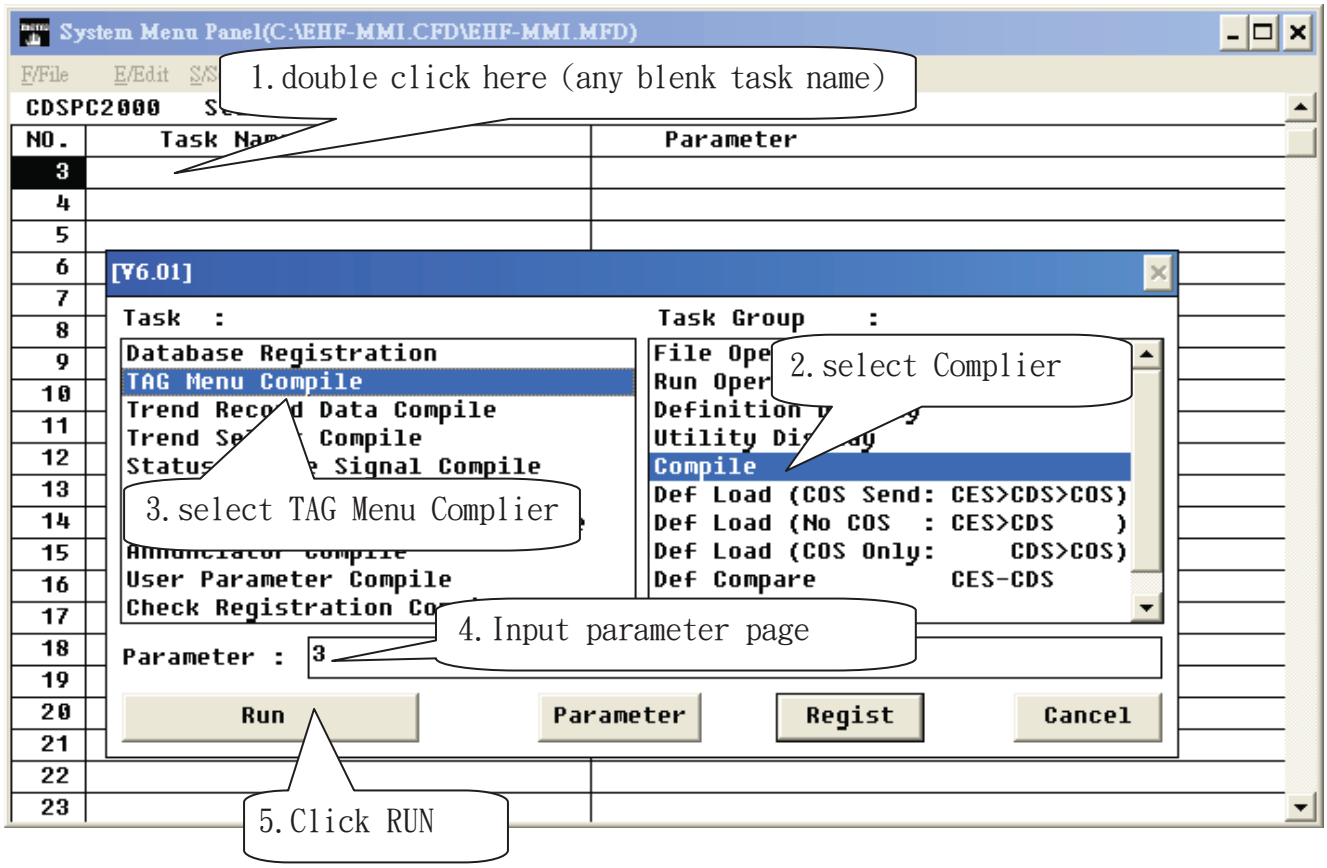
TAGMENU means TAG list (menu), one page of TAGMENU has 8 GROUP, and one group has 16 LOOP , when you LOADING parameter by PAGE.



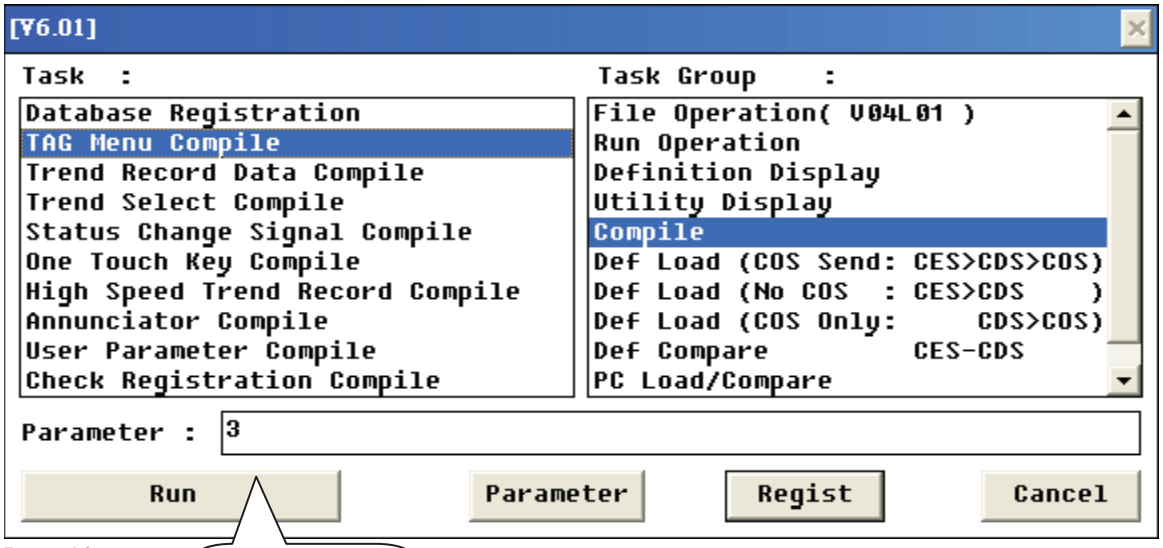
8 groups in the same PAGE have the same page name.



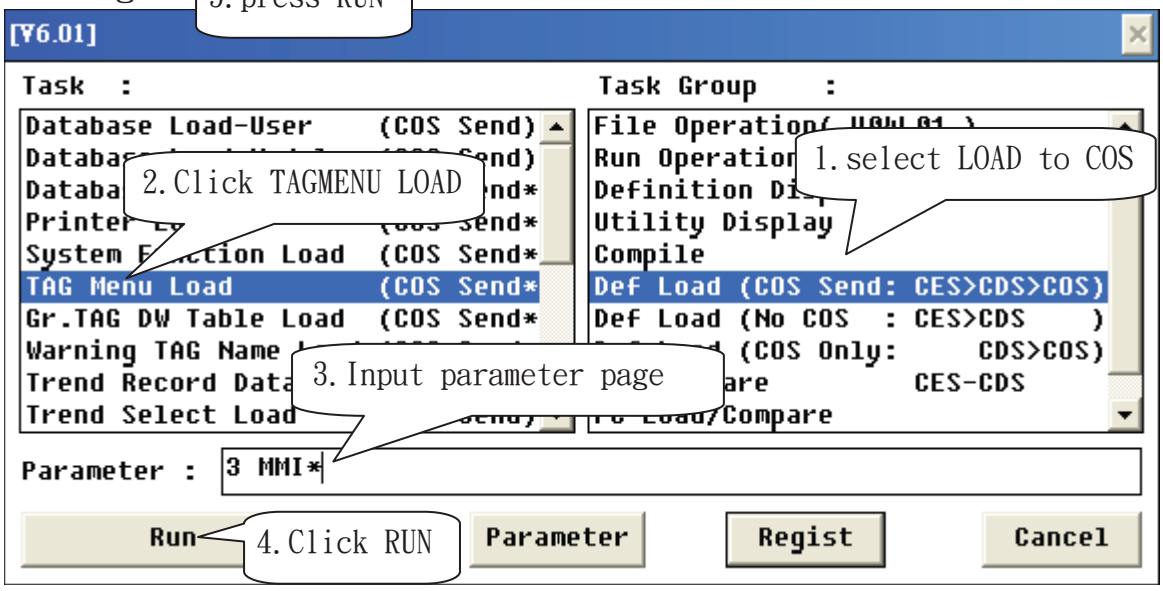

ES50 Shell
T/Tool F/File E/Edit D/Display C/Operation
Drive a:
Drive b:
Drive c:
Drive d:
Drive e:
Drive f:
Drive c:
EHF-101.CFD
EHF-MMI.CFD
JPVC-101.CFD
JPVC-MMI.CFD
c:\EHF-MMI.CFD
CP1
EHF-MMI.1FD
CP2
EHF-MMI.2FD
CP3
EHF-MMI.3FD
CP4
EHF-MMI.4FD
FAI
EHF-MMI.FFD
PLT
EHF-MMI.GFD
LOG
EHF-MMI.LFD
MMI
EHF-MMI.MFD
TAG
EHF-MMI.UFD
c:\EHF-MMI.CFD\EHF-MMI.MFD
ALMLINK.PNO
ANNUN.AWO
AUTOLINK.AAO
BOARD.BAO
CHKREG.TAO
COMPUTER.FCO
GROUPING.GPO
MOTORDRV.MVO
PRINTER.PRO
STN
STATION.SNO
STATNAME.ENO
STATSIGN.ESO
SYSCON.SSO
menu
SYSMENU.MEO
SYSTEM.SFO
TAGMENU.TMO
TOUCHKEY.KEO
TRDHIGH.HTO
TRDSELE.TSO
TRDSET.TRO
USERPARM.FIO
USERVIEW.FVO
WAV_NO.WVO


System Menu Panel(C:\EHF-MMI.CFD\EHF-MMI.MFD)
1.double click here (any blenk task name)
CDSPC2000
NO.
Task Nam
Parameter
3
4
5
6
7
8
9
10
11
12
13
14
15
16
17
18
19
20
21
22
23
[V6.01]
Task :
Task Group :
Database Registration
TAG Menu Compile
Trend Reco d Data Compile
Trend Se Compile
Status Signal Compile
User Parameter Compile
Check Registration Co
File Ope
Run Oper
Definition
Utility Di ay
Compile
Def Load (COS Send: CES>CDS>COS)
Def Load (No COS : CES>CDS)
Def Load (COS Only: CDS>COS)
Def Compare CES-CDS
2.select Complier
3.select TAG Menu Complier
4.Input parameter page
Parameter : 3
Run
Parameter
Regist
Cancel
5.Click RUN

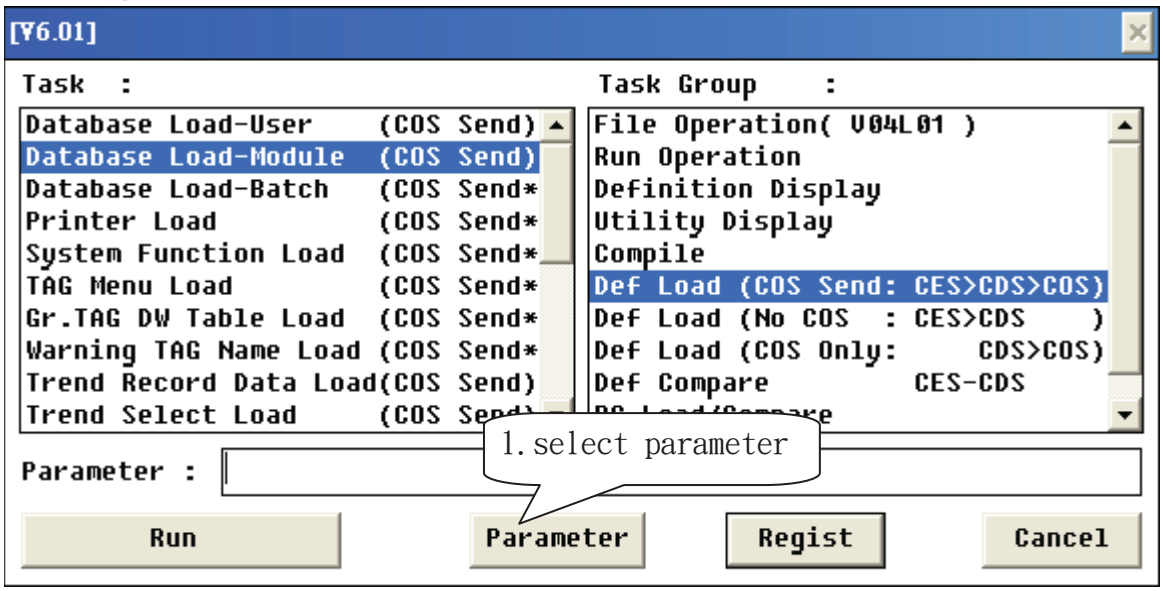
Compile

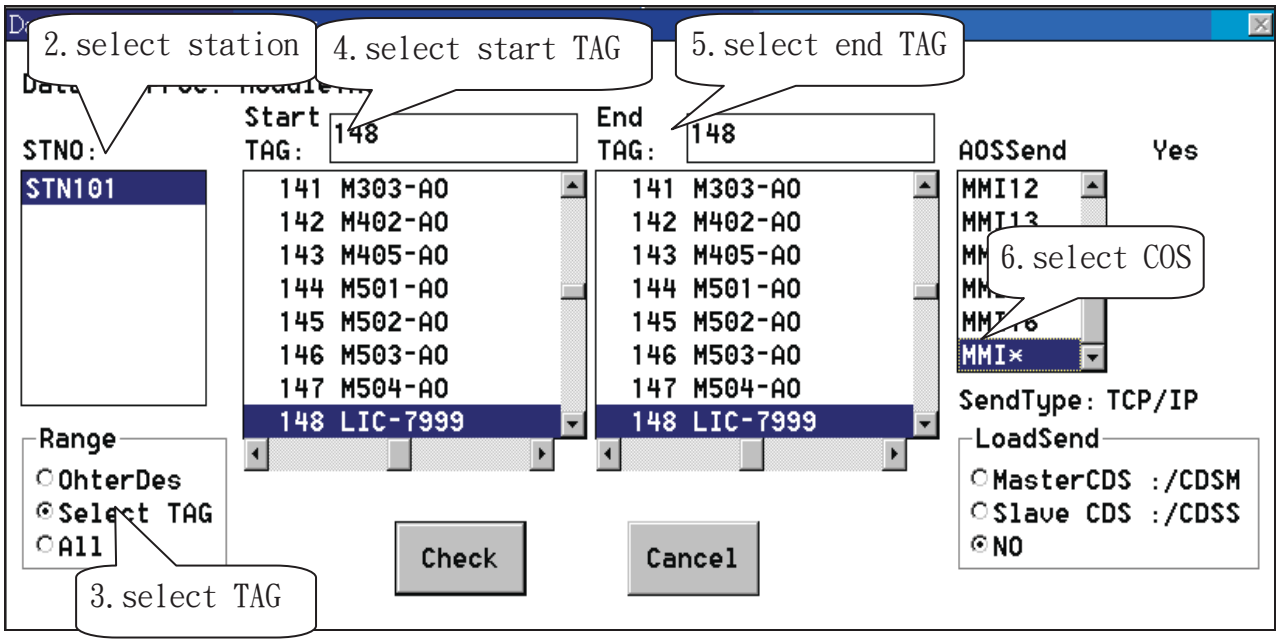


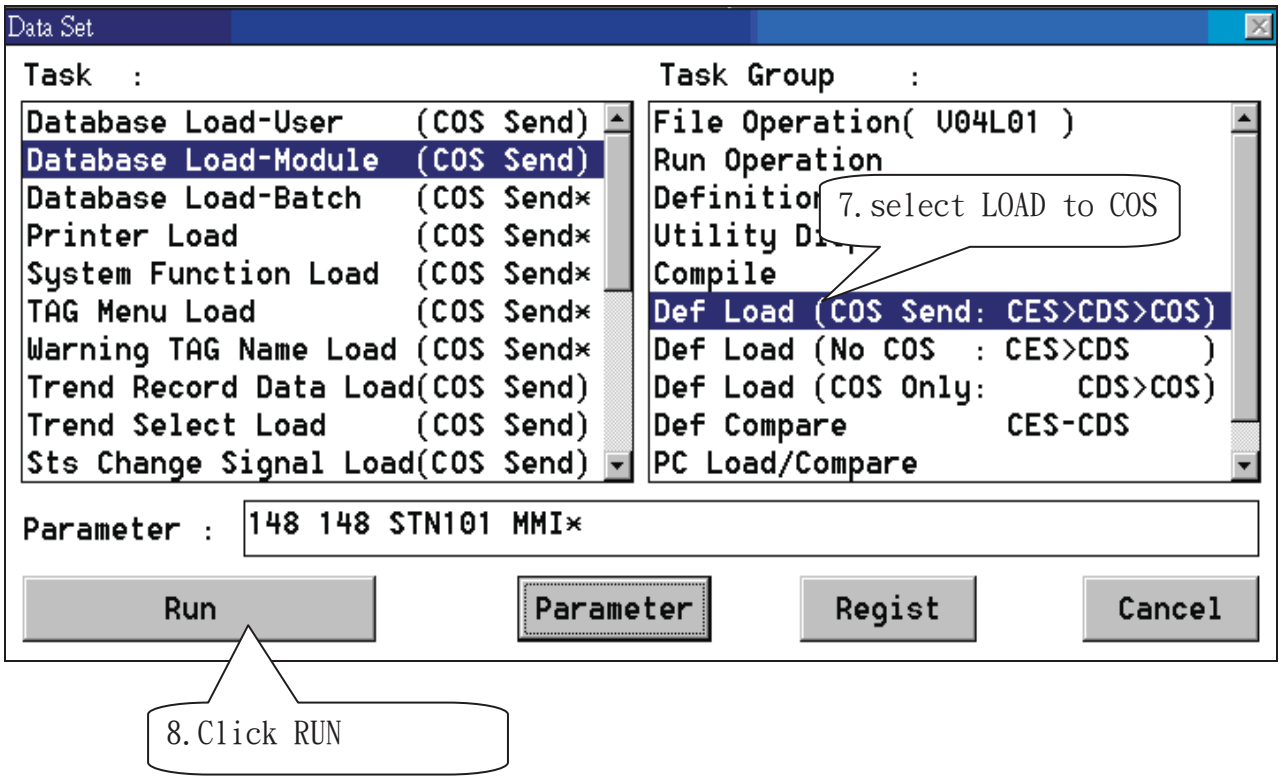
Loading



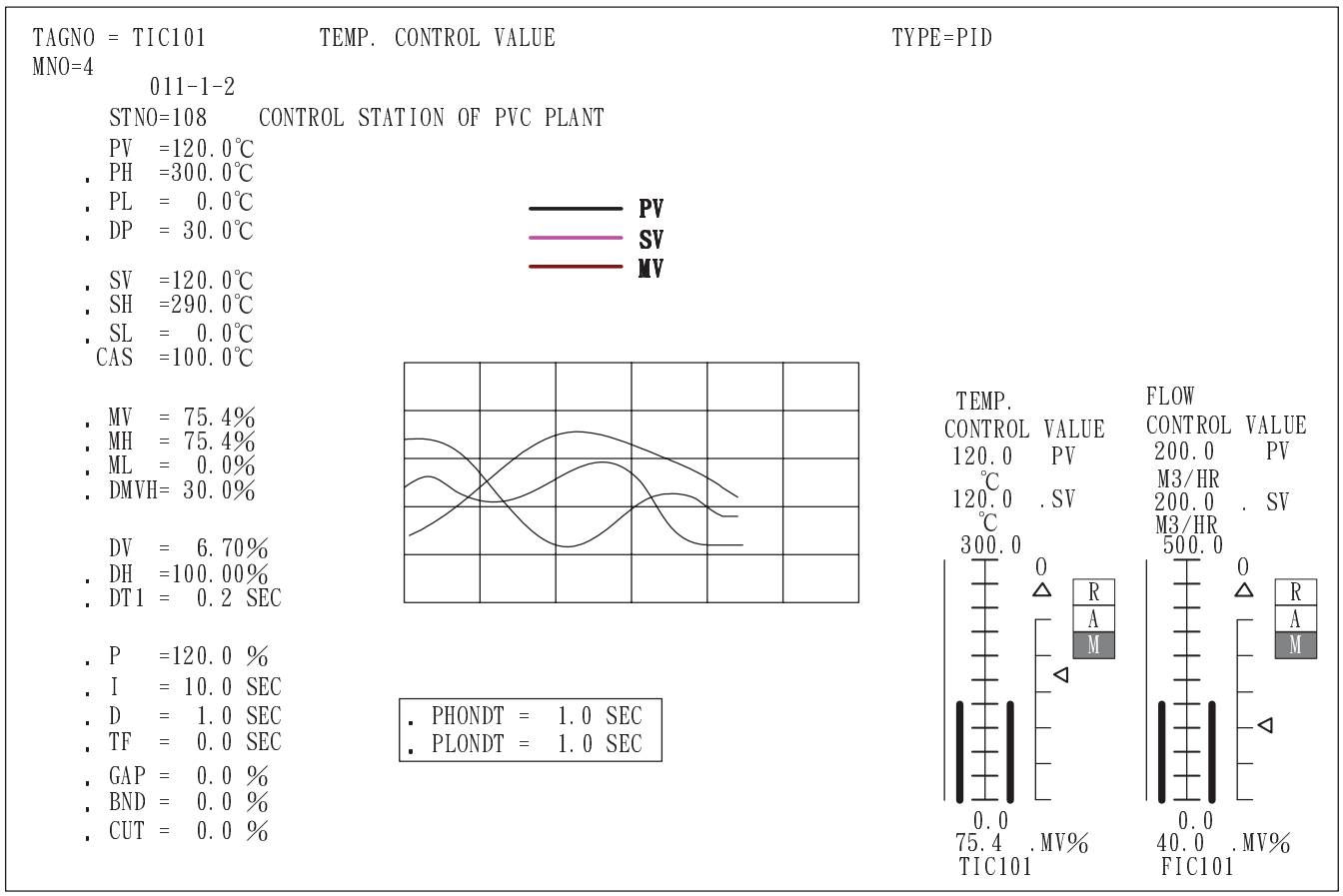
Loading DATABASE LOAD-MODULE LOAD to CDS




2. select station
4. select start TAG
5. select end TAG
STNO:
Start TAG:
148
End TAG:
148
AOSSend Yes
STN101
141 M303-AO
142 M402-AO
143 M405-AO
144 M501-AO
145 M502-AO
146 M503-AO
147 M504-AO
148 LIC-7999
MMI12
MMI*
6. select COS
SendType: TCP/IP
Range
OhterDes
Select TAG
All
3. select TAG
Check
Cancel
LoadSend
MasterCDS :/CDSM
Slave CDS :/CDSS
NO


Data Set
Task :
Task Group :
Database Load-User (COS Send)
Database Load-Module (COS Send)
Database Load-Batch (COS Send*
Printer Load (COS Send*
System Function Load (COS Send*
TAG Menu Load (COS Send*
Warning TAG Name Load (COS Send*
Trend Record Data Load(COS Send)
Trend Select Load (COS Send)
Sts Change Signal Load(COS Send)
File Operation(V04L01)
Run Operation
Compile
7. select LOAD to COS
Def Load (COS Send: CES>CDS>COS)
Def Load (No COS : CES>CDS)
Def Load (COS Only: CDS>COS)
Def Compare CES-CDS
PC Load/Compare
Parameter :
148 148 STN101 MMI*
Run
Parameter
Regist
Cancel
8. Click RUN

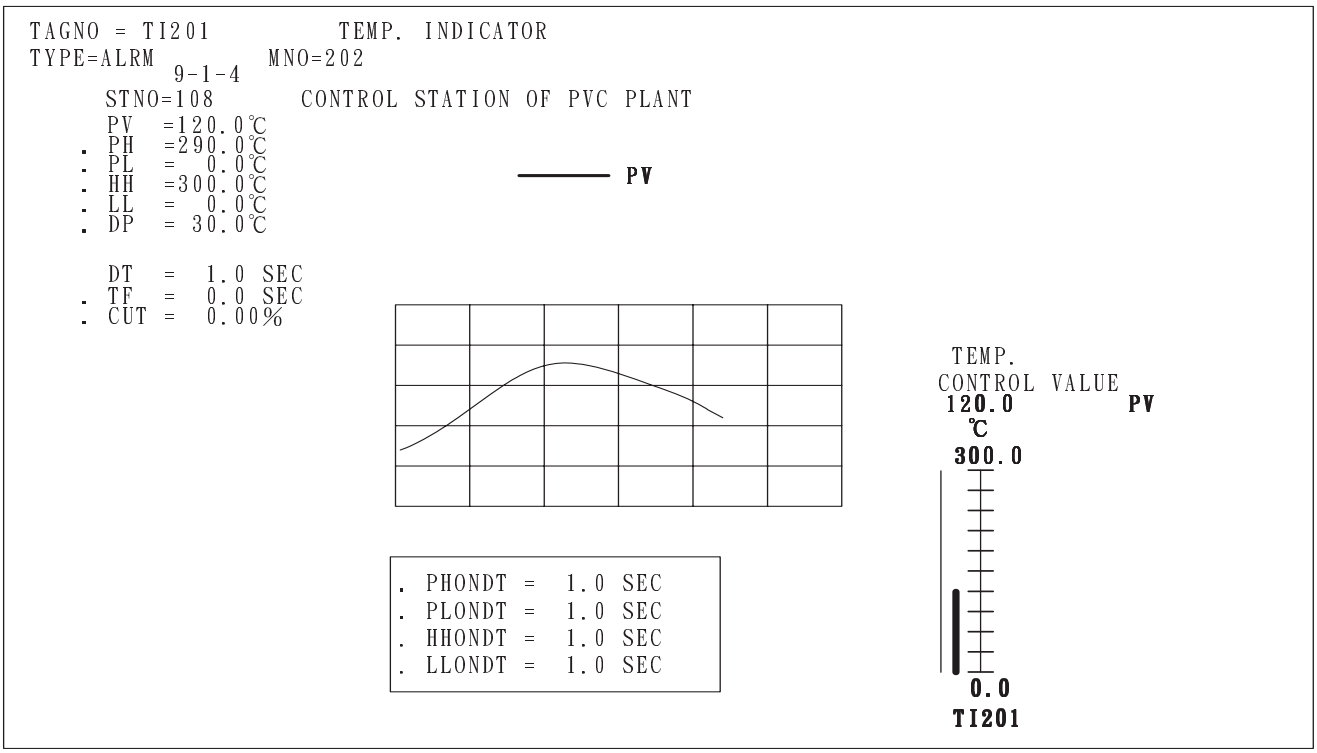
PID PARAMETER PANEL OF LOOP PANEL



abbreviation	Meaning	set up	Setting range		Unit	Remarks
			MIN	MAX		
PV	Press variable	——	——			
PH	Press variable upper limit	○	Within specified range of engineering valve			PL≦PH
PL	Press variable lower limit	○	Within specified range of engineering valve			
DP	dPV absolute value upper limit	○	0.0	327.67	%	
SV	Set point variable	○	In module dimensions range			SL≦SV≦SH
SH	Set point variable upper limit	○	In module dimensions range			SL≦SH
SL	Set point variable lower limit	○	In module dimensions range			SL≦SH
CAS	Remote set point variable	——	——			
MV	Manipulated output variable	○	-25.0	125.0	%	ML≦MV≦MH
MH	Manipulated output upper limit	○	-25.0	125.0	%	ML≦MH
ML	Manipulated output lower limit		-25.0	125.0	%	ML≦MH
DMVH	Manipulated output variable change ratio upper limit	○	0.0	100.0	%	

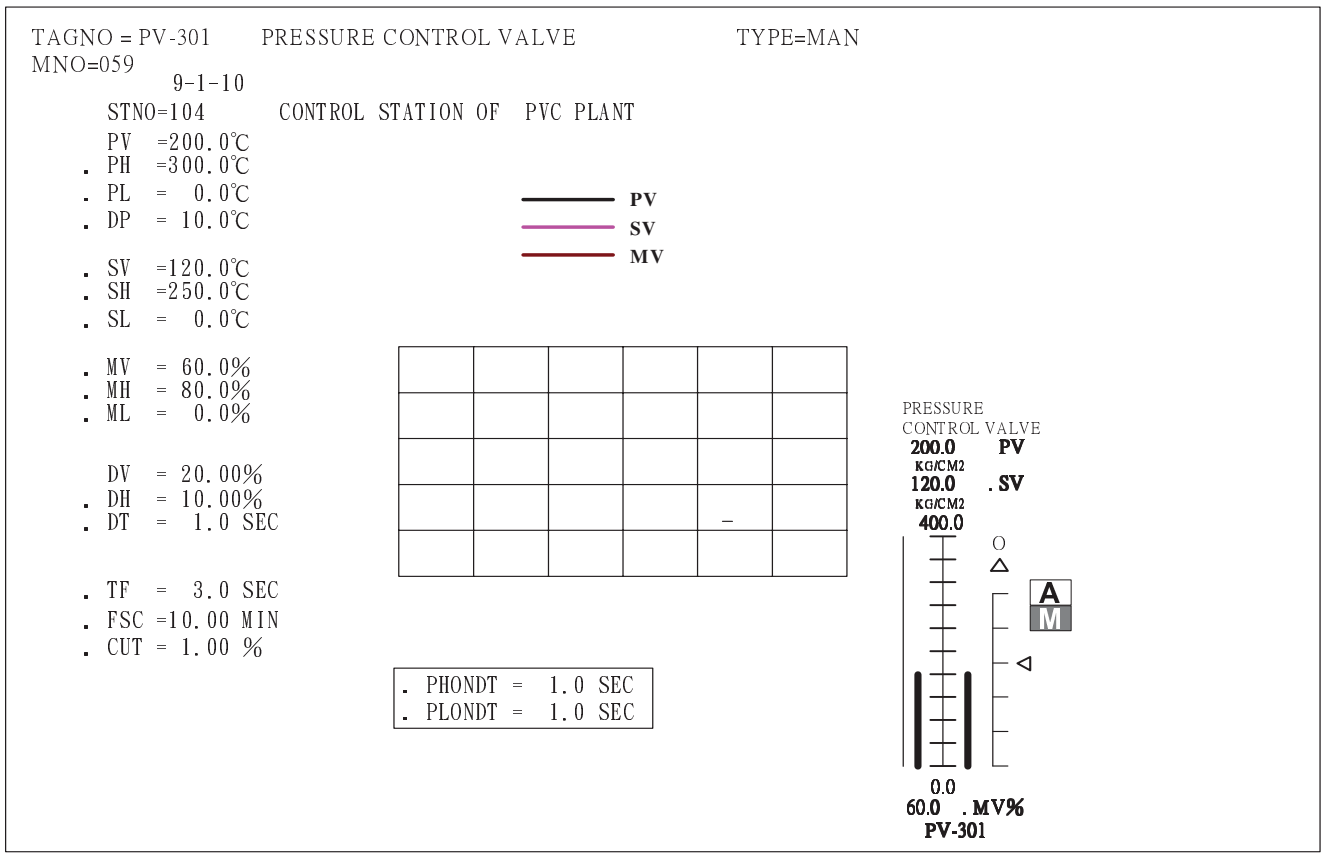
abbreviation	Meaning	set up	Setting range		Unit	Remarks
			MIN	MAX		
DV	Deviation	——	——			
DH	Deviation absolute value upper limit	○	0.00	100.00	%	
DT1	Sampling time	○	0.2	3276.6	SEC	
P	Proportional band	○	0.2	655.3	%	
I	Integration time	○	0.2	3276.6	SEC	SL≦SV≦SH
D	Differentiation time	○	0.0	900.0	SEC	SL≦SH
TF	Filter time constant	○	0.0	900.0	SEC	SL≦SH
GAP	Dead band	○	0.0	100.0	%	
BND	Vertical dead band	○	0.0	100.0	%	
CUT	Low input cut value	○	-25.00	327.67	%	
PHONDT	PH on-delay timer set value	○	0.0	51.0	SEC	
PLONDT	PL on-delay timer set value	○	0.0	51.0	SEC	

ALRM PARAMETER PANEL OF LOOP PANEL



abbreviation	Meaning	set up	Setting range		Unit	Remarks
			MIN	MAX		
PV	Press variable	——	——			
PH	Press variable upper limit	○	Within specified range of engineering valve			PL≦PH≦HH
PL	Press variable lower limit	○	Within specified range of engineering valve			LL≦PL≦PH
HH	Press variable upper limit		Within specified range of engineering valve			PH≦HH
LL	Press variable lower limit		Within specified range of engineering valve			LL≦PL
DP	dPV absolute value upper limit	○	0.0	327.67	%	
DT	Sampling time	——	——		SEC	
TF	Filter time constant	○	0.0	900.0	SEC	
CUT	Low input cut value	○	-25.00	327.67	%	
PHONDT	PH on-delay timer set value	○	0.0	51.0	SEC	
PLONDT	PL on-delay timer set value	○	0.0	51.0	SEC	
HHONDT	HH on-delay timer set value	○	0.0	51.0	SEC	
LLONDT	LL on-delay timer set value	○	0.0	51.0	SEC	

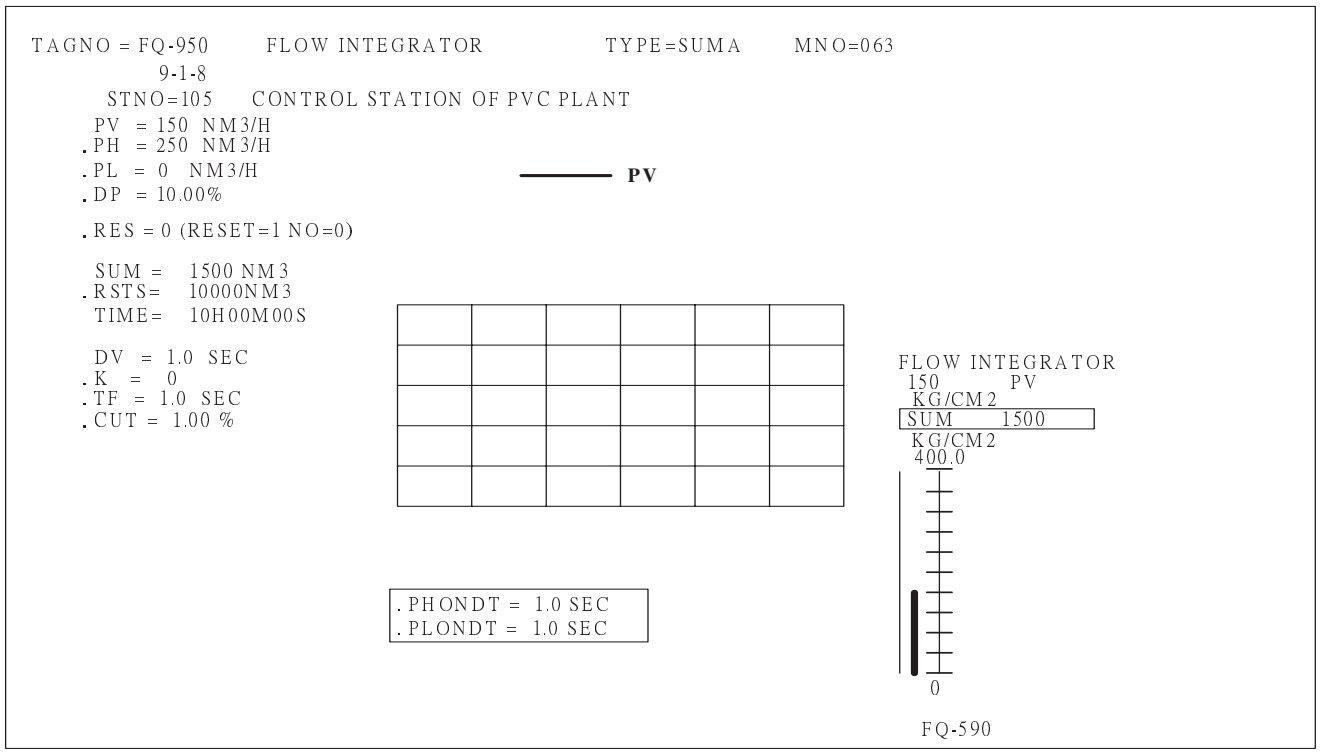
MAN PARAMETER PANEL OF LOOP PANEL



abbreviation	Meaning	set up	Setting range MIN	Setting range MAX	Unit	Remarks
PV	Press variable	— —	——			
PH	Press variable upper limit	○	Within specified range of engineering valve			PL≦PH
PL	Press variable lower limit	○	Within specified range of engineering valve			
DP	dPV absolute value upper limit	○	0.0	327.67	%	
SV	Set point variable	○	In module dimensions range			SL≦SV≦SH
SH	Set point variable upper limit	○	In module dimensions range			SL≦SH
SL	Set point variable lower limit	○	In module dimensions range			SL≦SH
MV	Manipulated output variable	○	-25.0	125.0	%	ML≦MV≦MH
MH	Manipulated output upper limit	○	-25.0	125.0	%	ML≦MH
ML	Manipulated output lower limit		-25.0	125.0	%	ML≦MH
DMVH	Manipulated output variable change ratio upper limit	○	0.0	100.0	%	

abbreviation	Meaning	set up	Setting range		Unit	Remarks
			MIN	MAX		
DV	Deviation	——	——			
DH	Deviation absolute value upper limit	○	0.00	100.00	%	
DT1	Sampling time	○	0.2	3276.6	SEC	
TF	Filter time constant	○	0.0	900.0	SEC	SL≦SH
FSC	Full stroke travel time	○	0.00	327.67	MIN	
CUT	Low input cut value	○	-25.00	327.67	%	
PHONDT	PH on-delay timer set value	○	0.0	51.0	SEC	
PLONDT	PL on-delay timer set value	○	0.0	51.0	SEC	

SUMA PARAMETER PANEL OF LOOP PANEL

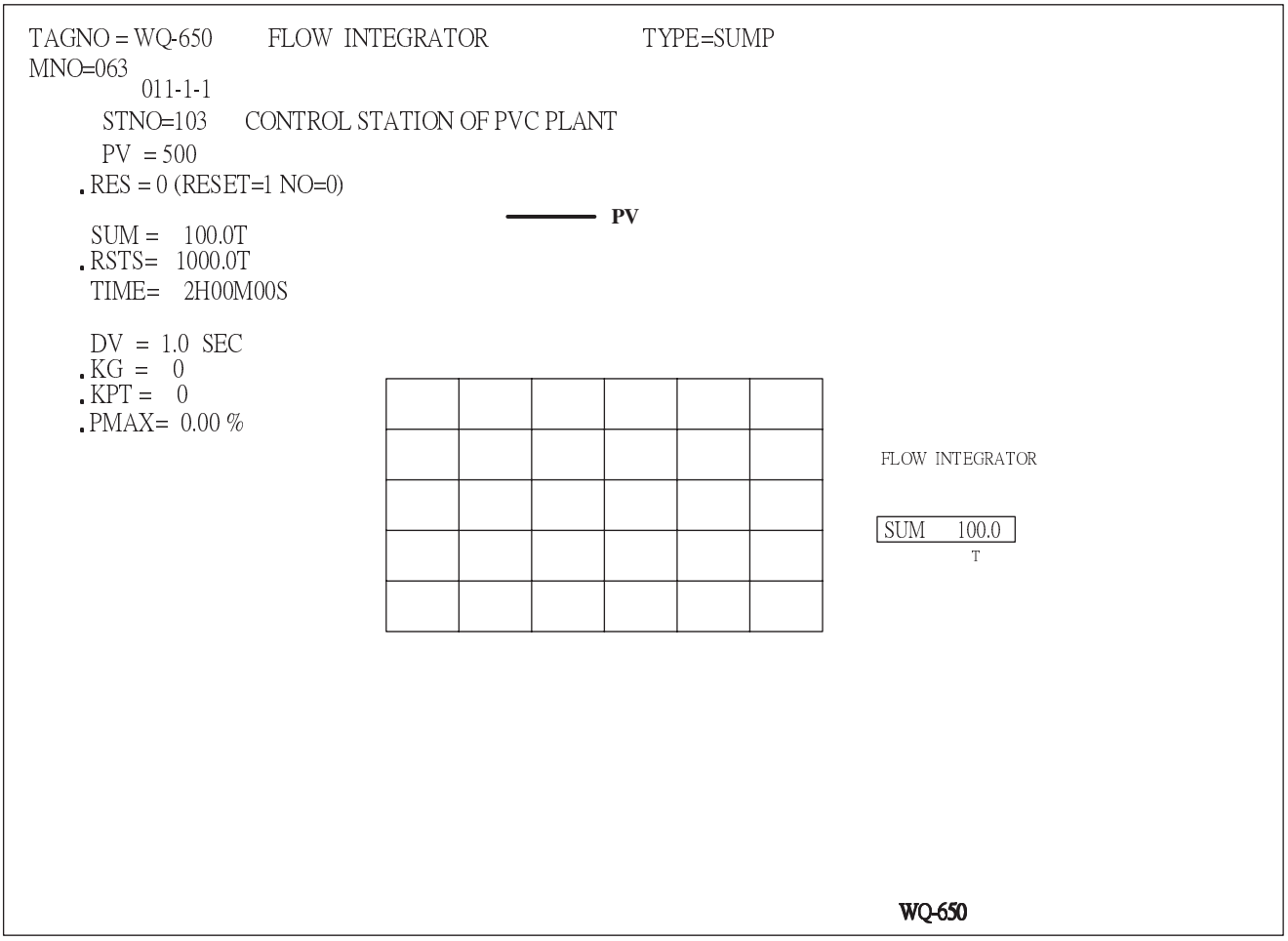


abbreviation	Meaning	Set up	Setting range MIN	Setting range MAX	Unit	Remarks
PV	Press variable	—	——		*1	
PH	Press variable upper limit	○	Within specified range of engineering valve		*1	PL≦PH
PL	Press variable lower limit	○	Within specified range of engineering valve		*1	
DP	dPV absolute value upper limit	○	0.0	327.67	%	
RES	Reset of integrated value and integrated time	○	1 when reset		——	
SUM	Integrated value	—	——		*1	
RSTS	Reset value	○	0	2147483647	*1	
TIME	Integrated time	—	——		*2	
DT	Integrated time interval	—	——		SEC	
K	Integration unit conversion efficiency	○	0	32767	%	
TF	Filter time constant	○	0.0	900.0	SEC	SL≦SH
CUT	Low input cut value	○	-25.00	327.67	%	
PHONDT	PH on-delay timer set value	○	0.0	51.0	SEC	
PLONDT	PL on-delay timer set value	○	0.0	51.0	SEC	

*1 :PV UNIT

*2 : ****H**M**S

SUMP PARAMETER PANEL OF LOOP PANEL

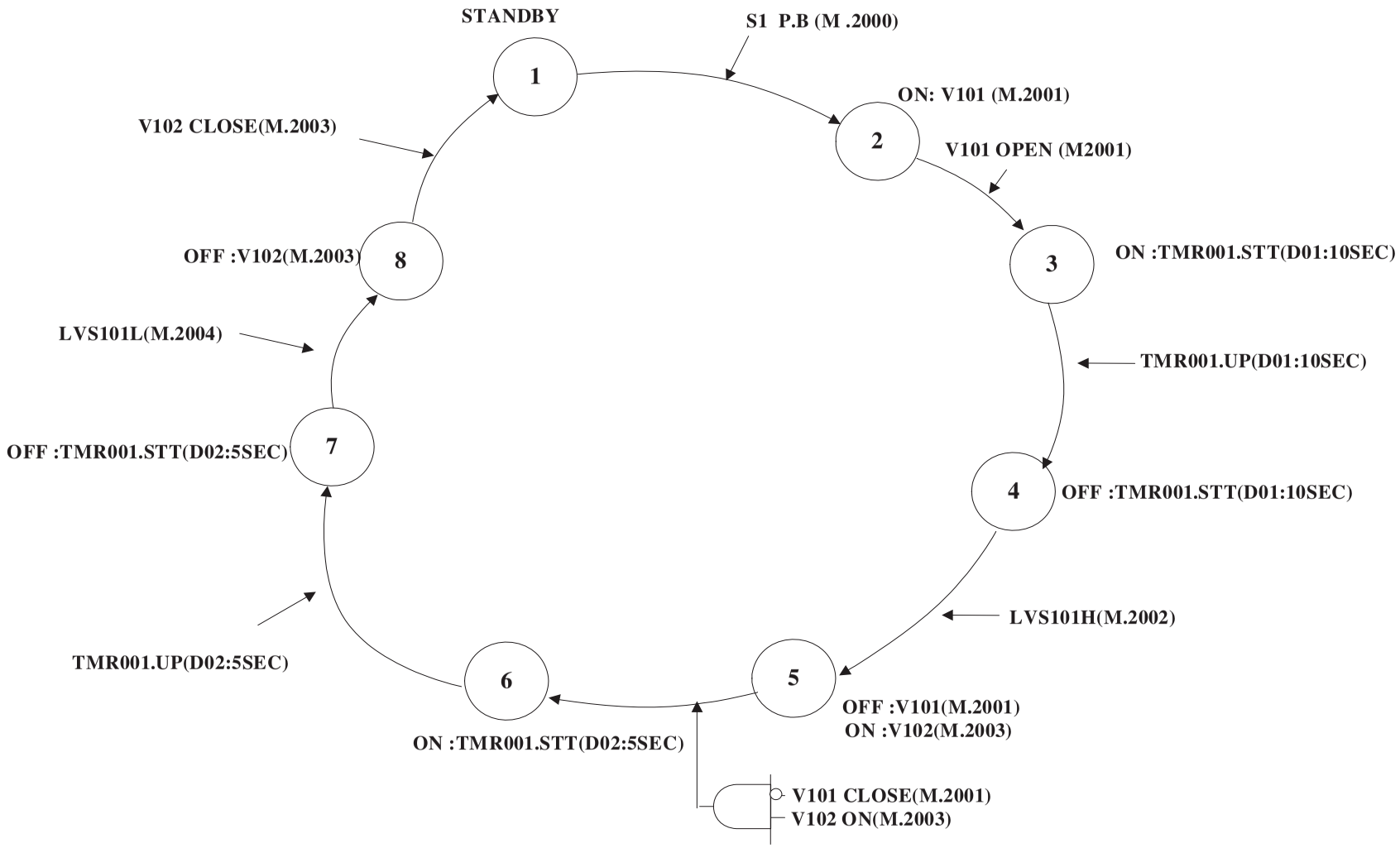


abbreviation	Meaning	Set up	Setting range		Unit	Remarks
			MIN	MAX		
PV	Press variable	—	——		*1	
RES	Reset of integrated value and integrated time	◯	1 when reset		——	
SUM	Integrated value	—	——		*1	
RSTS	Reset value	◯	0	2147483647	*1	
TIME	Integrated time	—	——		*2	
DT	Integrated time interval	—	——		SEC	
KG	Scaling factor numeric part	◯	1	32767	——	
KPT	Scaling factor exponent part	◯	0	4	——	
PMAX	1second maximum integration uint	◯	0.01	327.67	%	

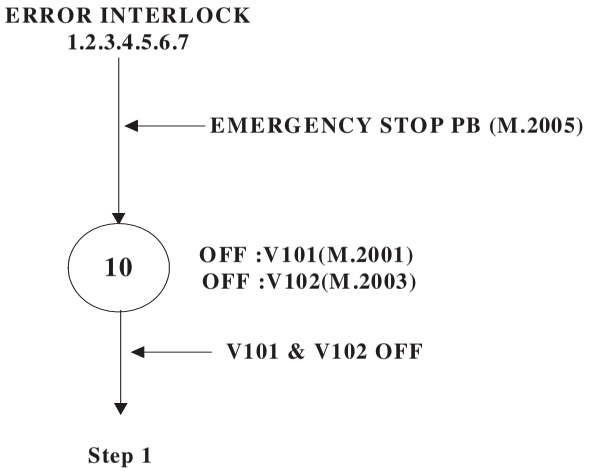
*1 :PV UNIT

*2 : ****H**M**S

8---SEQUENCE CONTROL 1



Revision			Title		Dwg.No.	VisioDocument	1/1
					Ref.	STN108 -- SEQ064	



Revision				Title		Dwg.No.	VisioDocument	1/1
						Ref.	STN108 -- SEQ064	

TYPE1:

COMMENT		OUTPUT TAG NO.	INTERLOCK TAG NO.	INTERLOCK STS	SOURCE STEP NO.	01	02	03	04	05	06	07	08	09	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32
					STANDBY																																
V-101 ON	01	M.2001																																		
V-102 ON	02	M.2003																																		
TMR001.STT	03	TMR001.STT						D01			D02																										
	04																																			
	05																																			
	06																																			
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	31																																			
	32																																			
					DESTINATION STEP NO.	2	3	4	5	6	7	8	1		1																						
					SOURCE STEP NO.	01	02	03	04	05	06	07	08	09	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32

8--SEQUENCE CONTROL 3

Revision				

Title: Z-106

Dwg.No.: VisioDocument 1/1

Ref.: STN108 -- SEQ064

TYPE2:

COMMENT		OUTPUT TAG NO .	INTERLOCK TAG NO .	STS	STANDBY	01	02	03	04	05	06	07	08	09	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32
					SOURCE STEP NO .	01	02	03	04	05	06	07	08	09	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32
V-101 ON	01	M.2001			NOP	S	NOP	NOP	R	NOP	NOP	NOP		R																						
V-102 ON	02	M.2003			NOP	NOP	NOP	NOP	S	NOP	NOP	R		R																						
TMR001.STT	03	TMR 001 .STT					D01			D02																										
	04																																		
	05																																		
	06																																		
	07																																		
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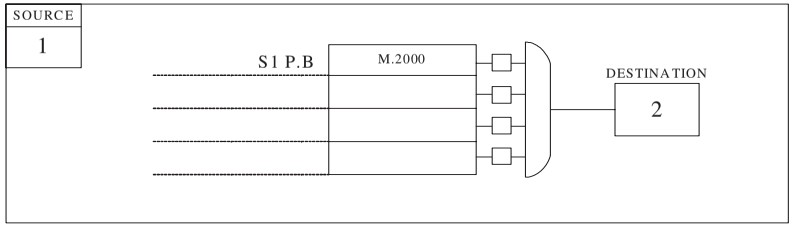
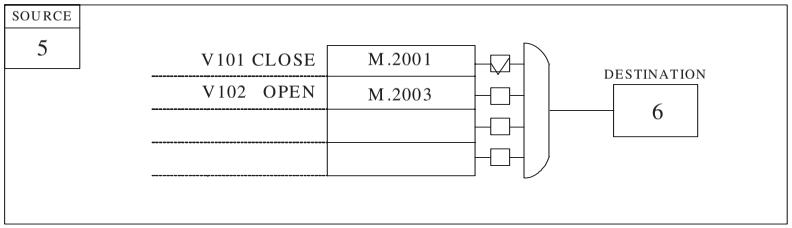
	30																																		
	31																																		
	32																																		
					DESTINATION	2	3	4	5	6	7	8	1		1																						
					STEP NO .																																
					SOURCE STEP NO .	01	02	03	04	05	06	07	08	09	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32

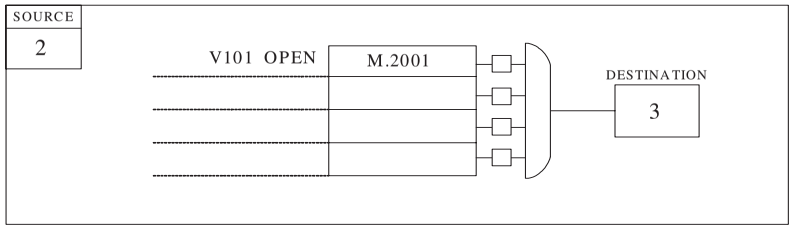
Revision					Title		Dwg No.	VisioDocument	1/1
						Z-106	Ref.	STN 108 -- SEQ 064	

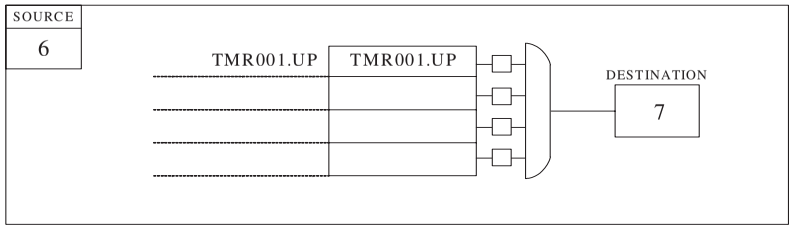
8--SEQUENCE CONTROL 4

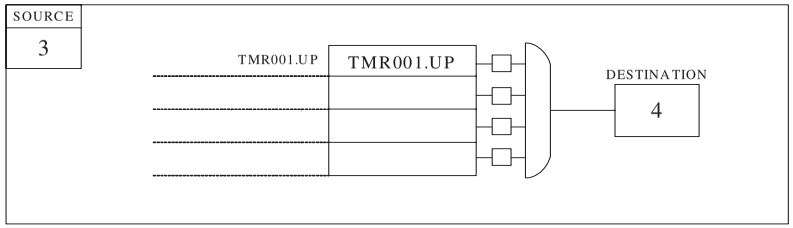
8---SEQUENCE CONTROL 5

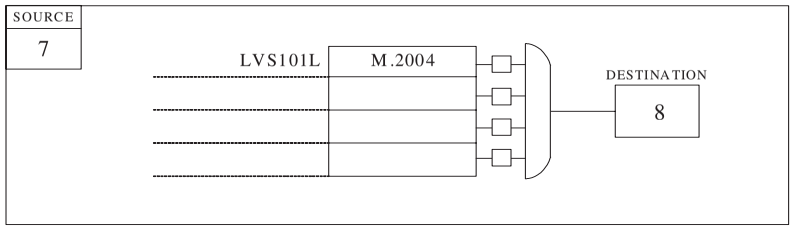


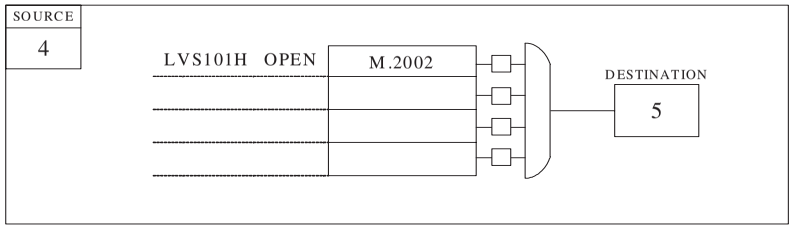


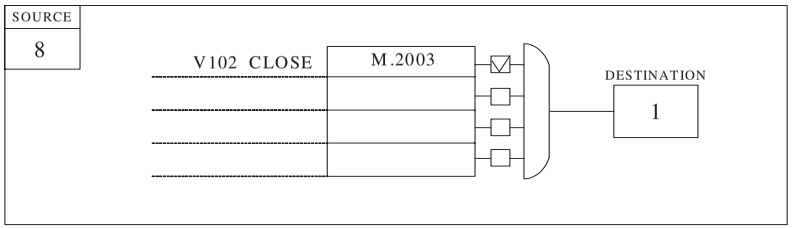












Revision					Title		Dwg.No.	VisioDocument	1/1
							Ref.	STN108 -- SEQ064	

8---SEQUENCE CONTROL 6

NO	COMMENT	InterLock TAG	Sts	Proc	01	02	03	04	05	06	07	08	09	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32
01	EMERGENCY STOP	M.2005	ON	10		○	○	○	○	○	○	○																								
02																																				
03																																				
04																																				
05																																				
06																																				
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29																																				
30																																				
31																																				
32																																				

Revision					Title		Dwg.No.	VisioDocument	1/1
							Ref.	STN108 --- SEQ064	

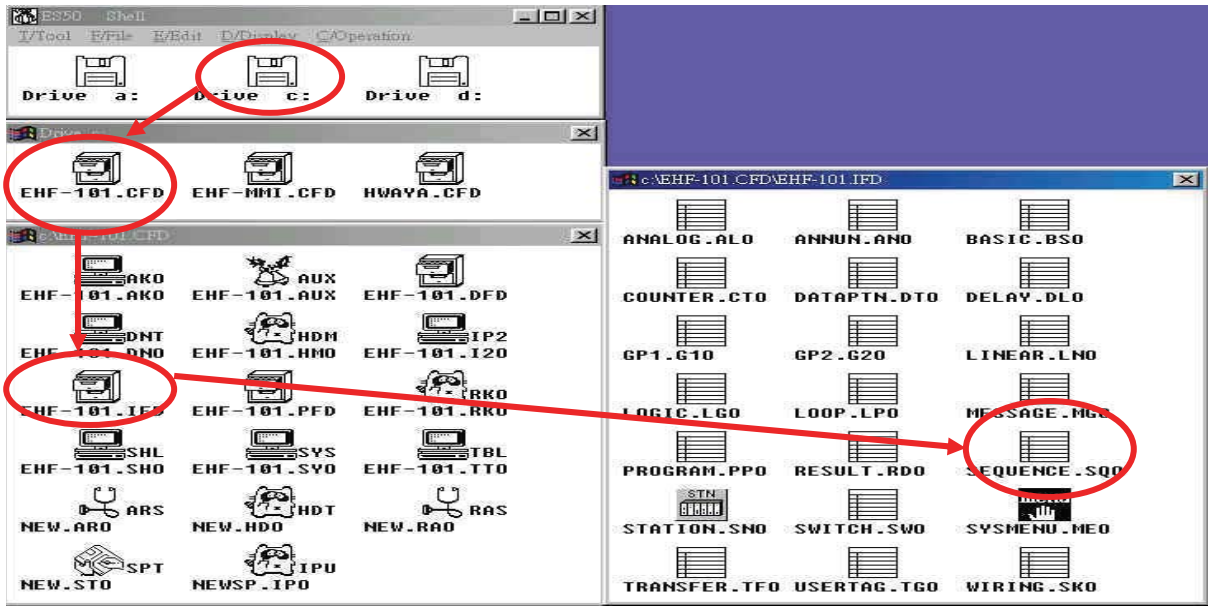
8---SEQUENCE CONTROL 7

NO	Description	Comment	Type	Scale	Unit	Set Range	Data
01	TMR001	D1	TMR				0H00M10S
02	TMR001	D2	TMR				0H00M05S
03							
04							
05							
06							
07							
08							
09							
10							
11							
12							
13							
14							
15							
16							
17							
18							
19							
20							
21							
22							
23							
24							
25							
26							
27							
28							
29							
30							
31							
32							

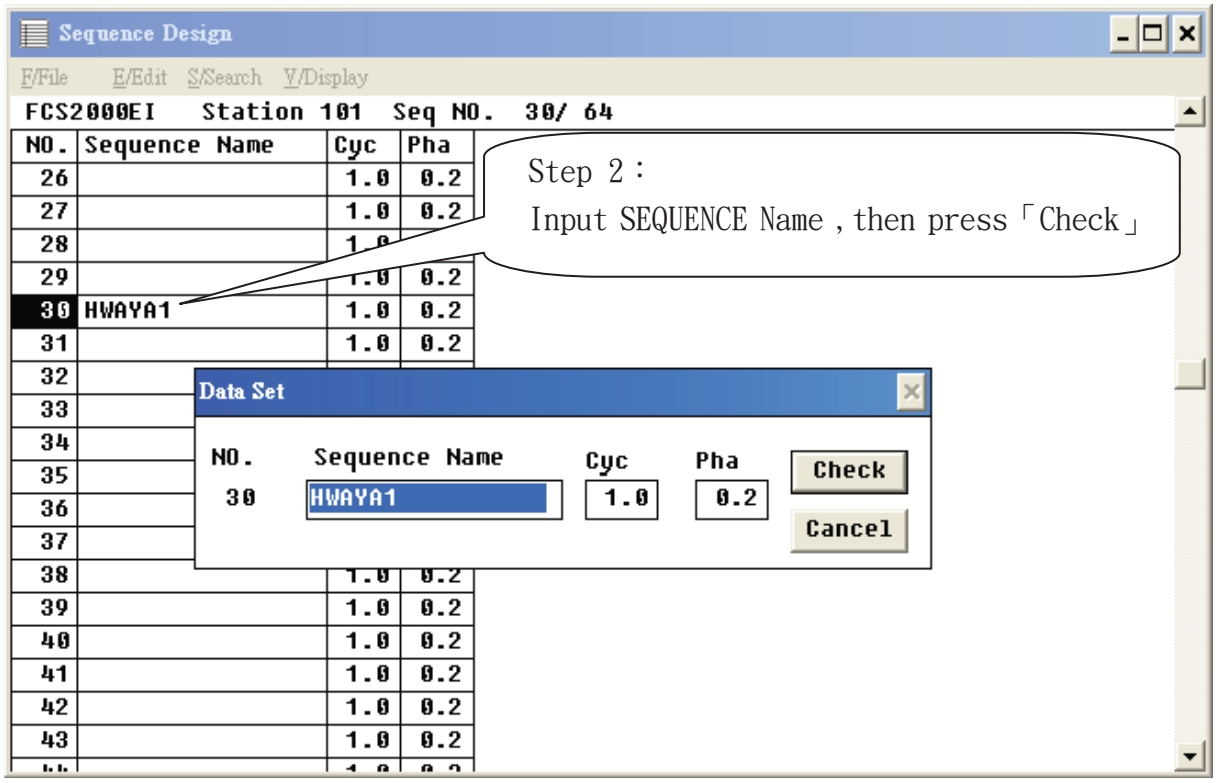
Revision					Title		Dwg.No.	VisioDocument	1/1
							Ref.	STN108 --- SEQ064	

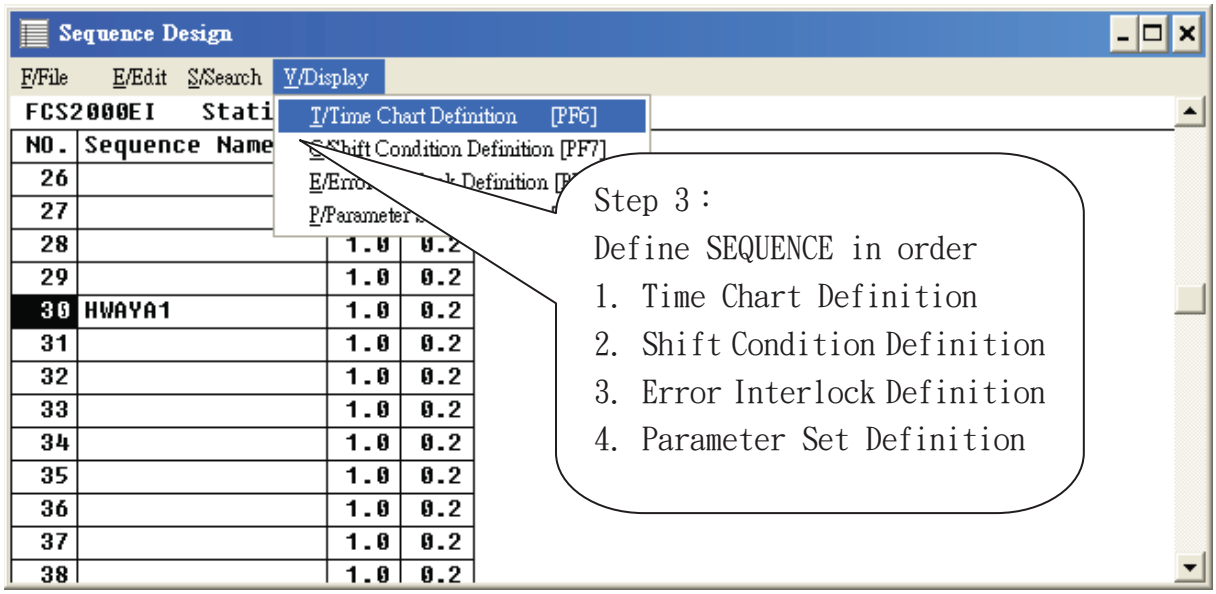
SEQUENCE

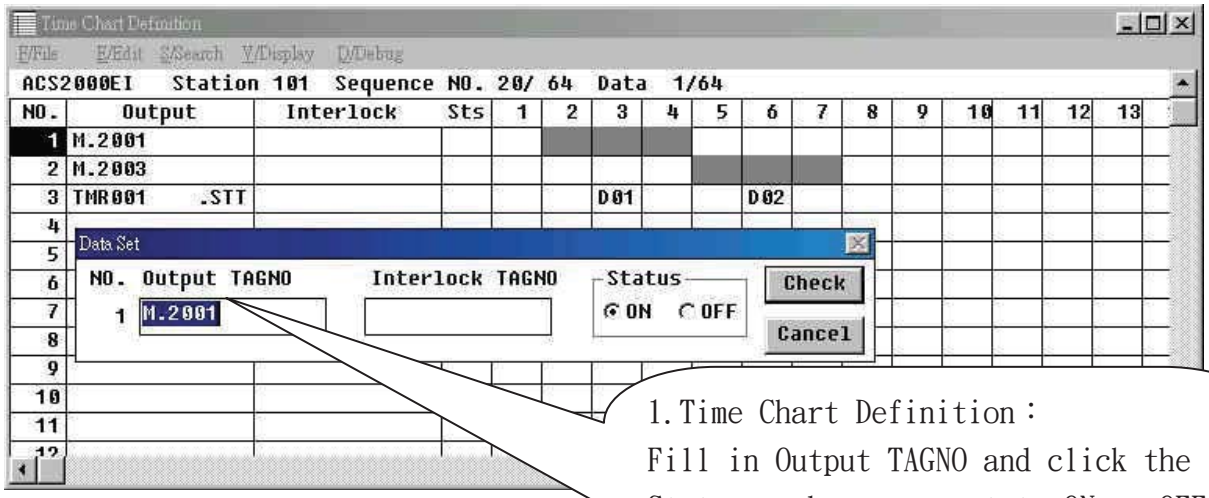
Setting up the function of sequence control
You may follow the steps step by step to set up sequence.

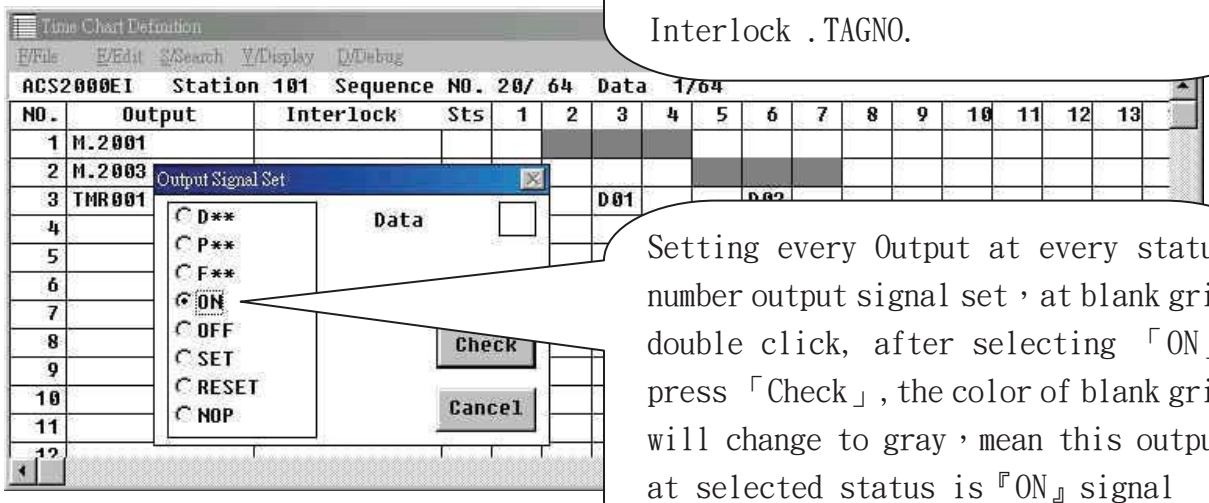


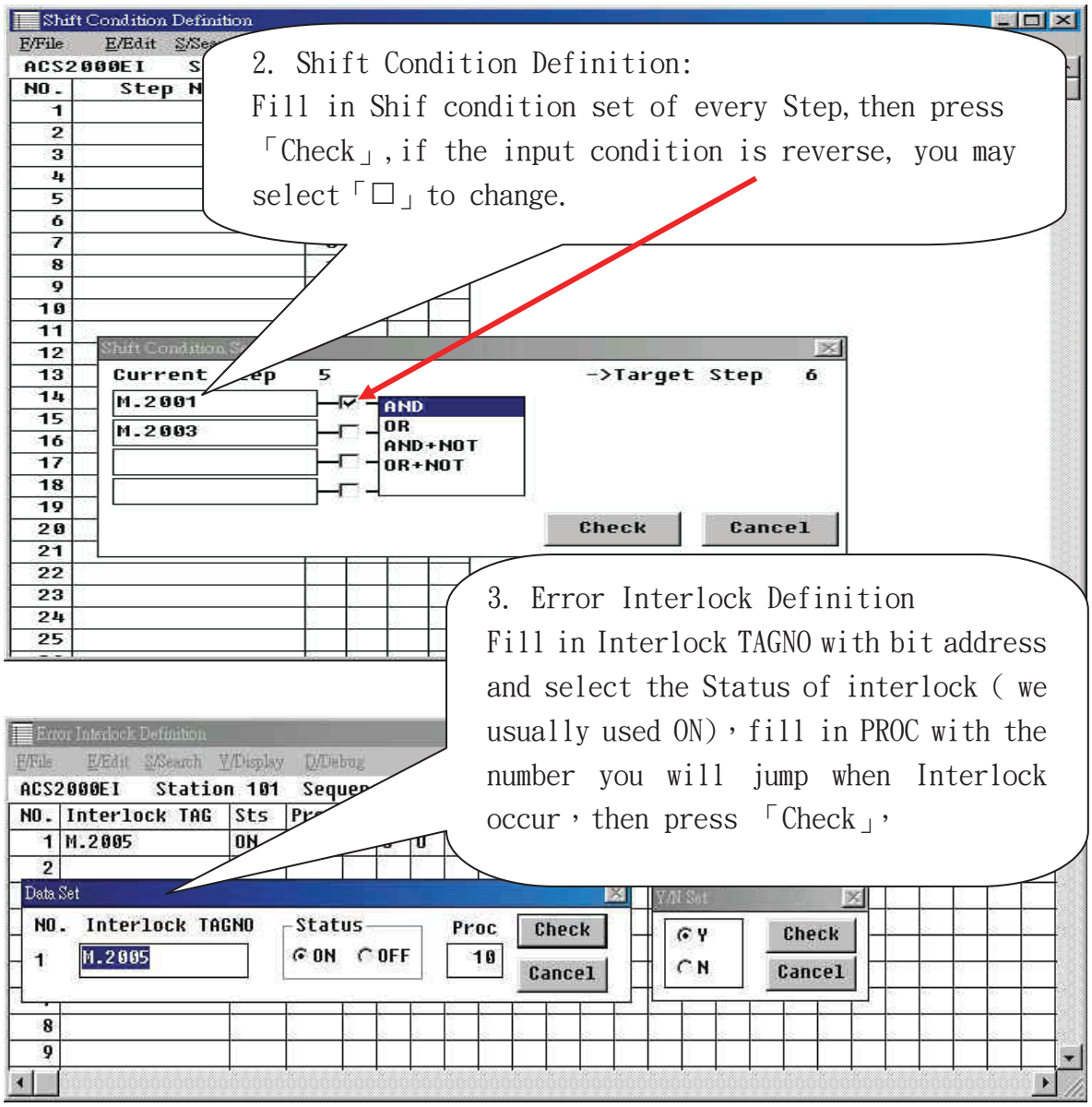
Step1：OPEN Drive c:→EHF-101.CFD→EHF-101.IFD→Double click SEQUENCE.SQO

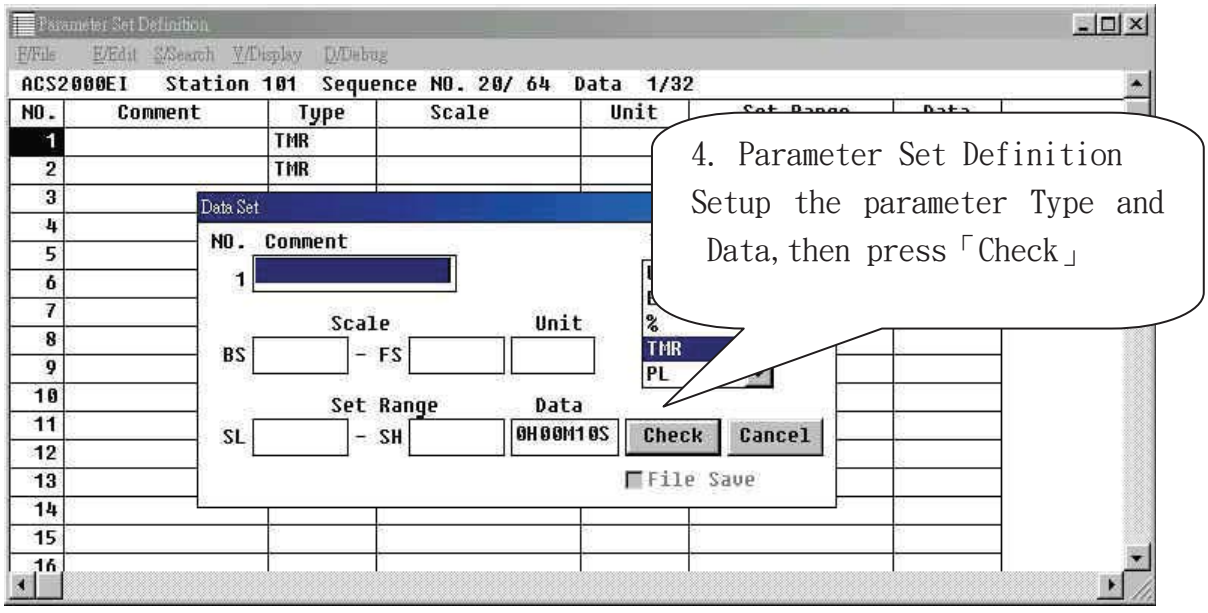


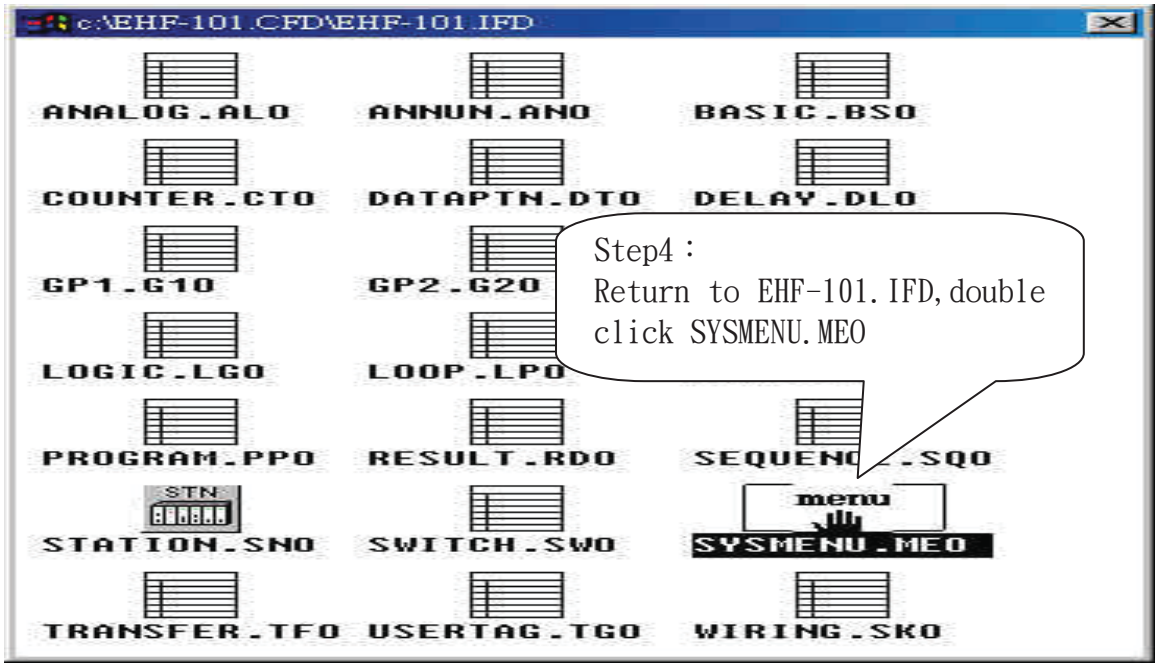

Sequence Design
F/File E/Edit S/Search V/Display
T/Time Chart Definition [PF6]
S/Shift Condition Definition [PF7]
P/Parameter
FCS2000EI Stati
NO. Sequence Name
26
27
28
29
30 HWAYA1
31
32
33
34
35
36
37
38
1.0 0.2
Step 3：
Define SEQUENCE in order
1. Time Chart Definition
2. Shift Condition Definition
3. Error Interlock Definition
4. Parameter Set Definition


Time Chart Definition
F/File E/Edit S/Search V/Display D/Debug
ACS2000EI Station 101 Sequence NO. 20/ 64 Data 1/64
NO. Output Interlock Sts 1 2 3 4 5 6 7 8 9 10 11 12 13
1 M.2001
2 M.2003
3 TMR001 .STT
D01
D02
Data Set
NO. Output TAGNO
Interlock TAGNO
Status
ON OFF
Check
Cancel
1.Time Chart Definition：
Fill in Output TAGNO and click the Status number you want to ON or OFF，then press「Check」，if this output TAG has Interlock，also fill in Interlock .TAGNO.


Time Chart Definition
F/File E/Edit S/Search V/Display D/Debug
ACS2000EI Station 101 Sequence NO. 20/ 64 Data 1/64
NO. Output Interlock Sts 1 2 3 4 5 6 7 8 9 10 11 12 13
1 M.2001
2 M.2003
3 TMR001
D01
Output Signal Set
D** P** F** ON OFF SET RESET NOP
Data
Check
Cancel
Setting every Output at every status number output signal set，at blank grid double click, after selecting 「ON」 press「Check」, the color of blank grid will change to gray，mean this output at selected status is『ON』signal

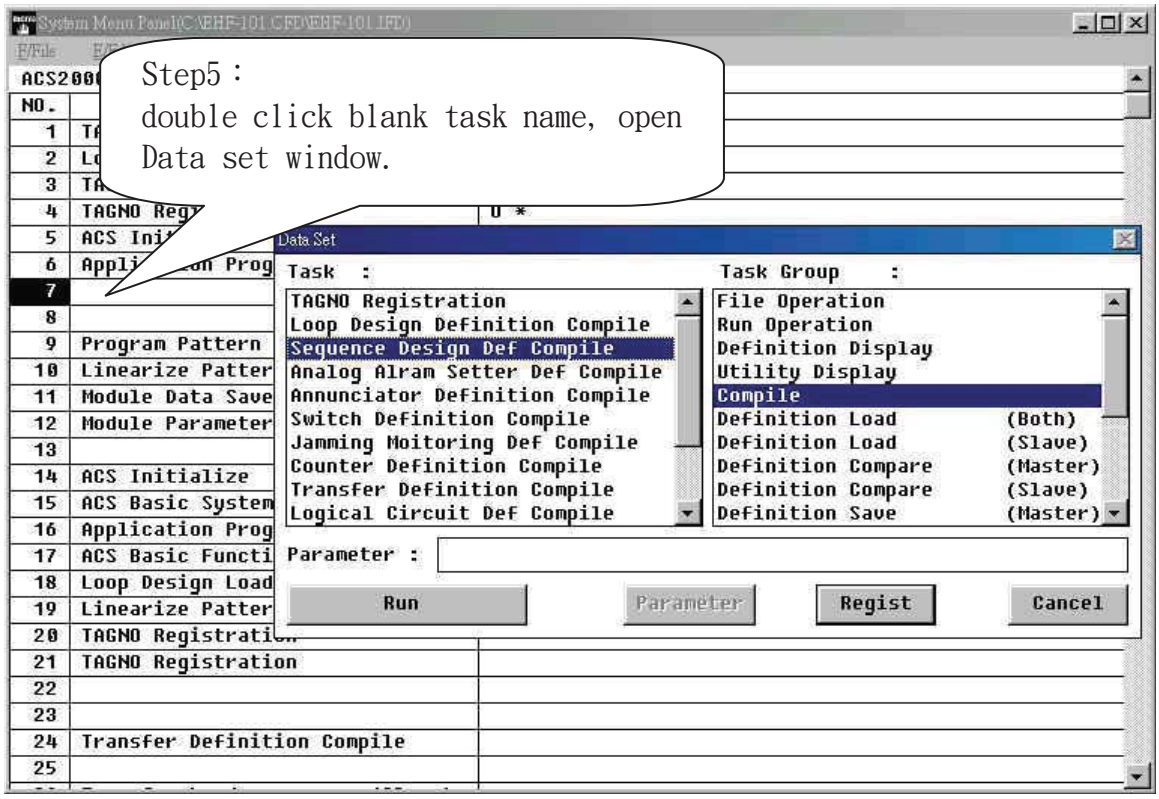

Shift Condition Definition
F/File E/Edit S/Search
ACS2000EI S
NO. Step N
Current Step 5
->Target Step 6
M.2001
M.2003
AND
OR
AND+NOT
OR+NOT
Check
Cancel
2. Shift Condition Definition:
Fill in Shif condition set of every Step, then press 「Check」, if the input condition is reverse, you may select「☐」to change.
Error Interlock Definition
F/File E/Edit S/Search V/Display D/Debug
ACS2000EI Station 101 Sequen
NO. Interlock TAG Sts
1 M.2005 ON
2
Data Set
NO. Interlock TAGNO
Status
ON OFF
Proc
10
Check
Cancel
Y/N Set
Y
N
Check
Cancel
3. Error Interlock Definition
Fill in Interlock TAGNO with bit address and select the Status of interlock (we usually used ON)，fill in PROC with the number you will jump when Interlock occur，then press 「Check」，

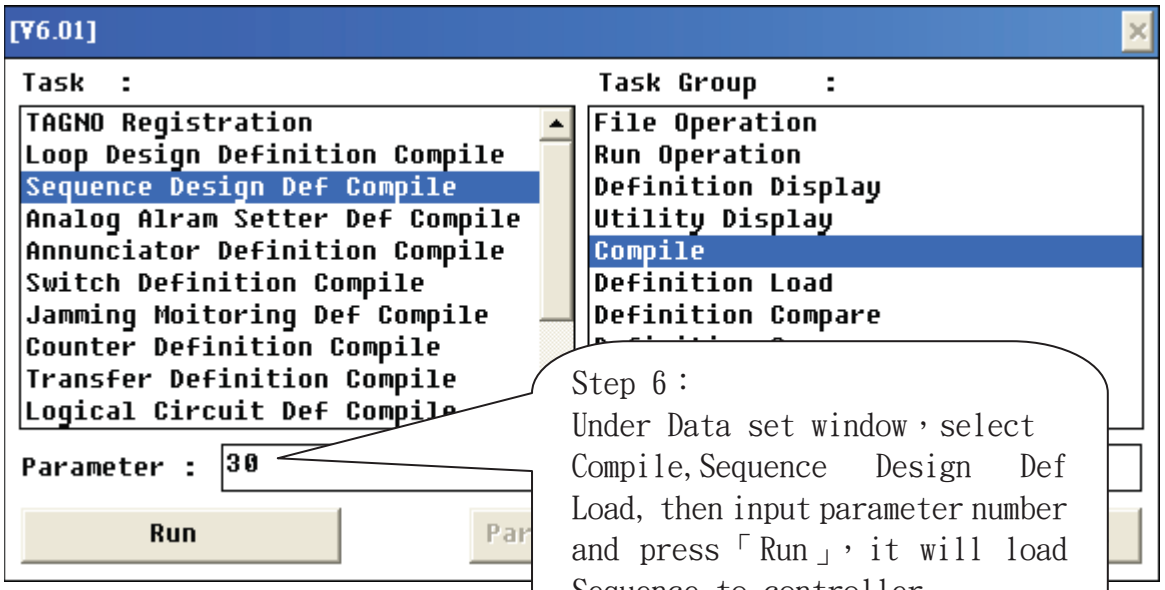

Parameter Set Definition
F/File E/Edit S/Search V/Display D/Debug
ACS2000EI Station 101 Sequence NO. 20/ 64 Data 1/32
NO. Comment Type Scale Unit
1 TMR
2 TMR
Data Set
NO. Comment
1
Scale
Unit
BS - FS
Set Range
Data
SL - SH
0H00M10S
Check
Cancel
File Save
%
TMR
PL
4. Parameter Set Definition
Setup the parameter Type and Data, then press「Check」

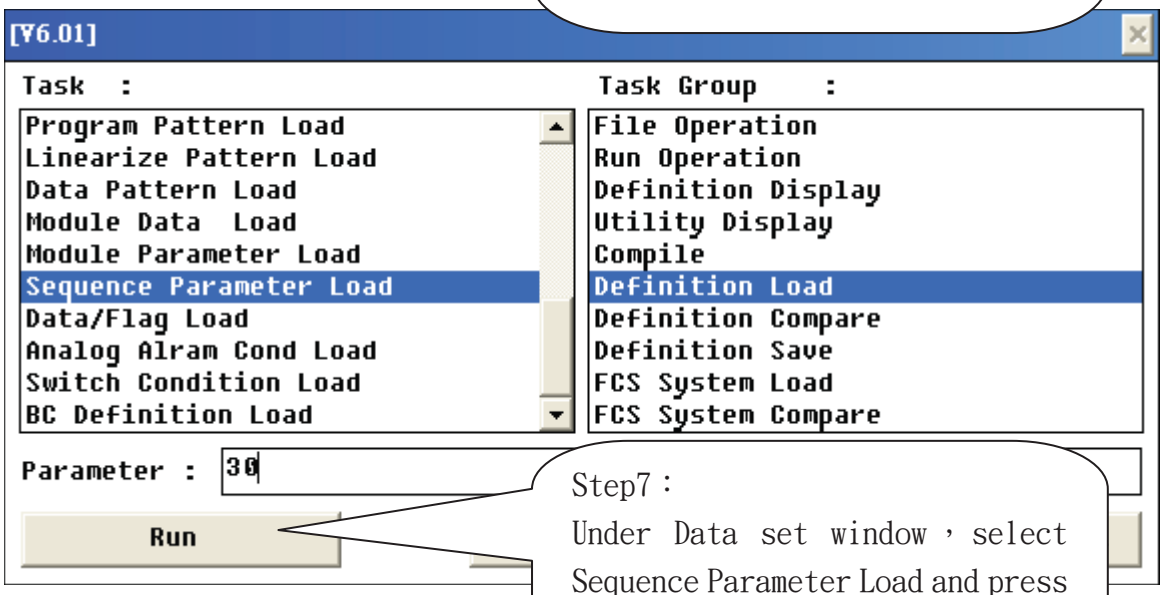


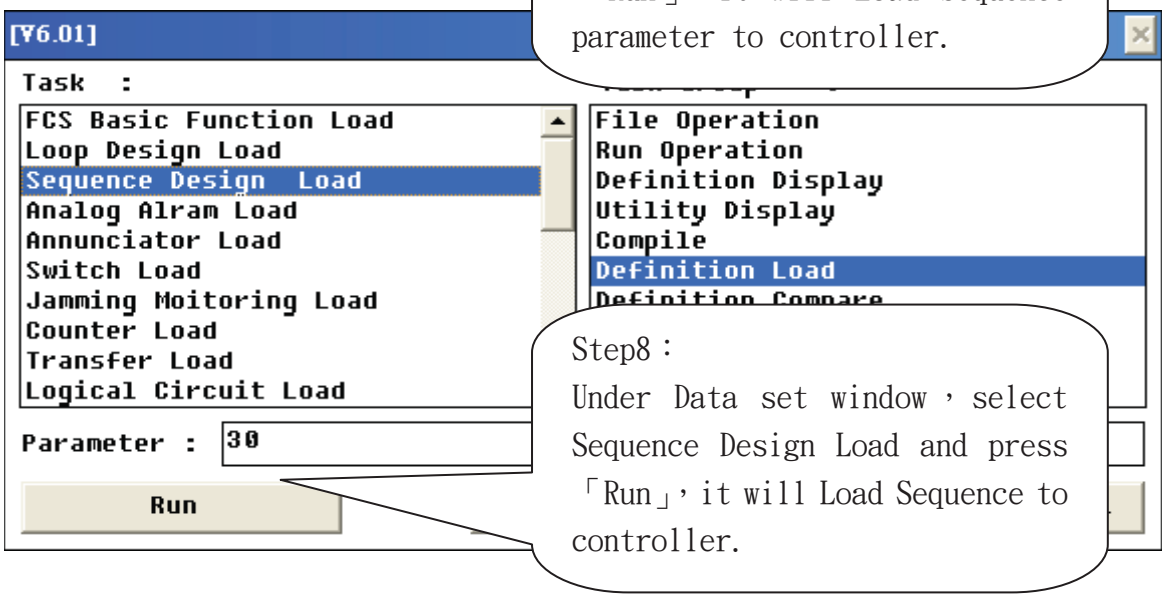
Finally compile the sequence and load the sequence to your control station.

See the follow setting step by step.

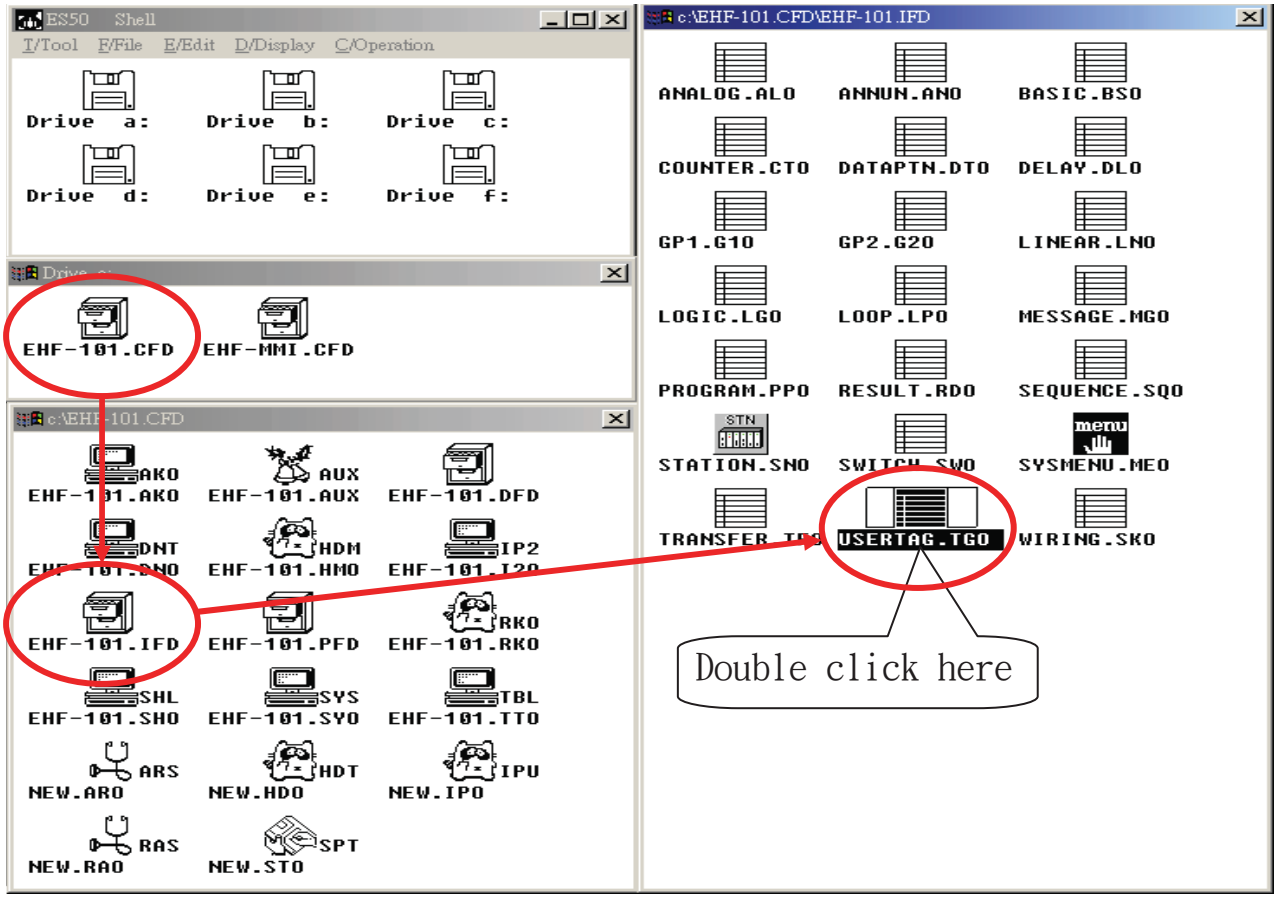



[V6.01]
Task :
Task Group :
TAGNO Registration
Loop Design Definition Compile
Sequence Design Def Compile
Analog Alram Setter Def Compile
Annunciator Definition Compile
Switch Definition Compile
Jamming Moitoring Def Compile
Counter Definition Compile
Transfer Definition Compile
Logical Circuit Def Compile
File Operation
Run Operation
Definition Display
Utility Display
Compile
Definition Load
Definition Compare
Parameter : 30
Run
Step 6：
Under Data set window，select Compile, Sequence Design Def Load, then input parameter number and press「Run」, it will load Sequence to controller.

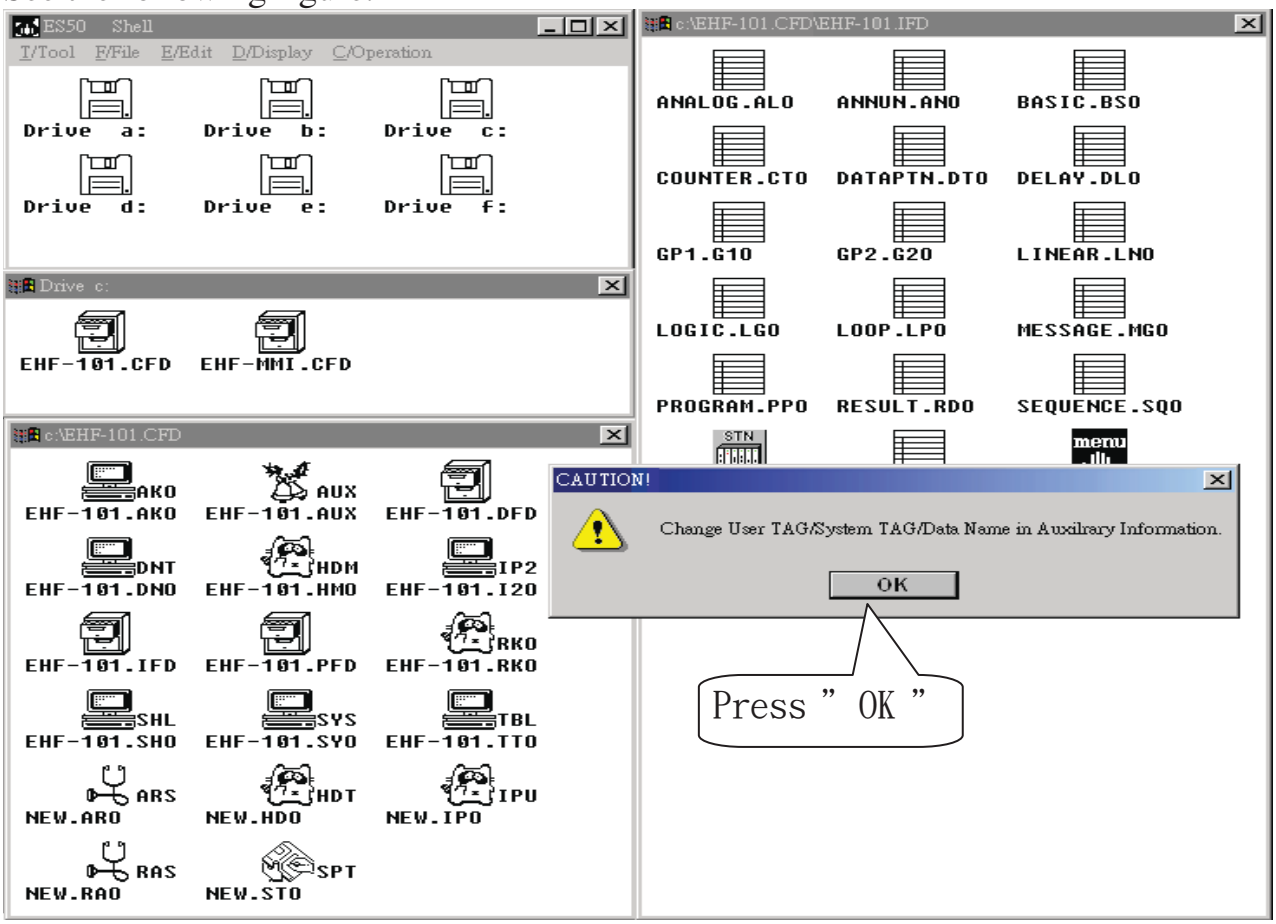

[V6.01]
Task :
Task Group :
Program Pattern Load
Linearize Pattern Load
Data Pattern Load
Module Data Load
Module Parameter Load
Sequence Parameter Load
Data/Flag Load
Analog Alram Cond Load
Switch Condition Load
BC Definition Load
File Operation
Run Operation
Definition Display
Utility Display
Compile
Definition Load
Definition Compare
Definition Save
FCS System Load
FCS System Compare
Parameter : 30
Run
Step7：
Under Data set window，select Sequence Parameter Load and press「Run」, it will Load Sequence parameter to controller.

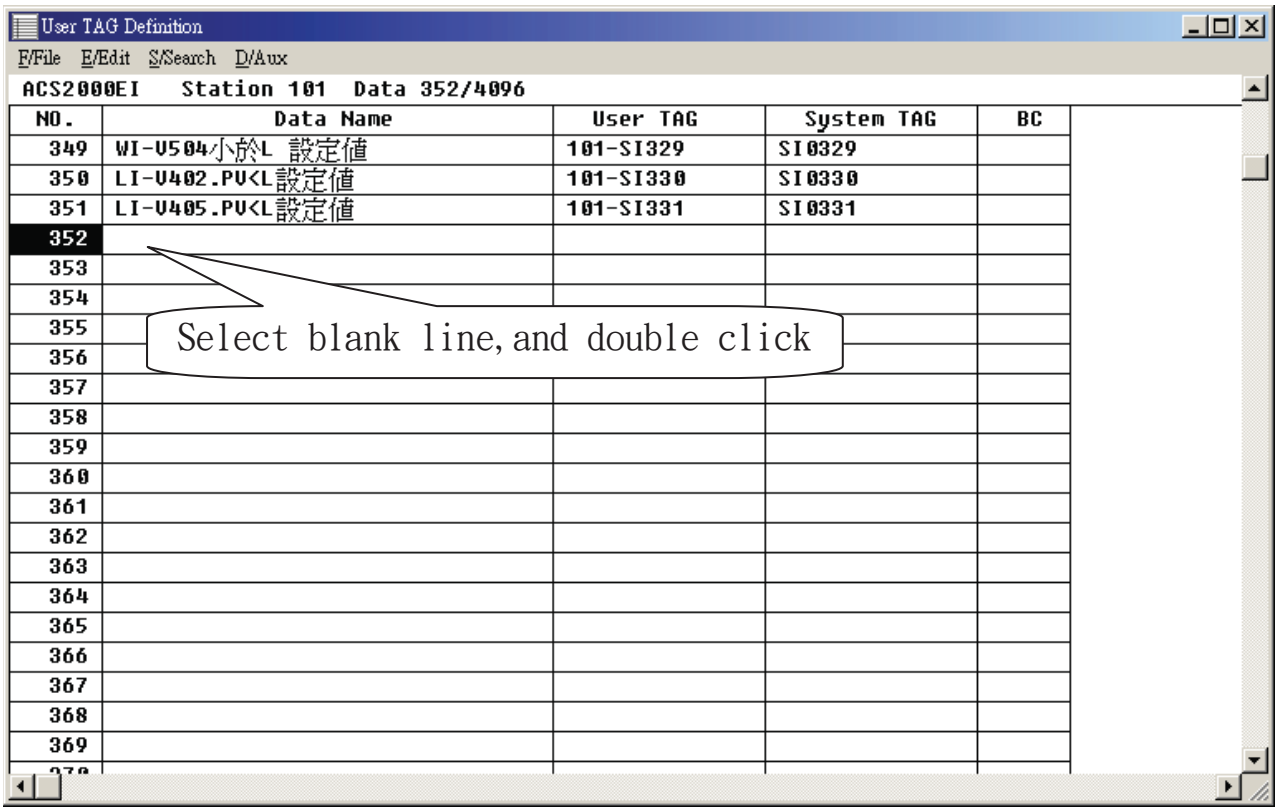

[V6.01]
Task :
FCS Basic Function Load
Loop Design Load
Sequence Design Load
Analog Alram Load
Annunciator Load
Switch Load
Jamming Moitoring Load
Counter Load
Transfer Load
Logical Circuit Load
File Operation
Run Operation
Definition Display
Utility Display
Compile
Definition Load
Parameter : 30
Run
Step8：
Under Data set window，select Sequence Design Load and press「Run」, it will Load Sequence to controller.

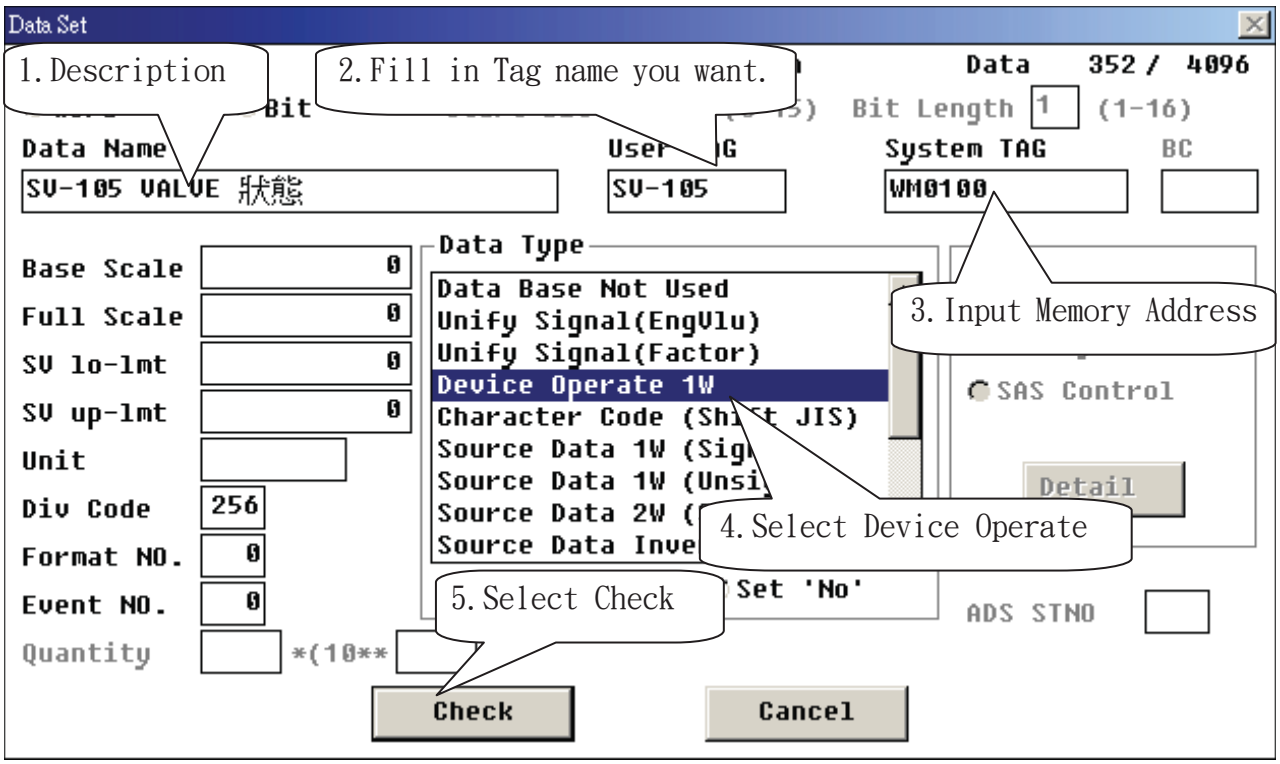
CREATE USER TAG



See the following figure.

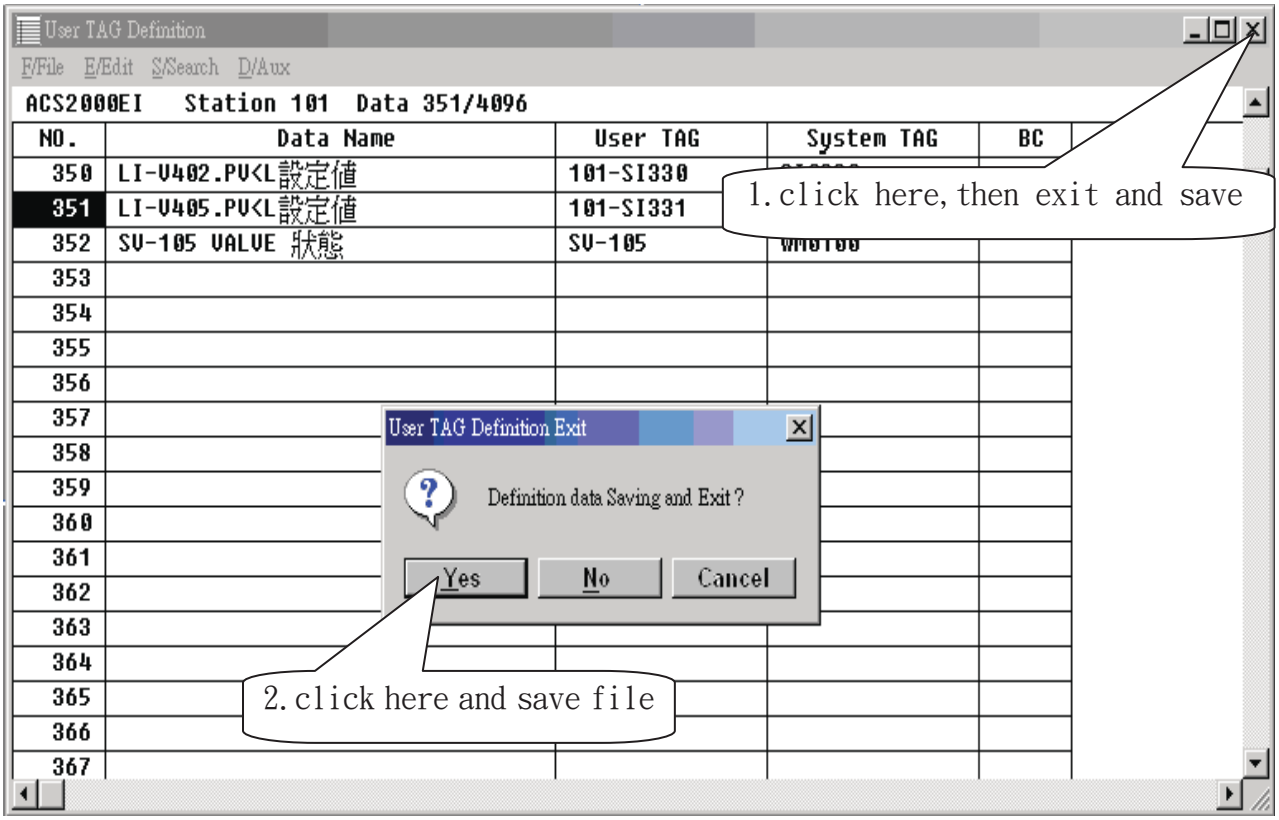


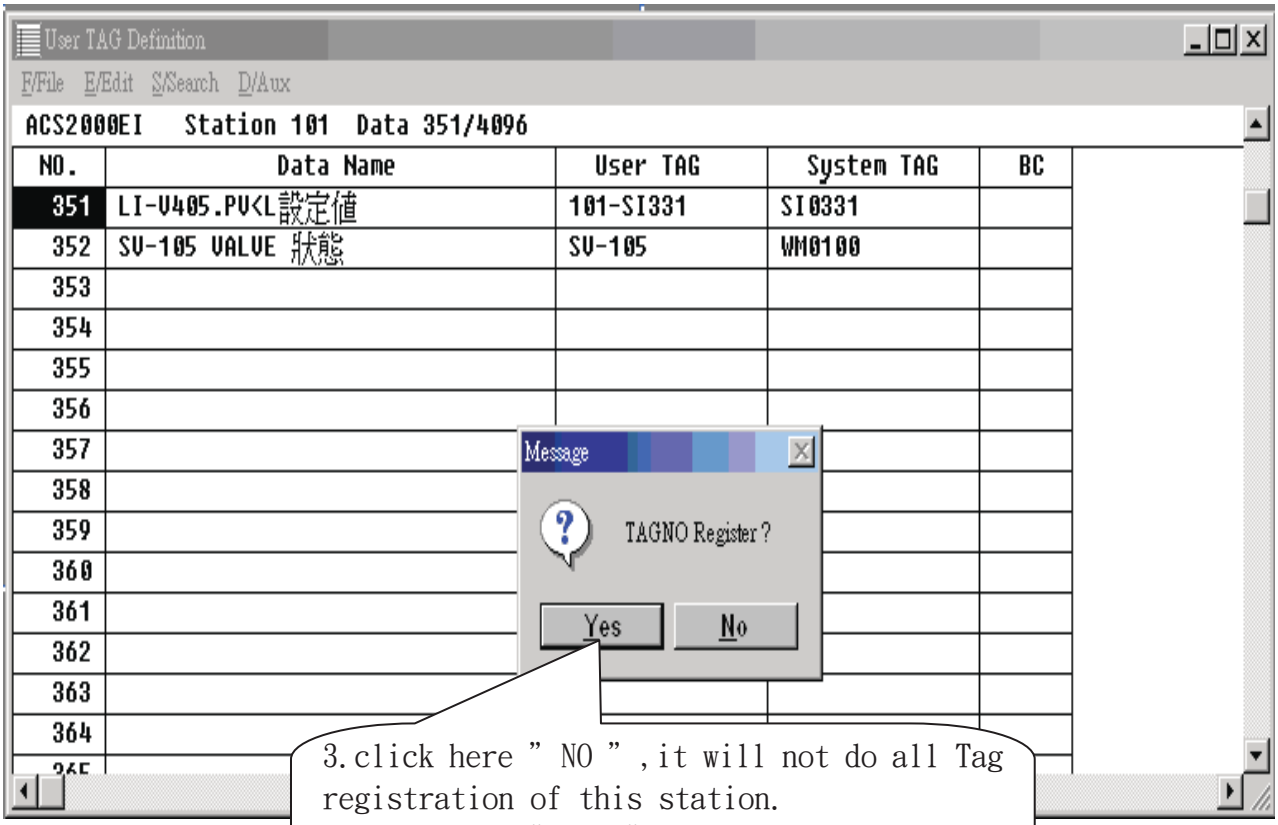


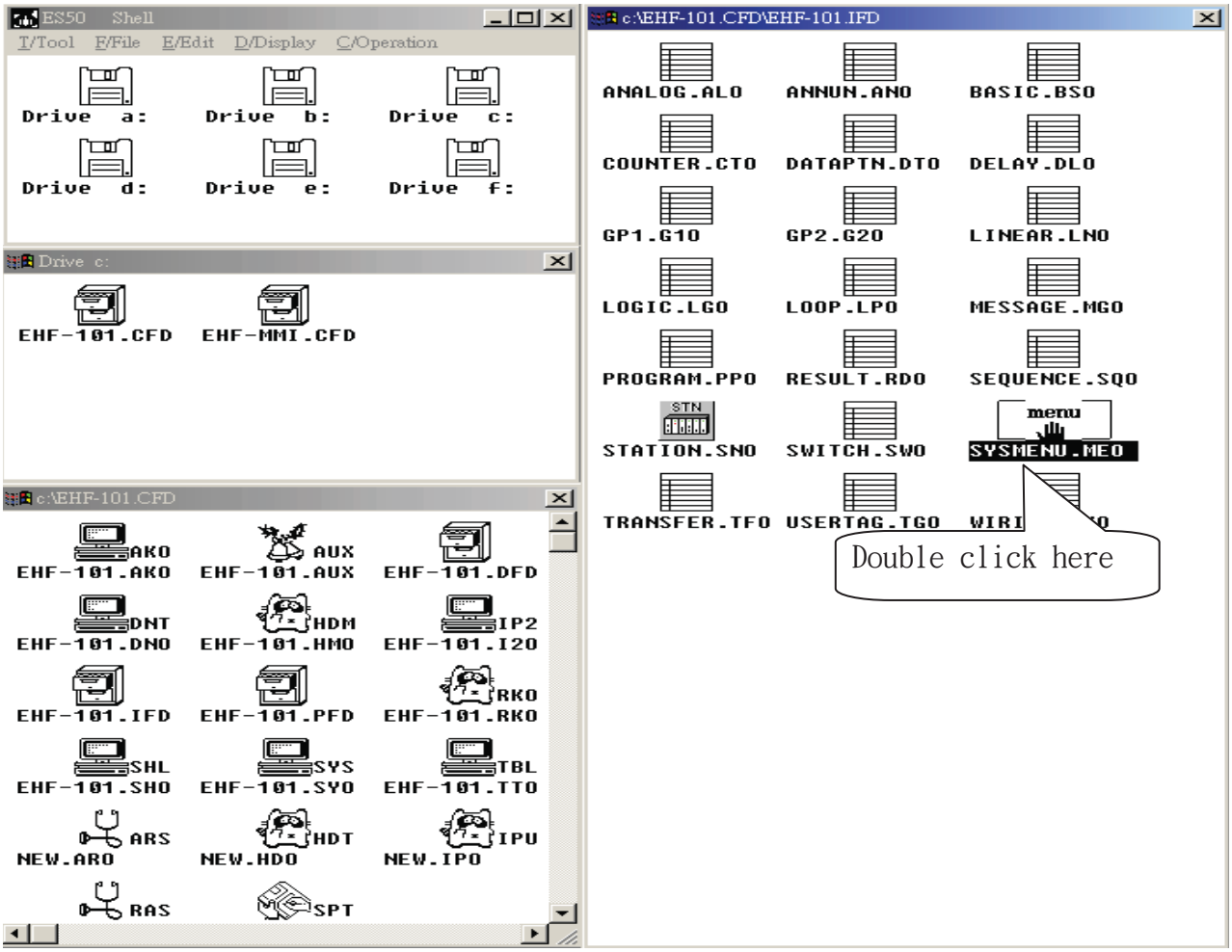


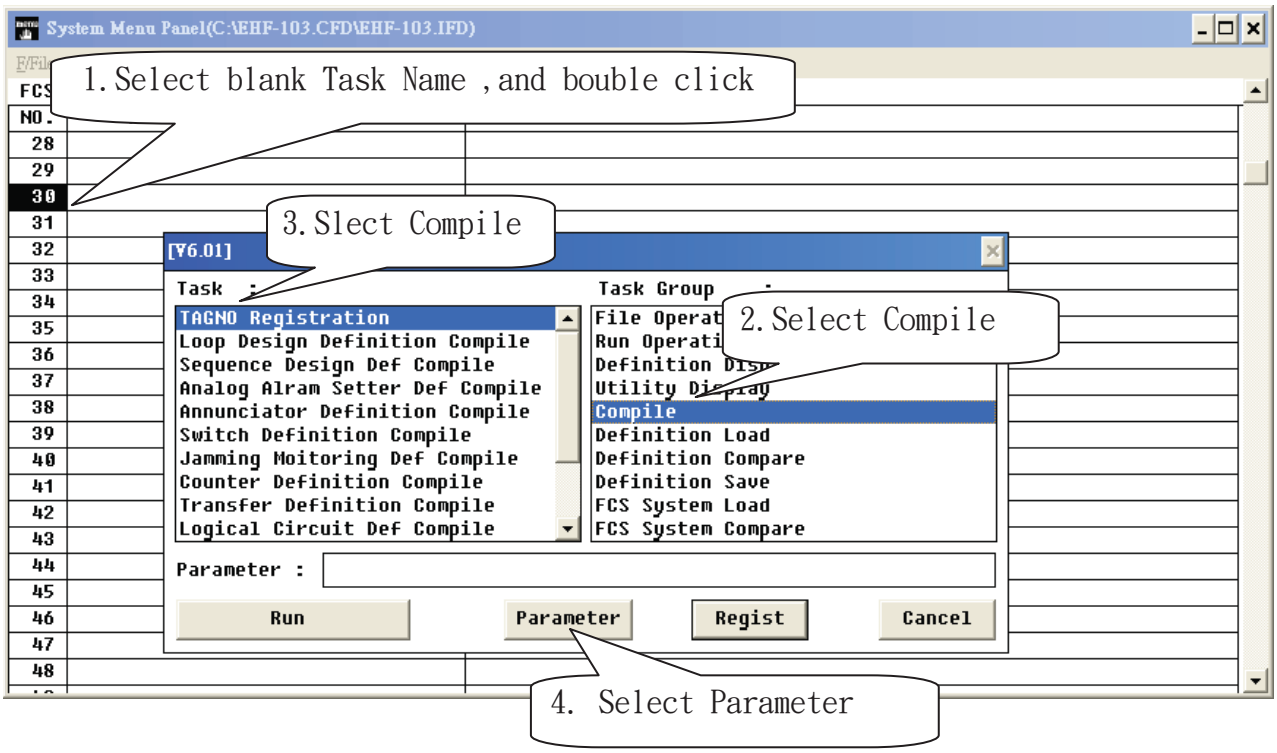
Select **SET 'YES'** means can change the value in MMI TAG panel under SUPPORT MENU on AOS.

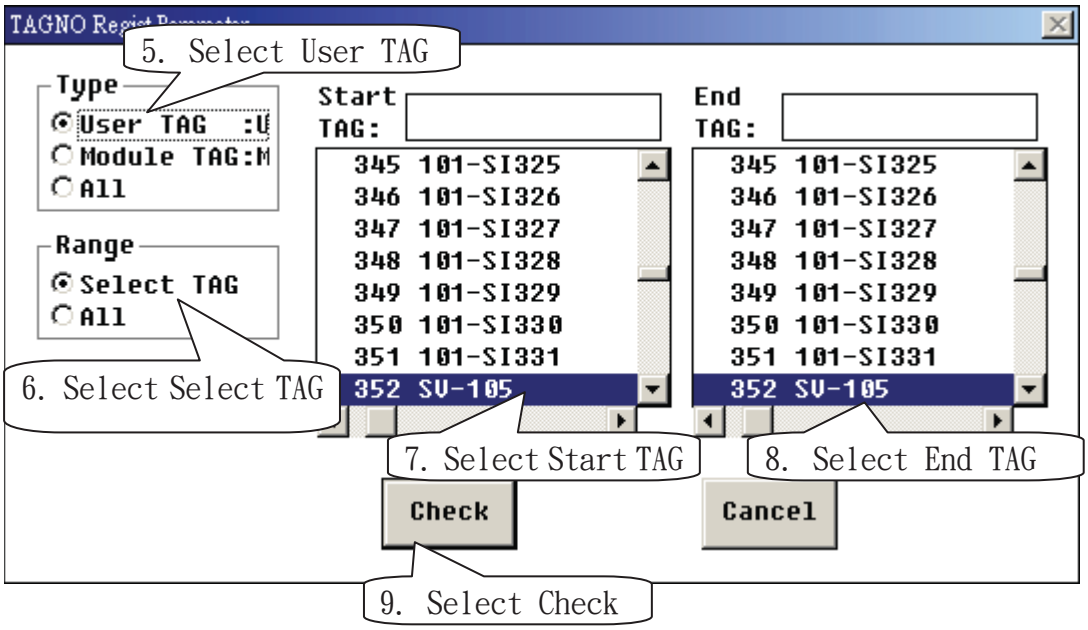
Select **SET 'NO'** means can not change the value in MMI TAG panel under SUPPORT MENU on AOS.

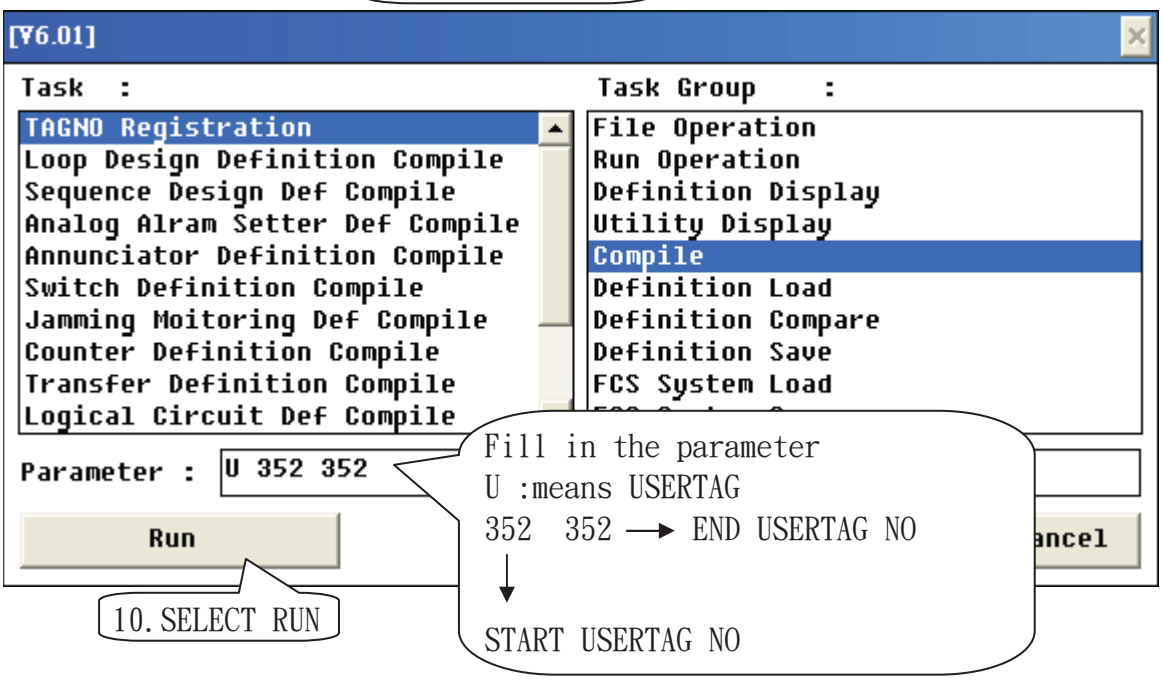

User TAG Definition
F/File E/Edit S/Search D/Aux
ACS2000EI Station 101 Data 351/4096
NO. Data Name User TAG System TAG BC
350 LI-V402.PV<L設定値 101-SI330
351 LI-V405.PV<L設定値 101-SI331
352 SV-105 VALVE 狀態 SV-105
1.click here, then exit and save
User TAG Definition Exit
Definition data Saving and Exit ?
Yes No Cancel
2.click here and save file

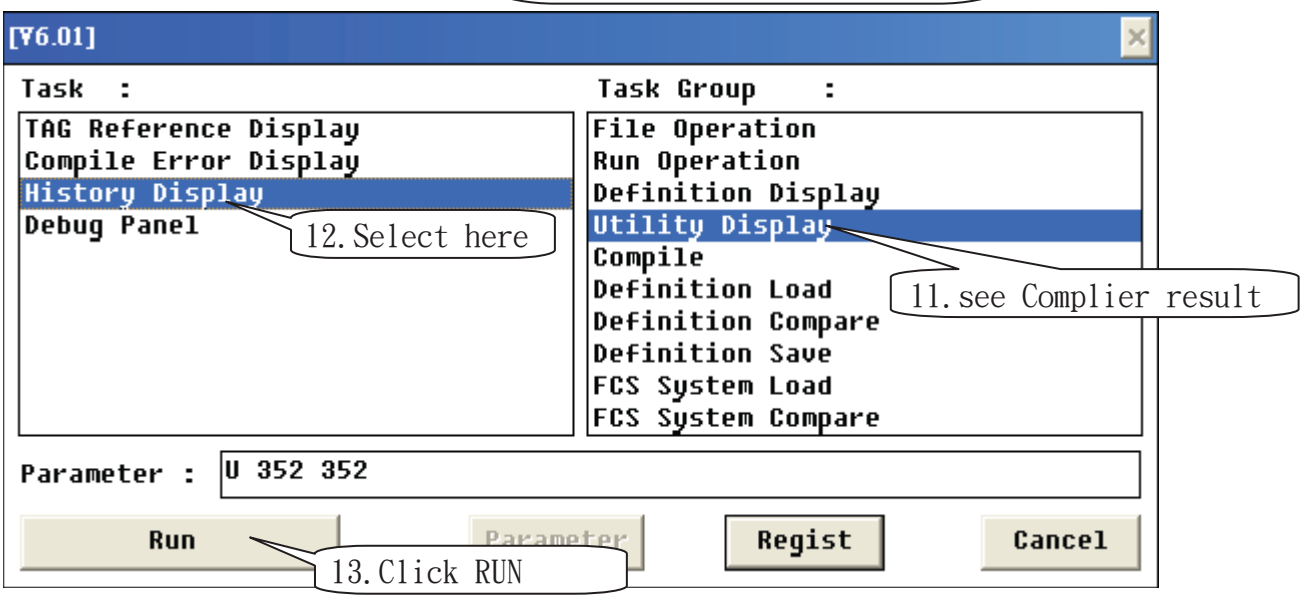

User TAG Definition
F/File E/Edit S/Search D/Aux
ACS2000EI Station 101 Data 351/4096
NO. Data Name User TAG System TAG BC
351 LI-V405.PV<L設定値 101-SI331 SI0331
352 SV-105 VALVE 狀態 SV-105 WM0100
Message
TAGNO Register ?
Yes No
3.click here " NO ", it will not do all Tag registration of this station.
If you click " Yes ", it will do all Tag registration of this station.

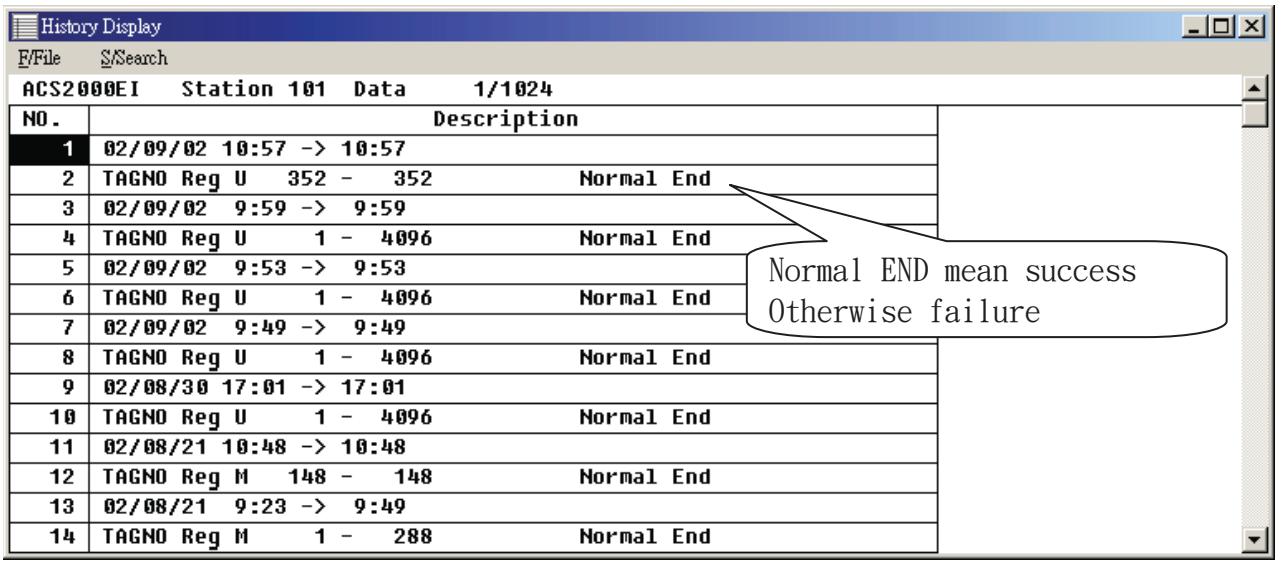

ES50 Shell
T/Tool F/File E/Edit D/Display C/Operation
Drive a:
Drive b:
Drive c:
Drive d:
Drive e:
Drive f:
Drive c:
EHF-101.CFD
EHF-MMI.CFD
c:\EHF-101.CFD
EHF-101.AKO
EHF-101.AUX
EHF-101.DFD
EHF-101.DNO
EHF-101.HMO
EHF-101.I2O
EHF-101.IFD
EHF-101.PFD
EHF-101.RKO
EHF-101.SHO
EHF-101.SYO
EHF-101.TTO
NEW.ARO
NEW.HDO
NEW.IPO
c:\EHF-101.CFD\EHF-101.IFD
ANALOG.ALO
ANNUN.ANO
BASIC.BSO
COUNTER.CTO
DATAPTN.DTO
DELAY.DLO
GP1.G1O
GP2.G2O
LINEAR.LNO
LOGIC.LGO
LOOP.LPO
MESSAGE.MGO
PROGRAM.PPO
RESULT.RDO
SEQUENCE.SQO
STATION.SNO
SWITCH.SWO
SYSMENU.MEO
TRANSFER.TFO
USERTAG.TGO
Double click here


System Menu Panel(C:\EHF-103.CFD\EHF-103.IFD)
1.Select blank Task Name ,and bouble click
3.Slect Compile
2.Select Compile
[V6.01]
Task :
Task Group :
TAGNO Registration
Loop Design Definition Compile
Sequence Design Def Compile
Analog Alram Setter Def Compile
Annunciator Definition Compile
Switch Definition Compile
Jamming Moitoring Def Compile
Counter Definition Compile
Transfer Definition Compile
Logical Circuit Def Compile
File Operat
Run Operati
Definition Dis
Utility Display
Compile
Definition Load
Definition Compare
Definition Save
FCS System Load
FCS System Compare
Parameter :
Run
Parameter
Regist
Cancel
4. Select Parameter

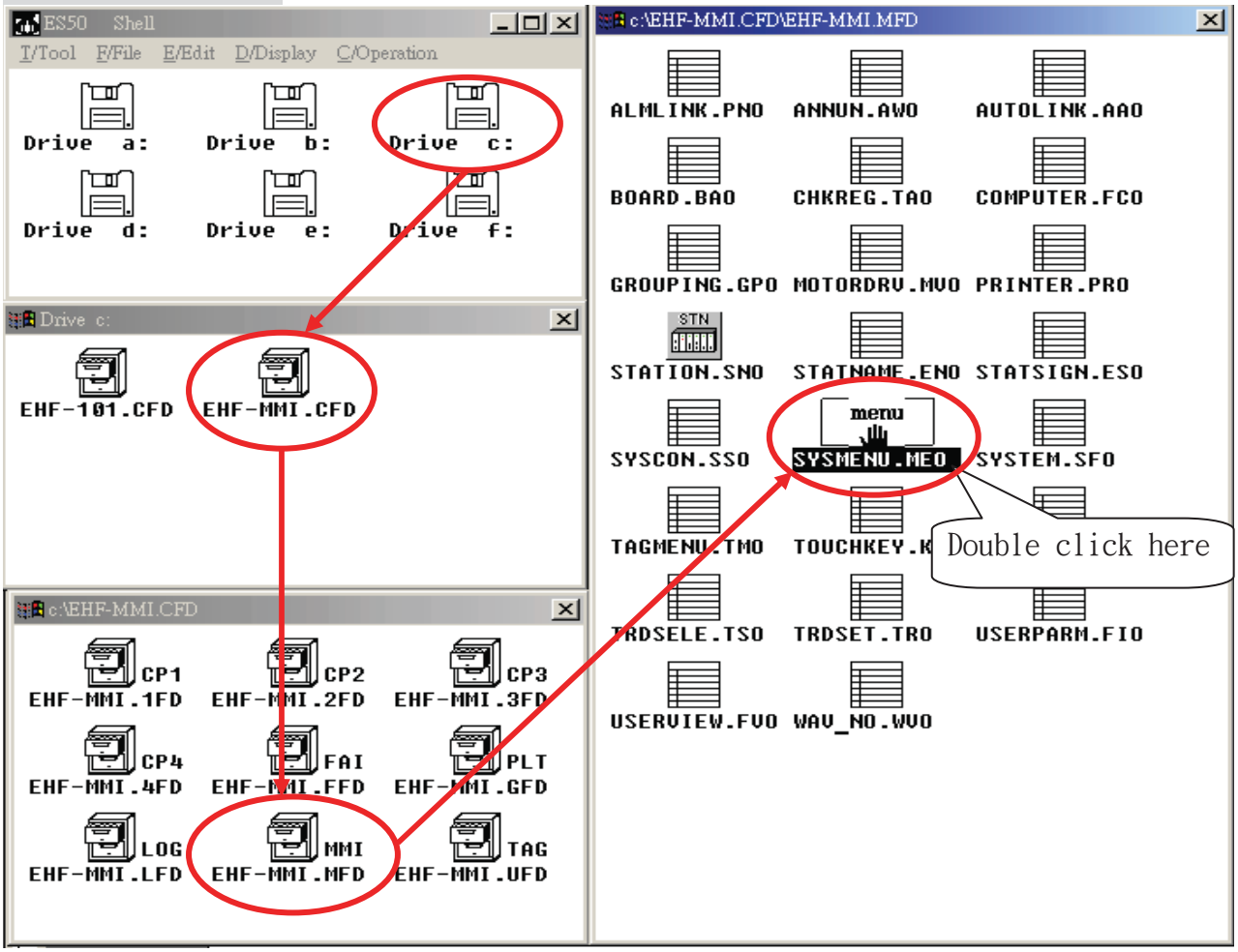

TAGNO Regist Parameter
5. Select User TAG
Type
User TAG :U
Module TAG:M
All
Range
Select TAG
All
Start TAG:
End TAG:
345 101-SI325
346 101-SI326
347 101-SI327
348 101-SI328
349 101-SI329
350 101-SI330
351 101-SI331
352 SV-105
6. Select Select TAG
7. Select Start TAG
8. Select End TAG
Check
Cancel
9. Select Check

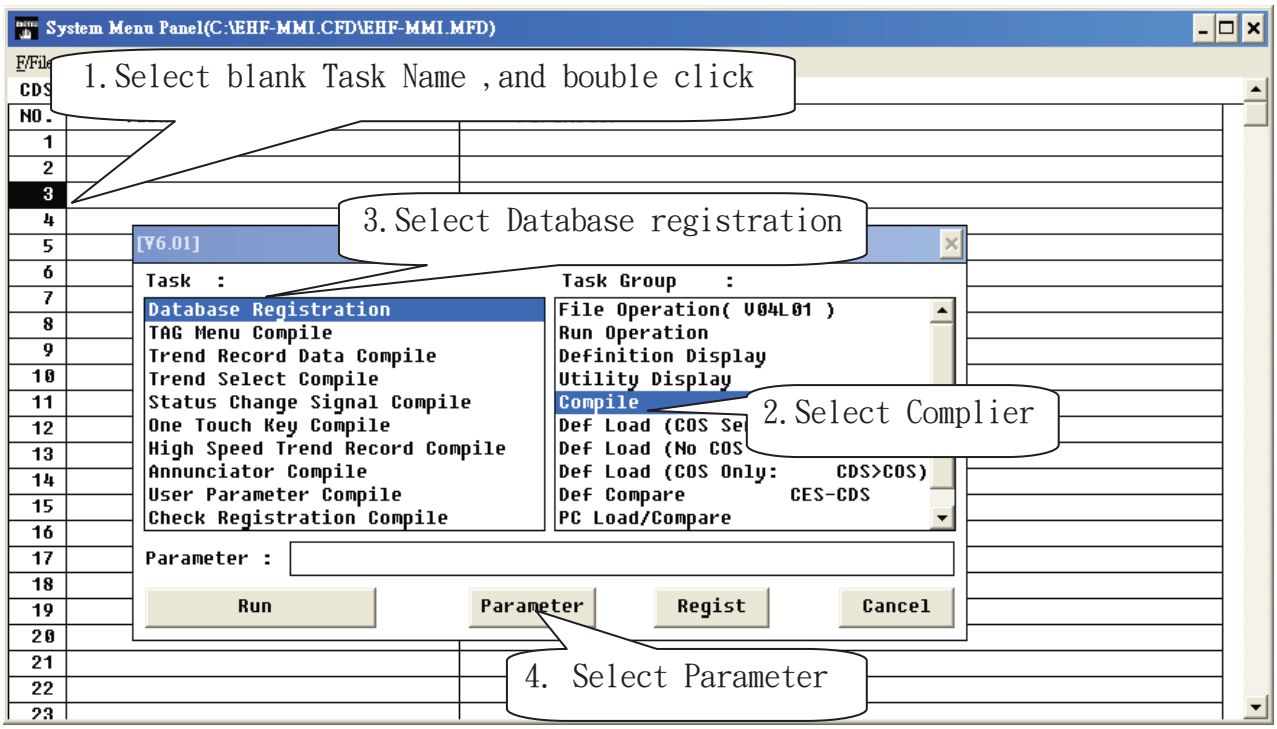

[V6.01]
Task :
Task Group :
TAGNO Registration
Loop Design Definition Compile
Sequence Design Def Compile
Analog Alram Setter Def Compile
Annunciator Definition Compile
Switch Definition Compile
Jamming Moitoring Def Compile
Counter Definition Compile
Transfer Definition Compile
Logical Circuit Def Compile
File Operation
Run Operation
Definition Display
Utility Display
Compile
Definition Load
Definition Compare
Definition Save
FCS System Load
Parameter :
U 352 352
Run
10. SELECT RUN
Fill in the parameter
U :means USERTAG
352 352 → END USERTAG NO
START USERTAG NO

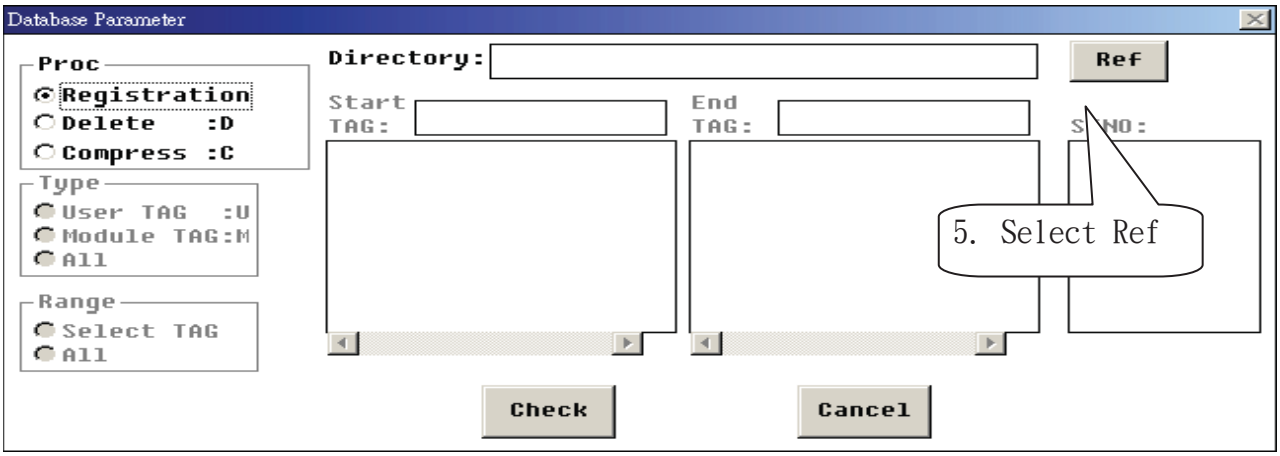

[V6.01]
Task :
Task Group :
TAG Reference Display
Compile Error Display
History Display
Debug Panel
File Operation
Run Operation
Definition Display
Utility Display
Compile
Definition Load
Definition Compare
Definition Save
FCS System Load
FCS System Compare
12. Select here
11. see Complier result
Parameter :
U 352 352
Run
Parameter
Regist
Cancel
13. Click RUN

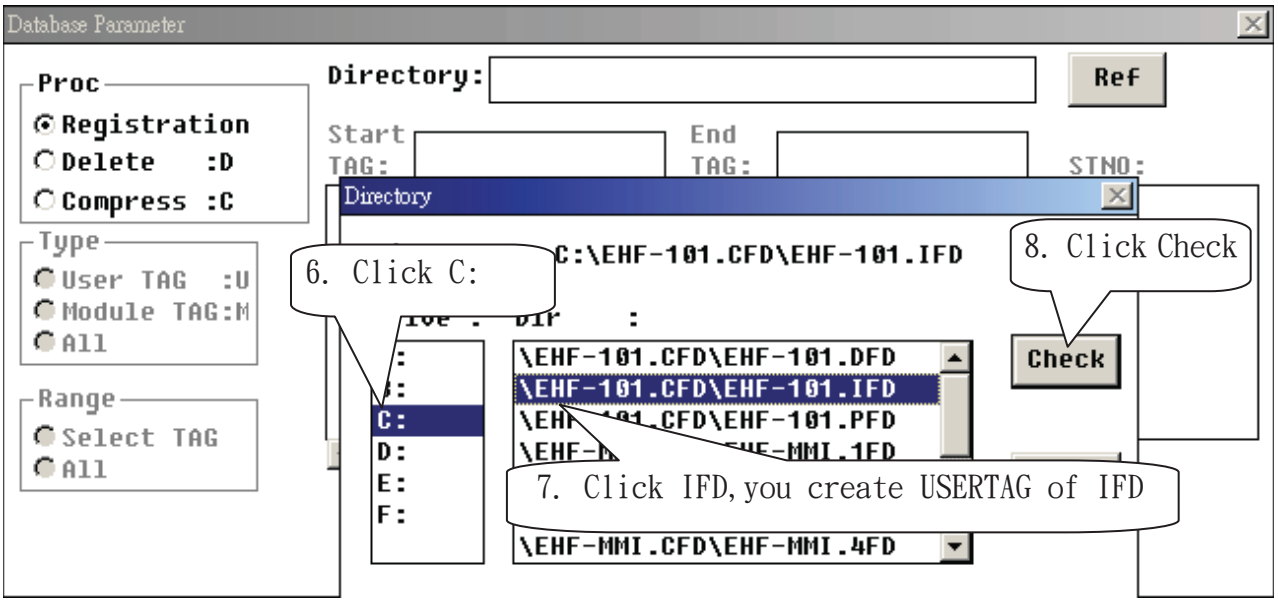

History Display
F/File S/Search
ACS2000EI Station 101 Data 1/1024
NO. Description
1 02/09/02 10:57 -> 10:57
2 TAGNO Reg U 352 - 352 Normal End
3 02/09/02 9:59 -> 9:59
4 TAGNO Reg U 1 - 4096 Normal End
5 02/09/02 9:53 -> 9:53
6 TAGNO Reg U 1 - 4096 Normal End
7 02/09/02 9:49 -> 9:49
8 TAGNO Reg U 1 - 4096 Normal End
9 02/08/30 17:01 -> 17:01
10 TAGNO Reg U 1 - 4096 Normal End
11 02/08/21 10:48 -> 10:48
12 TAGNO Reg M 148 - 148 Normal End
13 02/08/21 9:23 -> 9:49
14 TAGNO Reg M 1 - 288 Normal End
Normal END mean success
Otherwise failure

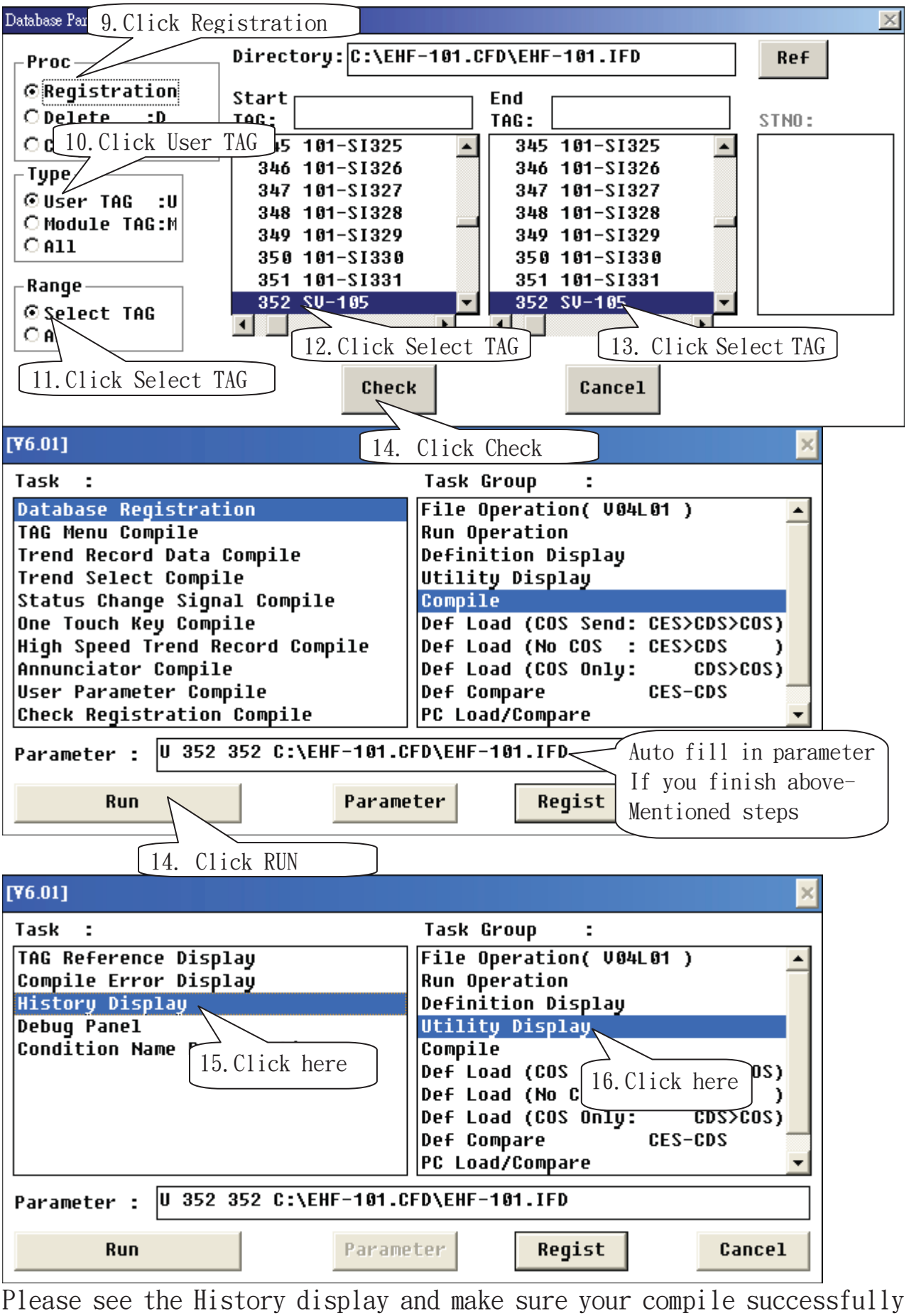
Move to MMI


ES50 Shell
T/Tool F/File E/Edit D/Display C/Operation
Drive a:
Drive b:
Drive c:
Drive d:
Drive e:
Drive f:
Drive c:
EHF-101.CFD
EHF-MMI.CFD
c:\EHF-MMI.CFD
CP1 EHF-MMI.1FD
CP2 EHF-MMI.2FD
CP3 EHF-MMI.3FD
CP4 EHF-MMI.4FD
FAI EHF-MMI.FFD
PLT EHF-MMI.GFD
LOG EHF-MMI.LFD
MMI EHF-MMI.MFD
TAG EHF-MMI.UFD
c:\EHF-MMI.CFD\EHF-MMI.MFD
ALMLINK.PNO
ANNUN.AWO
AUTOLINK.AAO
BOARD.BAO
CHKREG.TAO
COMPUTER.FCO
GROUPING.GPO
MOTORDRV.MVO
PRINTER.PRO
STN
STATION.SNO
STATNAME.ENO
STATSIGN.ESO
SYSCON.SSO
menu
SYSMENU.MEO
SYSTEM.SFO
TAGMENU.TMO
TOUCHKEY.K
TRDSELE.TSO
TRDSET.TRO
USERPARM.FIO
USERVIEW.FVO
WAV_NO.WVO
Double click here

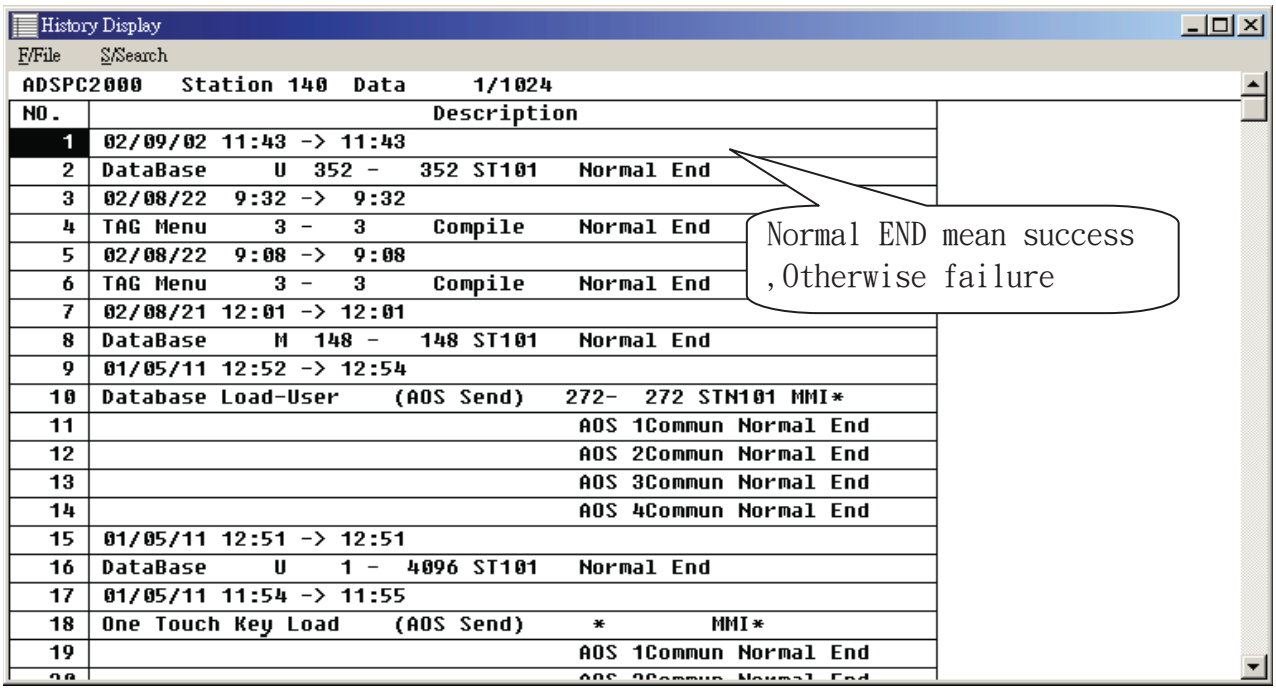

System Menu Panel(C:\EHF-MMI.CFD\EHF-MMI.MFD)
1. Select blank Task Name , and bouble click
3. Select Database registration
[V6.01]
Task :
Task Group :
Database Registration
TAG Menu Compile
Trend Record Data Compile
Trend Select Compile
Status Change Signal Compile
One Touch Key Compile
High Speed Trend Record Compile
Annunciator Compile
User Parameter Compile
Check Registration Compile
File Operation(V04L01)
Run Operation
Definition Display
Utility Display
Compile
Def Load (COS Se
Def Load (No COS
Def Load (COS Only: CDS>COS)
Def Compare CES-CDS
PC Load/Compare
2. Select Complier
Parameter :
Run
Parameter
Regist
Cancel
4. Select Parameter

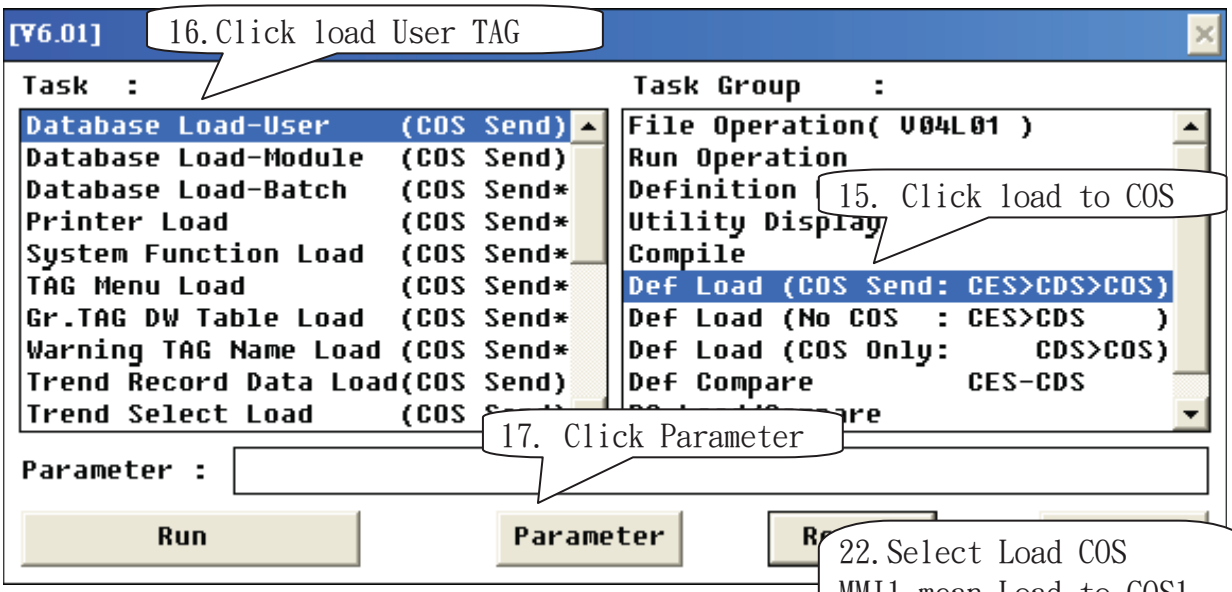

Database Parameter
Proc
Registration
Delete :D
Compress :C
Type
User TAG :U
Module TAG:M
All
Range
Select TAG
All
Directory:
Ref
Start TAG:
End TAG:
STNO:
5. Select Ref
Check
Cancel

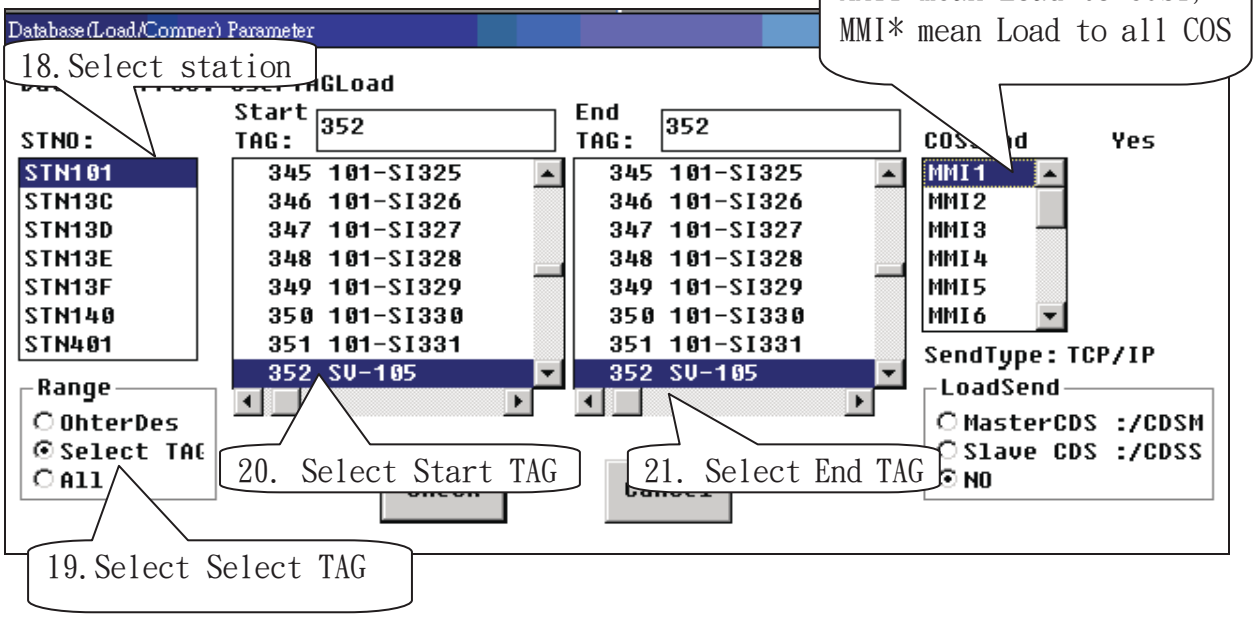

Database Parameter
Proc
Registration
Delete :D
Compress :C
Type
User TAG :U
Module TAG:M
All
Range
Select TAG
All
Directory:
Ref
Start TAG:
End TAG:
STNO:
Directory
C:\EHF-101.CFD\EHF-101.IFD
6. Click C:
8. Click Check
C:
D:
E:
F:
\EHF-101.CFD\EHF-101.DFD
\EHF-101.CFD\EHF-101.IFD
\EHF-101.CFD\EHF-101.PFD
\EHF-MMI.CFD\EHF-MMI.4FD
Check
7. Click IFD, you create USERTAG of IFD

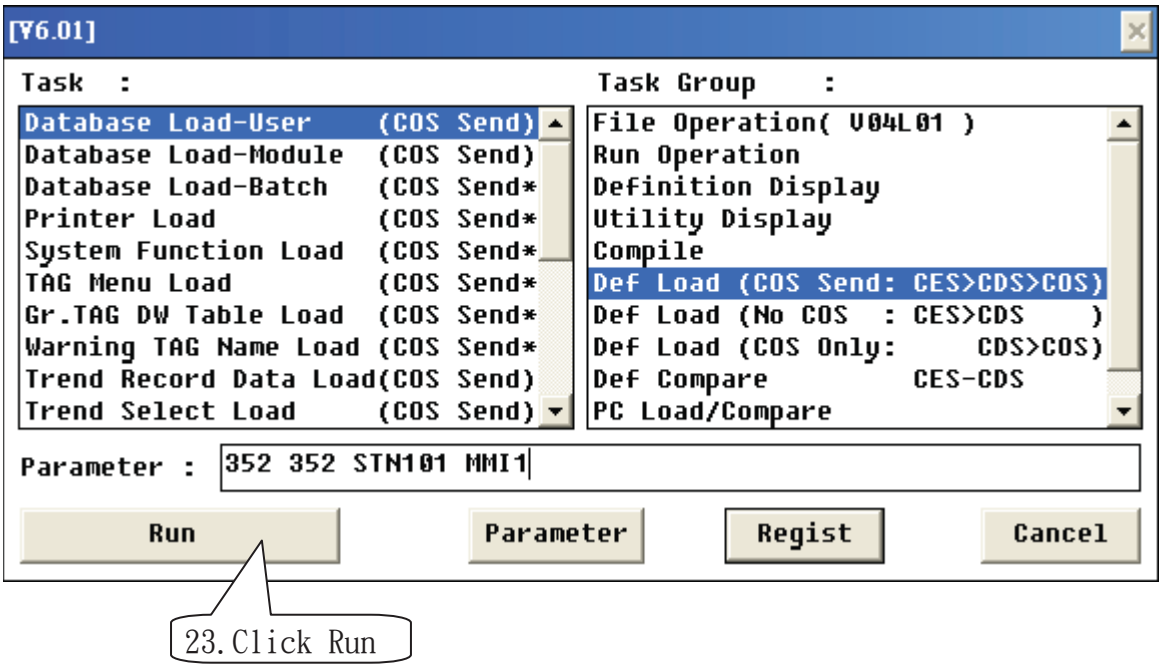


Please see the History display and make sure your compile successfully above-Mentioned steps.


History Display
F/File S/Search
ADSPC2000 Station 140 Data 1/1024
NO. Description
1 02/09/02 11:43 -> 11:43
2 DataBase U 352 - 352 ST101 Normal End
3 02/08/22 9:32 -> 9:32
4 TAG Menu 3 - 3 Compile Normal End
5 02/08/22 9:08 -> 9:08
6 TAG Menu 3 - 3 Compile Normal End
7 02/08/21 12:01 -> 12:01
8 DataBase M 148 - 148 ST101 Normal End
9 01/05/11 12:52 -> 12:54
10 Database Load-User (AOS Send) 272- 272 STN101 MMI*
11 AOS 1Commun Normal End
12 AOS 2Commun Normal End
13 AOS 3Commun Normal End
14 AOS 4Commun Normal End
15 01/05/11 12:51 -> 12:51
16 DataBase U 1 - 4096 ST101 Normal End
17 01/05/11 11:54 -> 11:55
18 One Touch Key Load (AOS Send) * MMI*
19 AOS 1Commun Normal End
Normal END mean success ,Otherwise failure

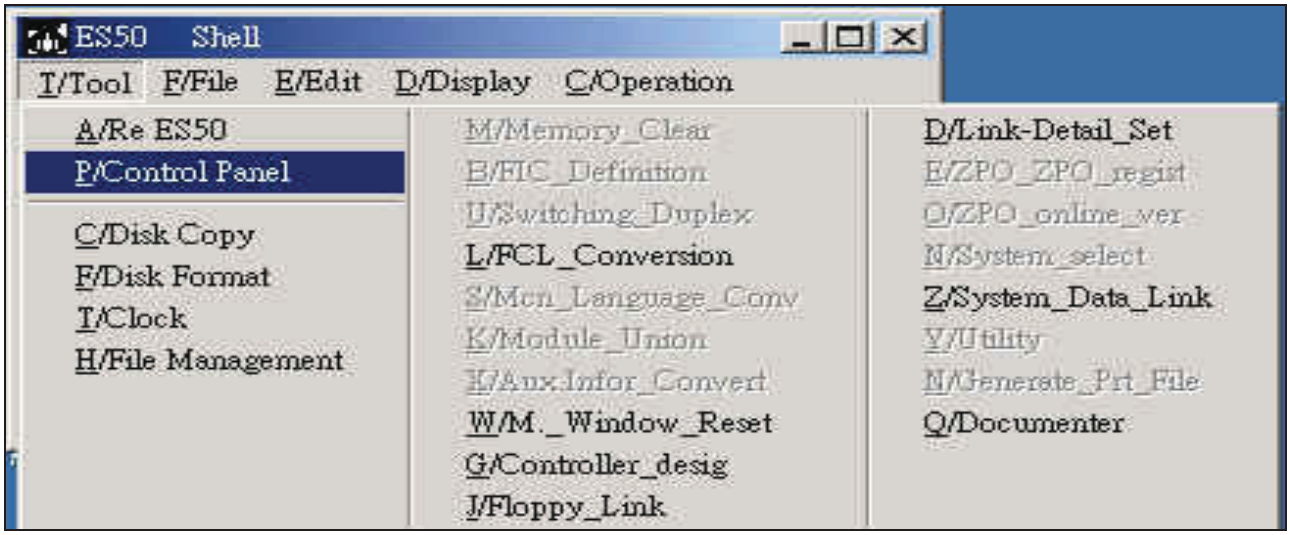

[V6.01]
16. Click load User TAG
Task :
Task Group :
Database Load-User (COS Send)
Database Load-Module (COS Send)
Database Load-Batch (COS Send*
Printer Load (COS Send*
System Function Load (COS Send*
TAG Menu Load (COS Send*
Gr.TAG DW Table Load (COS Send*
Warning TAG Name Load (COS Send*
Trend Record Data Load(COS Send)
Trend Select Load (COS
File Operation(V04L01)
Run Operation
Definition
Utility Display
Compile
Def Load (COS Send: CES>CDS>COS)
Def Load (No COS : CES>CDS)
Def Load (COS Only: CDS>COS)
Def Compare CES-CDS
15. Click load to COS
17. Click Parameter
Parameter :
Run
Parameter

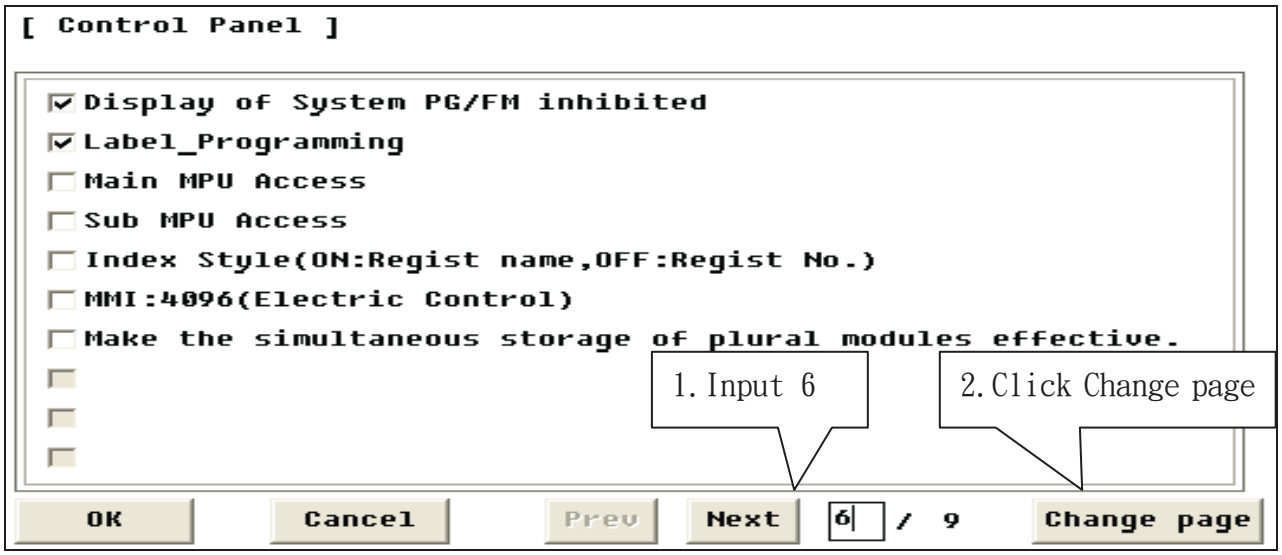

Database(Load/Comper) Parameter
18. Select station
22. Select Load COS
MMI1 mean Load to COS1,
MMI* mean Load to all COS
Start TAG: 352
End TAG: 352
STNO:
STN101
STN13C
STN13D
STN13E
STN13F
STN140
STN401
345 101-SI325
346 101-SI326
347 101-SI327
348 101-SI328
349 101-SI329
350 101-SI330
351 101-SI331
352 SV-105
MMI1
MMI2
MMI3
MMI4
MMI5
MMI6
Yes
SendType: TCP/IP
Range
OhterDes
Select TAG
All
LoadSend
MasterCDS :/CDSM
Slave CDS :/CDSS
NO
20. Select Start TAG
21. Select End TAG
19. Select Select TAG



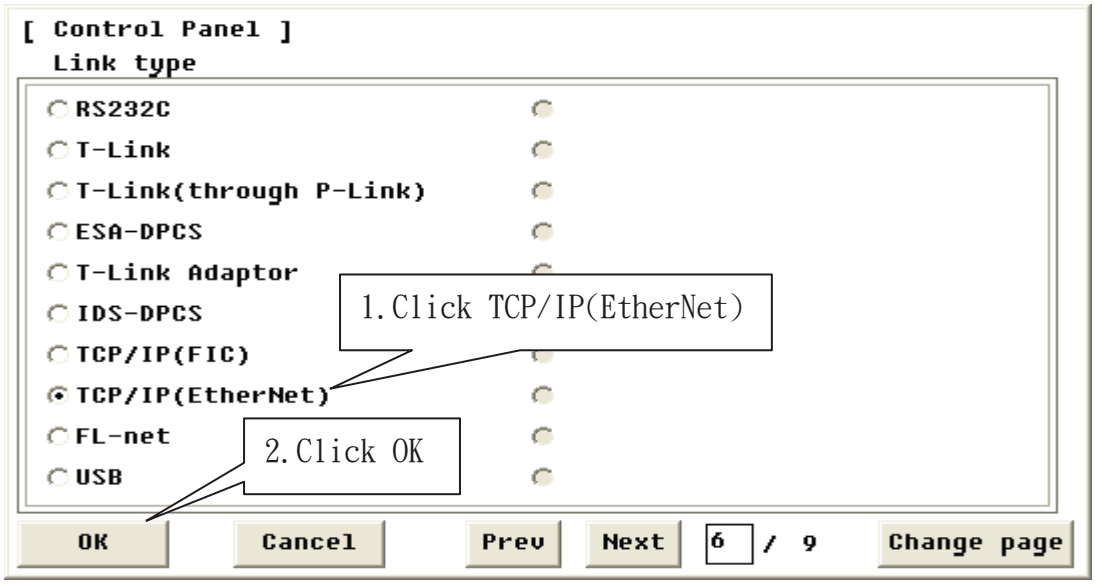
After Loading if you want to make sure,you may check your USERTAG existed in MMI TAG panel under SUPPORT MENU on COS.Please do not stay in this windown too long.

DEBUG Function:

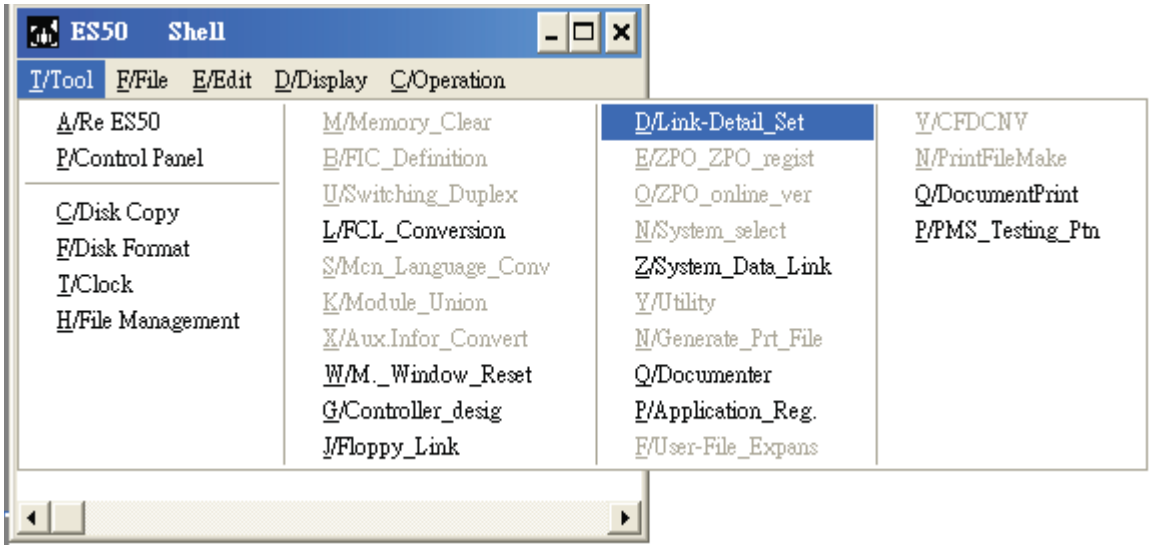


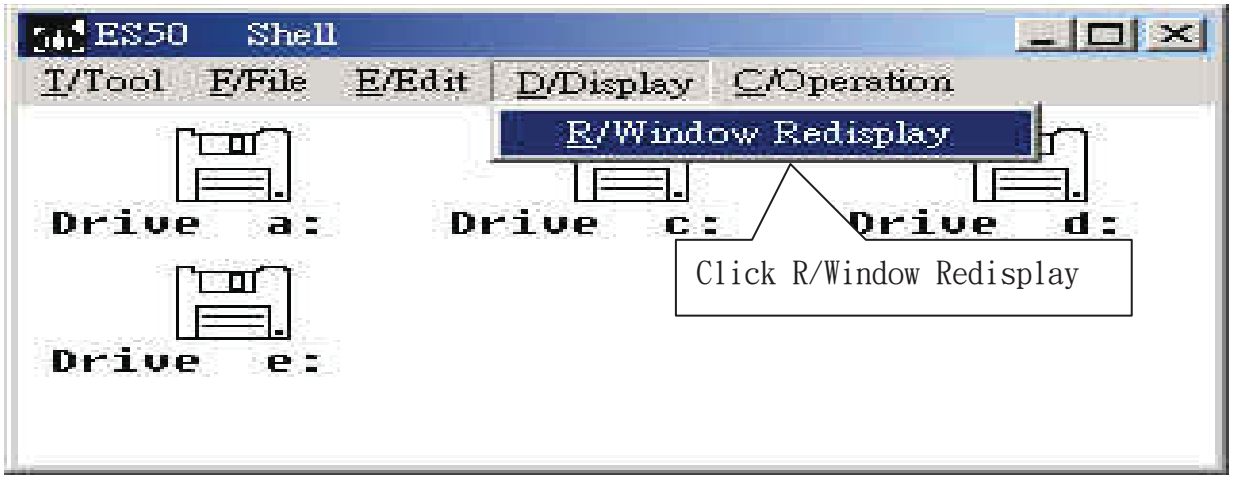


We usually use TCP/IP(EthernetNet) link to control station for FCS.

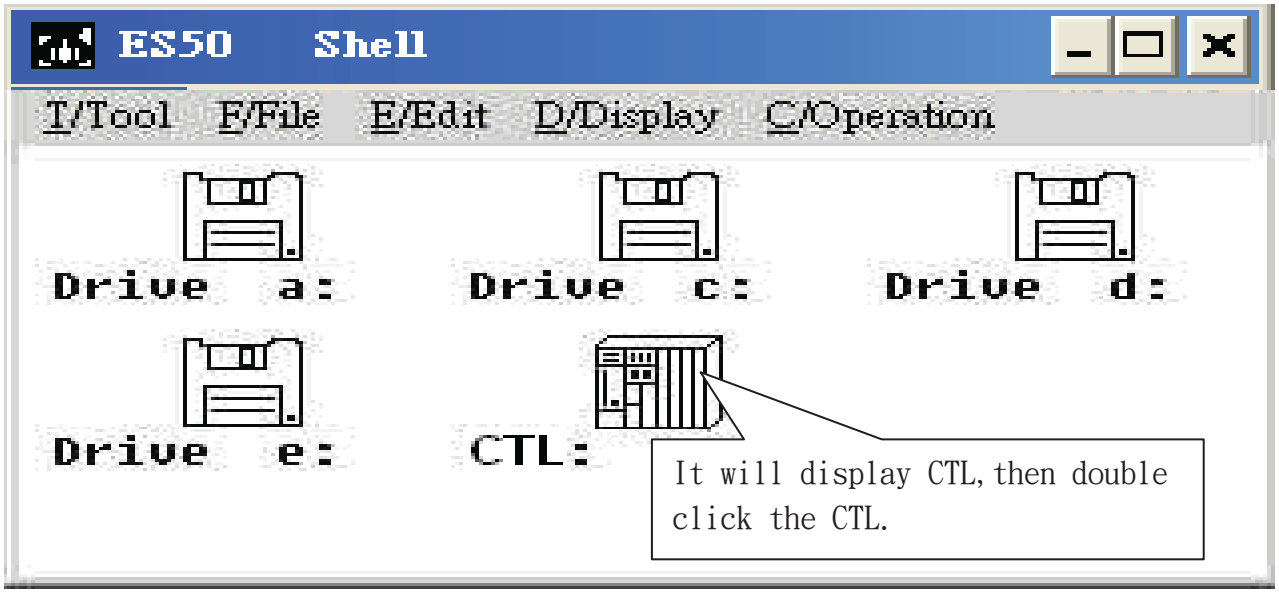


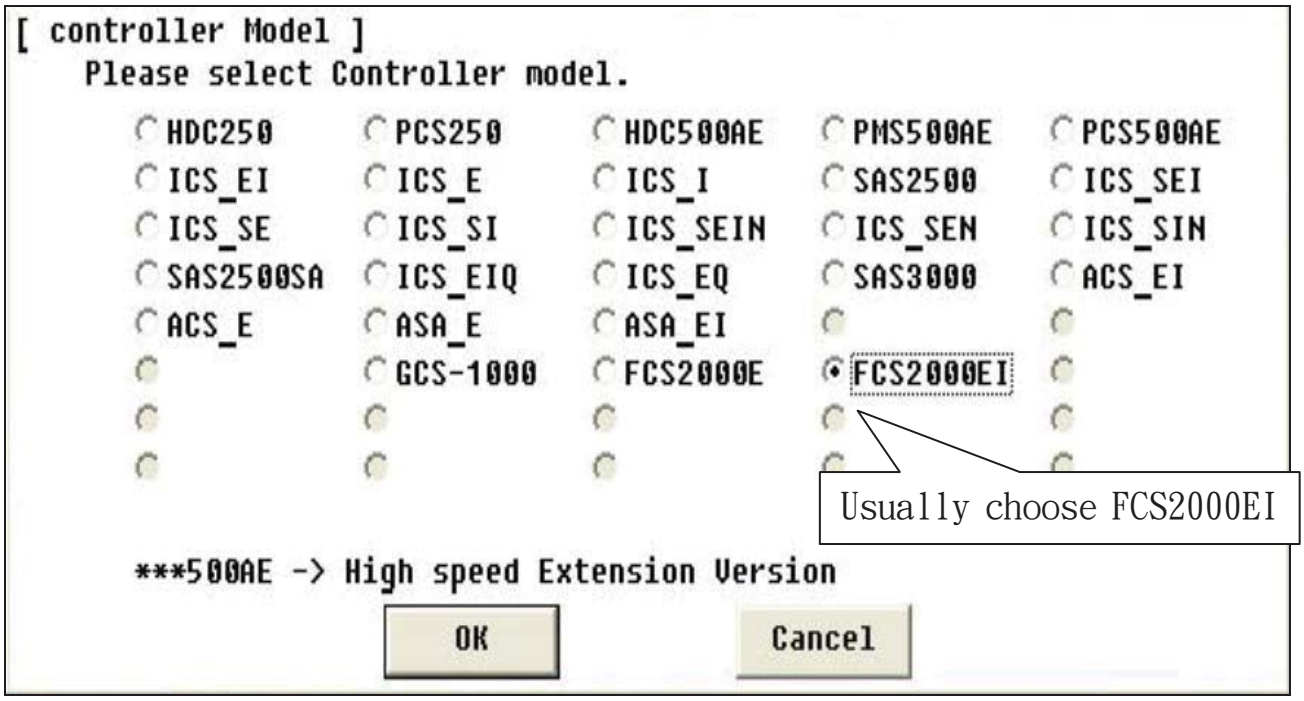
Select T/Tool➔D/Link-Detail_Set

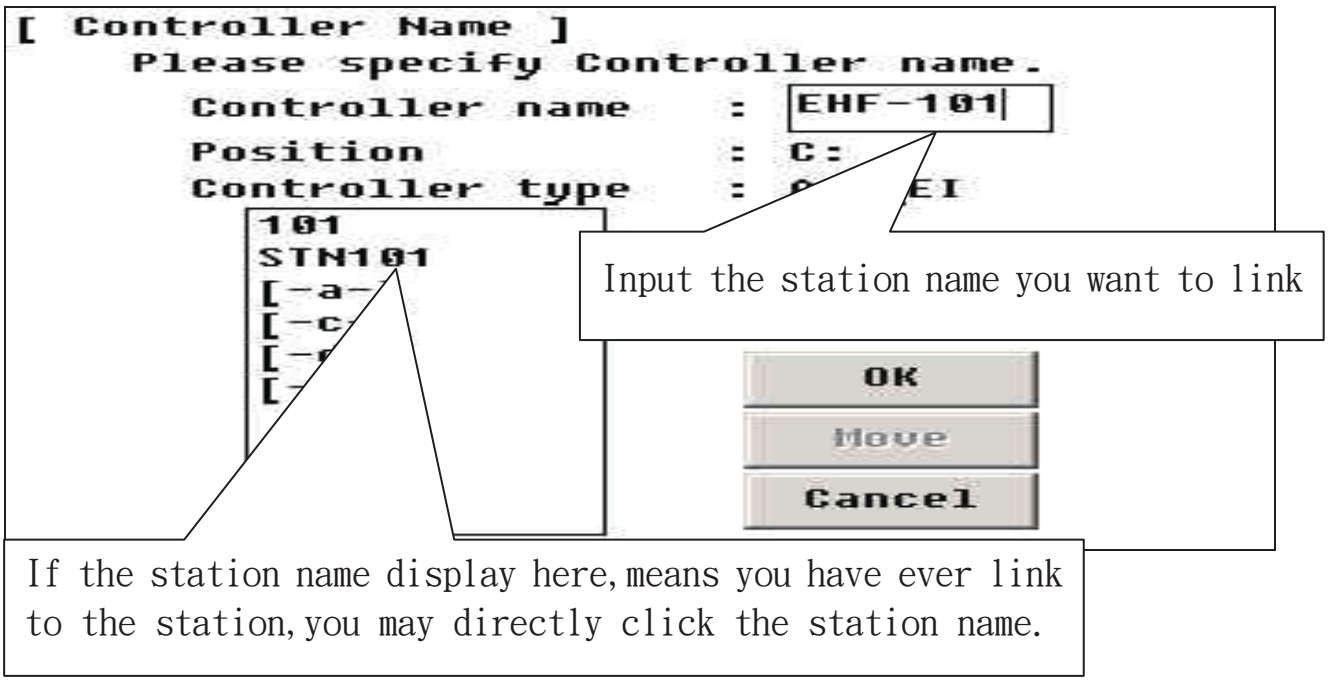




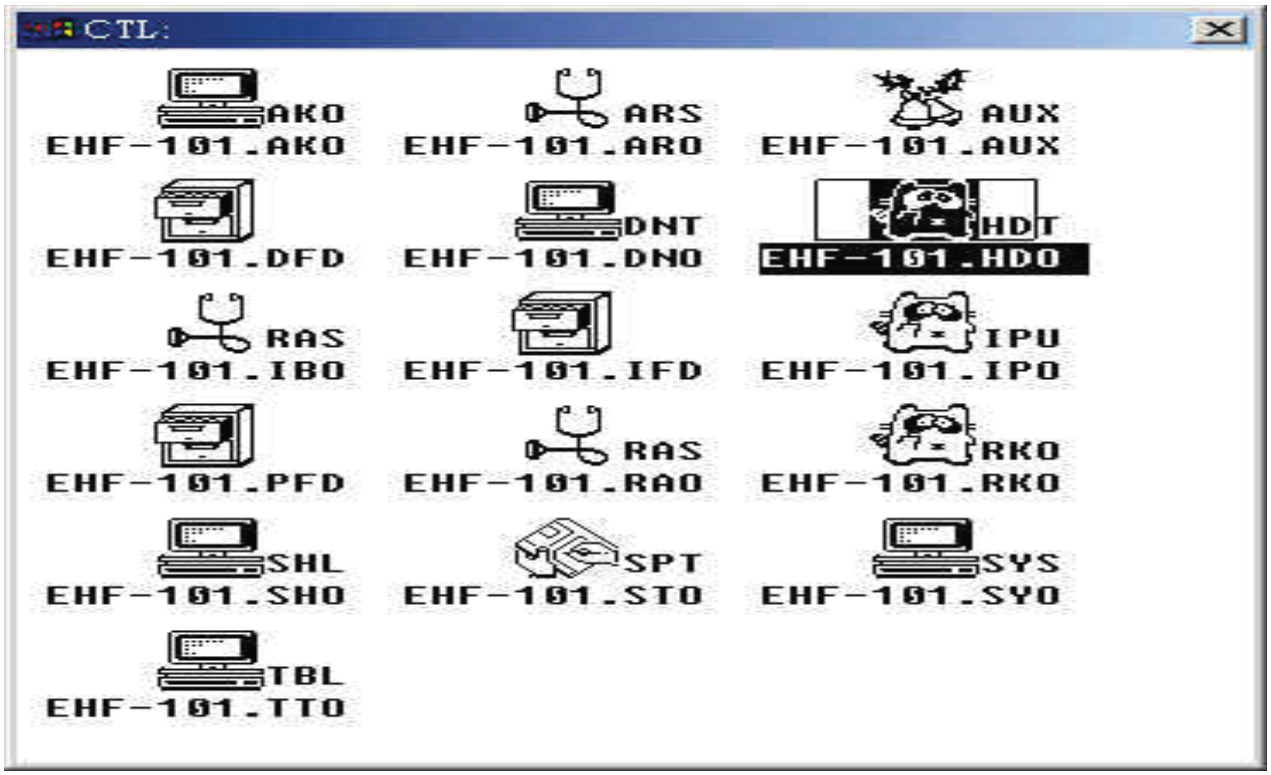
For the controller of CX system we usually select TCP/IP(EtherNet) to communicate, after running R/Window Redisplay will display the following.

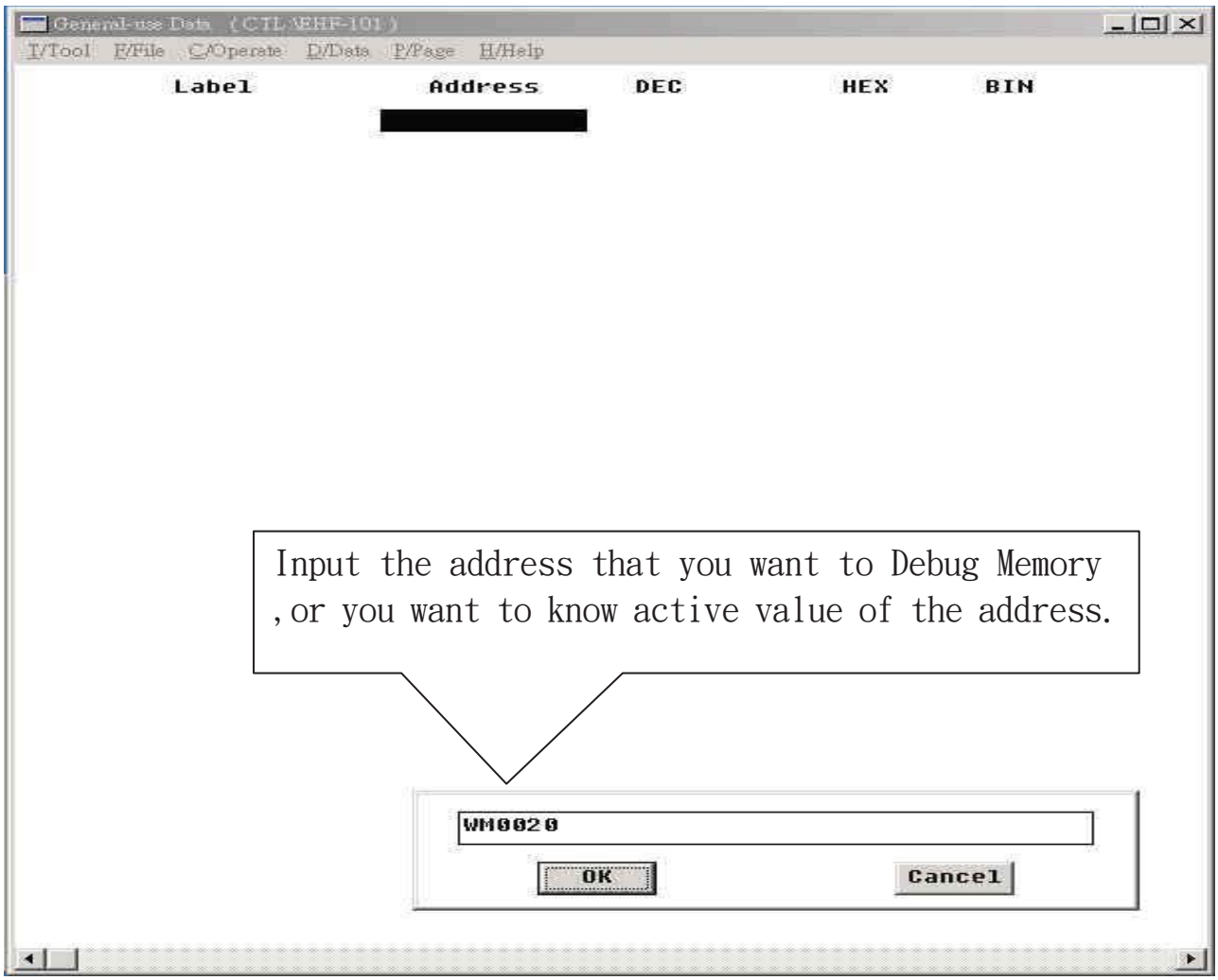


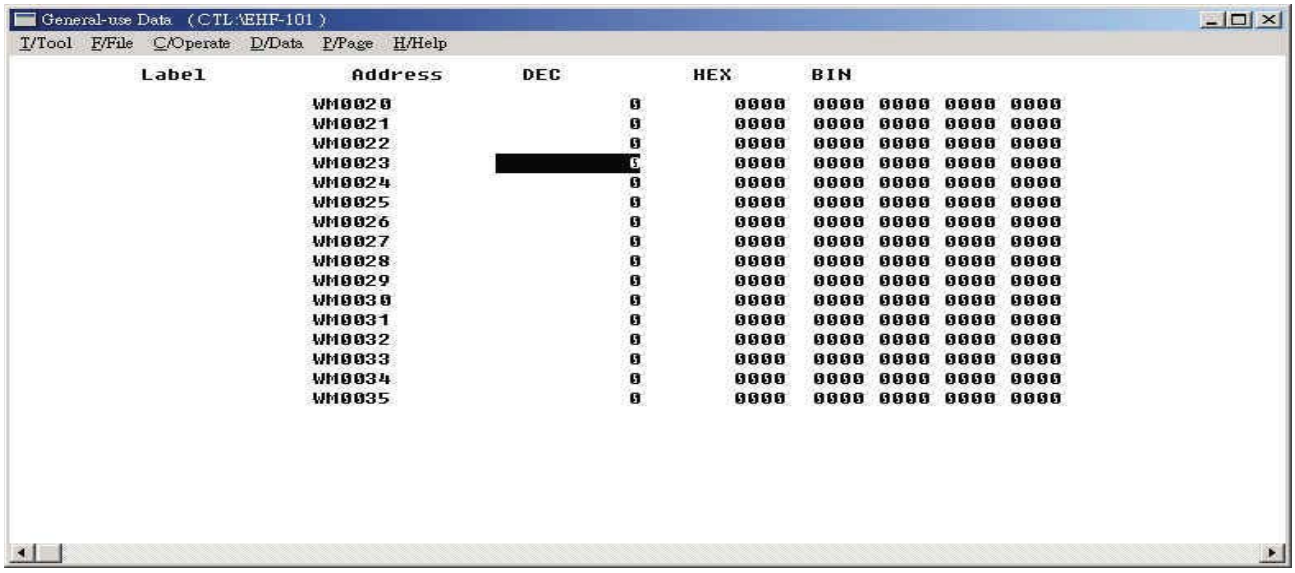

[controller Model]
Please select Controller model.
HDC250
PCS250
HDC500AE
PMS500AE
PCS500AE
ICS_EI
ICS_E
ICS_I
SAS2500
ICS_SEI
ICS_SE
ICS_SI
ICS_SEIN
ICS_SEN
ICS_SIN
SAS2500SA
ICS_EIQ
ICS_EQ
SAS3000
ACS_EI
ACS_E
ASA_E
ASA_EI
GCS-1000
FCS2000E
FCS2000EI
Usually choose FCS2000EI
***500AE -> High speed Extension Version
OK
Cancel

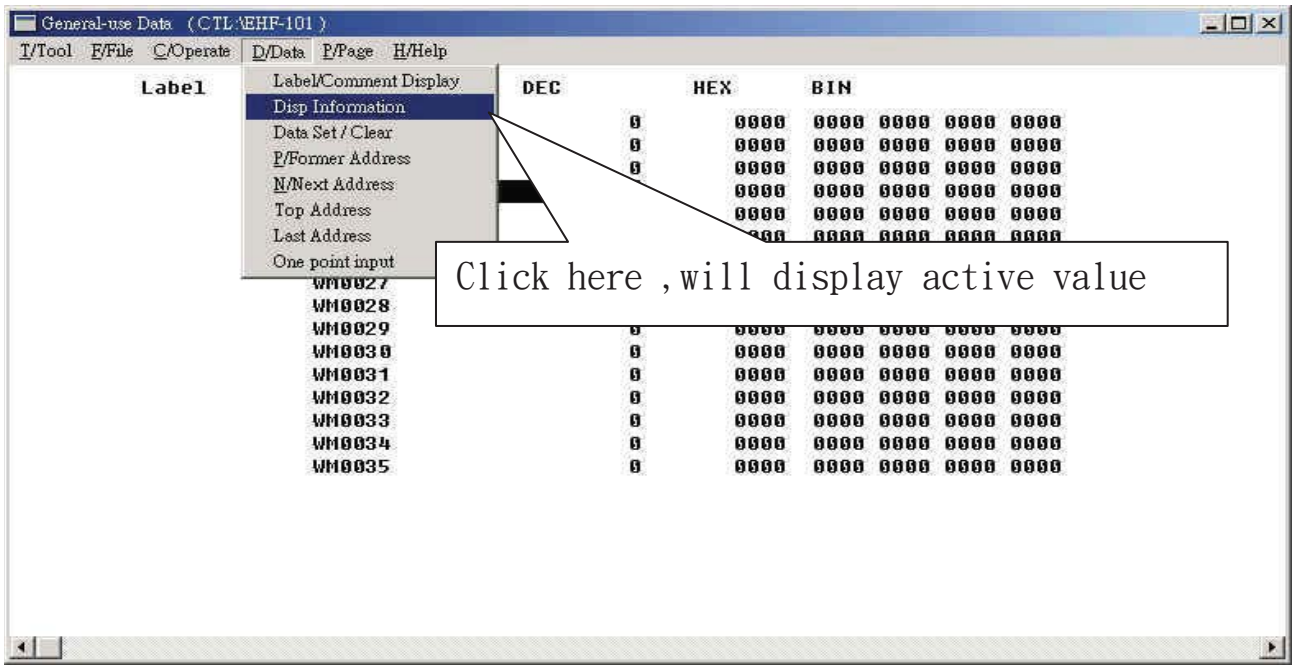

[Controller Name]
Please specify Controller name.
Controller name : EHF-101
Position : C:
Controller type
101
STN101
Input the station name you want to link
OK
Move
Cancel
If the station name display here, means you have ever link to the station, you may directly click the station name.

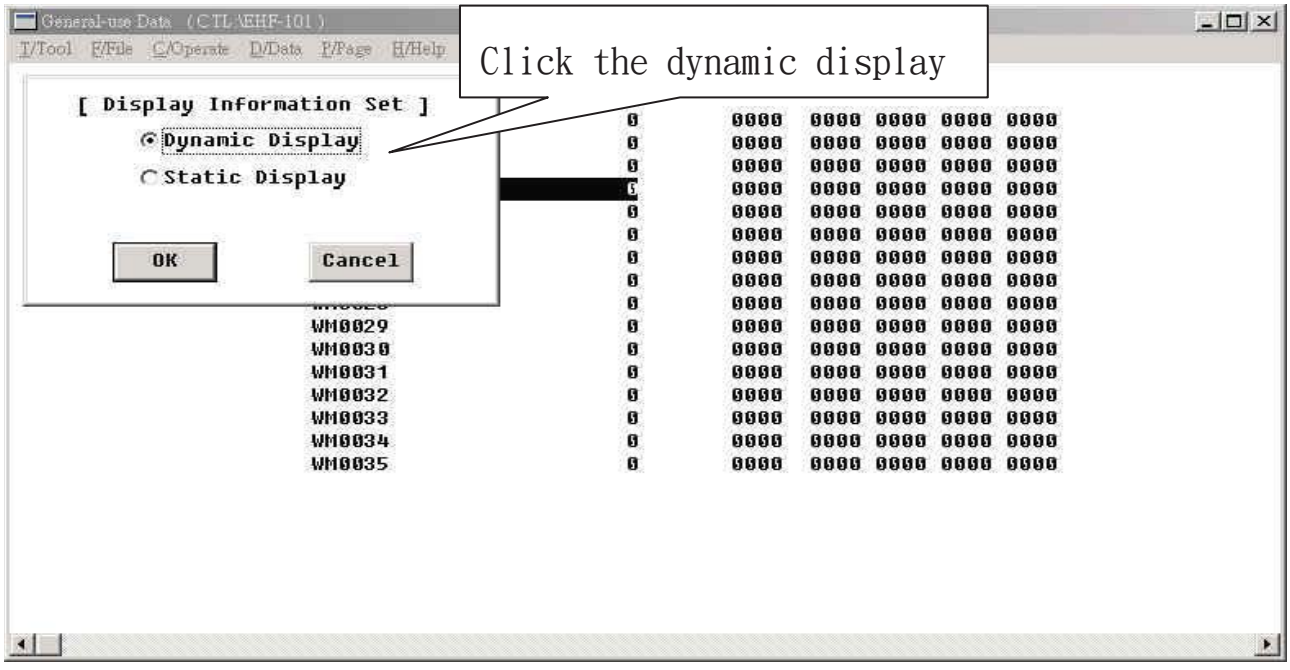
[Label Format Set]
Extended label MAX. User TAGs 4 * 1024-Point(4-10)
14-character label
9-character label
OK
Cancel

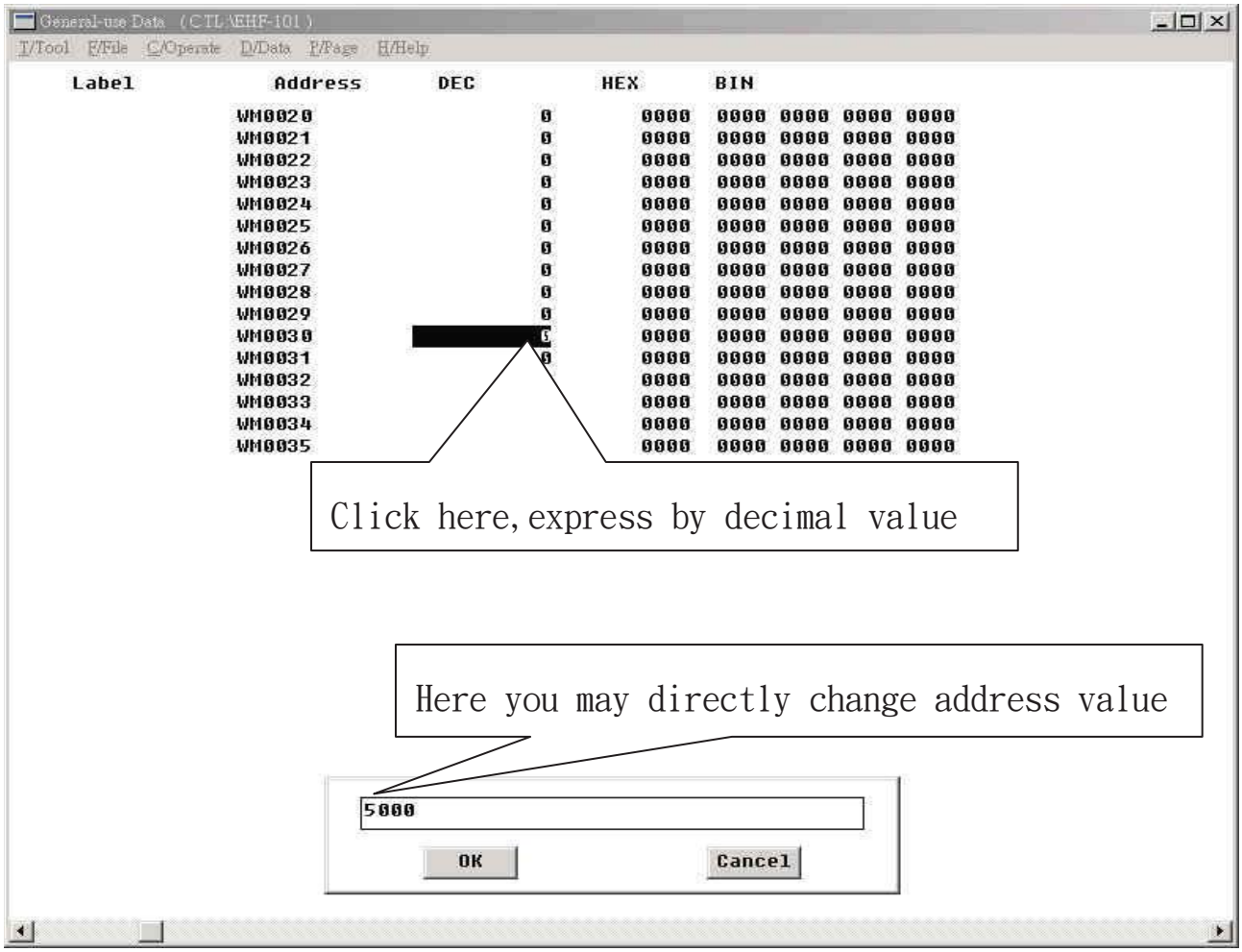

CTL:
AKO
EHF-101.AKO
ARS
EHF-101.ARO
AUX
EHF-101.AUX
EHF-101.DFD
DNT
EHF-101.DNO
HDT
EHF-101.HDO
RAS
EHF-101.IBO
EHF-101.IFD
IPU
EHF-101.IPO
EHF-101.PFD
RAS
EHF-101.RAO
RKO
EHF-101.RKO
SHL
EHF-101.SHO
SPT
EHF-101.STO
SYS
EHF-101.SYO
TBL
EHF-101.TTO


General-use Data (CTL\EHF-101)
T/Tool F/File C/Operate D/Data P/Page H/Help
Label
Address
DEC
HEX
BIN
Input the address that you want to Debug Memory ,or you want to know active value of the address.
WM0020
OK
Cancel

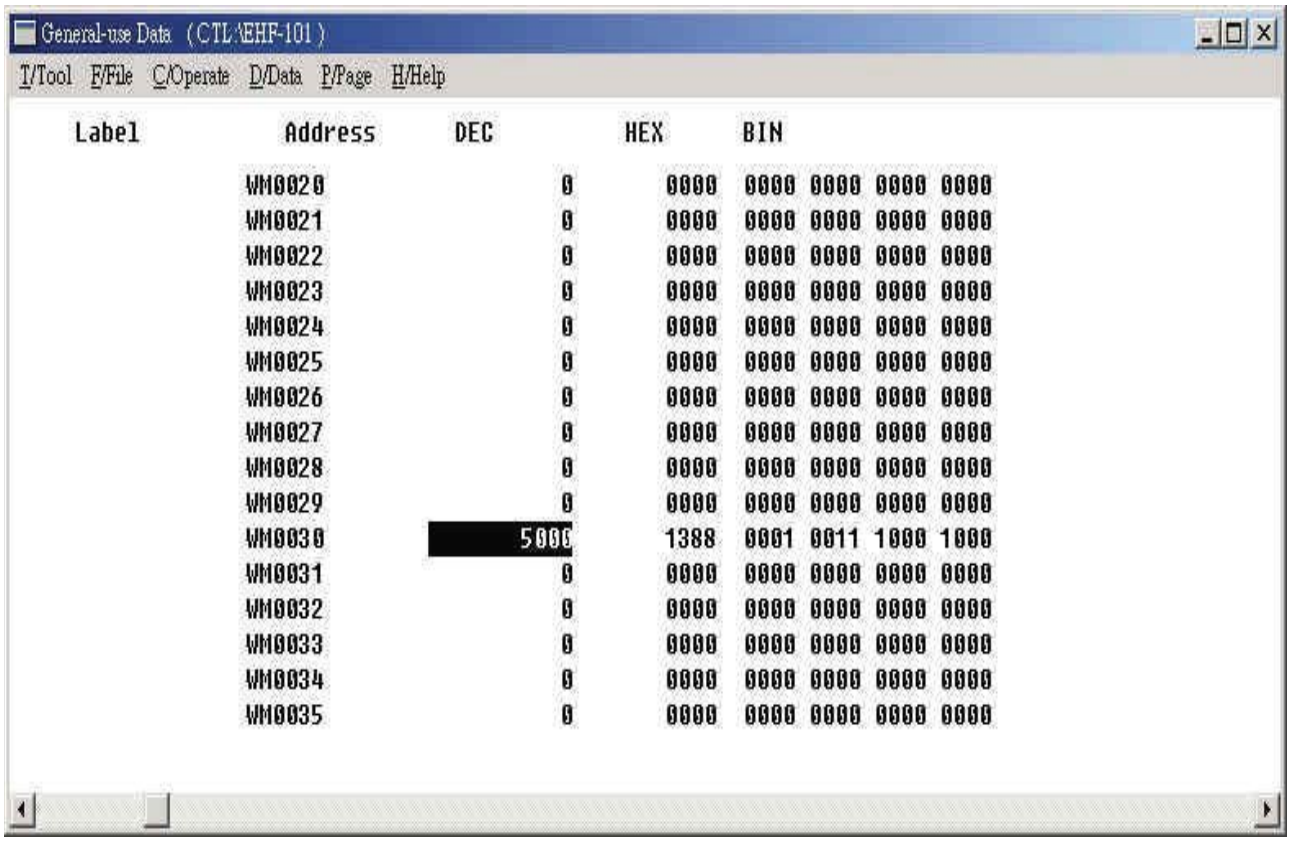

General-use Data (CTL:\EHF-101)
T/Tool F/File C/Operate D/Data P/Page H/Help
Label Address DEC HEX BIN
WM0020 0 0000 0000 0000 0000 0000
WM0021 0 0000 0000 0000 0000 0000
WM0022 0 0000 0000 0000 0000 0000
WM0023 0 0000 0000 0000 0000 0000
WM0024 0 0000 0000 0000 0000 0000
WM0025 0 0000 0000 0000 0000 0000
WM0026 0 0000 0000 0000 0000 0000
WM0027 0 0000 0000 0000 0000 0000
WM0028 0 0000 0000 0000 0000 0000
WM0029 0 0000 0000 0000 0000 0000
WM0030 0 0000 0000 0000 0000 0000
WM0031 0 0000 0000 0000 0000 0000
WM0032 0 0000 0000 0000 0000 0000
WM0033 0 0000 0000 0000 0000 0000
WM0034 0 0000 0000 0000 0000 0000
WM0035 0 0000 0000 0000 0000 0000

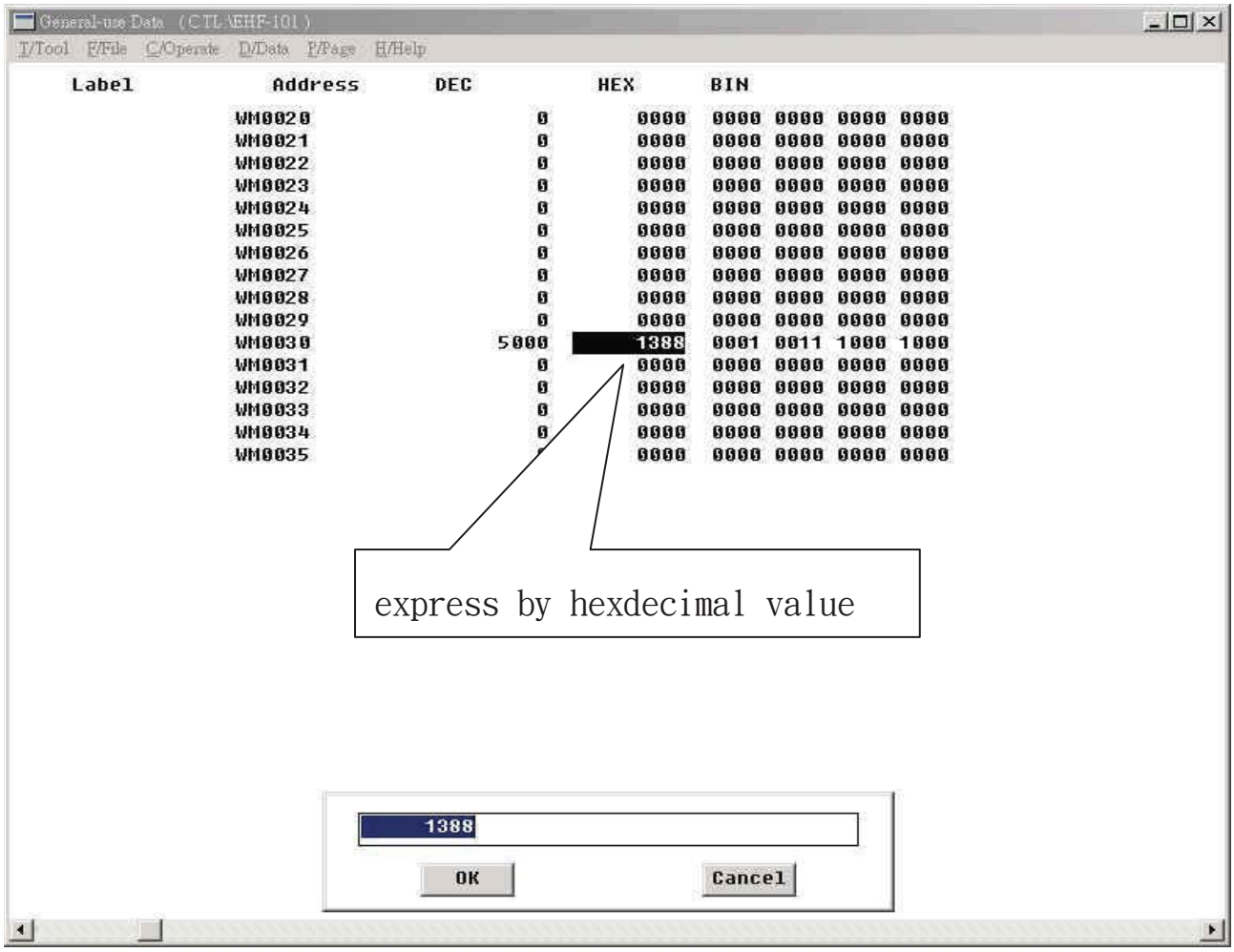

General-use Data (CTL:\EHF-101)
T/Tool F/File C/Operate D/Data P/Page H/Help
Label/Comment Display
Disp Information
Data Set / Clear
P/Former Address
N/Next Address
Top Address
Last Address
One point input
Label DEC HEX BIN
Click here ,will display active value
WM0028
WM0029
WM0030
WM0031
WM0032
WM0033
WM0034
WM0035

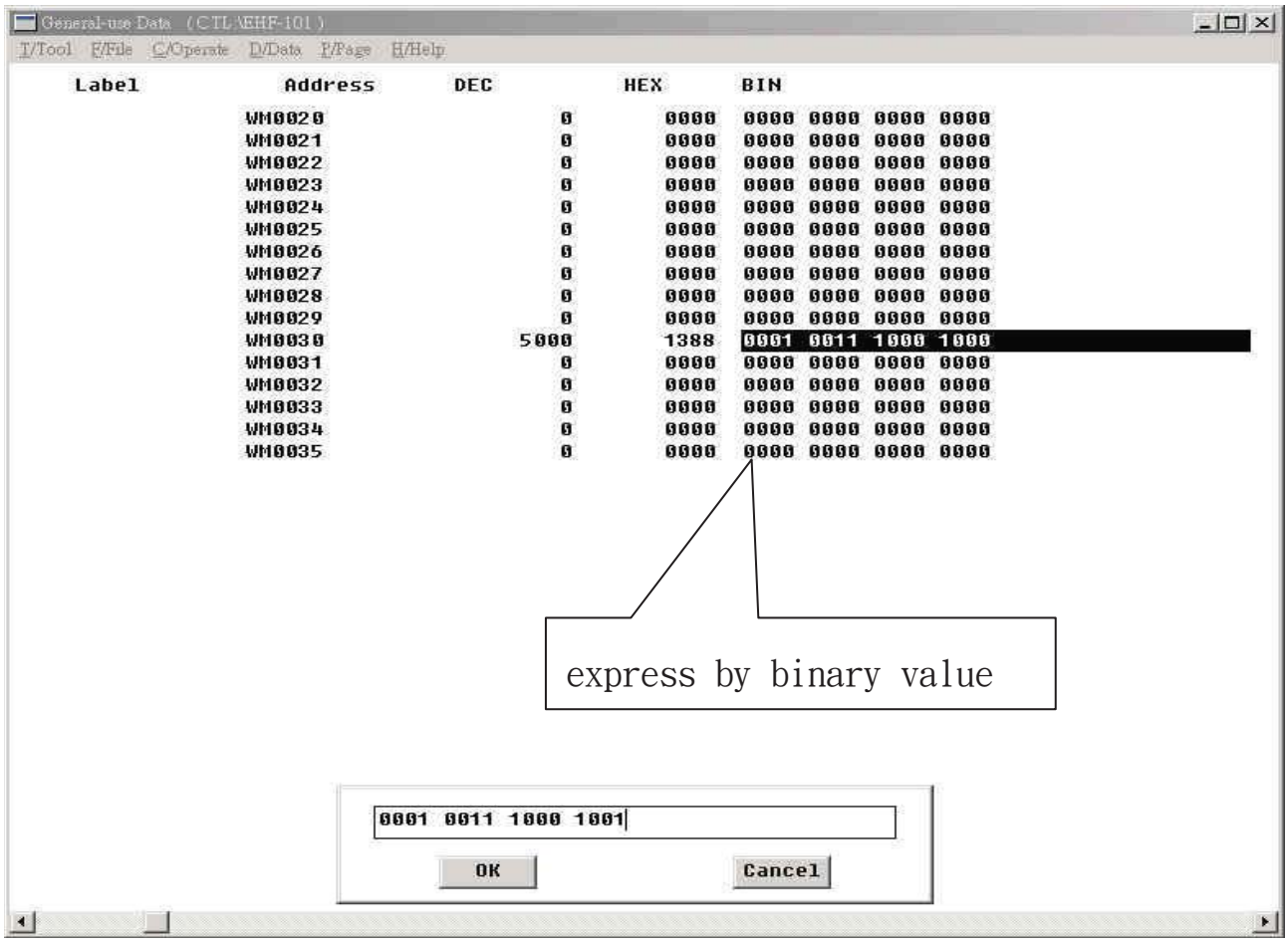

General-use Data (CTL:\EHF-101)
T/Tool F/File C/Operate D/Data P/Page H/Help
Click the dynamic display
[Display Information Set]
Dynamic Display
Static Display
OK
Cancel
WM0029
WM0030
WM0031
WM0032
WM0033
WM0034
WM0035



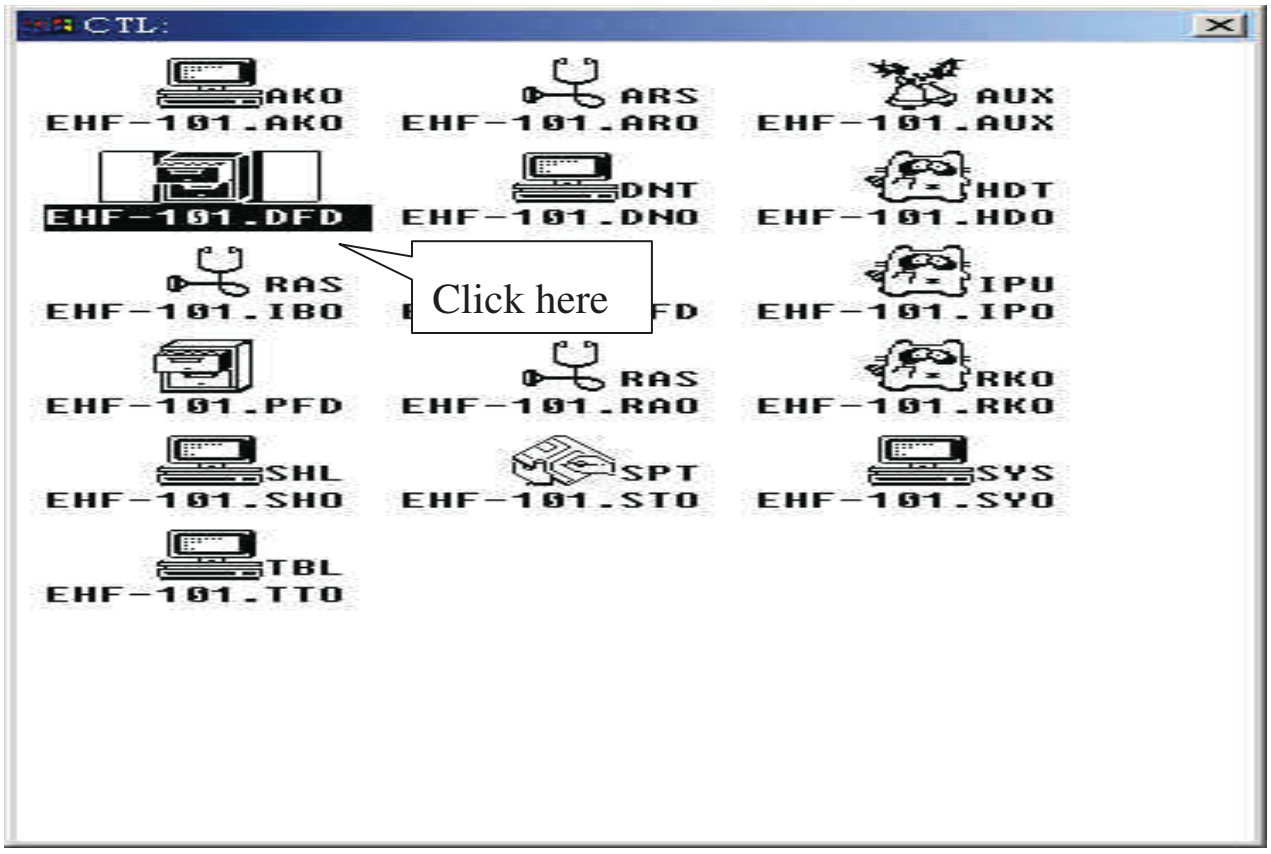
See the result as the following



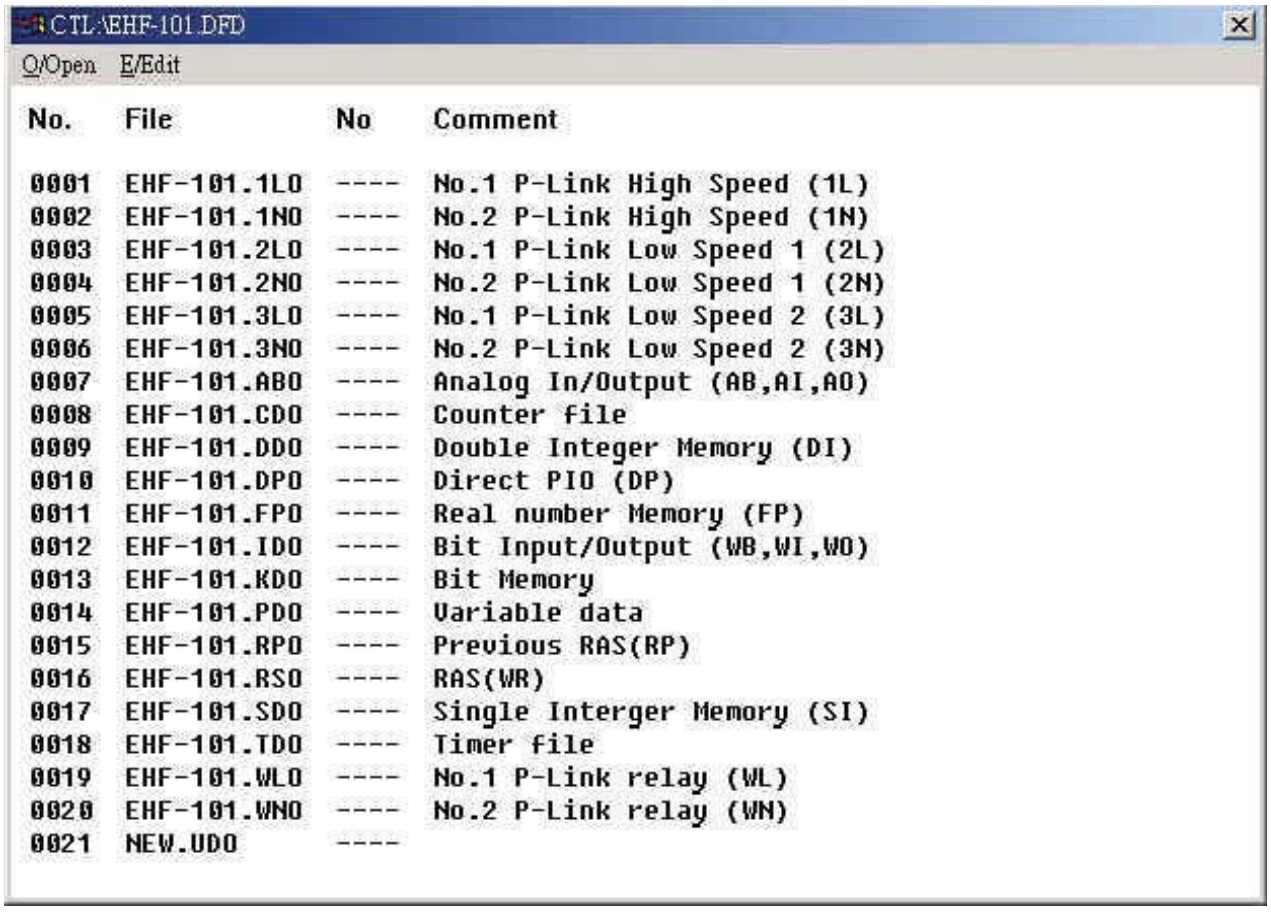

General-use Data (CTL\EHF-101)
T/Tool F/File C/Operate D/Data P/Page H/Help
Label Address DEC HEX BIN
WM0020 0 0000 0000 0000 0000 0000
WM0021 0 0000 0000 0000 0000 0000
WM0022 0 0000 0000 0000 0000 0000
WM0023 0 0000 0000 0000 0000 0000
WM0024 0 0000 0000 0000 0000 0000
WM0025 0 0000 0000 0000 0000 0000
WM0026 0 0000 0000 0000 0000 0000
WM0027 0 0000 0000 0000 0000 0000
WM0028 0 0000 0000 0000 0000 0000
WM0029 0 0000 0000 0000 0000 0000
WM0030 5000 1388 0001 0011 1000 1000
WM0031 0 0000 0000 0000 0000 0000
WM0032 0 0000 0000 0000 0000 0000
WM0033 0 0000 0000 0000 0000 0000
WM0034 0 0000 0000 0000 0000 0000
WM0035 0000 0000 0000 0000 0000
express by hexdecimal value
1388
OK
Cancel


General-use Data (CTL\EHF-101)
T/Tool F/File C/Operate D/Data P/Page H/Help
Label Address DEC HEX BIN
WM0020 0 0000 0000 0000 0000 0000
WM0021 0 0000 0000 0000 0000 0000
WM0022 0 0000 0000 0000 0000 0000
WM0023 0 0000 0000 0000 0000 0000
WM0024 0 0000 0000 0000 0000 0000
WM0025 0 0000 0000 0000 0000 0000
WM0026 0 0000 0000 0000 0000 0000
WM0027 0 0000 0000 0000 0000 0000
WM0028 0 0000 0000 0000 0000 0000
WM0029 0 0000 0000 0000 0000 0000
WM0030 5000 1388 0001 0011 1000 1000
WM0031 0 0000 0000 0000 0000 0000
WM0032 0 0000 0000 0000 0000 0000
WM0033 0 0000 0000 0000 0000 0000
WM0034 0 0000 0000 0000 0000 0000
WM0035 0 0000 0000 0000 0000 0000
express by binary value
0001 0011 1000 1001
OK
Cancel

You may choose EDF-101,DFD of CTL instead of EDF-101,HDO of CTL

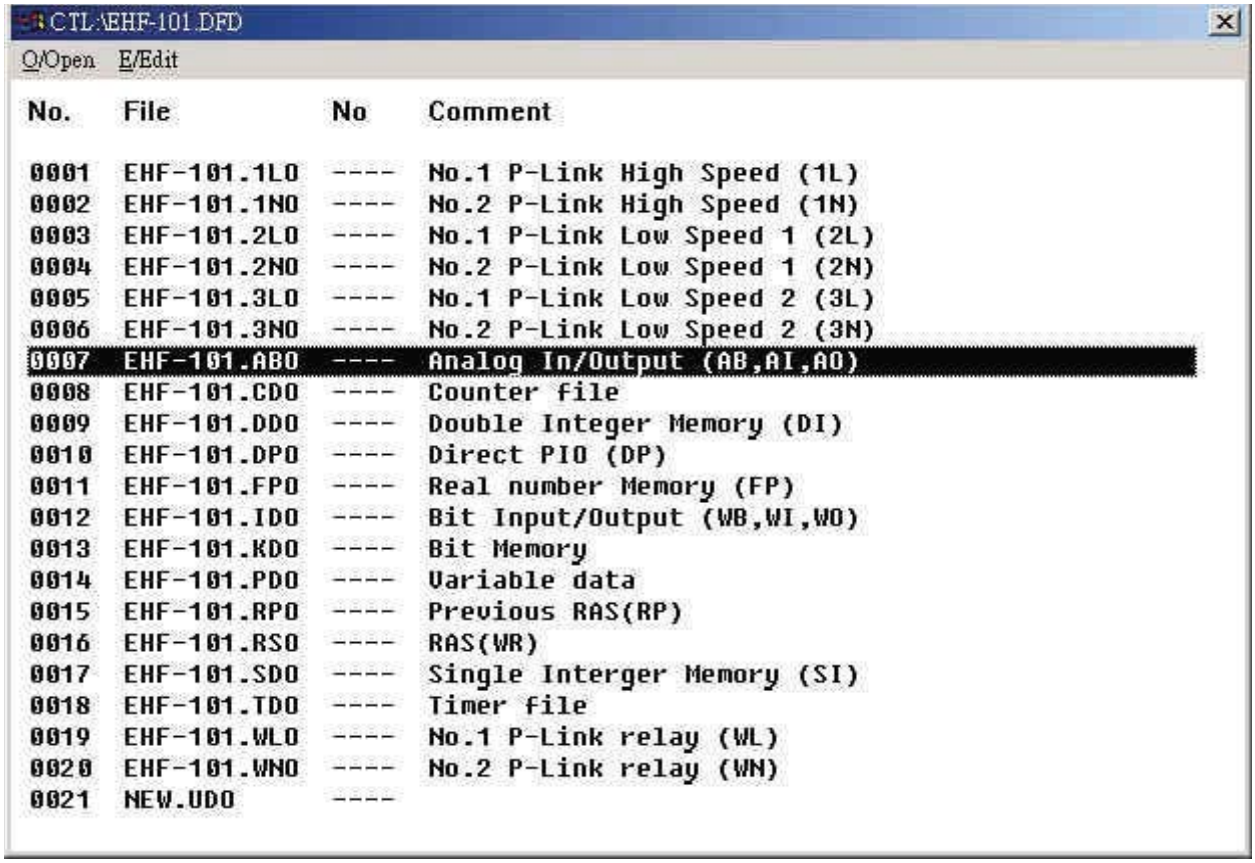


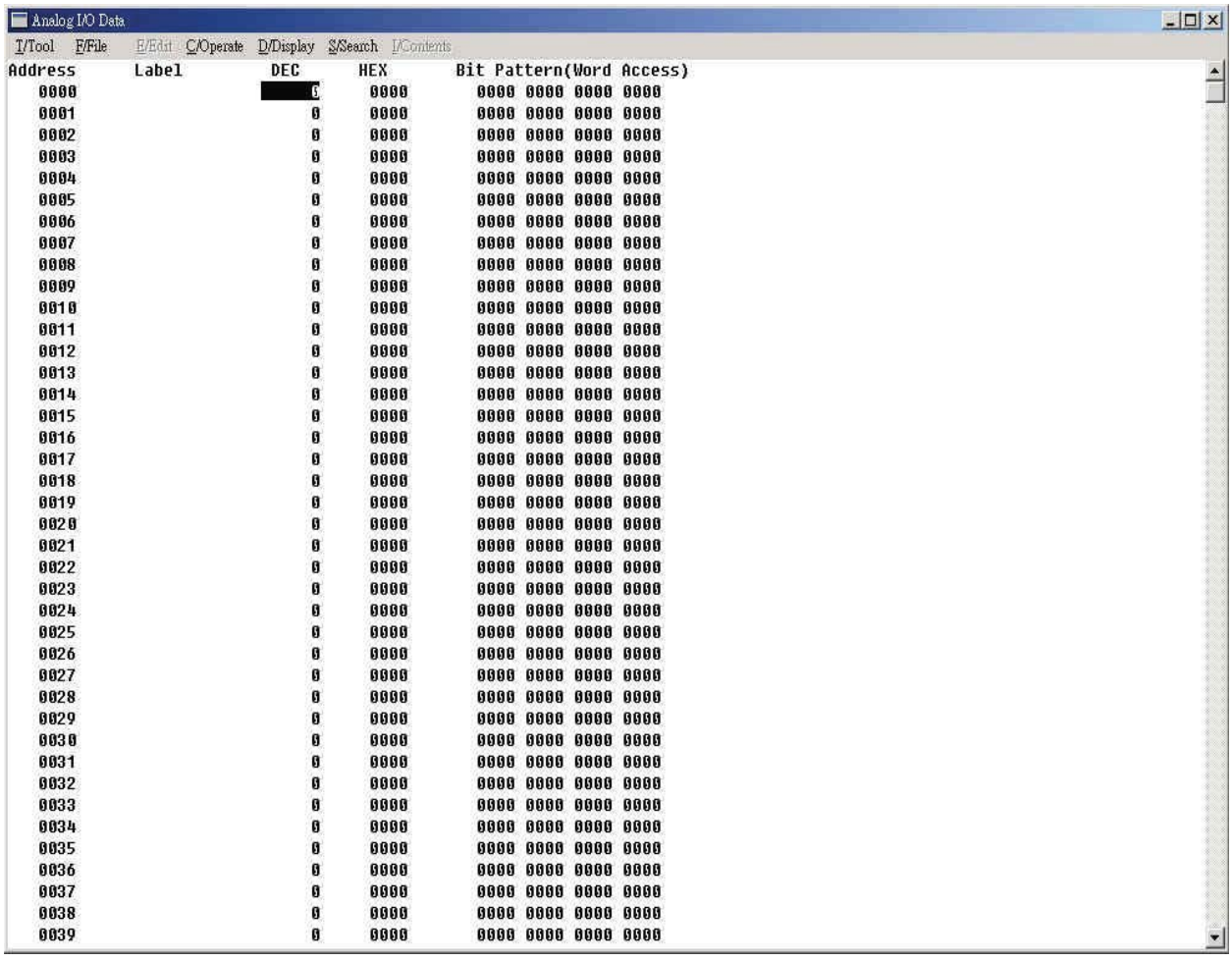
EDF-101,DFD separate memory into different files.

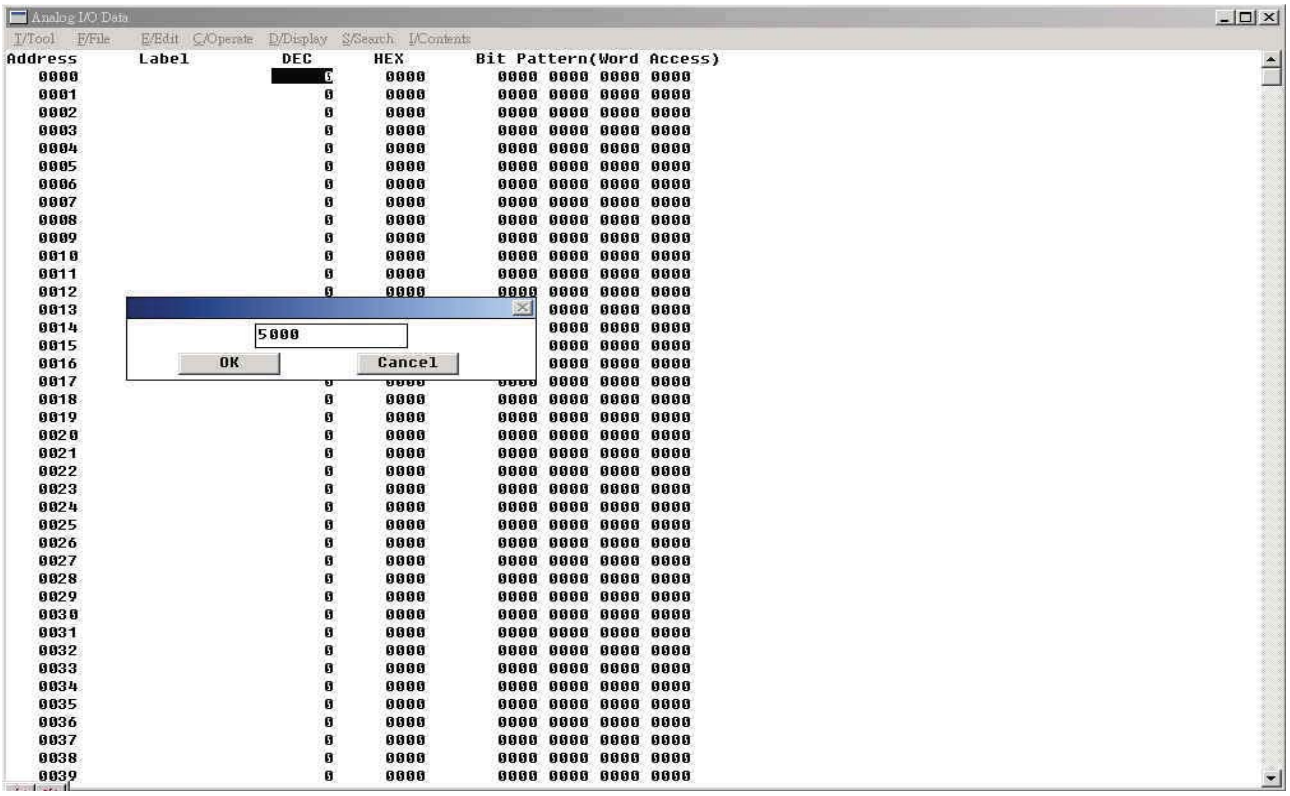


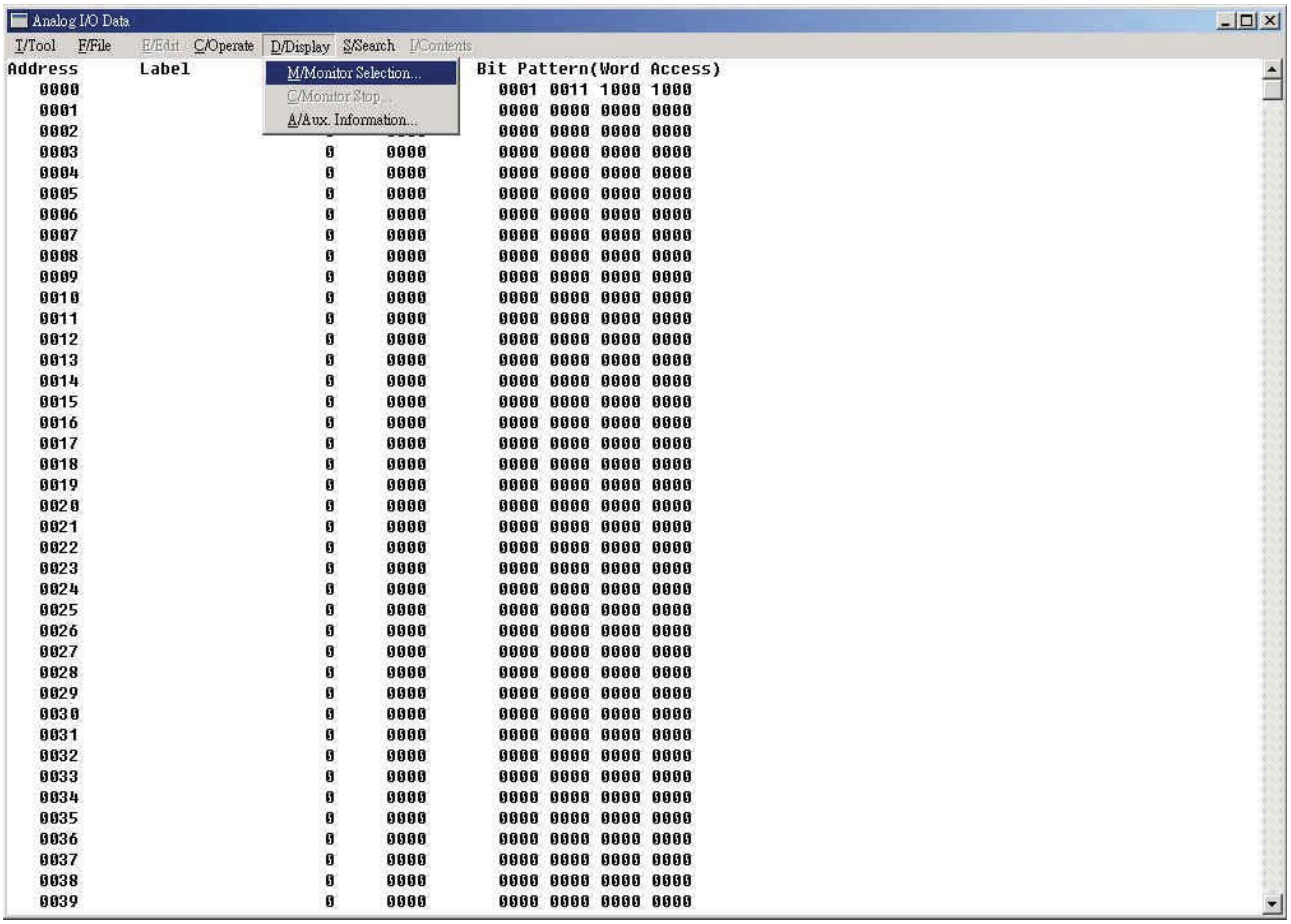
No.	File	No	Comment
0001	EHF-101.1LO	----	No.1 P-Link High Speed (1L)
0002	EHF-101.1NO	----	No.2 P-Link High Speed (1N)
0003	EHF-101.2LO	----	No.1 P-Link Low Speed 1 (2L)
0004	EHF-101.2NO	----	No.2 P-Link Low Speed 1 (2N)
0005	EHF-101.3LO	----	No.1 P-Link Low Speed 2 (3L)
0006	EHF-101.3NO	----	No.2 P-Link Low Speed 2 (3N)
0007	EHF-101.ABO	----	Analog In/Output (AB,AI,AO)
0008	EHF-101.CDO	----	Counter file
0009	EHF-101.DDO	----	Double Integer Memory (DI)
0010	EHF-101.DPO	----	Direct PIO (DP)
0011	EHF-101.FPO	----	Real number Memory (FP)
0012	EHF-101.IDO	----	Bit Input/Output (WB,WI,WO)
0013	EHF-101.KDO	----	Bit Memory
0014	EHF-101.PDO	----	Variable data
0015	EHF-101.RPO	----	Previous RAS(RP)
0016	EHF-101.RSO	----	RAS(WR)
0017	EHF-101.SDO	----	Single Interger Memory (SI)
0018	EHF-101.TDO	----	Timer file
0019	EHF-101.WLO	----	No.1 P-Link relay (WL)
0020	EHF-101.WNO	----	No.2 P-Link relay (WN)
0021	NEW.UDO	----	

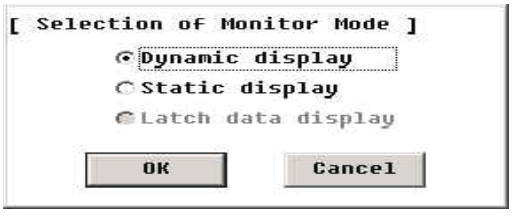
You may choose the file you want to debug.





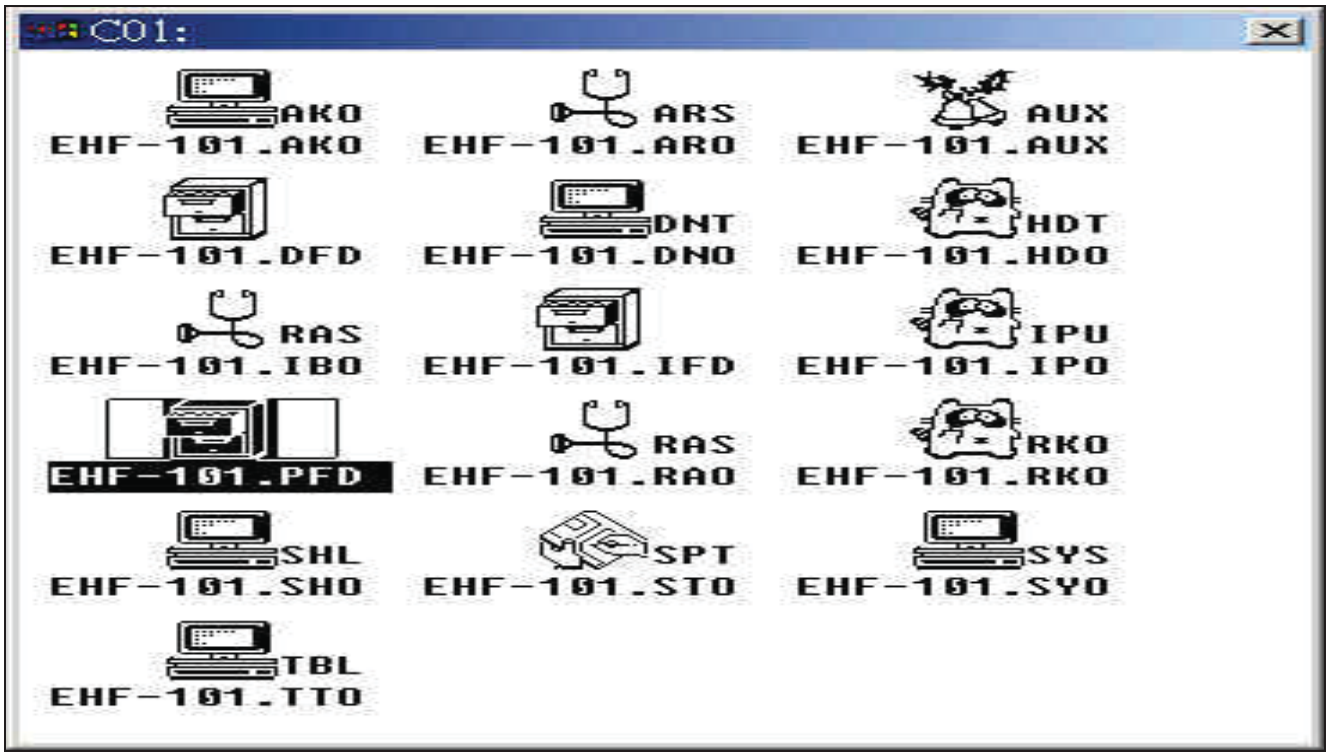

Analog I/O Data
I/Tool F/File E/Edit C/Operate D/Display S/Search I/Contents
Address Label DEC HEX Bit Pattern(Word Access)
5000
OK
Cancel

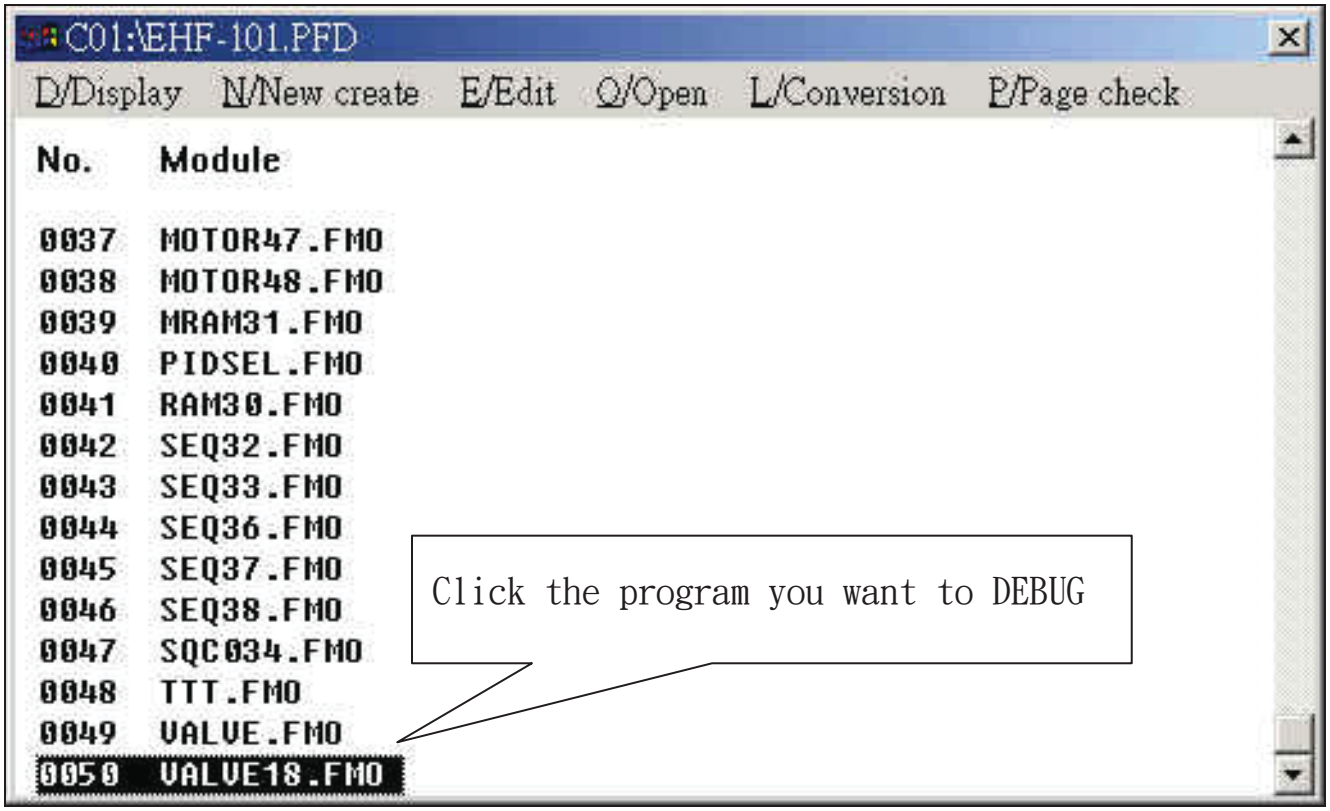

Analog I/O Data
I/Tool F/File E/Edit C/Operate D/Display S/Search I/Contents
M/Monitor Selection...
C/Monitor Stop...
A/Aux. Information...
Address Label Bit Pattern(Word Access)

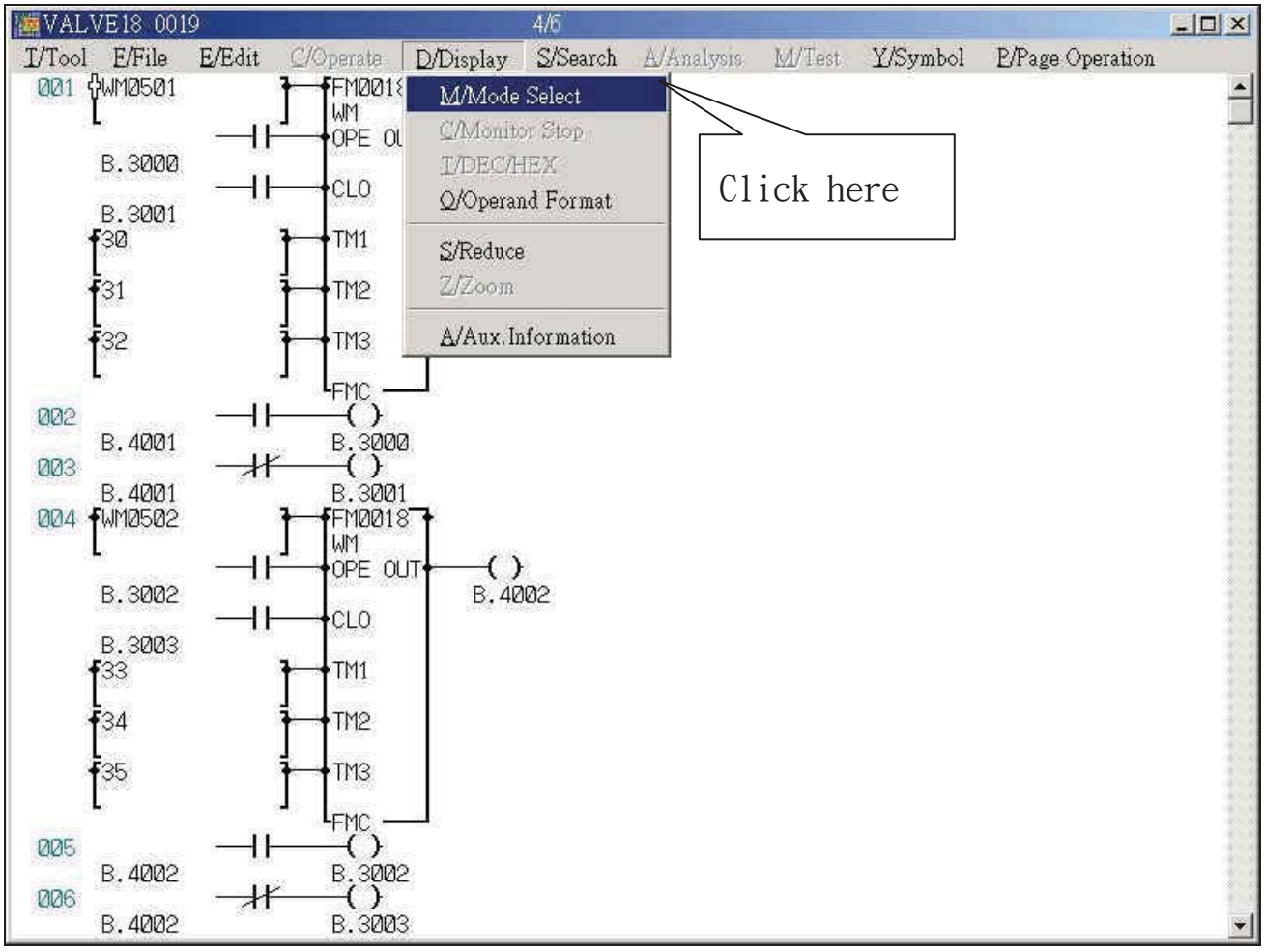

[Selection of Monitor Mode]
Dynamic display
Static display
Latch data display
OK
Cancel

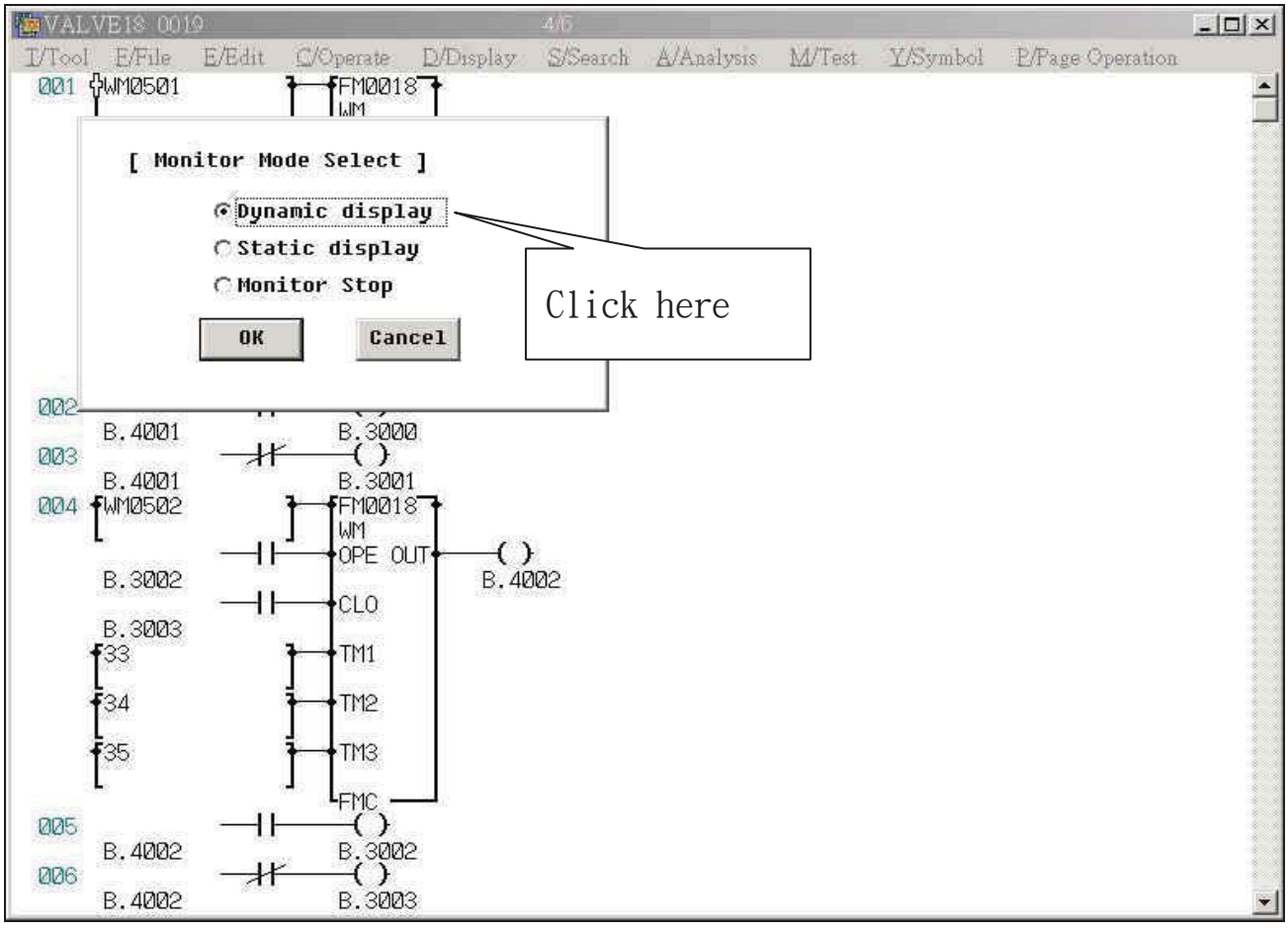
PROGRAM DEBUG:

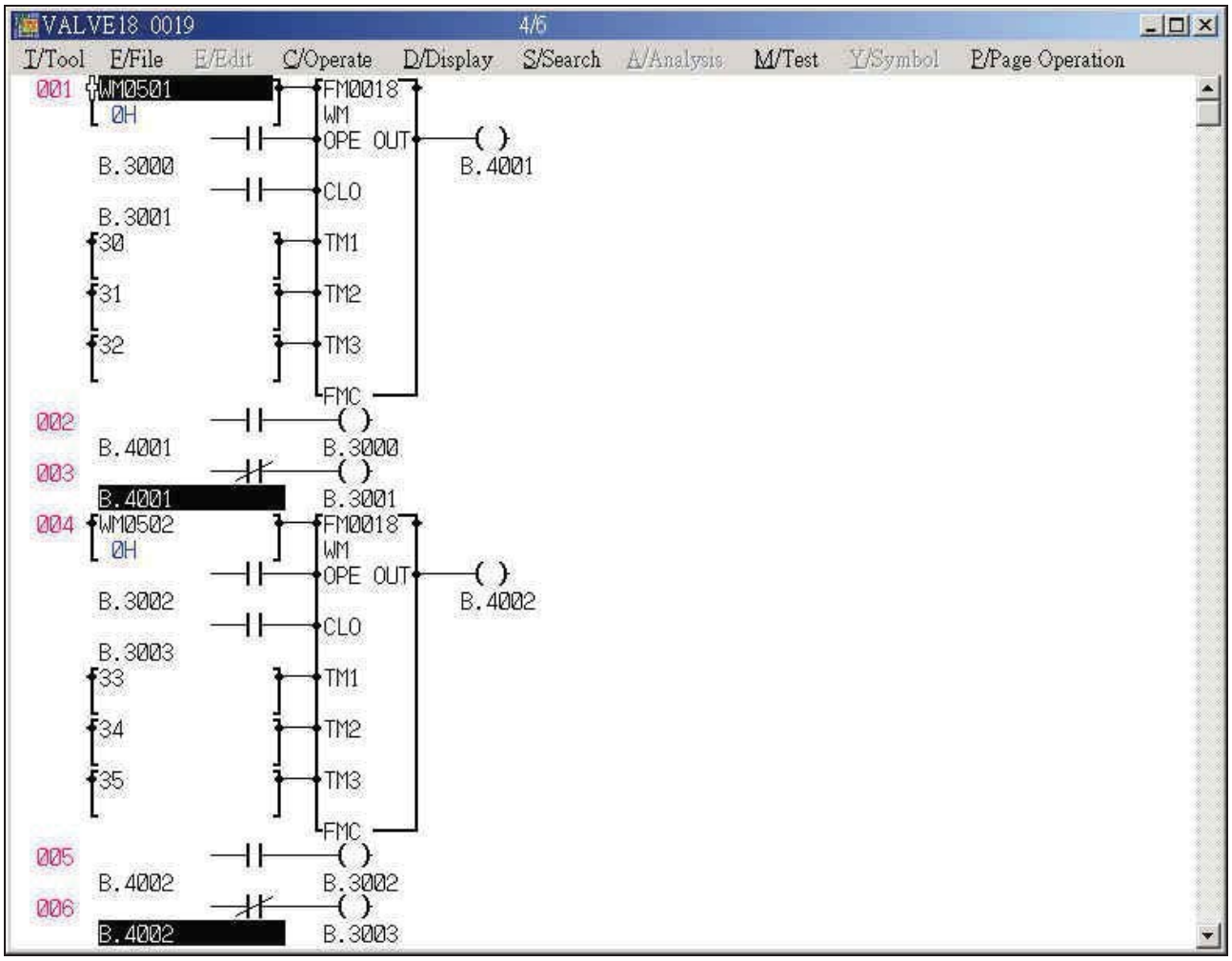
OPEN FPD in C01 (or CTL)

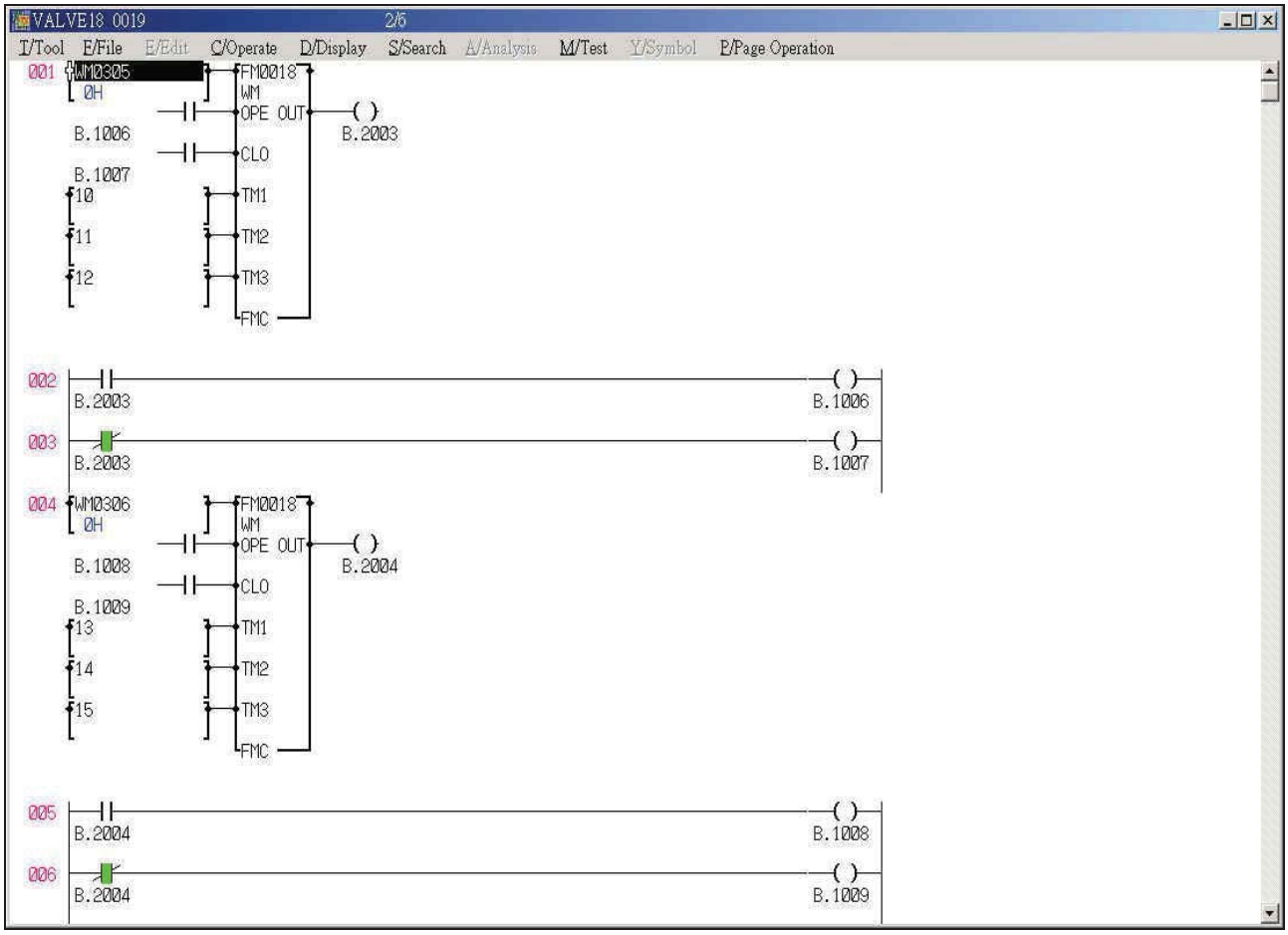


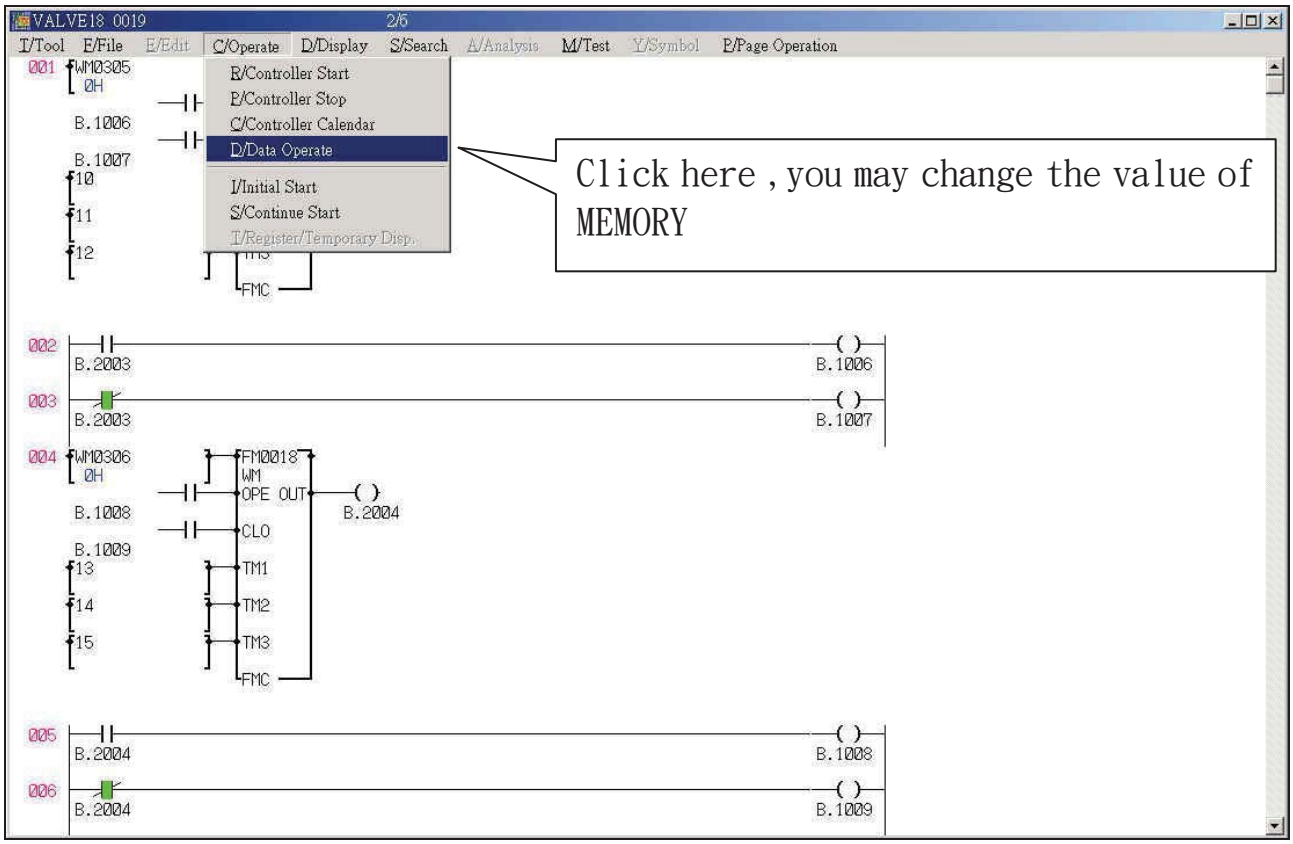


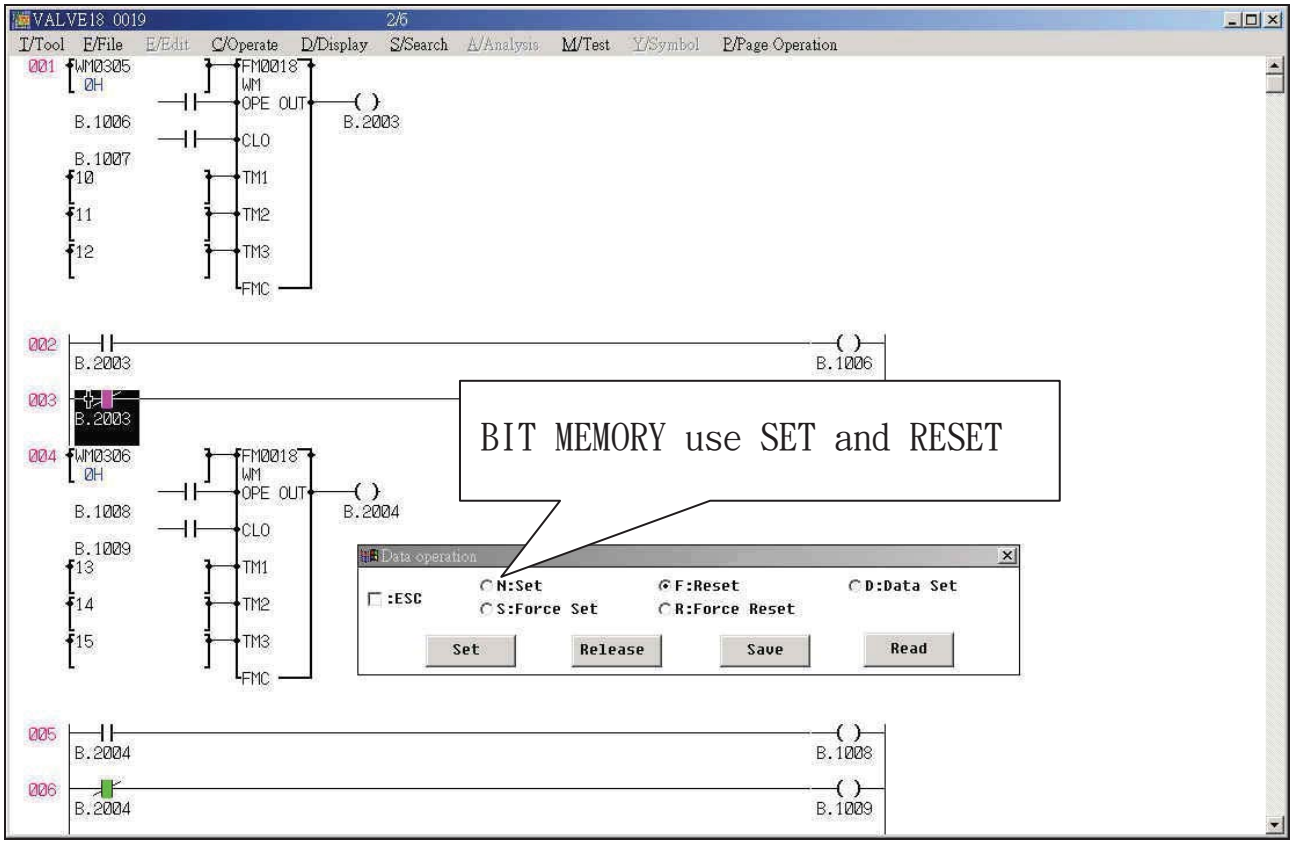

VALVE18 0019
4/6
T/Tool E/File E/Edit C/Operate D/Display S/Search A/Analysis M/Test Y/Symbol P/Page Operation
M/Mode Select
C/Monitor Stop
T/DEC/HEX
O/Operand Format
S/Reduce
Z/Zoom
A/Aux.Information
Click here

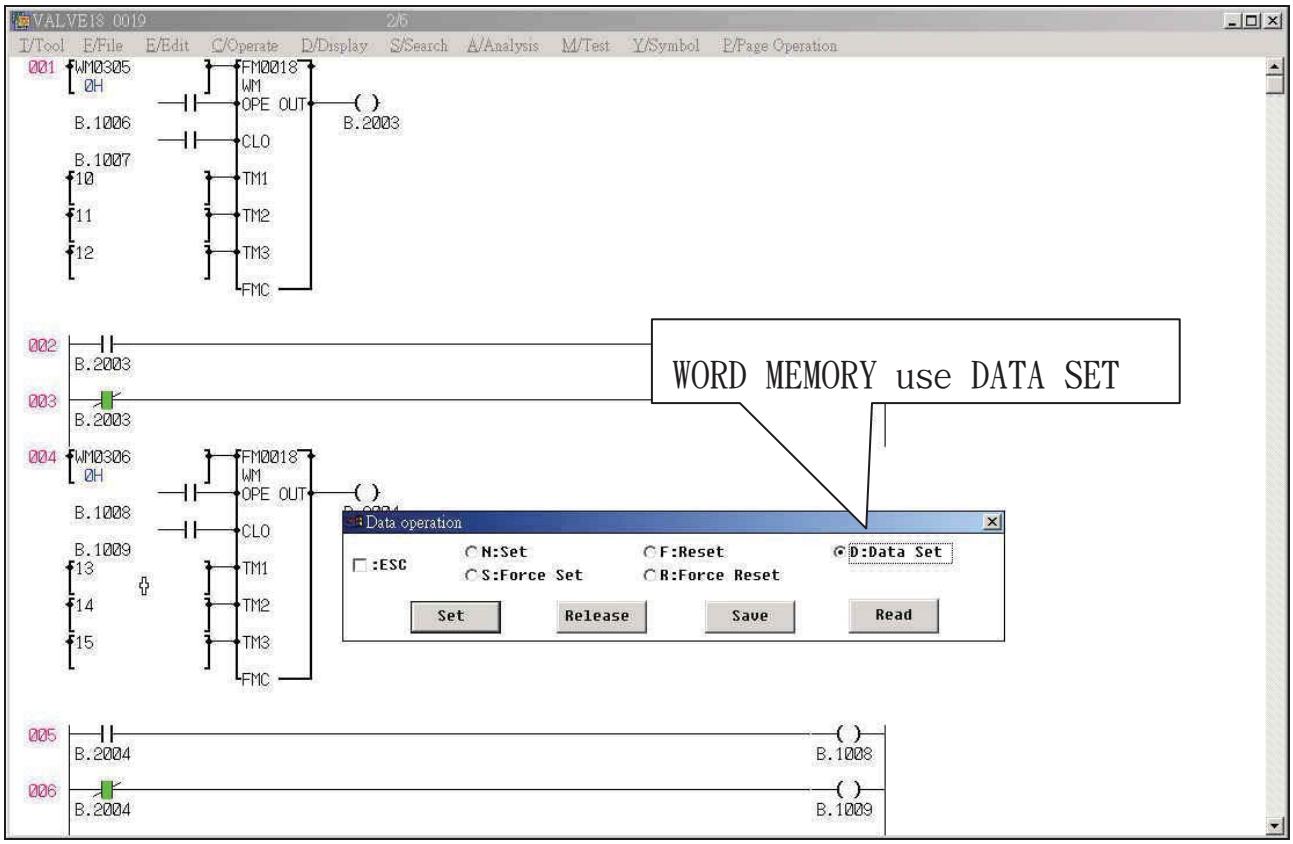

VALVE18 0019
4/6
T/Tool E/File E/Edit C/Operate D/Display S/Search A/Analysis M/Test Y/Symbol P/Page Operation
[Monitor Mode Select]
Dynamic display
Static display
Monitor Stop
OK
Cancel
Click here

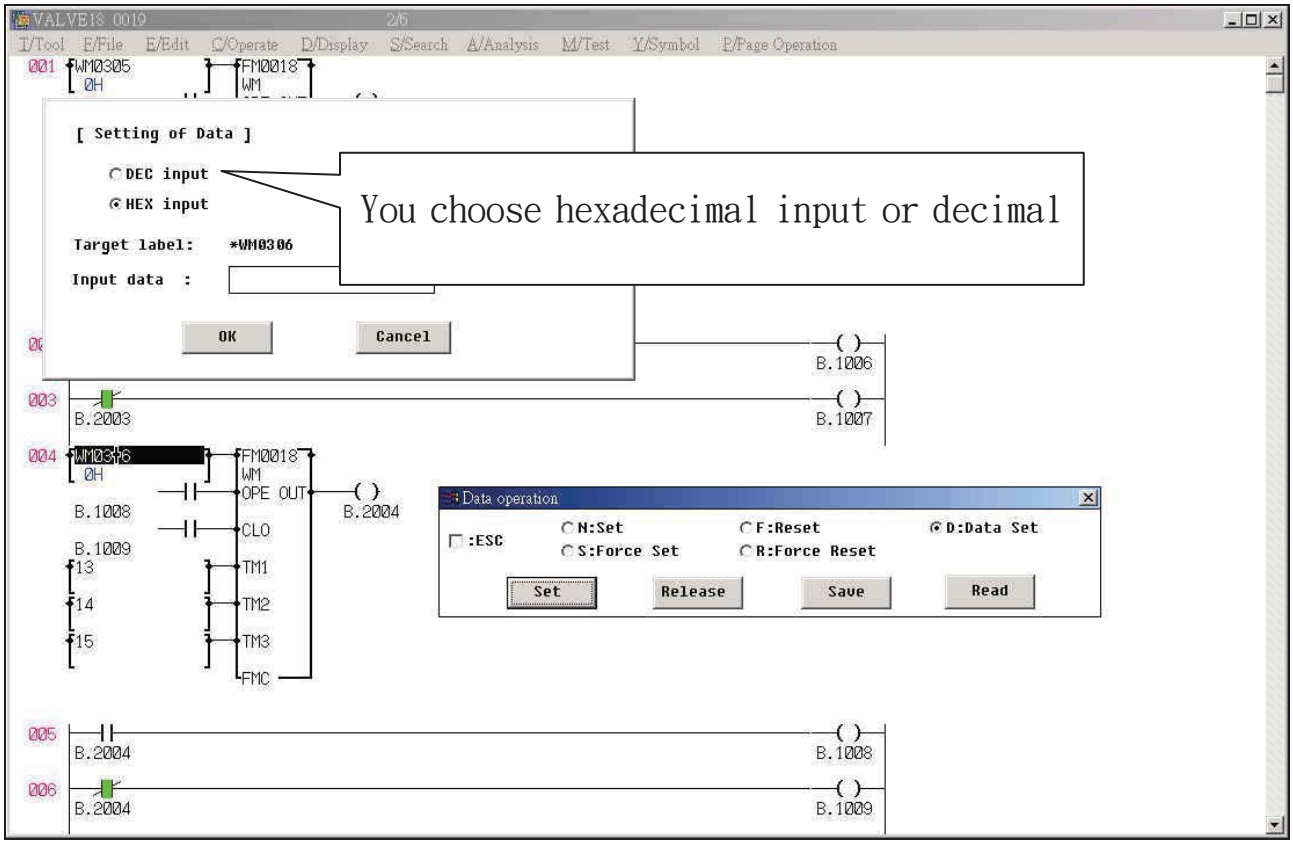

VALVE18 0019
4/5
I/Tool E/File E/Edit C/Operate D/Display S/Search A/Analysis M/Test Y/Symbol P/Page Operation
001 WM0501
0H
FM0018
WM
OPE OUT
B.3000
B.4001
CLO
B.3001
30
TM1
31
TM2
32
TM3
FMC
002
B.4001
B.3000
003
B.4001
B.3001
004 WM0502
0H
FM0018
WM
OPE OUT
B.3002
B.4002
CLO
B.3003
33
TM1
34
TM2
35
TM3
FMC
005
B.4002
B.3002
006
B.4002
B.3003


VALVE18 0019
2/5
I/Tool E/File E/Edit C/Operate D/Display S/Search A/Analysis M/Test Y/Symbol P/Page Operation
001 WM0305
0H
FM0018
WM
OPE OUT
B.1006
B.2003
CLO
B.1007
10
TM1
11
TM2
12
TM3
FMC
002
B.2003
B.1006
003
B.2003
B.1007
004 WM0306
0H
FM0018
WM
OPE OUT
B.1008
B.2004
CLO
B.1009
13
TM1
14
TM2
15
TM3
FMC
005
B.2004
B.1008
006
B.2004
B.1009


VALVE18 0019
2/5
T/Tool E/File E/Edit C/Operate D/Display S/Search A/Analysis M/Test Y/Symbol P/Page Operation
R/Controller Start
P/Controller Stop
C/Controller Calendar
D/Data Operate
I/Initial Start
S/Continue Start
T/Register/Temporary Disp.
Click here , you may change the value of MEMORY


VALVE18 0019
2/5
T/Tool E/File E/Edit C/Operate D/Display S/Search A/Analysis M/Test Y/Symbol P/Page Operation
BIT MEMORY use SET and RESET
Data operation
:ESC
N:Set
S:Force Set
F:Reset
R:Force Reset
D:Data Set
Set
Release
Save
Read

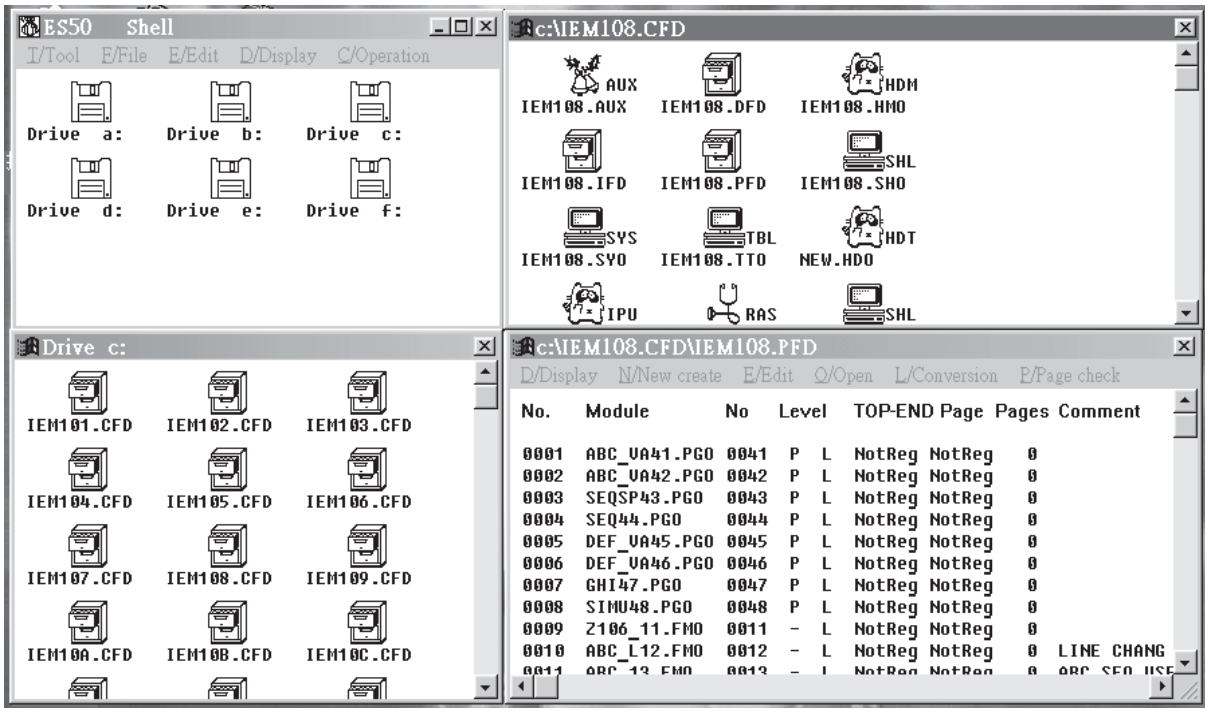

VALVE18 0019
2/6
T/Tool E/File E/Edit C/Operate D/Display S/Search A/Analysis M/Test Y/Symbol P/Page Operation
001 WM0305 0H
FM0018 WM OPE OUT
B.2003
B.1006
CLO
B.1007
10
11
12
TM1
TM2
TM3
FMC
002 B.2003
003 B.2003
WORD MEMORY use DATA SET
004 WM0306 0H
B.1008
B.1009
13
14
15
Data operation
:ESC
N:Set
S:Force Set
F:Reset
R:Force Reset
D:Data Set
Set
Release
Save
Read
005 B.2004
B.1008
006 B.2004
B.1009


VALVE18 0019
2/6
T/Tool E/File E/Edit C/Operate D/Display S/Search A/Analysis M/Test Y/Symbol P/Page Operation
[Setting of Data]
DEC input
HEX input
You choose hexadecimal input or decimal
Target label: *WM0306
Input data :
OK
Cancel
B.1006
B.1007
B.2003
B.2004
Data operation
:ESC
N:Set
S:Force Set
F:Reset
R:Force Reset
D:Data Set
Set
Release
Save
Read
B.1008
B.1009
B.2004
B.1008
B.2004
B.1009

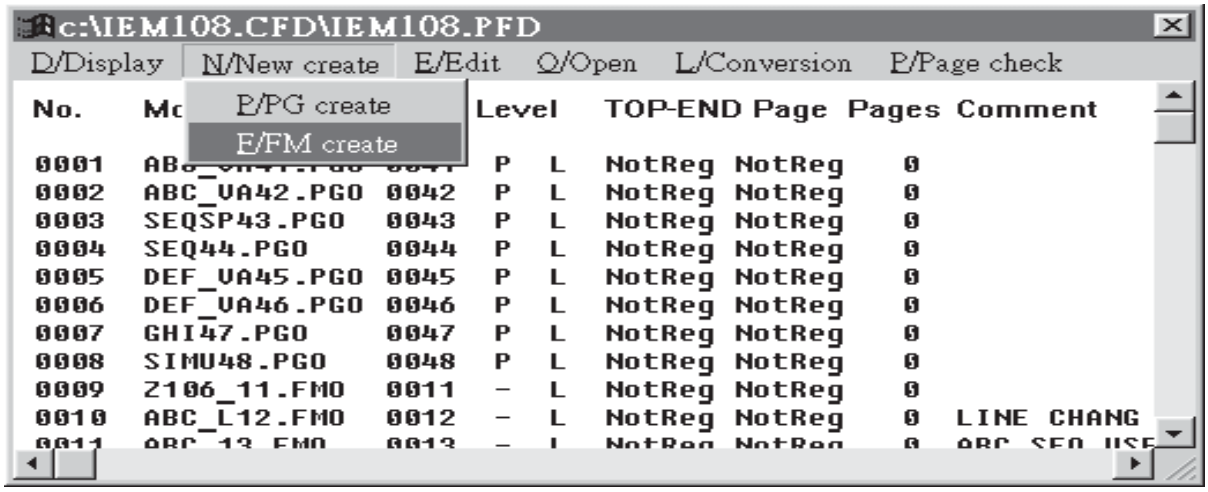
FM & PG PROGRAM

1. First you need to select FM(Function module) (XX.FMO) or PG (XX.PGO) use FUNCTION BLOCK and LADDER (representative format) to write the program.
   FM need to be call by GP1,GP2,but PG do' t need.
2. After finish writeing program it need to do MACHINE LANGUAGE CONVER and make sure no error.
3. Next step LOADING the program AES(PC) to ACS-2000 comunication by FL-net or TCP/IP(EtherNet)

From C:\XXXXX.CFD\XXXXX.PFD double clicl PFD,and create new program.

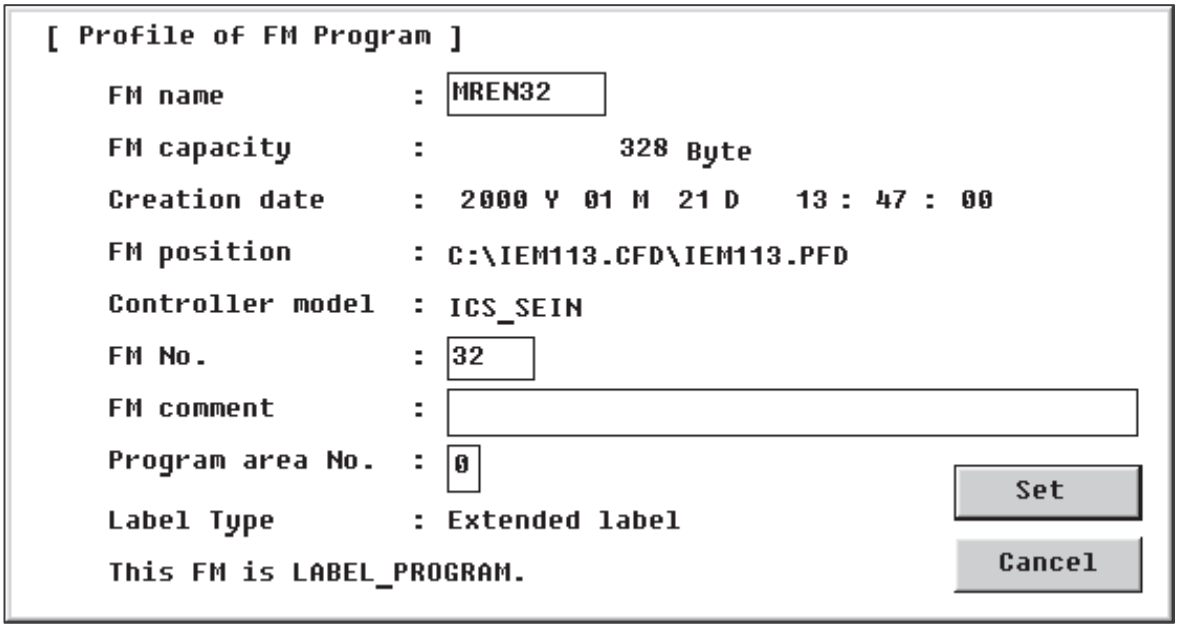


Choose FM (XX.FMO) or PG (XX.PGO),usually choose FM FOR ACS_EI TYPE.



Select toolbar F/File ➔ B/PROFILE ,then will display the following figure.
Assign FM NAME,FM NO (PG NAME,PG NO) ,FM program need to define GP1,or GP2 and describe in the following.

THE RANGE OF PG & FM NUMBER



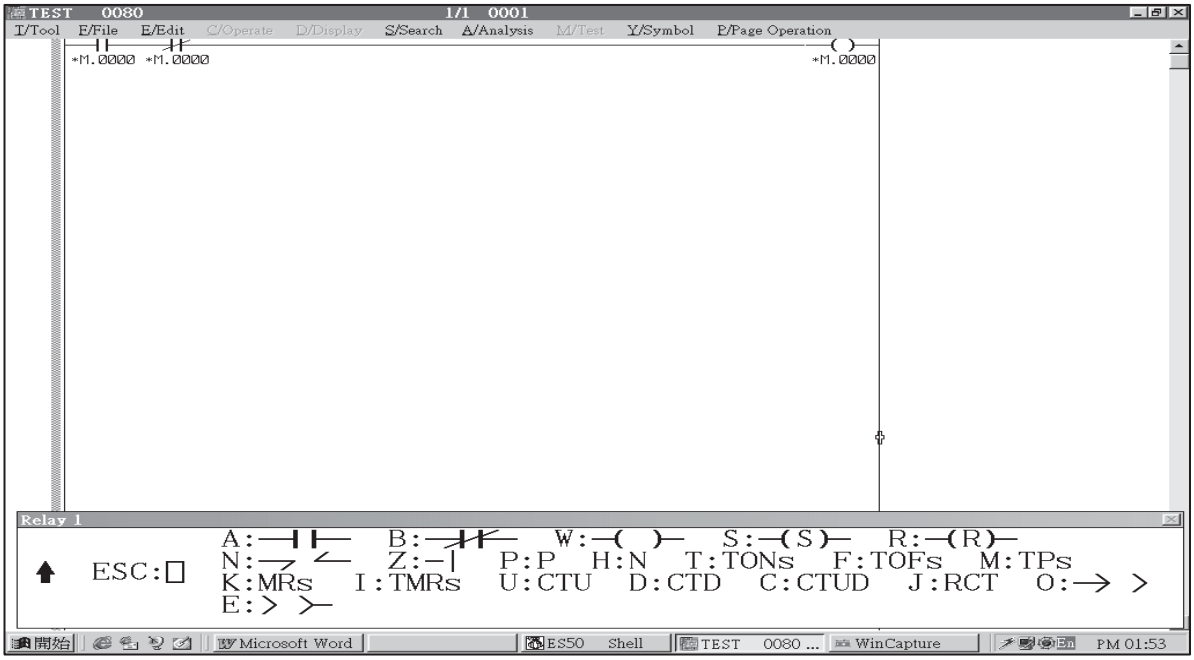
PG : LEVEL priority 0 > 1 > 2 > 3 > 4 > P
LEVEL 0 : system use

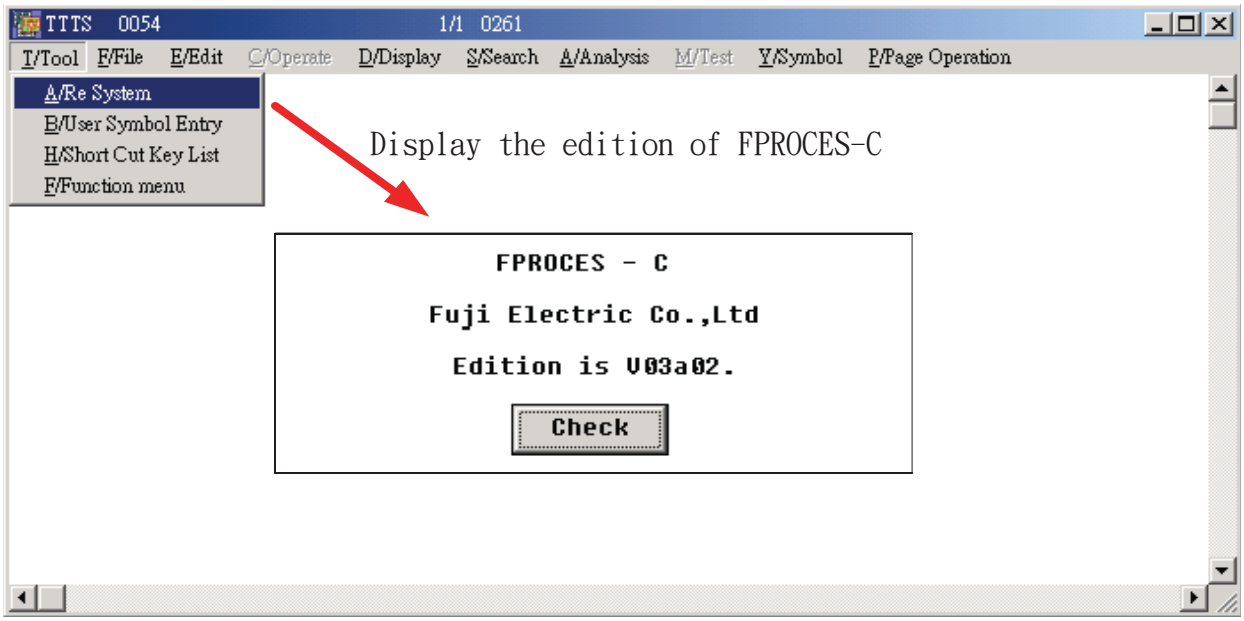
Level	E-Type	EI-Type
P	0 ~ 49(100ms)	34 ~ 49 (100ms)
4	50 ~ 81(10ms)	74 ~ 81 (10ms)
3	82 ~ 89(10ms)	88 ~ 89 (10ms)
2	90 ~ 94(10ms)	
1	95 ~ 99(1ms)	

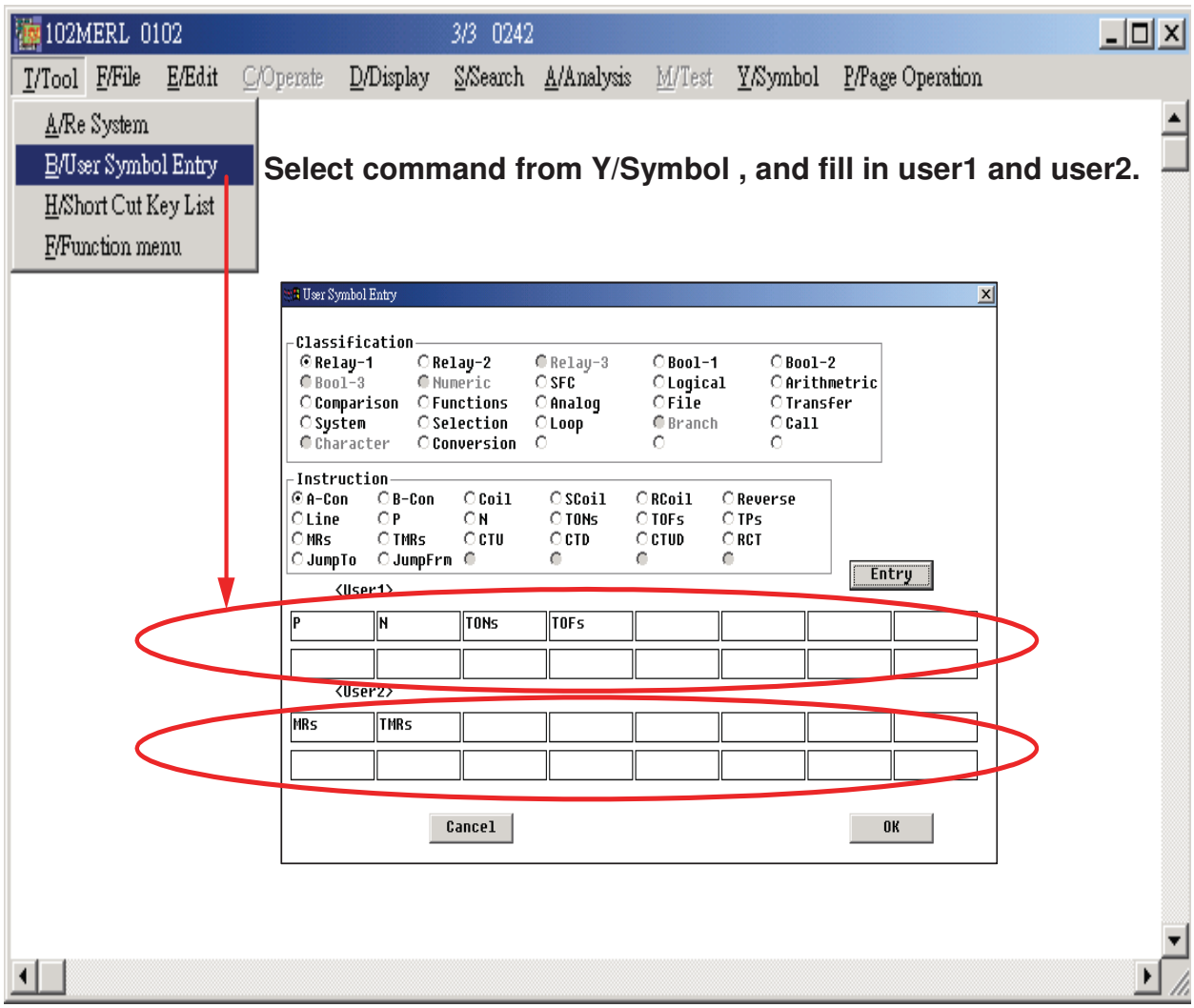
P level : Cyclic Program
1 - 4 Level : Fixed Period Interrupt Program
FM : may use the number of USER range,otherwise you may cover WAFER or SYSTEM program.

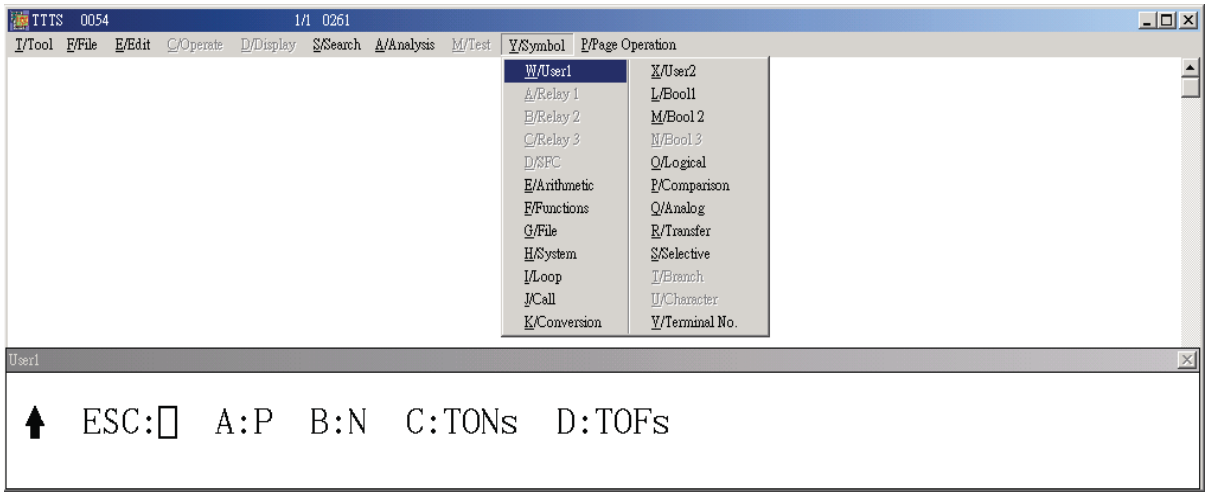
User	1 ~ 383
Wafer	384 ~ 447
User	448 ~ 479
System	480 ~ 1024

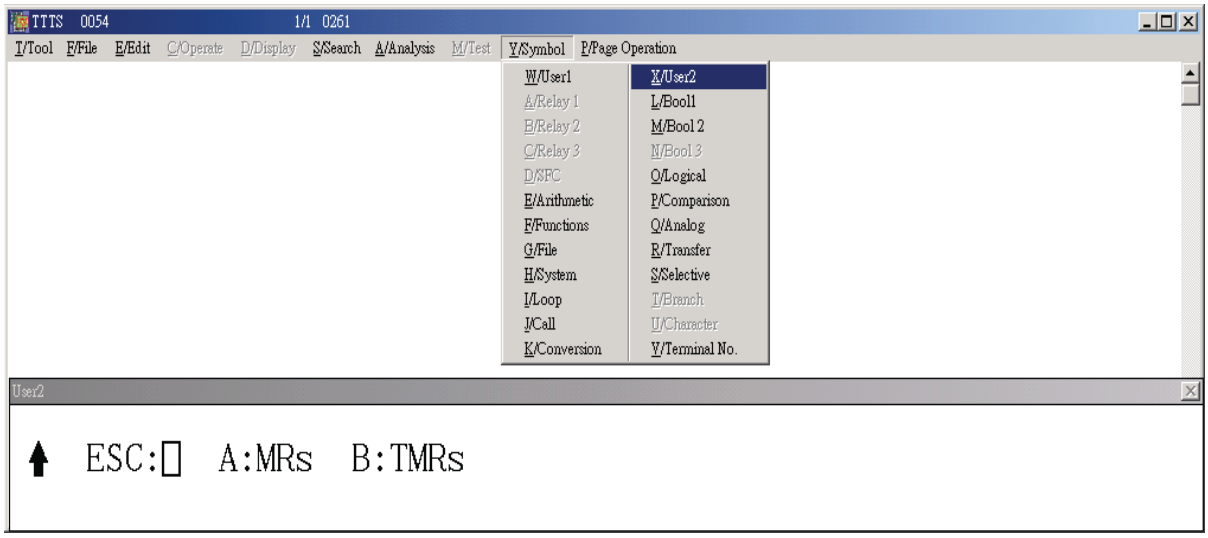
Start to write program,and SAVE before you close the program.
Function block ,and LADDER format,please refer to ACS-2500 EI-WAFER SYSTEM.

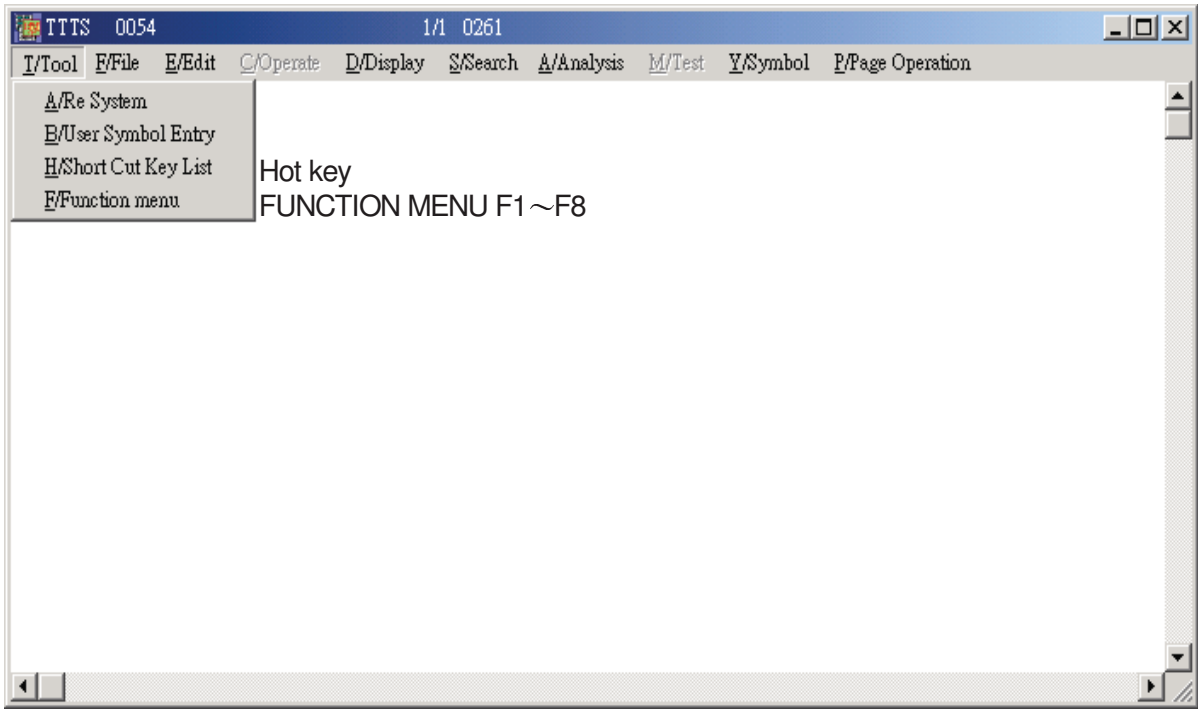


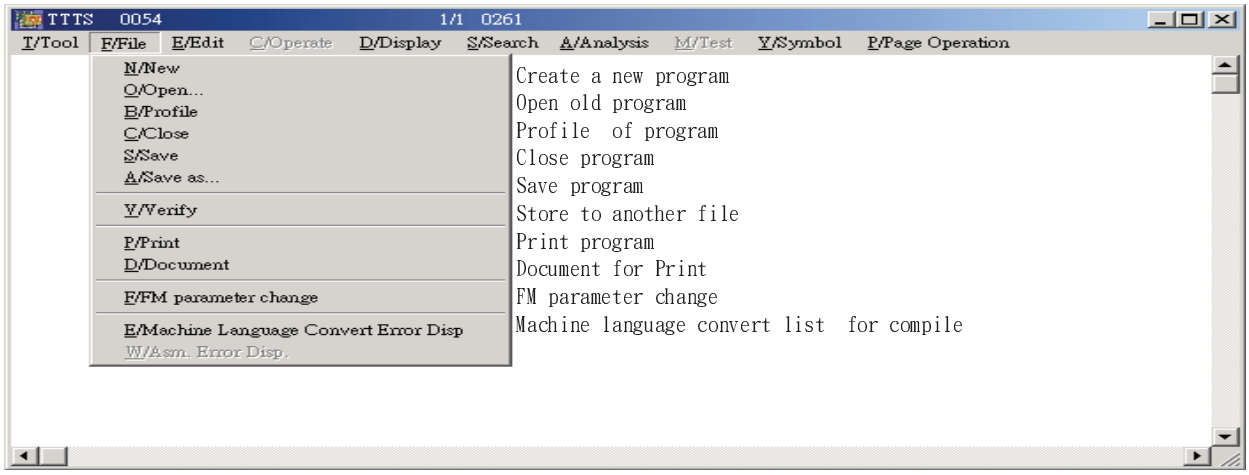

TTTS 0054 1/1 0261
T/Tool F/File E/Edit C/Operate D/Display S/Search A/Analysis M/Test Y/Symbol P/Page Operation
A/Re System
B/User Symbol Entry
H/Short Cut Key List
F/Function menu
Display the edition of FPROCES-C
FPROCES - C
Fuji Electric Co.,Ltd
Edition is V03a02.
Check

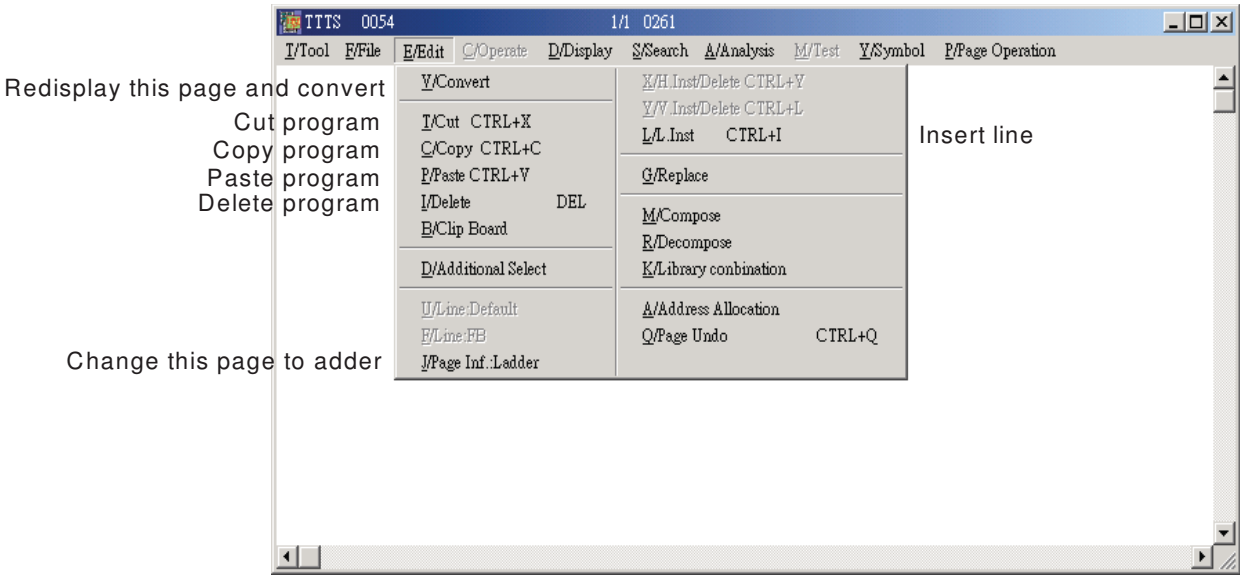

102MERL 0102 3/3 0242
T/Tool F/File E/Edit C/Operate D/Display S/Search A/Analysis M/Test Y/Symbol P/Page Operation
A/Re System
B/User Symbol Entry
H/Short Cut Key List
F/Function menu
Select command from Y/Symbol , and fill in user1 and user2.
User Symbol Entry
Classification
Relay-1 Relay-2 Relay-3 Bool-1 Bool-2
Bool-3 Numeric SFC Logical Arithmetric
Comparison Functions Analog File Transfer
System Selection Loop Branch Call
Character Conversion
Instruction
A-Con B-Con Coil SCoil RCoil Reverse
Line P N TONs TOFs TPs
MRs TMRs CTU CTD CTUD RCT
JumpTo JumpFrm
Entry
<User1>
P N TONs TOFs
<User2>
MRs TMRs
Cancel
OK

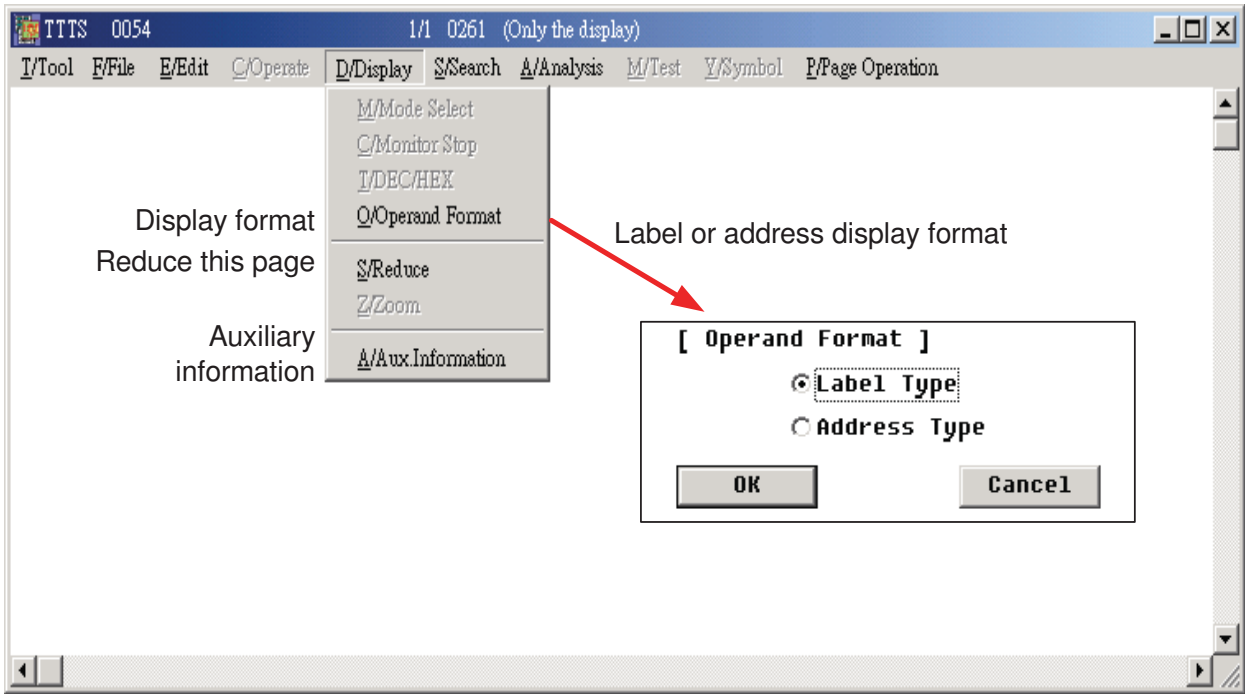

TTTS 0054 1/1 0261
T/Tool F/File E/Edit C/Operate D/Display S/Search A/Analysis M/Test Y/Symbol P/Page Operation
W/User1
A/Relay 1
B/Relay 2
C/Relay 3
D/SFC
E/Arithmetic
F/Functions
G/File
H/System
I/Loop
J/Call
K/Conversion
X/User2
L/Bool1
M/Bool 2
N/Bool 3
Q/Logical
P/Comparison
Q/Analog
R/Transfer
S/Selective
T/Branch
U/Character
V/Terminal No.
User1
ESC: A:P B:N C:TONs D:TOFs

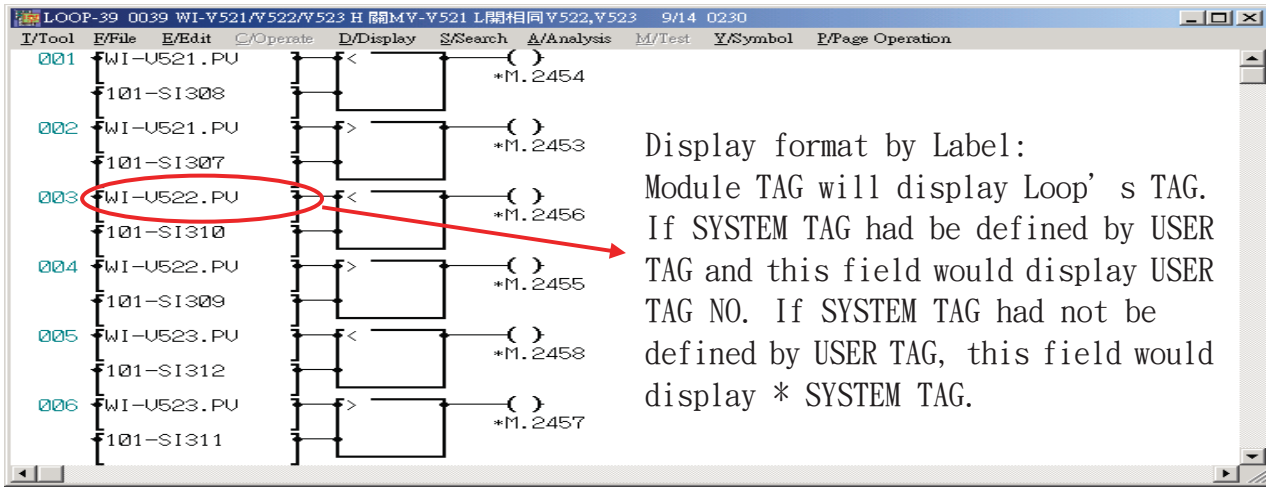

TTTS 0054 1/1 0261
T/Tool F/File E/Edit C/Operate D/Display S/Search A/Analysis M/Test Y/Symbol P/Page Operation
W/User1
A/Relay 1
B/Relay 2
C/Relay 3
D/SFC
E/Arithmetic
F/Functions
G/File
H/System
I/Loop
J/Call
K/Conversion
X/User2
L/Bool1
M/Bool 2
N/Bool 3
Q/Logical
P/Comparison
Q/Analog
R/Transfer
S/Selective
T/Branch
U/Character
V/Terminal No.
User2
ESC: A:MRs B:TMRs

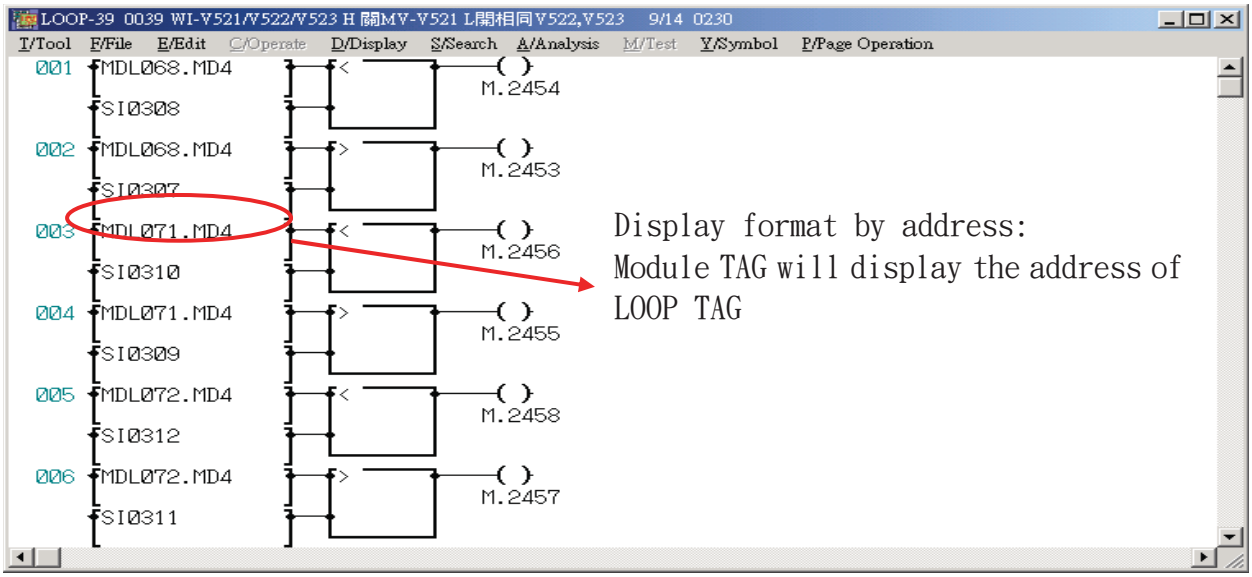

TTTS 0054 1/1 0261
T/Tool F/File E/Edit C/Operate D/Display S/Search A/Analysis M/Test Y/Symbol P/Page Operation
A/Re System
B/User Symbol Entry
H/Short Cut Key List
F/Function menu
Hot key
FUNCTION MENU F1～F8

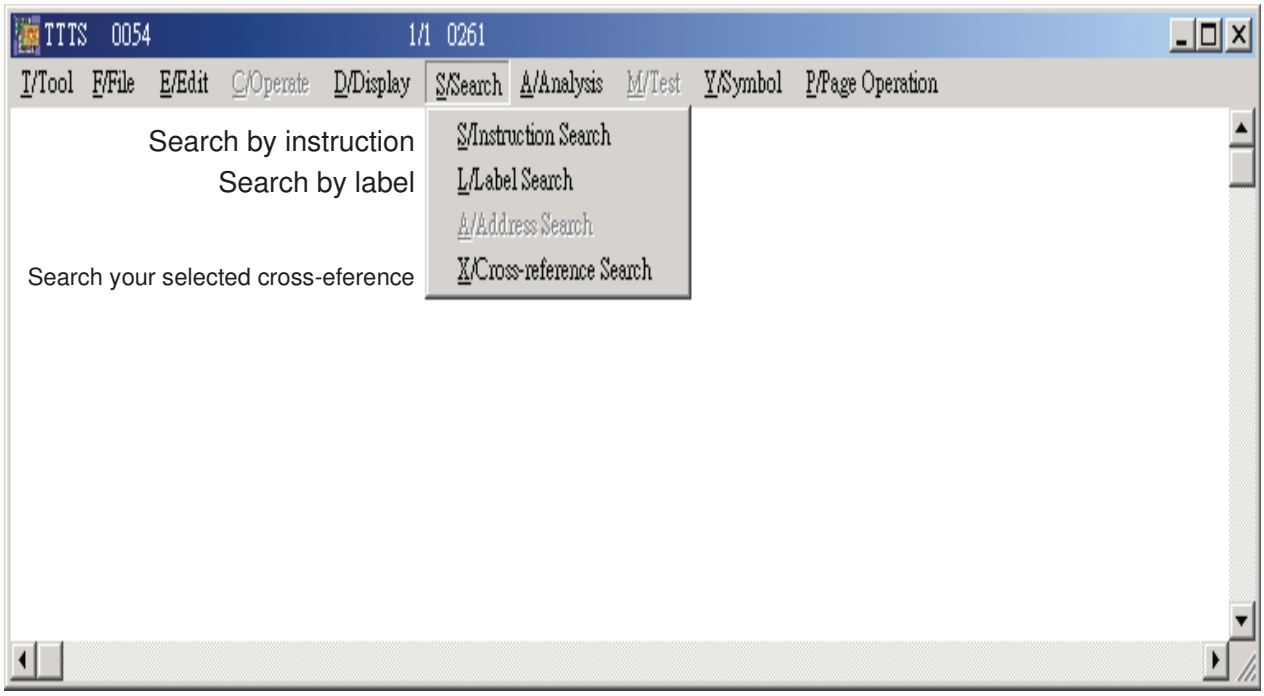

TTTS 0054 1/1 0261
T/Tool F/File E/Edit C/Operate D/Display S/Search A/Analysis M/Test Y/Symbol P/Page Operation
N/New
O/Open...
B/Profile
C/Close
S/Save
A/Save as...
V/Verify
P/Print
D/Document
F/FM parameter change
E/Machine Language Convert Error Disp
W/Asm. Error Disp.
Create a new program
Open old program
Profile of program
Close program
Save program
Store to another file
Print program
Document for Print
FM parameter change
Machine language convert list for compile

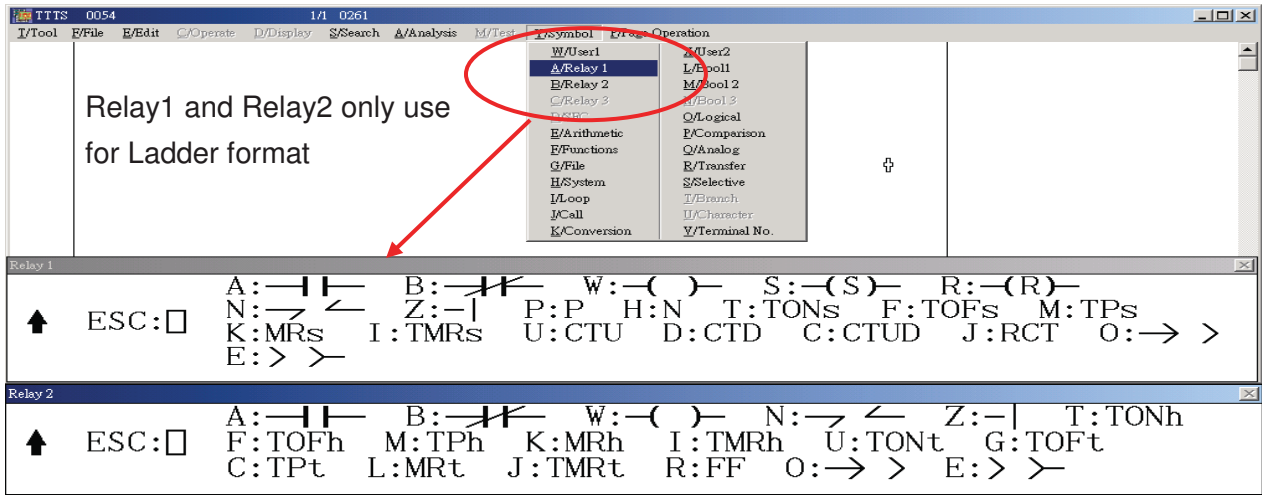

Redisplay this page and convert
Cut program
Copy program
Paste program
Delete program
Change this page to adder
Insert line
TTTS 0054 1/1 0261
T/Tool F/File E/Edit C/Operate D/Display S/Search A/Analysis M/Test Y/Symbol P/Page Operation
V/Convert
T/Cut CTRL+X
C/Copy CTRL+C
P/Paste CTRL+V
I/Delete DEL
B/Clip Board
D/Additional Select
U/Line:Default
F/Line:FB
J/Page Inf.:Ladder
K/H.Inst/Delete CTRL+Y
Y/V.Inst/Delete CTRL+L
L/L.Inst CTRL+I
G/Replace
M/Compose
R/Decompose
K/Library conbination
A/Address Allocation
Q/Page Undo CTRL+Q

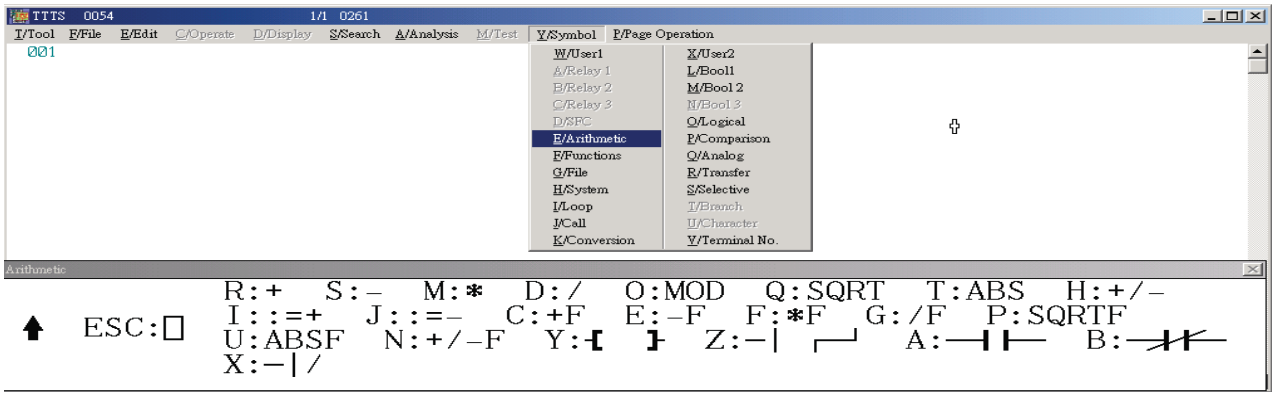

TTTS 0054 1/1 0261 (Only the display)
T/Tool F/File E/Edit C/Operate D/Display S/Search A/Analysis M/Test Y/Symbol P/Page Operation
Display format
Reduce this page
Auxiliary information
M/Mode Select
C/Monitor Stop
T/DEC/HEX
O/Operand Format
S/Reduce
Z/Zoom
A/Aux.Information
Label or address display format
[Operand Format]
Label Type
Address Type
OK
Cancel

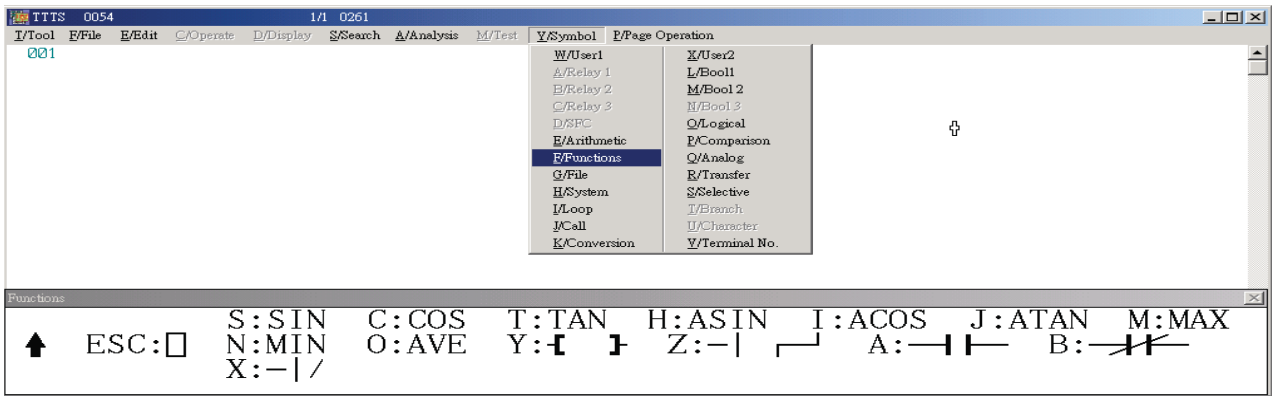

LOOP-39 0039 WI-V521/V522/V523 H 開MV-V521 L開相同V522,V523 9/14 0230
T/Tool F/File E/Edit C/Operate D/Display S/Search A/Analysis M/Test Y/Symbol P/Page Operation
001 WI-U521.PU
101-SI308
*M.2454
002 WI-U521.PU
101-SI307
*M.2453
003 WI-U522.PU
101-SI310
*M.2456
004 WI-U522.PU
101-SI309
*M.2455
005 WI-U523.PU
101-SI312
*M.2458
006 WI-U523.PU
101-SI311
*M.2457
Display format by Label:
Module TAG will display Loop's TAG. If SYSTEM TAG had be defined by USER TAG and this field would display USER TAG NO. If SYSTEM TAG had not be defined by USER TAG, this field would display * SYSTEM TAG.

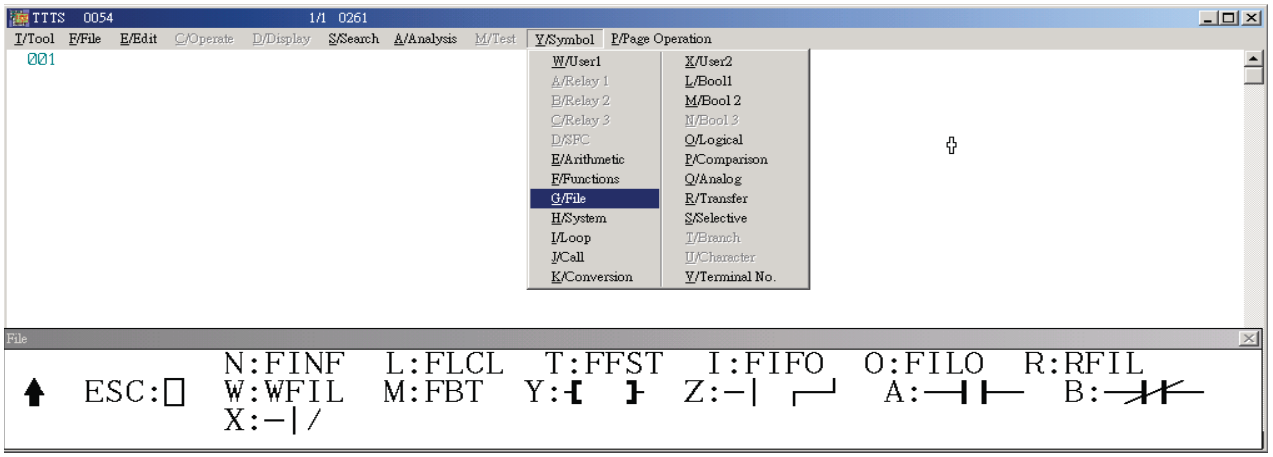

LOOP-39 0039 WI-V521/V522/V523 H 開MV-V521 L開相同V522,V523 9/14 0230
T/Tool F/File E/Edit C/Operate D/Display S/Search A/Analysis M/Test Y/Symbol P/Page Operation
001 MDL068.MD4
SI0308
M.2454
002 MDL068.MD4
SI0307
M.2453
003 MDL071.MD4
SI0310
M.2456
004 MDL071.MD4
SI0309
M.2455
005 MDL072.MD4
SI0312
M.2458
006 MDL072.MD4
SI0311
M.2457
Display format by address:
Module TAG will display the address of LOOP TAG

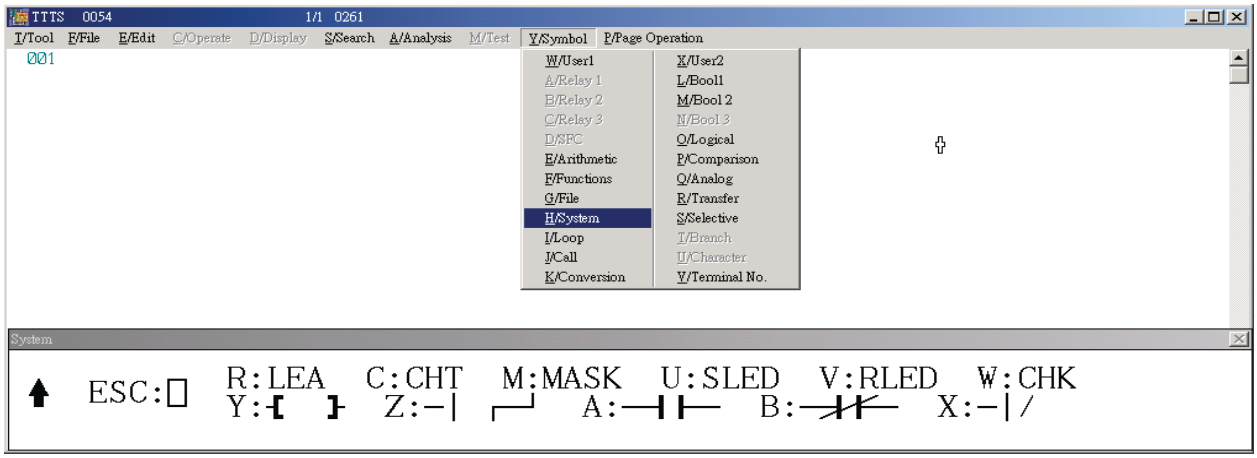

TTTS 0054 1/1 0261
T/Tool F/File E/Edit C/Operate D/Display S/Search A/Analysis M/Test Y/Symbol P/Page Operation
S/Instruction Search
L/Label Search
A/Address Search
X/Cross-reference Search
Search by instruction
Search by label
Search your selected cross-eference


Relay1 and Relay2 only use
for Ladder format
Relay 1
ESC:□
A: B: W: S:(S) R:(R)
N: Z:-| P:P H:N T:TONs F:TOFs M:TPs
K:MRs I:TMRs U:CTU D:CTD C:CTUD J:RCT O:→ >
E:> >
Relay 2
ESC:□
A: B: W: N: Z:-| T:TONh
F:TOFh M:TPh K:MRh I:TMRh U:TONt G:TOFt
C:TPt L:MRt J:TMRt R:FF O:→ > E:> >


Arithmetic
ESC:□
R:+ S:- M:* D:/ O:MOD Q:SQRT T:ABS H:+/-
I::=+ J::=- C:+F E:-F F:*F G:/F P:SQRTF
U:ABSF N:+/-F Y:[ ] Z:-| A: B:
X:-|/


Functions
ESC:□
S:SIN C:COS T:TAN H:ASIN I:ACOS J:ATAN M:MAX
N:MIN O:AVE Y:[ ] Z:-| A: B:
X:-|/


File
ESC:□
N:FINF L:FLCL T:FFST I:FIFO O:FILO R:RFIL
W:WFIL M:FBT Y:[ ] Z:-| A: B:
X:-|/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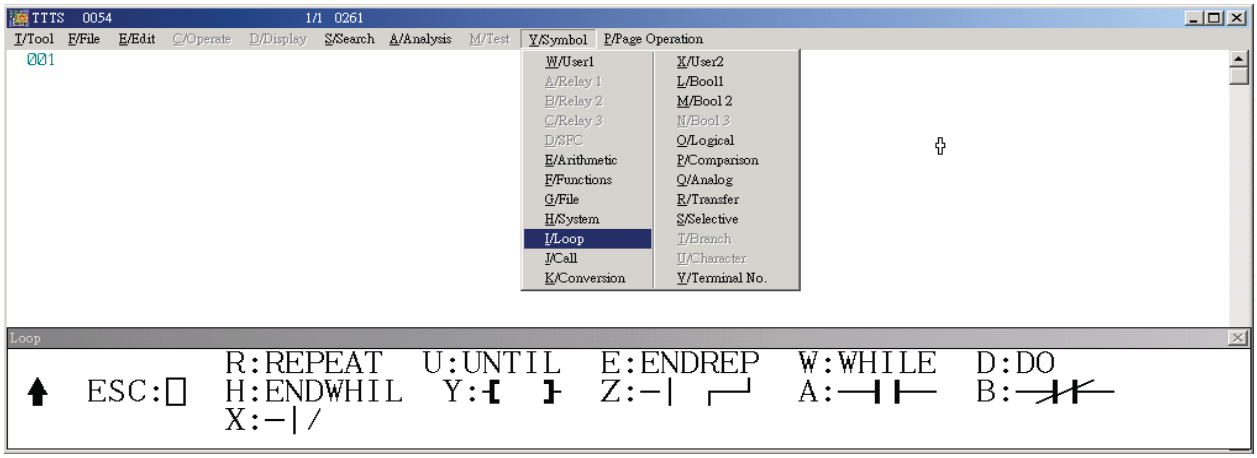
T/Tool F/File E/Edit C/Operate D/Display S/Search A/Analysis M/Test Y/Symbol P/Page Operation
001
W/User1 X/User2
A/Relay 1 L/Bool1
B/Relay 2 M/Bool 2
C/Relay 3 N/Bool 3
D/SFC O/Logical
E/Arithmetic P/Comparison
F/Functions Q/Analog
G/File R/Transfer
H/System S/Selective
I/Loop T/Branch
J/Call U/Character
K/Conversion V/Terminal No.
System
ESC:□ R:LEA C:CHT M:MASK U:SLED V:RLED W:CHK
Y:[ ] Z:-| A:-| |- B:-|/|- X:-|/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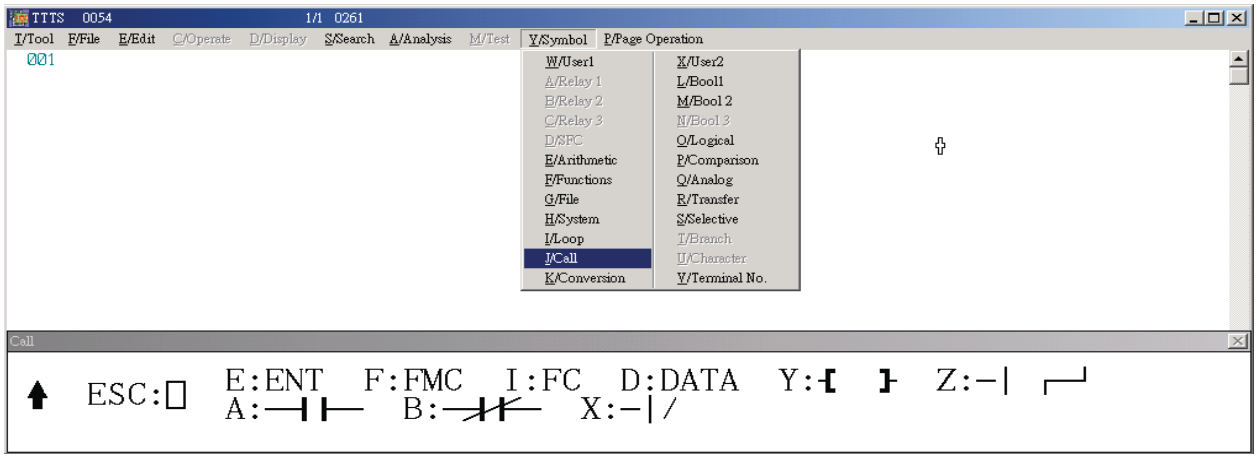
001
W/User1 X/User2
A/Relay 1 L/Bool1
B/Relay 2 M/Bool 2
C/Relay 3 N/Bool 3
D/SFC O/Logical
E/Arithmetic P/Comparison
F/Functions Q/Analog
G/File R/Transfer
H/System S/Selective
I/Loop T/Branch
J/Call U/Character
K/Conversion V/Terminal No.
Loop
ESC:□ R:REPEAT U:UNTIL E:ENDREP W:WHILE D:DO
H:ENDWHIL Y:[ ] Z:-| A:-| |- B:-|/|-
X:-|/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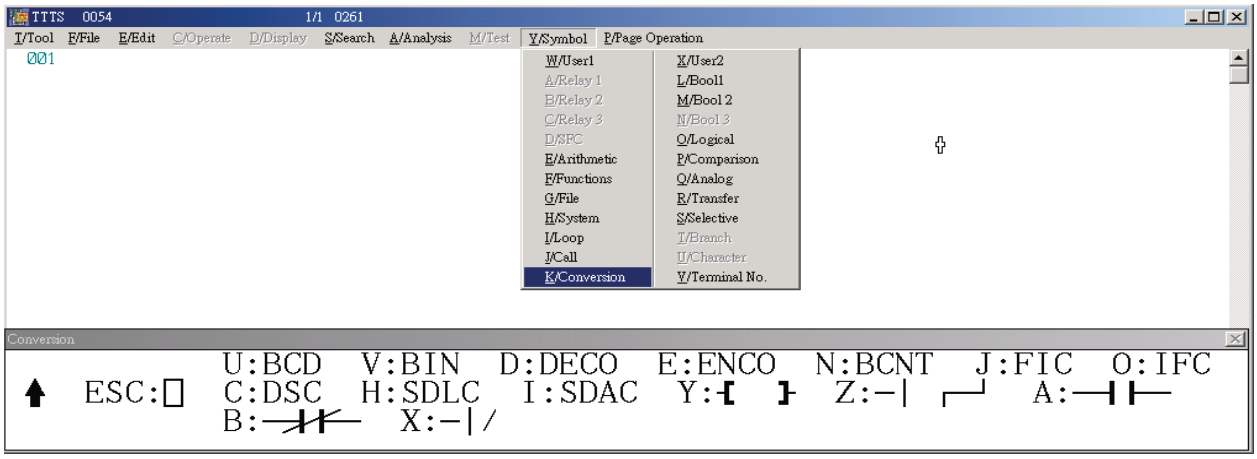
T/Tool F/File E/Edit C/Operate D/Display S/Search A/Analysis M/Test Y/Symbol P/Page Operation
001
W/User1 X/User2
A/Relay 1 L/Bool1
B/Relay 2 M/Bool 2
C/Relay 3 N/Bool 3
D/SFC O/Logical
E/Arithmetic P/Comparison
F/Functions Q/Analog
G/File R/Transfer
H/System S/Selective
I/Loop T/Branch
J/Call U/Character
K/Conversion V/Terminal No.
Call
ESC:□ E:ENT F:FMC I:FC D:DATA Y:[ ] Z:-|
A:-| |- B:-|/|- X:-|/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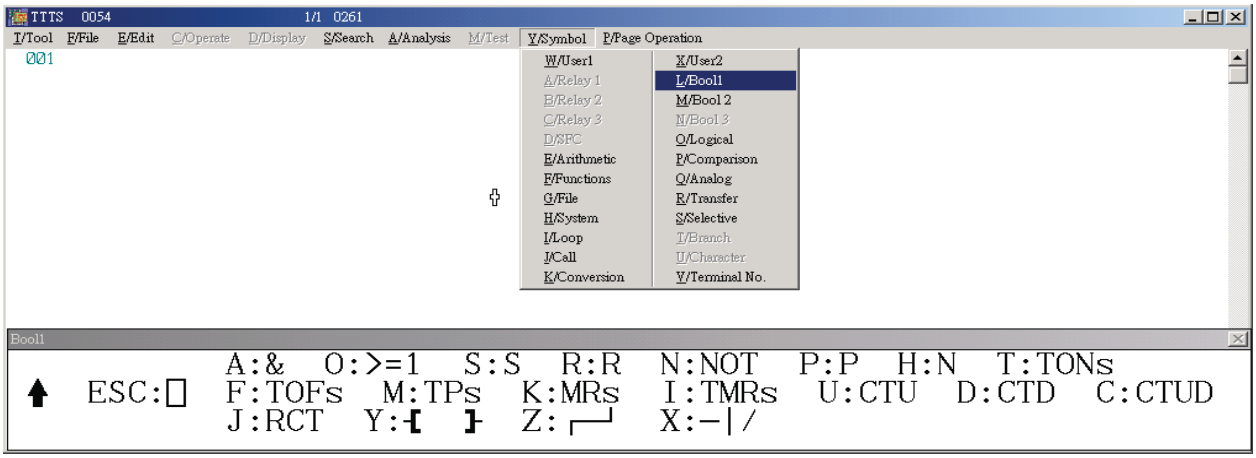
A/Relay 1 L/Bool1
B/Relay 2 M/Bool 2
C/Relay 3 N/Bool 3
D/SFC O/Logical
E/Arithmetic P/Comparison
F/Functions Q/Analog
G/File R/Transfer
H/System S/Selective
I/Loop T/Branch
J/Call U/Character
K/Conversion V/Terminal No.
Conversion
ESC:□ U:BCD V:BIN D:DECO E:ENCO N:BCNT J:FIC O:IFC
C:DSC H:SDLC I:SDAC Y:[ ] Z:-| A:-| |-
B:-|/|- X:-|/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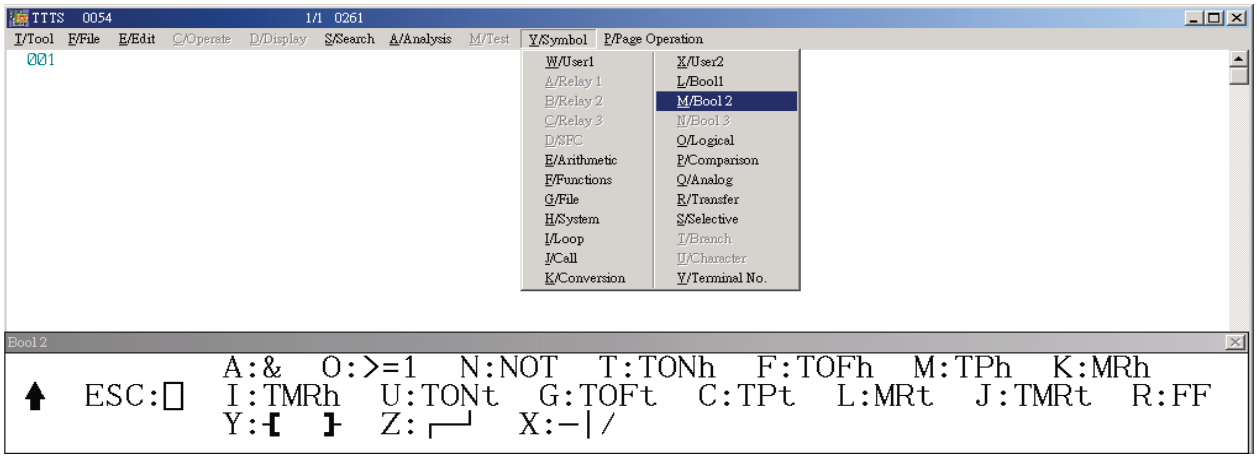
B/Relay 2 M/Bool 2
C/Relay 3 N/Bool 3
D/SFC O/Logical
E/Arithmetic P/Comparison
F/Functions Q/Analog
G/File R/Transfer
H/System S/Selective
I/Loop T/Branch
J/Call U/Character
K/Conversion V/Terminal No.
Bool1
ESC:
A:& O:>=1 S:S R:R N:NOT P:P H:N T:TONs
F:TOFs M:TPs K:MRs I:TMRs U:CTU D:CTD C:CTUD
J:RCT Y:[ ] Z: X:-|/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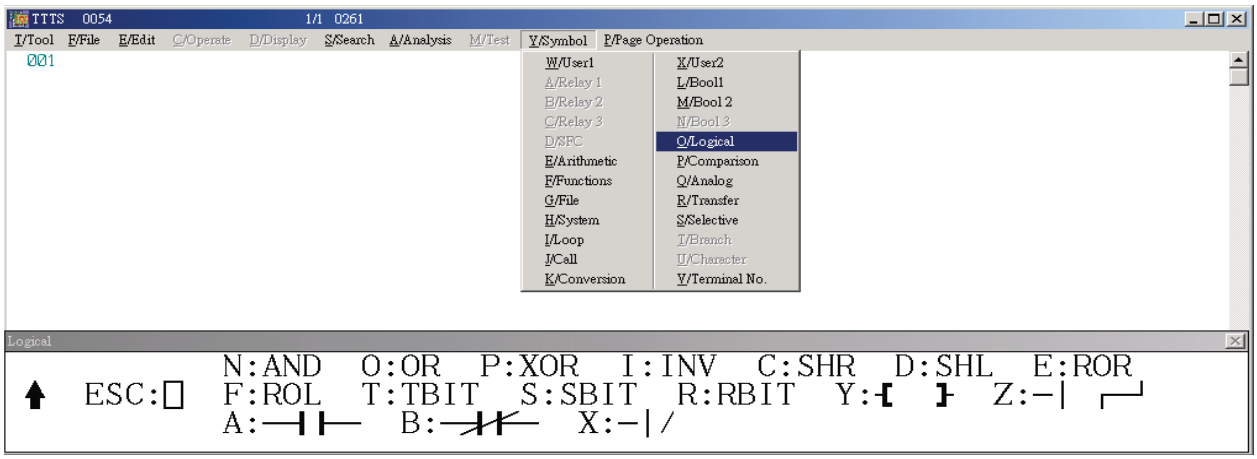
A/Relay 1 L/Bool1
B/Relay 2 M/Bool 2
C/Relay 3 N/Bool 3
D/SFC O/Logical
E/Arithmetic P/Comparison
F/Functions Q/Analog
G/File R/Transfer
H/System S/Selective
I/Loop T/Branch
J/Call U/Character
K/Conversion V/Terminal No.
Bool 2
ESC:
A:& O:>=1 N:NOT T:TONh F:TOFh M:TPh K:MRh
I:TMRh U:TONt G:TOFt C:TPt L:MRt J:TMRt R:FF
Y:[ ] Z: X:-|/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A/Relay 1 L/Bool1
B/Relay 2 M/Bool 2
C/Relay 3 N/Bool 3
D/SFC O/Logical
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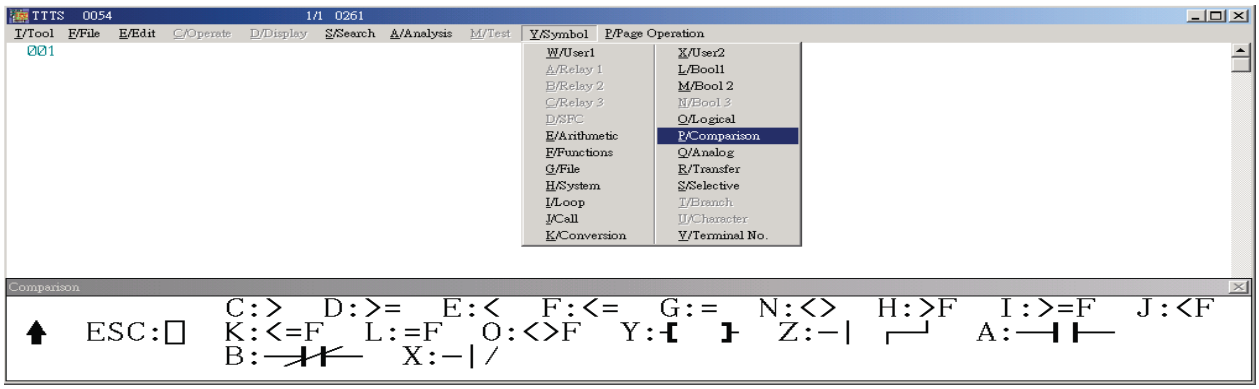
F/Functions Q/Analog
G/File R/Transfer
H/System S/Selective
I/Loop T/Branch
J/Call U/Character
K/Conversion V/Terminal No.
Logical
ESC:
N:AND O:OR P:XOR I:INV C:SHR D:SHL E:ROR
F:ROL T:TBIT S:SBIT R:RBIT Y:[ ] Z:-| |
A:-| |- B:-|/|- X:-|/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W/User1 X/User2
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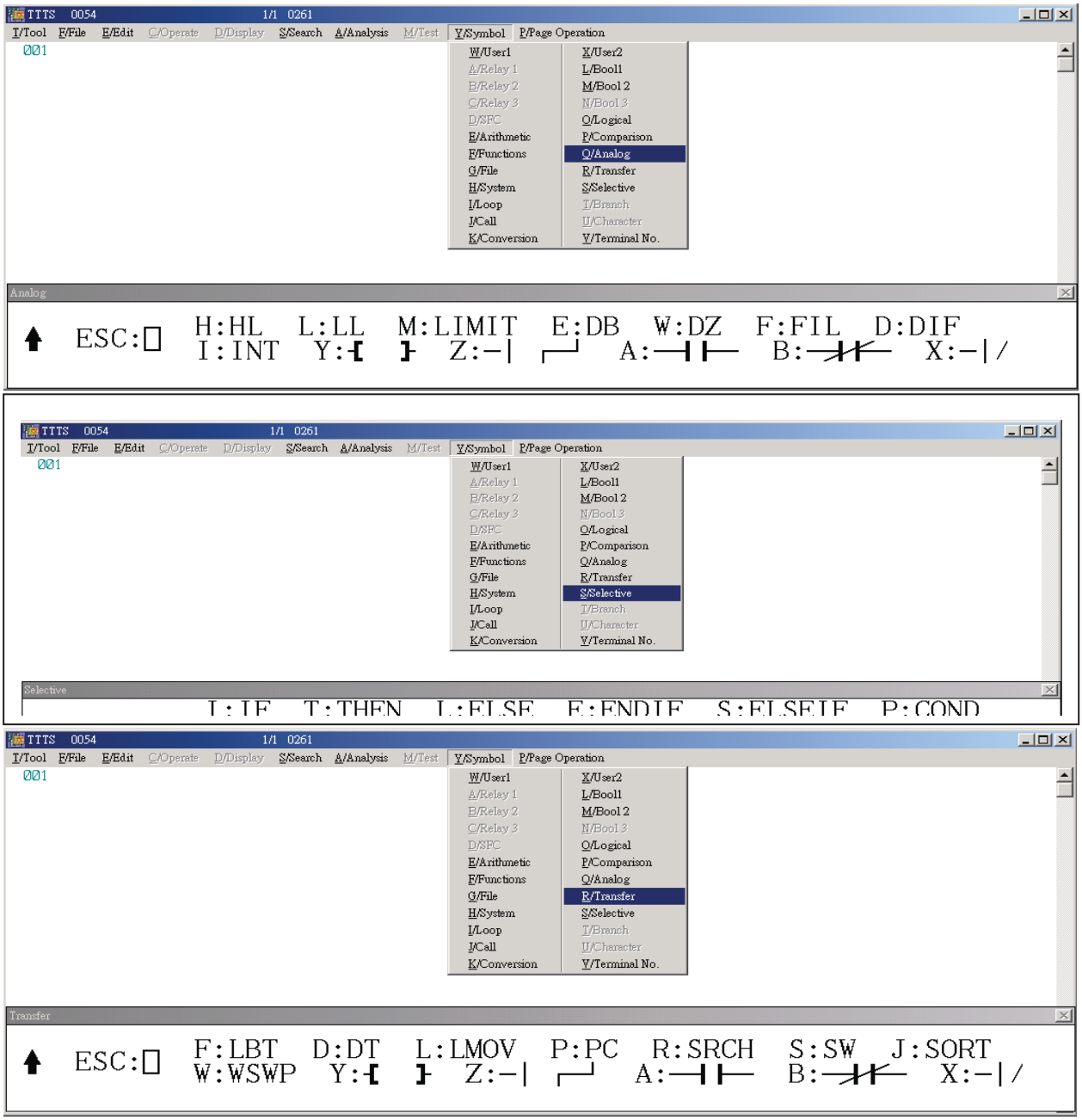
B/Relay 2 M/Bool 2
C/Relay 3 N/Bool 3
D/SFC O/Logical
E/Arithmetic P/Comparison
F/Functions Q/Analog
G/File R/Transfer
H/System S/Selective
I/Loop T/Branch
J/Call U/Character
K/Conversion V/Terminal No.
Comparison
ESC:
C:> D:>= E:< F:<= G:= N:<> H:>F I:>=F J:<F
K:<=F L:=F O:<>F Y:[ ] Z:-| | A:-| |-
B:-|/|- X:-|/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T/Tool F/File E/Edit C/Operate D/Display S/Search A/Analysis M/Test Y/Symbol P/Page Operation
001
W/User1 X/User2
A/Relay 1 L/Bool1
B/Relay 2 M/Bool 2
C/Relay 3 N/Bool 3
D/SFC O/Logical
E/Arithmetic P/Comparison
F/Functions Q/Analog
G/File R/Transfer
H/System S/Selective
I/Loop T/Branch
J/Call U/Character
K/Conversion V/Terminal No.
Analog
ESC:
H:HL L:LL M:LIMIT E:DB W:DZ F:FIL D:DIF
I:INT Y: Z:-| A: B: X:-|/
TTTS 0054 1/1 0261
T/Tool F/File E/Edit C/Operate D/Display S/Search A/Analysis M/Test Y/Symbol P/Page Operation
001
W/User1 X/User2
A/Relay 1 L/Bool1
B/Relay 2 M/Bool 2
C/Relay 3 N/Bool 3
D/SFC O/Logical
E/Arithmetic P/Comparison
F/Functions Q/Analog
G/File R/Transfer
H/System S/Selective
I/Loop T/Branch
J/Call U/Character
K/Conversion V/Terminal No.
Selective
I:IF T:THEN L:ELSE E:ENDIF S:ELSEIF P:COND
TTTS 0054 1/1 0261
T/Tool F/File E/Edit C/Operate D/Display S/Search A/Analysis M/Test Y/Symbol P/Page Operation
001
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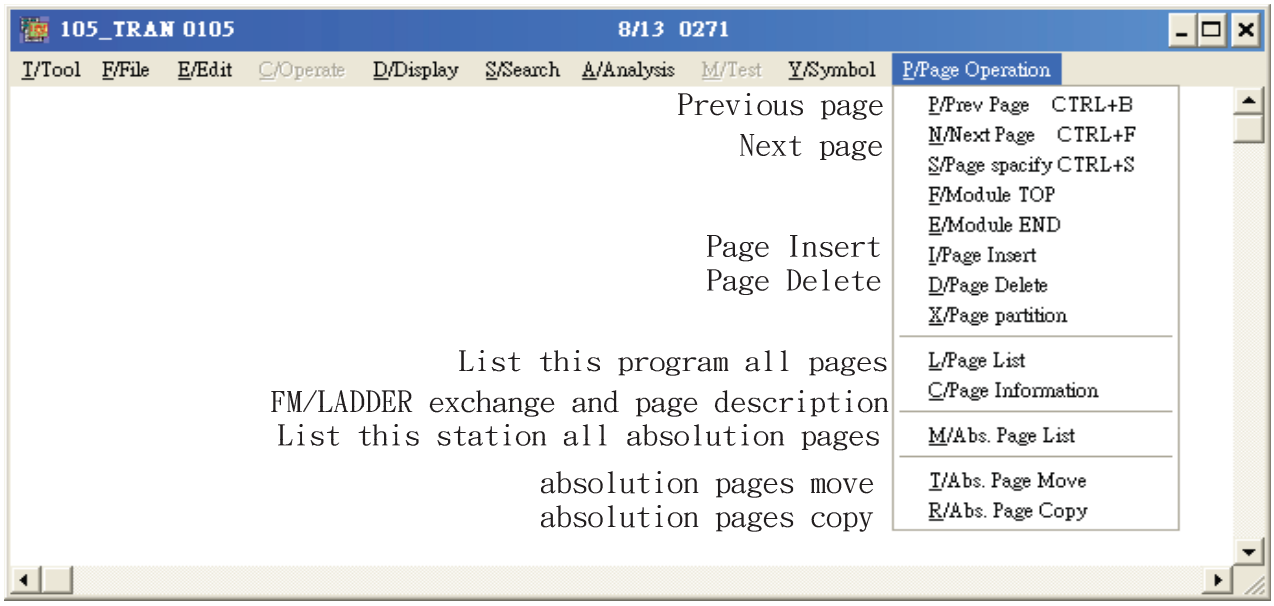
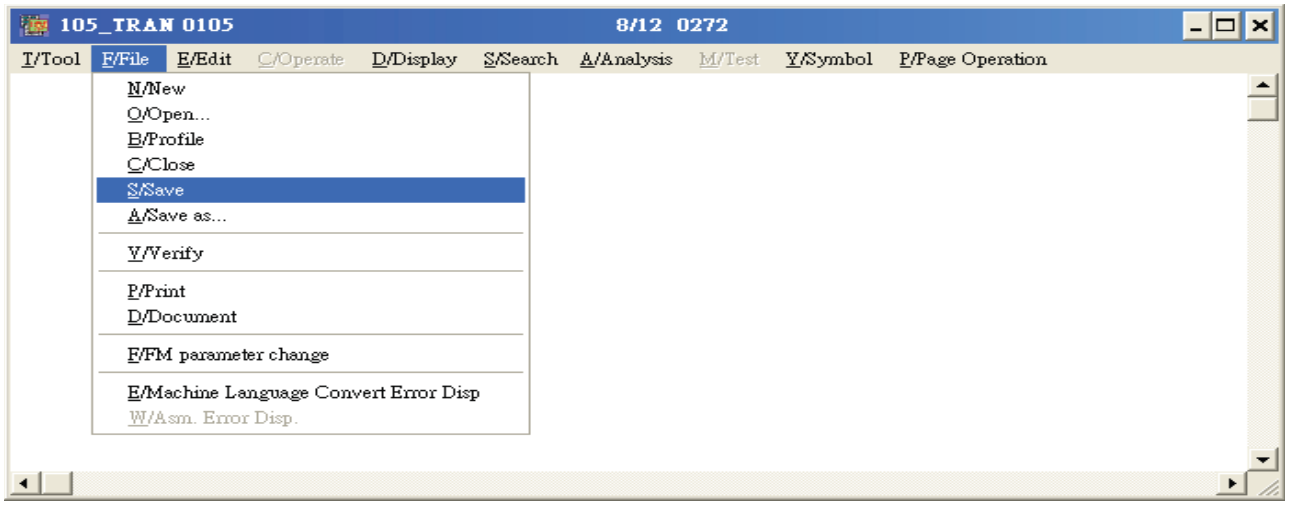
B/Relay 2 M/Bool 2
C/Relay 3 N/Bool 3
D/SFC O/Logical
E/Arithmetic P/Comparison
F/Functions Q/Analog
G/File R/Transfer
H/System S/Selective
I/Loop T/Branch
J/Call U/Character
K/Conversion V/Terminal No.
Transfer
ESC:
F:LBT D:DT L:LMOV P:PC R:SRCH S:SW J:SORT
W:WSWP Y: Z:-| A: B: X:-|/

Page change/Insert page/FM and LADDER exchange/Page Description



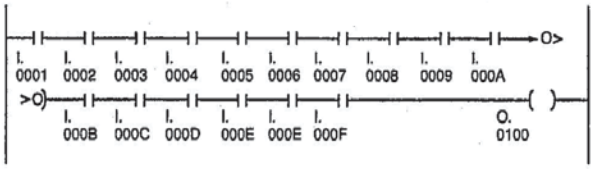
Before you exit this program, please save this program.



Exit this program

[1]	NO contact, NC contact, Output coil	Functions
Symbol	—| |— : NO Contact I. 0000 —|/|— : NC Contact I. 0001 —( )— : Output coil O. 0160	① Indicates the NO contact, NC contact, and output coil. A symbol, an identifier, and an address function as a contact or an output coil. ② If the contacts of the same address are used or if output coils are used as contacts, the number of contacts employed is not limited. ③ Limited item of one circuit. Within 192 words.

Symbol	Name

[2]	Returning	Functions
Symbol	—→N> : Return source >N>— : Return destination (N: 0~9)	① A series circuit exceeding 11 contacts can be returned by assigning match numbers. ② Ten match numbers (N) 0 to 9 can be used. If the number exceeds 9, it is used from 0 again on another page, and it can be used repeatedly by optional times. [Example] 

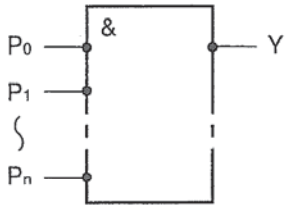
Symbol	Name

Note) For returning the sequence circuit, set the numbers of the return source and return destination to be 1-to-n or n-to-1 in one circuit.

If a 1-to-n circuit or an n-to-1 circuit is prepared, a "breakage" failure occurs in case of "conversion", "write", etc.

[3]	AND	&	Functions

Symbol

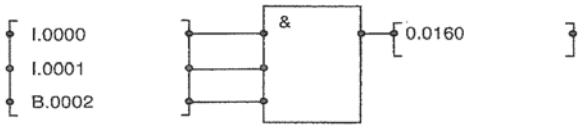


Symbol	Name
Y	Output
P0	Input
⟆	⟆
Pn	Input

• $P_0 \wedge P_1 \wedge$ ------ $\wedge P_n \rightarrow Y$

a. Ladder diagram

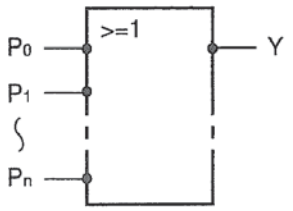
b. FB diagram



A circuit is limited by 10 contacts + 1 coil in the serial direction, and a serial circuit exceeding 10 contacts inclusive can be returned by assigning match numbers. (Ten match numbers 0 to 9 are applicable)

[4]	OR	> = 1	Functions

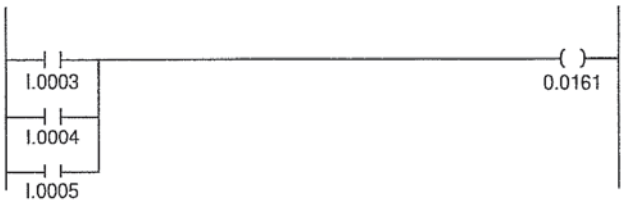
Symbol



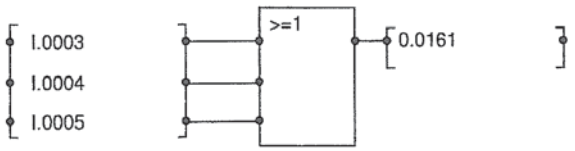
Symbol	Name
Y	Output
P0	Input
⟆	⟆
Pn	Input

• $P_0 \vee P_1 \vee$ ------ $\vee P_n \rightarrow Y$

a. Ladder diagram

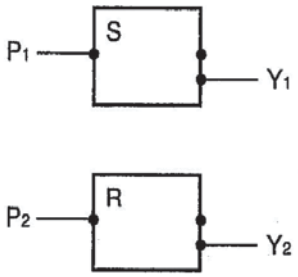


b. FB diagram



[5]	SET, RESET	S, R

Symbol



Symbol	Name
Y1	Set coil (output)
P1	Set input
Y2	Reset coil (output)
P2	Reset input

Functions

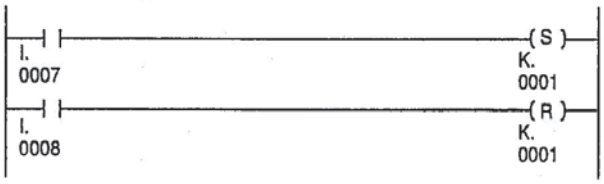
- When set input (P1) turns on, set coil (Y1) turns on, and this set coil (Y1) remains turned on even if set input (P1) turns off. The set status is reset when reset input (P2) is turned on.

① Assign the same address to a set of set coil and reset coil.

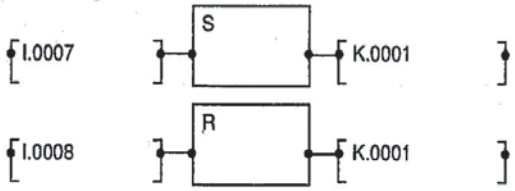
② Other circuits may exist between the set circuit and the reset circuit.

③ The set coil and reset coil of the same address can be assigned by optional times in a program.

a. Ladder diagram

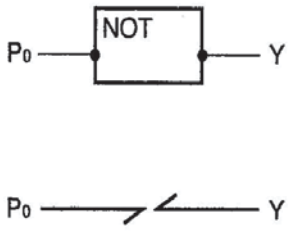


b. FB diagram



[6]	INVERSE	NOT

Symbol

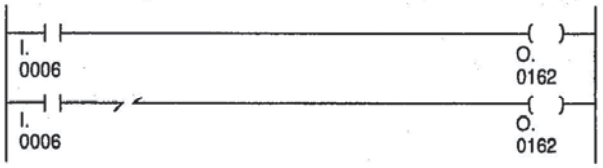


Symbol	Name
Y	Output
P0	Input

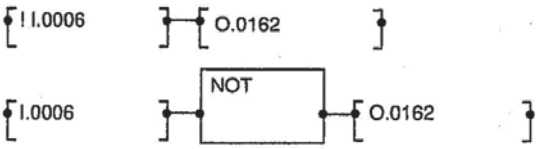
Functions

- Inverses a logical value of the logical circuit on the left side of the symbol, and transfers its result to the right side of the symbol.

a. Ladder diagram



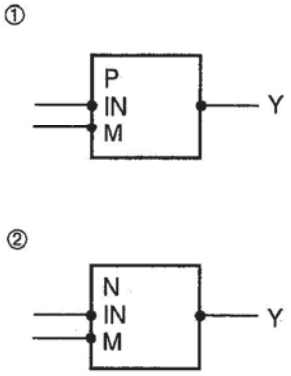
b. FB diagram



Note) Symbol "!" in FB diagram shows an NC contact.

[7]	① Rise-time differential ② Fall-time differential	P N	Functions

Symbol

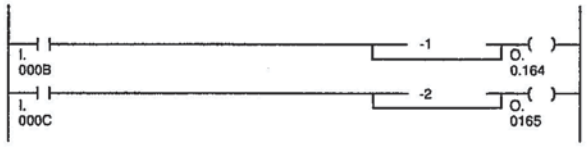


Symbol	Name
Y	Output
IN	Input
M	Work area for bit operation (0 ~ 1023)

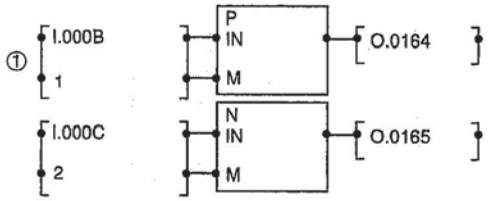
① Turns on an output by one scan only at the rise time (0 → 1) of an input signal.

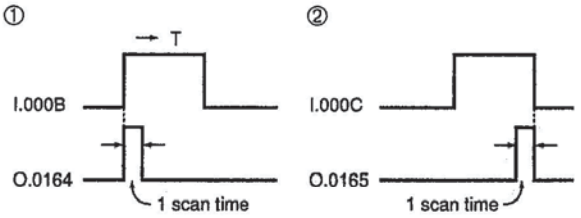
② Turns on an output by one scan only at the fall-time (1 → 0) of an input signal.

a. Ladder diagram



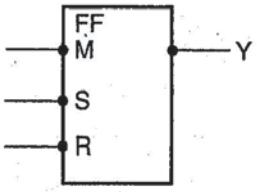
b. FB diagram





[8]	Flip-flop	FF	Functions

Symbol

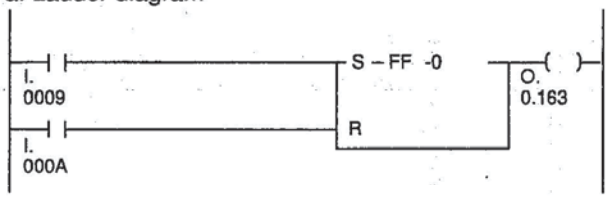


Symbol	Name
Y	Output
M	Work area for bit operation (0 ~ 1023)
S	Set input
R	Reset input

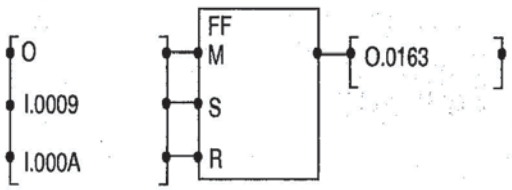
S	R	Y
0	0	Last value
0	1	0
1	0	1
1	1	0

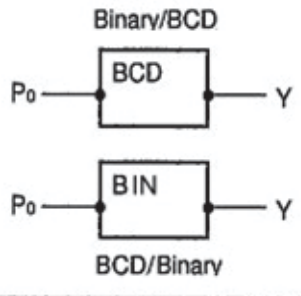
- Composes set and rest functions as a relay (Reset preferential circuit).

a. Ladder diagram



b. FB diagram



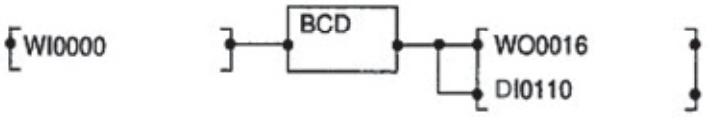
[1]	Binary/BCD conversion	BCD	Symbol	Name
Symbol			Y	Output
			Po	Input

Functions

- Converts binary data input (Po) into BCD, and loads its resutls into output (Y).

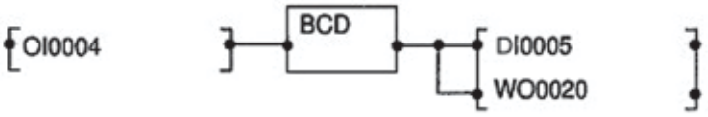
① Conversion from single-precision accuracy integer to BCD

If an output exceeds BCD 4 digits (7999 ~ F999), it is locked to the maximum value (or minimum value).



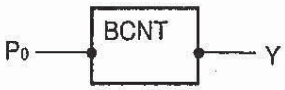
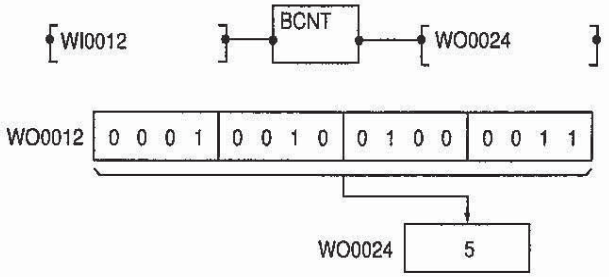
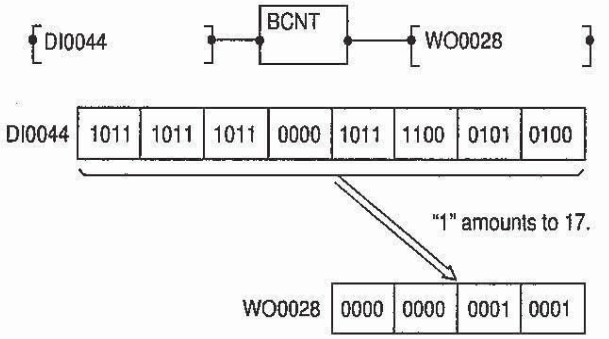
WI0000	BCD conversion	WO0016	DI0110	
0001	→	0001	00000001	(Correct value)
1000	→	4096	00004096	(Correct value)
1F3E	→	7998	00007998	(Correct value)
1F3F	→	7999	00007999	(Correct value)
1F40	→	7999	00007999	(Overflow)
7FFF	→	7999	00007999	(Overflow)
FFFF	→	8001	00008001	(Correct value)
F000	→	C096	0000C096	(Correct value)
E0C2	→	F998	0000F998	(Correct value)
E0C1	→	F999	0000F999	(Correct value)
E0C0	→	F999	0000F999	(Underflow)
8000	→	F999	0000F999	(Underflow)

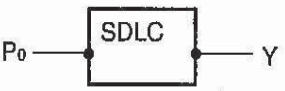
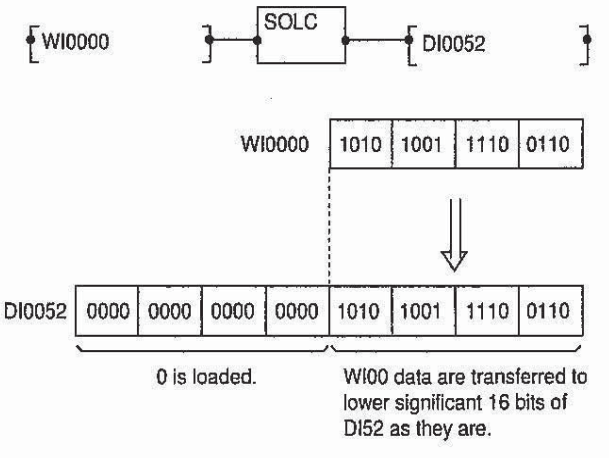
② Conversion from double-precision integer to BCD



DI0004	BCD conversion	DI0005		WO0020	
00000001	→	00000001	(Correct value)	0001	Lower significant 4 digits only are displayed.
04C4B3FE	→	79999998	(Correct value)	9998	
04C4B3FF	→	79999999	(Correct value)	9999	
04C4V400	→	79999999	(Overflow)	9999	
7FFFFFFF	→	79999999	(Overflow)	9999	
FFFFFFFF	→	80000001	(Correct value)	0001	
80000000	→	F9999999	(Underflow)	9999	

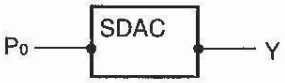
Note) If bit 0 (sign bit) of I/O is "1", it represents a negative number.

[5]	1 bit count	BCNT	Functions
Symbol			• Counts No. of bits of input (P0) "1", and loads it to output (Y). ① Count of single-precision integer bit "1"  ② Count of double-precision integer bit "1" 
Symbol	Name		
Y	Output		
P0	Input		

[6]	Single-precision integer/ double-precision integer logical conversion	SDLC	Functions
Symbol			• Converts single-precision integer of input (P0) logically into double-precision integer, and loads it to output (Y). Higher significant 16 bits of output turn to "0". 
Symbol	Name		
Y	Output		
P0	Input		

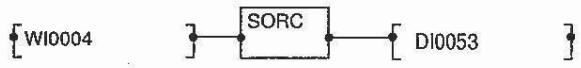
[7]	Single-precision integer/double-precision integer arithmetic conversion	SDAC	Functions

Symbol

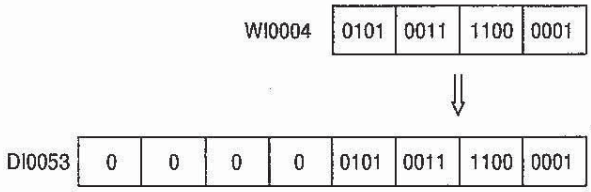


Symbol	Name
Y	Output
P0	Input

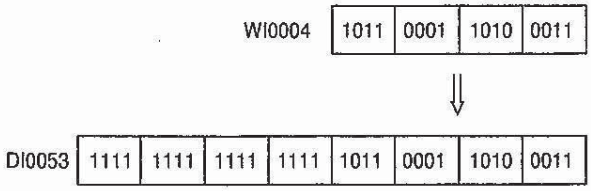
- Converts the single-precision integer of input (P0) into double-precision integer arithmetically, and loads it to output (Y).



① In case of positive

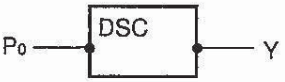


② In case of negative



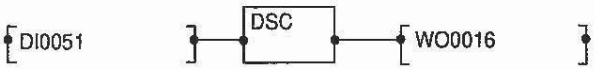
[8]	Double-precision integer/single-precision integer arithmetic conversion	DSC	Functions

Symbol

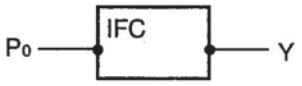
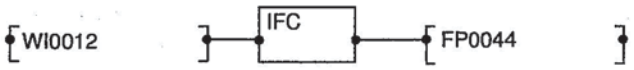


Symbol	Name
Y	Output
P0	Input

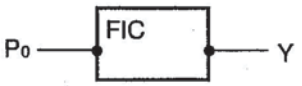
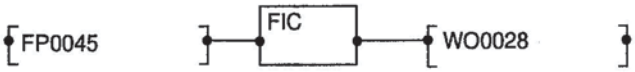
- Converts the single-precision integer of input (P0) into double-precision integer arithmetically, and loads it to output (Y). Locks the converted output to the maximum value/minimum value if it overflows/underflows.



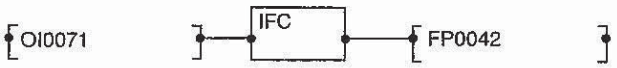
DI0051		WO0016	
00007FFE	→	7FFE	
00007FFF	→	7FFF	
00008000	→	7FFF	(Overflow)
7FFFFFFF	→	7FFF	(Overflow)
80000000	→	8000	(Underflow)
FFFFFFFF	→	FFFF	
0000FFFF	→	7FFF	(Overflow)
8000FFFF	→	8000	(Underflow)
FFFF8000	→	8000	
FFFF8001	→	8001	
FFFFFFFE	→	FFFE	

[9]	Single-precision integer/floating-point conversion	IFC	Functions
Symbol			• Converts the single-precision integer of input (P0) into double-precision integer arithmetically, and loads it to output (Y).  WI0012 FP0044 24560 → 2.456000E+04

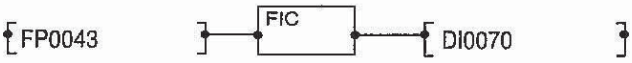
Symbol	Name
Y	Output
P0	Input

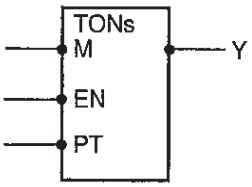
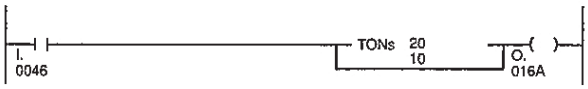
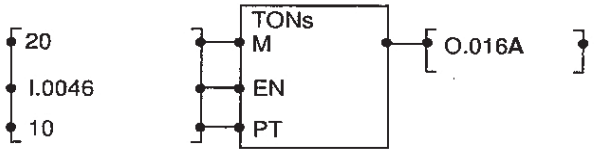
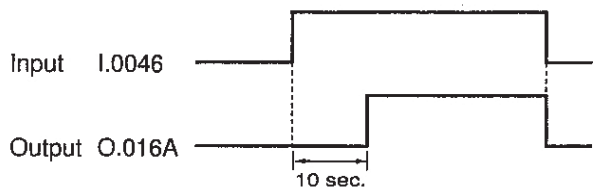
[10]	Floating point/single-precision integer conversion	FIC	Functions
Symbol			• Converts floating point of input (P0) into single-precision integer, and loads it to output (Y). Locks the converted value to the maximum value (or minimum value), if it overflows (or underflows). Rounding off (counting fraction over .5 and over as one and disregarding the reset)applies to decimal numbers.  FP0045 WO0028 3.276600E+04 → 7FFE 3.276700E+04 → 7FFF 3.276800E+04 → 7FFF (Overflow) -3.276800E+04 → 8000 -1.000000E+01 → FFF4

Symbol	Name
Y	Output
P0	Input

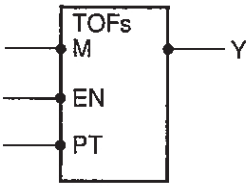
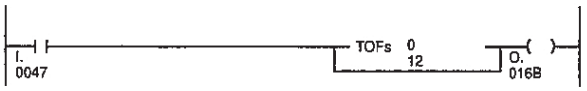
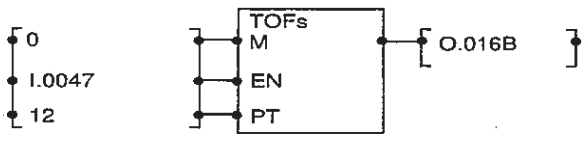
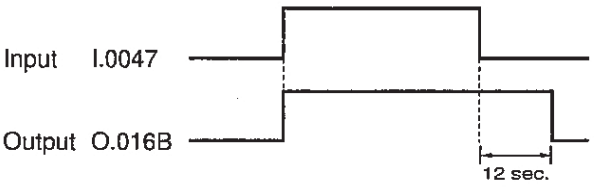
[11]	Double-precision integer/floating point conversion	IFC	Functions
Symbol			• Converts a double-precision integer of input (P0) into a floating point numeric, and loads it to output (Y).  DI0071 FP0042 7FFFFFFF → 2.147484E+09 80000000 → -2.147484E+09 FFFFFFFF → -1.000000E+00

Symbol	Name
Y	Output
P0	Input

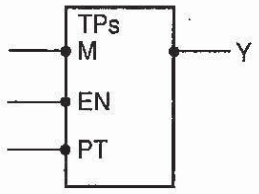
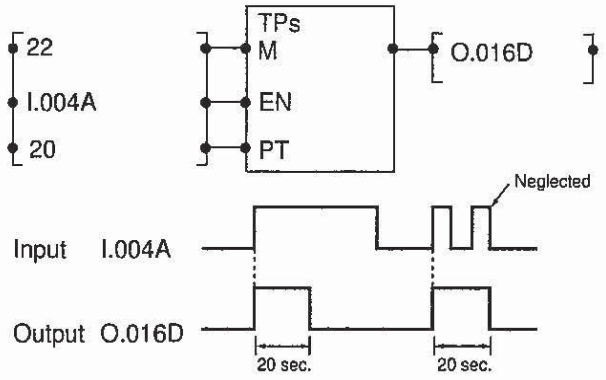
[12]	Floating point/double-precision integer conversion	FIC	Functions
Symbol			• Converts a floating point of numeric of input (P0) into a double-precision integer, and loads it to output (Y). Locks the converted value to maximum value (or minimum value), if it overflows (or underflows).  FP0043 DI0070 2.147483E+09 → 7FFFFD80 2.147484E+09 → 7FFFFFFF 2.147485E+09 → 7FFFFFFF (Overflow) -2.147485E+09 → 80000000 (Underflow) -2.147484E+09 → 80000000 -2.147843E+09 → 80000280 Correct values are shown below. $2.147483647 \times 10^9 = (7FFFFFFF)_{16}$ $-2.147483648 \times 10^9 = (80000000)_{16}$

[1]	On-delay timer	TON	Functions
Symbol	 • Timer base TONs : 1 sec. TONh : 0.1 sec. TONt : 0.01 sec.		• If the input turns on, the on-delay timer starts functioning, and the output turns on when the timer reaches its set time. a. Ladder diagram  b. FB diagram  

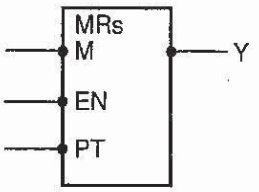
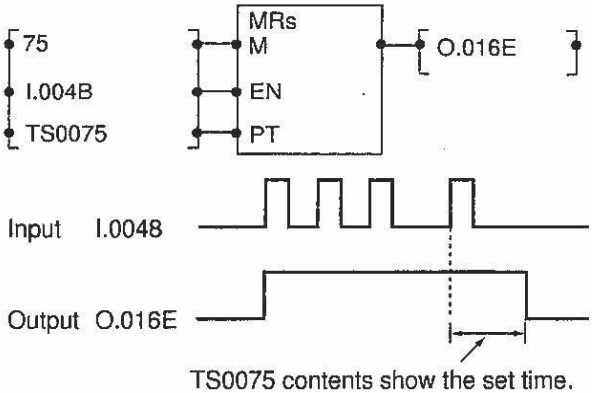
Symbol	Name
Y	Output
M	Timer No.
EN	Input
PT	Set time

[2]	Off-delay timer	TOF	Functions
Symbol	 • Timer base TONs : 1 sec. TONh : 0.1 sec. TONt : 0.01 sec.		• When the inptu turns off, the off-delay timer starts functioning, and the output turns off when the timer reaches its set time. a. Ladder diagram  b. FB diagram  

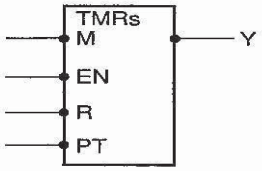
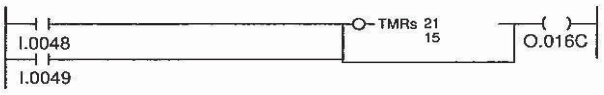
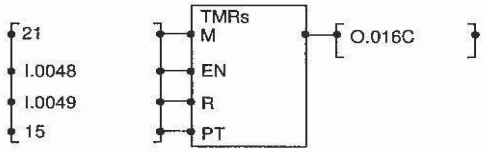
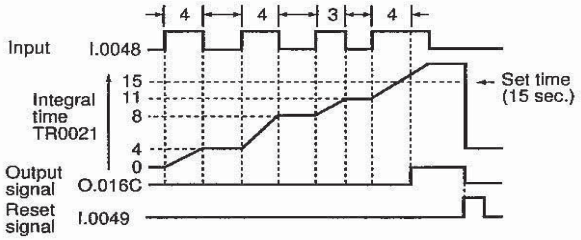
Symbol	Name
Y	Output
M	Timer No.
EN	Input
PT	Set time

[3]	Monostable timer	TP	Functions
Symbol	 • Timer base TPs : 1 sec. TPh : 0.1 sec. TPt : 0.01 sec.		① An output is turned on by the first input turn-on signal, and turns off after a set time. ② Even if the input signal is continued during the timer operation, it is neglected, and the output turns off when the timer reaches its preset tiime. a. Ladder diagram b. FB diagram 

Symbol	Name
Y	Output
M	Timer No.
EN	Input
PT	Set time

[4]	Retriggerable monostable timer	MR	Functions
Symbol	 • Timer base MRs : 1 sec. MRh : 0.1 sec. MRt : 0.01 sec.		① The timer starts functioning at the last input turn-on signal, and the output remains turned on until the timer reaches its set time. ② The timer starts again if the input signal turns on again during timer operation. a. Ladder diagram b. FB diagram 

Symbol	Name
Y	Output
M	Timer No.
EN	Input
PT	Set time

[5]	Integrating timer	TMR	Functions
Symbol	 • Timer base TMRs : 1 sec. TMRh : 0.1 sec. TMRt : 0.01 sec.		① Integrates the "ON" time when the input signal turns on, and turns on the output when the timer reaches the set time. ② Sets the integral time to 0, and turns off the output when the reset signal turns on. a. Ladder diagram  b. FB diagram  

Symbol	Name
Y	Output
M	Timer No.
EN	Input
R	Reset signal
PT	Set time

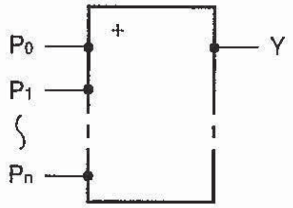
Name	Symbol	Timer No.	Set value memory	Present value memory	TImer relay	Set time
On-delay timer	TONs	0000	TS0000	TR0000	T. 0000	Suffix s: 1 sec. ~65535 sec.
	TONh					
	TONt					
Off-delay timer	TOFs					
	TOFh					
	TOFt					
Monostable timer	TPs	↓	↓	↓	↓	Suffix h: 0.1 sec. ~6553.5 sec.
	TPh					
	TPt					
Retriggerable monostable timer	MRs	1023	TS1023	TR1023	T. 1023	Suffix t: 0.01 sec. ~655.35 sec.
	MRh					
	MRt					
Integrating timer	TMRs					
	TMRh					
	TMRt					

Note) Each area of timer No. 0 to 1023 is shared by 5 types of timers.

- Time setting method

a. Method of writing into the timer instruction.....The time setting is fixed.

b. Method of directly writing into the set value momery
by program loader or another instruction .....The time setting is variable.

[1]	Addition	+	Symbol	Name
Symbol			Y	Output
			P0	Input
			⟆	⟆
			Pn	Input

Functions

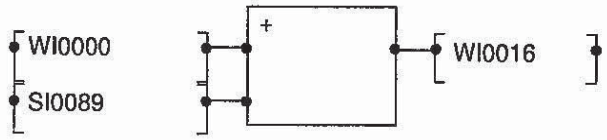
- P0 + P1 + ····· +Pn → Y .....

  Whether input/output is a single-precision integer or a double-precision integer depends upon the assignment of operands.

① Addition of single-precision integer

a. Ladder diagram

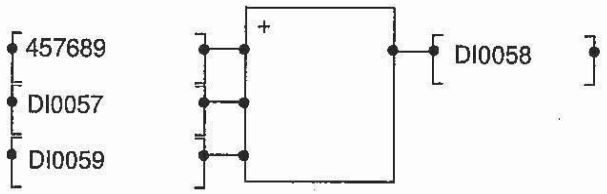
b. FB diagram



WI0000		0110	1010	1101	0111
SI0089	+	0100	0010	0101	1001
WO0016		1010	1101	0011	0000
		(A)	(D)	(3)	(0)

6AD7 ····· Data at WI0000
+) 4259 ····· Data at SI0089
AD30 ····· Data at WO0016

② Addition of double-precision integer

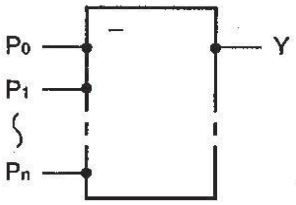


Loads (80BD9) 16 into DI0058 when DI0057 data is (I000) 16 or when DI0059 data is (I0000) 16.

(457689) 10 + (1000) 16 + (10000) 16

= (457689) 10 + (4096) 10 + (65536) 10

= (80BD9) 16

[2]	Substraction	–	Symbol	Name
Symbol			Y	Output
			P0	Input
			∫	∫
			Pn	Input

Functions

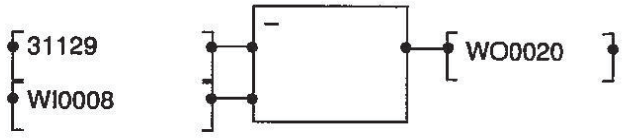
• P0 - P1 - ····· - Pn → Y .....

Whether input/output is a single-precision integer or a double-precision integer depends upon the assignment of operands.

① Substraction of single-precision integer

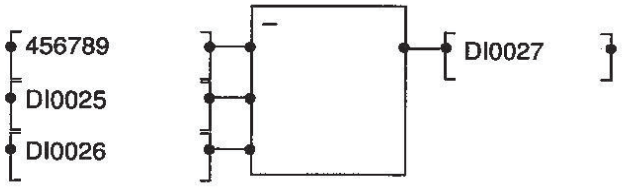
a. Ladder diagram

b. FB diagram



7999H	0111	1001	1001	1001	7999 ····· (7999) 16
WI0008 +	0010	0010	0011	0101	–) 2235 ····· Data at WI0008
WO0020	0101	0111	0110	0100	5764 ····· Data at WO0020
	(5)	(7)	(6)	(4)	

② Substraction of double-precision integer



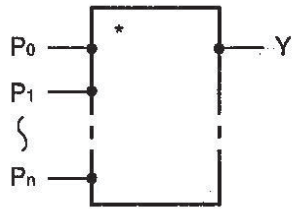
Loads (6E755) 16 into DI0027 when DI0025 data is (I000) 18 or when DI0056 data is (I00) 16.

(456789) 10 - (1000) 16 - (100) 16

= (452437) 10 - (4096) 10 - (256) 10

= (452437) 10

= (6E755) 16

[3]	Multiplication	*	Symbol	Name
Symbol			Y	Output
			P0	Input
			⟆	⟆
			Pn	Input

Functions

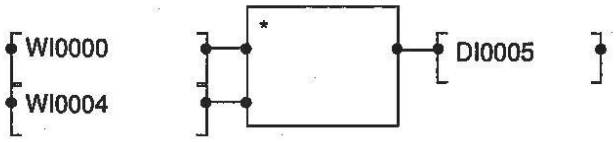
- $P_0 \times P_1 \times \cdots\cdots \times P_n \rightarrow Y$ .....

Whether input/output is a single-precision integer or a double-precision integer depends upon the assignment of operands.

① Multiplication of single-precision integer

a. Ladder diagram

b. FB diagram



Loads $(48B9A00)_{16}$ into DI0005 when WI0000 data is $(1560)_{16}$ or when WI0004 data is $(3670)_{16}$.

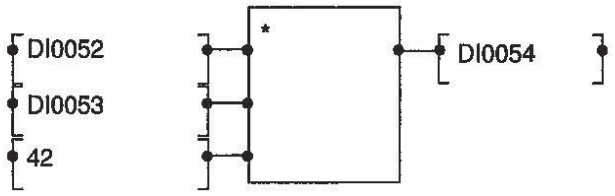
$(1560)_{16} \times (3670)_{16}$

$= (5472)_{10} \times (13037)_{10}$

$= (76257792)_{10}$

$= (48B9A00)_{16}$

② Multiplication of double-precision integer



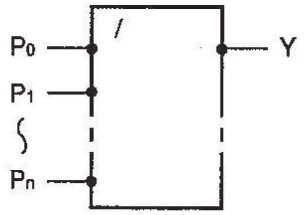
Loads $(401640)_{16}$ into DI0054 when DI0052 data is $(1000)_{10}$ or when DI0053 data is $(100)_{10}$.

$(3E8)_{16} \times (64)_{16} \times (42)_{10}$

$= (1000)_{10} \times (100)_{10} \times (42)_{10}$

$= (4200000)_{10}$

$= (401640)_{16}$

[4]	Division	/	Symbol	Name
Symbo			Y	Output
			P0	Input
			∫	∫
			Pn	Input

Functions

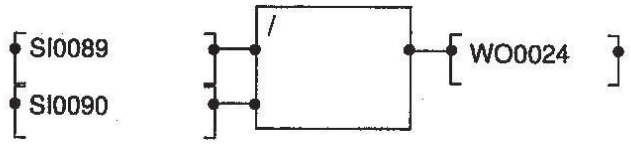
- P0 ÷P1 ÷····· ÷ Pn → Y .....

An integer only is output, and its remainder is truncated
Whether input/output is a single-precision integer or a double-precision integer depends upon the assignment of operands.

① Division of single-precision integer

a. Ladder diagram

b. FB diagram

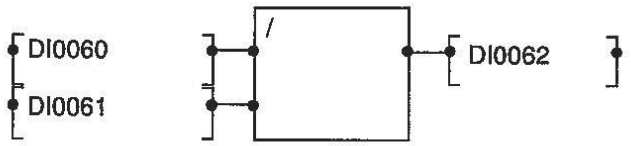


Outputs (BB) 16 to WO0024 when S10089 data is (2328)16 or SI0090 data is (30) 16.

(2328) 16 ÷ (30) 16

= (9000) 10 ÷ (48) 10 = 187.5 = (187) 10 = (BB) 16

② Division of double-precision integer



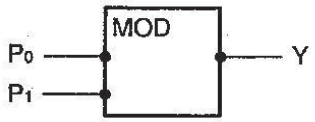
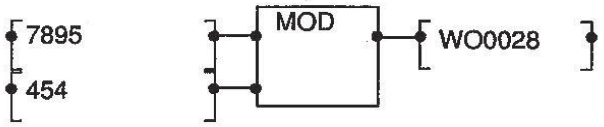
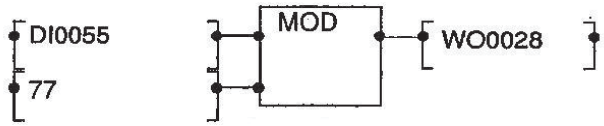
Outputs (A) 16 to DI0062 when DI0060 data is (3E8) 16 or when DI0061 data is (64)16.

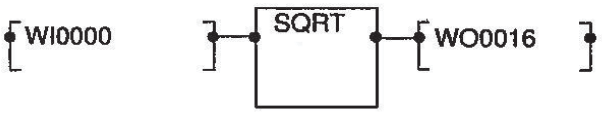
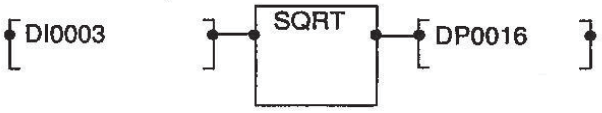
(3E8) 16 ÷ (64) 16

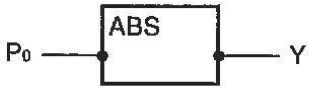
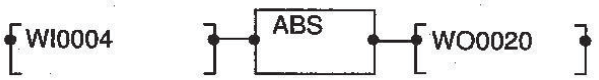
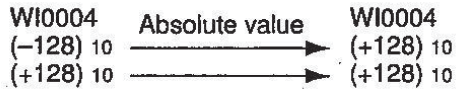
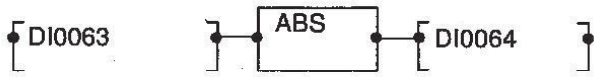
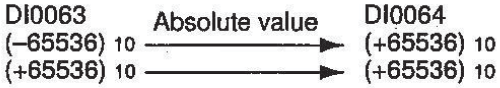
= (1000) 10÷ (100) 10

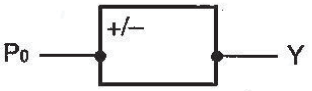
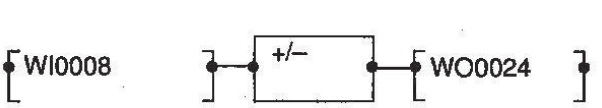
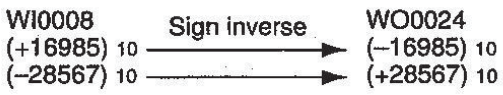
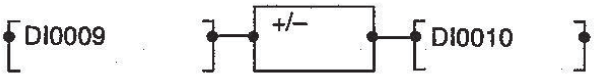
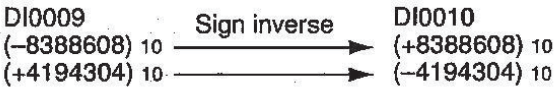
=(10) 10

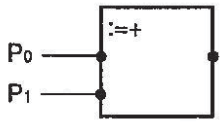
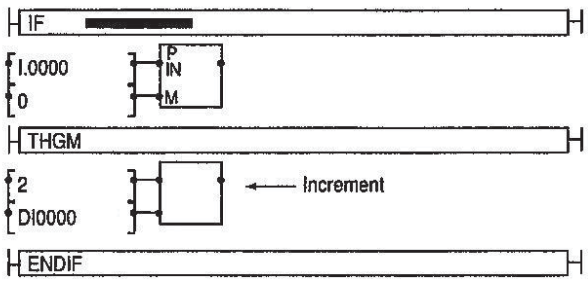
= (A) 16

[5]	Remainder	MOD	Functions
Symbol			• Remainder of P0 ÷ P1 → Y ① Remainder of division of signle-precision integer  7895 ÷ 454 = 17 remiander 117 ↓ Discard ↓ (B1) 16 Loads (B1) 16 into WO0028. ② Remainder of double-precision integer  (I000) 16 ÷ (77) 10 = (4096) 10 ÷ (77) 10 = 53 remainter 15 ↓ Discard ↓ (F) Loads (F) 16 into WO028 when data of DI0055 is (1000) 16.
Symbol	Name		
Y	Output		
P0	Input		
P1	Input		

[6]	Square root	SQRT	Functions
Symbol			• Loads square root of input (P0) to output (Y). ① Square root of signle-precision integer  $\sqrt{1000}$ = 31.622 = (32) 10 = (20) 16 Loads (20) 16 into WO016 when data of WI0000 is (3E8) 16. ② Square root of double-precision integer  (10000) 16 = (65536) 10 $\sqrt{65536}$ = (256) 10 = (100) 16 Loads (100) 16 into DP0016 when data of DI0003 is (10000) 16.
Symbol	Name		
Y	Output		
P0	Input		

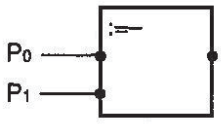
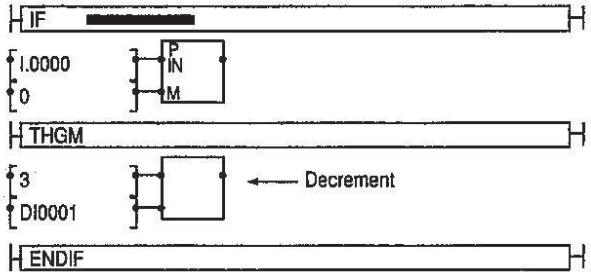
[7]	Absolute value	ABS	Functions
Symbol			• Loads an absolute value of input (P0) to output (Y). ① Absolute value of signle-precision integer   ② Absolute value of double-precision integer  
Symbol	Name		
Y	Output		
P0	Input		
			Note) Obtaines a compliment when the input is (-), converts it to (+), and outputs it. the output remains unchanged if the input is (+).

[8]	Sign inverse	+/–	Functions
Symbol			• Inverses sign (+/-) of input (P0), and loads its result to output (Y). ① Sign inverse of signle-precision integer   ② Sign inverse of double-precision integer  
Symbol	Name		
Y	Output		
P0	Input		
			Note) Obtains one's compliment of input (P0), and loads it to output (Y).

[9]	Increment	:=+	Functions
Symbol			• Adds addition vlaue (P0) to the value of input (P1), and loads the result to input (P1).  (Data at address DI0000) + 2 ⟶ Load address to DI0000

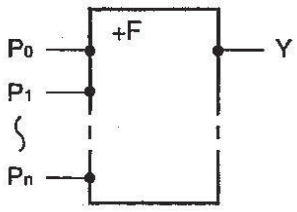
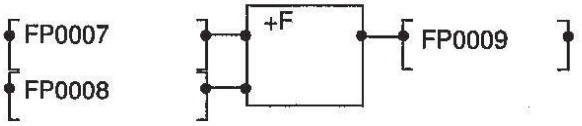
Symbol	Name
Y	Output
P0	Input

Note) Since the increment reaches a saturated value soon after it is executed normally, the increment is set to function only when the rise-time differential functions in the example.

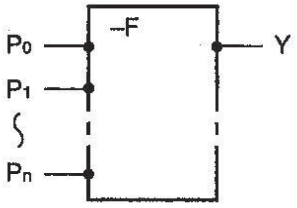
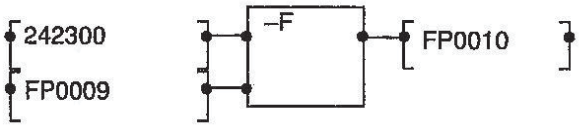
[10]	Decrement	:=-	Functions
Symbol			• Substracts subtraction value (P0) from input (P1), and loads the result to input (P1).  (Data at address DI0001) – 3 ⟶ Load address to DI0001

Symbol	Name
Y	Output
P0	Input

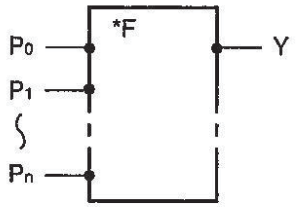
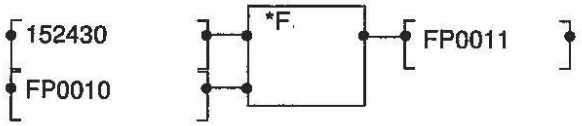
Note) Since the decrement is executed normally, data at address DI0001 become 0 soon the decrement functions only when the rise-time differential functions in this example. Accordingly, -3 is added to data at address DI001, each time I. 0001 turns on.

[11]	Floating point addition	+F	Functions
Symbol			• $P_0 + P_1 + \cdots + P \rightarrow Y$ Adds flosting point numerics together, and loads the result to output (Y).  Adds "data at FP0007" and "data at FP0008" and loads the result to FP0009 in the example. Symbol Name Y Output P_0 Input ⟆ ⟆ P_n Input

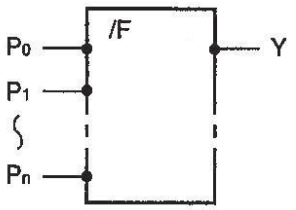
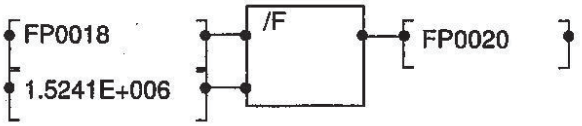
Symbol	Name
Y	Output
P_0	Input
⟆	⟆
Pn	Input

[12]	Floating point addition	-F	Functions
Symbol			• $P_0 - P_1 - \cdots - P_n \rightarrow Y$ Subtracts floating point numerics together, and loads the result to output (Y).  Subtracts "data of FP0009" from "2,423 × 10^5", and loads the result of FP0010.

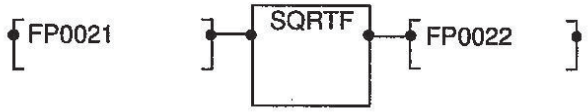
Symbol	Name
Y	Output
P_0	Input
⟆	⟆
Pn	Input

[13]	Floating point multiplication	*F	Functions
Symbol			• P0 × P1 × ··· × Pn → Y Multiples floating point numerics together, and loads the result to output (Y).  In this example, multiplies (1.5243×10^{-5}) by (data of FP0010), and loads the result to FP0011.

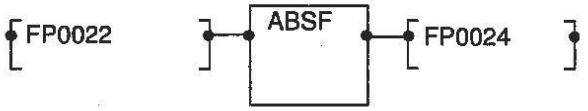
Symbol	Name
Y	Output
P0	Input
⟆	⟆
Pn	Input

[14]	Floating point division	/F	Functions
Symbol			• P0 ÷P1 ÷ ··· ÷Pn → Y Divides floating point numerics together, and loads the result to output (Y).  In this example, divides (data of FP0018) from (1.5241×10^{6}), and loads the result of FP0020.

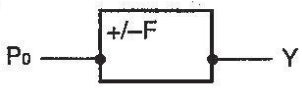
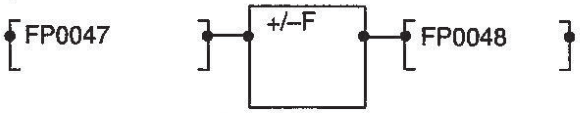
Symbol	Name

[15]	Floating point square root	SQRTF	Functions
Symbol			• Loads the square root of floating point input (P0) to output (Y).  In this example, $\sqrt{\text{(data at FP0021)}}$ → Load to FP0022

Symbol	Name
Y	Output
P0	Input

[16]	Floating point absolute value	ABSF	Functions
Symbol			• Loads an absolute value of floating point input (P0) to output (Y).  In this example, | (data at FP0023) | → Load to FP0024 −1.200000E + 05 → 1.200000E + 05

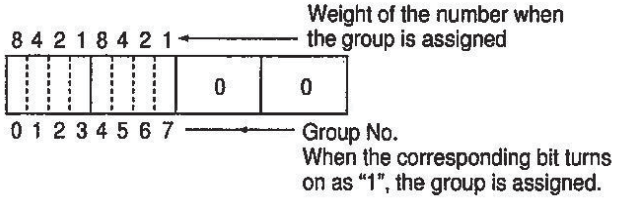
Symbol	Name
Y	Output
P0	Input

[17]	Floating point sign inverse	+/–F	Functions
Symbol			• Inverses sign (+/-) of floating point input (Po), and loads the result to output (Y).  In this example Data at FP0047 Sign inverse Data at FP0048 1.53×10^{15} → 1.53×10^{15} -1.26×10^{6} → 1.26×10^{6}

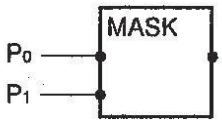
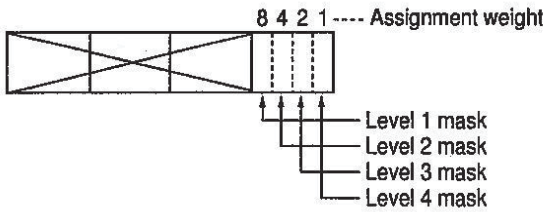
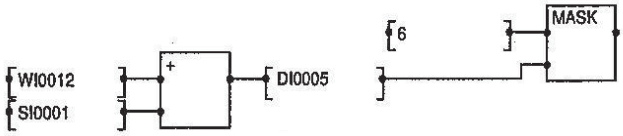
Symbol	Name
Y	Output
Po	Input

[1]	Load effective address	LEA	Functions
Symbol			① Loads the execution address of the area shown by input (Po) to output (Y). ② Either word address or bit address is applicable to input (Po).

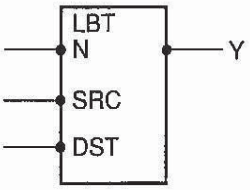
Symbol	Name
Y	Output
Po	Input

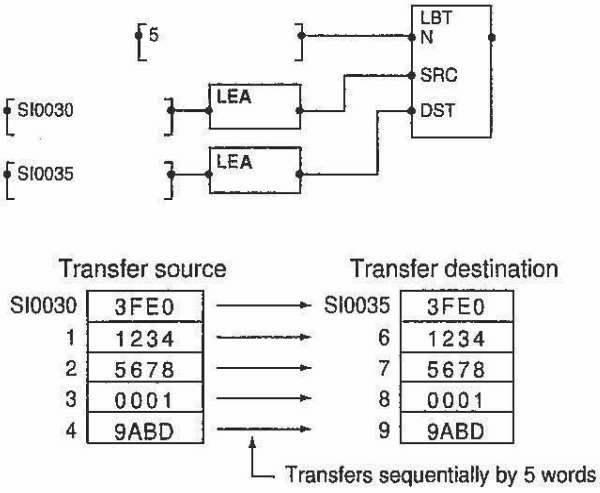
[2]	Check and transfer	CHT	Functions
Symbol			① Refreshes only a group assigned by input (Po) of the I/O area employed in an application program during the execution of this instruction. ② Assigns the group setting by CHT group No. 0 to 7 of PIO entry in system definition. ③ Method of assigning the group of CHT instruction  0C000H --- (C = 8 + 4) --------- Group 0 and 1 are assigned. 0FF00H --- (F = 8 + 4 + 2 + 1) --- All groups 0 to 7 are assigned. Indicates a hexadecimal number. 0 is added to the head when the numeric head is started with A-F.

Symbol	Name
Y	Output (Omissible)
Po	CH group No.

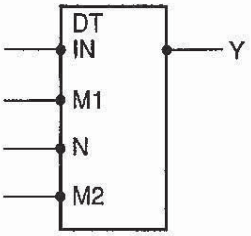
[3]	Program mask	MASK	Functions
Symbol			• If a level higher than the executing program level is assigned by the weight of interrupt level bit pattern (P0) in the following figure, the program interrupt at the assigned level is blocked when the program within the range assigned by the interrupt masked processor (P1) is running. Interrupt level bit pattern (P0)   In case of the example, the interrupt at interrupt program levels 2 and 3 is blocked during the execution of the left addition (+) instruction.

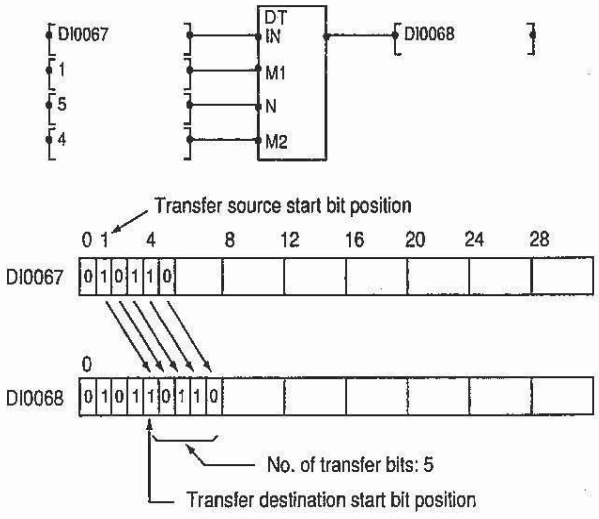
Symbol	Name
P0	Interrupt level bit pattern
P1	Interrupt masked processor

[1]	Logical block transfer	LBT	Functions
Symbol			• Transfers bit pattern data from the transfer source address (SRC) to the transfer destination address (DST) by the transfer size (N).

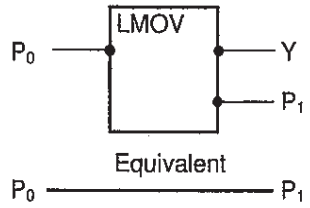
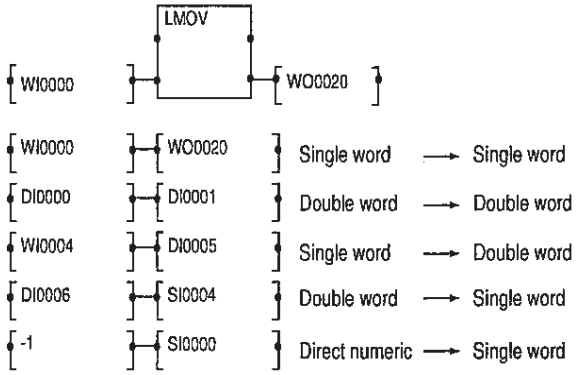


Symbol	Name
Y	Result (omissible)
N	Transfer size
SRC	Transfer source address
DST	Transfer destination address

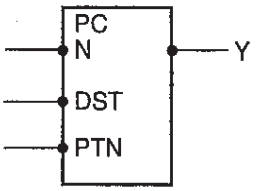
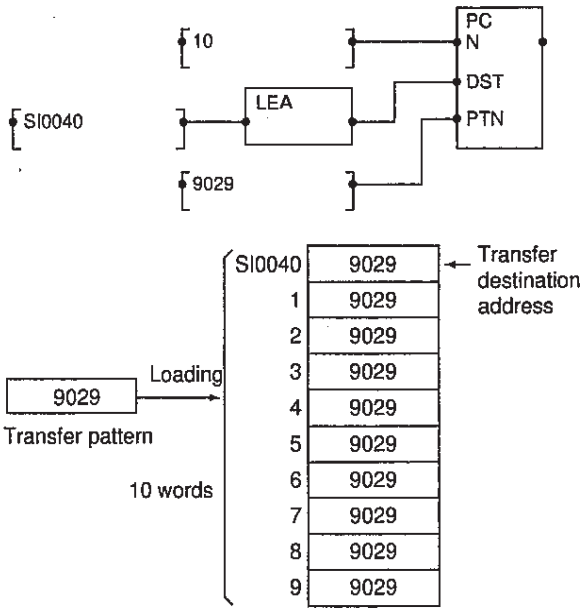
[2]	Digit transfer	DT	Functions
Symbol			① Transfers bit pattern data from the start bit position (M1) of the transfer source of input (IN) to the start bit position (M2) of the transfer destination of output (Y) and succeeding positions by the number of transfer bits (N).



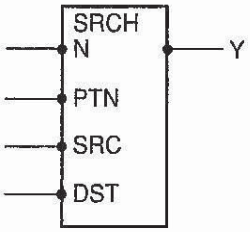
Symbol	Name
Y	Output
IN	Input
M1	Transfer source start bit position
N	No. of transfer bit position
M2	Transfer source start bit position

[3]	Logical transfer (substitution)	LMOV	Functions
Symbol			• Substitutes data of input (P0) to substitution destination (P1). ① When a data of double words is substituted into a signal word, its lower significant 4 digits are substituted, but its higher significant 4 digits are not substituted. ② When a data of a single word is substituted into double words, it is substituted to the lower significant 4 digits, and 0 is substituted to higher significant 4 digits. 

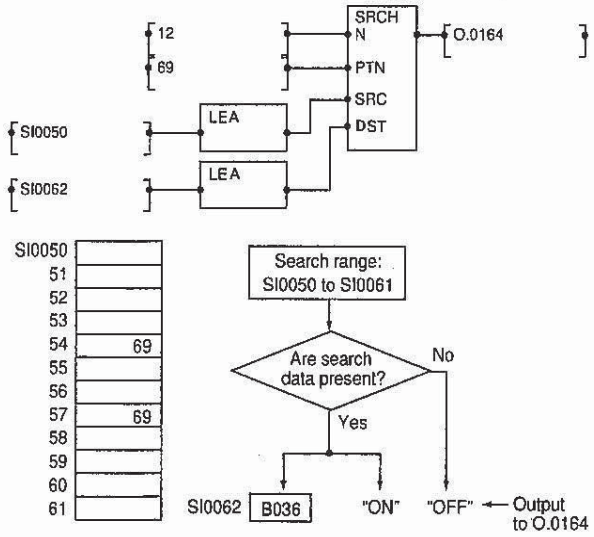
Symbol	Name
Y	Result (omissible)
P0	Input
P1	Substitution destination

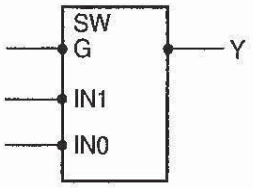
[4]	Pattern clear	PC	Functions
Symbol			• Loads transfer patterns into the area of transfer size (N) from the transfer destination address (DST). 

Symbol	Name
Y	Result
N	Transfer size
DST	Transfer destination address
PTN	Transfer pattern

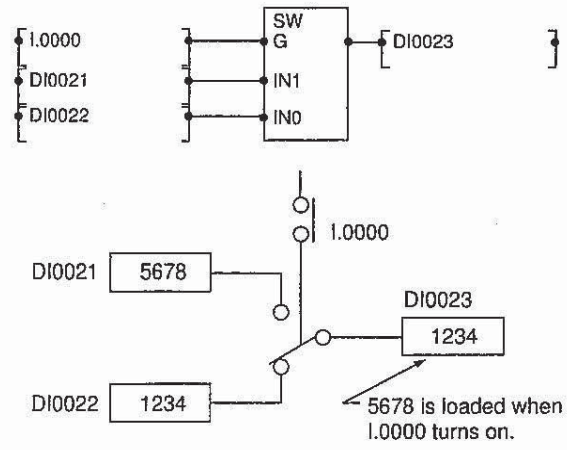
[5]	Search	SRCH	Functions
Symbol			• Searches data having the same contents as those of search data (PTN) in the area where the search start address (SRC) meets the size (N), and outputs whether data exist or not to output relay (Y). Loads the absolute address of detection memory to the detection point loading address (DST). Loads the first address, if many patterns are equal.

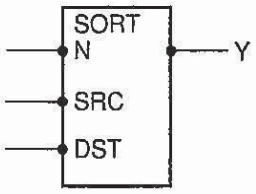
Symbol	Name
Y	Output relay
N	Size
PTN	Search data
SRC	Search start address
DST	Detection point loading address

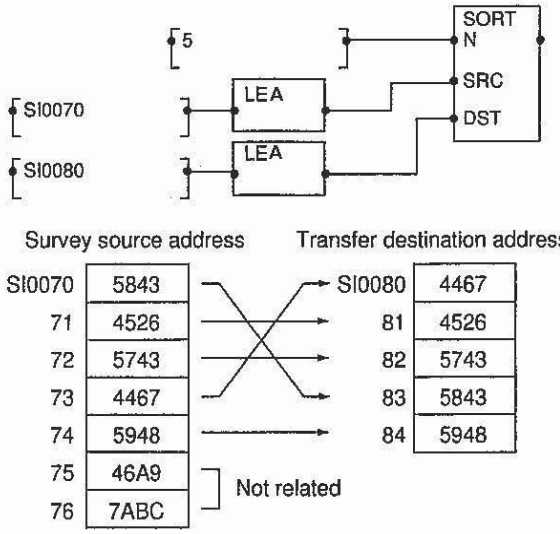


[6]	Switch	SW	Functions
Symbol			• Executes the following transfer according to the switch input conditions. Switch input "ON" : ON data (IN1) → Output (Y) Switch input "OFF" : OFF data (IN0) → Output (Y)

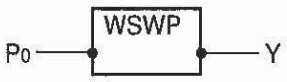
Symbol	Name
Y	Output
G	Switch input
IN1	ON data
IN0	OFF data

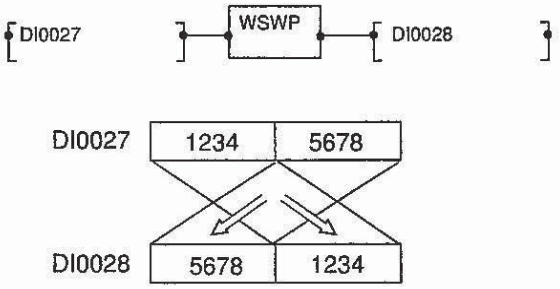


[7]	Sort	SORT	Functions
Symbol			• Rearanges data corresponding to size (N) from the survey source address (SRC) starting with smaller one, and loads the result from the transfer destination address into the area corresponding to size (N).

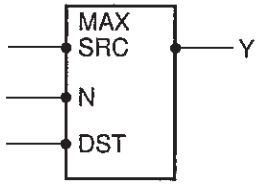
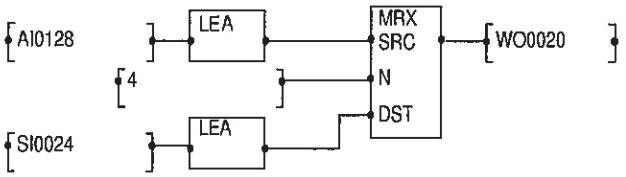


Symbol	Name
Y	Result (omissible)
N	Size
SRC	Survey source address
DST	Transfer detination address

[8]	Word swap	WSWP	Functions
Symbol			• Swaps high-order 16 bits and low-order 16 bits of input (P0) data from each other, and loads them to output (Y).



Symbol	Name
Y	Output
P0	Input

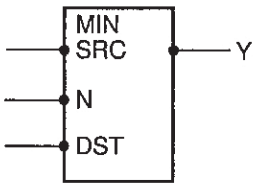
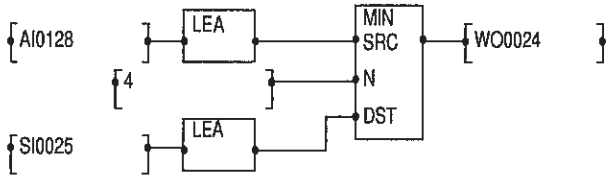
[1]	Maximam value detection	MAX	Functions
Symbol			• Detects the maximum value of data corresponding to size (N) from input data start address (SRC), loads its maximum value to output (Y), and also, loads the absolute value of the maximum value detection memory to the detection point loading address (DST). 

Address	Label	Decimal	
0128	AI0128	620	
0129	AI0129	1034	
0130	AI0130	1380	← Maximum value
0131	AI0131	558	

Searches the maximum value out of 4 words at address AI0128 to 0131, loads its maximum value (1380) to WO020, and also, loads the absolute address (682H) of the maximum value detection memory to SI0024.

Absolute address AI0130 → Absolute address 682 H

Symbol	Name
Y	Output
SRC	Input data start address
N	Size
DST	Detection point loading address

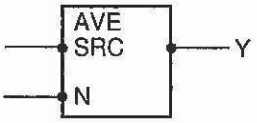
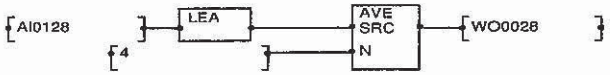
[2]	Minimum value detection	MIN	Functions
Symbol			• Detects the minimum value of data corresponding to size (N) from input data start address (SRC), loads its minimum value to output (Y), and also, loads the absolute value of the minimum value detection memory to the detection point loading address (DST). 

Address	Label	Decimal	
0128	AI0128	1537	
0129	AI0129	1097	← Minimum value
0130	AI0130	1120	
0131	AI0131	1644	

Searches the minimum value out of 4 words at address AI0128 to 0131, loads its maximum (1097) to WO024, and also, loads the absolute address (681H) of the minimum value detection memory to SI0025.

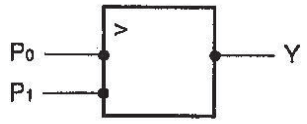
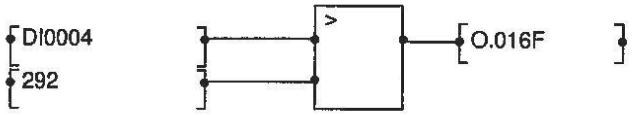
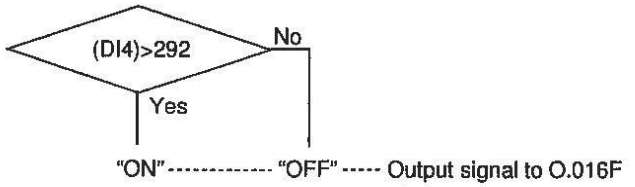
Absolute address AI0129 → Absolute address 681 H

Symbol	Name
Y	Output
SRC	Input data start address
N	Size
DST	Detection point loading address

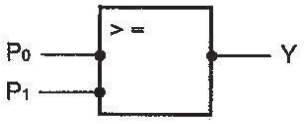
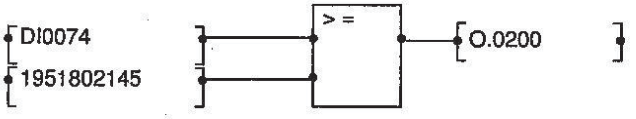
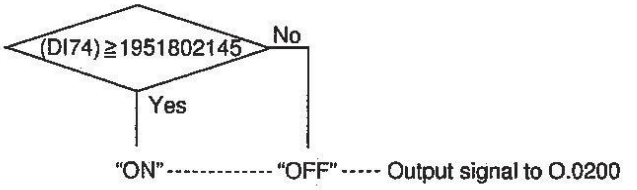
[3]	Average value calculation	AVE	Functions
Symbol			• Calculates an average value of data corresponding to size (N) from input data start address (SRC), and loads the average value to output (Y).  Address Label Decimal 0128 AI0128 112 0129 AI0129 214 0130 AI0130 82 0131 AI0131 291 Average value 174 Calculates an average value of 4 words at address AI0128 to 0131, and loads the average value (174) to WO0028.

Symbol	Name
Y	Output
SRC	Input data start address
N	Size

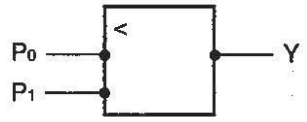
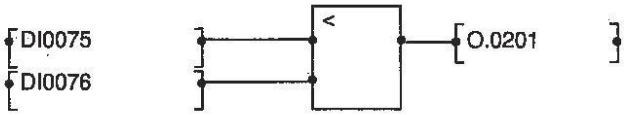
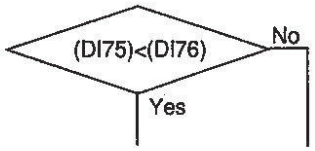
Note : The average value is 174.75, but 174 is loaded by truncating the decimal number.

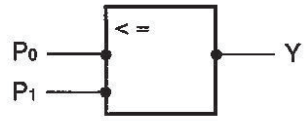
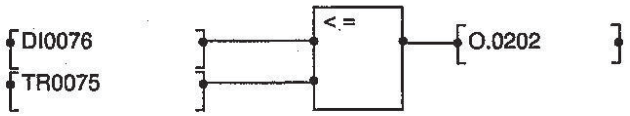
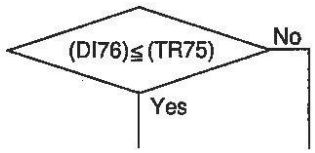
[1]	Comparison larger than	>	Functions
Symbol			• Compares input 1 (P0) with input 2 (P1), and outputs the result to output relay (Y). When input 1 > input 2 (condition is satisfied), the output relay turns on. When input 1 > input 2 (condition is not satisfied), the output relay turns off.   "ON" ---------- "OFF" ----- Output signal to O.016F In this example, an ON signal is output to 0.16F when "the data of DI4" is larger than "292". Note : Whether input/output is a single-precision integer or a double-precision integer depends upon the assignment of operands.

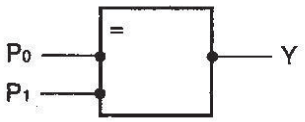
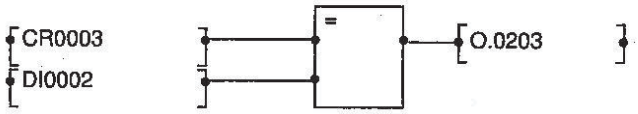
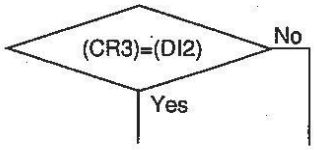
Symbol	Name
Y	Output
P0	Input 1
P1	Input 2

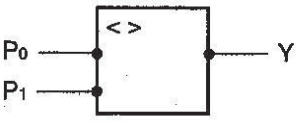
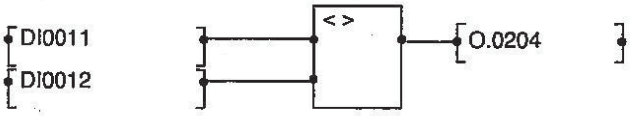
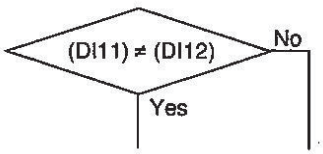
[2]	Comparison Larger than or equal to	> =	Functions
Symbol			• Compares input 1 (P0) with input 2 (P1), and outputs the result to output relay (Y). When input 1 ≥ input 2 (condition is satisfied), the output relay turns on. When input 1 ≥ input 2 (condition is not satisfied), the output relay turns off.   "ON" ---------- "OFF" ----- Output signal to O.0200 An ON signal is output to 0.0200 when "the data of DI74" is larger than or equal to "1951802145" in this example. Note : Whether input/output is a single-precision integer or a double-precision integer depends upon the assignment of operands.

Symbol	Name
Y	Output
P0	Input 1
P1	Input 2

[3]	Comparison Smaller than	<	Functions
Symbol			• Compares input 1 (P0) with input 2 (P1), and outputs the result to output relay (Y). When input 1 < input 2 (condition is satisfied), the output relay turns on. When input 1 < input 2 (condition is not satisfied), the output relay turns off.   "ON" ---------- "OFF" ----- Output signal to O.0201 An ON signal is output to O.0201 when "the data of DI75" is smaller than "the data of DI76" in the example.
Symbol	Name		
Y	Output		
P0	Input 1		
P1	Input 2		
			Note : Whether input/output is a single-precision integer or a double-precision integer depends upon the assignment of operands.

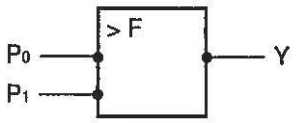
[4]	Comparison Smaller than or equal to	<=	Functions
Symbol			• Compares input 1 (P0) with input 2 (P1), and outputs the result to output relay (Y). When input 1 ≤ input 2 (condition is satisfied), the output relay turns on. When input 1 ≤ input 2 (condition is not satisfied), the output relay turns off.   "ON" ---------- "OFF" ----- Output signal to O.0202 An ON signal is output to O.0202 when "the data of DI76" is equal to or smaller than "the data of timer present value TR75" in the example.
Symbol	Name		
Y	Output		
P0	Input 1		
P1	Input 2		
			Note : Whether input/output is a single-precision integer or a double-precision integer depends upon the assignment of operands.

[5]	Comparison Equal to	=	Functions
Symbol			• Compares input 1 (P0) with input 2 (P1), and outputs the result to output relay (Y). When input 1 = input 2 (condition is satisfied), the output relay turns on. When input 1 = input 2 (condition is not satisfied), the output relay turns off.   "ON" ---------- "OFF" ----- Output signal to O.0203 An ON signal is output to O.0203 when "the data of counter present value CR3" is equal to "the data of DI2" in the exmaple. Note : Whether input/output is a single-precision integer or a double-precision integer depends upon the assignment of operands.
Symbol	Name		
Y	Output		
P0	Input 1		
P1	Input 2		

[6]	Comparison Unequal to	< >	Functions
Symbol			• Compares input 1 (P0) with input 2 (P1), and outputs the result to output relay (Y). When input 1 ≠ input 2 (condition is satisfied), the output relay turns on. When input 1 ≠ input 2 (condition is not satisfied), the output relay turns off.   "ON" ---------- "OFF" ----- Output signal to O.0204 An ON signal is output to O.0204 when "the data of DI11" is not equal to "the data of DI12" in the example. Note : Whether input/output is a single-precision integer or a double-precision integer depends upon the assignment of operands.
Symbol	Name		
Y	Output		
P0	Input 1		
P1	Input 2		

[7]	Floating point comparison Larger than	> F	Functions

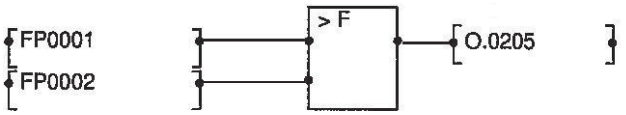
Symbol

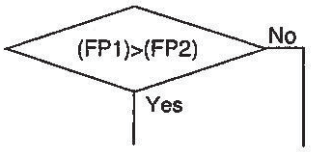


Symbol	Name
Y	Output
P0	Input 1
P1	Input 2

- Compares floating point input 1 (P0) with input 2 (P1), and outputs the result to output relay (Y).

  When input 1 > input 2 (condition is satisfied), the output relay turns on.
  When input 1 > input 2 (condition is not satisfied), the output relay turns off.

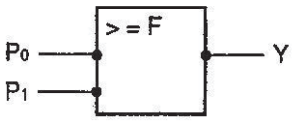




"ON" ---------- "OFF" ----- Output signal to O.0205

An ON signal is output to O.0205 when "the data of FP1" is larger than "the data of FP2" in the example.

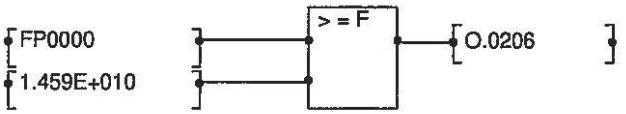
Symbol

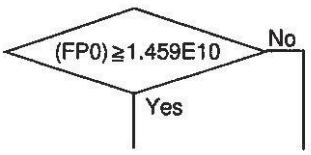


Symbol	Name
Y	Output
P0	Input 1
P1	Input 2

- Compares floating point input 1 (P0) with input 2 (P1), and outputs the result to output relay (Y).

  When input 1 ≥ input 2 (condition is satisfied), the output relay turns on.
  When input 1 ≥ input 2 (condition is not satisfied), the output relay turns off.



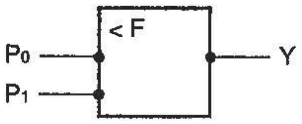


"ON" ---------- "OFF" ----- Output signal to O.0206

An ON signal is output to O.0206 when "the data of FP0" is larger than or equal to "1.459×10^{10}".

[9]	Floating point comparison Smaller than	< F	Functions

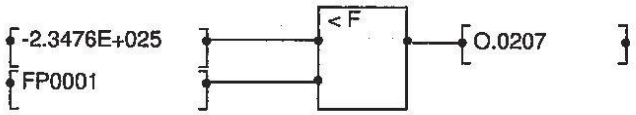
Symbol

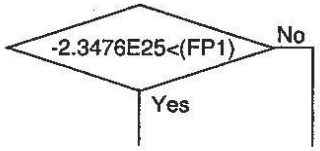


Symbol	Name
Y	Output
P0	Input 1
P1	Input 2

- Compares floating point input 1 (P0) with input 2 (P1), and outputs the result to output relay (Y).

  When input 1 < input 2 (condition is satisfied), the output relay turns on.
  When input 1 < input 2 (condition is not satisfied), the output relay turns off.



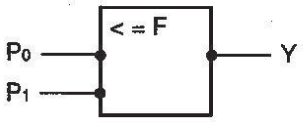


"ON" ---------- "OFF" ----- Output signal to O.0207

An ON signal is output to O.0207 when "-2.3476×10^{25}" is smaller than "the data of FP1" in the example.

[10]	Floating point Smaller than or equal to	< = F	Functions

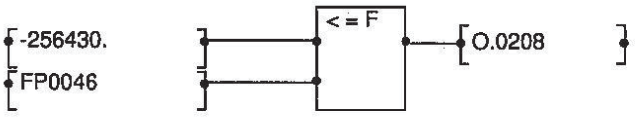
Symbol

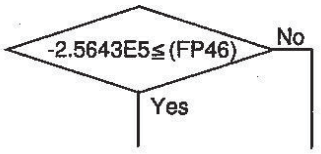


Symbol	Name
Y	Output
P0	Input 1
P1	Input 2

- Compares floating point input 1 (P0) with input 2 (P1), and outputs the result to output relay (Y).

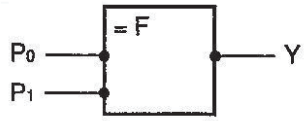
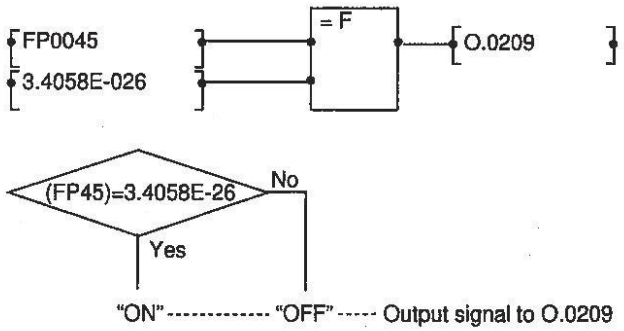
  When input 1 ≤ input 2 (condition is satisfied), the output relay turns on.
  When input 1 ≤ input 2 (condition is not satisfied), the output relay turns off.





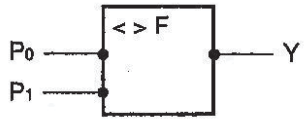
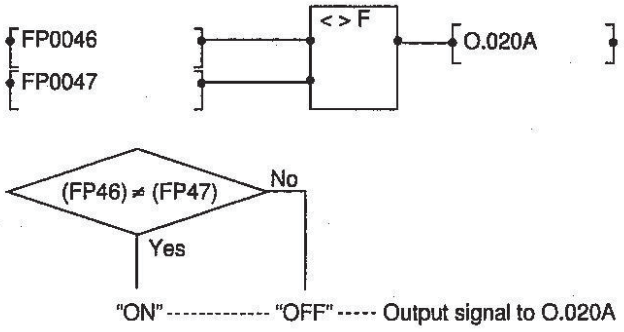
"ON" ---------- "OFF" ----- Output signal to O.0208

An ON signal is output to O.0208 when "-2.5643×10^{5}" is smaller than "the data of FP46" in the example.

[11]	Floating point Equal to	= F	Functions
Symbol			• Compares floating point input 1 (P0) with input 2 (P1), and outputs the result to output relay (Y). When input 1 = input 2 (condition is satisfied), the output relay turns on. When input 1 = input 2 (condition is not satisfied), the output relay turns off. 

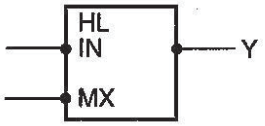
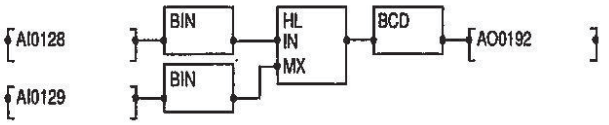
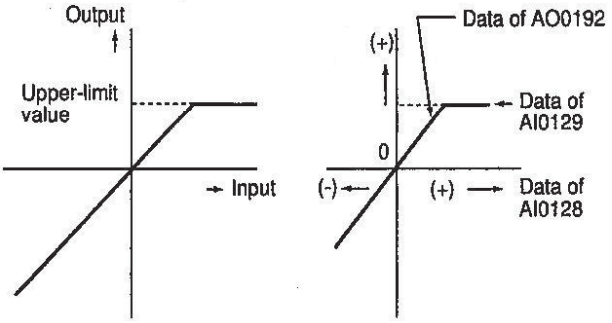
Symbol	Name
Y	Output
P0	Input 1
P1	Input 2

An ON signal is output to O.0209 when "the data of FP45" is equal to "3.4058×10^{-26}" in the example.

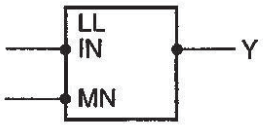
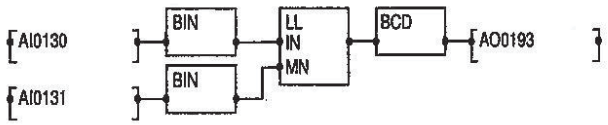
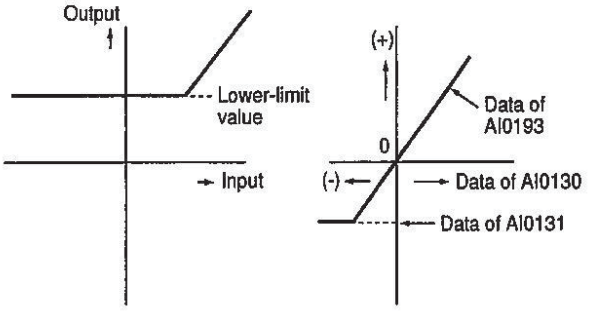
[12]	Floating point Unequal to	< > F	Functions
Symbol			• Compares floating point input 1 (P0) with input 2 (P1), and outputs the result to output relay (Y). When input 1 ≠ input 2 (condition is satisfied), the output relay turns on. When input 1 ≠ input 2 (condition is not satisfied), the output relay turns off. 

Symbol	Name
Y	Output
P0	Input 1
P1	Input 2

An ON signal is output to O.020A when "the data of FP46" is not equal to "the data of FP47" in the example.

[1]	Upper limit	HL	Functions
Symbol			• Limits the input data (IN) value by the upper-limit value (MX), and loads it to output (Y). Input (IN) < Upper limit value (MX) → Output (Y) : Input value Input (IN) ≥ Upper limit value (MX) → Output (Y) : Upper limit value  

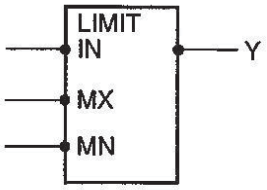
Symbol	Name
Y	Output
IN	Input data
MX	Upper imit value

[2]	Lower limit	LL	Functions
Symbol			• Limits the lower-limit value (MN) of input data (IN) value by the lower-limit value (MN), and loads it to output (Y). Input (IN) > Lower limit value (MN) → Output (Y) : Input value Input (IN) ≤ Lower limit value (MN) → Output (Y) : Lower limit value  

Symbol	Name
Y	Output
IN	Input data
MN	Lower limit value

[3]	Upper-limit/lower-limit	LIMIT

Symbol



Symbol	Name
Y	Output
IN	Input data
MX	Upper limit value
MN	Lower limit value

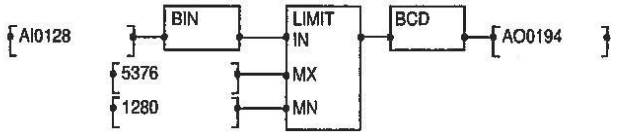
Functions

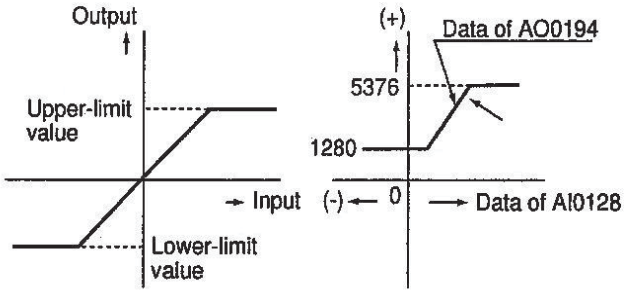
- Limits the input data (IN) value by the upper-limit value and lower-limit values, and loads it to output (Y).

Input (IN) ≥ Upper limit value (MX)
→ Output (Y) : Upper limit value

Upper limit value (MX) > Input (IN) > Lower limit value (MN)
→ Output (Y) : Input value

Lower limit value (MN) ≥ Input value (IN)
→ Output (Y) : Lower limit value

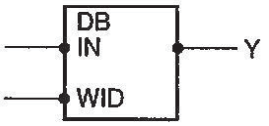




Note : If upper-limit value (MX) < lower-limit value (MN), the lower-llimit precedence over the other, and the upper-limit is neglected.

[4]	Dead band	DB

Symbol



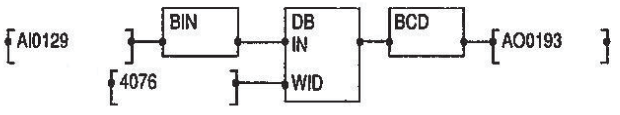
Symbol	Name
Y	Output
IN	Input data
WID	Dead band width

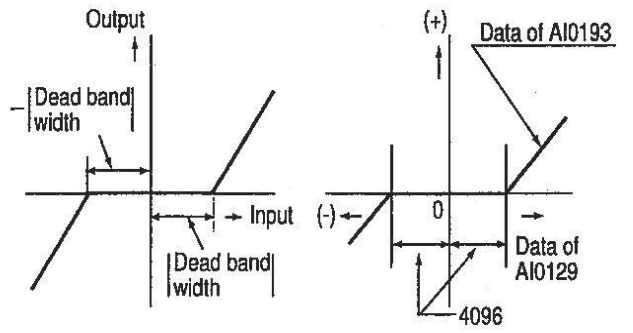
Functions

- Processes | WID | as a dead band width.

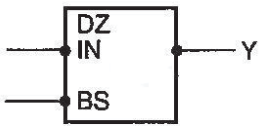
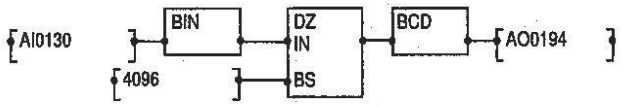
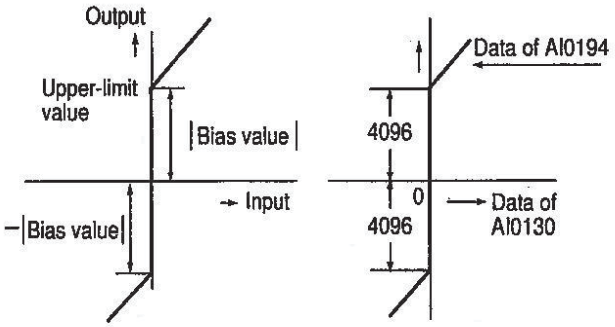
Input data (IN) > | WID |·····IN – | WID | → Output (Y)

Input data (IN) > – | WID |·····IN + | WID | → Output (Y)

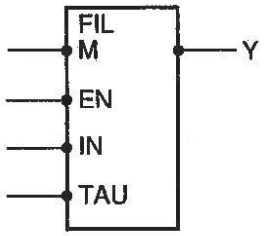
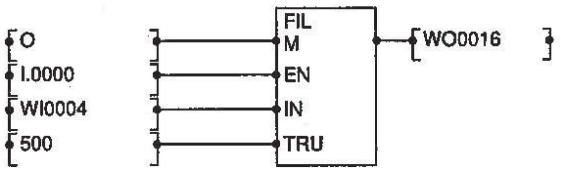
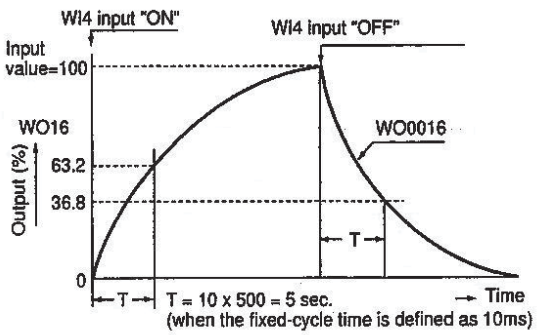




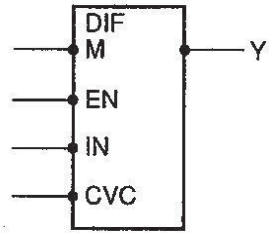
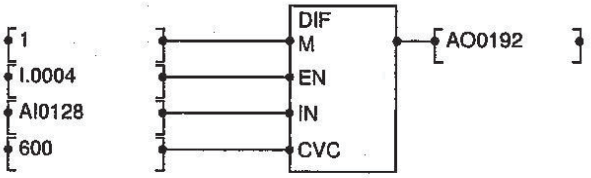
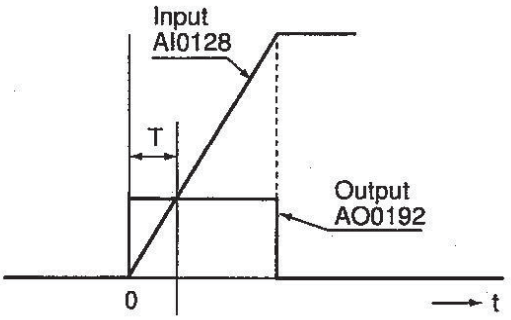
Note : The dead band width is limited to 0 or a positive value.

[5]	Bias	DZ	Functions
Symbol			• Processes | BS | as a bias value. Input data (IN) > 0 ····· IN + | BS | → Output (Y) Input data (IN) < 0 ····· IN – | BS | → Output (Y) Input data (IN) = 0 ····· 0 → Output (Y)   Note : The bias value is limited to 0 or positive value.

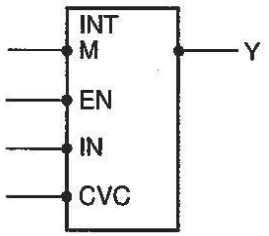
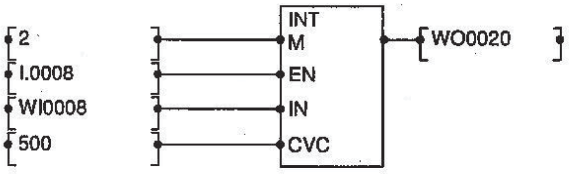
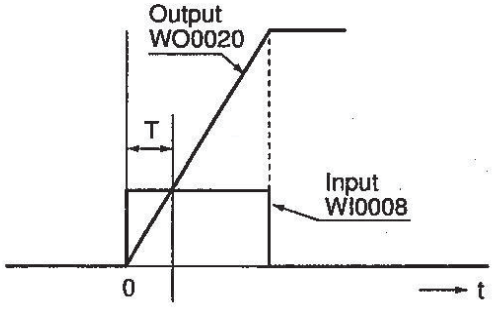
Symbol	Name
Y	Output
IN	Input data
BS	Bias value

[6]	Filter	FIL	Functions
Symbol			• Outputs input data (IN) after filtering. Uses fixed-cycle processing programs. $\text{Output} = \frac{\text{Input data} - \text{last output value}}{\text{Filter time constant / fixed-cycle time}} + \text{Last output value}$   The figure shows an example when WI4 is turned on and off with I.0 turned on.

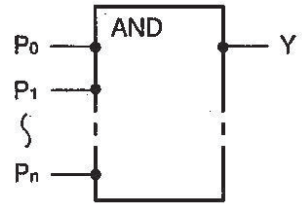
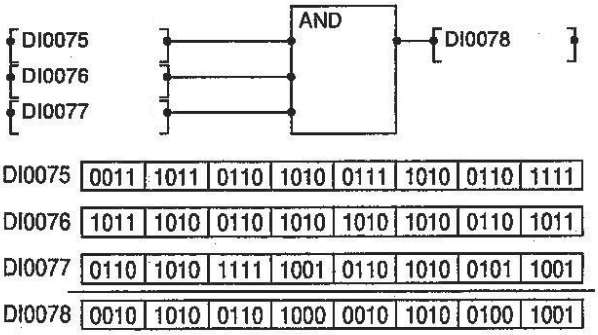
Symbol	Name
Y	Output
M	Analog work area
EN	Enable input
IN	Input data
TAU	Filter time constant/ΔT

[7]	Differential	DIF	Functions
Symbol			• Outputs input data (IN) after differentiating it. Uses fixed-cycle processing programs. Output = Input data − last input data / Fixed-cycle time / derivative time   Derivative time (T) = 10 ÷ 600=17 μ sec. ..... When the fixed cycle time is defined as 10ms.

Symbol	Name
Y	Output
M	Analog work area
EN	Enable input
IN	Input data
CVC	ΔT/derivative time

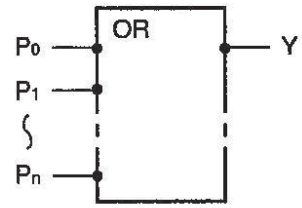
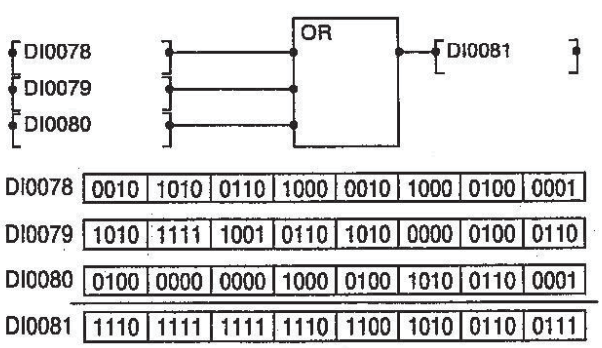
[8]	Integral	INT	Functions
Symbol			• Outputs input data (IN) after integrating it. Uses fixed cycle processing programs. Output = Input data / Integral time / fixed-cycle time + Last output value   Integral time (T) = 10 × 500 = 5 sec. ..... When the fixed cycled time is defined as 10ms.

Symbol	Name
Y	Output
M	Analog work area
EN	Enable input
IN	Input data
CVC	Integral time/ΔT

[1]	Logical product	AND	Functions
Symbol			• P0 ∧ P1 ∧ ..... ∧ Pn → Y Obtains a logical product of input (P0 ~ Pn) every bit, and loads the result to output (Y). 

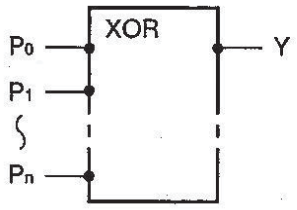
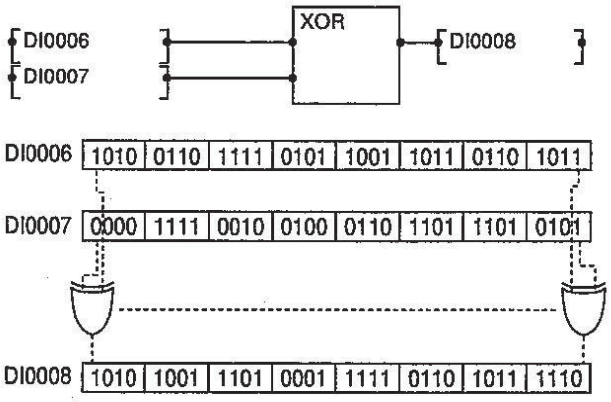
Symbol	Name
Y	Output
P0	Input
⟆	⟆
Pn	Input

Note: Whether input/output is a single-precision integer or a double-precision integer depends upon the assignment of operands.

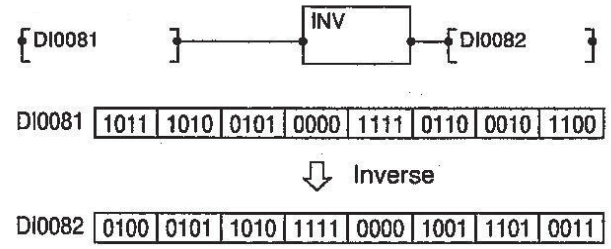
[2]	Logical sum	OR	Functions
Symbol			• P0 ∨ P1 ∨ ..... ∨ Pn → Y Obtains a logical sum of input (P0 ~ Pn) every bit, and loads the result to output (Y). 

Symbol	Name
Y	Output
P0	Input
⟆	⟆
Pn	Input

Note: Whether input/output is a single-precision integer or a double-precision integer depends upon the assignment of operands.

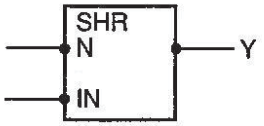
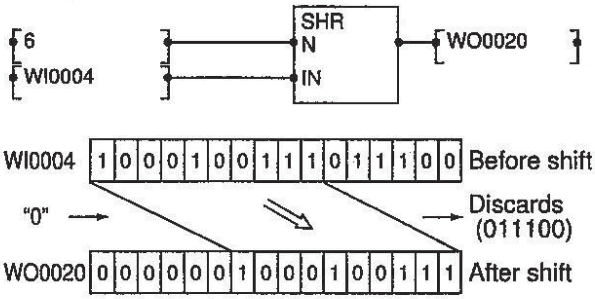
[3]	Exclusive logical sum	XOR	Functions
Symbol			• $P_0 \forall P_1 \forall \ldots\ldots \forall P_n \rightarrow Y$ Obtains a exclusive logical sum of input (P_0 ~ P_n) every bit, and loads the result to output (Y). 
Symbol	Name		
Y	Output		
P_0	Input		
≀	≀		
P_n	Input		

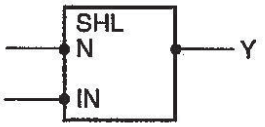
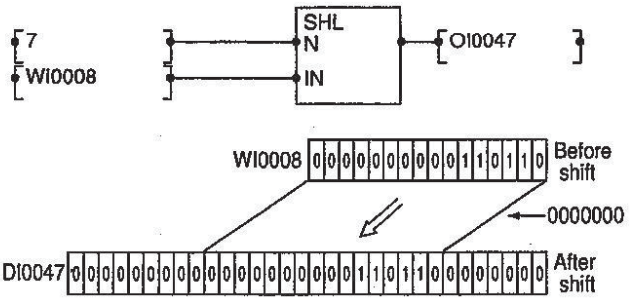
Note: Whether input/output is a single-precision integer or a double-precision integer depends upon the assignment of operands.

[4]	Inverse	INV	Functions
Symbol			• Inverses the logic of each bit of input (P_0) ("0" → "1", "1" → "0"), and loads the results to output (Y). 
Symbol	Name		
Y	Output		
P_0	Input		

Inverses the logic of each bit about DI0081 ("0" → "1", "1" → "0"), and loads the result to DI0082.

Note: Whether input/output is a single-precision integer or a double-precision integer depends upon the assignment of operands.

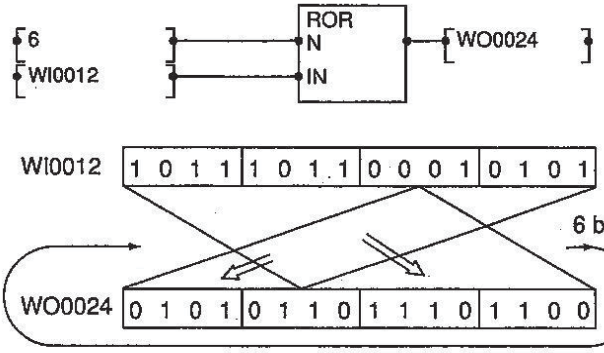
[5]	Shift right logical	SHR	Functions
Symbol			• Shifts shifted data (IN) rightward by the number of shift bits (N), and loads the result to output (Y). Loads "0" to the empty bits after shifting. 
Symbol	Name		
Y	Output		Shifts WI0004 rightward by 6 bits, and loads the result to WO0020.
N	No. of shift bits		
IN	Shifted data		
			Note: Whether shifted data and output are single-precision integers or double-precision integers depends upon the assignment of operands.

[6]	Shift left logical	SHL	Functions
Symbol			• Shifts shifted data (IN) leftward by No. of shift bits (N), and loads the result to output (Y). Loads "0" to the empty bits after shifting. 
Symbol	Name		
Y	Output		Shifts WI0008 leftward by 7 bits, and loads the result to DI0047.
N	No. of shift bits		
IN	Shifted data		
			Note: Whether input/output is a single-precision integer or a double-precision integer depends upon the assignment of operands.

[7]	Shift right circular	ROR	Functions

Symbol

- Rotates shifted data (IN) clockwise by No. of shift bits (N), and loads the result to output (Y).



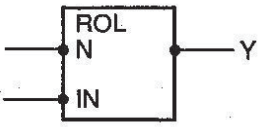
Symbol	Name
Y	Output
N	No. of shift bits
IN	Shifted data

Rotates WI0012 clockwise by 6 bits, and loads the result to W00024.

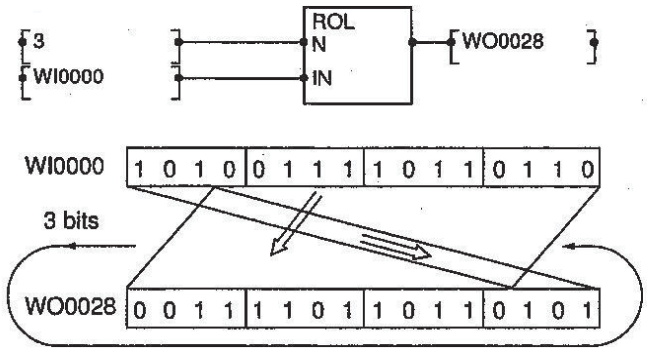
Note: Whether shifted data and output are single-precision integers or double-precision integers depends upon the assignment of operands.

[8]	Shift left circular	ROL	Functions

Symbol



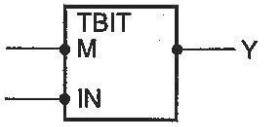
- Rotates shifted data (IN) counterclockwise by No. of shift data (N), and loads the result to output (Y).

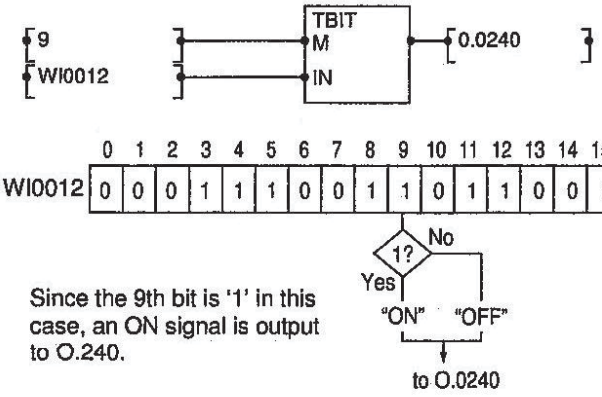


Symbol	Name
Y	Output
N	No. of shift bits
IN	Shifted data

Rotates WI0000 counterclockwise by 3 bits, and loads the result to WO0028.

Note: Whether input/output is a single-precision integer or a double-precision integer depends upon the assignment of operands.

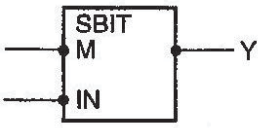
[9]	Test bit	TBIT	Functions
Symbol			• Checks whether test bit No. (M) of tested data (IN) is "1" or "0", and turns on the output relay (Y), if "1", or turns off the output relay (Y), if "0".

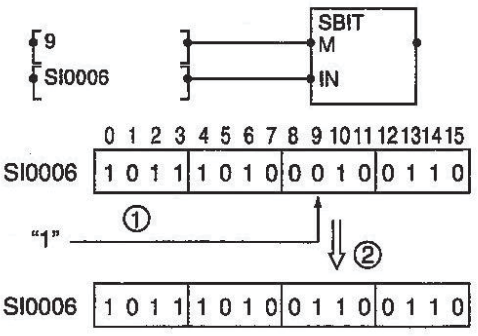


Since the 9th bit is '1' in this case, an ON signal is output to O.240.

Symbol	Name
Y	Output
M	Test bit No.
IN	Tested data

Note: Tested data and test bit No. are assigned by a single-precision integer (16 bits).

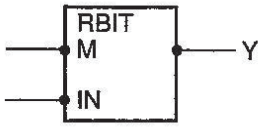
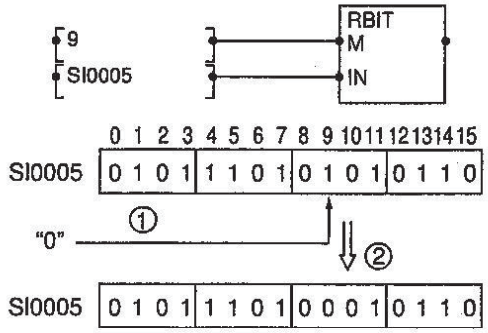
[10]	Set bit	SBIT	Functions
Symbol			• Sets "1" to set bit No. (M) of data (IN) to be set, and loads the set data to data (IN) to be set.



Sets "1" to the 9th bit of SI0006, and loads the result to SI0006.

Symbol	Name
Y	Output (Not used)
M	Set bit No.
IN	Data to be set

Note: Tested data and test bit No. are assigned by a single-precision integer (16 bits).

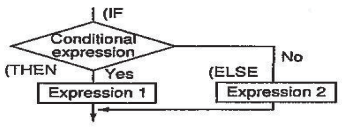
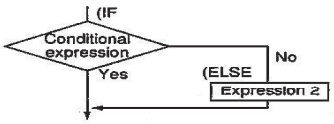
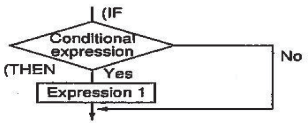
[11]	Reset bit	RBIT	Functions
Symbol			• Sets "0" to the reset bit No. (M) of the data to be reset (IN), and loads the set data to data (IN) to be reset. 

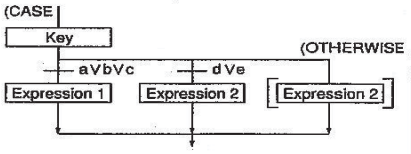
Symbol	Name
Y	Output (Not used)
M	Reset bit No.
IN	Data to be reset

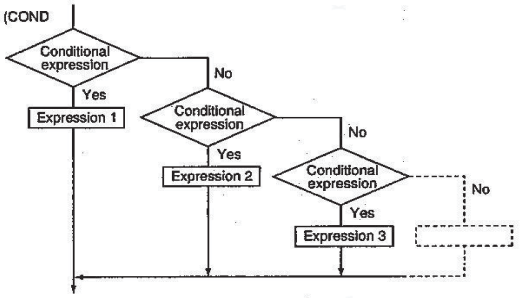
Sets "0" to the 9th bit of SI0005, and loads the result to SI0005.

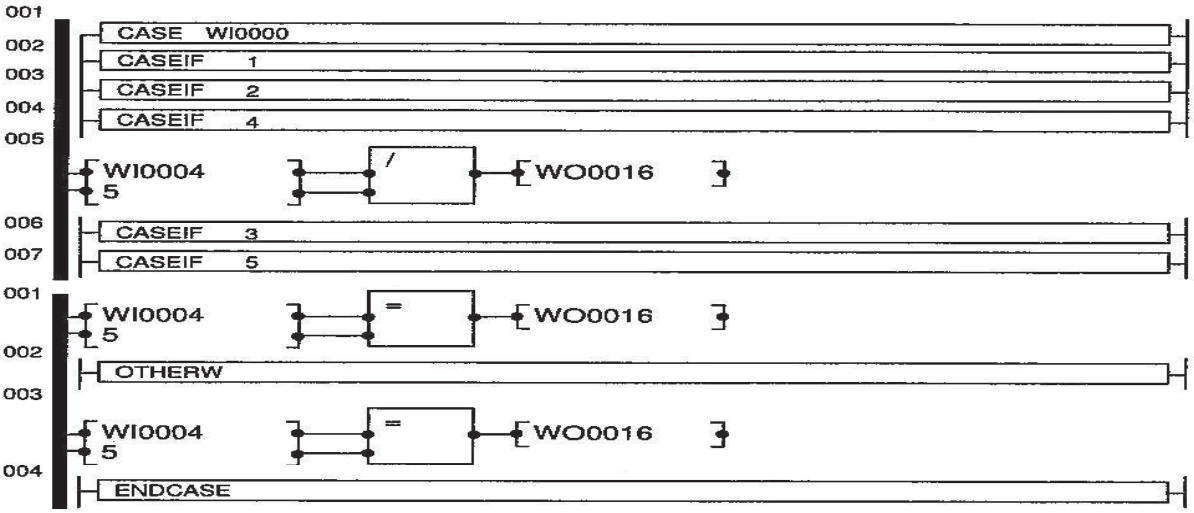
Note: The data to be reset and reset bit No. are assigned by single-precision integers.

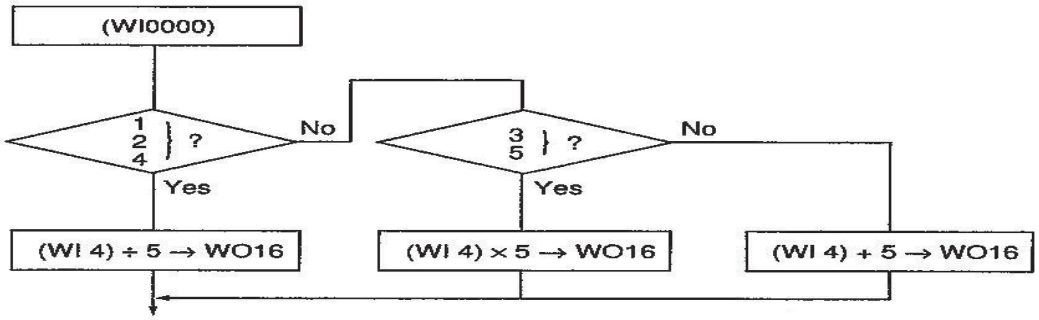
[1] Name	Description examples	Processing contents

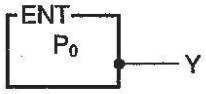
IF statement	i) Basic type
	① The conditional part indicates bit information
	IF | Condition
	Expression 1
	ELSE
	Expression 2
	ENDIF
	② Conditional part indicates an expression.
	IF | ▲
	Conditional expression
	THEN
	Expression 1
	ELSE
	Expression 2
	ENDIF
	ii) When THEN equation is absent
	③ The conditional part indicates bit information
	IF | Condition
	ELSE
	Expression 2
	ENDIF
	④ Conditional part indicates an expression
	IF | ▲
	Conditional expression
	ELSE
	Expression 2
	ENDIF
	iii) When ELSE equation is absent
	⑤ The conditional part indicates bit information
	IF | Condition
	Expression 2
	ENDIF
	⑥ Conditional part indicates an expression
	IF | ▲
	Conditional expression
	THEN
	Expression 1
	ENDIF
		i) Basic type
		 
		
		ii) When THEN equation is absent
		 
		
		iii) When ELSE equation is absent
		 
		
		Note) Condition ≠ 0 (true)
		= Expression 1
		(THEN processing)
		Condition = 0 (false)
		= Expression 2
		(ELSE processing)

[2] Name	Description examples	Processing contents
CASE statement	① Key CASE | Key CASEIF | a CASEIF | b CASEIF | c Expression 1 CASEIF | d CASEIF | e Expression 2 [OTHERWISE]* Expression 3 ENDCASE ② The key indicates an expression CASE | ▲ Key CASEIF | a CASEIF | b CASEIF | c Expression 1 CASEIF | d CASEIF | e Expression 2 [OTHERWISE]* Expression 3 ENDCASE Asterisked OTHERWISE is imissible.	 CASE statement function example i) Expression 1 is executed when a, b, or c coincides with (key). ii) Expression 2 is executed when d or e coincides with (key). iii) Expression 3 is executed when CASEIF value does not coincide with any key (OTHERWISE).

[3]	Name	Description examples	Processing contents
	COND statement	① The conditional part indicates an expression COND ▲ Conditional expression THEN Expression 1 ELSEIF ▲ Conditional expression THEN Expression 2 ELSEIF ▲ Conditional expression THEN Expression 3 ENDCOND	

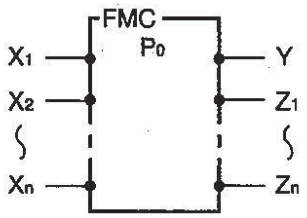
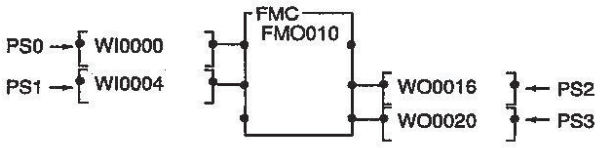
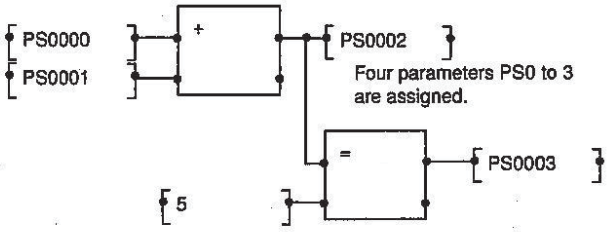




[1]	Program enter	ENT	Functions
Symbol			• By applying the ENT instruction to a user program, the other program can be started. ① Assignable program numbers are 50 to 94 (2 to 4 levels), on condition that these programs are not entered as a fixed-cycle interrupt program by system definition. ② The ENT instruction interrupts to run at the run point in a P level program. ③ The ENT instruction interrupts to run just before a program having a level lower than that of the program No. assigned by ENT is executed in case of a program of 4 level or higher.

Symbol	Name
Y	Result (omissible)
P0	Interrupt program No.

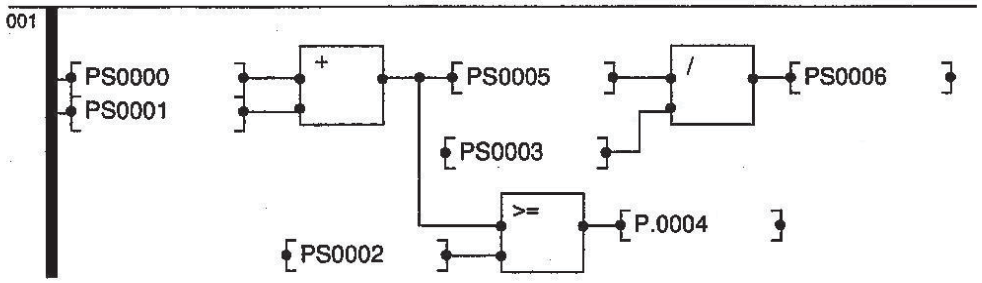
Program No. 50 of 4 level interrupts to run.

[2]	FM call	FMC	Functions
Symbol			• Call of FM of function module No. (P0) ① FM of the same No. can be repeatedly called for use. ② Other FM can be called in an FM (nesting). • Call of FM No. 10 (Written into an optional program)  • Program of FM No. 10 (To be prepared independently of program No.)  • After FM call, the parameter part runs at the following addresses PSO:WI0, PS1:WI4, PS2:WO16, PS3:WO20

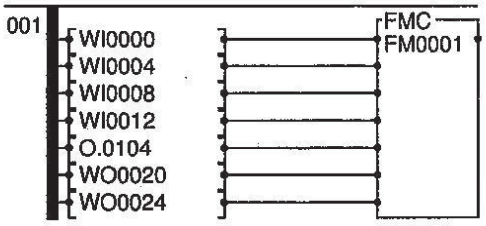
Symbol	Name
Y	Output (omissible)
P0	FM No. (0 to 511)
X1 to Xn	Input parameters (Addresses, data)
Z1 to Zn	Output parameters (Addresses, data)

(4) Function module of arithmetic operation circuit

① Entry of function module



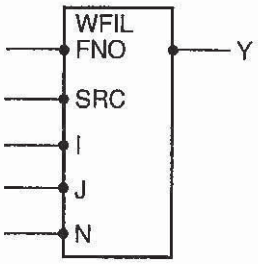
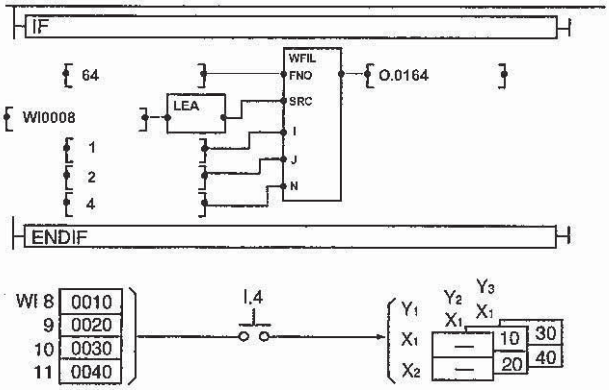
② Call of function module



The above program calls function module FM0001.

The upper right output (omitted in the above figure) of FM call (FMC) symbol shows the data of final operation register.

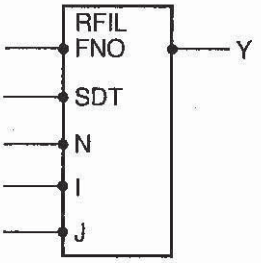
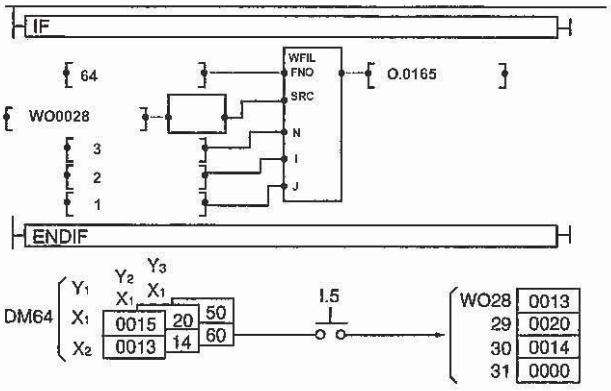
Note) Parameter numbers are sequentially assigned, starting with 0 from the upper left.

[5]	File write	WFIL	Functions
Symbol			• Transfers N words starting with SRC as the start address to X(I), Y(J) address and subsequent addresses of file No. (FNO). File pointer remains unchanged even after transfer. 

Symbol	Name
Y	Output (omissible)
FNO	File No. (36 ~ 109)
SRC	Write data start address
I	X-address of file
J	Y-address of file
N	No. of written words

File No. 64 is entered by user file—X size: 2, Y size: 3

For O.164 { "1": Could be read. "0": Could not be written or no file exists.

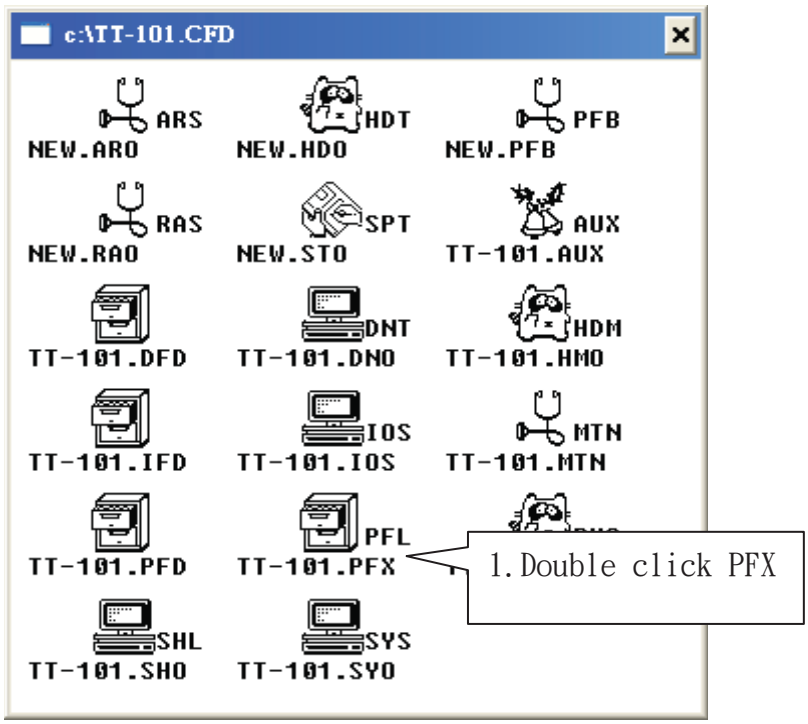
[6]	File read	RFIL	Functions
Symbol			• Transfers N words from X (I) and Y (J) addresses of file No. (FNO) to the area starting with DST as the start address. File pointer and file data remain unchanged even after transfer. This function is ineffective, if X, Y assignment or transfer area exceeds the file area. 

Symbol	Name
Y	Output (omissible)
FNO	File No. (36 ~ 109)
SRC	Start address of transfer destination
N	No. of read words
I	X-address of file
J	Y-address of file

File No. 64 is entered by user file—X size: 2, Y size: 3

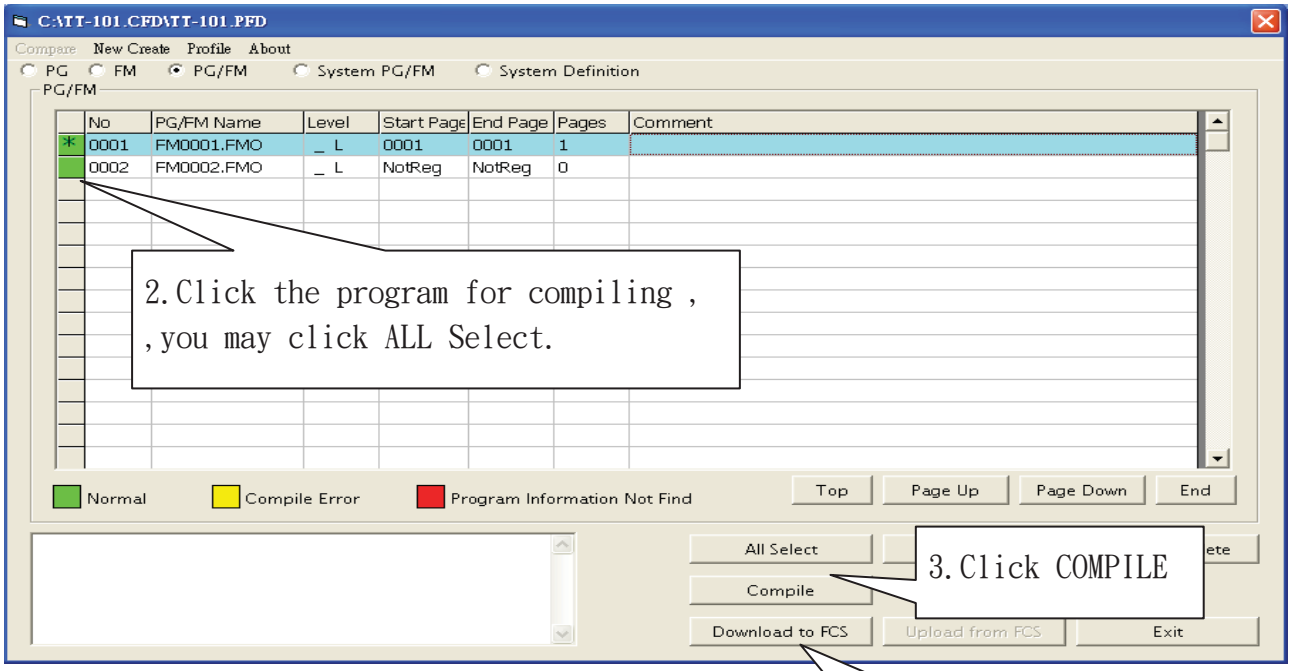
For O.165 { "1": Could be read. "0": Could not be written or no file exists.

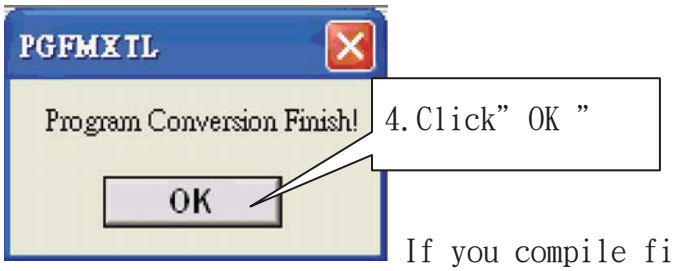
Compiling and loading in CX system program



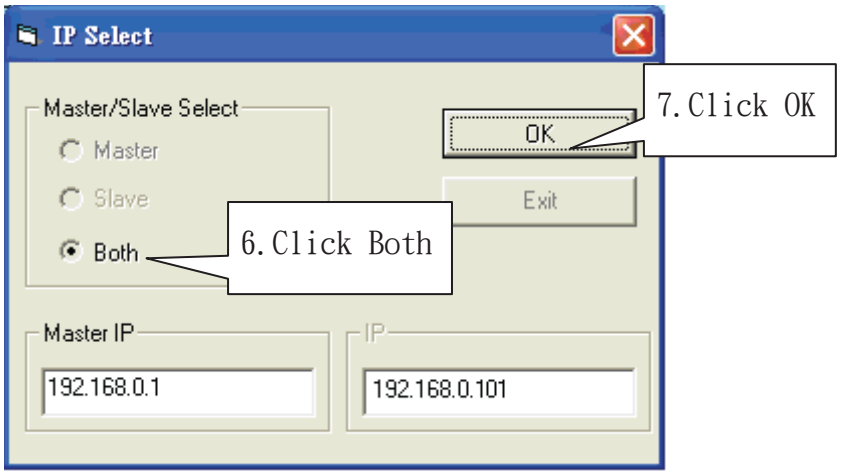
You select the program to Compile, following the steps.

1. Click Compile,after compile finish click ok, then click Download to DCS。
2. In IP Select Window select downoad to Master/Slave/Both, then click OK。



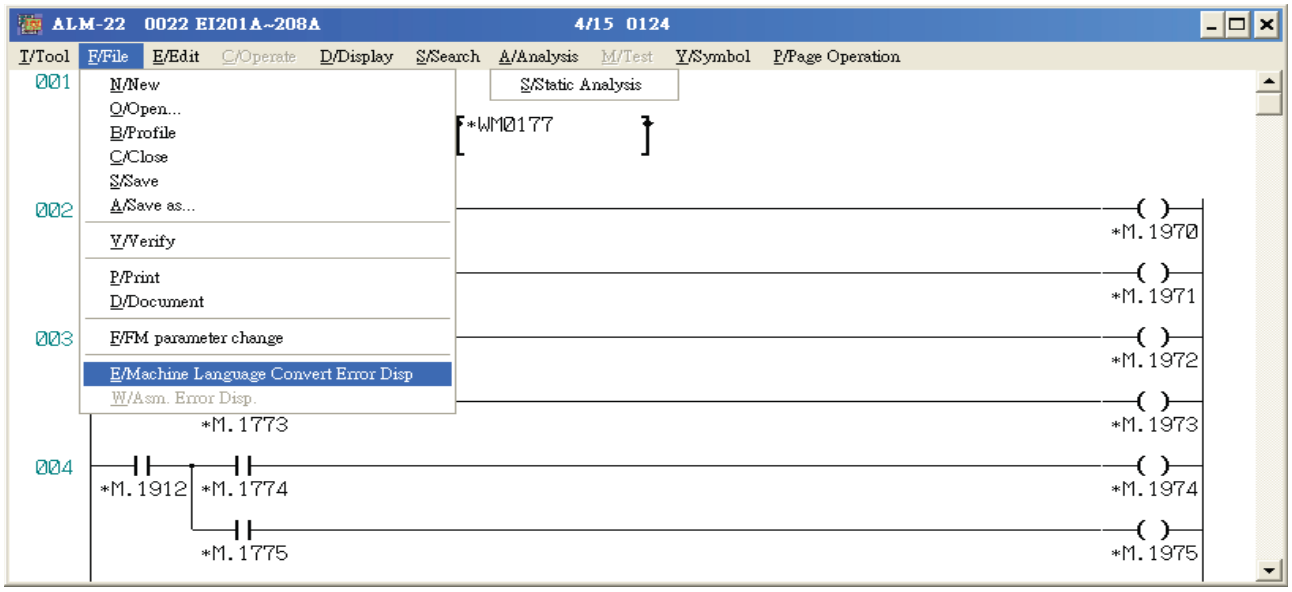


If you compile finish and no error, then click Download to FCS

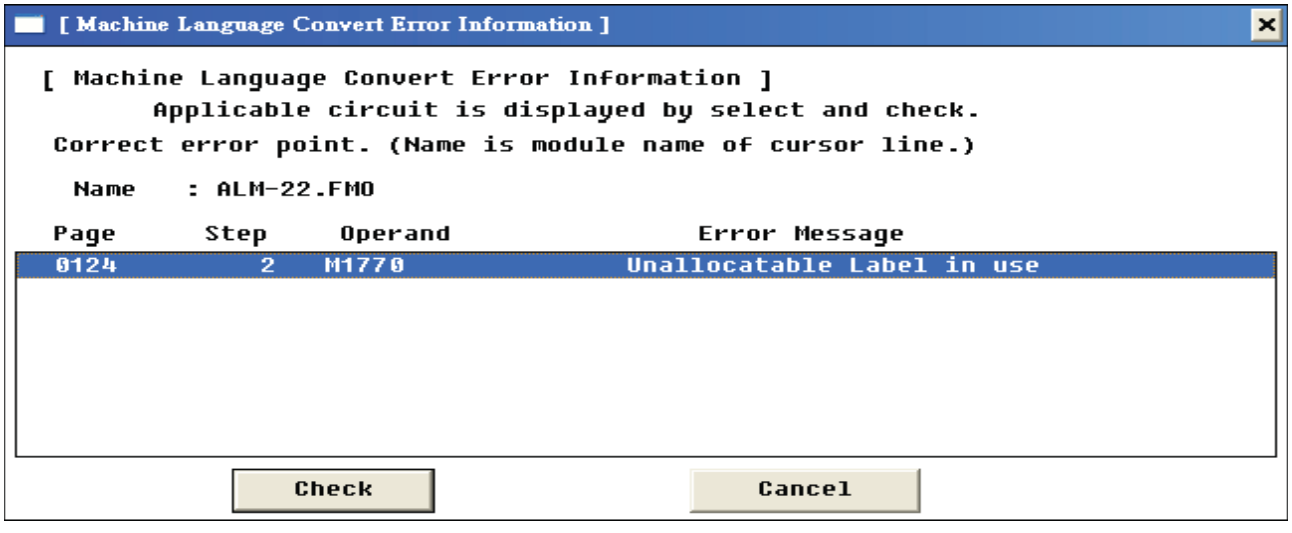


When Compile finish the mark change to green, it means Compile normally, yellow/red means error.

When yellow/red you should check your program,click F/File→E/Machine Language Convert Error Disp.

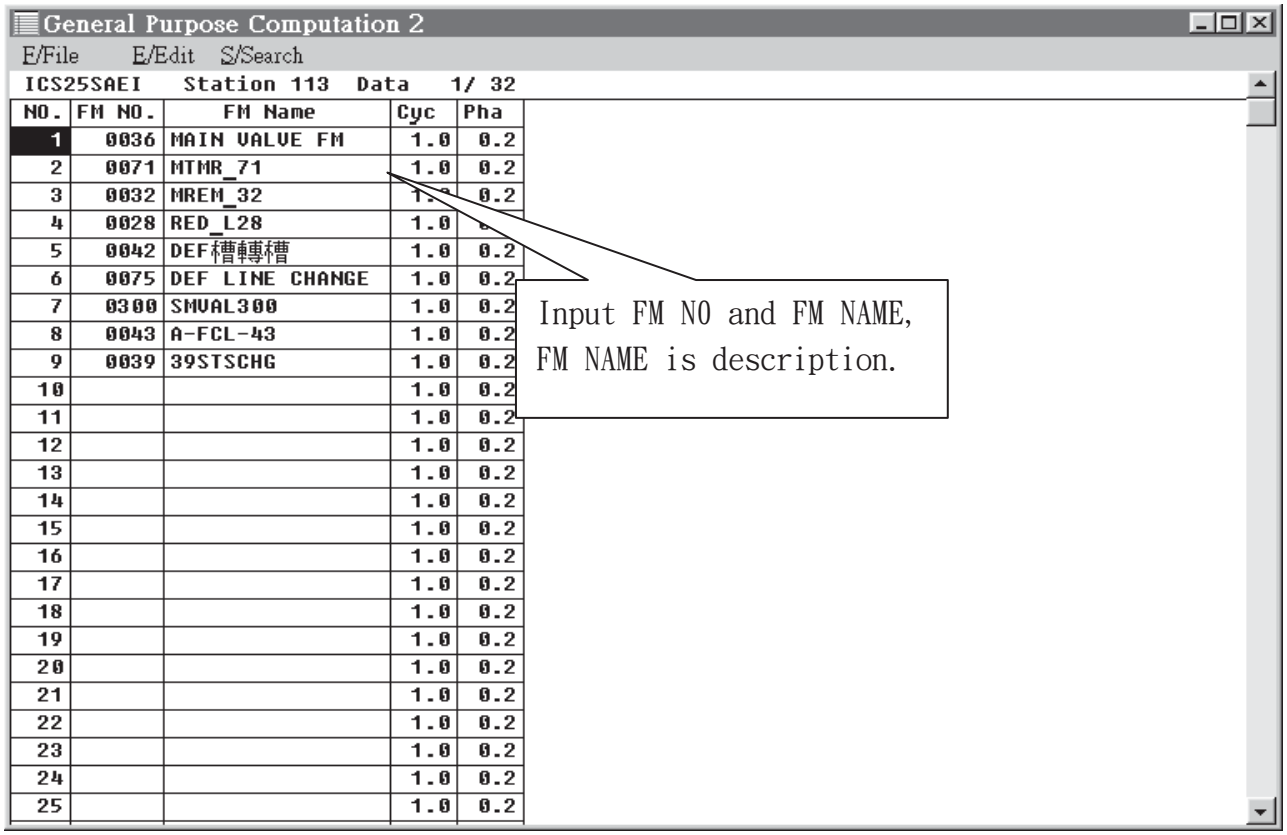


The following windown means page124, step 2 and lable M1770, it is wrong label, it need change to M.1770.



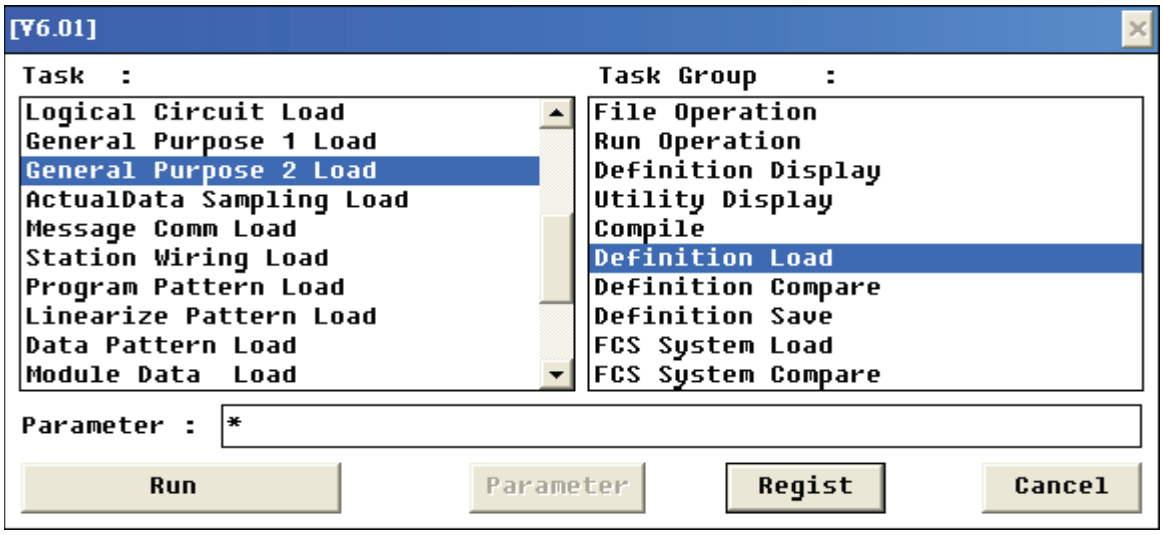
定義 GP2 :

FM program need define GP2, see the following windown.
Open C:\XXX.CFD\XXX.IFD and select GP2.G20 , input the FM number and FM name, then load GP2 to FCS.

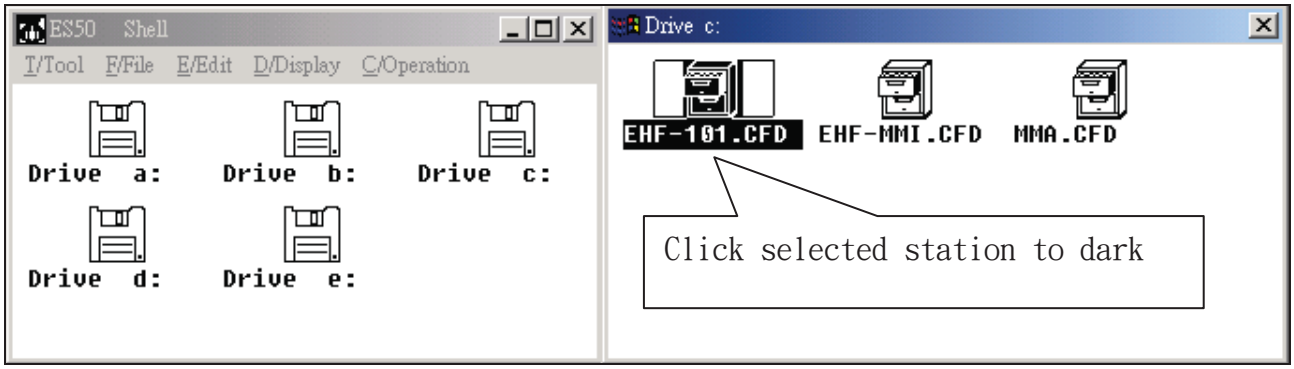


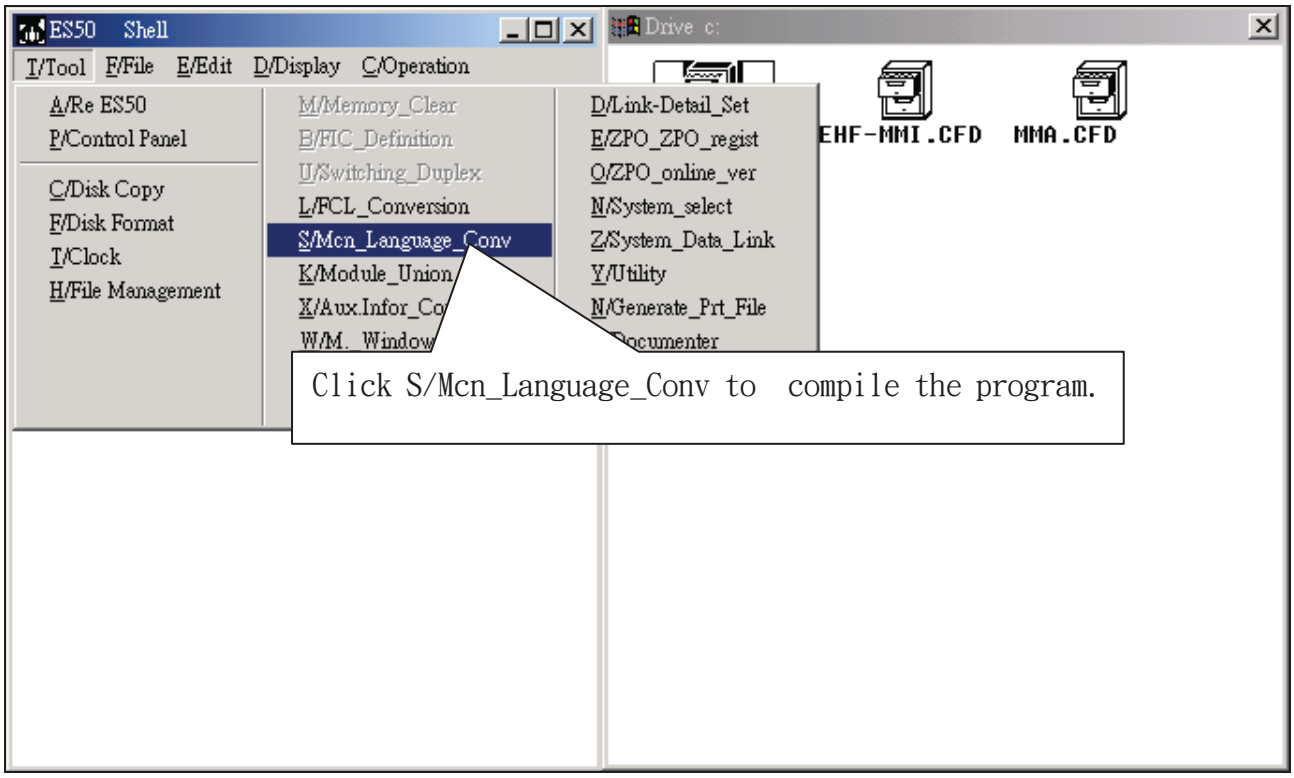
LOADING GP2 use XXX.CFD\XXX.IFD\SYSMENU.MEO

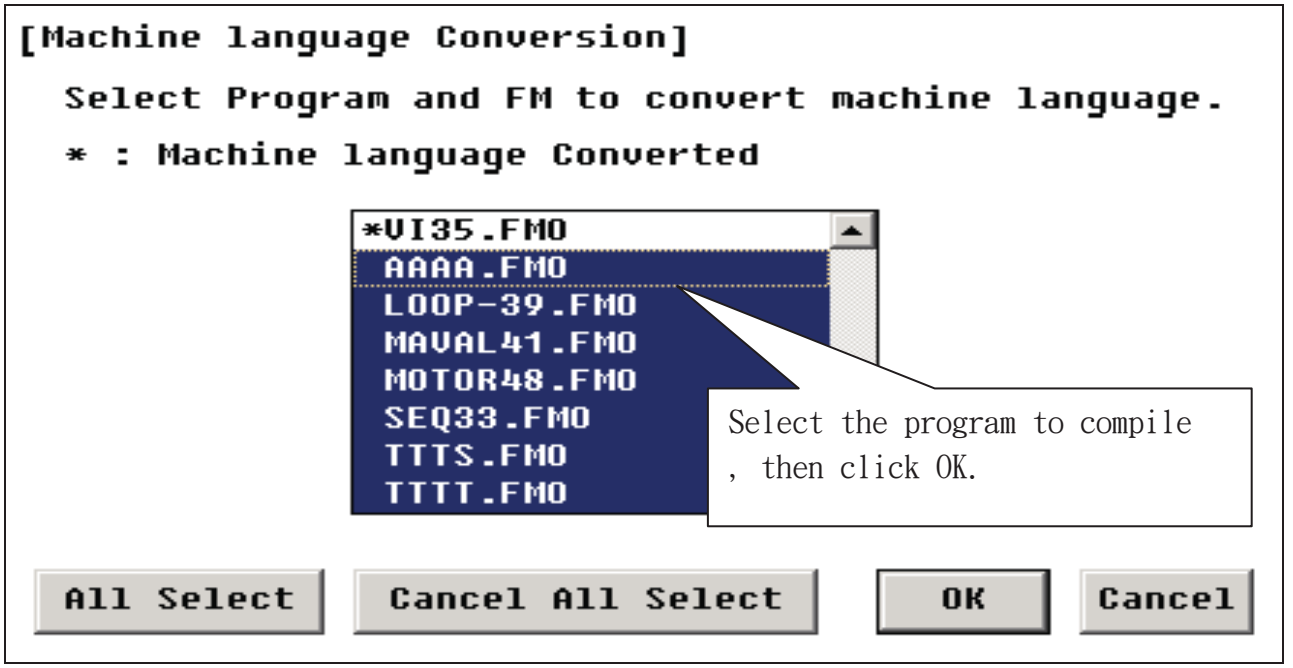
CX System For CX system we do not need open STATION.SNO and select COMMUNICATION.CX system communicate to FCS automatically, after your Compiling and loading to FCS.FM program running need call by GP2, the following train you how to Load GP2.

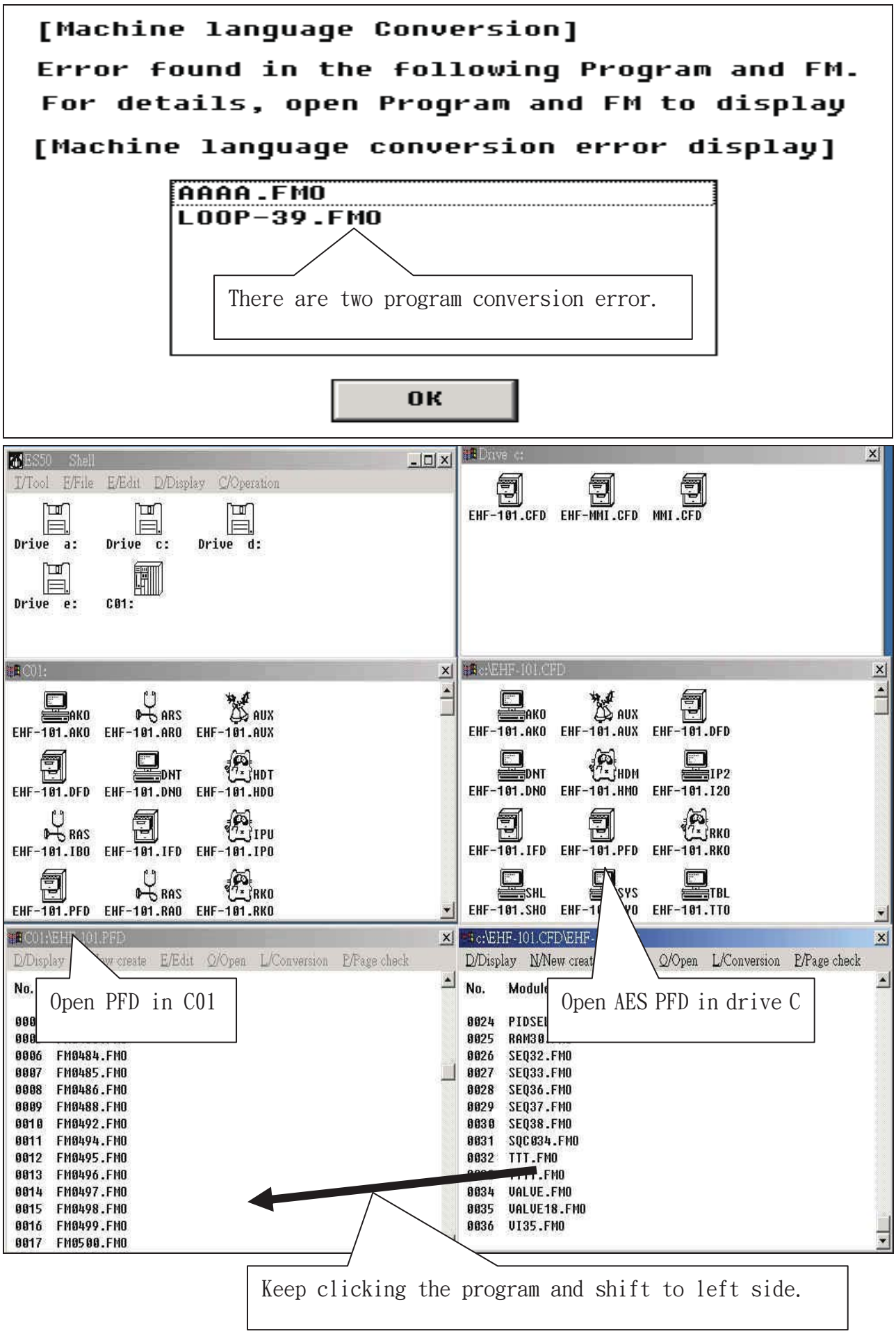


Compiling and loading in AX system program

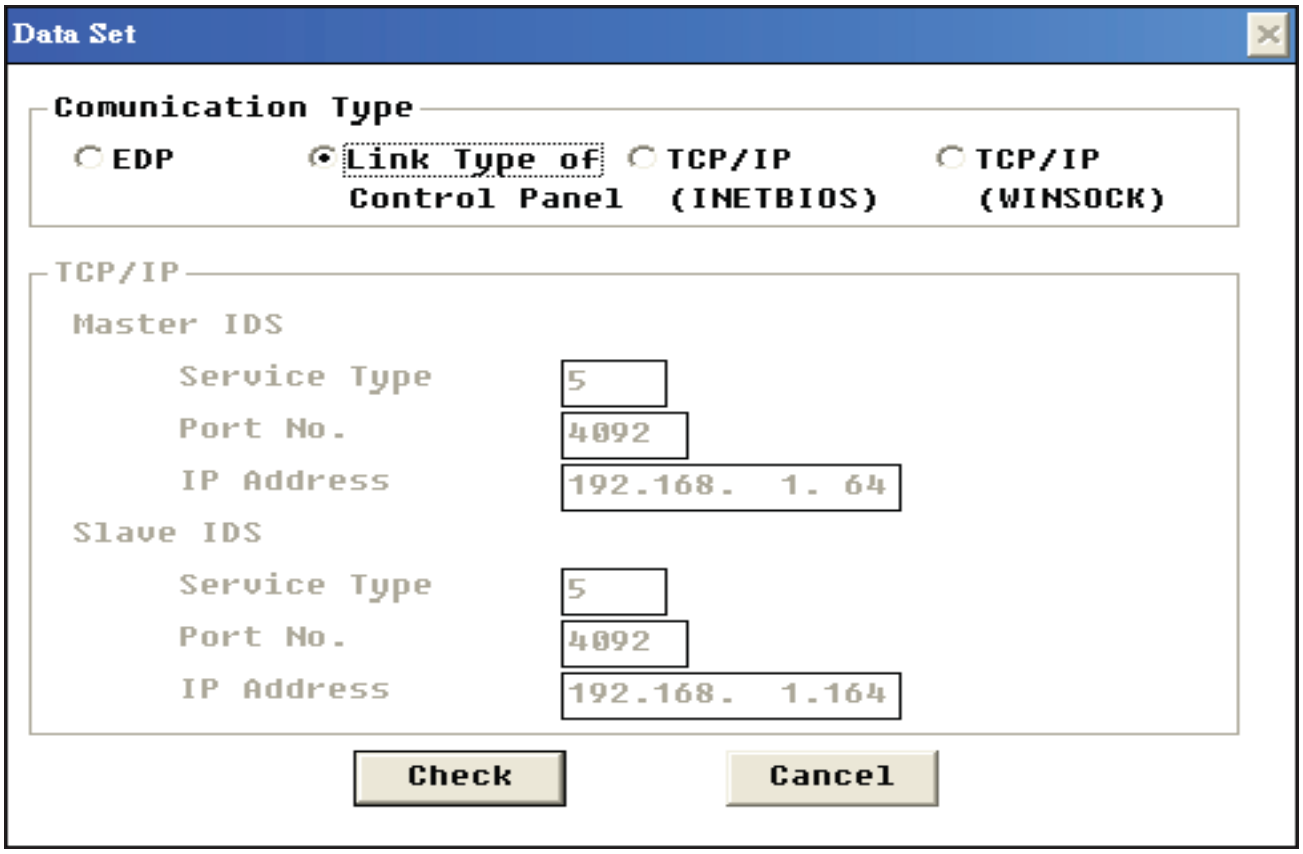




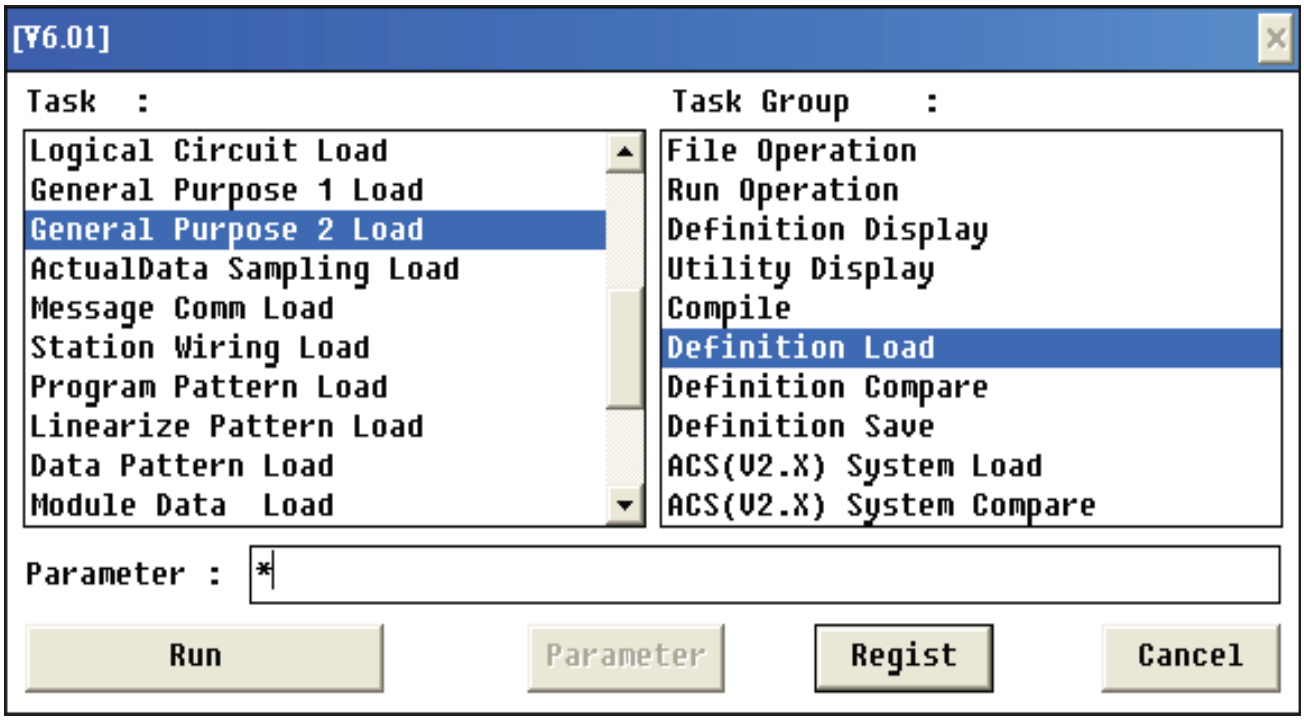



[Machine language Conversion]
Error found in the following Program and FM.
For details, open Program and FM to display
[Machine language conversion error display]
AAAA.FM0
LOOP-39.FM0
There are two program conversion error.
OK
Open PFD in C01
Open AES PFD in drive C
Keep clicking the program and shift to left side.

AX system For AX system you need select COMMUNICATION type in STATION.SNO. You may select Link Type of control panel or TCP/IP(WINSOCK).



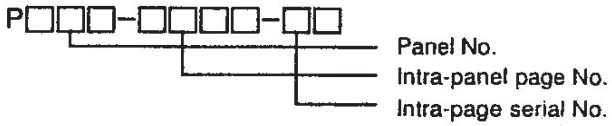
LOADING GP2 use XXX.CFD\XXX.IFD\SYSMENU.MEO, see the following.



APPENDIX

2.9 Panel No. List

The panel number is shown as table list in the following:



Panel No.		Intra-panel page No.	Intra-page serial No.
1	TAG menu panel	TAG menu page (1 to 135)	
2	Group panel	TAG menu page (1 to 135)	TAG menu page (1 to 8)
3	Loop panel	TAG menu page (1 to 135)	□□— Intra-TAG menu group No.(1 to 16) └— TAG menu group (1 to 8)
4	Plant panel	Plant page (1 to 1024)	
6	Trend panel	Trend page (1 to 256)	
7	High speed trend panel	Trend page (1 to 256)	
8	Historical message panel		
10	Historical alarm panel		
11	System condition panel (DPCS-F)	System condition page (1 to 6)	
14	System detail panel	DPCS-F line (1 to 3)+ physical station (01H~40H)	
15	Printer manage panel		
17	User parameter overview panel	User parameter overview No. (1)	
18	User parameter group panel	User parameter group page No. (1 to 8)	
19	User parameter panel	User parameter No. (1 to 8)	□□— Intra-group No.(1 to 8) └— Group(1 to 8)
20	System condition panel (FL-net)	System condition display page (1 to 6)	
21	Process value panel	Process value page (1 to 720)	
22	Run record panel	Run record page (1 to 256)	
23	Inspect Code panel	Inspect code page (1 to 4)	
24	Inspection Code panel	Inspection code page (1 to 4)	
25	Save trend panel (option)	Save trend page (1 to 32)	

Panel No.		Intra-panel page No.	Intra-page serial No.
26	Annunciator panel	Annunciator page (1 to 64)	
33	TAG menu select panel	TAG menu page (1 to 5)	
34	Plant menu panel	Plant menu page (1 to 32)	
35	Log menu panel	Log menu page (1 to 4)	
36	Trend menu panel	Trend menu page (1 to 8)	
41	Process value menu panel	Process value menu page(1 to 23)	
42	Run record menu panel	Run record menu page(1 to 8)	
43	Inspection code menu panel	Inspection code menu page (1 to 8)	
44	Save trend menu panel (option)	Save trend menu page (1 to 2)	
45	Annunciator page priority order call		
46	Annunciator panel page call		
57	MMI select TAG display panel		
59	Bit status display panel	Bit status signal page (1 to 3840)	

Note: For the panel linkage, enter the line number and physical station number in sequential numeric values.

(Examples: In case where the line number is 1 and physical station No. is 40H, enter 140).

Function Codes (1/2)

Keyboad. Touch. Virtual key: 0x00**
* The vertical is the value of the lower second digit, the horizontal of the lower first digit.

H ↓ L →

	0	1	2	3	4	5	6	7	8	9	A	B	C	D	E	F
0	TAG	GROUP	LOOP	PLANT	LOG	TREND	ALARM	ALM HIST	RUN HIST							
1	SYS DPC	SYS DETAIL	SYS P/PE	SYS P/PE/T	PRINT											
2																
3	SUPPT		MMI TAG	BIT STATUS	SYS FUNC	CON TROL		FILLE	MMI DETAIL							
4	ALM STOP	ALM CHECK	ALM CLR		PAGE	HOME	PAGE BACK	PAGE FWD	UNDO	REDO	DISP	GUIDE	DATA	CLR	CHECK	EXEC
5			PNL SEND		TOUCH ON/OFF	PNL CLR	PNL MEM		ONE TOUCH							
6	TAG SELECT	TREND MENU	PLANT MENU	LOG MENU												
7																
8																
9																
A																
B																
C																
D																
E	PV MENU	REC MENU	INS MENU						PV	RUN REC	INS-PECT					
F																

Note: These are function codes (parameters unused) with virtual keyboard (2)which can be used for plant panels.

Function Codes (2/2)

Plant panel: 0x02**

* The vertical is the value of the lower second digit, the horizontal of the lower first digit.

H ↓ L →

	0	1	2	3	4	5	6	7	8	9	A	B	C	D	E	F
0																
1																
2																
3																
4																
5																
6																
7																
8																
9																
A	ITV1 CALL	ITV2 CALL	ITV1 DW CALL	ITV2 DW CALL												
B																
C																
D																
E																
F																

Note: The function code in the virtual keyboard 2 that can be used in the Plant panel.
0x02A1, 0x02A1 has parameter. For the parameter specification method, please refer to virtual keyboard 2.

Instruction Manual

FORMOSA - FX

MICREX-VieW/H
HCI Support Edition

Formosa Plastics Corp.

INC-TN5A2098

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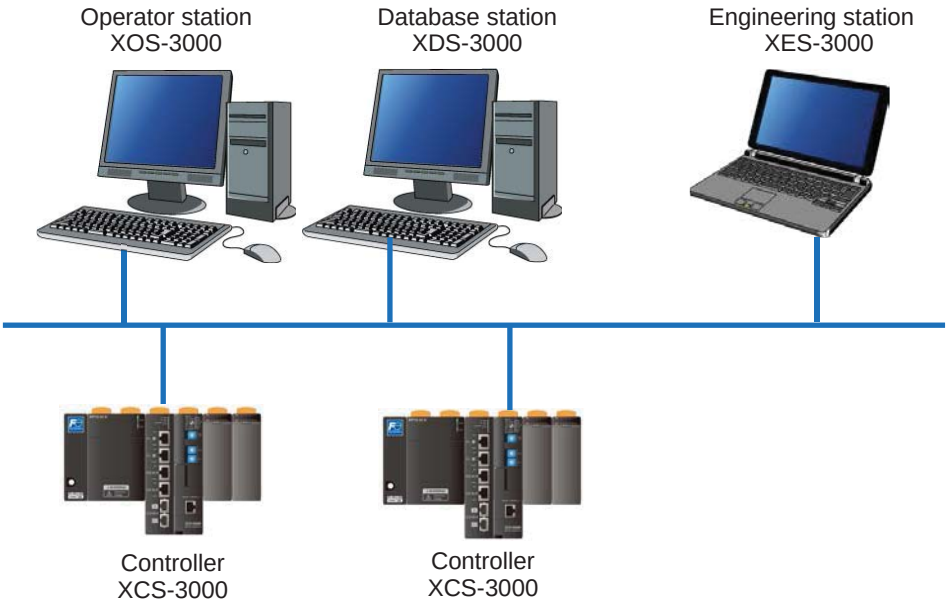
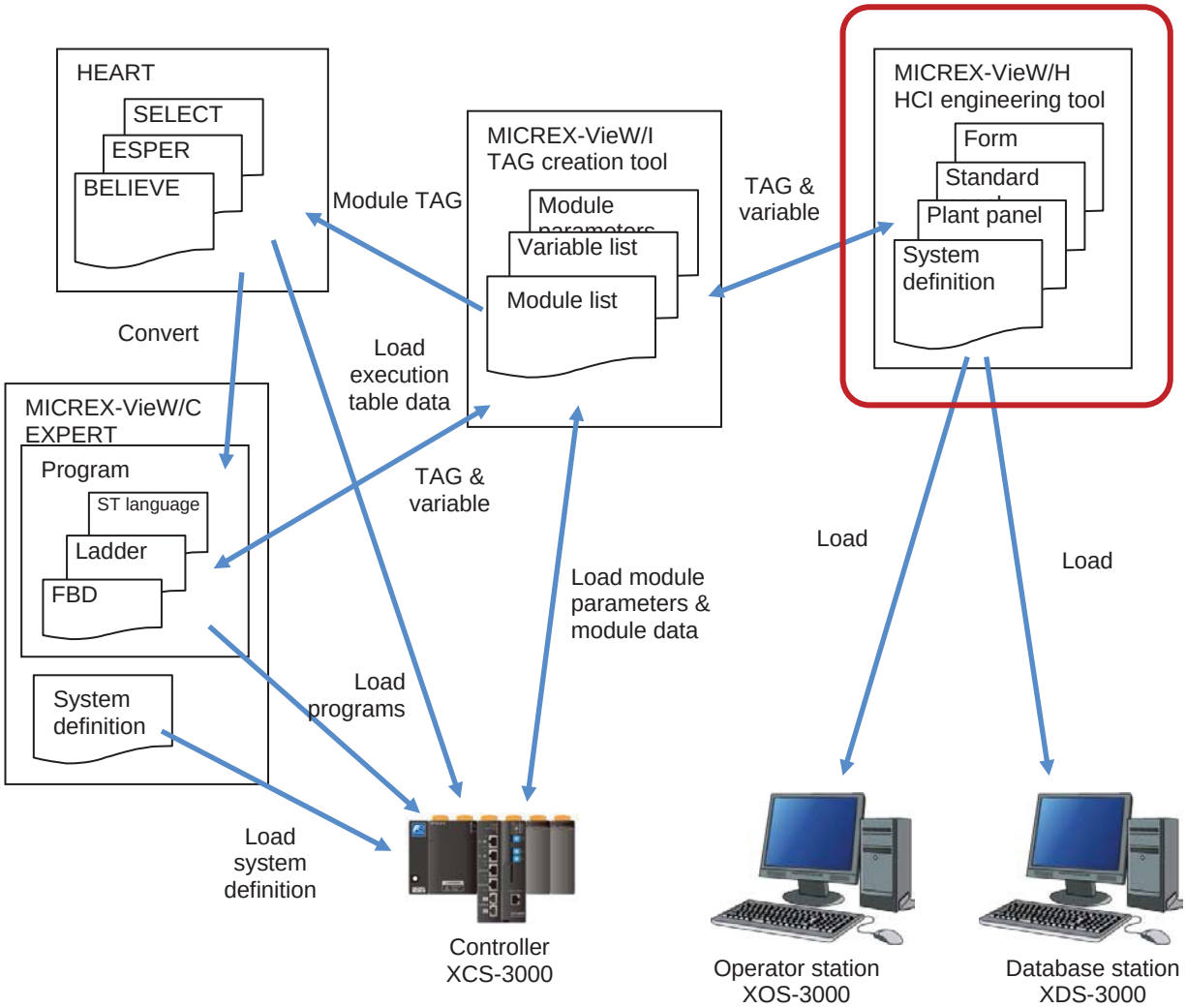
1. Overview

This chapter describes the function to define various settings for HCI engineering, the common control system.





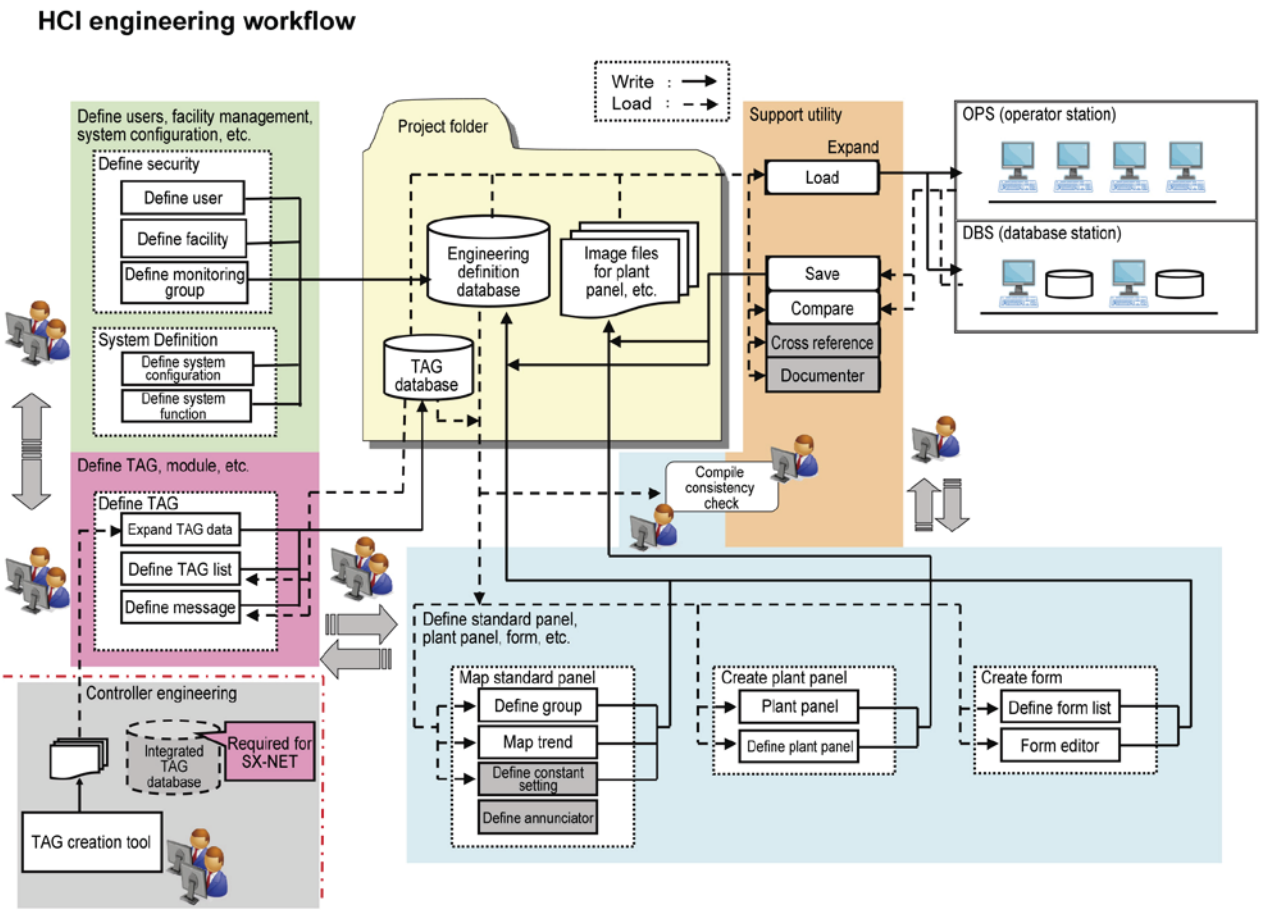
1.1 HCI Engineering Capability

The HCI engineering capability provides the functions below to engineer common control system.

Large category	Medium category	Small category	Remarks
Map standard panel	Define TAG menu	Define TAG menu panel group	Defines groups for TAG menu panel
	Define group	Define group in group panel	Defines groups in group panel
		Map group panel	Defines TAGs to be displayed in group panel
	Define trend, define high speed trend	Define trend record	Defines the collection cycle of data to be displayed in the trend panel and high speed trend panel
		Define trend panel group	Defines groups of the trend panel and high speed trend panel
		Map trend	Defines TAGs to be displayed in the trend panel and high speed trend panel
Define security	Define user		Maps user definitions and user group definitions
	Define facility category		Maps definitions of facility categories
	Define monitoring group		Maps definitions of user monitoring groups
Define system	Define system configuration		Defines system configuration
	Define common system	Operator station	Defines operations of operator station
		Database station	Defines operations of database station
		Common	Sets up language, display format, etc.
		Order of function menu display	Sets the order of standard menu display in HCI panel
		Set function display	Sets the order of display of toolbox and others
Support utility	Compile		Converts engineering data into online definitions
	Load		Distributes online definitions to each online equipment
	Save		Stores online definitions in engineering data
	Compare		Compares engineering data with online definitions

1.2 Engineering Workflow

The figure below shows an engineering workflow.



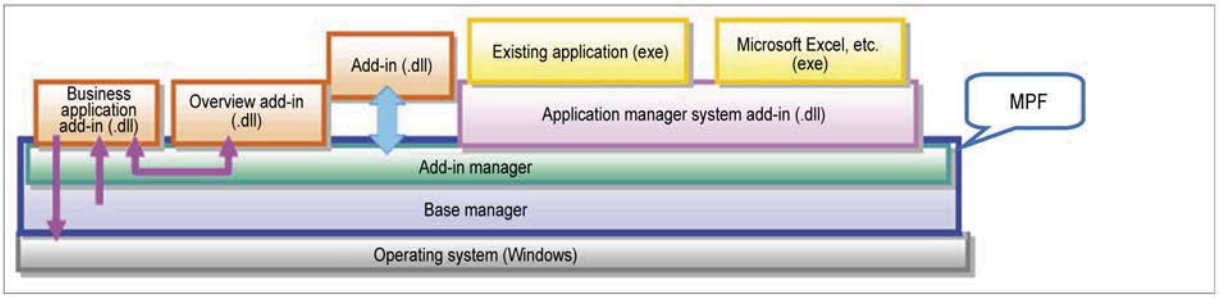
2. Software

This chapter describes the screen configuration for HCI engineering capability and the window functions.

2.1 Basic Construction

The HCI engineering capability is realized by “Multi panel framework” (hereinafter "MPF"), the HCI perform for common control system, and “Add-ins”, the program modules running MPF.

The figure below shows MPF (enclosed with blue frame) and add-ins (enclosed with red frame).

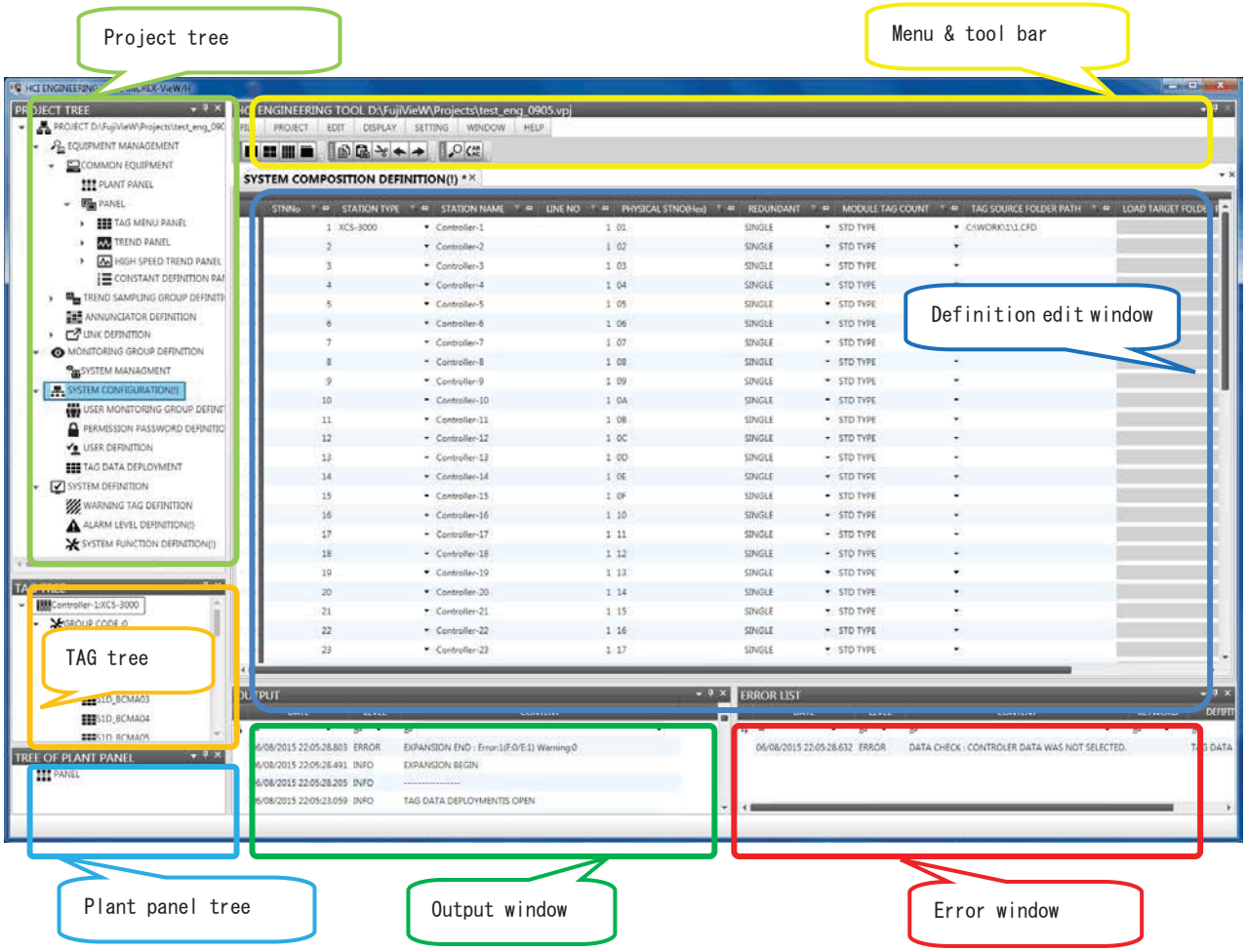


2.2 Screen Configuration

The table below lists components of the HCI engineering screen.

Panel	Window type	Description
Menu	Tool window	Specifies (create new or open) the objects of tool and displays menus for loading, saving, comparing objects.
Project tree	Tool window	Decides the definition to edit and displays a project tree for opening definition edit window.
TAG tree	Tool window	Displays the TAG tree for registration by dragging and dropping TAGs onto the project tree or definition edit window.
Plant panel tree	Tool window	Displays the plant panel tree for registration by dragging and dropping plant panels onto the project tree or definition edit window.
Definition edit window	Document window	Displays the window for editing definitions in table format.
Output window	Log viewer	Displays process statuses and results on the log viewer by using the log service of MPF main program.
Error window	Log viewer	Displays error details on the log viewer by using the log service of MPF main program.

The screen components are laid out as shown in the figure below. The panel layout can be changed.



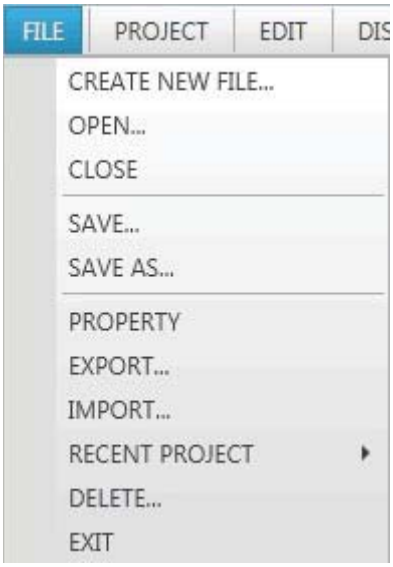
2.3 Menu

The table below lists menu items of the HCI engineering capability.

Menu	Item	Description
File	Creating New Project	Creates a new project file.
	Open	Opens an existing project.
	Close	Closes an open project.
	Save	Overwrites the open project data.
	Save As	Saves the open project data with another name.
	Export	Outputs the contents of the definition edit window displayed to external file (CSV format).
	Import	Imports external file (CSV format) into an open project.
	Recent Project	History of projects opened.
	Delete	Removes an existing project.
	Exit	Exits the HCI engineering tool.
Project	Compile	Displays the support utility panel.
	Cross Reference	Displays the support utility panel.
	Documenter	Displays the support utility panel.
	Load/Save/Compare/Change Online	Displays the support utility panel.
Edit	Copy	Copies the selected item. (Copies to clipboard.)
	Paste	Pastes the contents copied on the selected item. (Pastes clipboard.)
	Cut	Cuts the selected item. (Removes and copies to clipboard.)
	Undo	Puts previous operation back.
	Redo	Performs final operation undone again.
	Find	Performs a search under specified conditions in the selected range and outputs the result to the search result window.
	Replace	Performs the search above and replaces with new data. (select this in the search result window)
View	Tree	Displays or hides a tree window, such as "Project tree" and "TAG tree".
	Window	Displays or hides a window, such as "Output", "Error list", "Search result", "Cross reference", and "Compare result".
	Plant panel editor	Starts the Plant panel editor.
Settings	Settings	Displays and changes various settings for this software.
Window	<Opened window list>	Displays the opened window list in the pull-down menu to change windows.
Help	About	Displays the version information.

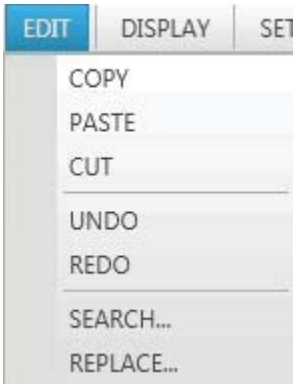
[Menu (top) and tool bar (bottom)]

[File menu]

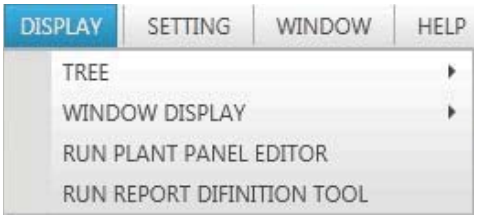


[Project menu]

[Edit menu]

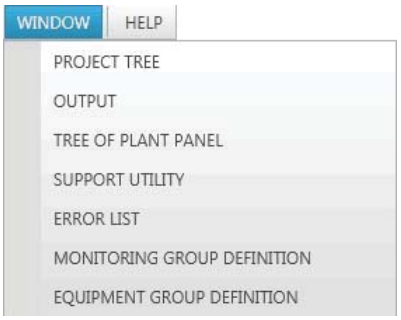


[View menu]



[Settings menu]

[Window menu]



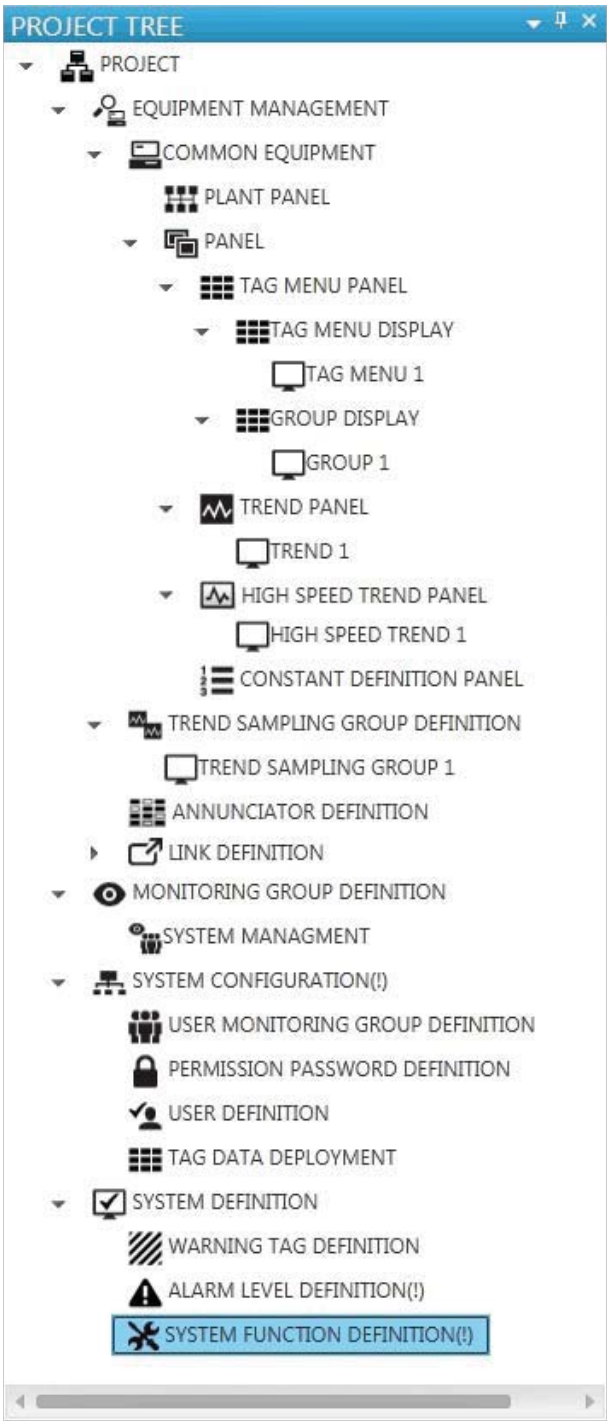
[Help menu]

2.4 Project Tree

The table below lists items on the project tree.

Large category	Medium category	Small category	Description
Facility management			Displays definition items for facilities in hierarchical structure.
	Facility category		Displays panels associated with facility categories in hierarchical structure.
	Panel	TAG menu panel	Displays groups associated with the TAG menu panel of the facility category.
		Group panel	Displays groups associated with the group panel of the facility category.
		Trend panel	Displays groups associated with the trend panel of the facility category.
		High speed trend panel	Displays groups associated with the high speed trend panel of the facility category.
	Define trend collection group		Defines the definition panel to set up trend collection groups.
Monitoring group management			Displays definition items for monitoring group in hierarchical structure.
	Monitoring group		Displays facility categories associated with the monitoring group in hierarchical structure.
System configuration			Displays the definition panel of the configuration of systems, such as XDS and XOS.
	Map user monitoring group		Displays the definition panel for associating users with monitoring groups.
	Authority password management		Displays the definition panel to set up a password for each system authority.
	Define user		Displays the definition panel to register users and set up passwords and authorities.
Define system	Trend view		Displays the definition panel to set up the display of the trend panel.
	Loop view		Displays the definition panel to set up the display of the loop panel.
	Tool box		Displays the definition panel to set up the tool box.
	Function menu display order		Displays the definition panel to set up the display of function menu.
	Alarm indicator		Displays the definition panel to set up the alarm indicators.
	Tag		Displays the definition panel to set up the tags.
	Language		Displays the definition panel to set up language.
	Expand TAG data		Displays the Expand TAG data panel.

[Project tree example]

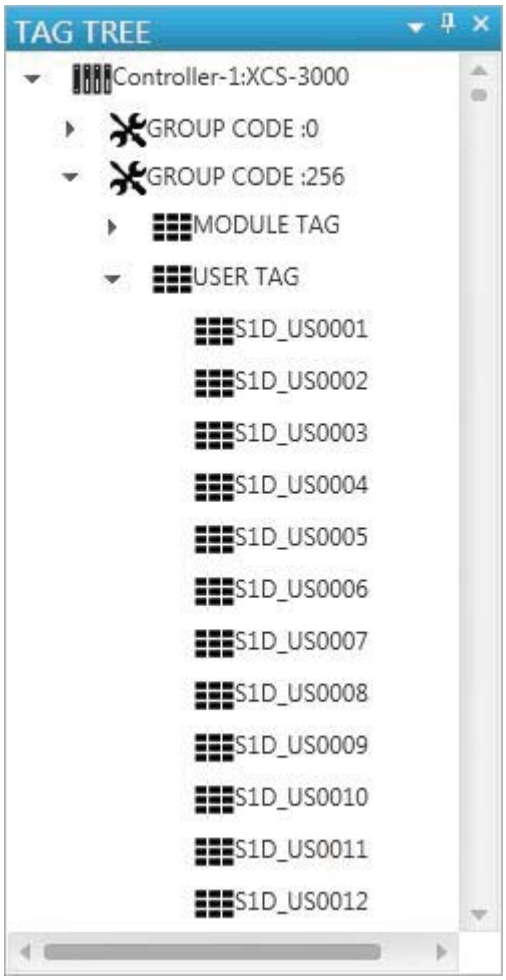


2.5 TAG Tree

The table below lists items of the TAG tree.

Large category	Medium category	Small category	Description
Controller			Displays the controller name mapped to a TAG. Also, a model name is displayed at the end of the controller name. (Example: "XX controller: model CODE")
	Category		Displays the category code of a TAG.
		TAG name	Displays the TAG name. Also, a tool tip displays remarks on the TAG.

[TAG tree example]

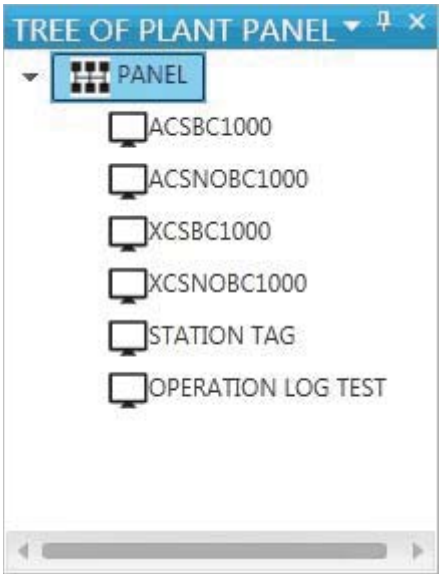


2.6 Plant Panel Tree

The table below lists items of the plant panel tree.

Large category	Medium category	Small category	Description
Panel management folder			Displays the folder (group) name for managing the plant panel.
	Plant panel		Displays the plant panel name.

[Plant panel tree example]



2.7 Tree Edit Operation (Common Specification)

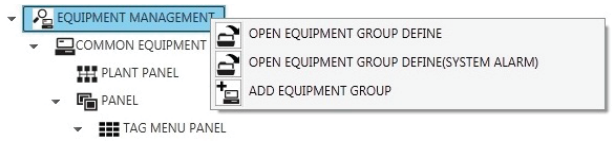
This section describes common operations for the edit function of the HCI engineering capability.

The table below lists the operations for the project tree.

Operation
Right-click to open the definition edit window from the pop-up menu.
Double-click to open the same window as that opened by the Open pop-up menu.

This section describes the pop-up menu of each node in the Project tree.

[Facility management]

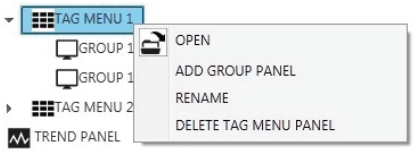


[Facility category]

[Plant panel]

[TAG menu panel]

[Group panel]



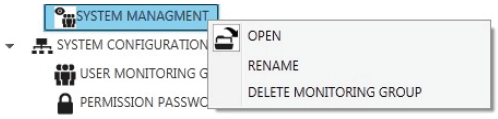
[Trend panel]

[High speed trend panel]

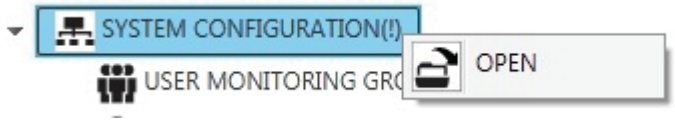
[Define trend collection group]

[Monitoring group management]

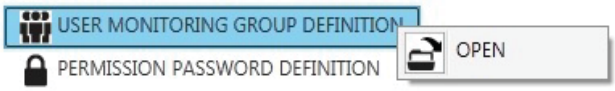
[System management]



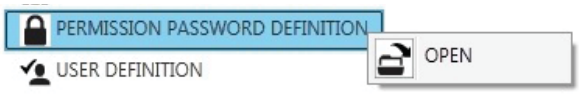
[System configuration]



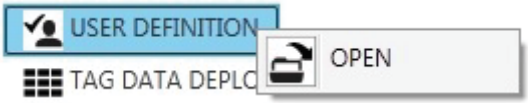
[Map user monitoring group]



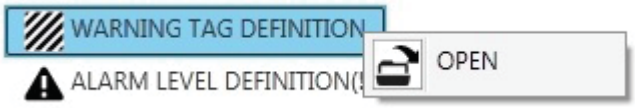
[Authority password management]



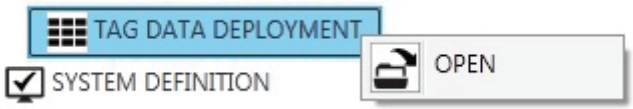
[Define user]



[Tag]



[Expand TAG data]



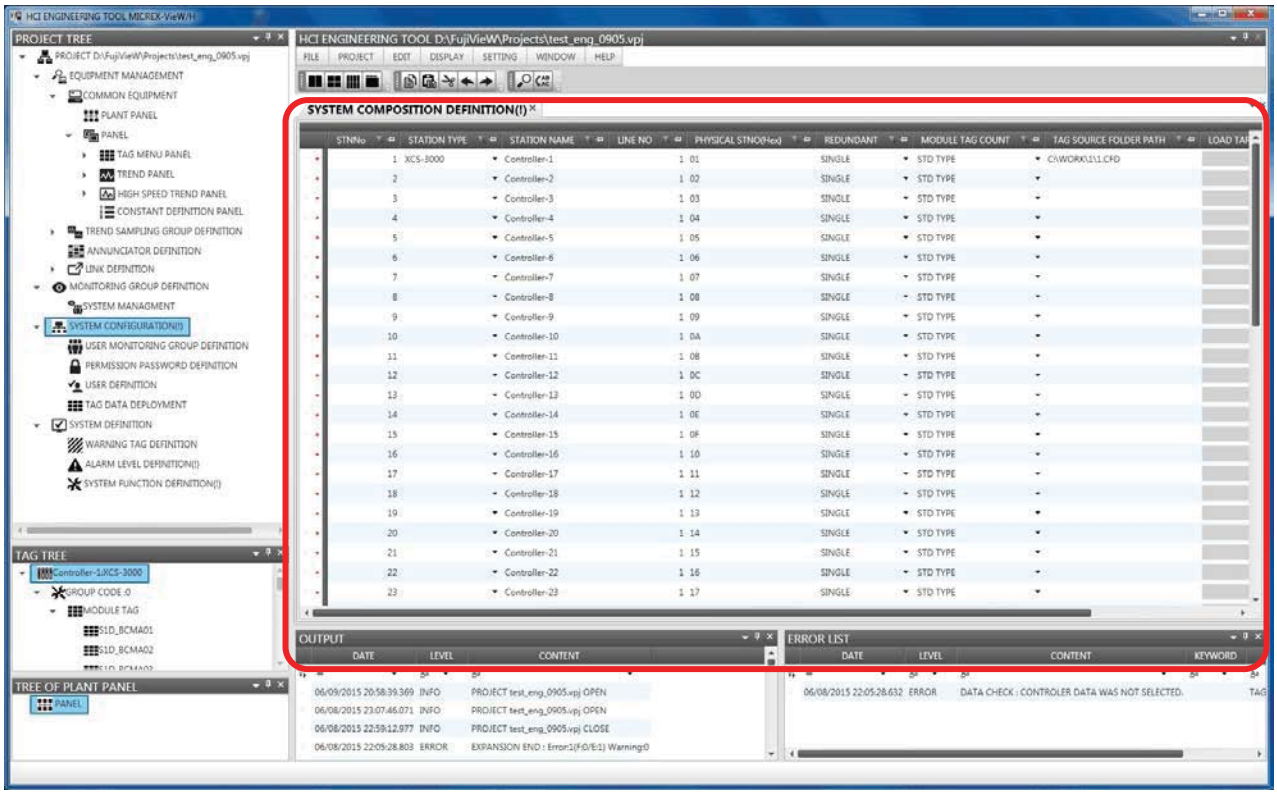
[Controller]

[Plant panel]

2.8 Definition Edit Window

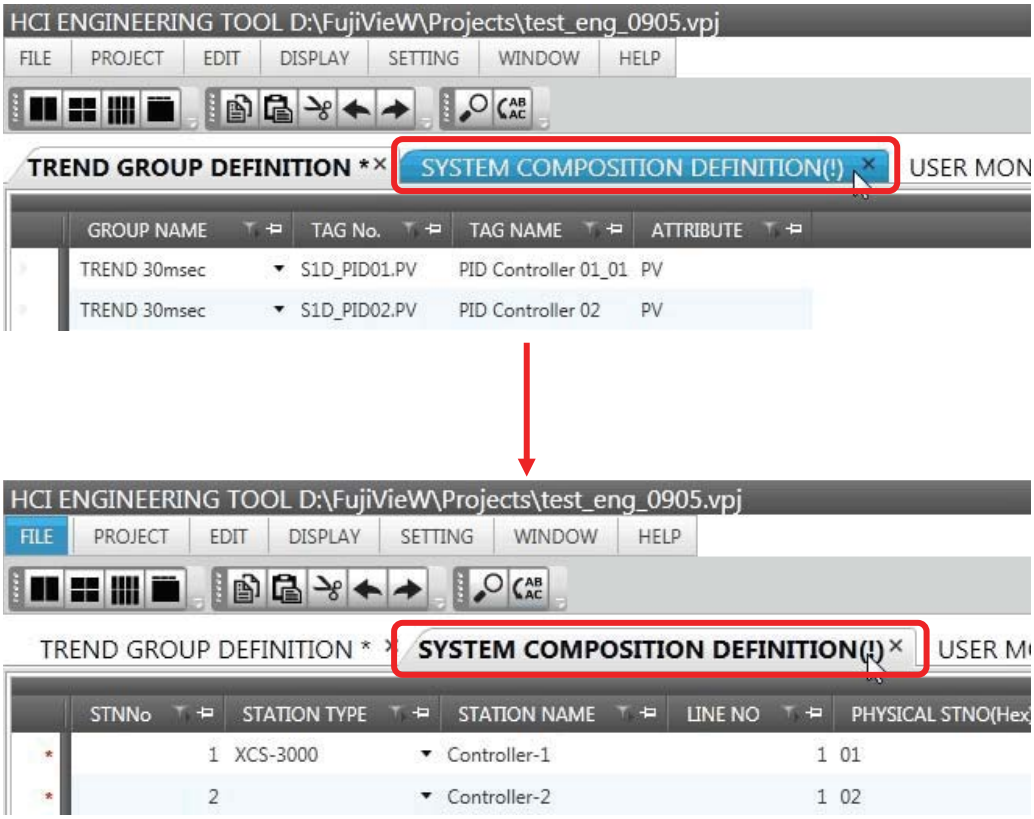
The definition edit window edits items selected in the Project tree or TAG tree.

[Definition edit window example]



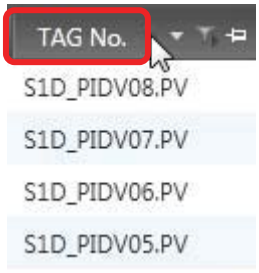
The definition edit window enables you to switch views by using a panel title shown in tab format.

[Switching example from “Define trend group” panel to “Define system configuration” panel]



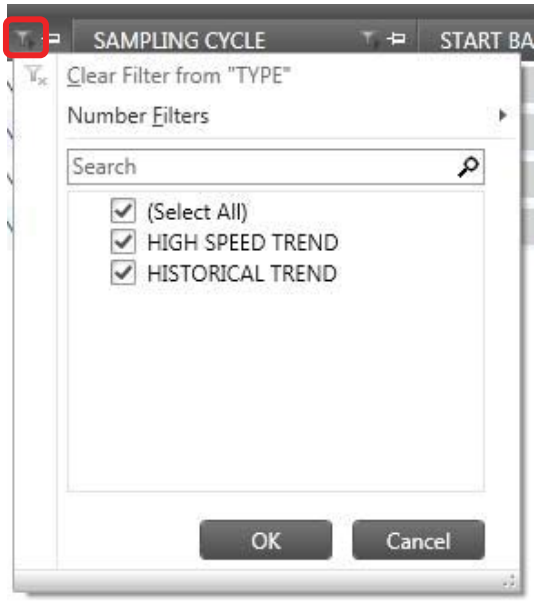
Sort

Click the title section to sort the data in ascending or descending order.



Narrow

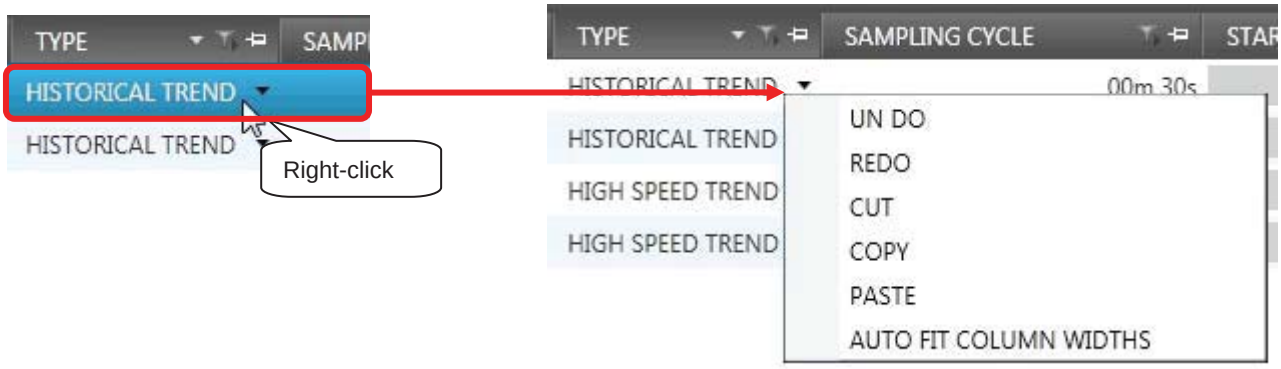
Click the funnel icon to narrow rows by the selected column's data.



Operate the pop-up menu to edit features common to the definition edit window.

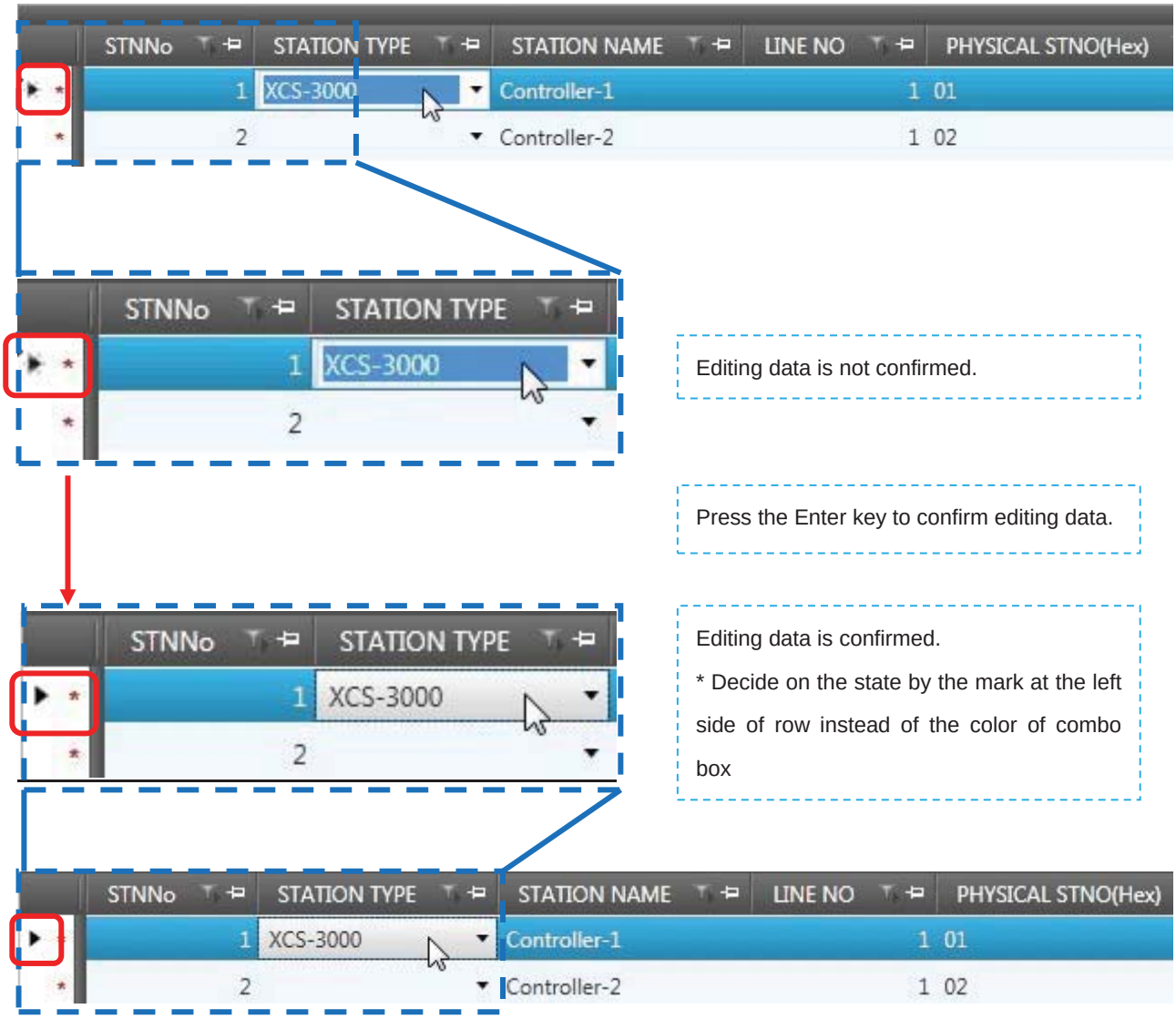
pop-up menu

Right-click an item to display a menu.



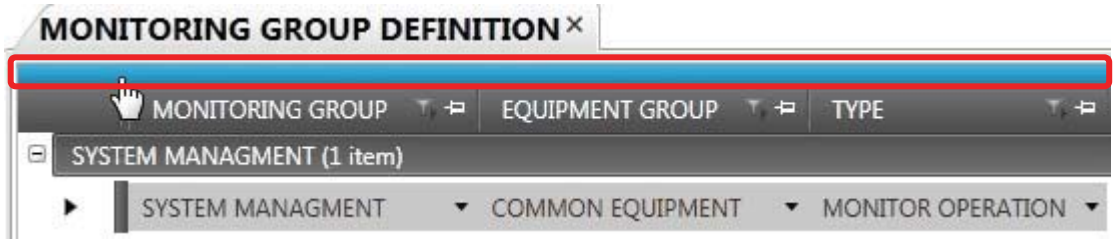
2.8.1 Editing State

When the mark ("*") shown below is displayed at the left side of row, the editing data is not confirmed. If the data is compiled under this condition, the data before editing is compiled.

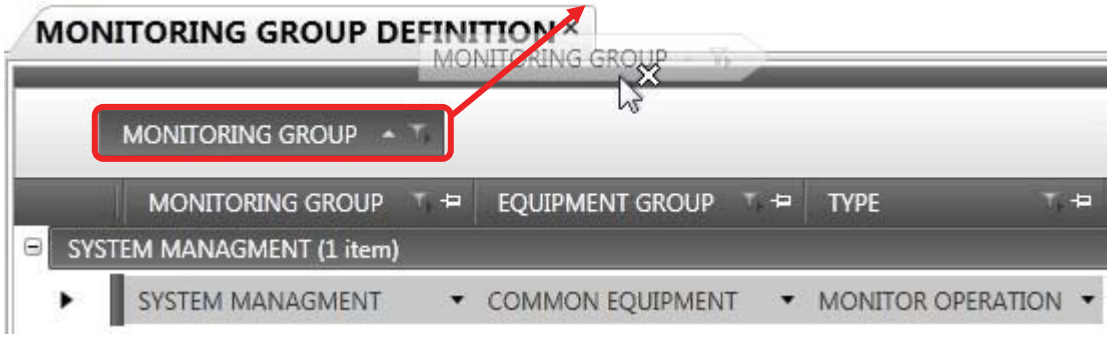


2.8.2 Adding Data Row

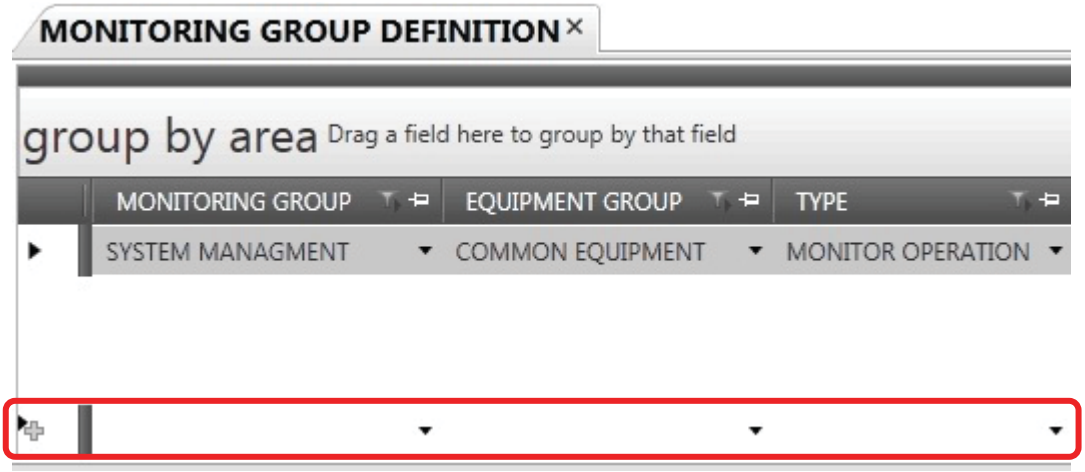
Click the part above the title row to display fields for grouping.



Drag and drop the field upward on the screen to clear the grouping.



The area for adding row is shown under ungrouped condition at the bottom of the screen.



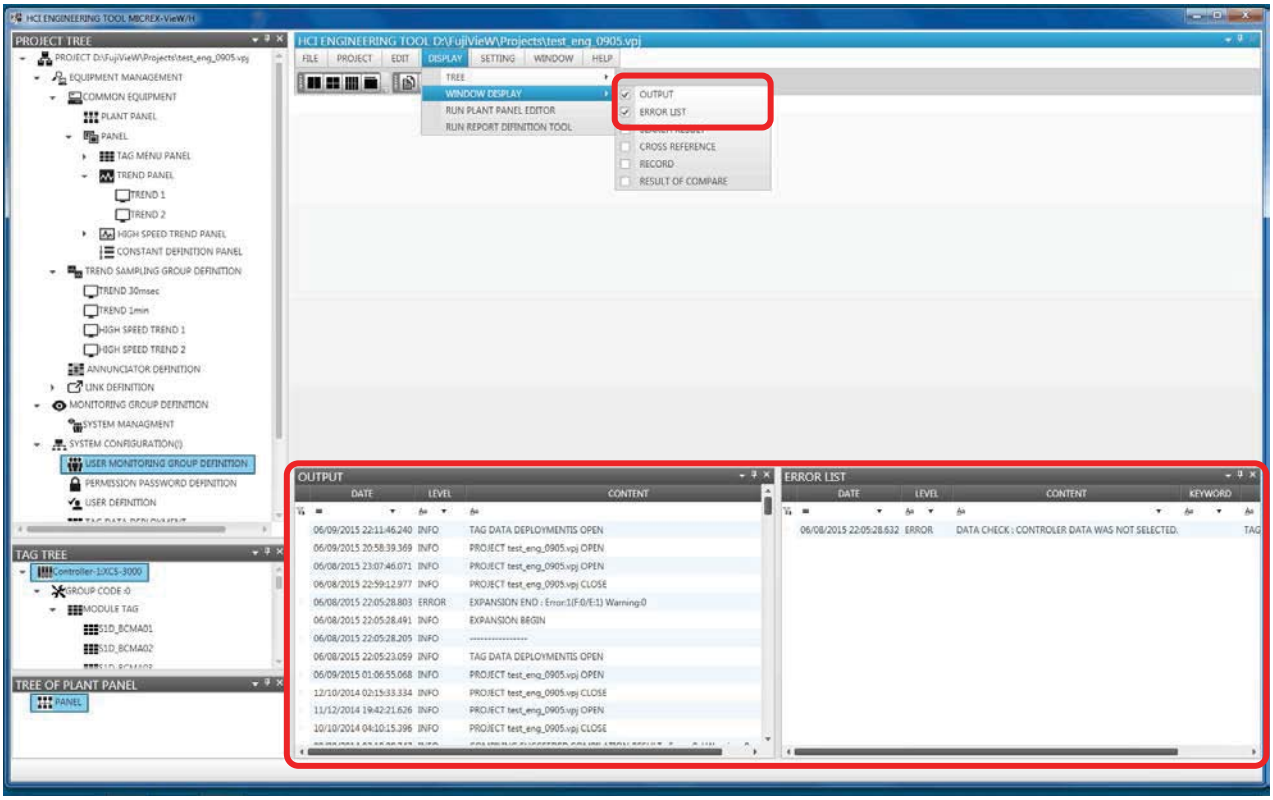
2.9 Output Window

This section describes the output window.

The table below lists types of output windows.

Type of output window	Description
Output	Outputs the process results at each function. Consistency check result during the compilation (check result, number of error cases)
Error list	When an error occurs during the compilation, for example, the details are output. Consistency check result (error details)
Search results	Outputs search results.
Record	Based on US FDA21 CFR Part 11, records the operations which may give an impact on the system and outputs them to this window.
Comparison result	Outputs comparison results (mismatch) as well as source data and destination data for comparison.

[Output window example]



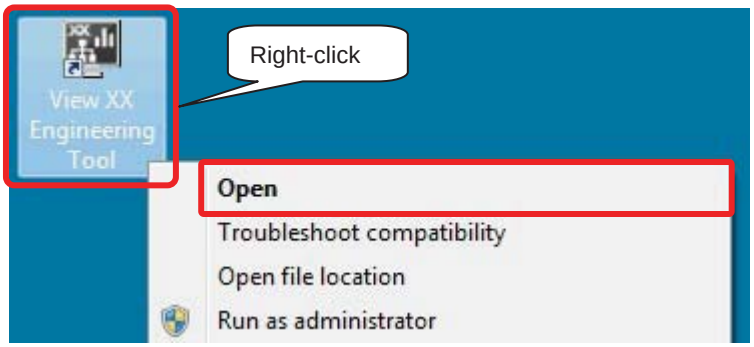
For information about the operation of output window, refer to “6.7 Record Function (Log Management)”.

3. Creating Project

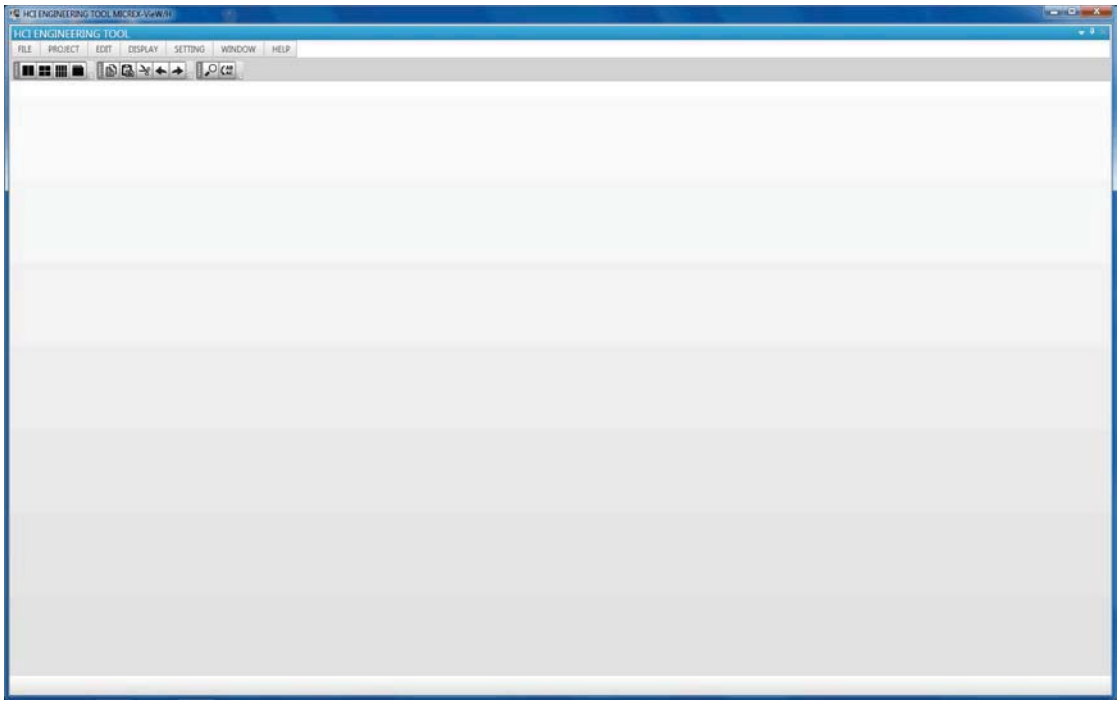
This chapter describes creation procedures of projects.

3.1 Creating New Project

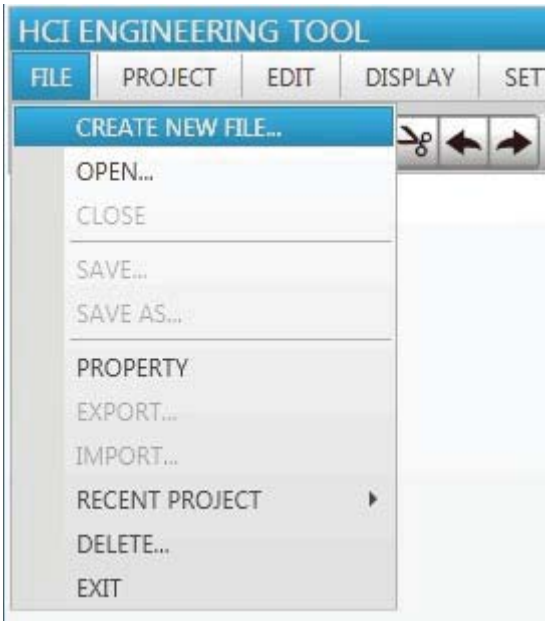
Right-click the [VieW XX Engineering Tool] icon on desktop and select [Open] from the pop-up menu.



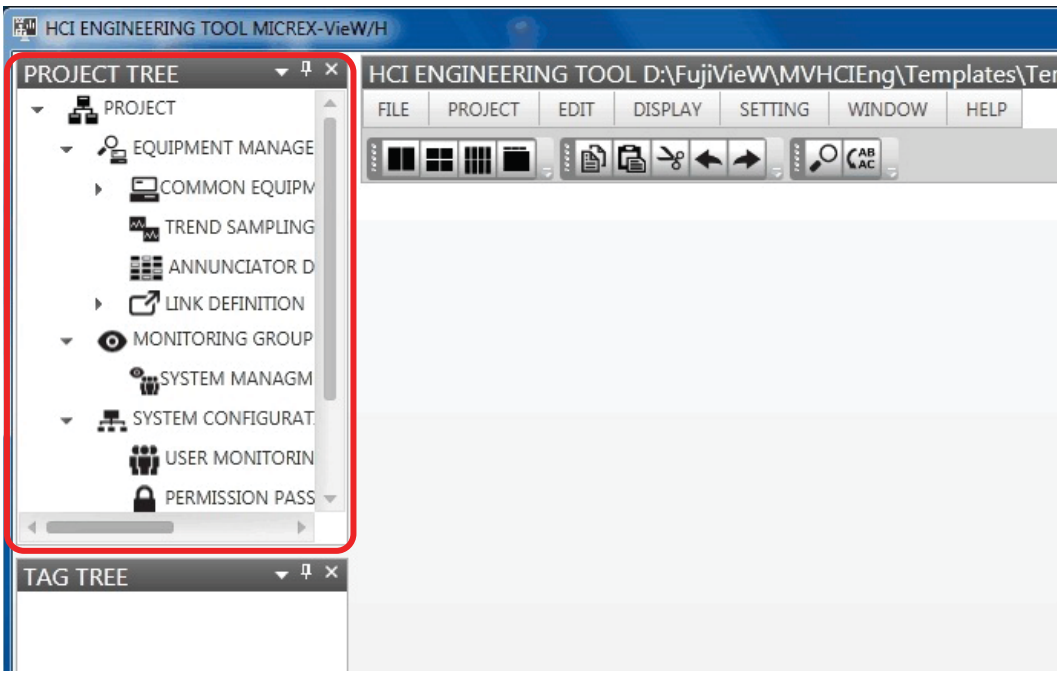
The HCI engineering tool starts.



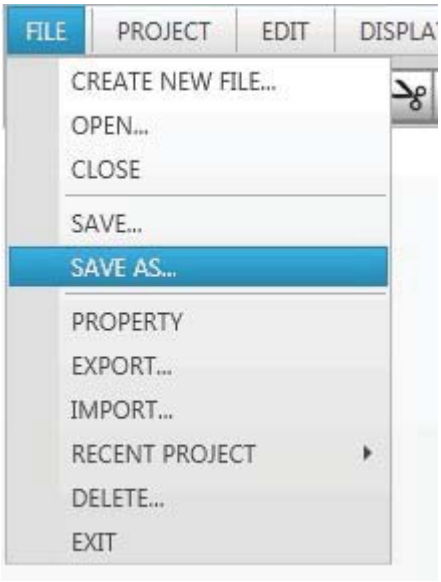
Select [New] in the File menu.



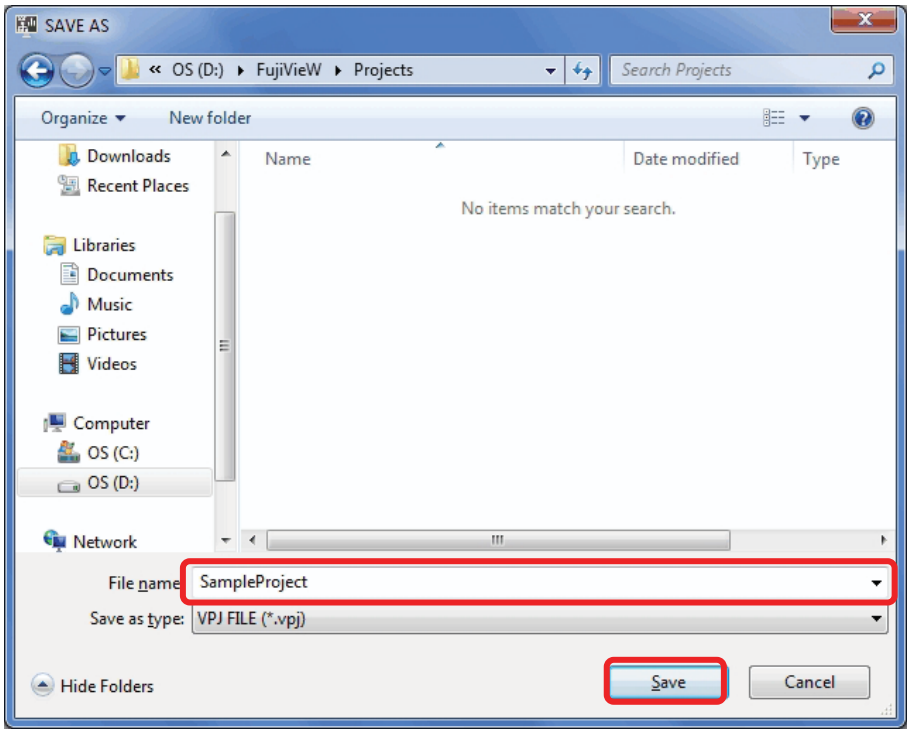
The project tree for a created project is displayed.



Select [Save As] in the File menu.



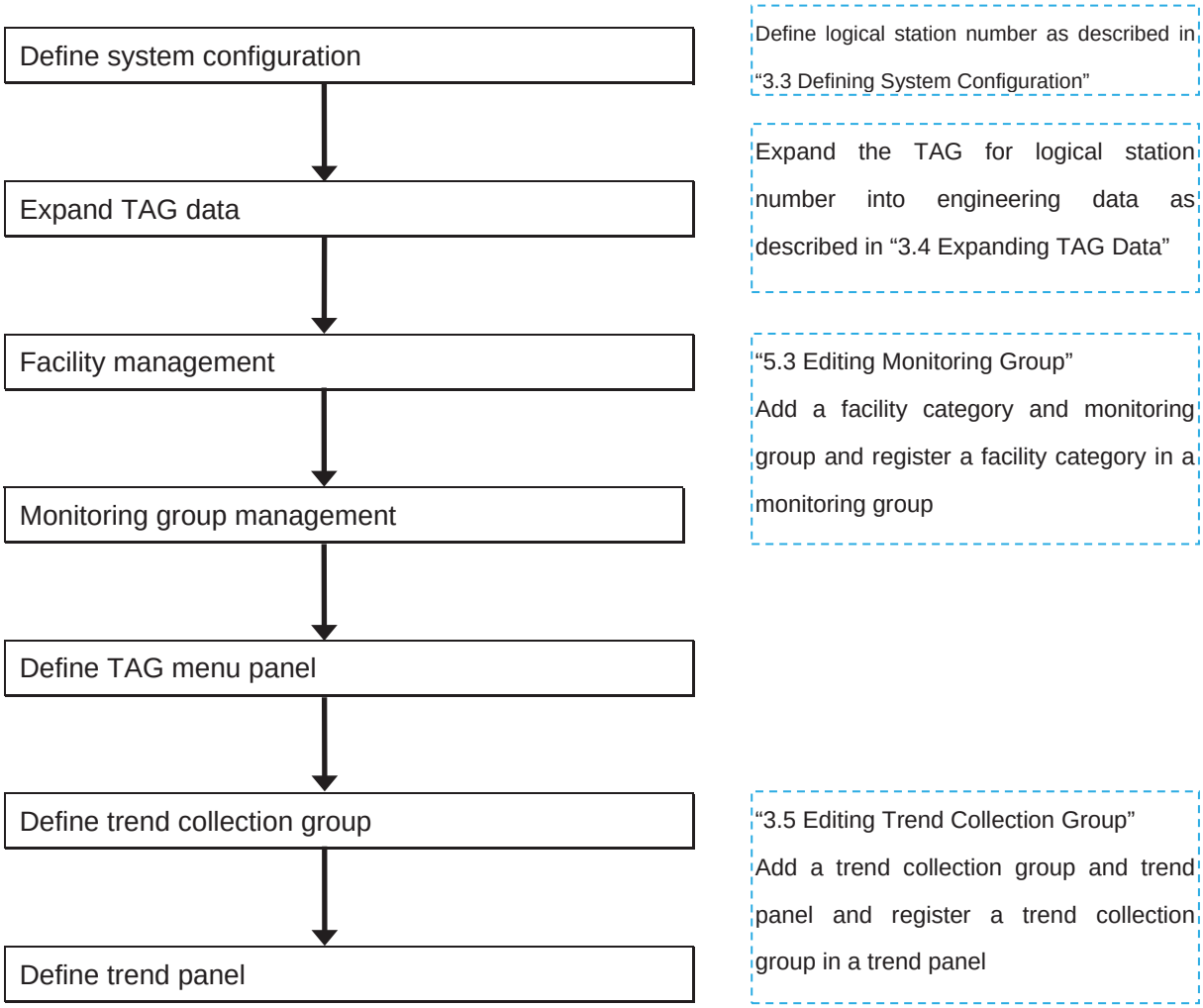
Enter a new project name on the Save As dialog and click the [Save] button to save the project.



The file name of the project tree is changed.

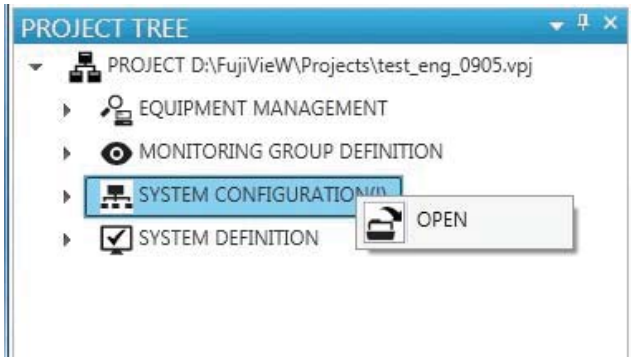
3.2 Definition Procedures

This section describes basic definition procedures for the HCI engineering.

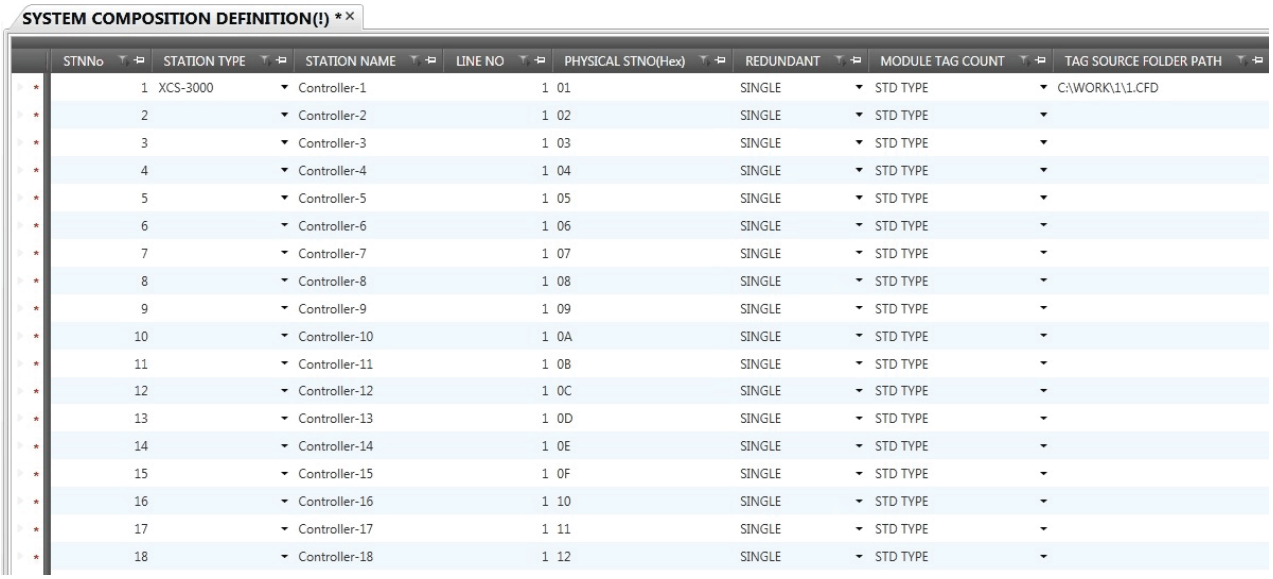


3.3 Defining System Configuration

Right-click [System configuration] in the Project tree and select [Open] from the pop-up menu.



The “Define system configuration” panel below opens.

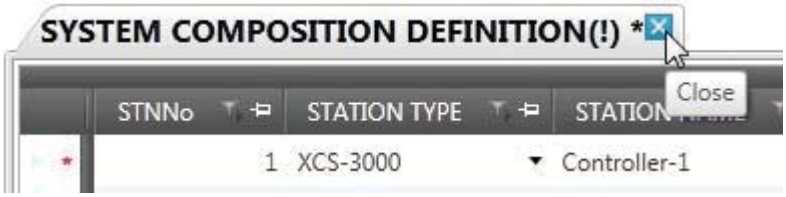


Set up necessary items by referring to the following table.

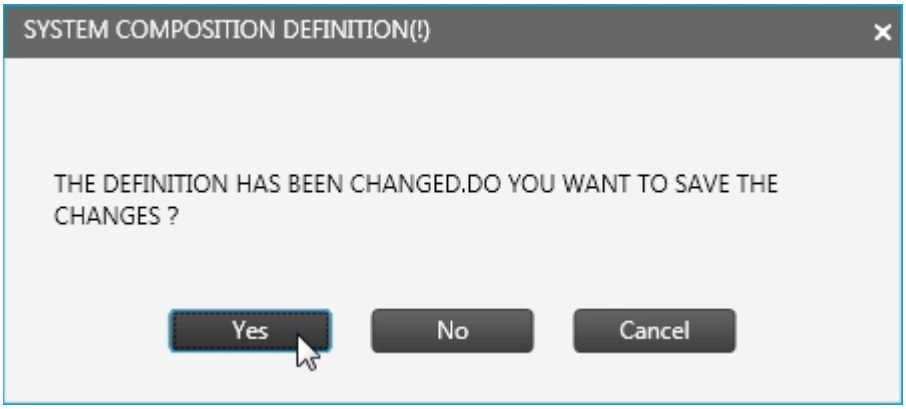
Item	Description	Type	Remarks
STNNo	Logical station number	Display only	1 to 30: Controller 40 to 55: XOS 64 (systems A, B): XDS
Model	Model name	Mandatory	Select ACS-2000, XCS-3000, XOS-3000, or XDS-3000. The row with the model selected is considered to be the valid station.
Station name	Name of station		Up to 32 characters. The system alarm is displayed with the station name.
Line number	Number of line	Display only	Fixed at one
Physical STNO	Physical station number	Mandatory	Enter hexadecimal number. The possible value range is between 01 and 40.
Source TAG folder path	Source folder path of TAG	Mandatory	Specify this only for controller. Specify a full path to CFD.
Destination folder path	Loading folder path of definition DB to XDS and XOS	Mandatory	Specify this only for XDS and XOS models. Specify a full path to "D:\FujiVieW\cnfig\dist" for XDS and XOS. In XDS and XOS, set the path above as shared path in advance.

When the definition edit is complete, confirm that the editing data has been confirmed. (refer to “2.8.1 Editing State”)

In order to save changes, click the [X] button on the tab in the “Define system configuration” panel.

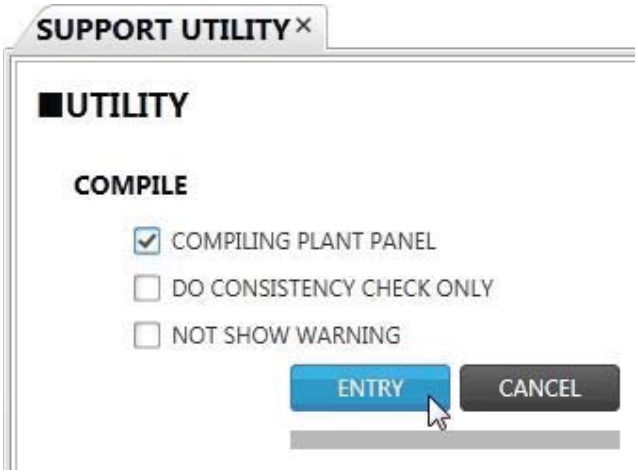


Since a confirmation panel is displayed, click [Yes] to save changes.



In order to reflect changes on other panels, select [Compile...] in the Project menu.

When the “Support utility” panel opens, click the [Run] button to compile.



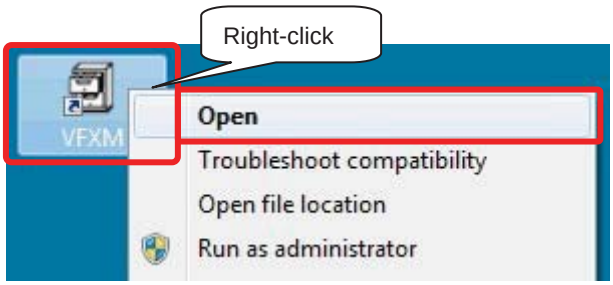
Since the compile result is shown on the Output window, you can confirm that the compile succeeded or failed.

OUTPUT
DATE
LEVEL
CONTENT
06/10/2015 14:56:56.196 INFO COMPILING SUCCEEDED.COMPILATION RESULT : Error: 0 / Warning: 0
06/10/2015 14:56:53.154 INFO CHECK INTEGRITY : CONSTANT SETTING DEFINITION

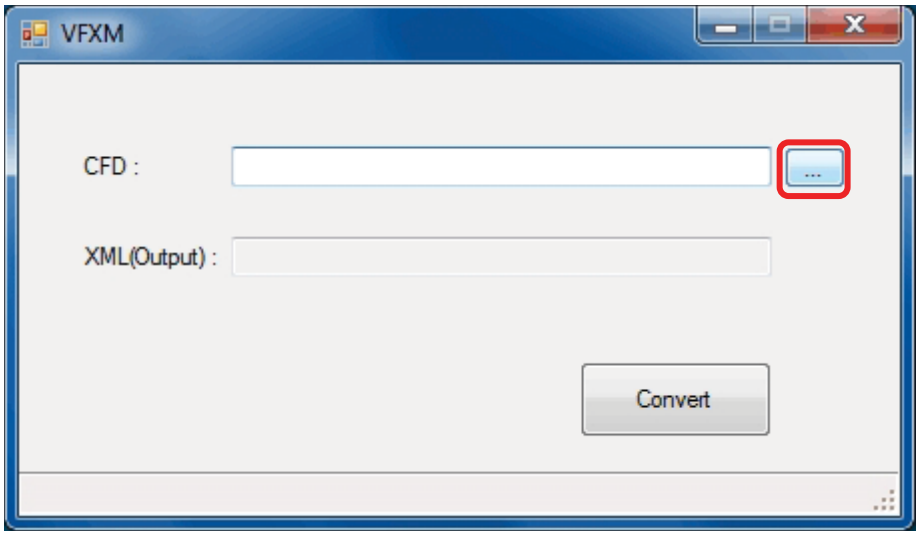
3.4 Expanding TAG Data

As the preparation for performing the definition with the HCI engineering tool, convert TAGs in order to import TAGs for ACS defined in FPROCES. Because XCS loads the file itself output by the TAG definition tool, XCS does not need this operation.

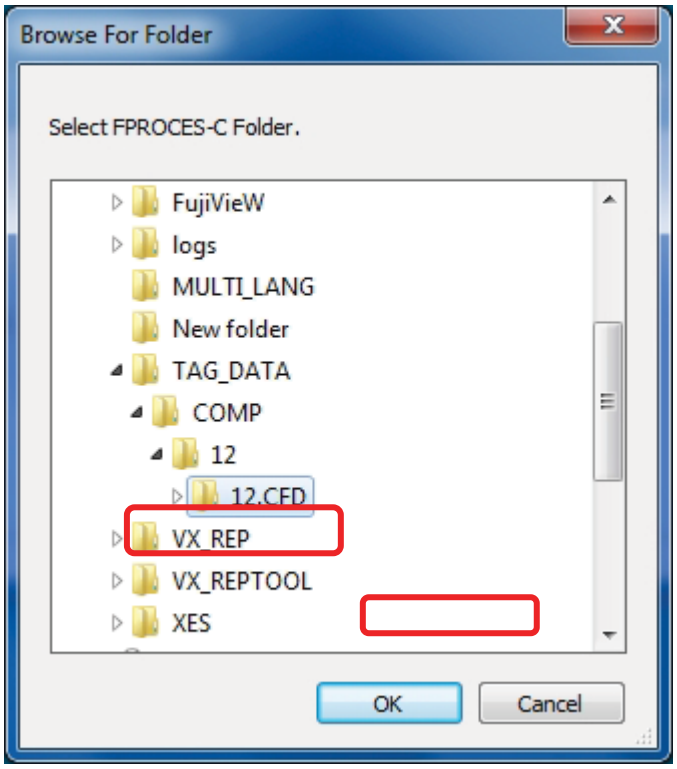
The VFXM tool performs this TAG conversion. Right-click the [VFXM] icon on desktop and select [Open] from the pop-up menu to start VFXM.



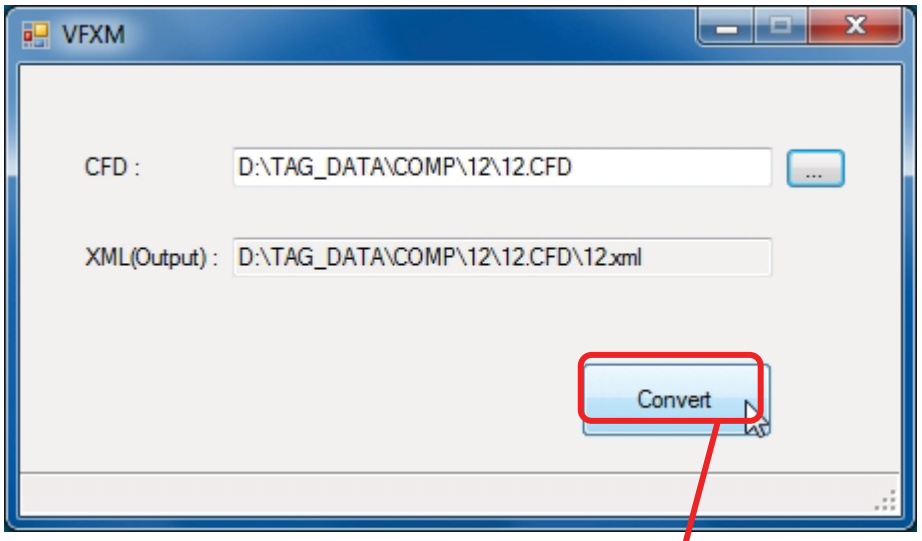
When the VFXM tool starts, click the [...] button for specifying a source file.



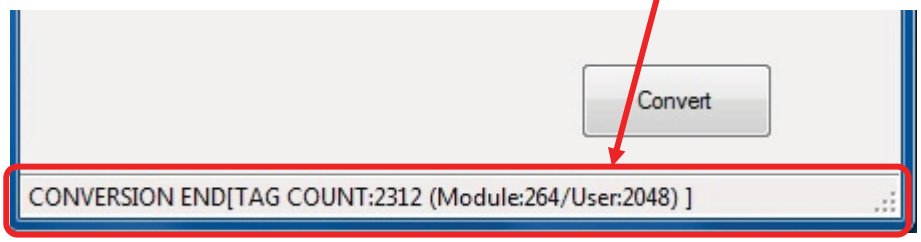
Specify a CFD folder for FPROCES on the “Browse folders” dialog and click the [OK] button.



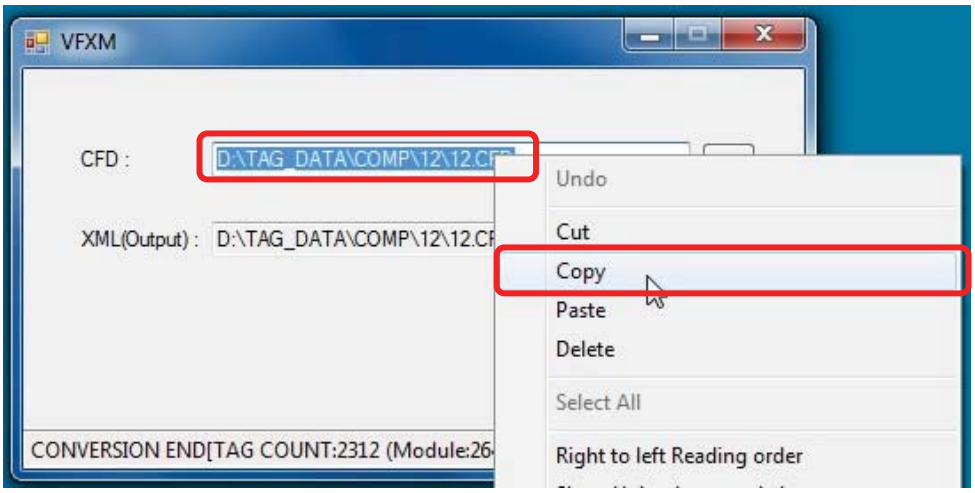
Click the [Convert] button to perform the conversion.



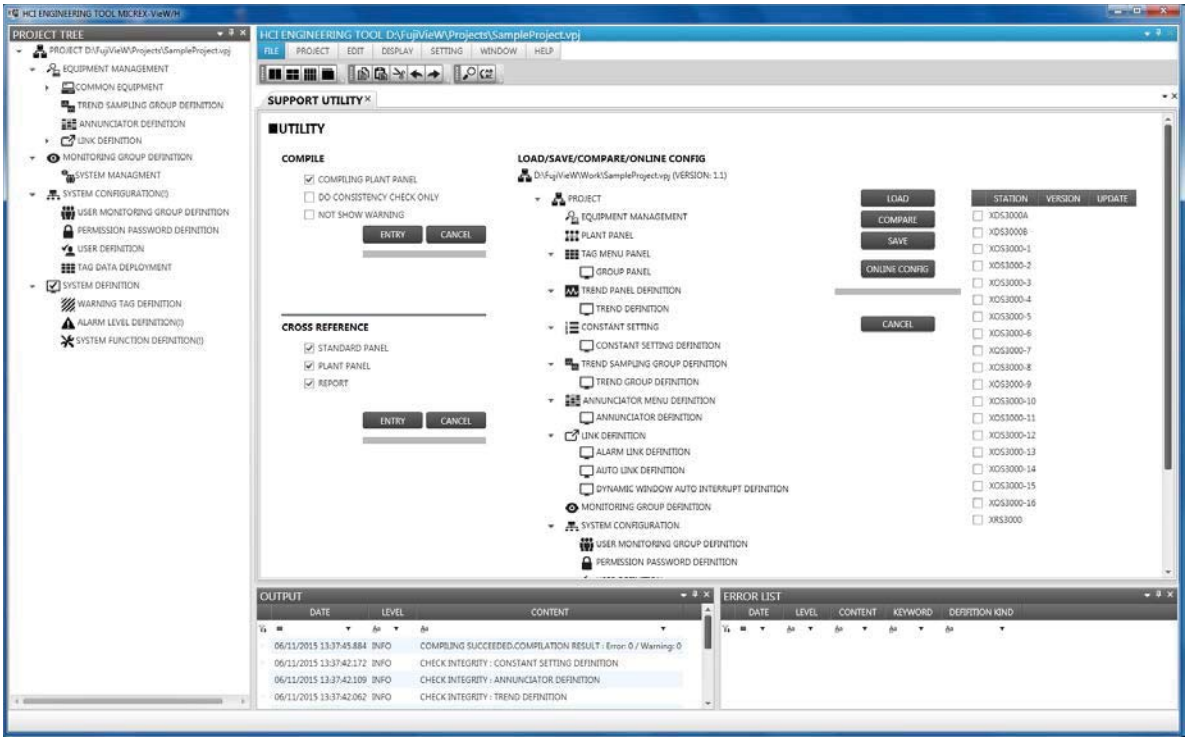
The conversion result is shown at the bottom of the VFXM tool panel.



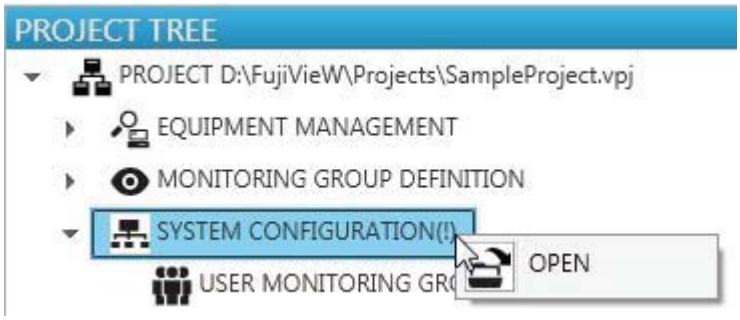
Select and right-click the text in the [Source CFD] field and select [Copy] from the pop-up menu.



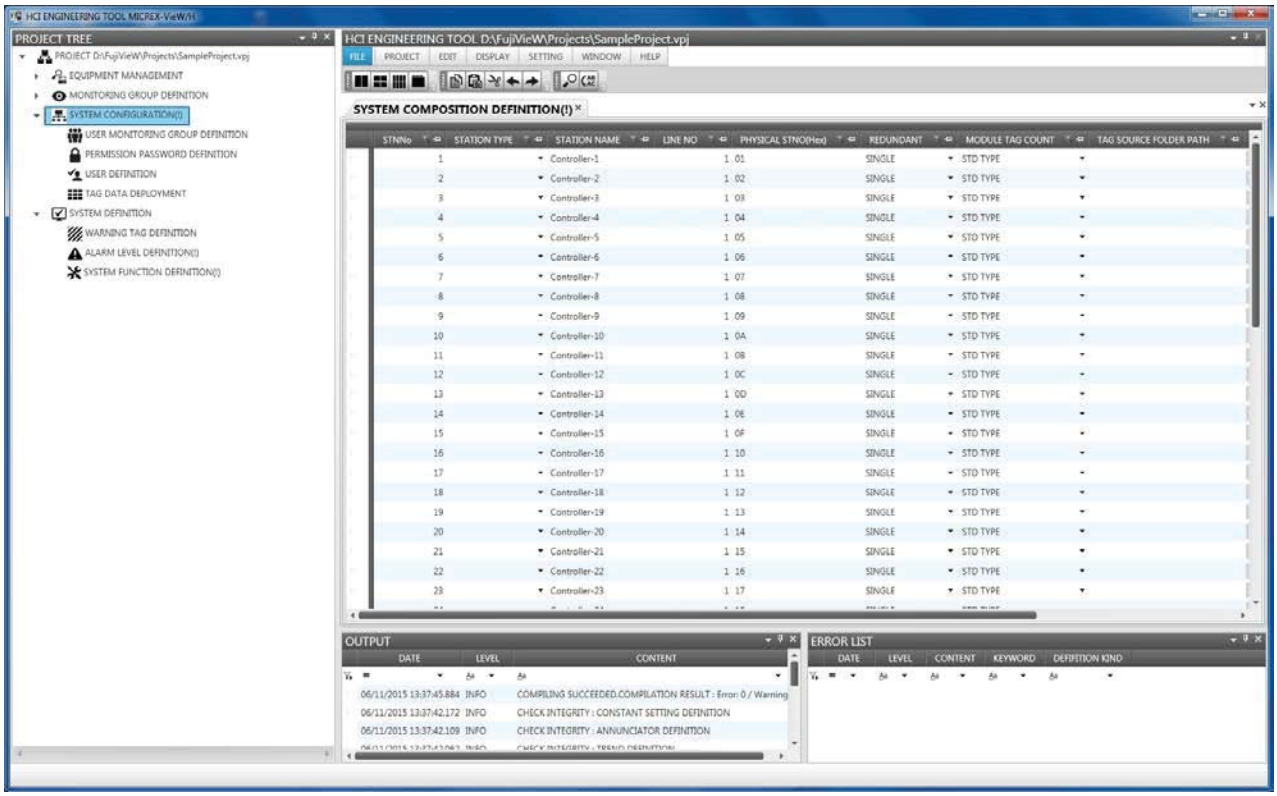
Now, perform the operations in the HCI engineering tool.



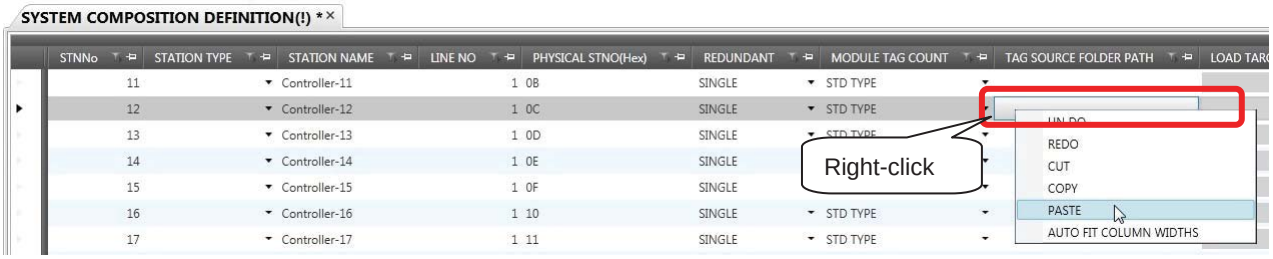
Right-click [System configuration] in the Project tree and select [Open] from the pop-up menu.



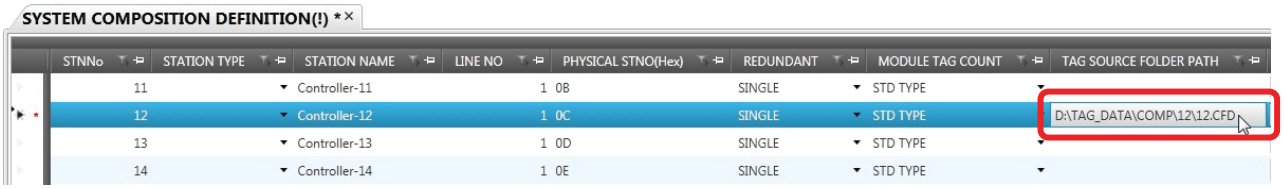
The “Define system configuration” panel opens.



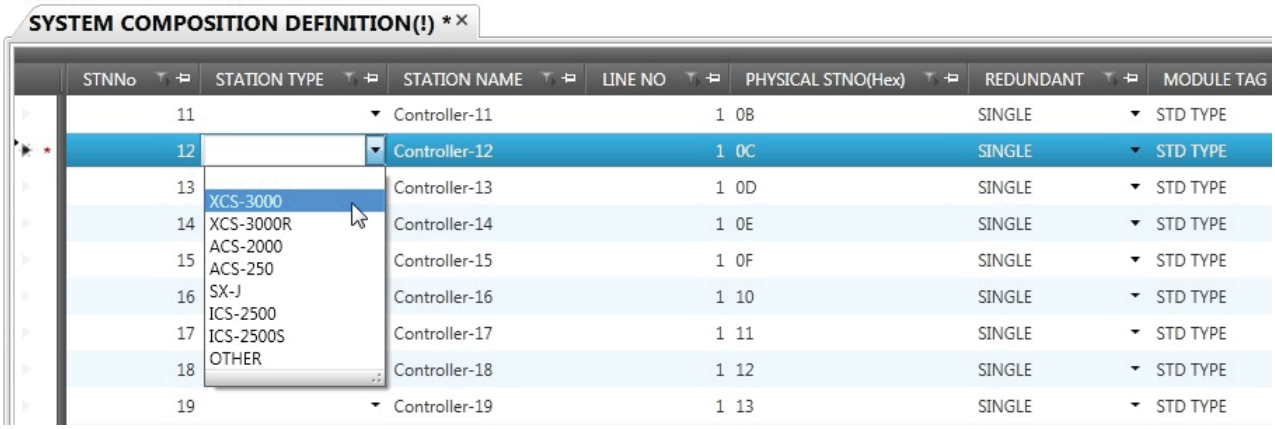
When the “Define system configuration” panel opened, right-click [Source TAG folder path] for corresponding controller and select [Paste] from the pop-up menu.



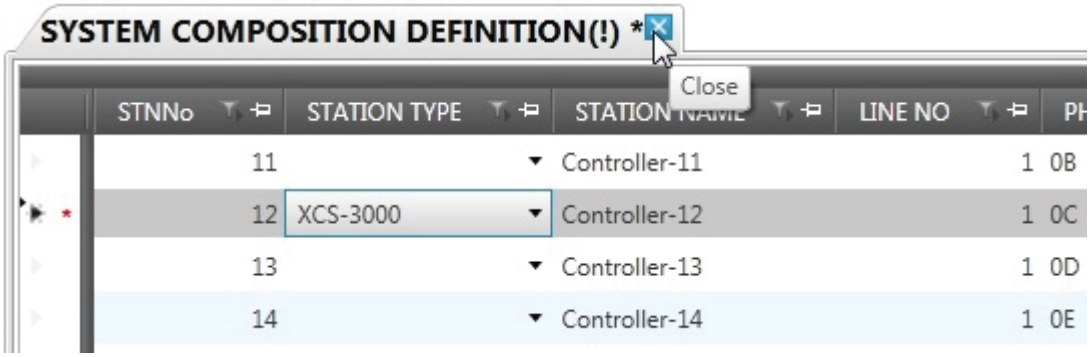
Confirm that the text in the [Source CFD] field in the VFXM tool panel was pasted.



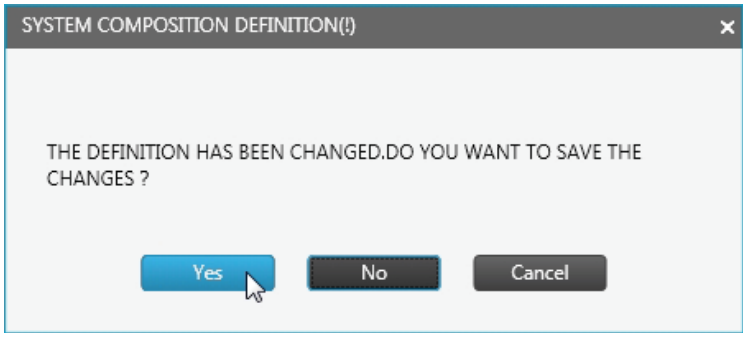
Click [▼] in the [Model] field and select a model from the combo box.



When the definition edit is complete, confirm that the editing data has been confirmed. (refer to “2.8.1 Editing State”) In order to save changes, click the [X] button on the tab in the “Define system configuration” panel.

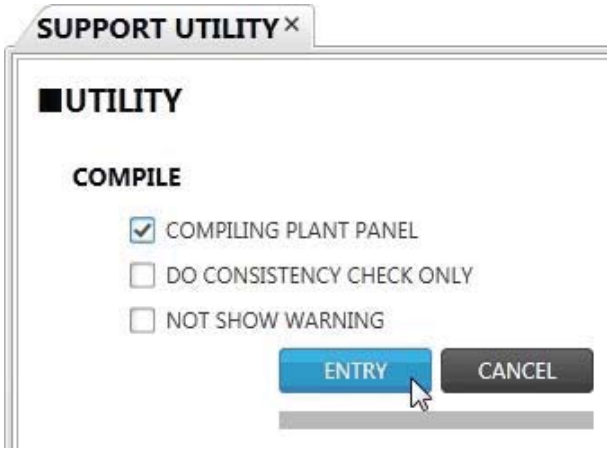


Since a confirmation panel is displayed, click [Yes] to save changes.

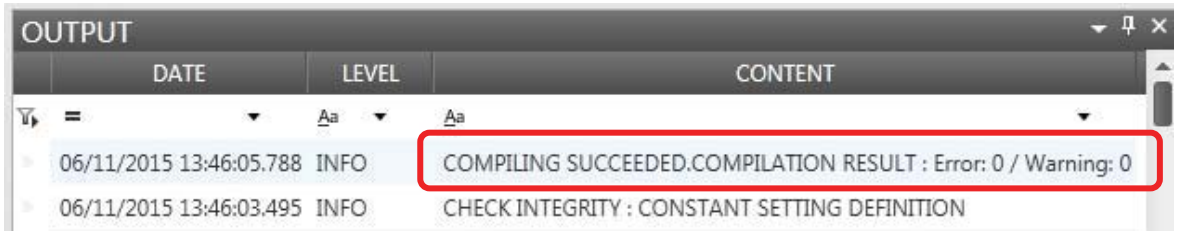


In order to reflect the change on other panels, select [Compile...] in the Project menu.

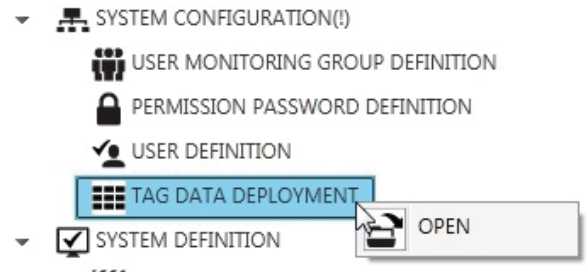
When the “Support utility” panel opens, click the [Run] button to compile.



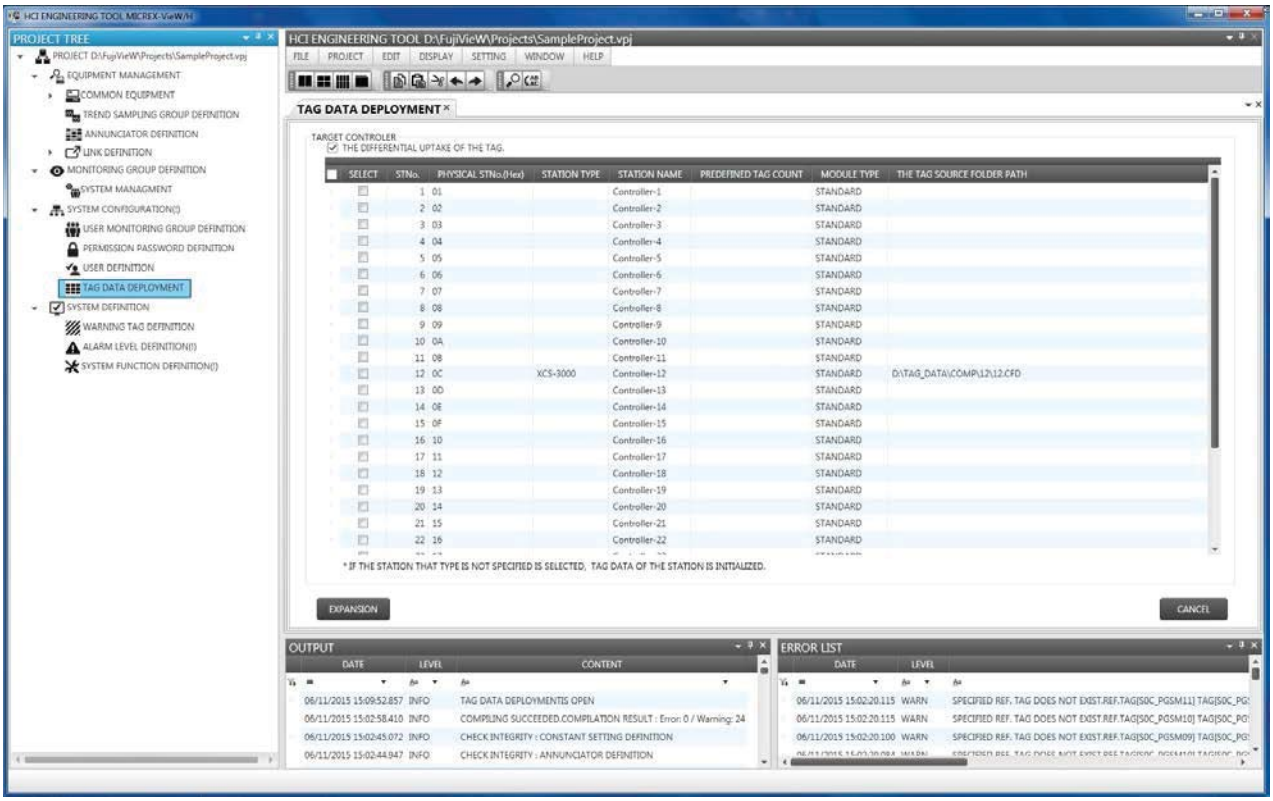
Since the compile result is shown on the Output window, you can confirm that the compile succeeded or failed.



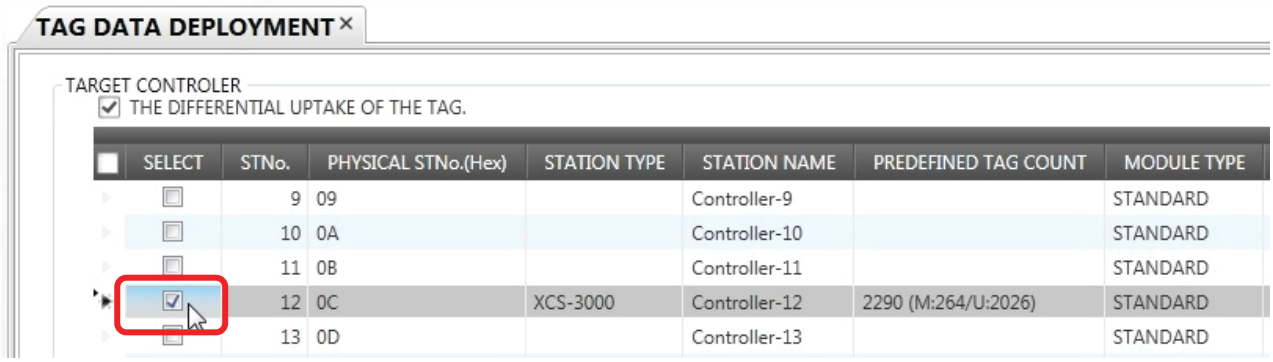
Right-click the [Expand TAG data] in the Project tree and select [Open] from the pop-up menu.



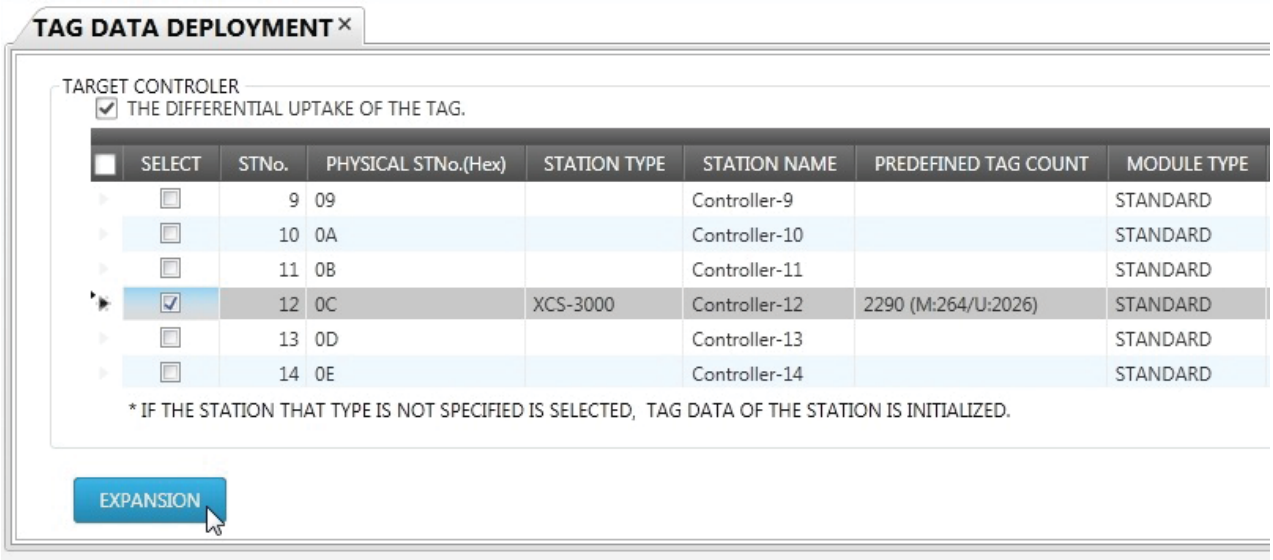
The “Expand TAG data” panel opens.



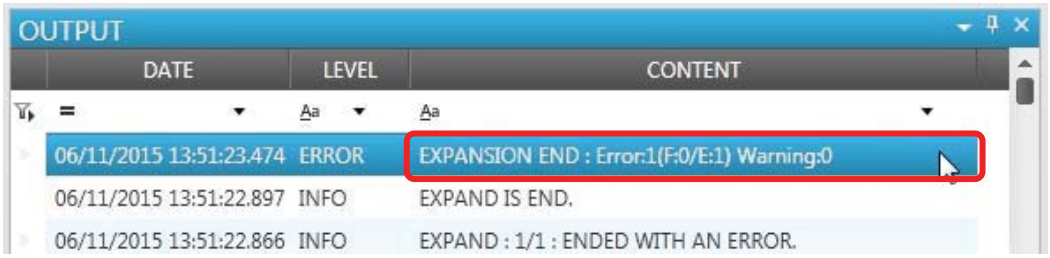
Select the [Select] checkbox for the controller whose “Source TAG folder path” was just specified on the “Expand TAG data” panel.



Click the [Expand] button.

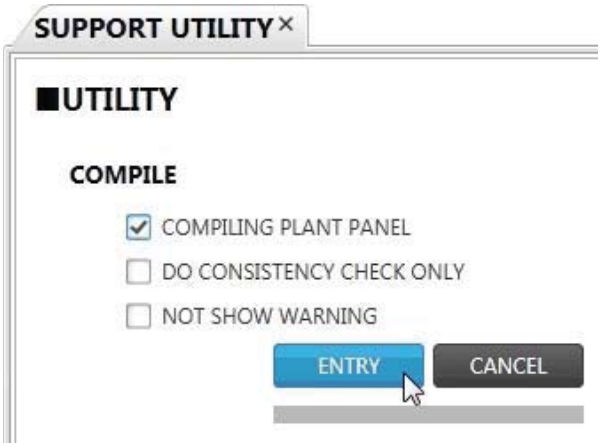


Confirm on the Output window that the expansion of TAG data was complete.

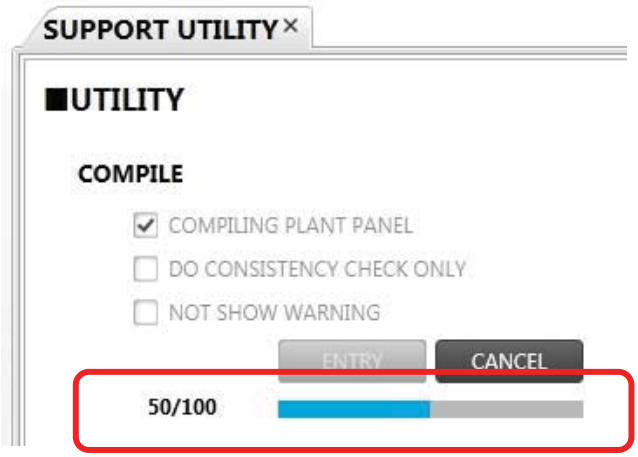


In order to reflect changes on other panels, select [Compile...] in the Project menu.

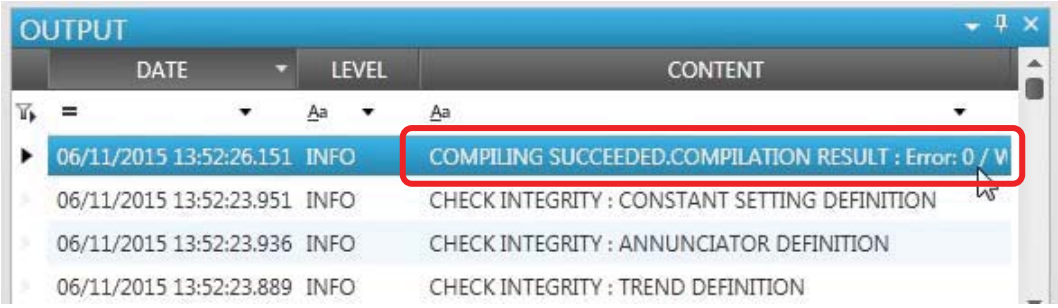
When the “Support utility” panel opens, click the [Run] button to compile. You can confirm the compile result on the Output window.



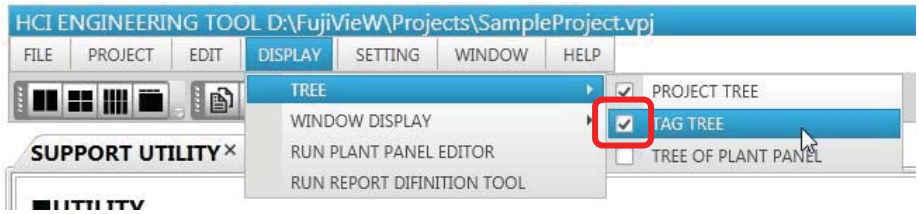
The compile takes some time while expanding TAG data in proportion to the TAG count. The progress status of compile is shown on the screen.



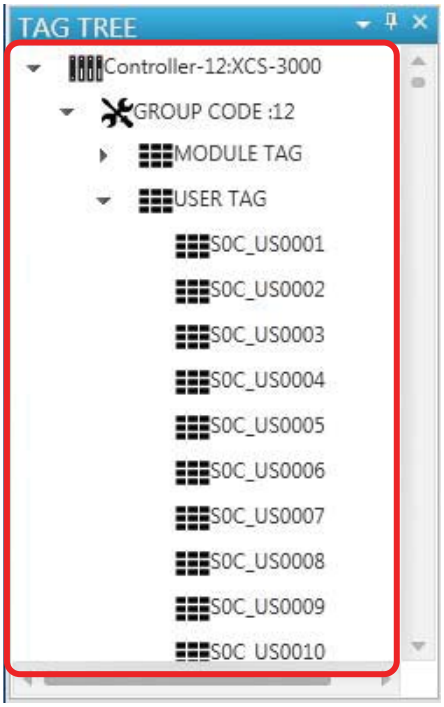
Since the compile result is shown on the Output window, you can confirm that the compile succeeded or failed.



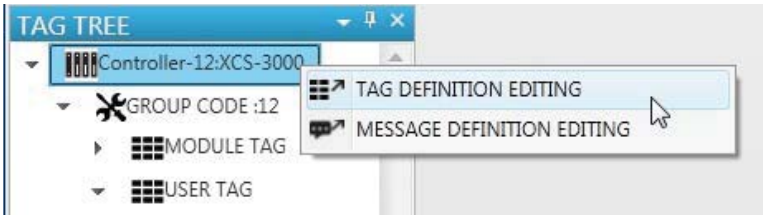
Select [Tree] and [TAG tree] in order on the View menu.



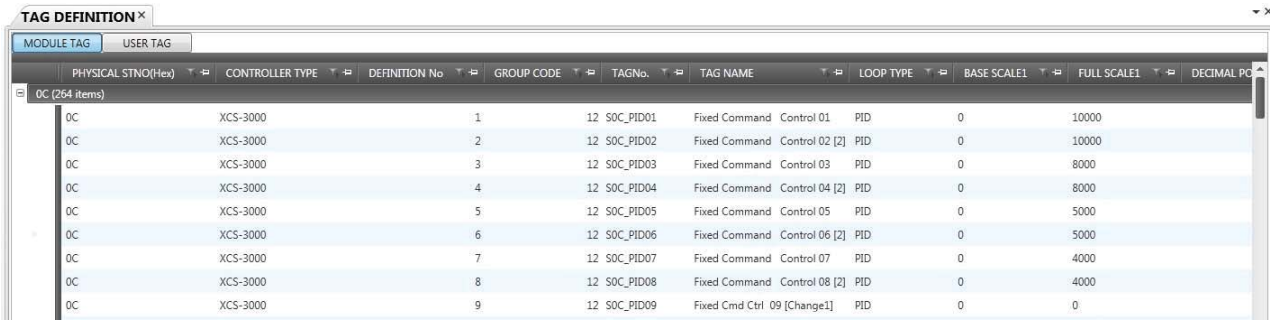
Confirm that the TAGs are imported into the TAG tree.



Right-click [Controller], [Category code], [Module TAG], [User TAG], or [TAG] and select [Edit TAG definition] from the pop-up menu.



The “TAG definition” panel opens.



When the data in the [Loop type] field is “MOV***”, a remark and color setting are required.

When the definition is changed, make sure to make a compilation again.

Initial item

Item Name	Description	Type	Remarks
Station No.	Logical station No where the TAGs are defined	Display only	
Model	Station model where the TAGs are defined	Display only	
Def No.	Definition No of a TAG	Display only	
TAGNO	TAG	Display only	
TAG name	Name of a TAG	Display only	
Description	Description of a TAG	Display only	

Module TAG item

Item Name	Description	Type	Remarks
Station No.	Logical station No where the TAGs are defined	Display only	
Model	Station model where the TAGs are defined	Display only	
Def No.	Definition No of a TAG	Display only	
TAG type	Type of a TAG	Display only	10: Module tag 20: User tag
TAGNO	TAG	Display only	
TAG name	Name of a TAG	Display only	
Loop type	Loop type	Display only	
Base scale 1	Base scale 1	Display only	
Full scale 1	Full scale 1	Display only	
Decimal point 1	Decimal point information 1	Display only	
Unit 1	Unit information 1	Display only	
Base scale 2	Base scale 2	Display only	
Full scale 1	Full scale 2	Display only	
Decimal point 2	Decimal point information 2	Display only	
Unit 2	Unit information 2	Display only	
Reference TAG 1	Tag 1 referred to	Display only	
Reference TAG 2	Tag 2 referred to	Display only	
Reference TAG 3	Tag 3 referred to	Display only	
VALVE state	VALVE state	Display only	
MV state	MV state	Display only	
Limiter	Validity of limiter	Display only	
Release REM mode	Whether to release REM mode	Display only	
Release AUT mode	Whether to release AUT mode	Display only	
Release MAN mode	Whether to release MAN mode	Display only	
Valid alarm of 1st bit	Whether the 1st bit of alarm is valid	Display only	
Alarm of 1st bit	Alarm number of 1st bit	Display only	
Recovery message of 1st bit	Whether the recovery message of 1st bit of alarm is valid	Display only	
Valid alarm of 2nd bit	Whether the 2nd bit of alarm is valid	Display only	
Alarm of 2nd bit	Alarm number of 2nd bit	Display only	
Recovery message of 2nd bit	Whether the recovery message of 2nd bit of alarm is valid	Display only	
Valid alarm of 3rd bit	Whether the 3rd bit of alarm is valid	Display only	
Alarm of 3rd bit	Alarm number of 3rd bit	Display only	
Recovery message of 3rd bit	Whether the recovery message of 3rd bit of alarm is valid	Display only	
Valid alarm of 4th bit	Whether the 4th bit of alarm is valid	Display only	
Alarm of 4th bit	Alarm number of 4th bit	Display only	
Recovery message of 4th bit	Whether the recovery message of 4th bit of alarm is valid	Display only	
Valid alarm of 5th bit	Whether the 5th bit of alarm is valid	Display only	

Alarm of 5th bit	Alarm number of 5th bit	Display only	
Recovery message of 5th bit	Whether the recovery message of 5th bit of alarm is valid	Display only	
Valid alarm of 6th bit	Whether the 6th bit of alarm is valid	Display only	
Alarm of 6th bit	Alarm number of 6th bit	Display only	
Recovery message of 6th bit	Whether the recovery message of 6th bit of alarm is valid	Display only	
Valid alarm of 7th bit	Whether the 7th bit of alarm is valid	Display only	
Alarm of 7th bit	Alarm number of 7th bit	Display only	
Recovery message of 7th bit	Whether the recovery message of 7th bit of alarm is valid	Display only	
Valid alarm of 8th bit	Whether the 8th bit of alarm is valid	Display only	
Alarm of 8th bit	Alarm number of 8th bit	Display only	
Recovery message of 8th bit	Whether the recovery message of 8th bit of alarm is valid	Display only	
Valid alarm of 9th bit	Whether the 9th bit of alarm is valid	Display only	
Alarm of 9th bit	Alarm number of 9th bit	Display only	
Recovery message of 9th bit	Whether the recovery message of 9th bit of alarm is valid	Display only	
Valid alarm of 10th bit	Whether the 10th bit of alarm is valid	Display only	
Alarm of 10th bit	Alarm number of 10th bit	Display only	
Recovery message of 10th bit	Whether the recovery message of 10th bit of alarm is valid	Display only	
Valid alarm of 11th bit	Whether the 11th bit of alarm is valid	Display only	
Alarm of 11th bit	Alarm number of 11th bit	Display only	
Recovery message of 11th bit	Whether the recovery message of 11th bit of alarm is valid	Display only	
Valid alarm of 12th bit	Whether the 12th bit of alarm is valid	Display only	
Alarm of 12th bit	Alarm number of 12th bit	Display only	
Recovery message of 12th bit	Whether the recovery message of 12th bit of alarm is valid	Display only	
Valid alarm of 13th bit	Whether the 13th bit of alarm is valid	Display only	
Alarm of 13th bit	Alarm number of 13th bit	Display only	
Recovery message of 13th bit	Whether the recovery message of 13th bit of alarm is valid	Display only	
Valid alarm of 14th bit	Whether the 14th bit of alarm is valid	Display only	
Alarm of 14th bit	Alarm number of 14th bit	Display only	
Recovery message of 14th bit	Whether the recovery message of 14th bit of alarm is valid	Display only	
Valid alarm of 15th bit	Whether the 15th bit of alarm is valid	Display only	
Alarm of 15th bit	Alarm number of 15th bit	Display only	
Recovery message of 15th bit	Whether the recovery message of 15th bit of alarm is valid	Display only	
Valid alarm of 16th bit	Whether the 16th bit of alarm is valid	Display only	
Alarm of 16th bit	Alarm number of 16th bit	Display only	
Recovery message of 16th bit	Whether the recovery message of 16th bit of alarm is valid	Display only	
Remark name of open	Remarks upon open operation	*	
Remark name of stop	Remarks upon stop operation		
Remark name of close	Remarks upon close operation		
Color during full open	Color displayed during full open		

Color during stop	Color displayed during stop		
Color during full close	Color displayed during full close		
Opening color	Color displayed during opening		
Closing color	Color displayed during closing		
Power factor type	Power factor type	Display only	Loop type: Power factor only
Description	Description of a TAG	Display only	

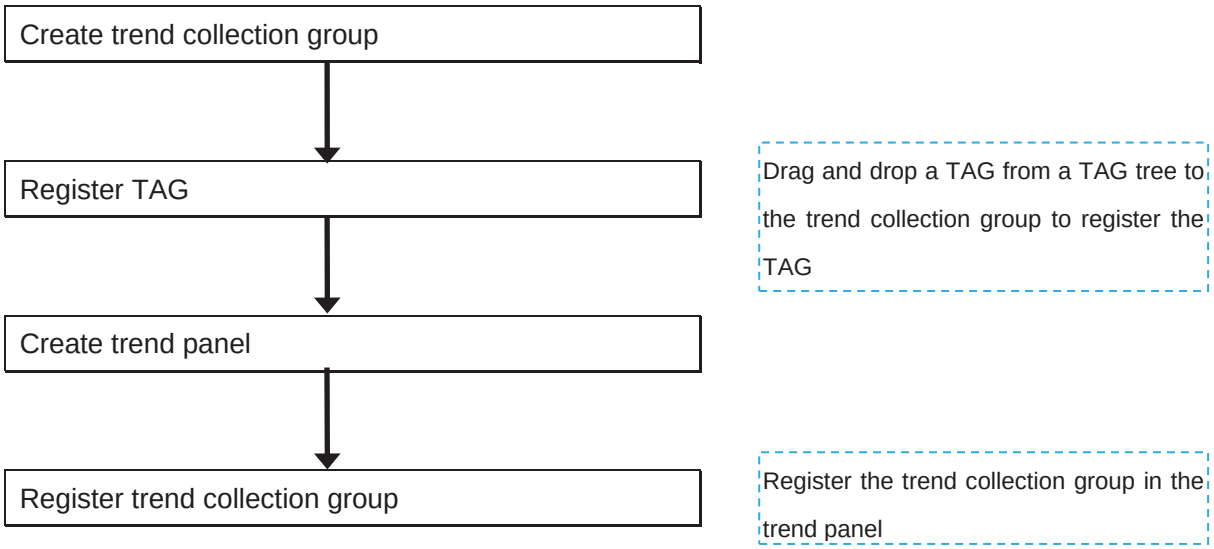
* Loop type: Only for “MOV***” (motor driven operating mechanism)

User TAG item

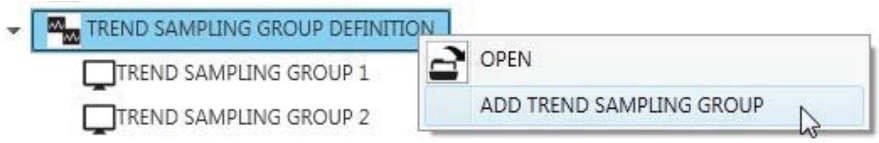
Item Name	Description	Type	Remarks
Station No.	Logical station No where the TAGs are defined	Display only	
Model	Station model where the TAGs are defined	Display only	
Def No.	Definition No of a TAG	Display only	
TAG type	Type of a TAG	Display only	10: Module tag 20: User tag
TAGNO	TAG	Display only	
TAG name	Name of a TAG	Display only	
Data type	Data type	Display only	
System tag No	TAGNO of system tag	Display only	
BC	Broadcast	Display only	
Setting	Whether writable or not	Display only	
Base scale	Base scale	Display only	
Full scale	Full scale	Display only	
Decimal point	Decimal point information	Display only	
Unit	Unit information	Display only	
Lower limit	Lower limit information of setting value	Display only	
Upper limit	Upper limit information of setting value	Display only	
Category code	Category code of a TAG	Display only	
Description	Description of a TAG	Display only	

3.5 Editing Trend Collection Group

This section describes procedures for editing trend collection groups.

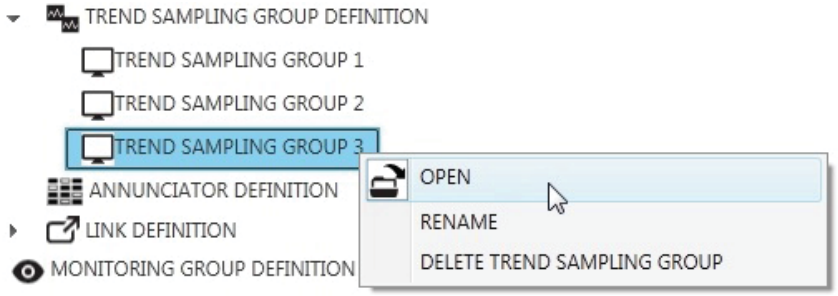


Right-click [Define trend collection group] in the Project tree and select [Add group] from the pop-up menu.

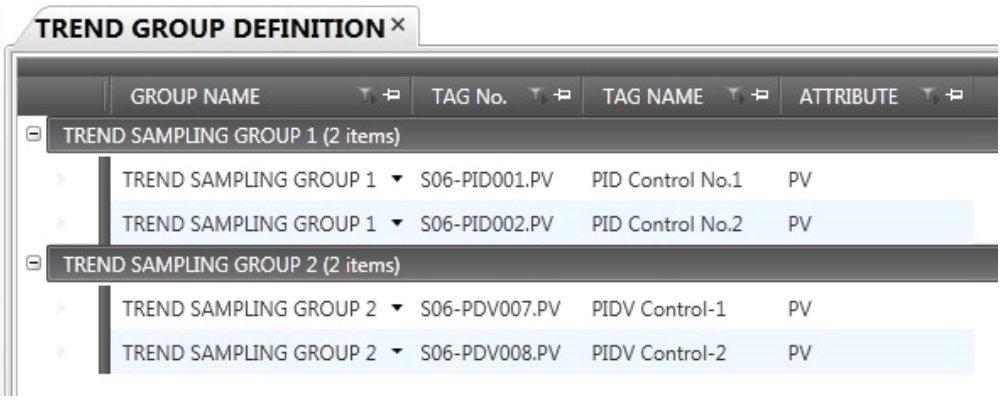


A new group is added.

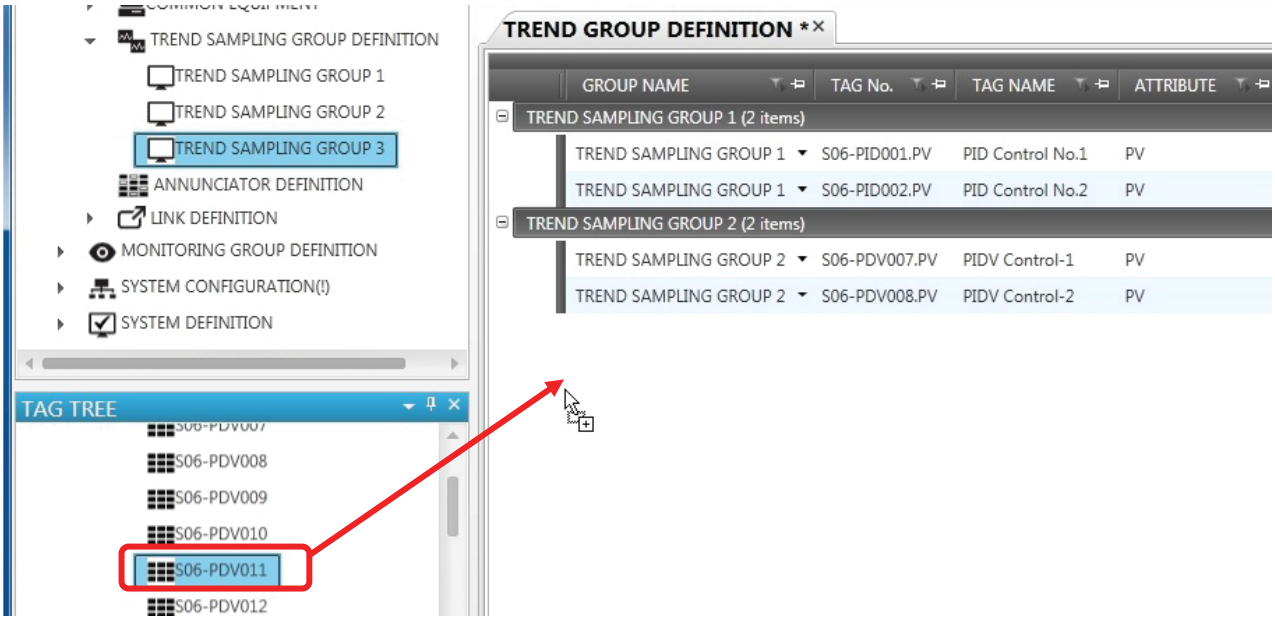
Right-click the group added and select [Open] from the pop-up menu.



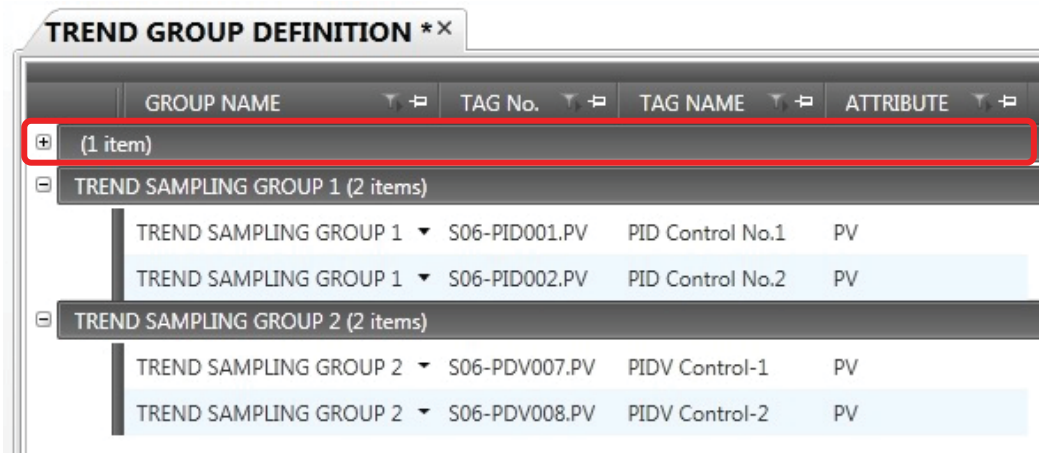
The “Define trend group” panel opens.



On the “Define trend group” panel, drag and drop a TAG or category code from the TAG tree to the “Define trend group” panel.

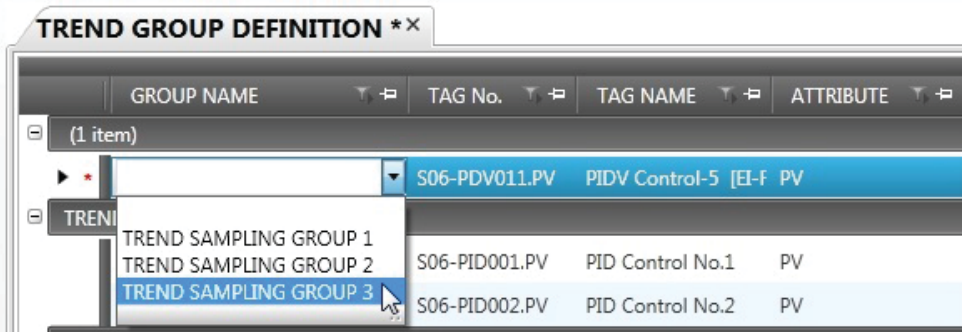


The TAG dragged and dropped is added.

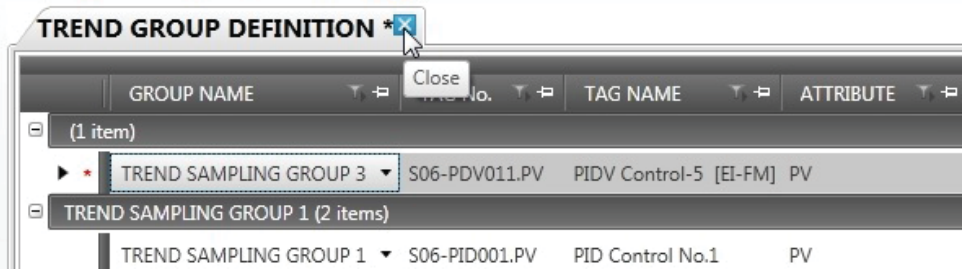


Click [+] to expand the row.

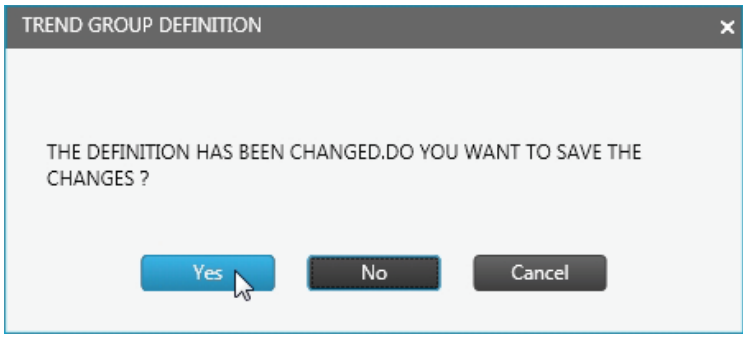
When the TAG is added, select a group name from the combo box.



When the definition edit is complete, confirm that the editing data has been confirmed. (refer to “2.8.1 Editing State”) In order to save changes, click the [X] button on the tab in the “Define trend group” panel.

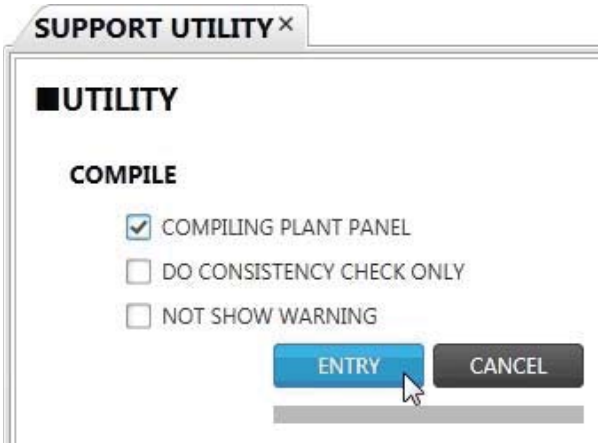


Since a confirmation panel is displayed, click [Yes] to save changes.

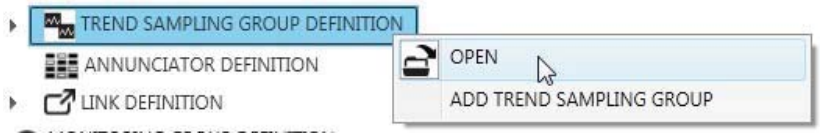


In order to reflect changes on other panels, select [Compile...] in the Project menu.

When the “Support utility” panel opens, click the [Run] button to compile. You can confirm the compile result on the Output window.



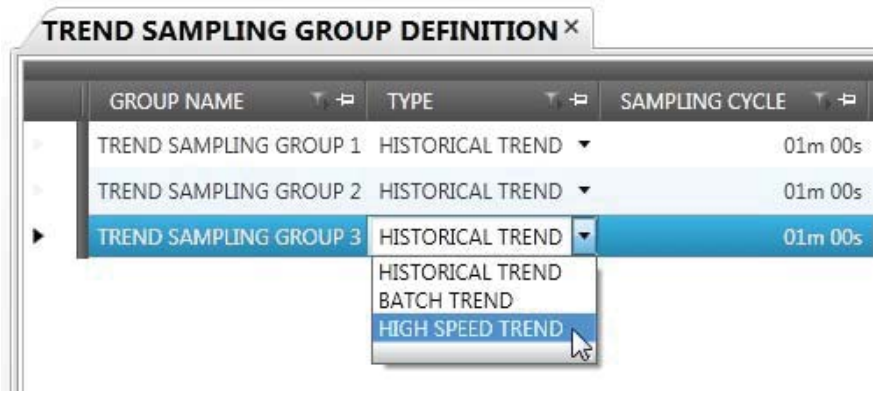
Right-click [Define trend collection group] in the Project tree and select [Open] from the pop-up menu.



The “Define trend collection group” panel opens.

On the “Define trend collection group” panel, select [Type] from the combo box.

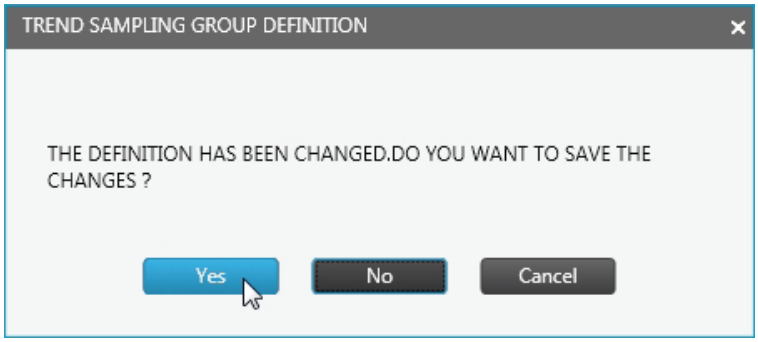
* Here, describes the case when a high speed trend is selected.



When the definition edit is complete, confirm that the editing data has been confirmed. (refer to “2.8.1 Editing State”)

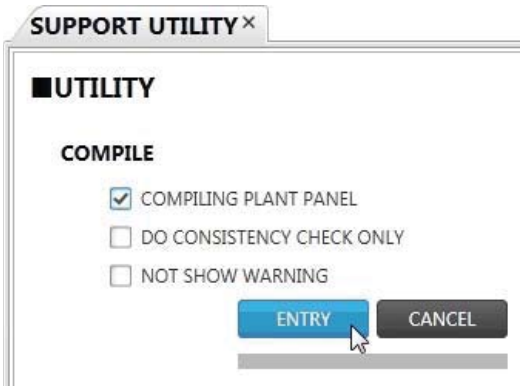
In order to save changes, click the [X] button on the tab in the “Define trend collection group” panel.

Since a confirmation panel is displayed, click [Yes] to save changes.



Select [Compile...] in the Project menu.

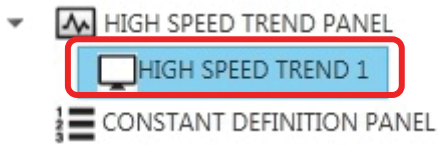
When the “Support utility” panel opens, click the [Run] button to compile. You can confirm the compile result on the Output window.



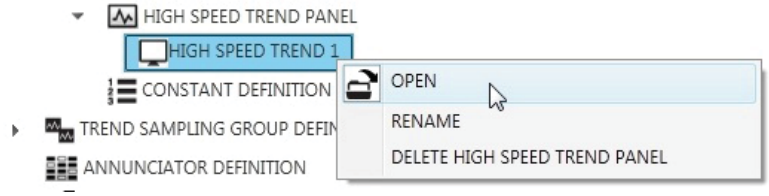
Right-click [High speed trend] in the Project tree and select [Add group] from the pop-up menu.

* When a historical trend is selected for [Type] on the “Define trend collection group” panel, right-click the [Trend] panel and select [Add group] from the pop-up menu.

A new group is added.

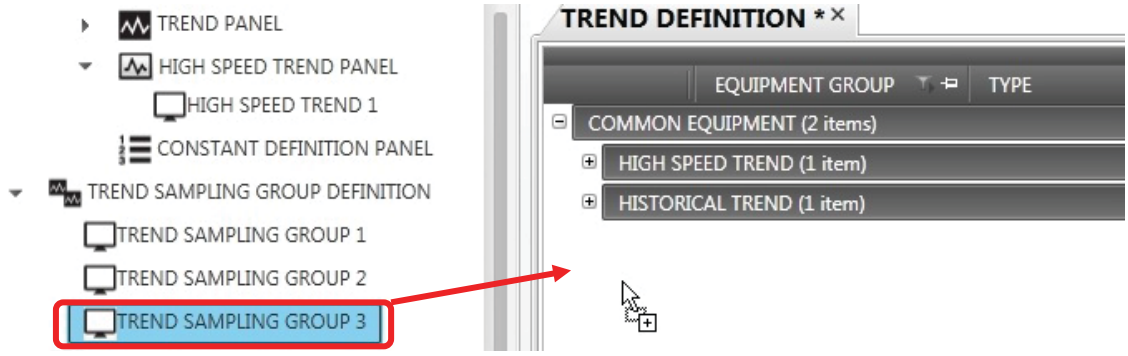


Right-click the group added and select [Open] from the pop-up menu.

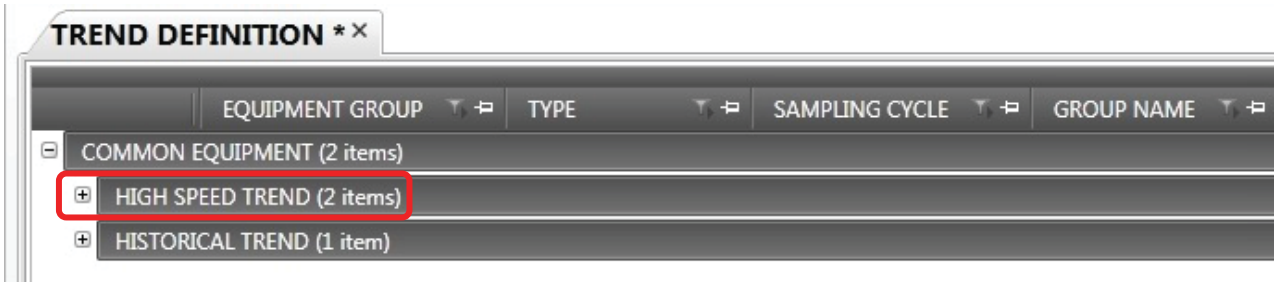


The “Define trend” panel opens.

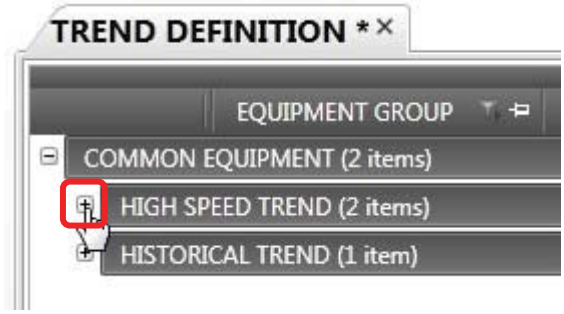
On the “Define trend” panel, drag and drop the trend collection group just defined from the Project tree to the “Define trend” panel.



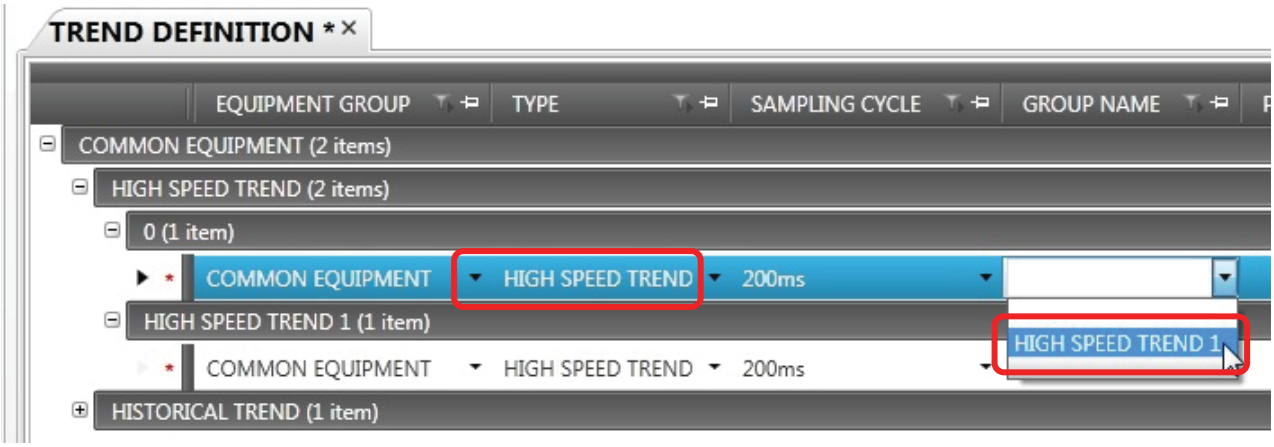
The trend collection group dragged and dropped is added.



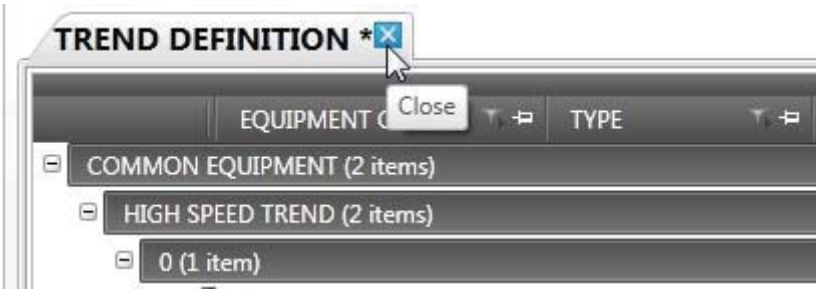
Click [+] to expand the row.



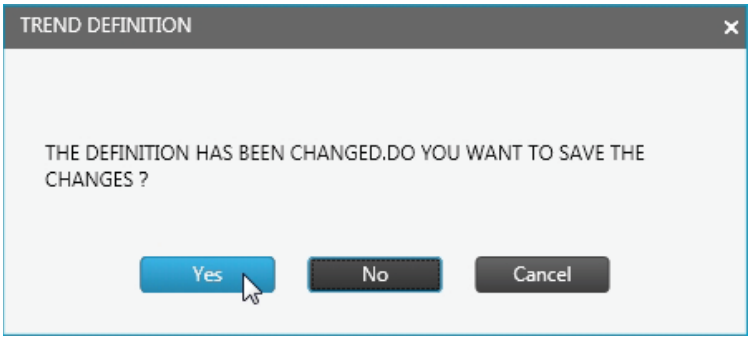
When the trend collection group is added, select the group just specified for opening the “Define trend” panel from the [Group name] combo box. Select appropriate type and pen No.



When the definition edit is complete, confirm that the editing data has been confirmed. (refer to “2.8.1Editing State”) In order to save changes, click the [X] button on the tab in the “Define trend” panel.

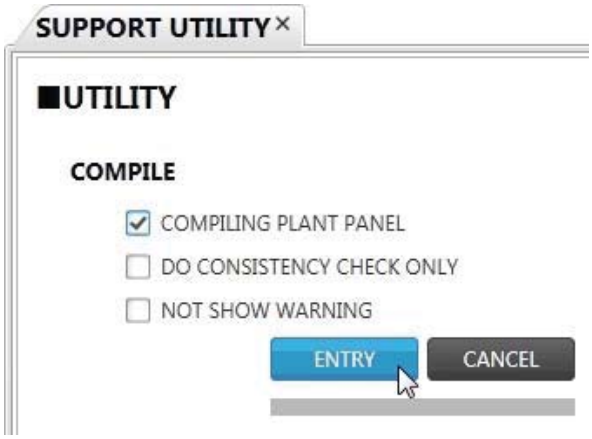


Since a confirmation panel is displayed, click [Yes] to save changes.



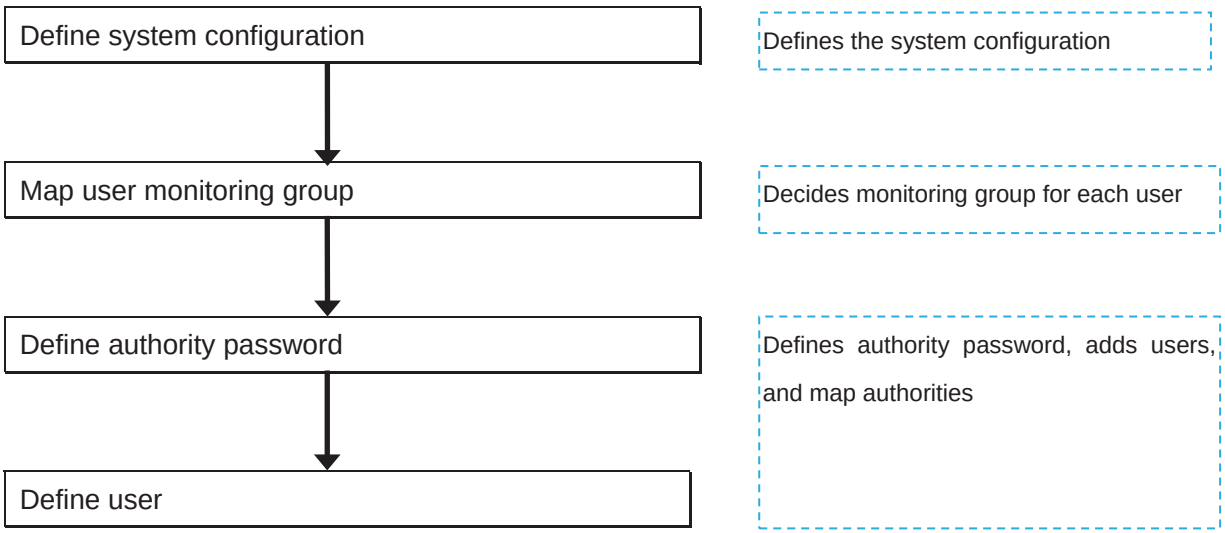
In order to reflect changes on other panels, select [Compile...] in the Project menu.

When the “Support utility” panel opens, click the [Run] button to compile. You can confirm the compile result on the Output window.

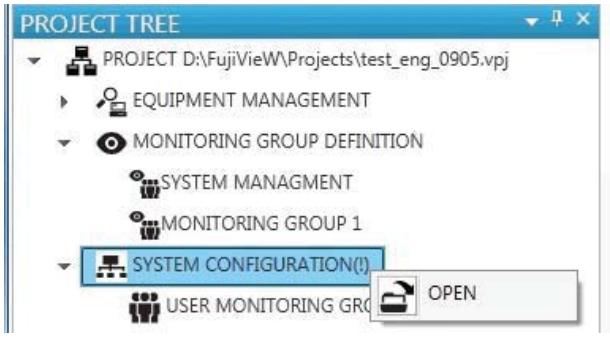


3.6 Editing System Configuration

This section describes the editing of system configuration, mapping of user monitoring group, defining of system's authority password, and editing of user definition.



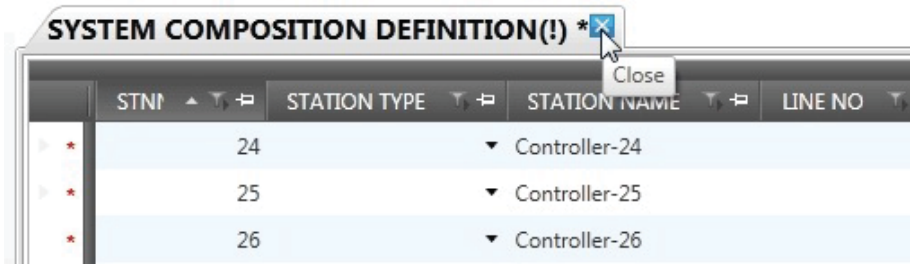
Right-click [System configuration] in the Project tree and select [Open] from the pop-up menu.



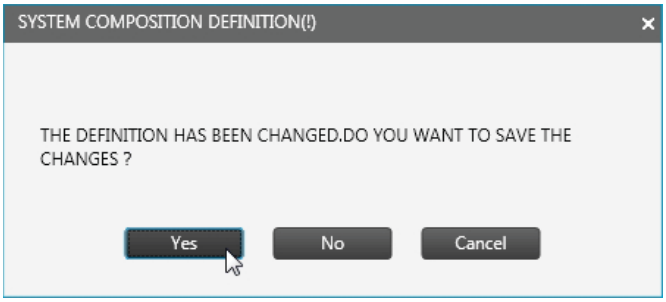
The “Define system configuration” panel opens.

On the “Define system configuration” panel, set up necessary items. For information about items, refer to “3.3 Defining System Configuration”.

When the definition edit is complete, confirm that the editing data has been confirmed. (refer to “2.8.1 Editing State”) In order to save changes, click the [X] button on the tab in the “Define system configuration” panel.

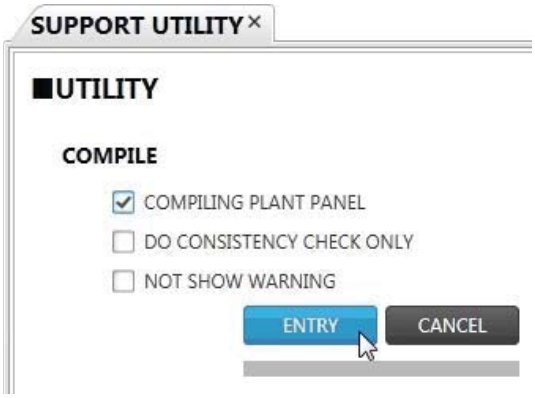


Since a confirmation panel is displayed, click [Yes] to save changes.

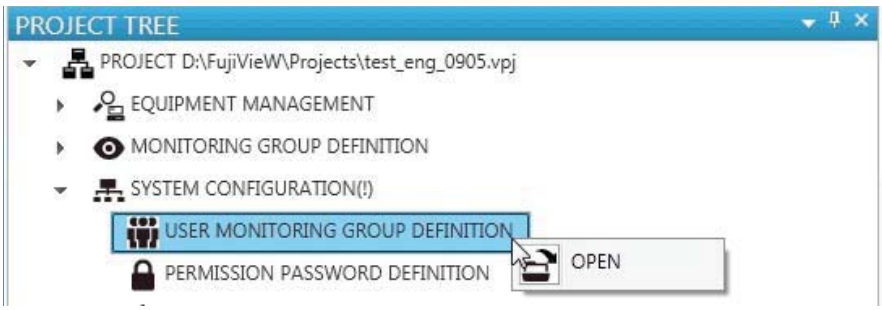


Select [Compile...] in the Project menu.

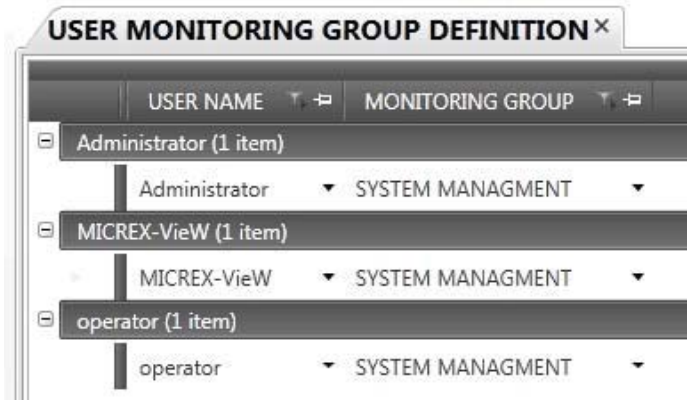
When the “Support utility” panel opens, click the [Run] button to compile. You can confirm the compile result on the Output window.



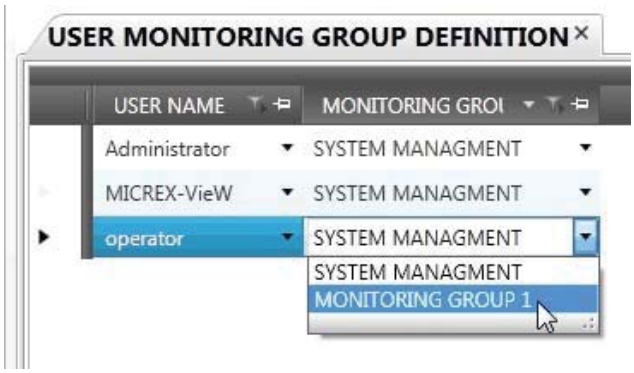
Right-click [Map user monitoring group] in the Project tree and select [Open] from the pop-up menu.



The “Map user monitoring group” panel opens.

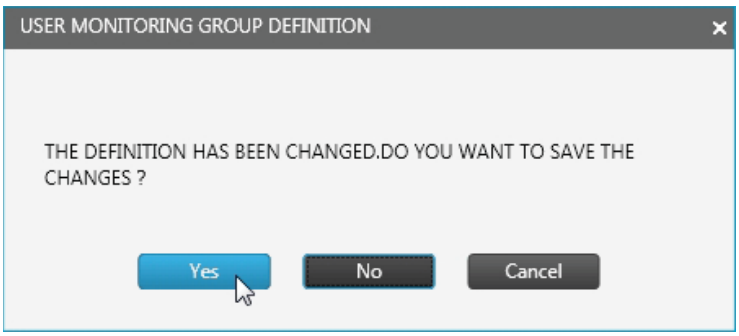


On the “Map user monitoring group” panel, select [Monitoring group]. For information about the monitoring group definition, refer to “5.3 Editing Monitoring Group”.



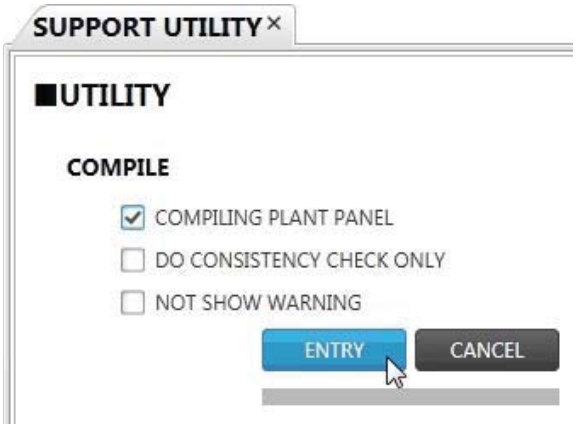
When the definition edit is complete, confirm that the editing data has been confirmed. (refer to “2.8.1 Editing State”) In order to save changes, click the [X] button on the tab in the “Map user monitoring group” panel.

Since a confirmation panel is displayed, click [Yes] to save changes.

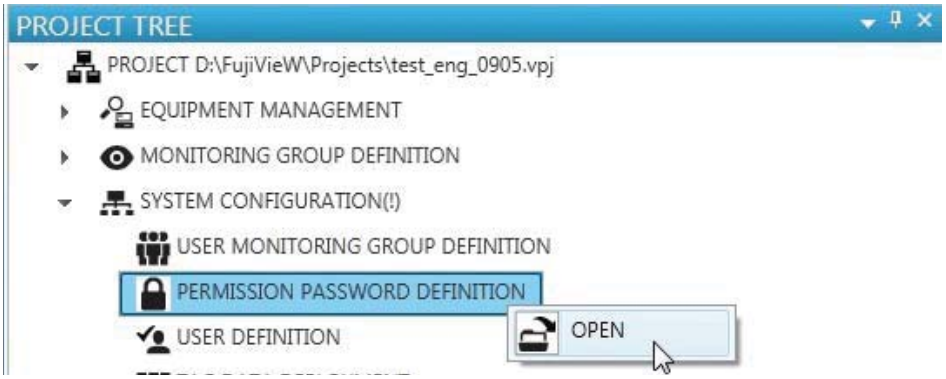


Select [Compile...] in the Project menu.

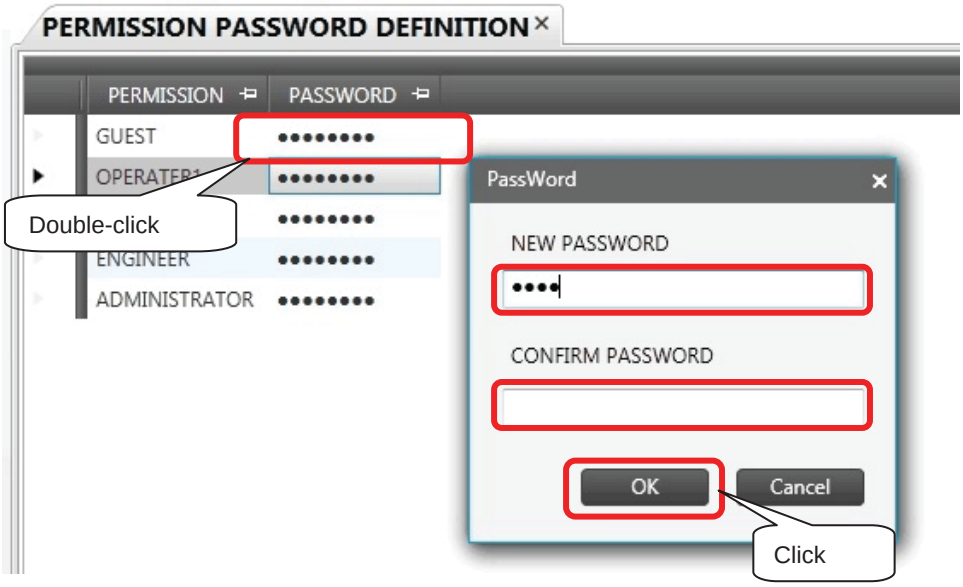
When the “Support utility” panel opens, click the [Run] button to compile. You can confirm the compile result on the Output window.



Right-click [Authority password management] in the Project tree and select [Open] from the pop-up menu.

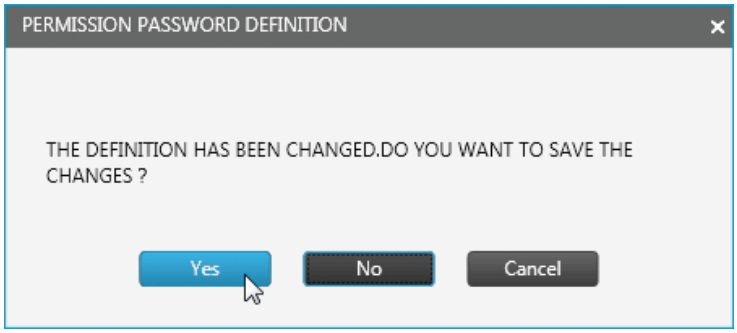


On the “Define authority password” panel, double-click “●●●●●●●●” in the [Password] field to open the Password dialog and set up the password.



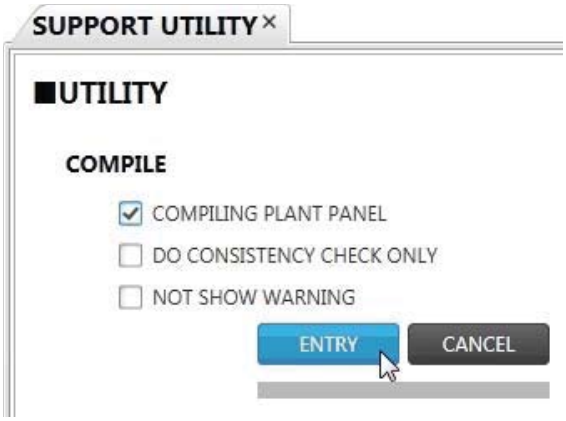
When the definition edit is complete, confirm that the editing data has been confirmed. (refer to “2.8.1Editing State”) In order to save changes, click the [X] button on the tab in the “Define authority password” panel.

Since a confirmation panel is displayed, click [Yes] to save changes.

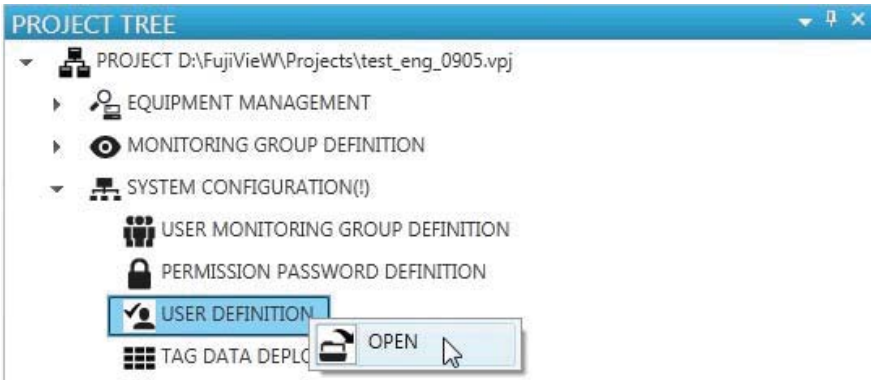


Select [Compile...] in the Project menu.

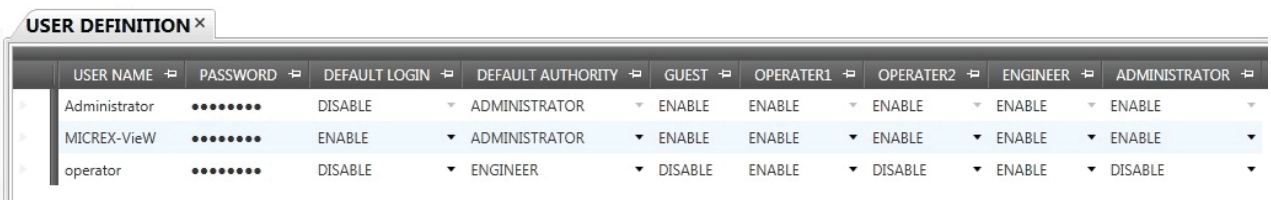
When the “Support utility” panel opens, click the [Run] button to compile. You can confirm the compile result on the Output window.



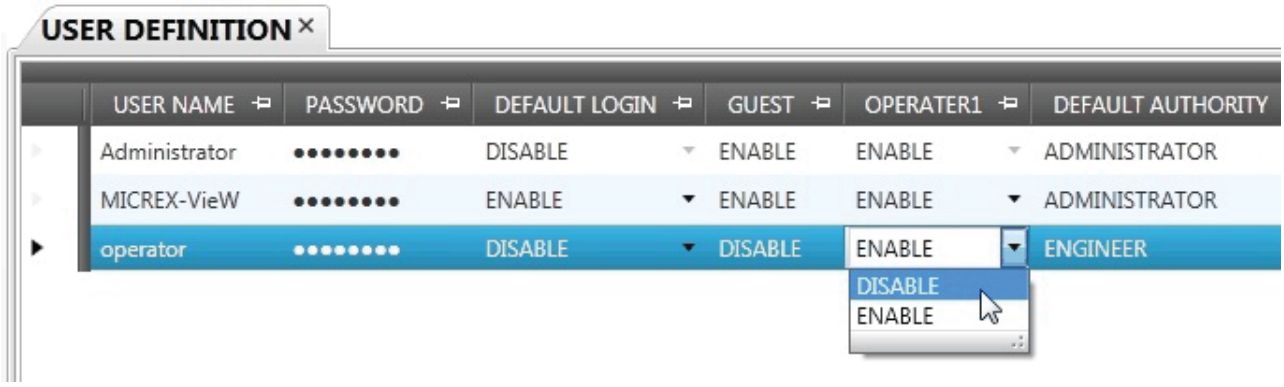
Right-click [Define user] in the Project tree and select [Open] from the pop-up menu.



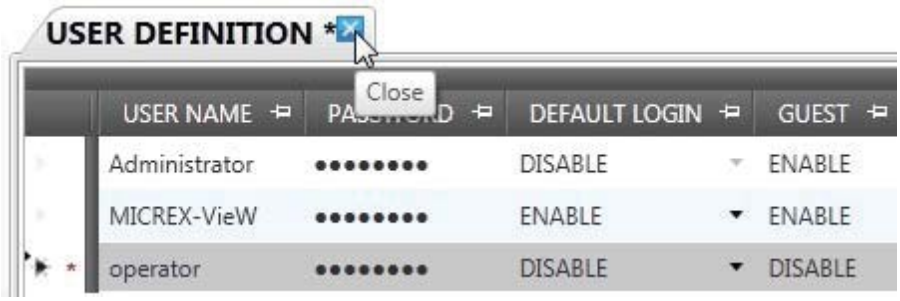
The “Define user” panel opens.



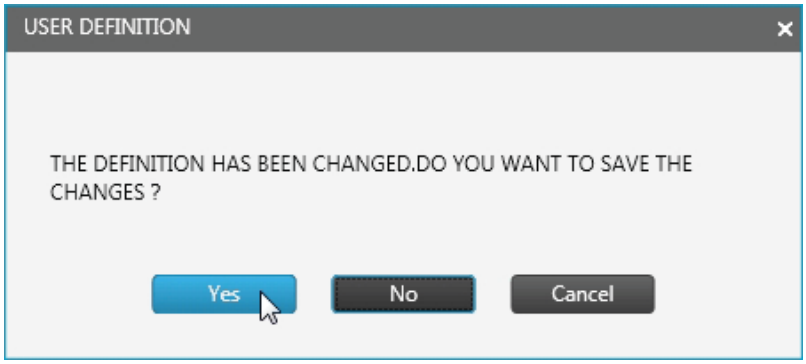
On the “Define user” panel, set up [Default login] authority, and enable/disable [Guest], [Operator], [Engineer], and [Administrator].



When the definition edit is complete, confirm that the editing data has been confirmed. (refer to “2.8.1 Editing State”) In order to save changes, click the [X] button on the tab in the “Define user” panel.

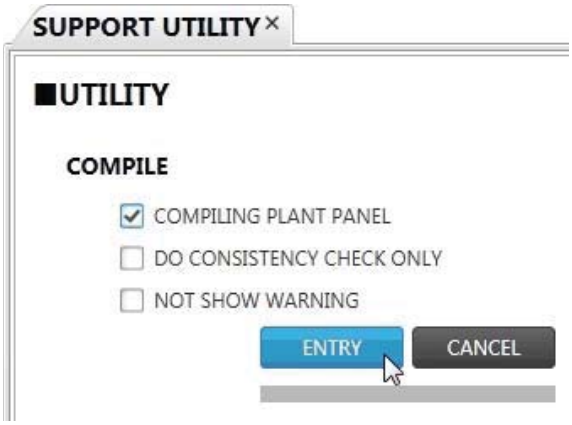


Since a confirmation panel is displayed, click [Yes] to save changes.



Select [Compile...] in the Project menu.

When the “Support utility” panel opens, click the [Run] button to compile. You can confirm the compile result on the Output window.



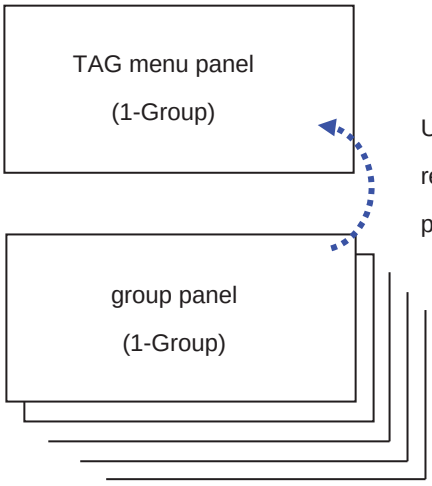
4. Mapping Standard Panel

This chapter describes the mapping functions for standard panel.

The table below lists the mapping functions for standard panel.

Function	Description
Define group	Registers TAGs and groups to be displayed in the group panel. Up to 16 TAGs can be registered per group and their display order can be designated.
Define trend	Registers TAGs and trend groups to be displayed in the trend panel. Up to 8 TAGs can be registered per group and their display order can be designated.

Relation between TAG menu panel and group panel



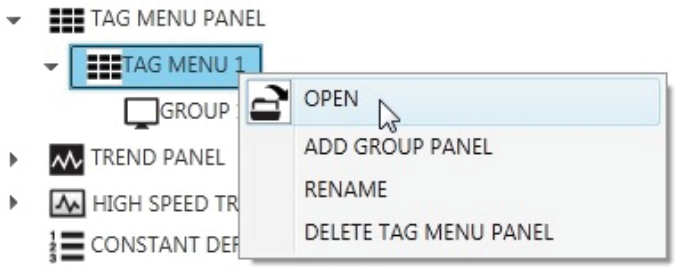
Up to 16 groups of the TAG menu can be registered in Group 1 of the TAG menu panel.

4.1 Define Group Panel

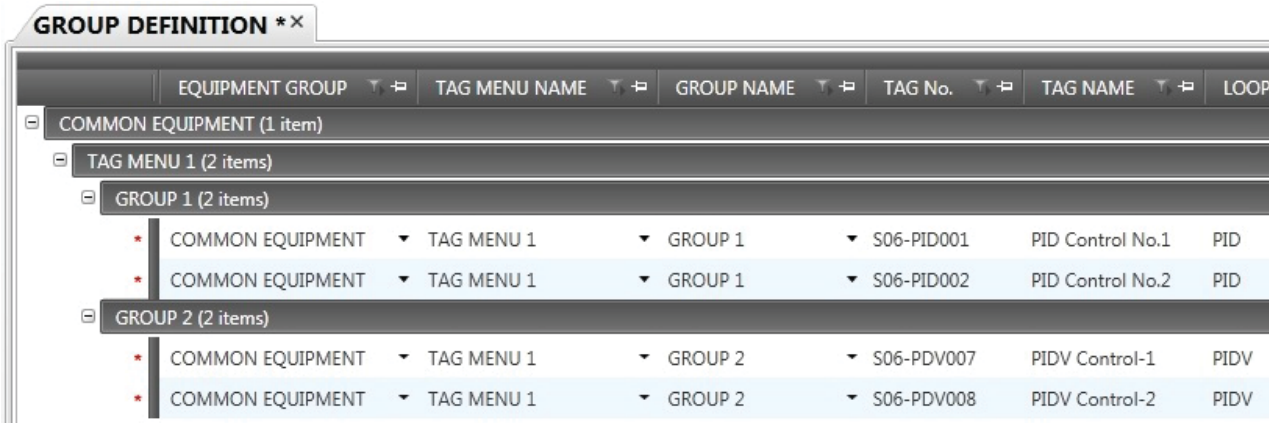
This section describes the functions of the Define group panel.

For information about the operational procedures of Define group panel, refer to “5.3 Editing Monitoring Group”.

Right-click [Group] in the Project tree and select [Open] from the pop-up menu.



The “Define group” panel opens.



Items of the “Define group” panel

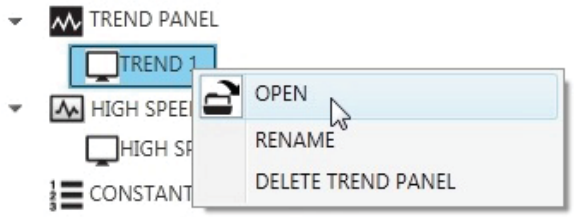
Item	Description	Type	Remarks
Facility category	Name of a facility	Mandatory	Select a facility for displaying groups.
Display order	Display order of TAGs in the group	Mandatory	
Group name	Name of group panel	Mandatory	
TAGNO	TAG to be displayed on the Group panel	Mandatory	Enabled to define module TAGs only. Up to 32.
TAG name	Name of a TAG	Display only	
Loop type	Loop type of a TAG	Display only	

4.2 Define Trend Panel

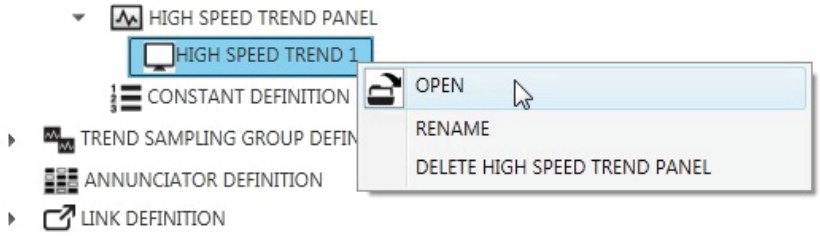
This section describes the functions of the Define trend panel.

For information about the operational procedures of Define trend panel, refer to “3.5Editing Trend Collection Group”.

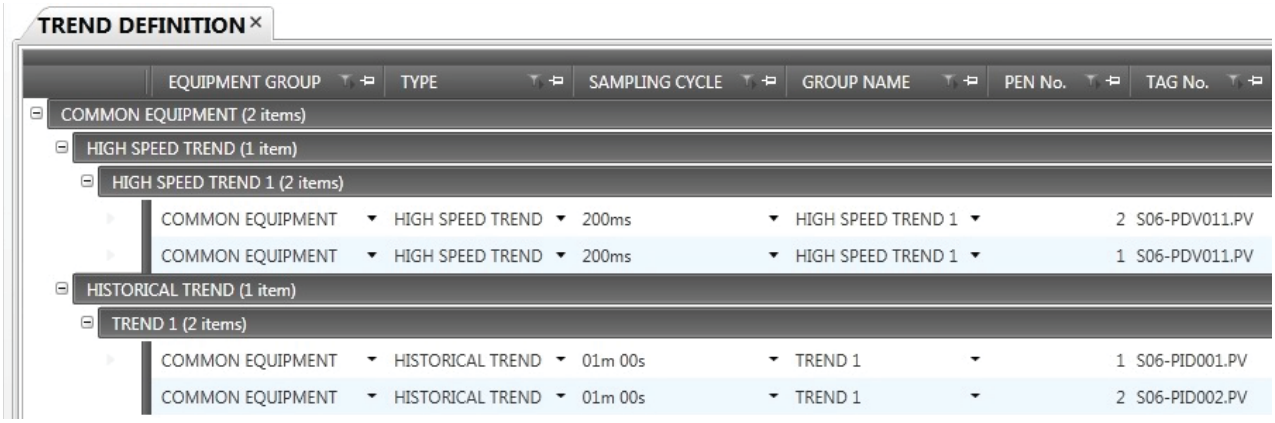
Click [Panel] > [Trend] in the Project tree, right-click a group, and select [Open] from the pop-up menu.



Or, right-click a group under [High speed trend] in the Project tree, and select [Open] from the pop-up menu.



The same “Define trend” panel opens in either procedures. Select either procedures according to the type of the trend collection group to be added.



Items of the “Define trend” panel

Item	Description	Type	Remarks
Facility category	Name of a facility	Mandatory	Select a facility for displaying the TAG menu.
Type	Type of a collection group	Mandatory	Historical trend or high speed trend
Collection cycle	Collection cycle defined in the collection group	Mandatory	
Group name	Name of the trend panel	Mandatory	
TAGNO	Collection cycle of TAGNO to be displayed in the trend panel	Mandatory	[TAG].[Attribute name] Enter TAGNO collected at the cycle selected in the collection cycle. Up to 8.

5. System Definition

This chapter describes the system definition function of the HCI engineering capability.

5.1 Setting Items for System Definition Function

The table below lists setting items for the system definition function.

Classification	Item	Description
System setting	Define system configuration	Sets up the system configuration
	Tag	Sets up the tag name
TAG menu panel	Define group in TAG menu panel	Sets up groups (groups in group panel) and details for displaying on the TAG menu panel
Group panel	Define group in group panel	Sets up groups and details (TAGs) for displaying on the group panel
Trend panel	Define group in trend panel	Sets up groups and details (TAGs) for displaying on the trend panel
	Trend collection cycle	Sets up trend collection groups and cycles
Define security panel	Set up trend view	Sets up display on the trend panel
	Set up monitoring group	Sets up monitoring groups
	Define user	Set up users
	Define password	Sets up password for each user authority
	Define facility category	Sets up facility category

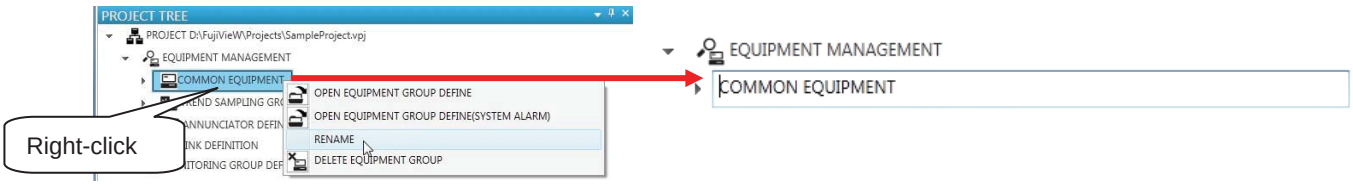
5.2 Input Pattern for System Definition Function

The table below lists setting items and input patterns for the system definition function.

Setting item example	Input pattern	Purpose
Facility category name, group name	Manual input through keyboard	Changes names
Source TAG folder path	Manual input through keyboard	Changes text
Pen No.	Manual input through keyboard Increase or decrease with a spin button	Inputs numbers
Collection cycle (_m_s), IP address (___.___.___.___)	Manual input through keyboard	Inputs formatted numbers
Facility category, group, model, operation comment, display color	Selection with a combo box	Decides a choice
Authority password (●●●●●●●●)	Manual input through keyboard	Inputs secret string
Display color	Selection with a combo box	Decides a choice

Project tree

[Change group name]

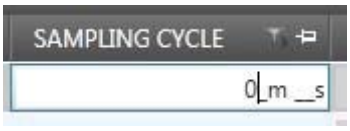


Definition edit window

[Change text]

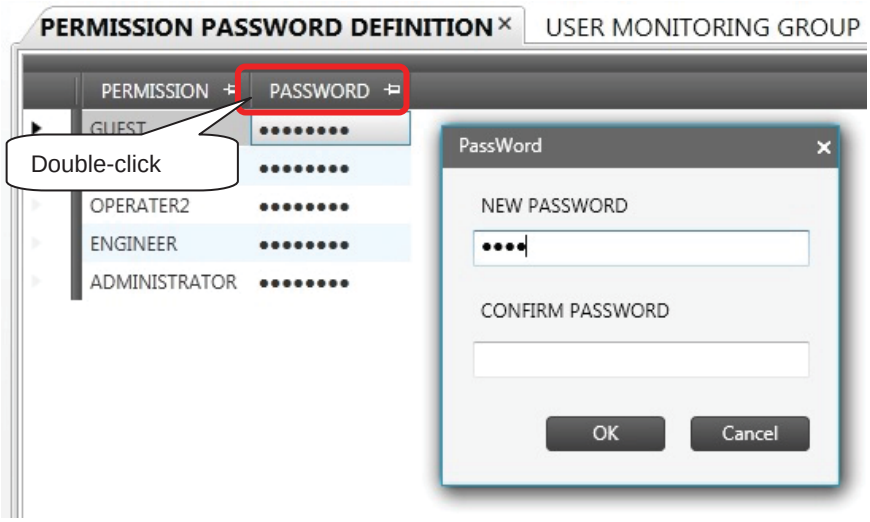
[Change number (spin button)]

[Change formatted number]

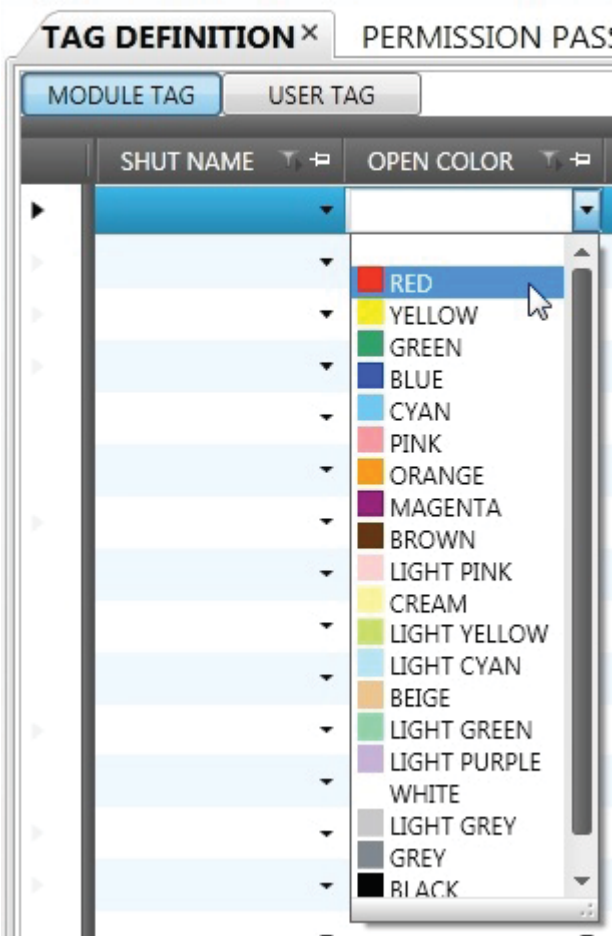


[Decide choice]

[Input secret string]

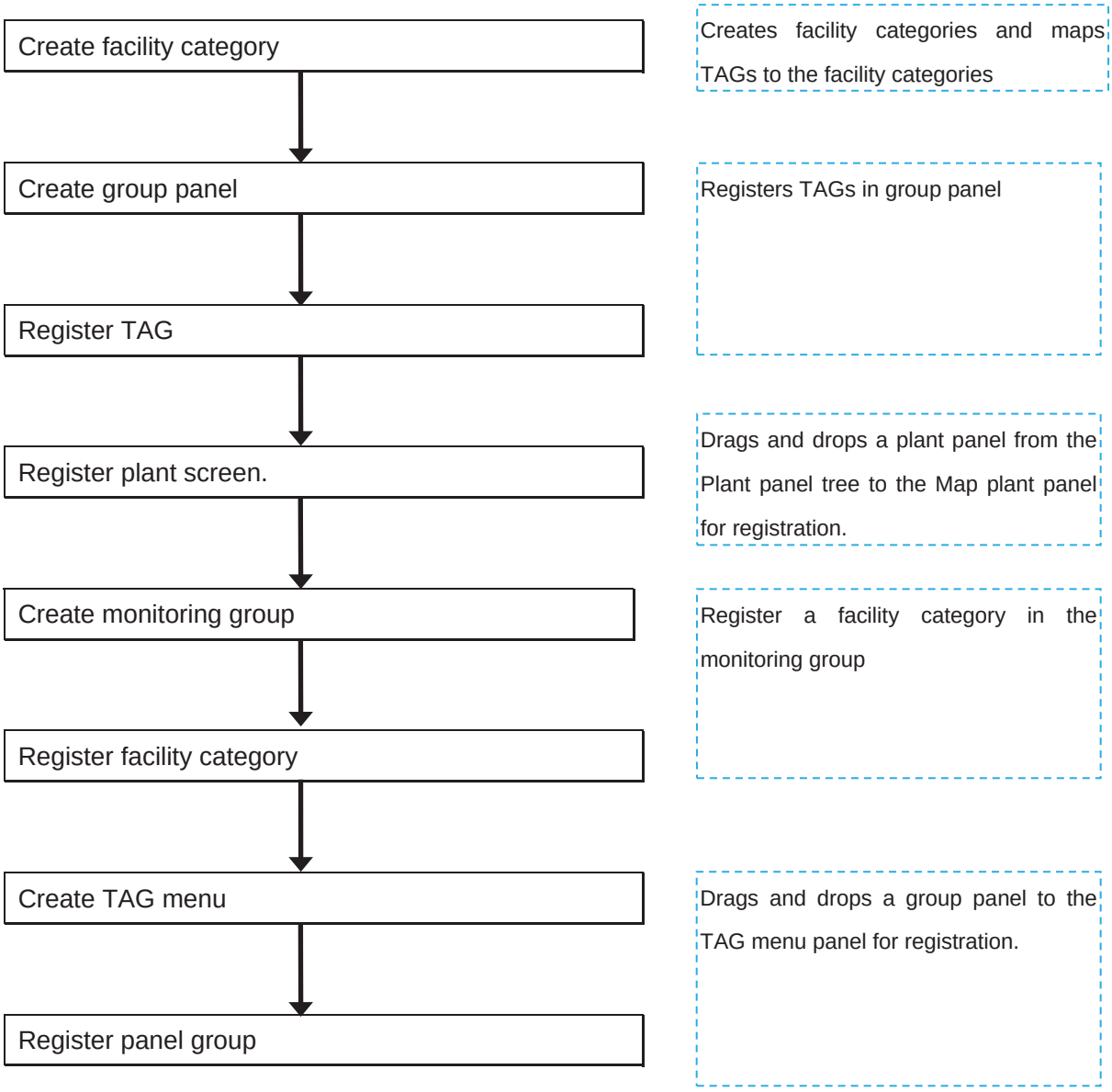


[Select color]

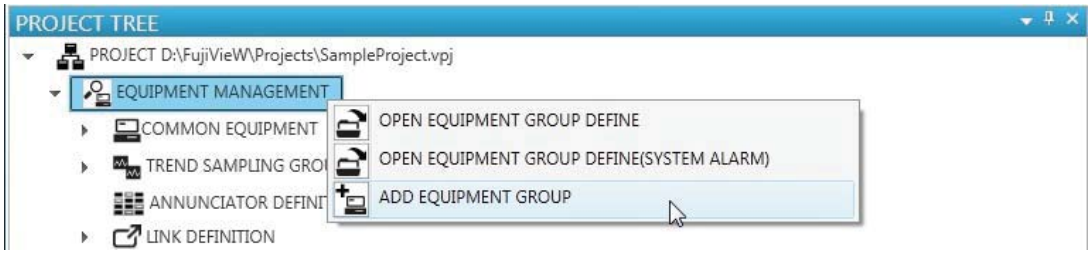


5.3 Editing Monitoring Group

This section describes the editing procedures of monitoring groups.

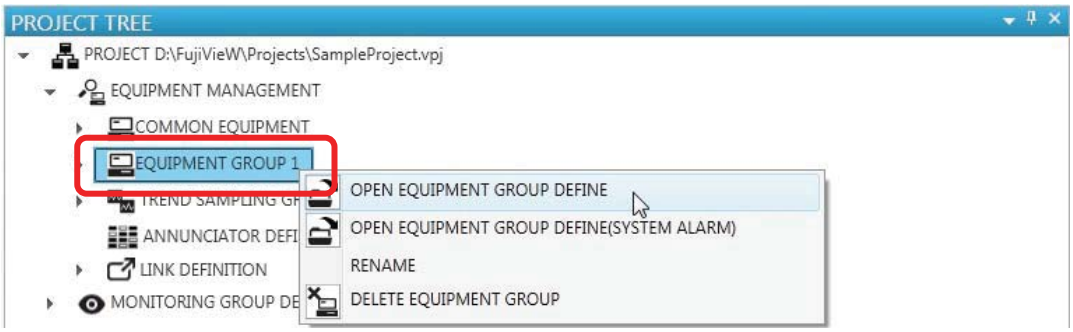


Right-click [Facility management] in the Project tree and select [Add category] from the pop-up menu.

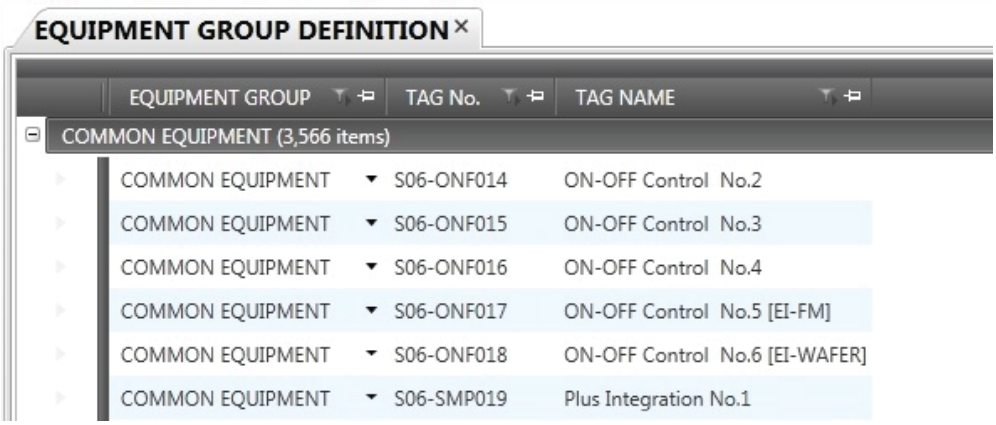


A facility category is added to the Project tree.

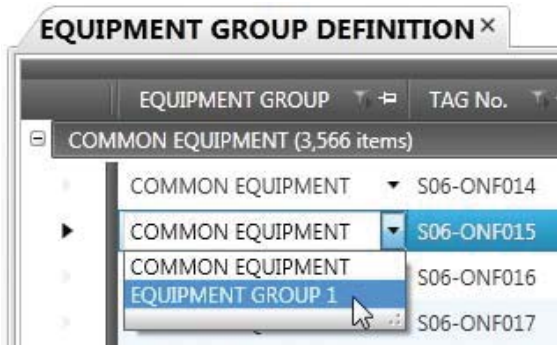
Right-click the facility category added and select [Open] from the pop-up menu.



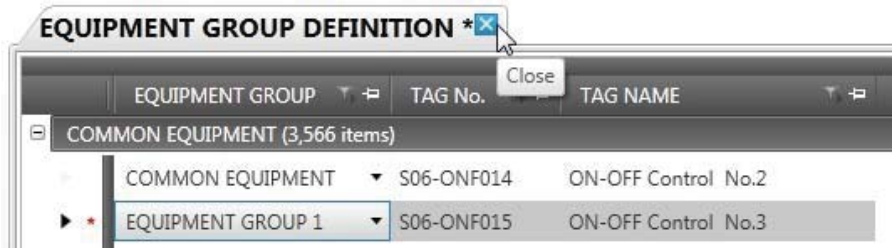
The “Define facility category” panel opens.



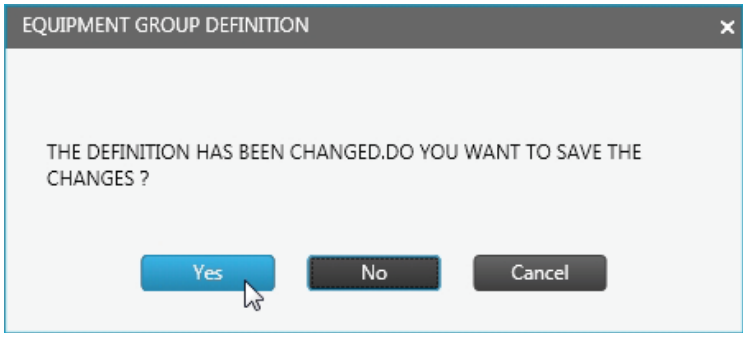
On the “Define facility category” panel, select the added facility category in the [Facility category] field of the TAG to be monitored.



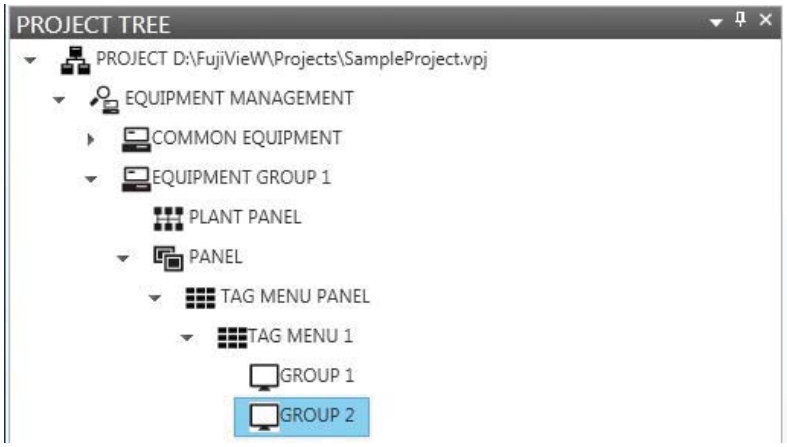
When the definition edit is complete, confirm that the editing data has been confirmed. (refer to “2.8.1 Editing State”) In order to save changes, click the [X] button on the tab in the “Define facility category” panel.



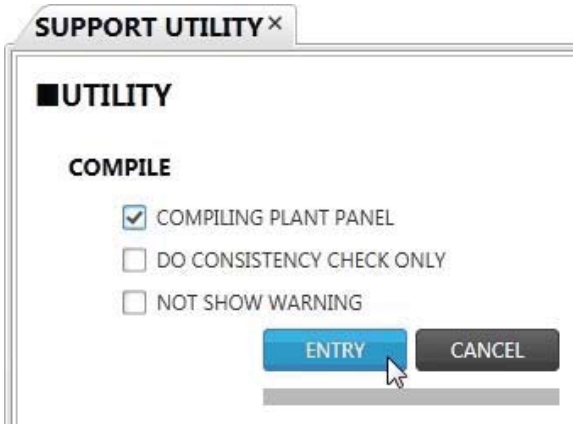
Since a confirmation panel is displayed, click [Yes] to save changes.



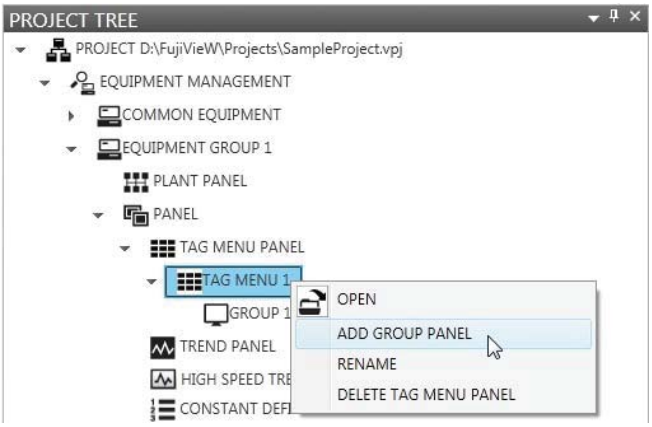
In order to reflect changes on other panels, select [Compile...] in the Project menu.



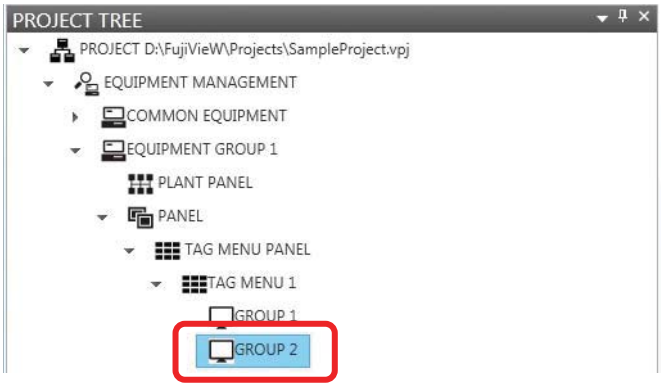
When the “Support utility” panel opens, click the [Run] button to compile. You can confirm the compile result on the Output window.



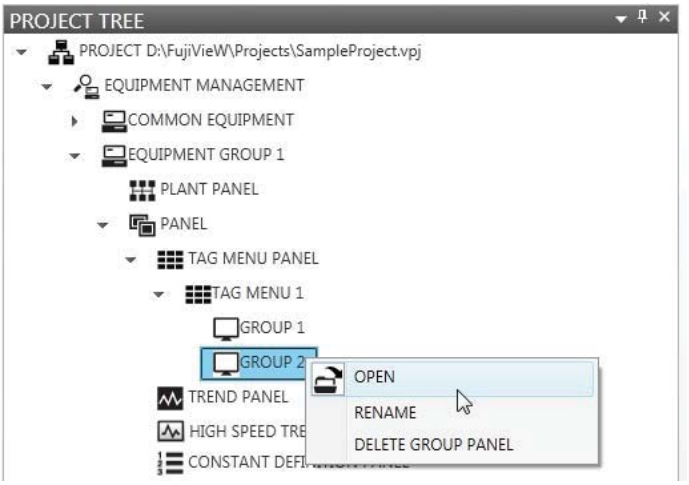
Right-click [Group panel] for the facility category added and select [Add group] from the pop-up menu.



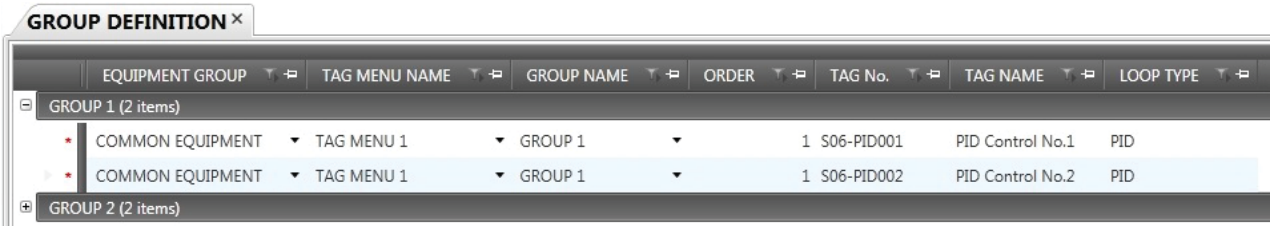
A new group is added.



Right-click the group added and select [Open] from the pop-up menu.

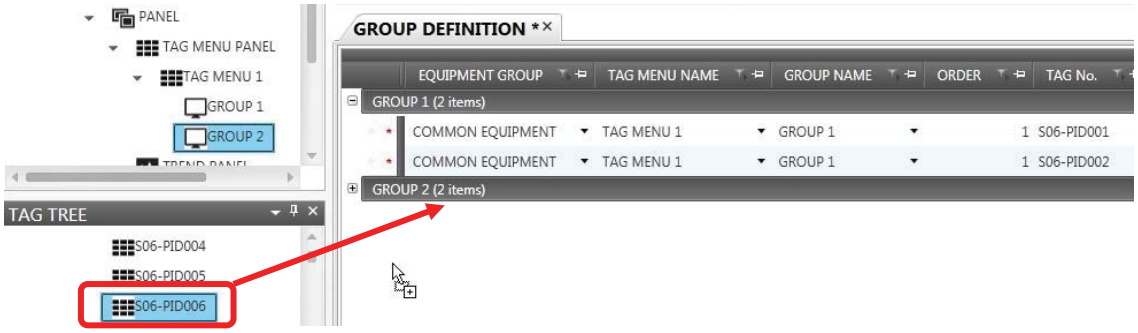


The “Define group” panel opens.

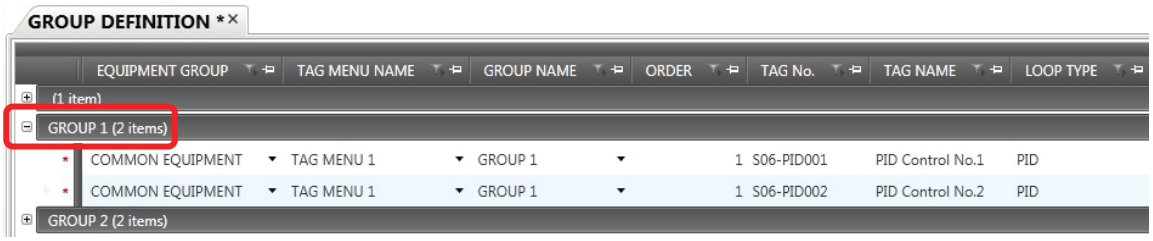


On the “Define group” panel, drag and drop a TAG or category code from the TAG tree to the “Define group” panel.

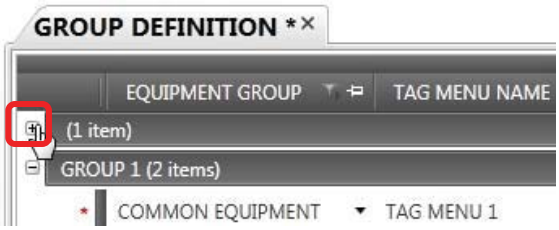
The example below indicates that a TAG is dragged and dropped.



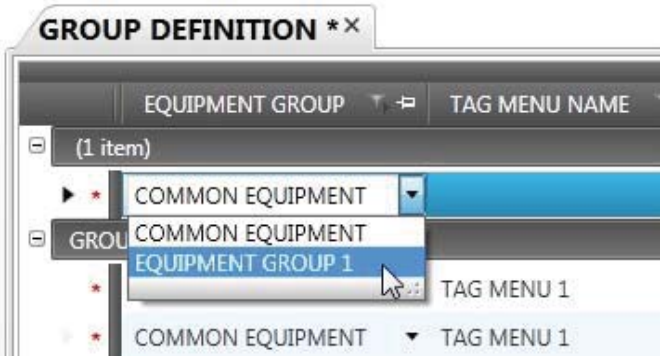
The TAG dragged and dropped is added.



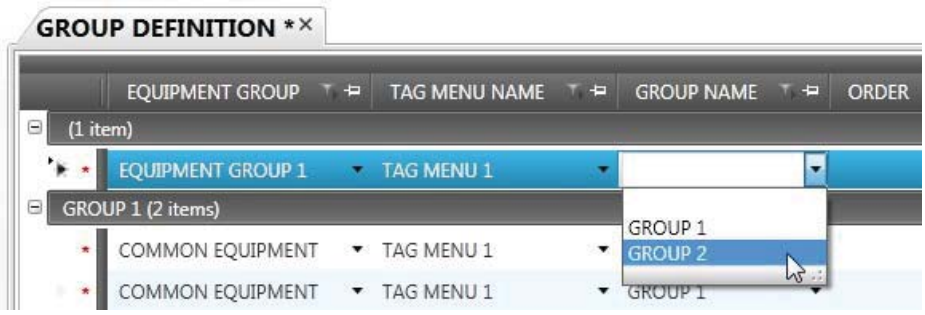
Click [+] to expand the row.



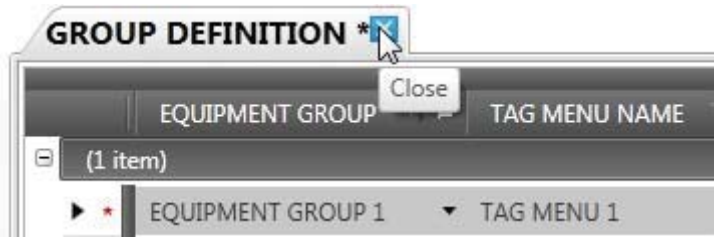
On the row where TAG was added, select the facility category added in the [Facility category] field.



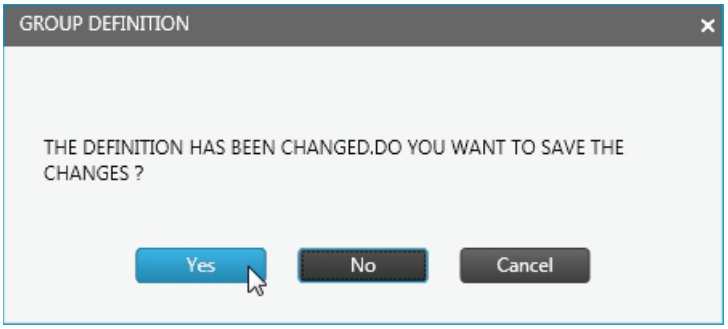
Select the group added in the [Group name] field.



When the definition edit is complete, confirm that the editing data has been confirmed. (refer to “2.8.1 Editing State”) In order to save changes, click the [X] button on the tab in the “Define group” panel.

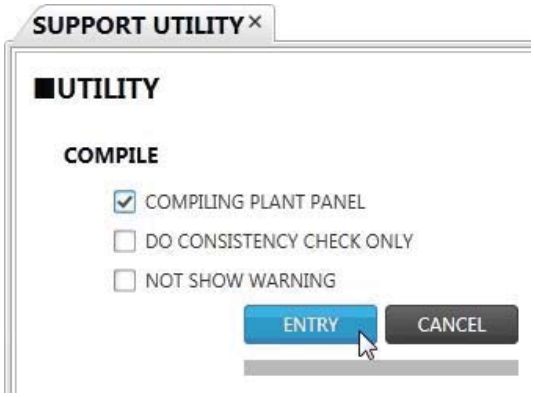


Since a confirmation panel is displayed, click [Yes] to save changes.

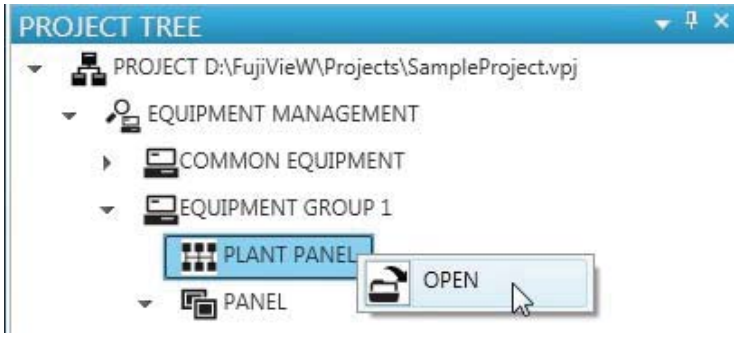


In order to reflect changes on other panels, select [Compile...] in the Project menu.

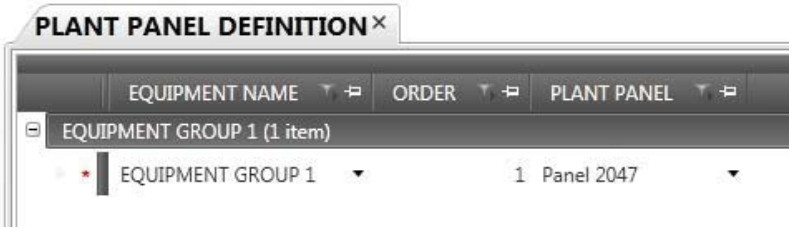
When the “Support utility” panel opens, click the [Run] button to compile. You can confirm the compile result on the Output window.



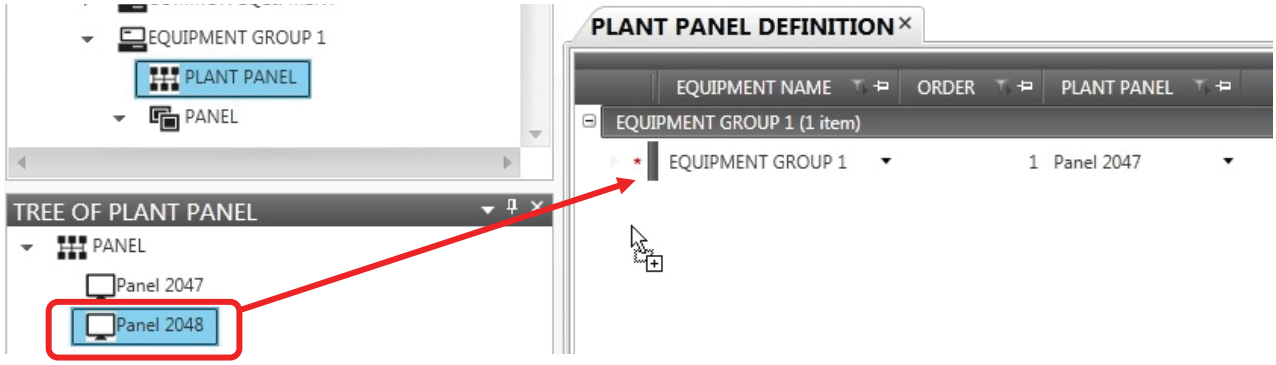
Right-click [Plant panel] for the facility category added and select [Open] from the pop-up menu.



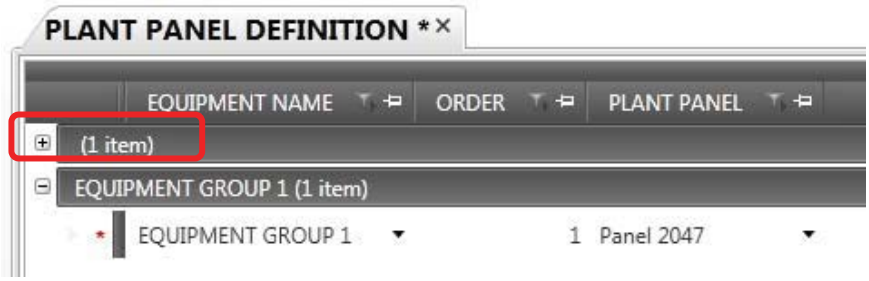
The “Map plant panel” panel opens.



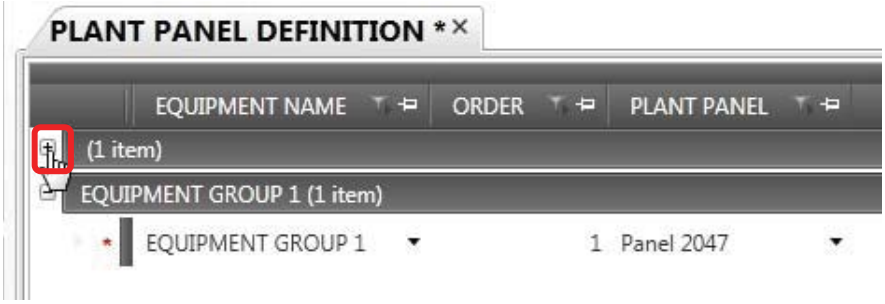
On the “Map plant panel” panel, drag and drop a plant panel from the Plant panel tree to the “Map plant panel” panel.



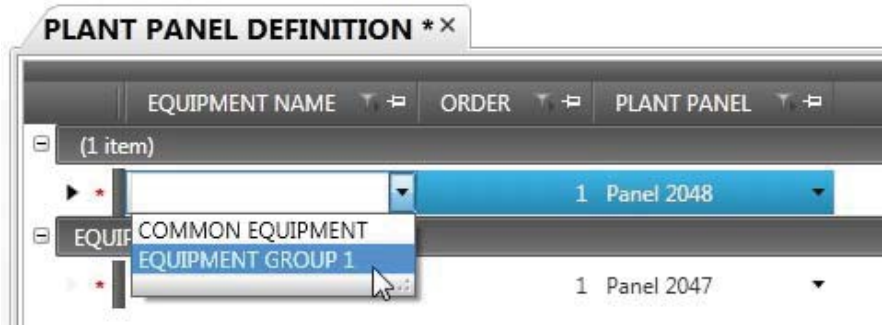
The plant panel dragged and dropped is added.



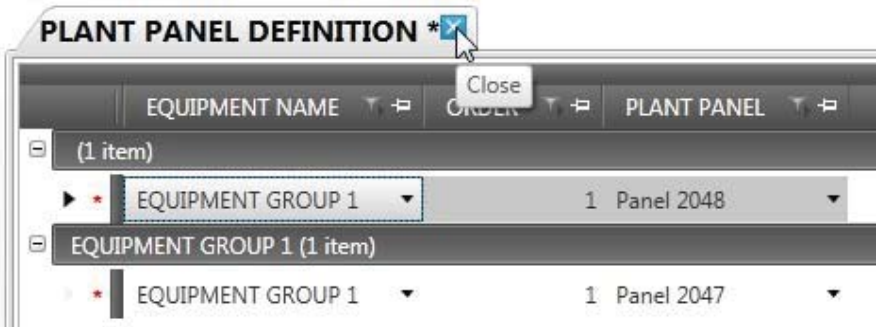
Click [+] to expand the row.



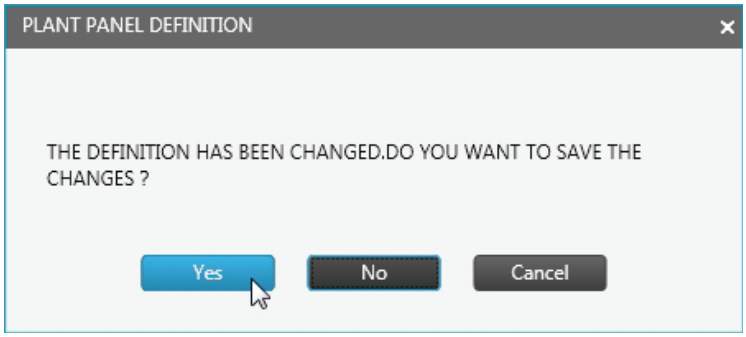
Select the facility category just added in the [Facility name] field.



When the definition edit is complete, confirm that the editing data has been confirmed. (refer to “2.8.1 Editing State”) In order to save changes, click the [X] button on the tab in the “Map plant panel” panel.

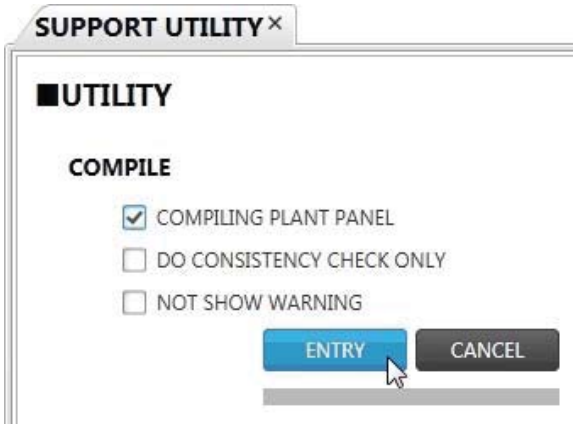


Since a confirmation panel is displayed, click [Yes] to save changes.

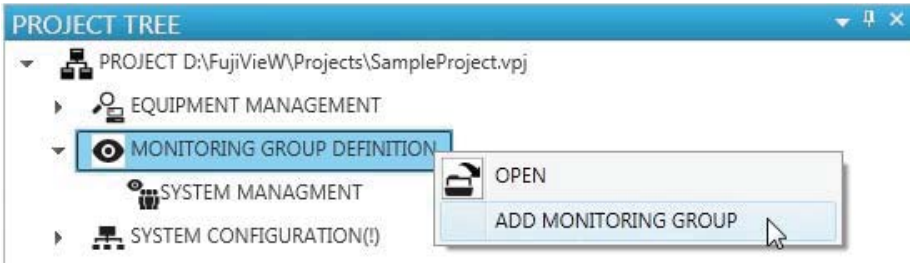


In order to reflect changes on other panels, select [Compile...] in the Project menu.

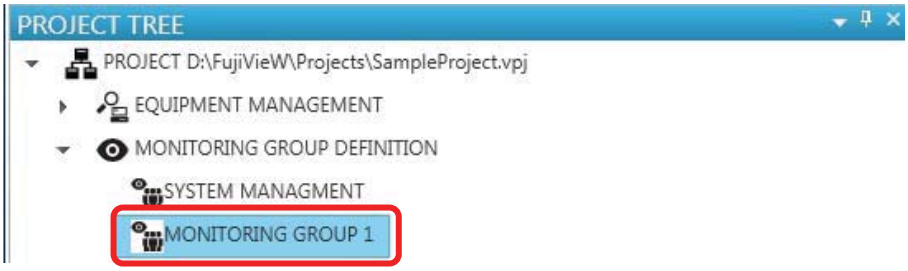
When the “Support utility” panel opens, click the [Run] button to compile. You can confirm the compile result on the Output window.



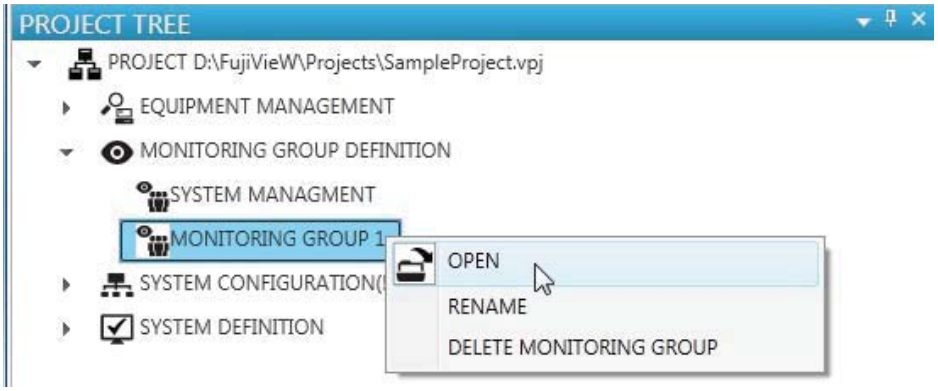
Right-click [Monitoring group management] and select [Add group] from the pop-up menu.



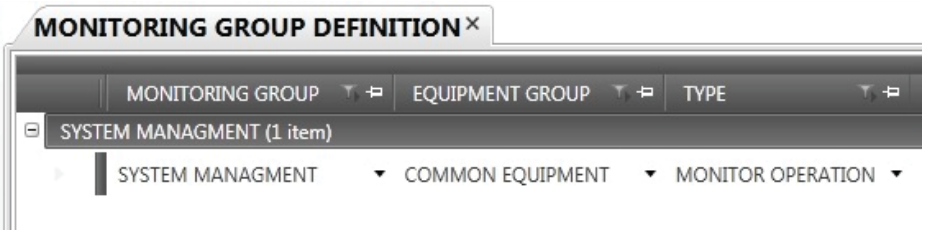
A new group is added.



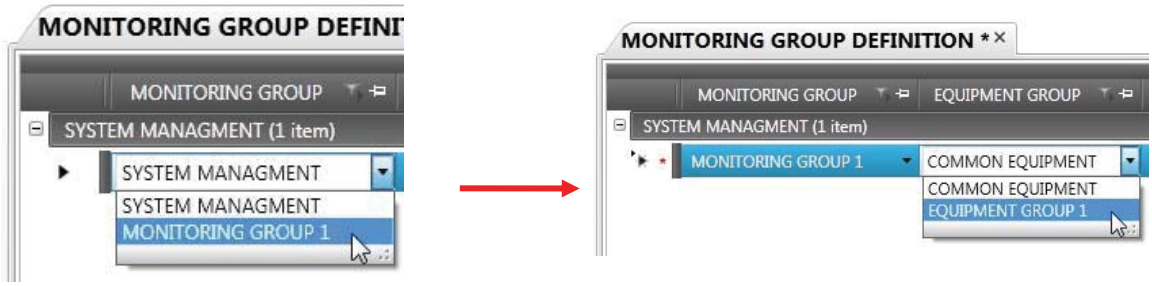
Right-click the monitoring group added and select [Open] from the pop-up menu.



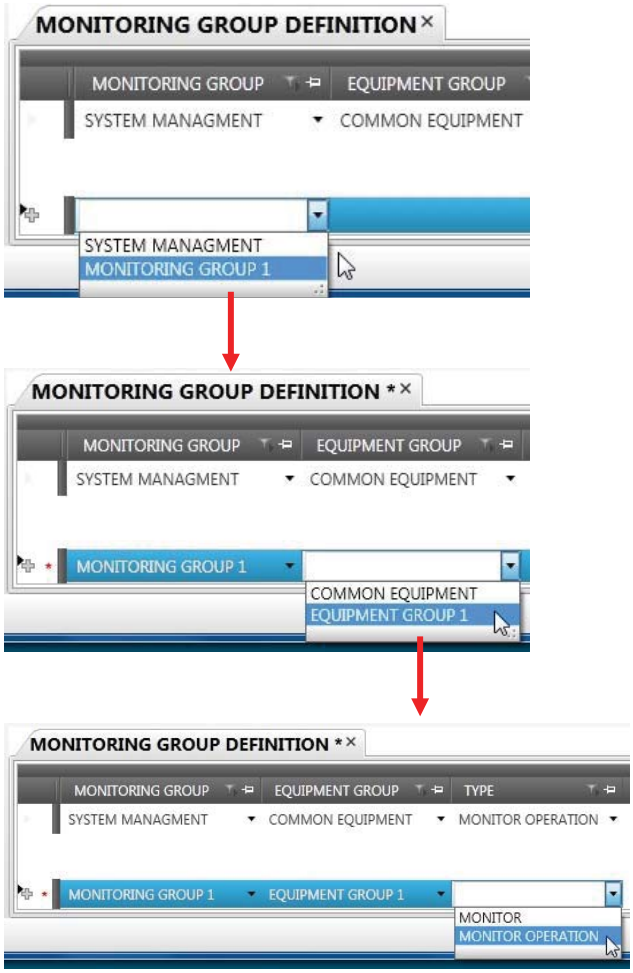
The “Define monitoring group” panel opens.



On the “Define monitoring group” panel, select the group just added in the [Monitoring group] field.*



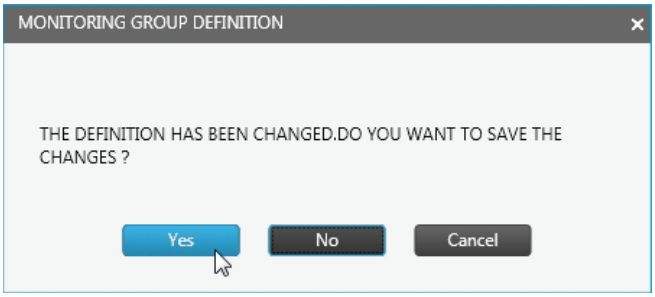
* To add a group definition while leaving the system management definition behind, select the group in the Add row area at the bottom in the Definition edit window. For information about the Add row area, refer to “2.8.2 Adding Data Row”.



Press the Enter key to confirm the addition of the row.

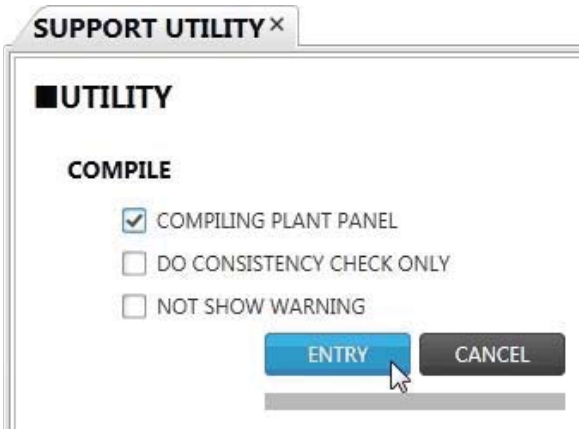
When the definition edit is complete, confirm that the editing data has been confirmed. (refer to “2.8.1 Editing State”) In order to save changes, click the [X] button on the tab in the “Define monitoring group” panel.

Since a confirmation panel is displayed, click [Yes] to save changes.

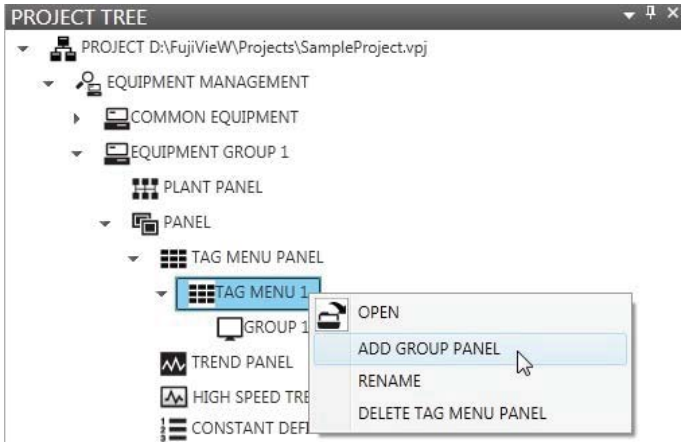


In order to reflect changes on other panels, select [Compile...] in the Project menu.

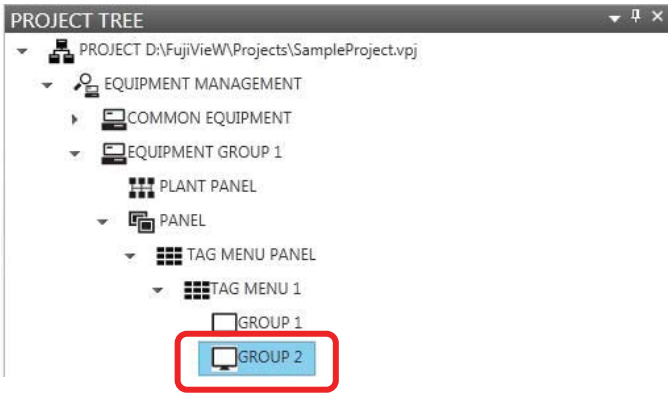
When the “Support utility” panel opens, click the [Run] button to compile. You can confirm the compile result on the Output window.



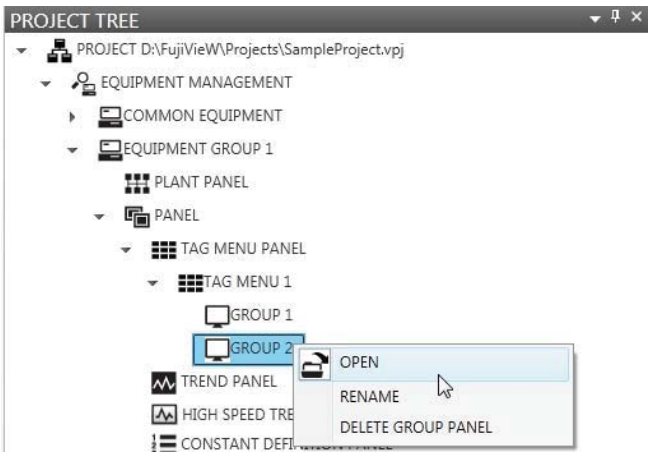
Right-click [Tag menu panel] in the Project tree and select [Add group] from the pop-up menu.



A new group is added.

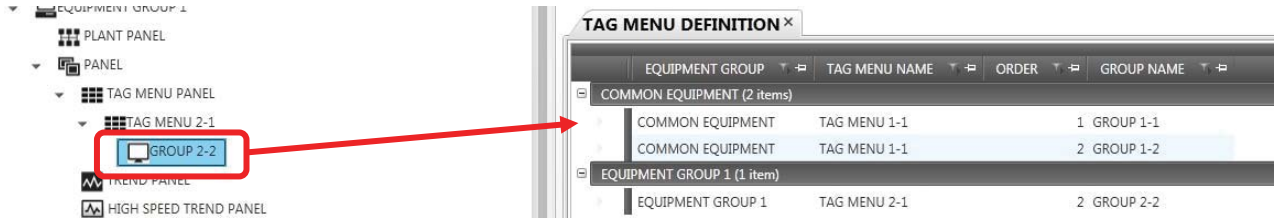


Right-click the group added and select [Open] from the pop-up menu.



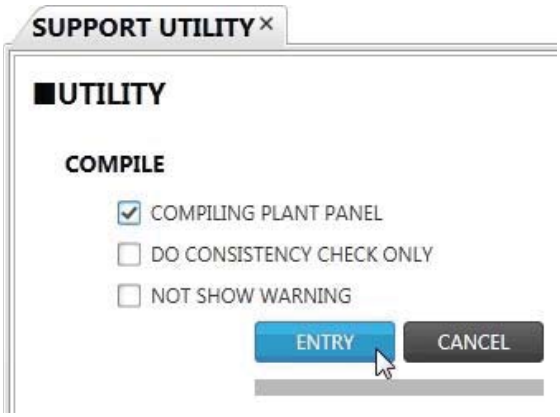
The “Define TAG menu” panel opens.

On the “Define TAG menu” panel, drag and drop the group just defined from the Project tree to the “Define TAG menu” panel.



In order to reflect changes on other panels, select [Compile...] in the Project menu.

When the “Support utility” panel opens, click the [Run] button to compile. You can confirm the compile result on the Output window.



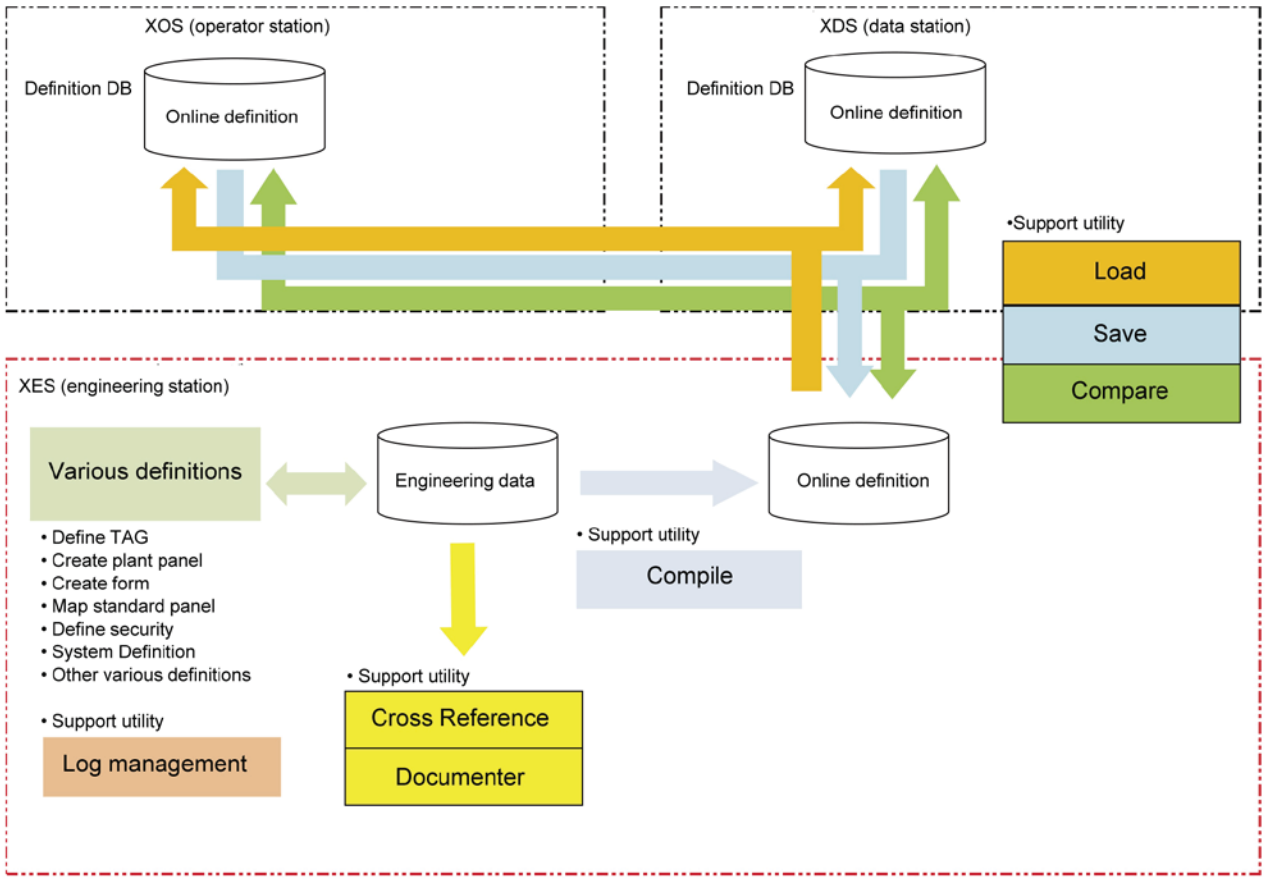
6. Support Utility

This chapter describes the functions listed in the table below for the support utility of the HCI engineering capability.

Function	Description
Compile	Converts engineering data into online definitions.
Cross Reference	Displays the usage of TAGs (usage panel, etc.)
Documenter	Outputs the documents (Excel files) from plant panels, form definitions, signal list (check sheet).
Load, save	Distributes online definitions to each equipment.
Compare	Compares online definitions with those at the destination equipment.
Log management	Tracks engineering tasks (traceability function)
Change online	Sends requests for online change to XOS and XDS.

6.1 Workflow

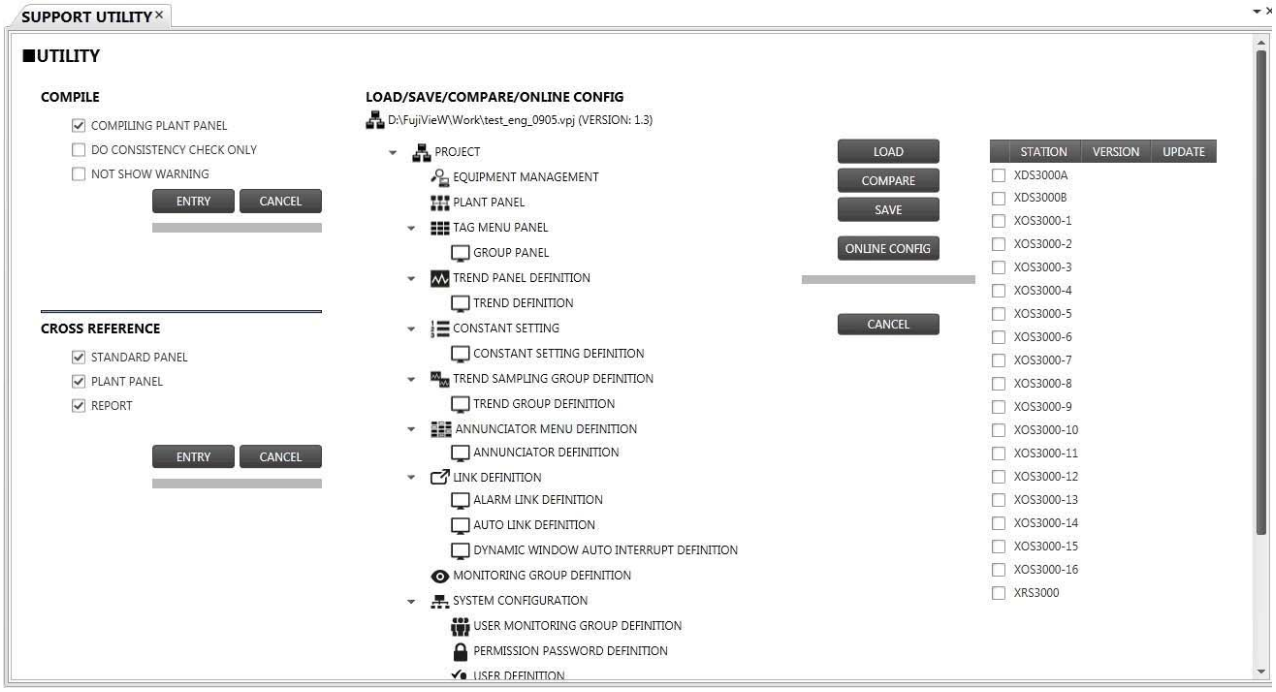
The figure below shows the workflow of the support utility.



6.2 User Interface

The support utility starts when [Compile...], [Cross reference], [Documenter], or [Load/Save/Compare/Change online] is selected in the Project menu.

Example of the Support utility panel



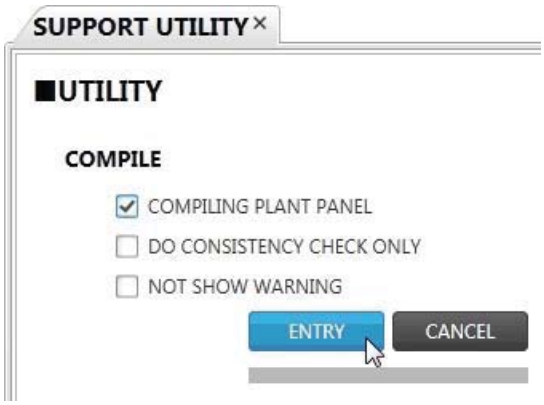
6.3 Compile

This function converts the engineering data into an object file in execution format. When a definition is changed and the change is saved in the definition edit window, make sure to make a compilation.

For the compile, the options listed in the table below can be specified with checkboxes.

Item	Description
Compile plant panel	Select this checkbox to compile a plant panel created in the Plant panel editor.
Consistency check only	Select this checkbox to check the consistency only.
Hide alarms	Select this checkbox to hide alarms.

Optional checkbox



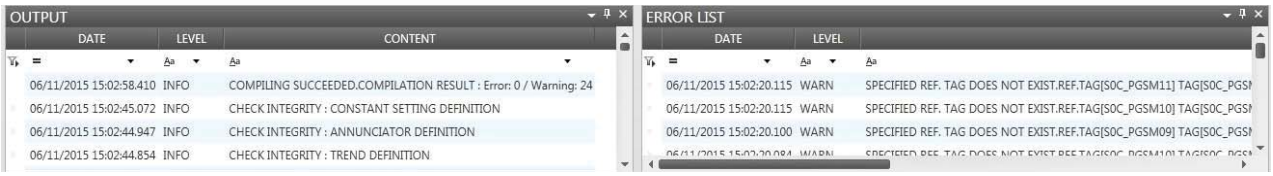
During compilation, the messages listed in the table below are output to the output window.

When to output	Message
Starts or ends compile process	Start, end, number of files to process, number of errors
During compilation	Name of file in progress, result for each file to process (numbers of errors and warnings)
Compile process ends	Total numbers of errors and warnings at the end of the process
Compile error occurs	Detailed information of the compile error (location, details of error and warning)
Starts or ends consistency check	Start, end, number of files to process, number of errors
Checking consistency	Name of file in progress, result for each file to process (numbers of errors and warnings)
Consistency check ends	Total numbers of errors and warnings at the end of the process
Consistency check error occurs	Detailed information of the consistency check error (location, details of error and warning)

When an error occurs in the compile, the messages listed in the table below are output to the error list window.

Output item	Message
Compile error	Syntax error, etc.
Consistency error	Undefined TAG used, etc.

Output window and error list window

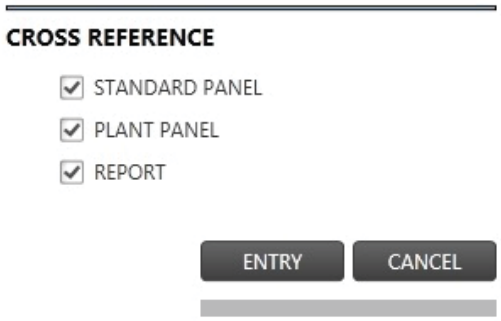


6.4 Cross Reference

This function is under development.

This function analyzes and displays the TAG usages based on the engineering data.

Specify target object for the cross reference with the following checkboxes.



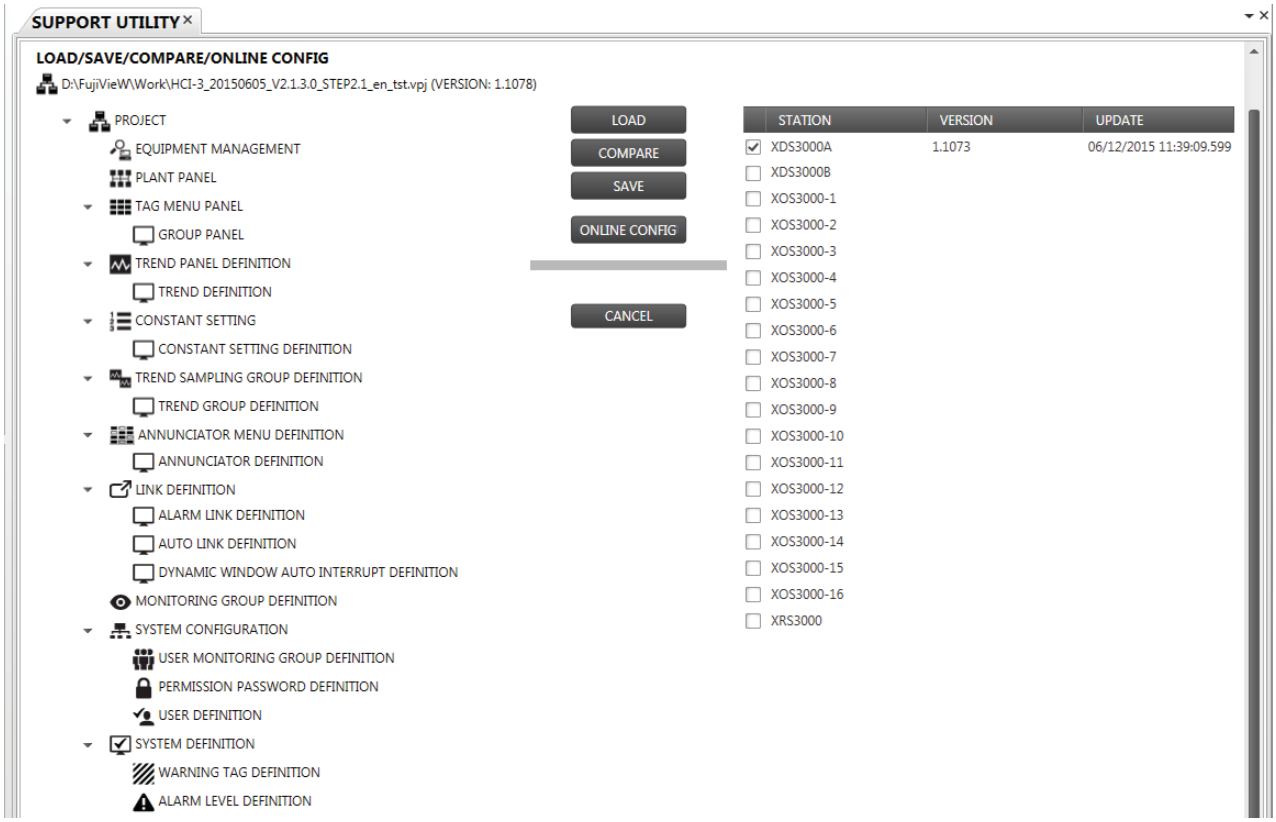
6.5 Documenter

This function is under development.

This function generates and displays documents from the engineering data.

6.6 Load/Save/Compare

This function expands files in execution format after the compile to each equipment, and collects or compares these files.



Load

Loads the definition file of the project opened by the support utility to the destination station.

Save

Saves the data from the specified source station to the definition file of the project opened by the support utility.

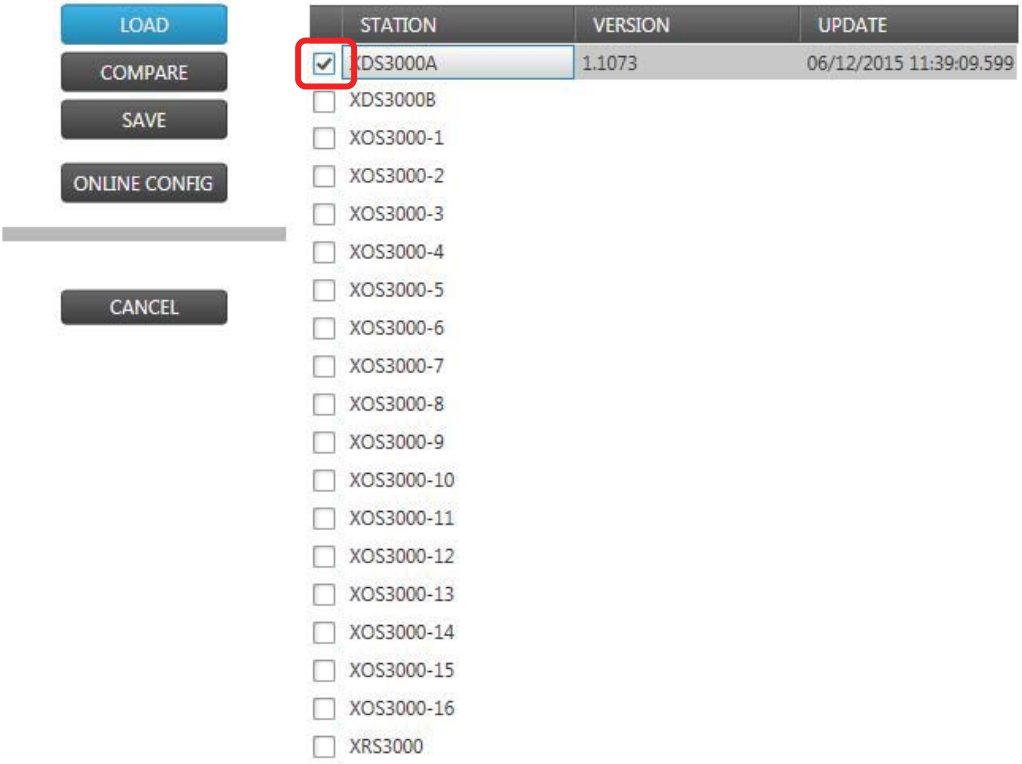
Compare

Compares the definition file of the project opened by the support utility with that in the specified station.

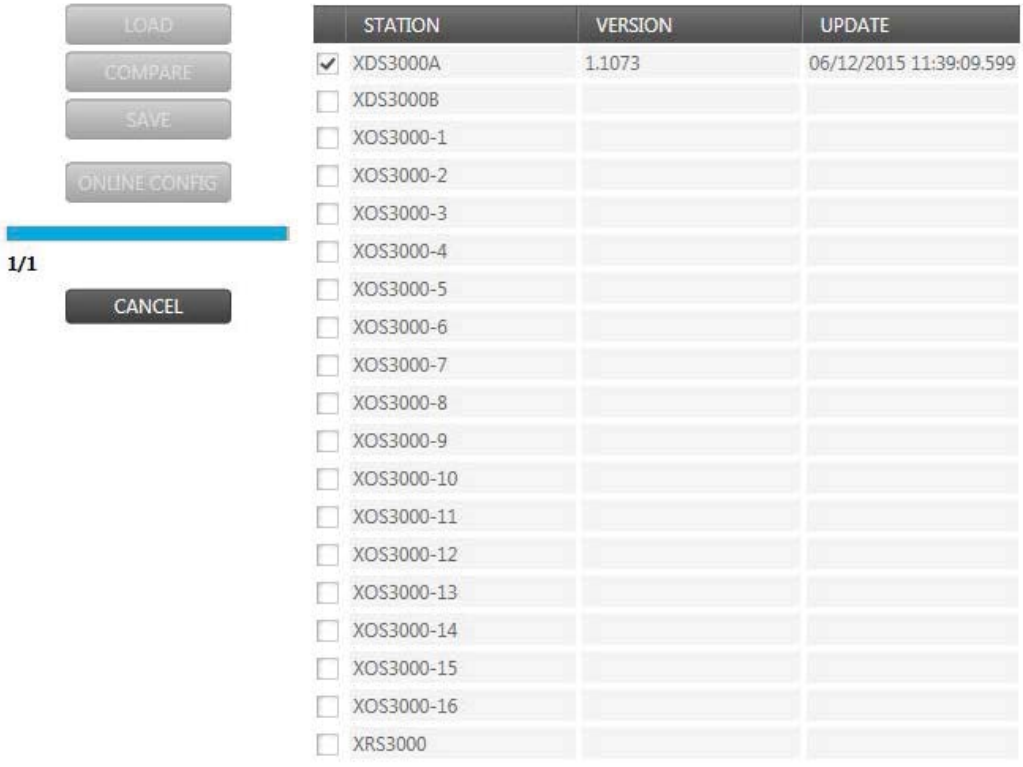
Change online

Asks the specified destination station to change the definition file (loaded) online on that station.

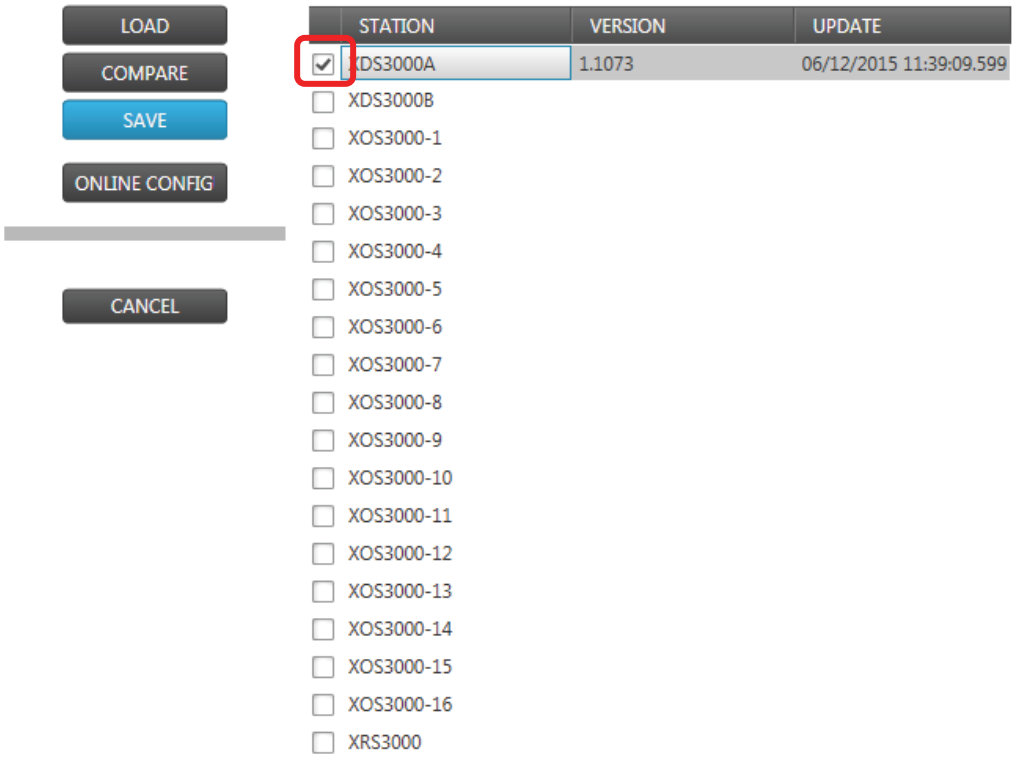
To load the definition file, select the checkbox for the target station and click the [Load] button.



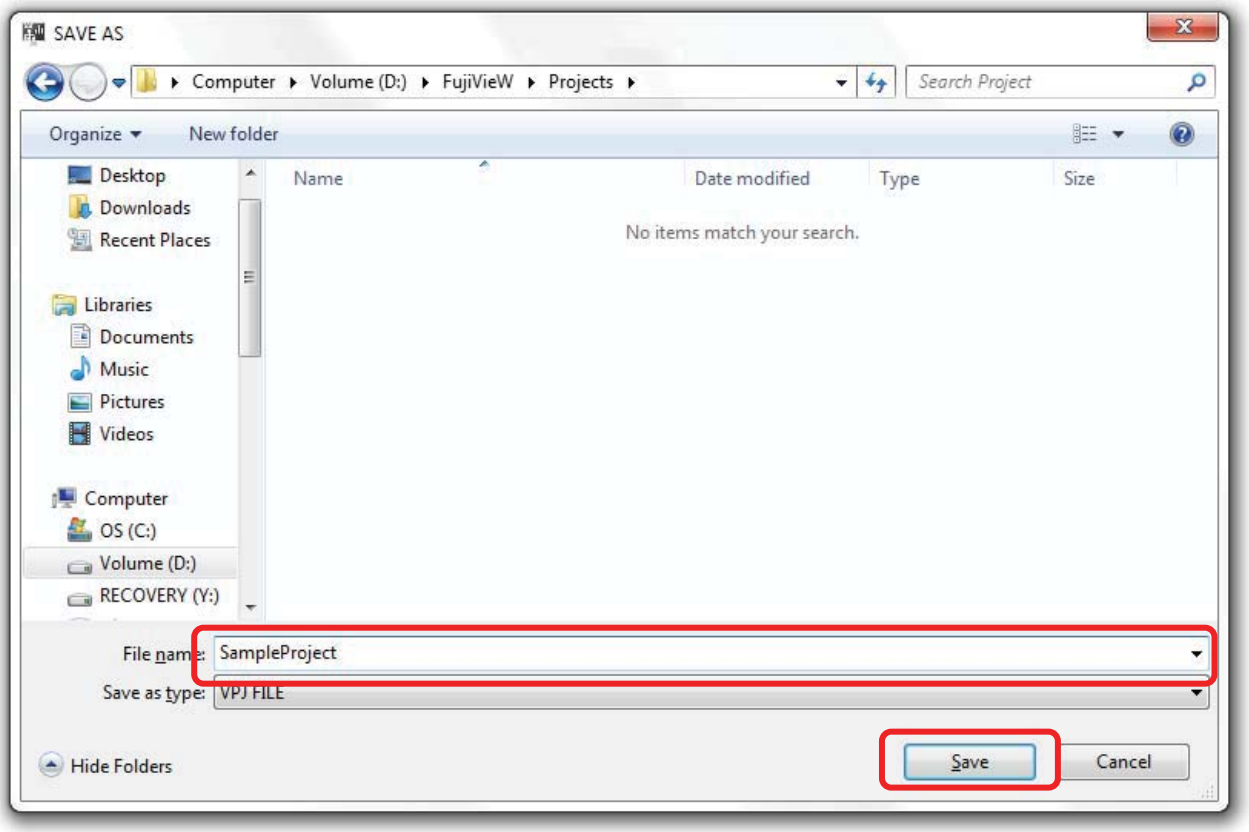
The loading starts.



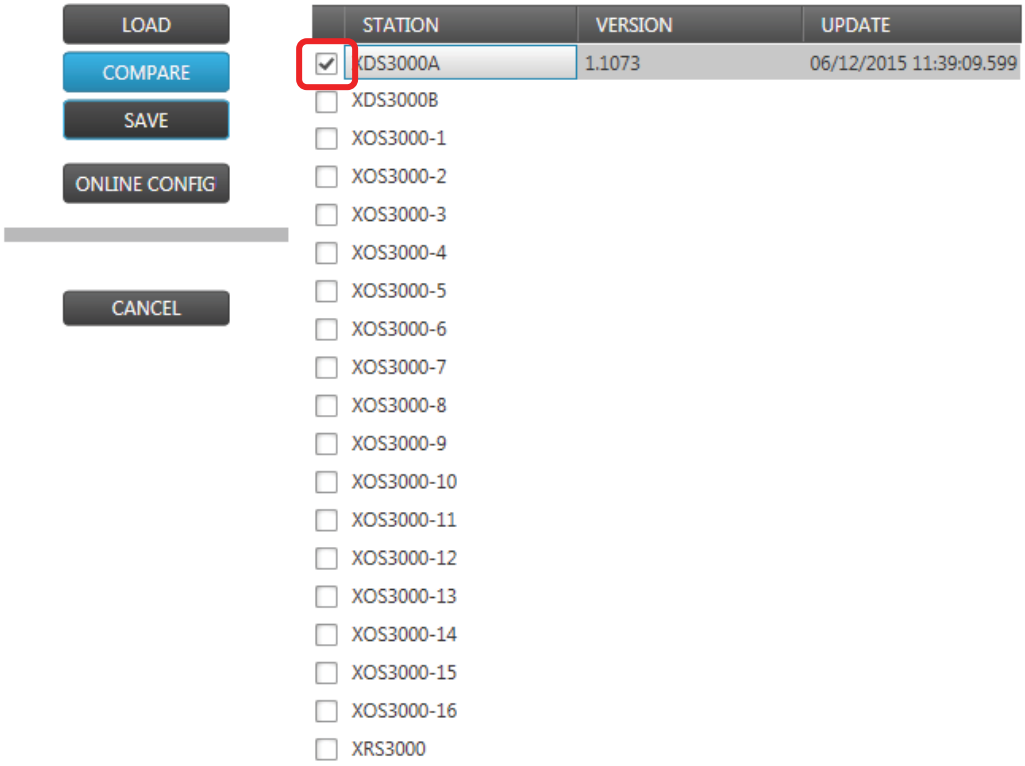
To save the definition file, select the checkbox for the destination station and click the [Save] button.



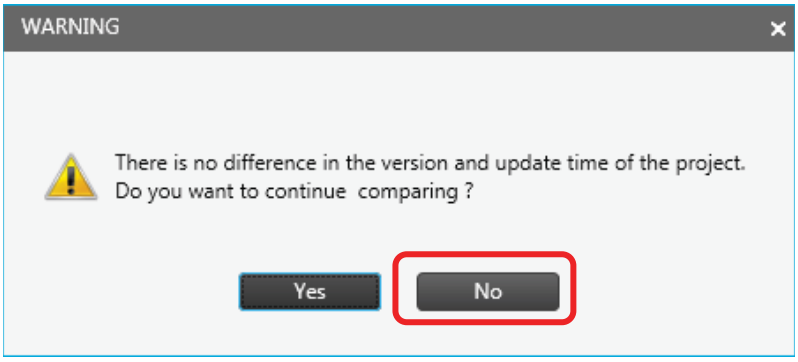
The dialog below appears, so save the project.



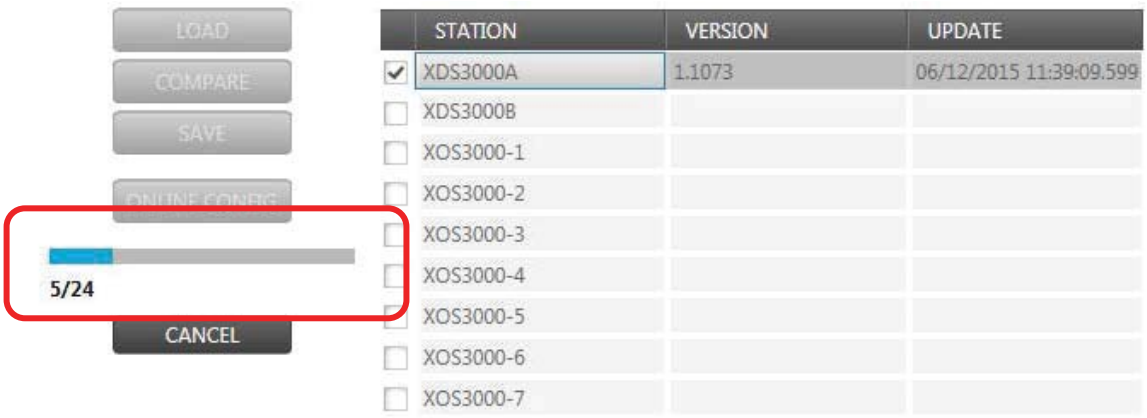
To compare the definition file, select the checkbox for the destination station and click the [Compare] button.



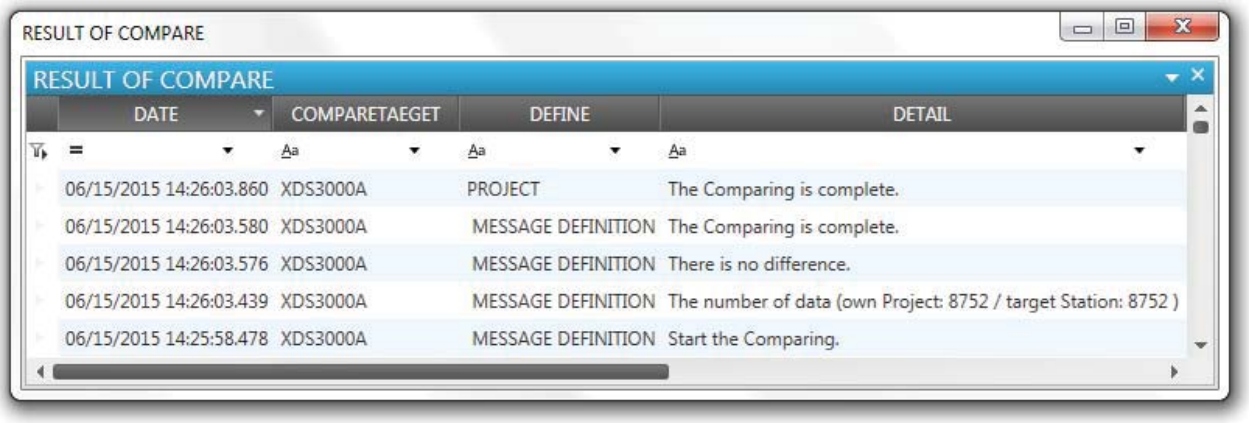
The dialog below may appear. Click [Yes] to continue.



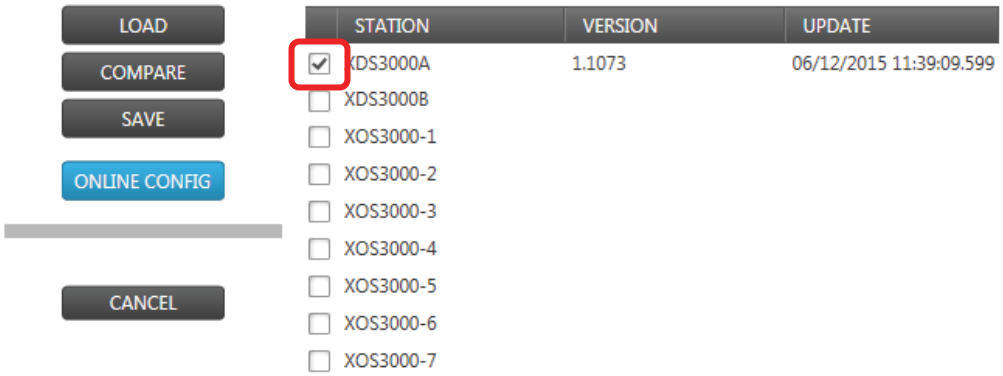
The comparison progress is displayed.



The operation result is output to the comparison result window.



To change the definition file online, select the checkbox for the destination station and click the [Change online] button.



The online change starts.

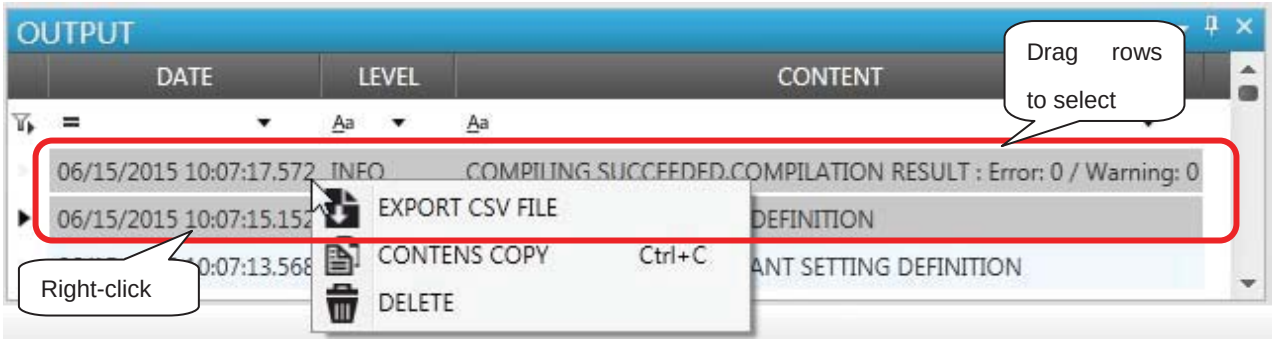
6.7 Record Function (Log Management)

This function records those operations which may give an impact on the system in order to track engineering operations (traceability).

As for the operations which may give an impact on the system, this function records the operations listed in the table below.

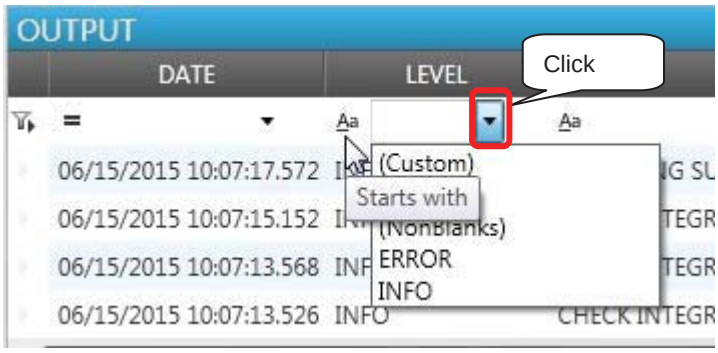
Item	Description
Support Utility, load	Records operations in order to load the engineering data into the system.
Support Utility, save	Saves the records of operations that may give an impact on the source code management in order to save the engineering data from the system.

The logs recorded are managed in the record window or output window.



To copy or clear the data, drag rows with a mouse to select.

Click the leading row of logs to narrow the data.



7. Appendix

7.1 Consistency Check

The table below lists details of the consistency check:

Type	Check element	Description	Limit value	Notification level
Relevant to TAG	Extraction of undefined TAG	Detects any TAGs that are not registered in TAG database but are used in the project.		Error
	Extraction of unused TAG	Detects those TAGs which are registered in TAG database but are not used in the project.		Warning
	Extraction of unregistered TAG	Detects those TAGs which are registered in TAG database but are not registered as the facility category.		Warning
	Inconsistency of definition of TAG database	Detects the inconsistency of device definition indicated in the TAG definition in TAG database.		Error
	Inconsistency of TAGNO and name	Detects that the numbers of characters of TAGNO and name do not exceed the limit.	Up to 10 characters for each TAG	Error
Relevant to facility category	Not defined	Detects those facility categories which are associated with monitoring group of the project but are not registered as facility categories for the project.		Error
	Not used	Detects those facility categories which are registered as facility categories for the project but are not associated with monitoring group of the project.		Warning
	Not registered	Detects no TAG is registered in facility category.		Warning
	Maximum	Checks that the number of registered facility categories does not exceed the maximum.	Up to 256	Error
Relevant to monitoring group	Not defined	Detects those groups which are associated with stations of the project but are not registered in monitoring group of the project.		Error
	Not used	Detects those monitoring groups which are registered in monitoring groups but are not associated with stations of the project.		Warning
	Not registered	Detects that no facility category is registered in monitoring group.		Warning
	Maximum of registration	Checks that the number of registered monitoring groups does not exceed the maximum.	Up to 16	Error
	Maximum number of registered facility categories	Checks that the number of facility categories registered in monitoring groups does not exceed the maximum.	Up to 255	Error
Relevant to system configuration	Not defined	Checks that XDS and XOS do not exist.		Warning
	Not used	Detects those stations which are registered in the system configuration but are not associated with monitoring groups of the project.		Warning
	Maximum number of XOSs	Checks that the number of registered XOSs does not exceed the maximum.	Up to 16	Warning
Relevant to TAG menu panel	Number of groups displayed in a page	Checks that the number of groups in the group panel that are mapped onto a page does not exceed the limit.	Up to 16	Error
Relevant to group panel	Number of TAGs in page	Checks that the number of TAGs mapped onto group panel does not exceed the limit of a page.	Up to 16	Error
	TAG type	Checks that all TAGs mapped onto group panel are registered and available.	Module TAG	Error
	Undefined TAG used	Detects any TAGs that are not registered in TAG database but are used.		Error
Relevant to constant setting panel	Number of TAGs in page	Checks that the number of TAGs mapped onto constant setting panel does not exceed the limit of a page.	Up to 16	Error
	TAG type	Checks that all TAGs mapped onto constant setting panel are registered and available.	User TAG, module TAG	Error
	Undefined TAG used	Detects any TAGs that are not registered in TAG database but are used.		Error
Relevant to trend panel	Number of TAGs in page	Checks that the number of TAGs mapped onto trend panel does not exceed the limit of a page.	Up to 8	Error
	TAG type	Checks that all TAGs mapped onto constant setting panel are registered and available.	Module TAG, user TAG on BC	Error
	Undefined TAG used	Detects any TAGs that are not registered in TAG database but are used.		Error
	Collection cycle of TAG in group	Checks that the collection cycles of TAGs registered in trend group are the same.		Error

Relevant to trend collection	Maximum number of collection cycles	Checks that the number of trend collection cycles does not exceed the maximum for each type.	Up to 16	Error
	Collection cycle	Checks that the trend collection cycle is in accordance with the restrictions (cycle range, setting unit)		Error
	Maximum number of collection groups	Checks that the number of trend collection groups does not exceed the maximum.	Up to 1024	Error
	Number of TAGs in collection group	Checks that the number of TAGs does not exceed the maximum definable in a group.	Up to 1024	Error
	Inconsistency of trend collection cycle	Checks that the trend collection cycle does not exceed the restrictions of consecutive trend and batch trend.	1 sec to 3600 sec (60 min)	Error
	Maximum number of collection groups	Checks that the number of trend collection groups does not exceed the maximum.	Up to 32	Error
	Number of TAGs in collection group	Checks that the number of TAGs does not exceed the maximum definable in a group.	Up to 256	Error
	Inconsistency of trend collection cycle	Checks that the trend collection cycle does not exceed the restrictions of high speed trend.	200 ms to 1000 ms (1 sec)	Error
	Condition for batch trend	Checks that the start condition for batch trend is set properly.	Bit TAG	Error
Relevant to user	Maximum number of registered users	Checks that the number of registered users does not exceed the maximum.	Up to 1024	

Formosa Plastics Corp.

Electronics Group

100, SWEI-KUAN RD., JEN-WU DISTRICT,
KAOHSIUNG CITY, TAIWAN, R.O.C

Instruction Manual

FORMOSA - FX

MICREX-VieW/H
Drawing Engineering Section

Formosa Plastics Corp.

INC-TN5A2099-E

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1. Drawing Engineering Specification

1.1 Overview

The HCI engineering tool has the following functions for creating plant panels.

(1) Plant panel editor.....Creates plant panels and dynamic windows.

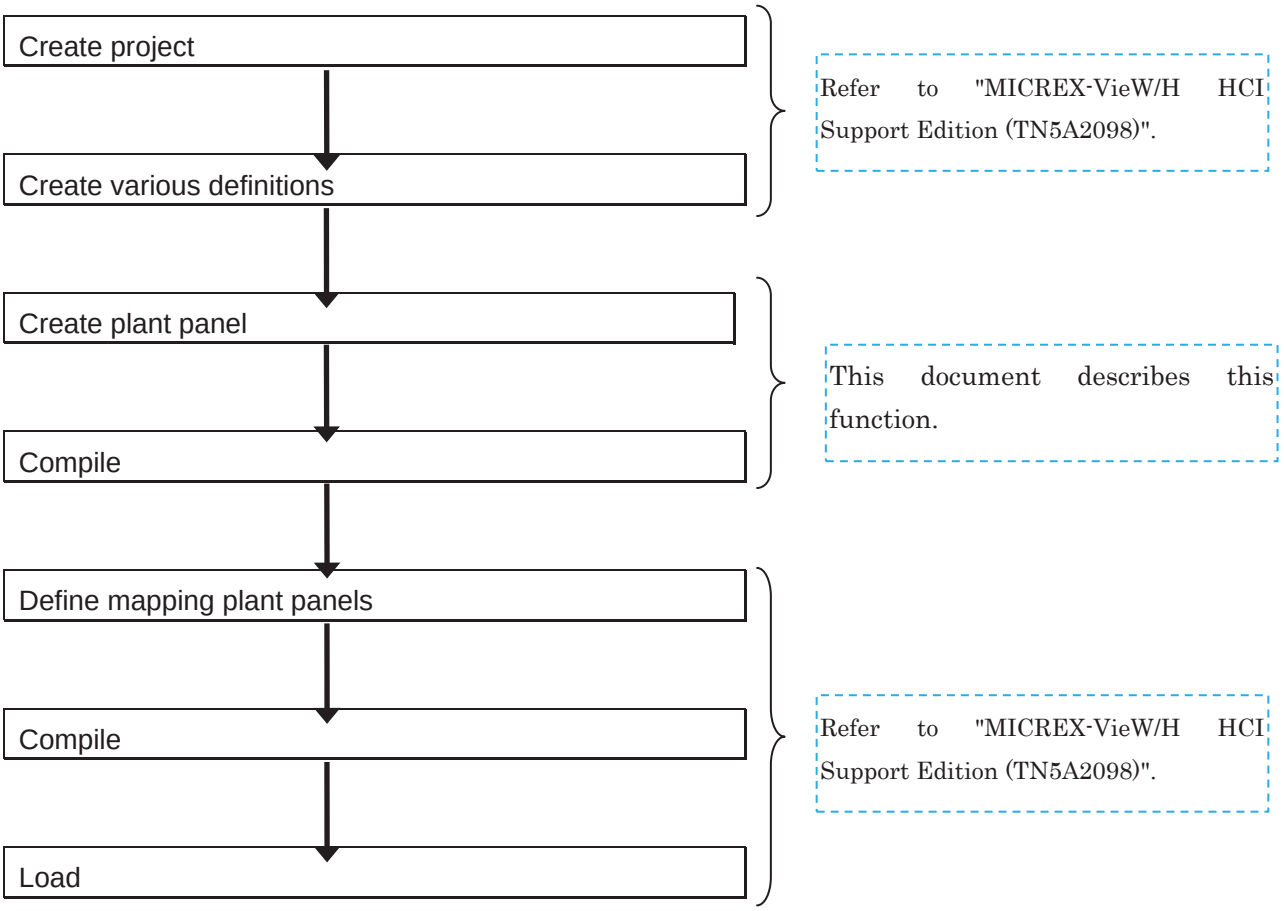
(2) Segment editor .....Creates segments composed of graphics.

1.2 Function List

NO.	Panel name	Overview	Quantity
1	Plant panel editor	Creates plant panels and dynamic windows. (still picture (graphic), operational picture (variable), entry object (operation))	2048 panels
2	Segment editor	Creates segments composed of graphics. The editor can be started from the plant panel editor.	1024 segments

1.3 Definition Procedures

This section describes definition procedures for preparing plant panels.



1.4 Basic Concept

1.4.1 About Quantity

A system scale file can manage quantity for supporting every model.

* For information about definable quantity on each definition screen, refer to "P.1-4 Table 1–1 Scale of definition screen".

1.4.2 Text Display

The text is shown as follows:

	Double-byte text	Single-byte text
24 dot character	24 dot / 24 dot	24 dot / 12 dot
16 dot character	16 dot / 16 dot	16 dot / 8 dot
8 dot character		8 dot / 8 dot
32 dot character	32 dot / 32 dot	32 dot / 16 dot
48 dot character	48 dot / 48 dot	48 dot / 24 dot
64 dot character	64 dot / 64 dot	64 dot / 32 dot
96 dot character	96 dot / 96 dot	96 dot / 48 dot

Figure 1–1 Font size

Each text is shown in each frame displayed for each type of dot. The actual size of text is different according to PC models.

1.4.3 How to Draw Screen

The whole screen may not be displayed according to each panel (high-resolution, medium-resolution, etc.) and PC. In this case, scroll the screen (by using a scroll bar) to define the screen.

1.5 Scale of Definition Screen

Table 1–1 Scale of definition screen

Screen	Quantity	Name	Item	Remarks
Plant panel (still pictures + operational pictures)	*1: 2048 panels	Up to 32 single-byte texts (double-byte and single-byte texts can be mixed)	Number of variable objects: 1024 per panel Number of entries: 512 per panel	
Segment	1024	Up to 8 single-byte texts	—	

* 1: The total number of plant panels and dynamic windows.

Note that individual touch entry and multiple touch entry are supposed to be two per command.

Note 1: Some commands may not be defined with 1024 panels.

This causes a problem of file management. If internal definable area is exhausted, an error message is shown.

Also, if definable area is running out room, a warning is displayed.

2. How to Operate Plant Panel Editor

2.1 Overview

2.1.1 Overview

XES enables you to prepare plant panels with a mouse (keyboard).

2.1.2 How to Operate Mouse

① Click
Push the button with a finger and then release the finger soon.

② Double-click
Click the button twice consecutively.

③ Drag
Move the mouse while holding the mouse button.

2.1.3 Basic Operations

① To specify a point (starting point, apexes, etc.), left-click the point.

② To confirm (register) a graphic, left-click the graphic.
(Except for lines and polygons. Double-click them with the left button)

③ To cancel creating a graphic, right-click the graphic.

④ To pick up (select) a graphic, left-click the graphic.

⑤ To move or copy a graphic, left-drag the graphic.

⑥ When displaying a window including a dialog box, the Run button, return key, and ENTER key confirm an operation and the Cancel key and the ESC key cancels the operation.

2.1.4 Relationship between Keyboard and Mouse

The table below lists how the keyboard can be used for performing the same operations as the mouse.

However, the keyboard is available for use only when the window is active.

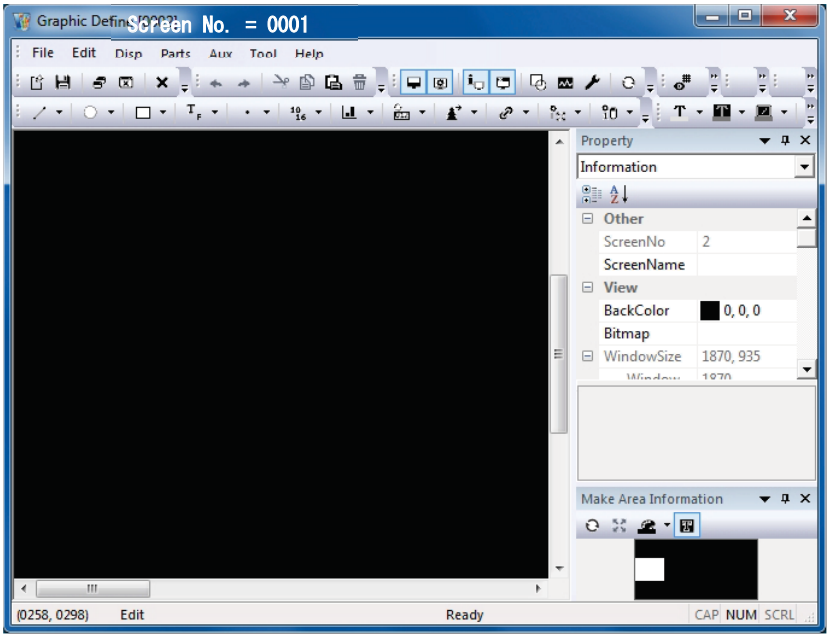
Mouse	Keyboard
Left-click	Return key, or Enter key
Left double-click	Shift + Return key, or Shift + Enter key
Right-click	Esc key
Pick	Ctrl + Return key, or Ctrl + Enter key
Move mouse (move cursor)	・←↑↓→ key: Move the cursor by one dot ・Shift + ←↑↓→ key: Move the cursor by ten dots
Right-drag	Select an object with Ctrl + Return key or Ctrl + Enter key and press ←↑↓→ key to move it while pressing the Ctrl key. (Press an arrow key while pressing the Shift + Ctrl key to move the object by ten dots)

Note 1) Besides, there are some operations that are possible only with the mouse. For details, see the remarks field in the item list in 2.2.4 Menu Bar.

Note 2) The operations with the keyboard are valid only in the drawing engineering tool.

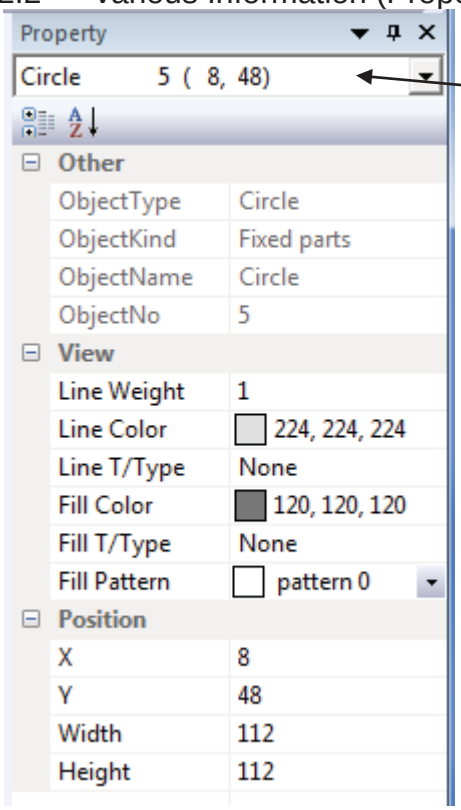
2.2 Display Items and Descriptions

2.2.1 Screen as a Whole



The screen number is shown on the title bar.

2.2.2 Various Information (Properties)



Object selected

Detailed definition information

* The details are described in how to draw each object.

2.2.3 Cursor

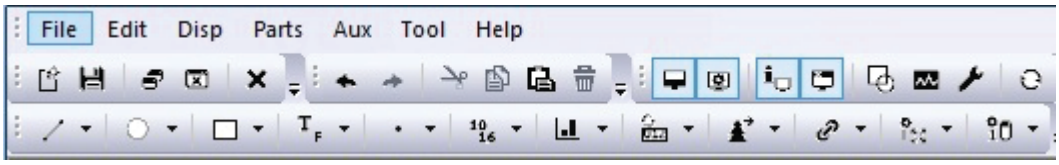
: Cursor for creation

: Cursor for moving

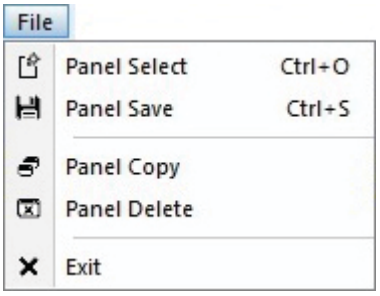
: Cursor for change and search

: Cursor for enlarging

2.2.4 Menu Bar

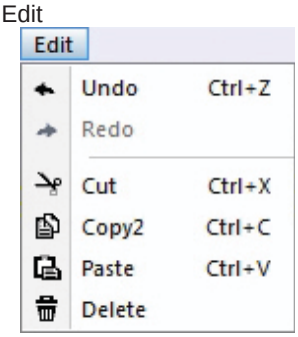


① Screen



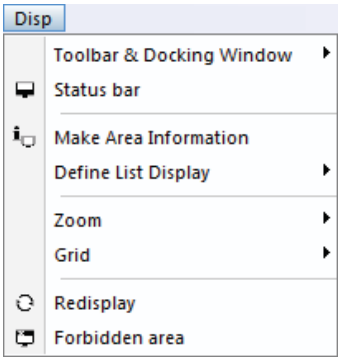
Item	Description	Remarks
Panel Select	Registers a screen name and type and selects each screen number.	
Panel Save	Saves the screen created.	
Panel Copy	Copies a screen to another screen number.	
Panel Delete	Deletes a screen.	
Exit	Exits the screen creation.	

② Edit

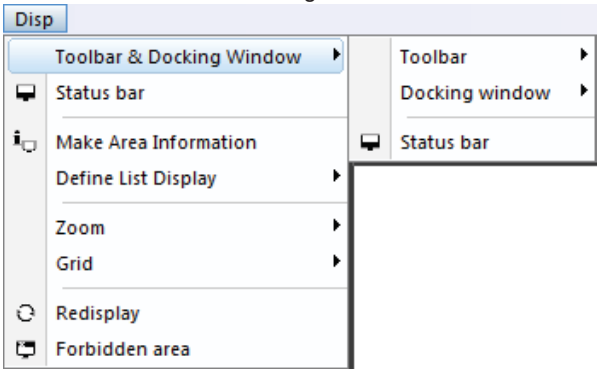


Item	Description	Remarks
Undo	Changes an operation back.	
Redo	Reverses an operation undone.	
Cut	Cuts an object selected. (Cut the object to the clipboard)	
Copy	Copies an object selected.	
Paste	Pastes the object cut or copied.	
Delete	Deletes an object selected.	

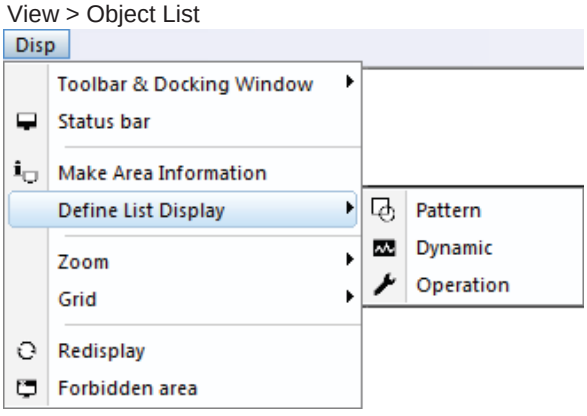
③ View



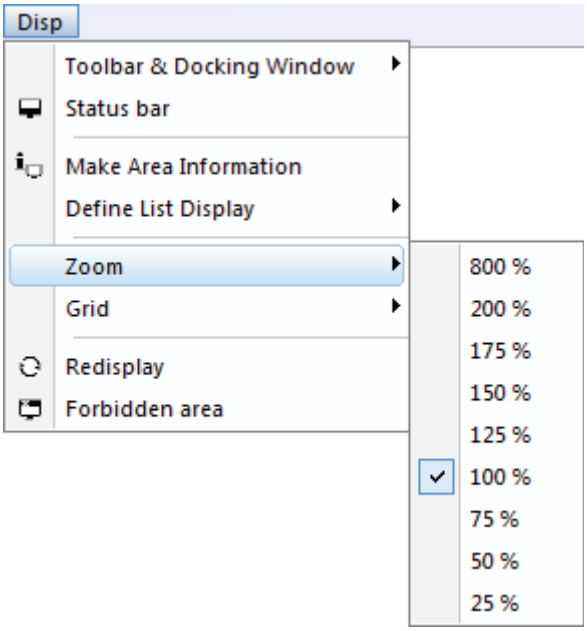
③-1 View > Tool Bar and Docking Window



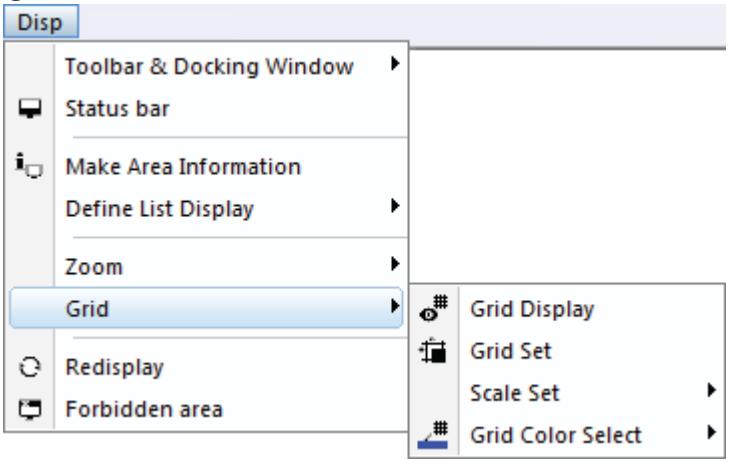
③-2 View > Object List



③-3 View > Zoom



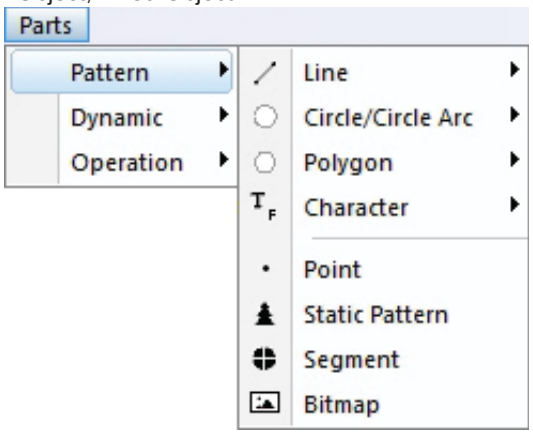
③-4 View > Grid



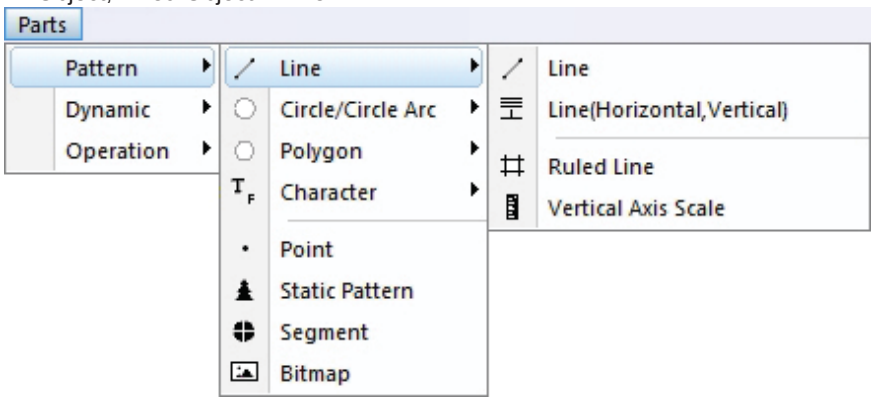
Item		Description	Remarks
Tool Bar and Docking Window	Standard	Displays the standard tool bar.	
	Object	Displays the object tool bar.	
	Property	Displays the property window.	
Status Bar		An item description is shown at the bottom of the screen.	
Display Area Information		Displays the information of the whole and area displayed.	
Object List	Fixed Object	Displays still picture information.	
	Variable Object	Displays variable object information.	
	Operation Object	Displays operation object information.	
Zoom	% designation	Displays at expansion & reduction ratio (800% - 25%).	
Grid	Display Grids	Displays the grids.	
	Align to Grid	Aligns a created object to grids.	
	Grid Spacing	Set the grid spacing.	
	Grid Color	Set the grid color.	
Refresh		Refreshes all objects.	
Forbidden area		Shows the forbidden area.	

④ Object

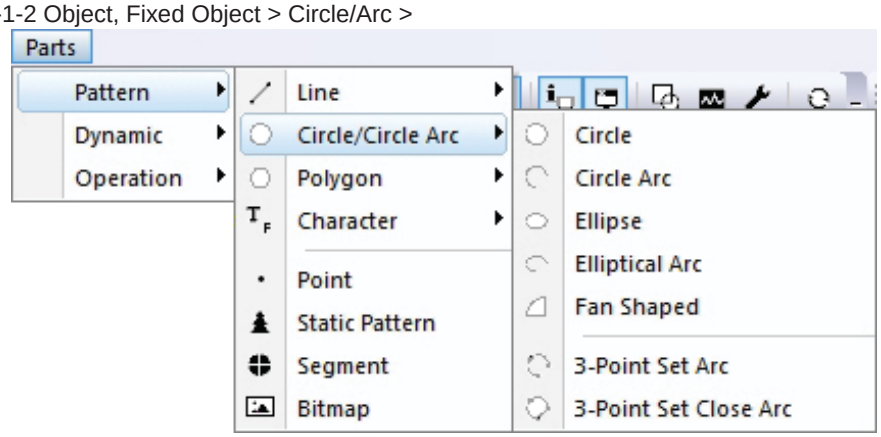
④-1 Object, Fixed Object >



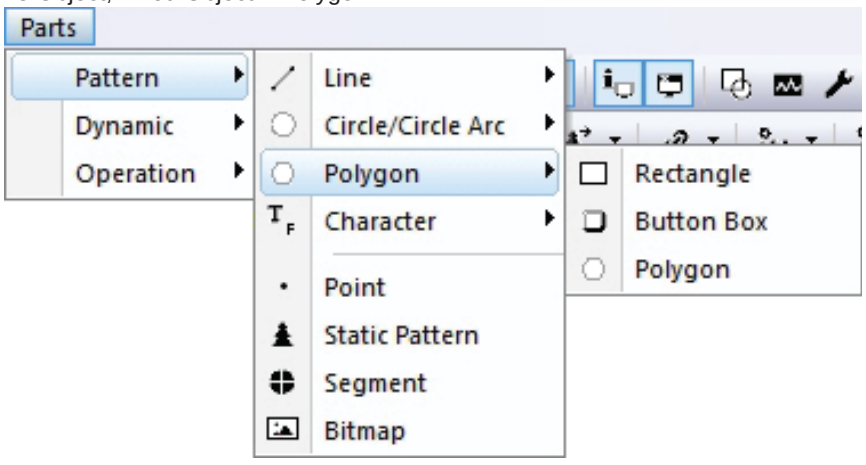
④-1-1 Object, Fixed Object > Line >



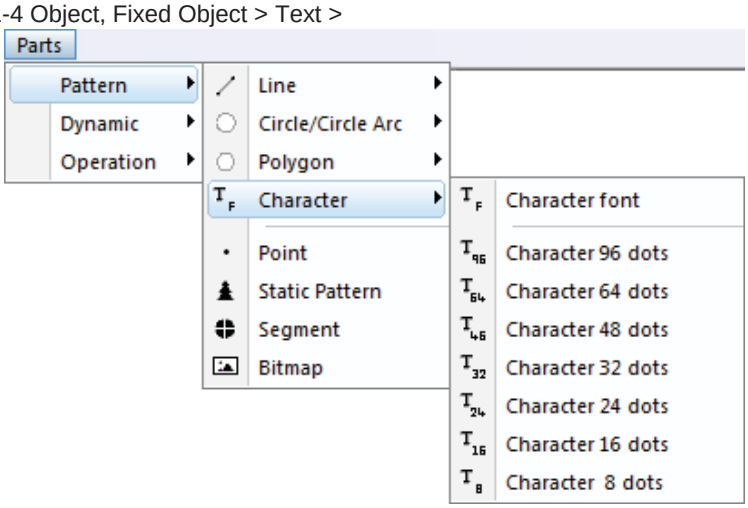
④-1-2 Object, Fixed Object > Circle/Arc >



④-1-3 Object, Fixed Object > Polygon >

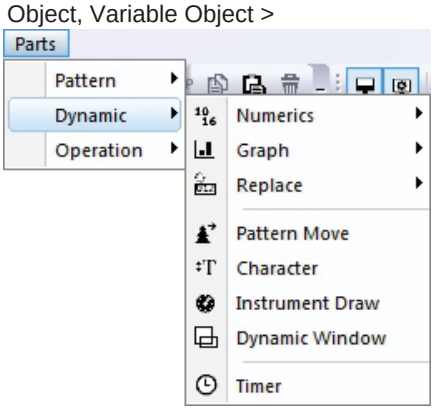


④-1-4 Object, Fixed Object > Text >

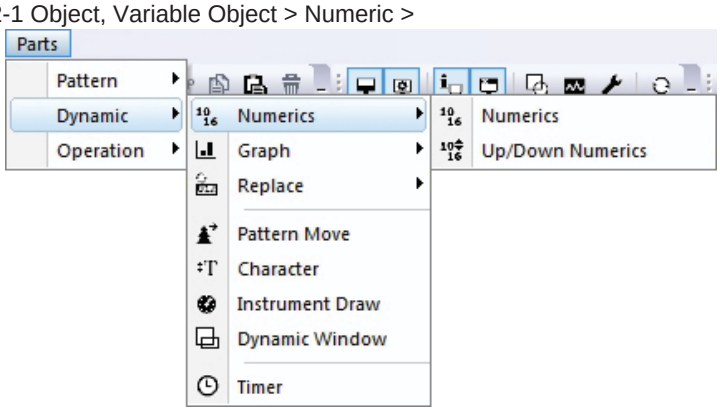


Item			Description	Remarks
Fixed Object	Line	Line	Draw a polygonal line segment.	
		Line (horizontal, vertical)	Draw multiple (horizontal/vertical) line segments.	
	Circle/Arc	Circle	Draw the graphic by specifying a center and radius.	
		Arc	Draw the graphic by specifying a center, radius, starting point and ending point.	
		Ellipse	Draw the graphic by specifying diagonal points.	
		Elliptic Arc	Draw the graphic by specifying a center and long and short radii.	
		Sector	Draw the graphic by specifying a center, radius, starting point and ending point.	
	Polygon	Rectangle	Draw the graphic by specifying diagonal points.	
		Button Type Rectangle	Draw the graphic by specifying diagonal points.	
		Polygon	Draw a polygonal line segment. Starting and ending points are connected.	
	Text	96 Dot Character	Write characters by specifying the starting point. The standard function on each PC is used to input characters.	
		64 Dot Character		
		48 Dot Character		
		32 Dot Character		
		24 Dot Character		
		16 Dot Character		
		8 Dot Character		
	Fixed Pattern		Input a fixed pattern registered by specifying the starting point.	Only the mouse operation is available for use in this window.
	Segment		Input a segment registered by specifying the starting point.	
	Bitmap		Input a picture file stored in the project name > PLT > BITMAP directory by specifying the starting point.	

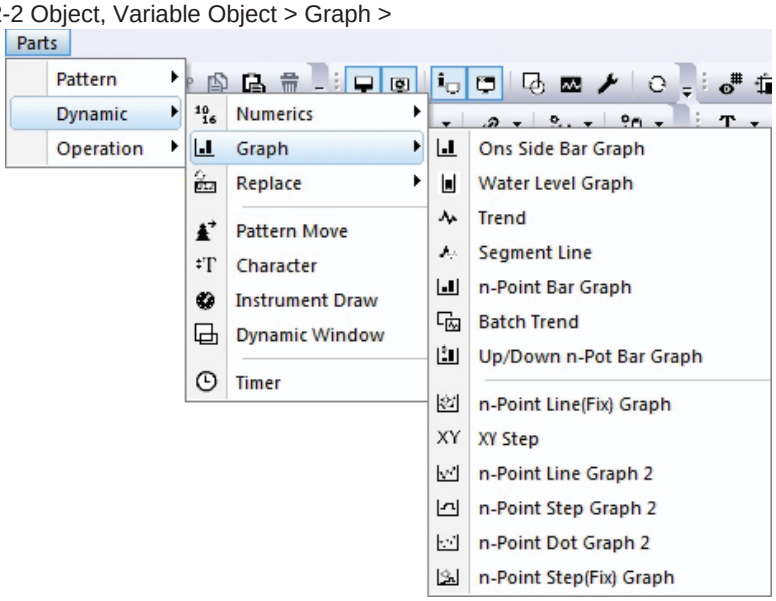
④-2 Object, Variable Object >



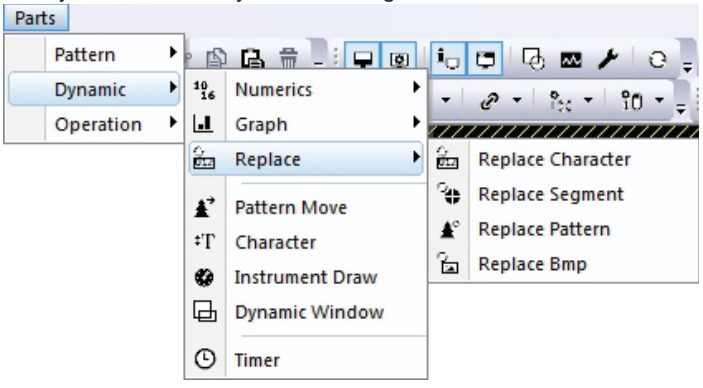
④-2-1 Object, Variable Object > Numeric >



④-2-2 Object, Variable Object > Graph >

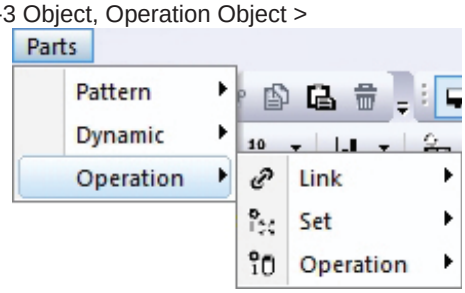


④-2-3 Object, Variable Object > Exchange >

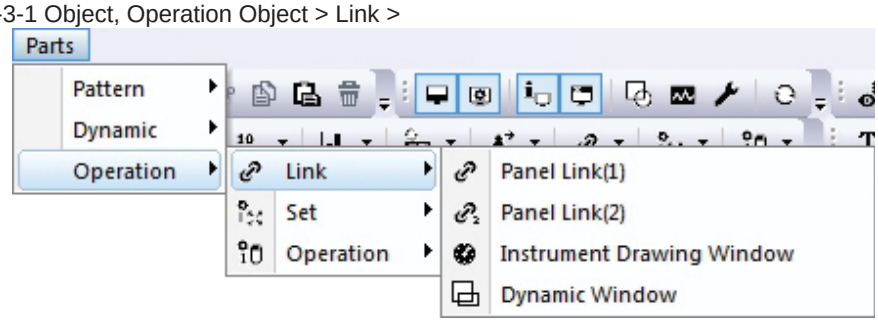


Item			Description	Remarks
Variable Object	Numeric	Numeric	Displays the specified value in decimal or hexadecimal format.	
		Limiter Numeric	Number with upper and lower limits.	
	Graph	Bar Graph	Displays specified values with rectangle graphs.	
		Picture Graph	Displays specified values with graphs of other than rectangles (tank pictures, for example).	
		Trend	Displays specified range of values with historical and real trend graphs.	
		Line	Displays specified range of specified values with polygonal line and then displays previous values with clearing color.	
		Prot Bar Graph	Displays specified multiple values with rectangle bars.	
		Batch Trend	Displays specified range of values with batch trend graph.	
		Prot Limiter Bar Graph	This prot bar graph has upper and lower limits.	
	Exchange	Exchange Text	Exchanges the text specified.	
		Exchange Segment	Exchanges the segment of the text specified.	
		Exchange Bitmap	Exchanges the bitmap specified.	
	Move Pattern		Move specified range of patterns by specifying values.	
	Text		Displays character code column from the specified place.	
	Meter Diagram		Displays a meter diagram at the specified position.	
	Dynamic Window		Displays a dynamic window at the specified position.	

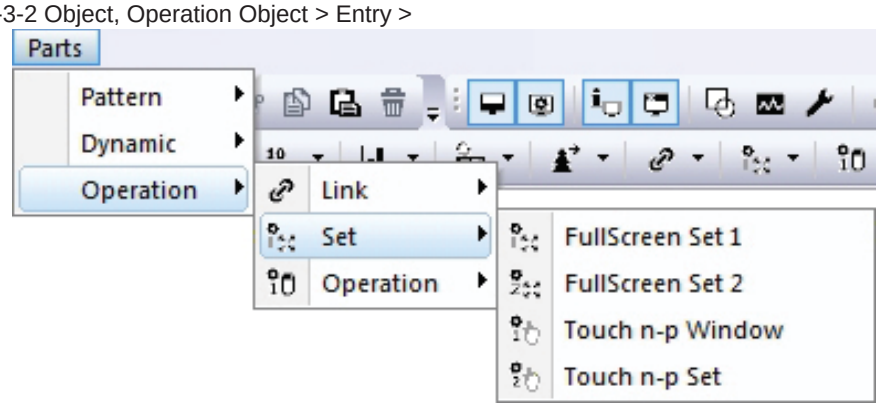
④-3 Object, Operation Object >



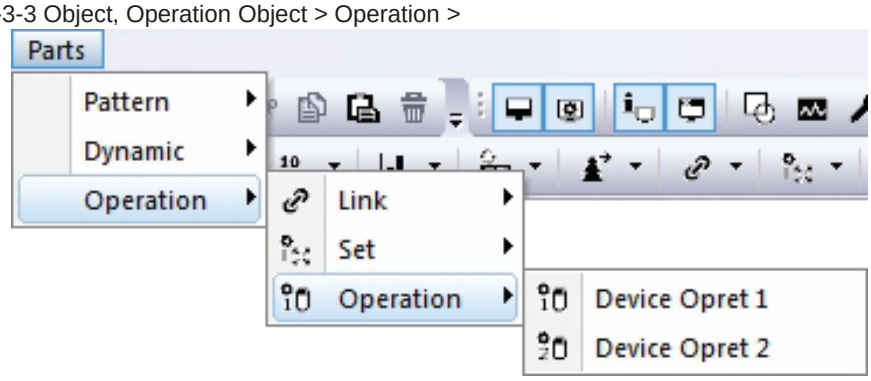
④-3-1 Object, Operation Object > Link >



④-3-2 Object, Operation Object > Entry >

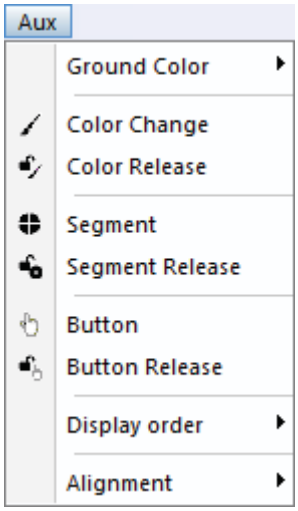


④-3-3 Object, Operation Object > Operation >

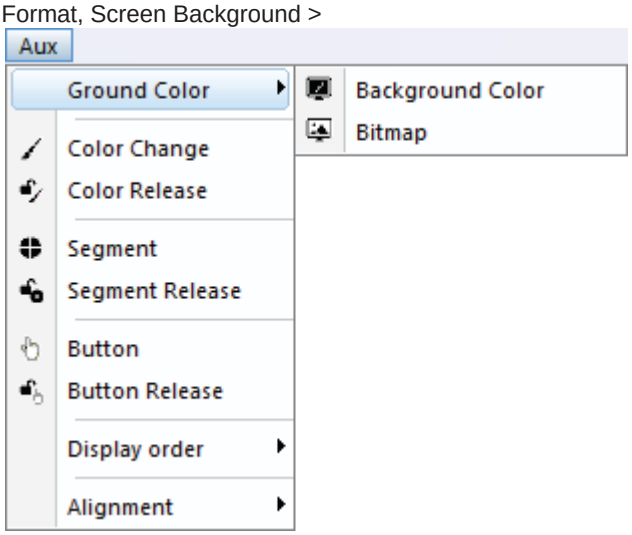


Item			Description	Remarks
Operation Object	Link	Link Screens (Single Action)	Link a screen to another one with single touch operation.	
		Link Screens (Two Actions)	Link a screen to the screen registered in the window displayed with a touch operation.	
		Meter Diagram Window	Calls up a meter diagram window at the specified position.	
		Dynamic Window	Calls up a dynamic window at the specified position.	
	Entry	Full Screen Entry 1	Set numeric value under locked condition. Full screen type.	
		Full Screen Entry 2	Set numeric value under locked condition. Full screen type.	
		Multiple Touch Point Window	Set numeric value under locked condition. Key-in area type	
		Multiple Touch Point Entry	Set numeric value under locked condition. Key-in area type	
	Operation	Device Operation 1	Set a mode under locked condition. Key-in area type.	
		Device Operation 2	Set a mode under locked condition. Full screen type.	

⑤ Format

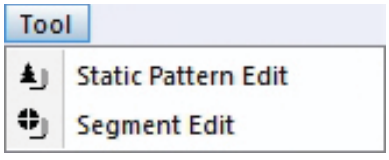


⑤-1 Format, Screen Background >



Item		Description	Remarks
Screen Background	Background Color	Specify a background color (one color) of the screen.	
	Bitmap	Specify a background bitmap of the screen.	
Change Color		Enables to change the color of fixed object.	
Clear Change Color		Puts the color changed back to original.	
Segment		Segments multiple fixed objects.	
Clear Segment		Puts the object segmented back to original.	
Create Button		Creates a button from an operation frame or rectangle.	
Clear Button		Puts the button object back to original.	

⑥ Tool



Item	Description	Remarks
Pattern Editor	Creates fixed patterns.	
Segment Editor	Creates segments.	

Item type list

- Figure * The item (1) indicates standard or extended screen and the item (2) indicates dynamic window.

ITEM TYPE	YES / NO		ITEM TYPE	YES/ NO		ITEM TYPE	YES/ NO	
	(1)	(2)		(1)	(2)		(1)	(2)
Line	Yes	Yes	Polygon	Yes	Yes	32 dot character	Yes	Yes
Line (horizontal, vertical)	Yes	Yes	Sector	Yes	Yes	24 dot character	Yes	Yes
Circle	Yes	Yes	Button Type Rectangle	Yes	Yes	16 Dot Character	Yes	Yes
Arc	Yes	Yes	Point	Yes	Yes	8 Dot Character	Yes	Yes
Rectangle	Yes	Yes	96 Dot Character	Yes	Yes	Fixed Pattern	No	No
Ellipse	Yes	Yes	64 Dot Character	Yes	Yes	Segment	Yes	Yes
Elliptic Arc	Yes	Yes	48 Dot Character	Yes	Yes			

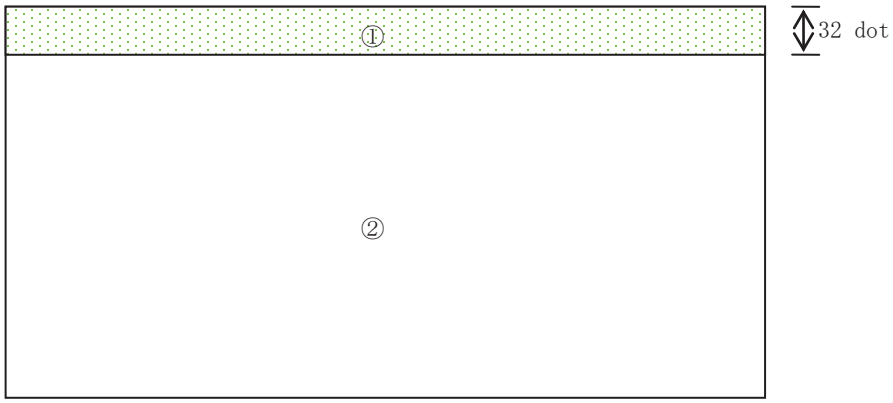
- Variable * The item (1) indicates standard or extended screen and the item (2) indicates dynamic window.

ITEM TYPE	YES / NO		ITEM TYPE	YES/ NO		ITEM TYPE	YES/ NO	
	(1)	(2)		(1)	(2)		(1)	(2)
Numeric	Yes	Yes	Trend	Yes	Yes	Batch Trend	Yes	Yes
Change Color	No	No	Move Pattern	Yes	Yes	Meter Diagram	Yes	No
Exchange	Yes	Yes	Line	Yes	Yes	Dynamic Window	Yes	No
Bar Graph	Yes	Yes	Text	Yes	Yes	Prot Limiter Bar Graph	Yes	Yes
Picture Graph	Yes	Yes	Prot Bar Graph	Yes	Yes	Limiter Numeric	Yes	Yes

Operation * The item (1) indicates standard or extended screen and the item (2) indicates dynamic window.

ITEM TYPE	YES/	NO	ITEM TYPE	YES/	NO	ITEM TYPE	YES/	NO
	(1)	(2)		(1)	(2)		(1)	(2)
Link Screens	Yes	Yes	Full Screen Entry 1	Yes	Yes	Multiple Touch Point Entry	Yes	Yes
Meter Diagram Window	Yes	Yes	Full Screen Entry 2	Yes	Yes	Device Operation 1	Yes	Yes
Dynamic Window	Yes	Yes	Multiple Touch Point Window	Yes	No	Device Operation 2	Yes	Yes

2.2.5 Drawing Area



①Nondrawable area

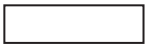
②Drawable area

＊: Do not draw any graphic in the ① nondrawable area.

(It is the system area. The part of graphic in the nondrawable area is not shown in the plant panel.)

In interactive drawing, the cursor can move only in ② drawable area.

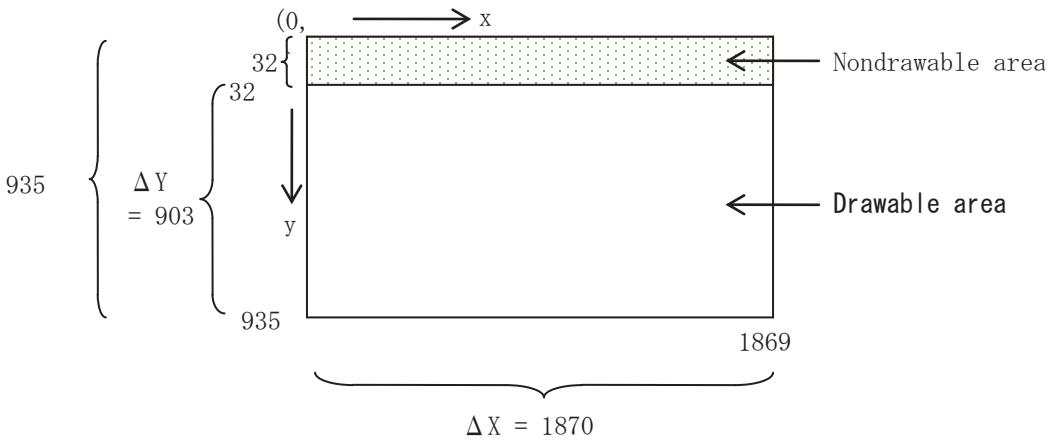
Next, the figures below show drawing area for each screen type (standard screen, extended screen (*1), and dynamic window).

The drawable area is indicated by ΔY. (unit: dot) 

Also, the area in green is nondrawable area.

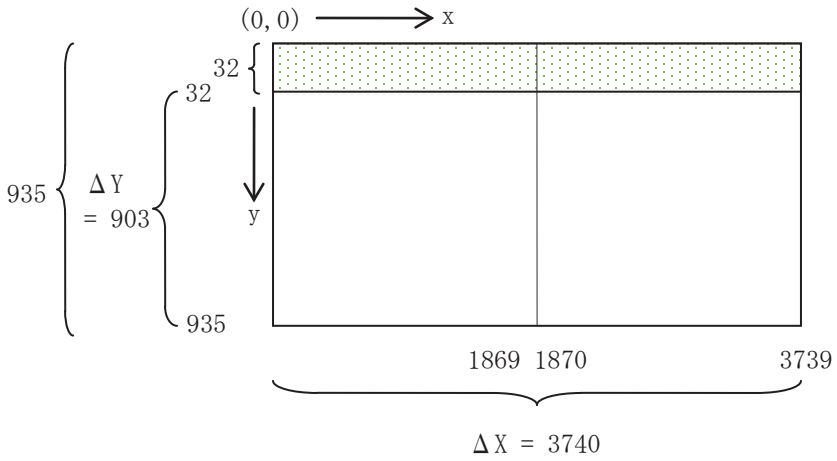
Note 1: The extended screen types are seven: extended (horizontal double), extended (horizontal triple), extended (horizontal four times), extended (vertical double), extended (vertical triple), extended (vertical four times), and extended (four times).

(1) Standard screen

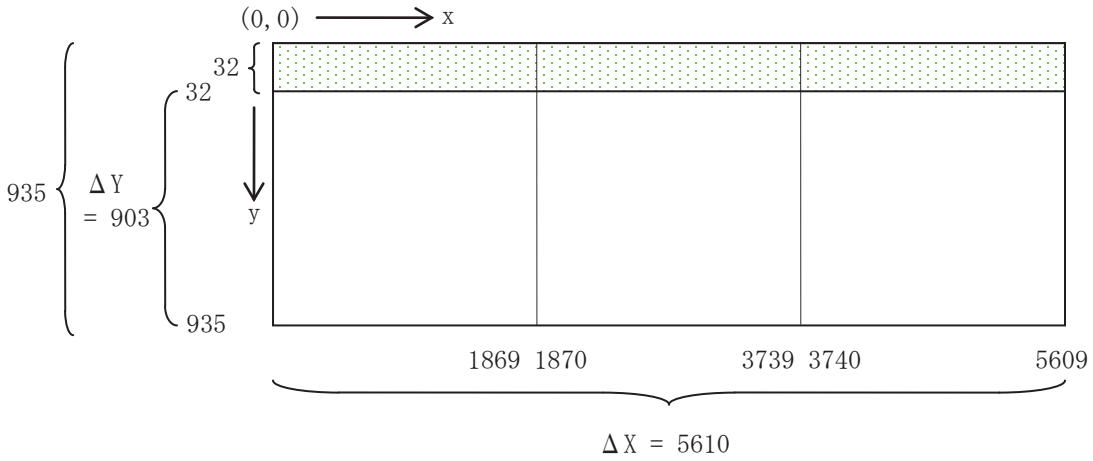


(2) Extend screen

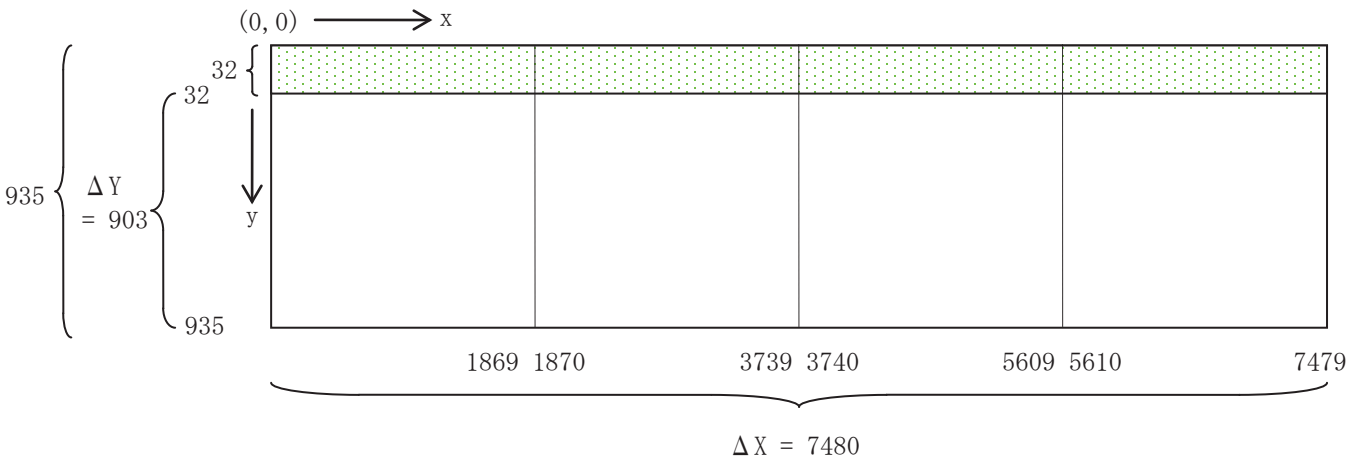
① Extend (horizontal double)



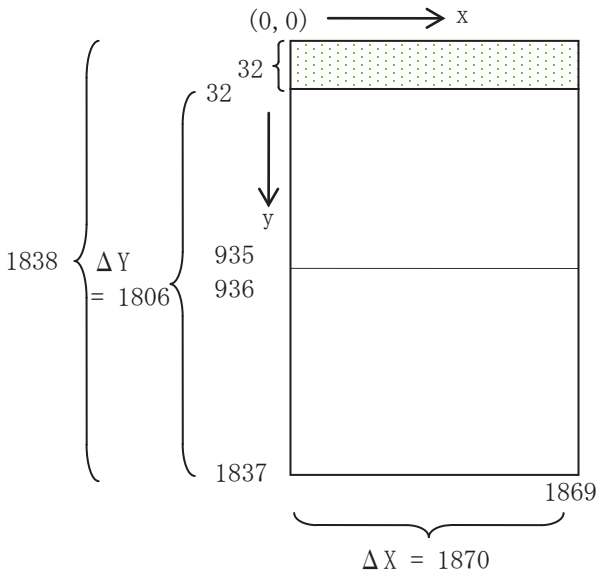
② Extend (horizontal triple)



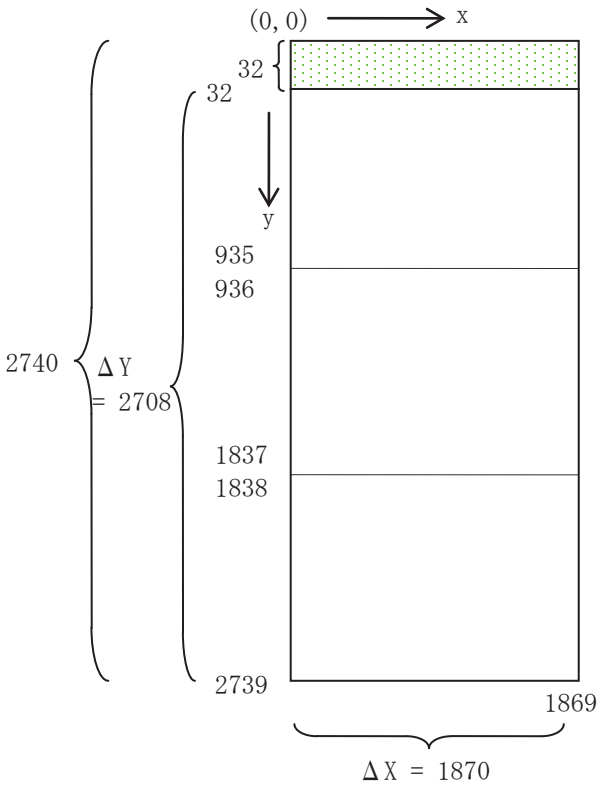
③ Extend (four times)



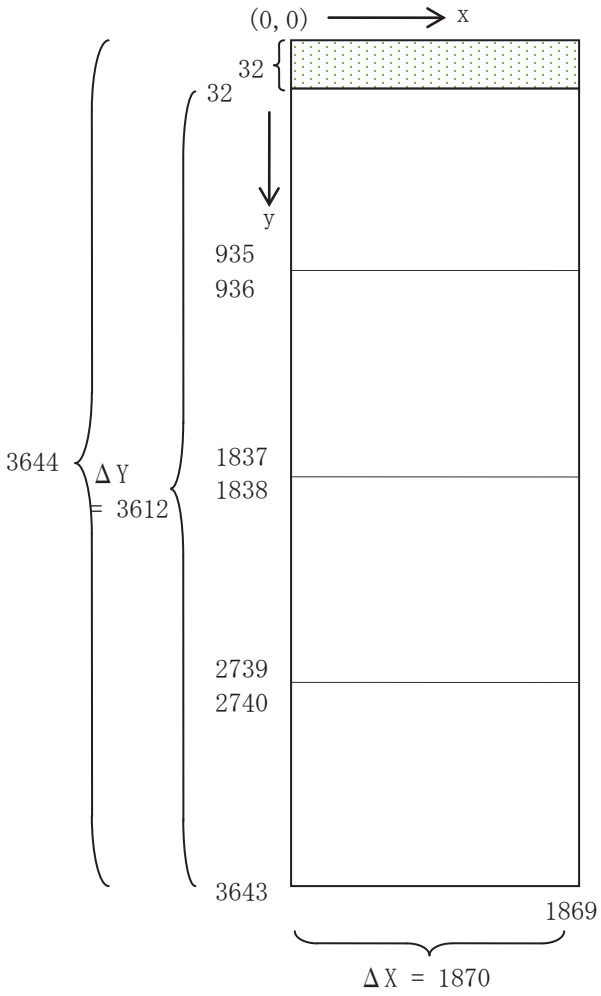
④ Extend (vertical double)



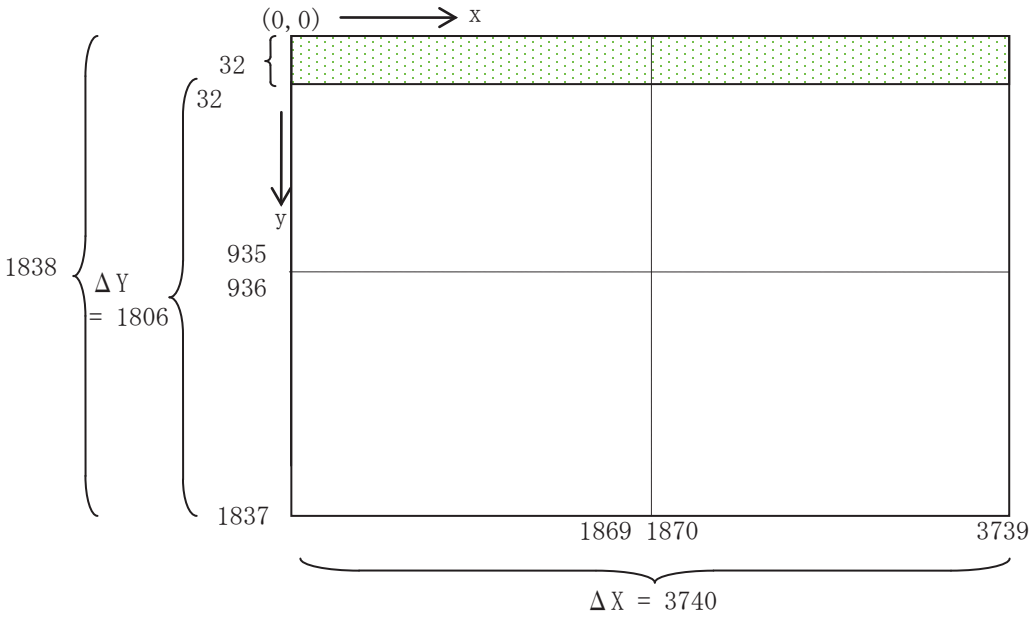
⑤ Extend (vertical triple)



⑥ Extend (vertical four times)

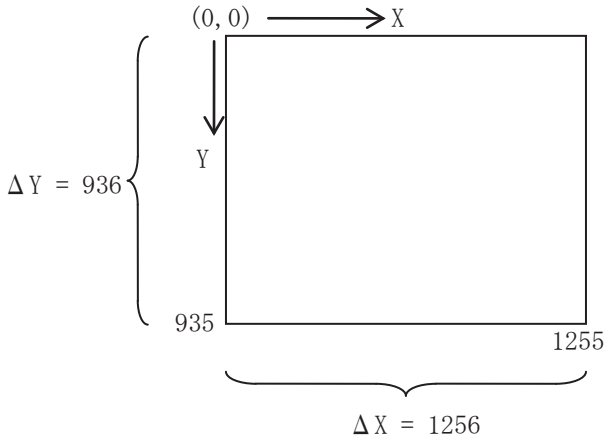


⑦ Extend (four times)



(2) Dynamic window

- The figure below shows a dynamic window of maximum size.



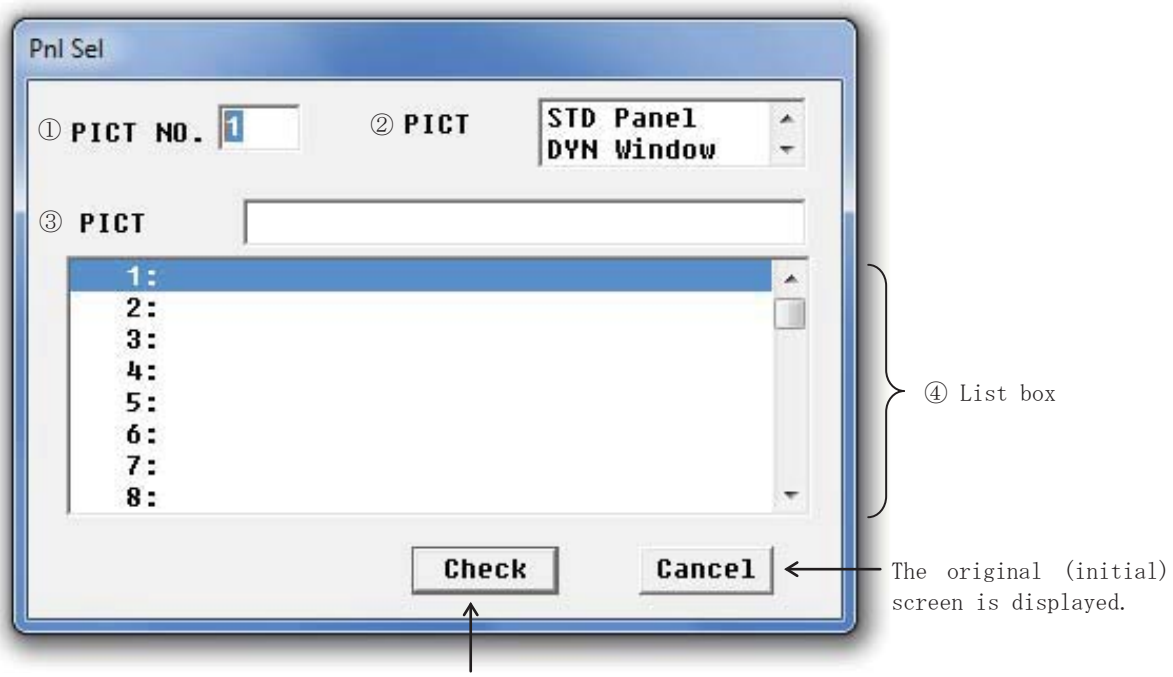
Note: The size of dynamic window can be defined at the time of drawing and the size ranges as follows:

$$\begin{cases} 0 \leqq \Delta X \leqq 1256 \\ 0 \leqq \Delta Y \leqq 936 \end{cases}$$

2.3 How to Operate

2.3.1 How to Operate Files

(1) Selecting screen



① Screen number: Specify the screen (PICT) number to select or create. (For the number to specify, refer to 1. Drawing Engineering Specification, 1.5 Scale of Definition Screen.)

② Screen type: Select the size of the screen. Select one from: standard screen, extended (horizontal double), extended (horizontal triple), extended (horizontal four times), extended (vertical double), extended (vertical triple), extended (vertical four times), extended (four times), and dynamic window. (standard screen if not selected)

③ Screen name: Specify a screen name. (Up to 32 single-byte texts. Single-byte and double-byte texts can be mixed)

④ List box: Select the screen number from the list box.

* How to display the list box

- The list box displays screen numbers and names.
  Up to 2048 screens can be defined.

- Change the display in the list box with the scroll bar at the right side.

Note 1: The screen name entered while selecting the screen is saved in the disk after saving the screen. When the name is changed during the saving of screen, the name becomes valid.

Note 2: Click Confirm to open (display) a screen based on the screen number.

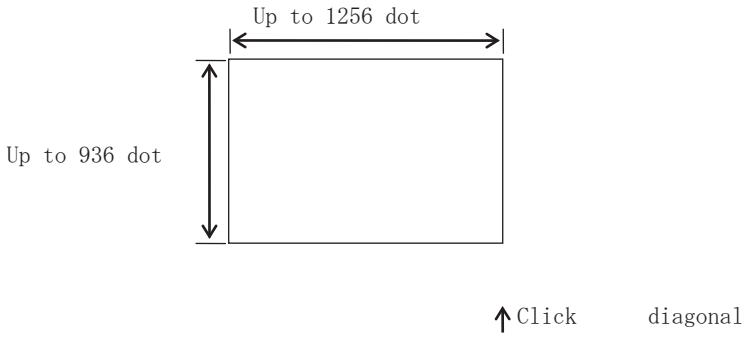
Note 3: If other than available range of numbers or other than numeric texts (alphabets and Kanjis, etc.) is entered, the screen number area is displayed inverted. Enter a correct number.

(a) Screen type: When creating a new dynamic window:

・- Creating a new dynamic window:

Other than File menu is shown grayed out.

First of all, use the mouse to set up the dynamic window frame. The frame size is up to 1256x936 dot. It is impossible to determine a frame beyond the maximum size.

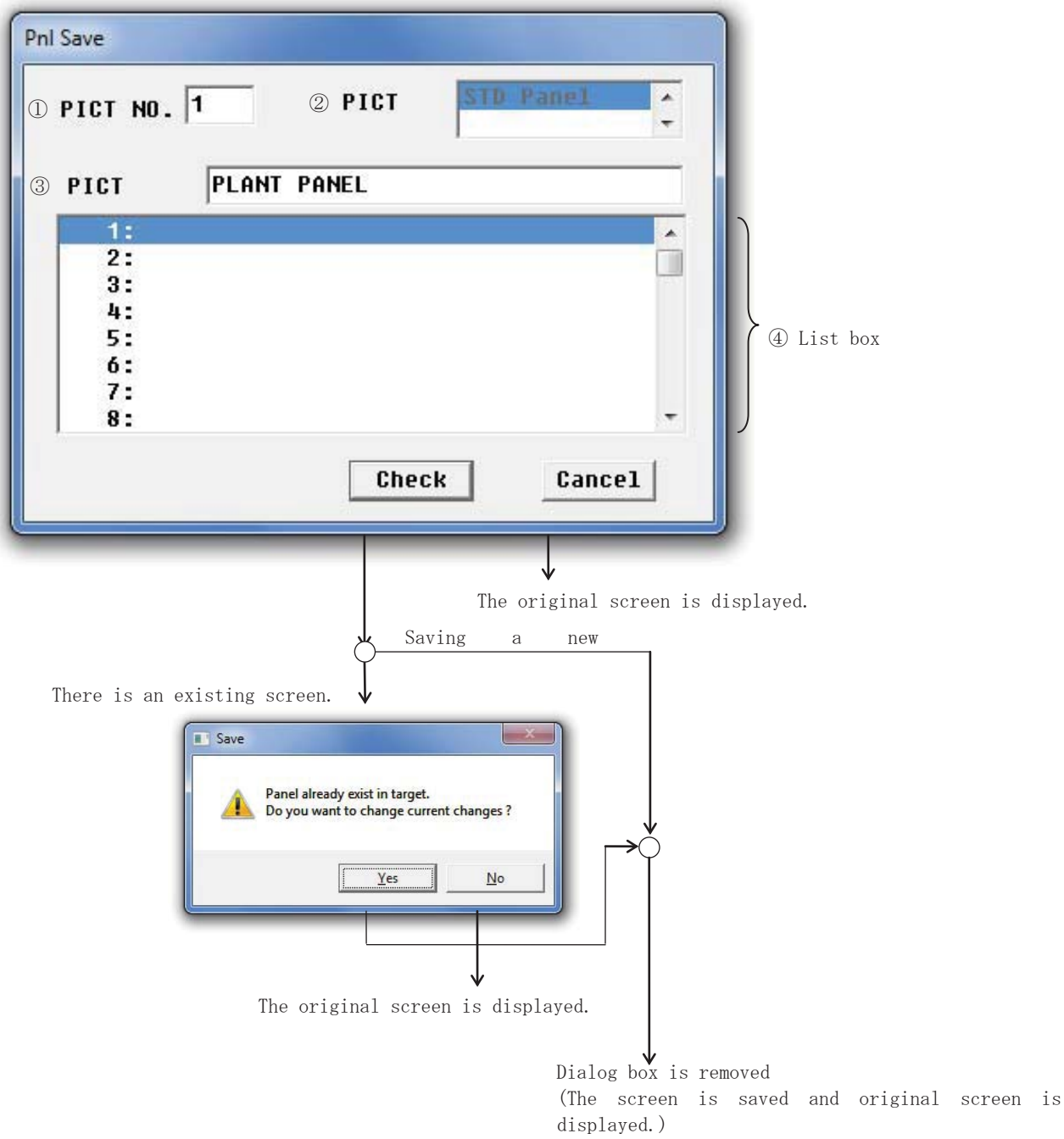


When the frame size is determined, all menus become available for use and the drawing is enabled similar to normal screens.

・- Creating an existing dynamic window:

The dynamic window frame cannot be moved, copied, or deleted. You can only modify existing one.

(2) Saving screen



① Screen number: Specify the screen (PICT) number to select or create. (For the number to specify, refer to 1. Drawing Engineering Specification, 1.5 Scale of Definition Screen.)

② Screen type: Select the size of the screen. Select one from: standard screen, extended (horizontal double), extended (horizontal triple), extended (horizontal four times), extended (vertical double), extended (vertical triple), extended (vertical four times), extended (four times), and dynamic window. (standard screen if not selected)

③ Screen name: Specify a screen name. (Up to 32 single-byte texts. Single-byte and double-byte texts can be mixed)

* When no screen name is entered, an error occurs.

④ List box: Select the screen number from the list box.

* How to display the list box

- The list box displays PICT numbers and names. [In case of panel basis, the panel number and name are displayed] Up to 128 screens can be displayed per page (list).
- Change the display in the list box with the scroll bar at the right side.

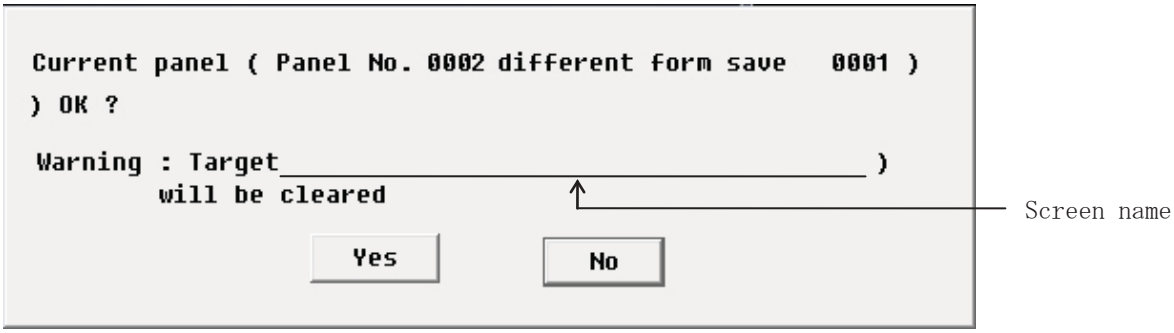
注１） The screen name entered while selecting the screen is saved in the disk after saving the screen. When the name is changed during the saving of screen, the name becomes valid.

注２） If other than available range of numbers or other than numeric texts (alphabets and Kanjis, etc.) is entered, the screen number area is displayed inverted. Enter a correct number.

注３） Power-off and no connection with cable may cause an external disk access error. Check the disk.

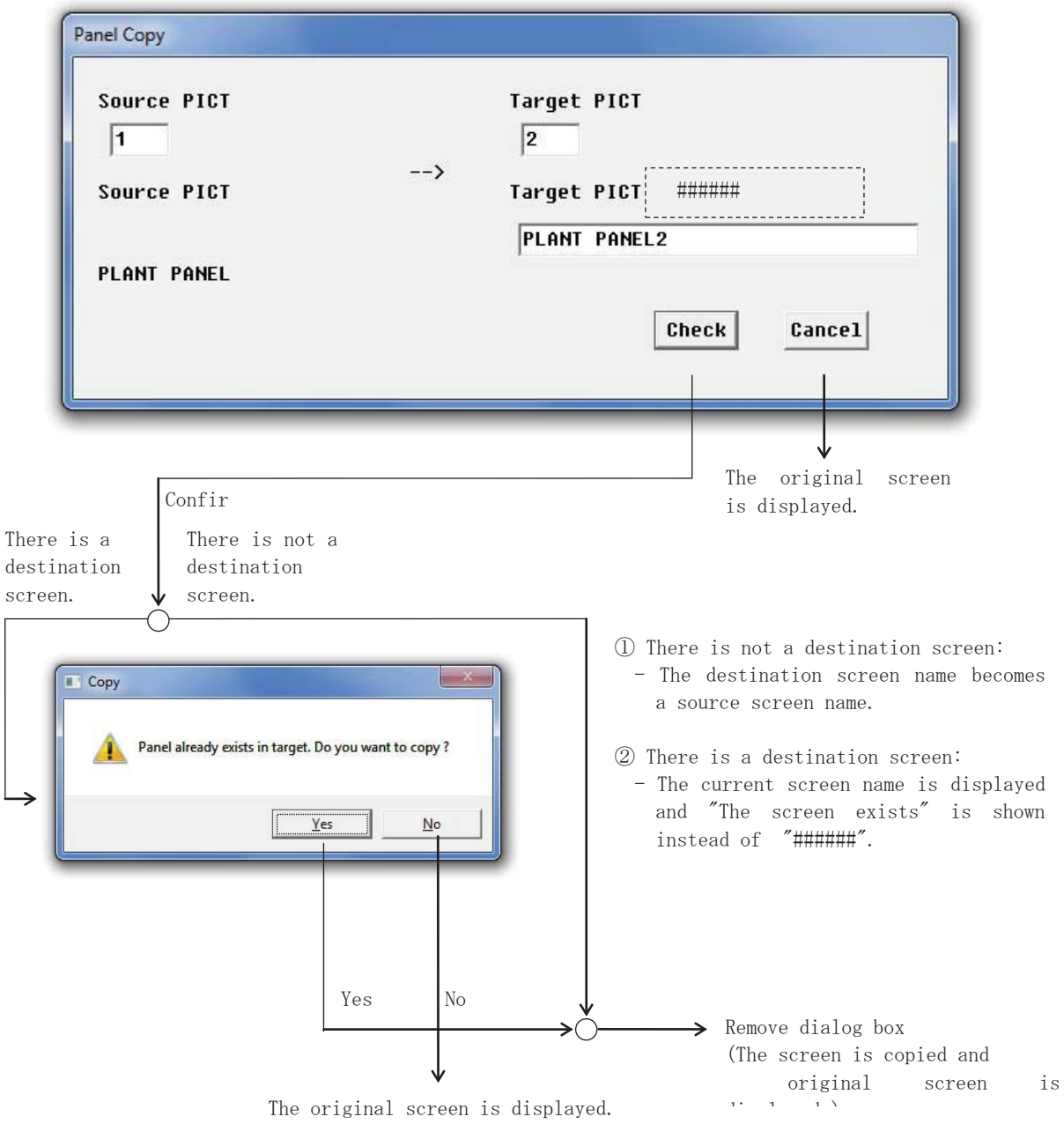
(warning message)

・The screen number is different between those during selecting and saving of screens:



・There is already a screen with the screen number:

(3) Copying screen

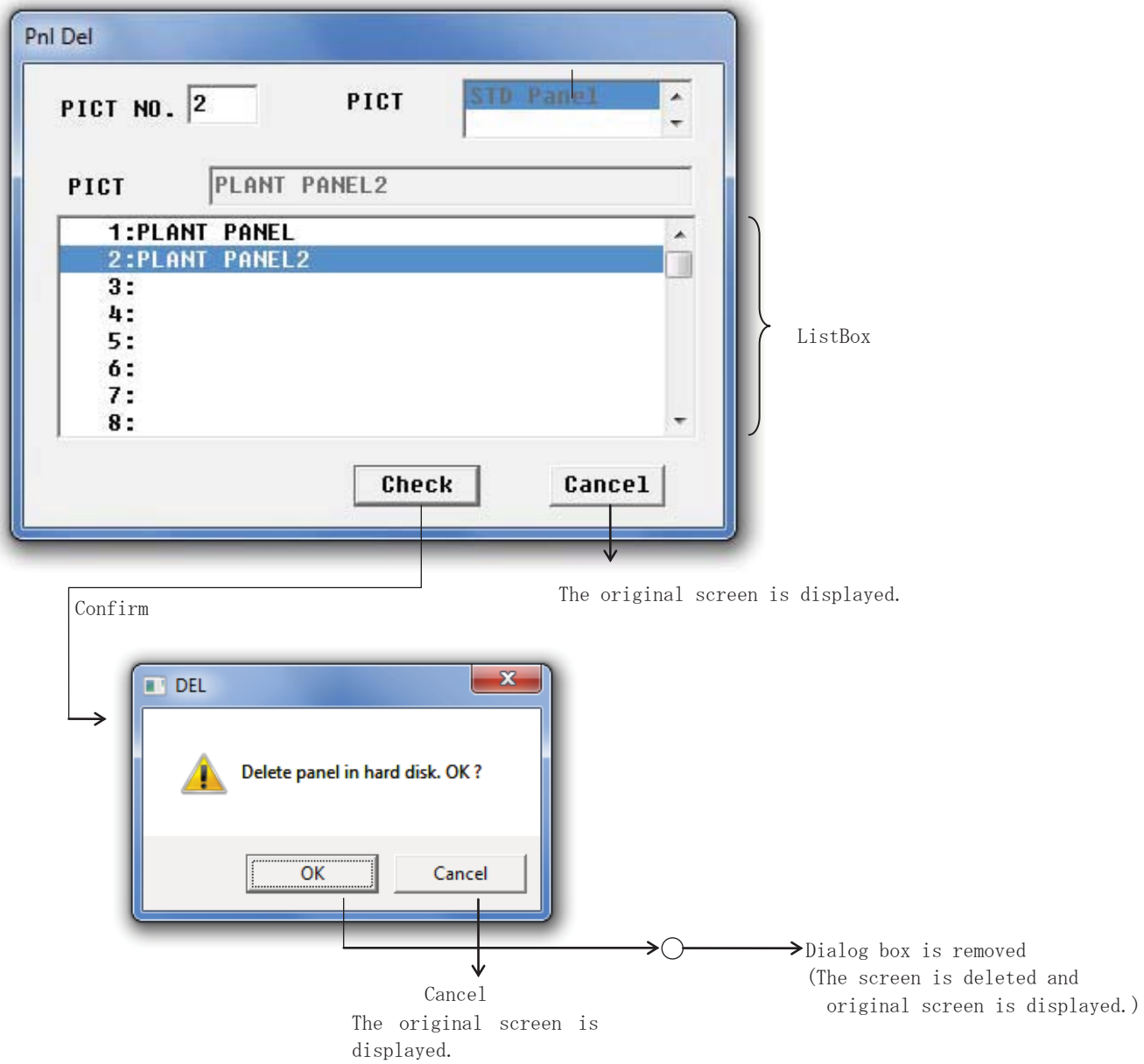


注１) When unavailable number (PICT unit: 1 to 2048) or other than numeric texts (alphabets and Kanjis, etc.) is entered into the source or destination screen number area, the area is displayed inverted. Enter a correct number.

注２) In case of panel basis, the panel number and name are displayed.

注３) Power-off and no connection with cable may cause an disk access error of external disk. Check the disk.

(4) Deleting screen



Note 1) Even if the number does not have any screen, the deletion is started.

Note 2) The name cannot be specified during deleting a screen.

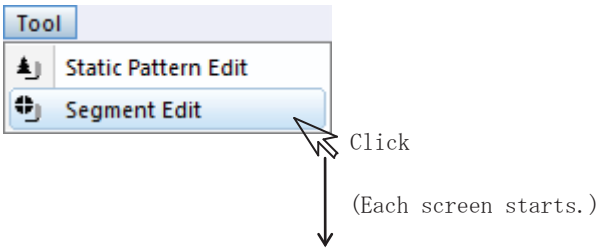
Note 3) If other than available range of numbers (PICT unit: 1 to 2048) or other than numeric texts (alphabets and Kanjis, etc.) is entered, the screen number area is displayed inverted. Enter a correct number.

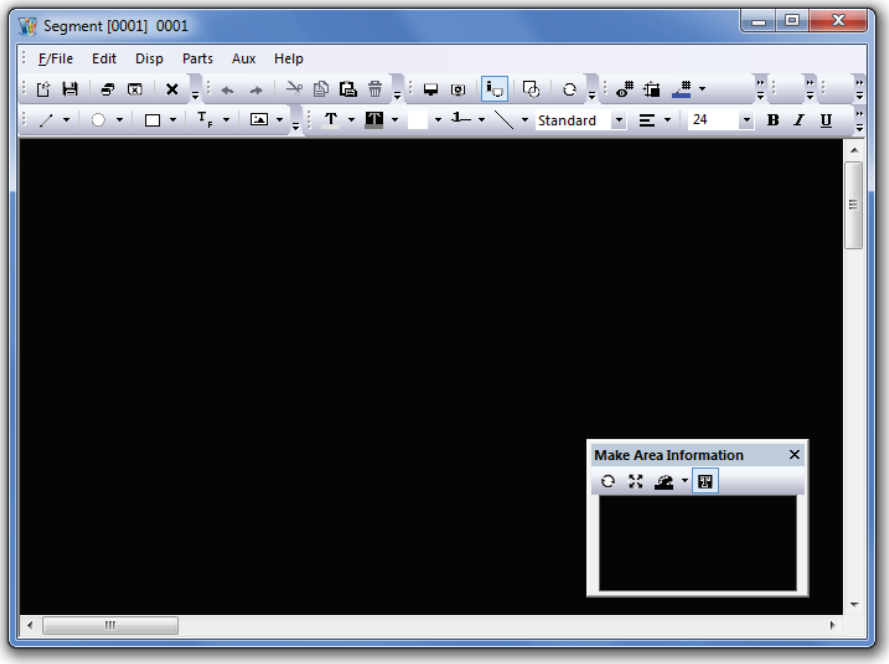
Note 4) Power-off and no connection with cable may cause an disk access error of external disk. Check the disk.

(5) Exit

Carries out the same process as saving and exit the interactive drawing.

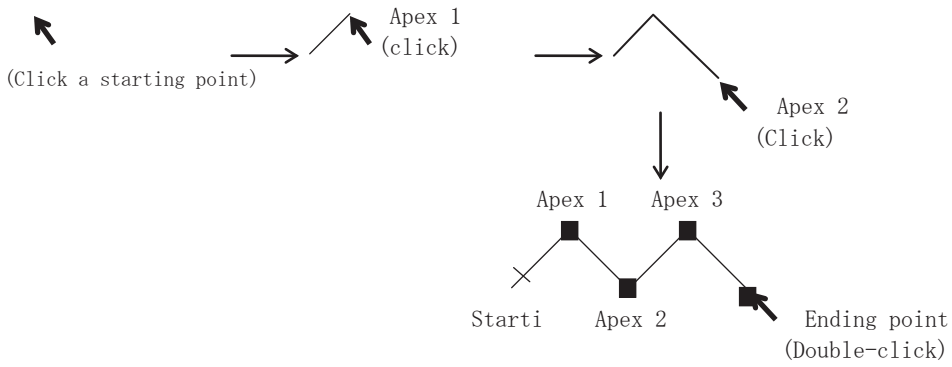
(6) Editing each screen



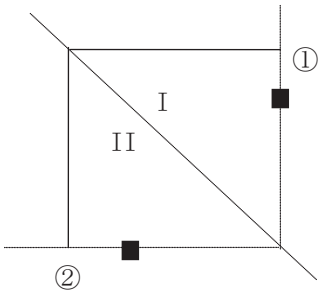


2.3.2 How to Operate Graphics

(1) Line: Draw a polygonal line segment.



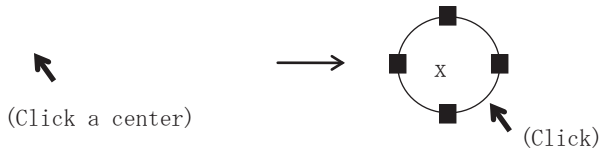
* The horizontal or vertical rubber band is shown automatically for horizontal or vertical line.



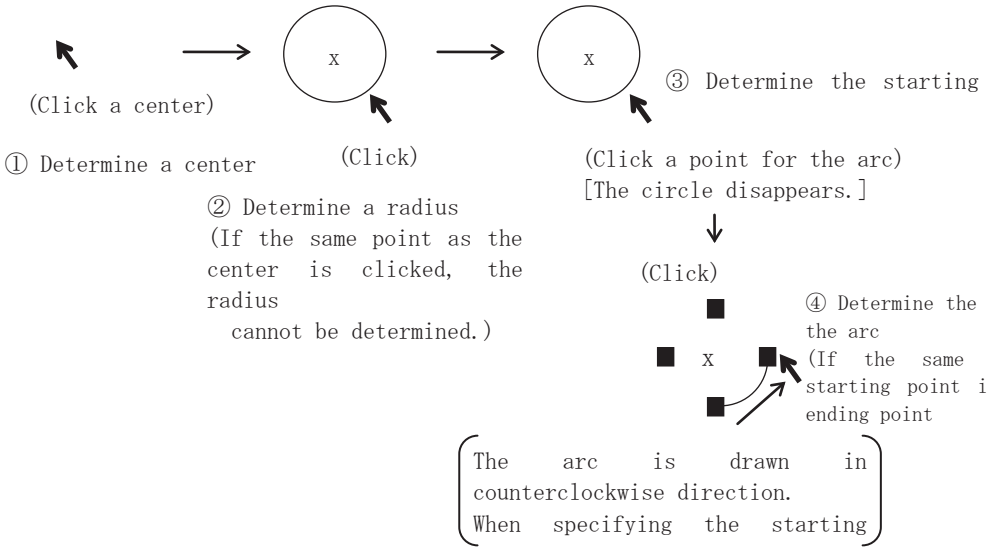
When the cursor is in the section I, the rubber band ① is shown.
Here, correct coordinate x is displayed in auxiliary information area.

When the cursor is in the section II, the rubber band ② is shown.
Here, correct coordinate y is displayed in auxiliary information area.

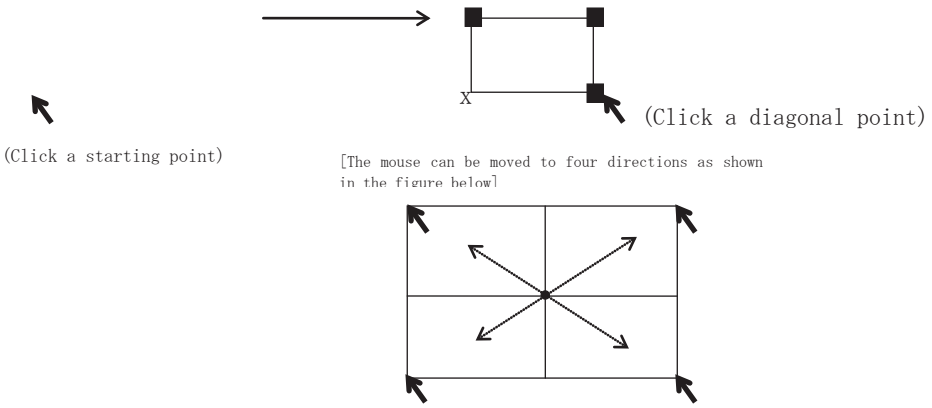
(2) Circle: Draw the graphic by specifying a center and radius.



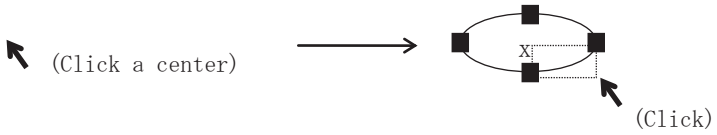
(3) Arc: Draw the graphic by specifying a center, radius, starting point and ending point.



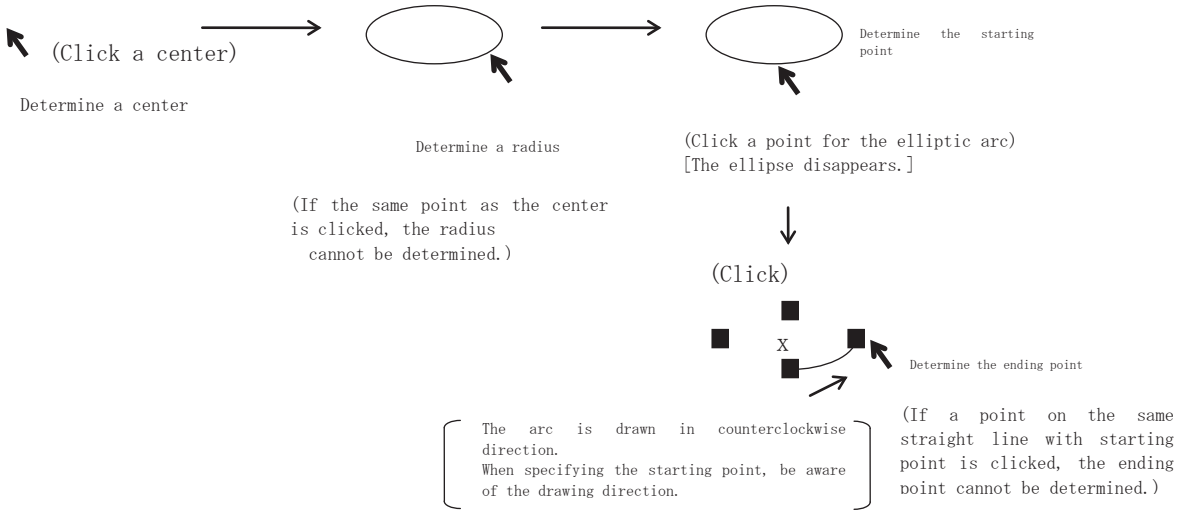
(4) Rectangle: Draw the graphic by specifying diagonal points.



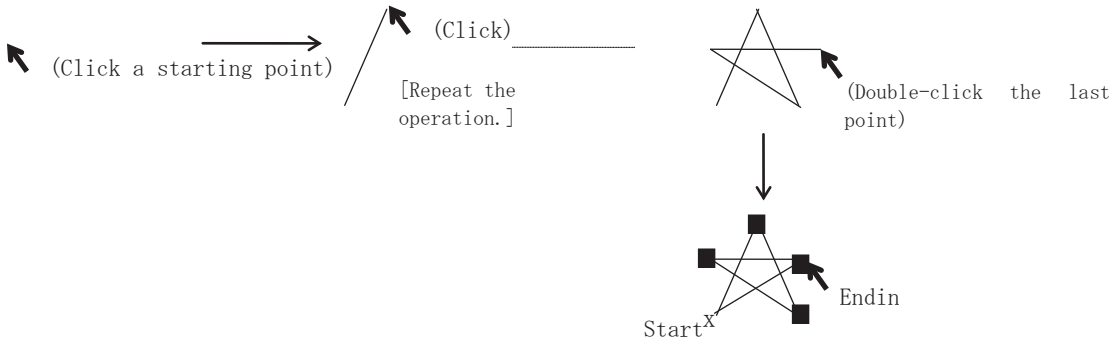
(5) Ellipse: Draw the graphic by specifying a center and long and short radii.



(6) Elliptic arc: Draw the graphic by specifying a center, long and short radii, and starting point and ending point.

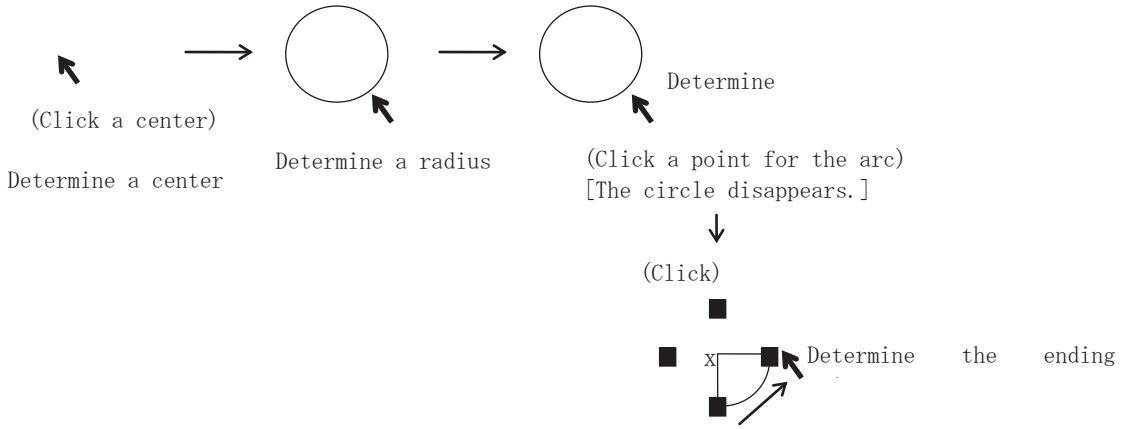


(7) Polygon: Draw a polygonal line segment. Starting and ending points are connected.



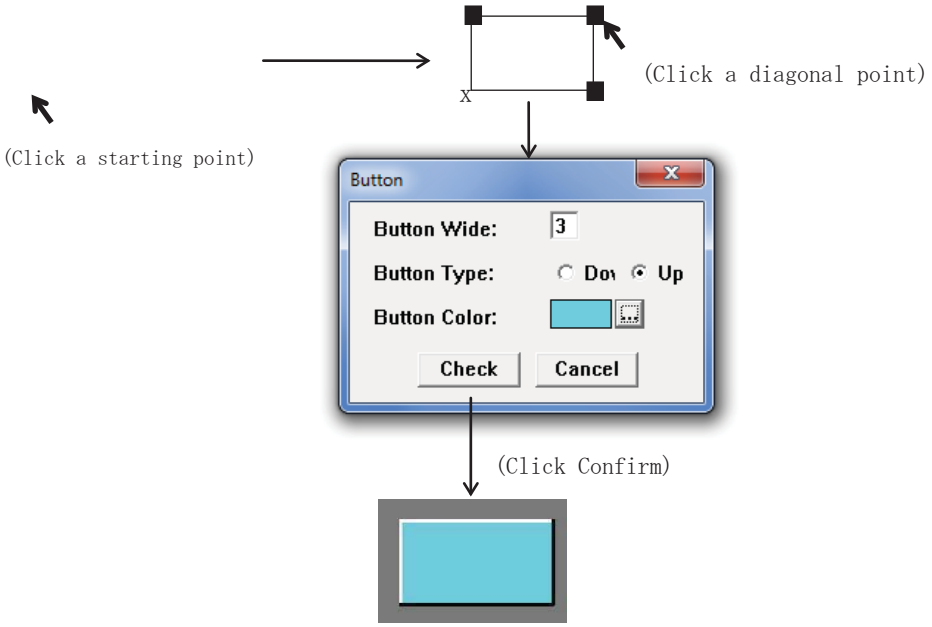
(The line connecting starting point and ending point is automatically drawn by double-clicking the last point.)

(8) Sector: Draw the graphic by specifying a center, radius, starting point and ending point.



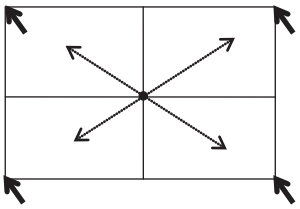
The arc is drawn in counterclockwise direction.
When specifying the starting point, be aware of the drawing direction.

(9) Button type rectangle: Draw the graphic by specifying diagonal points.



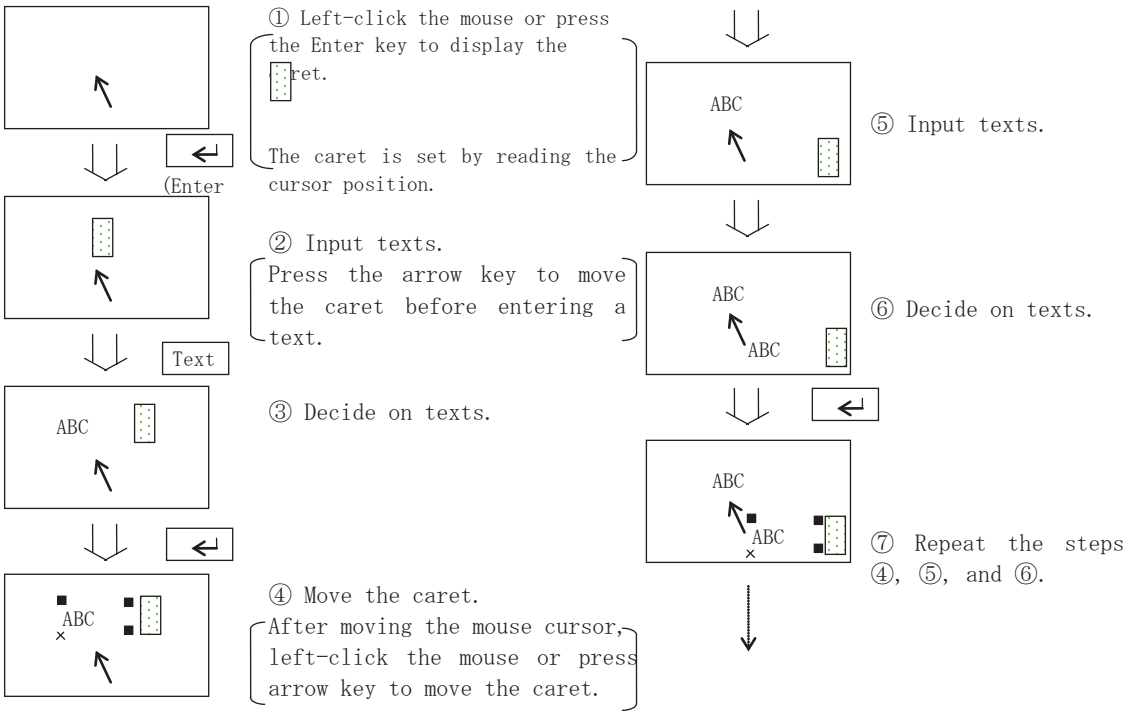
* In the "Button Type Rectangle" dialog box, the default for "Frame Width" and "Button Type" is one and convexed respectively.

[The mouse can be moved to four directions as shown in the figure

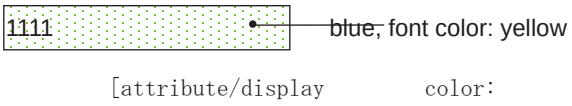


(10) Point: Draw the graphic by specifying a point.

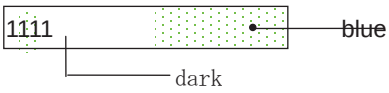
(11) Text (96, 64, 48, 32, 24, 16, 8 dots): Input the text by specifying a starting point.



* When inputting texts as shown in the figure below, select the background color.



- When the attribute/display color is yellow and background color is dark, the texts will be as shown in the figure below.



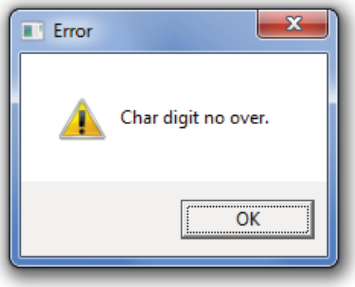
Note 1) The caret does not move even if the Enter key is pressed. Move the caret to desired place.

Note 2) Do not perform other operation (select a menu, for example) while inputting texts (until pressing the Enter key).

A wrong operation may occur. To perform other operation, press the Enter key to confirm texts or press the backspace key to delete all texts.

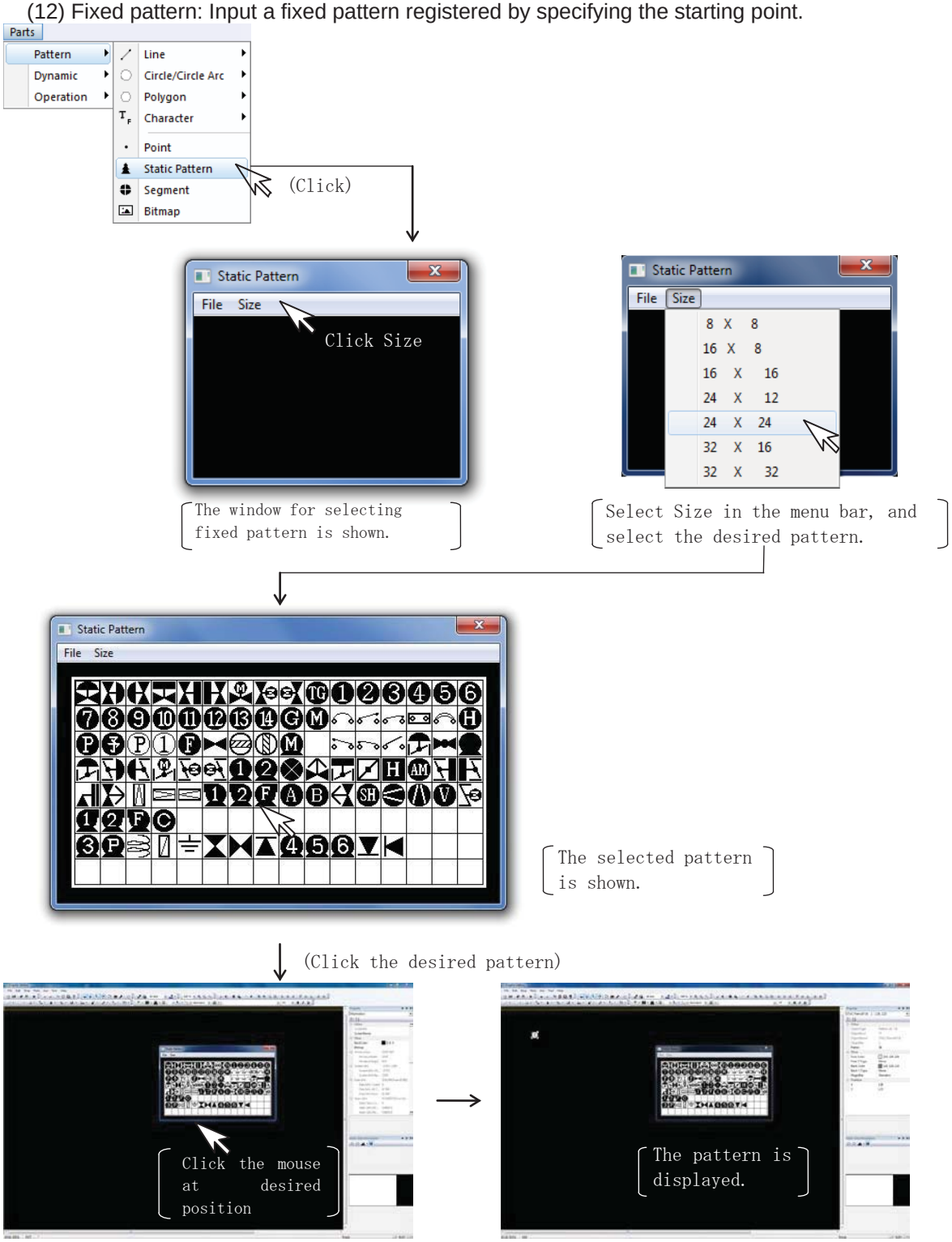
Note 3) The maximum number of characters should be 245 on a single-byte basis as long as the texts can be contained in the text area.

Note 4) If the input text exceeds the character area, an error message is shown as follows:



Note 5) The precedence of the input checks is: character area and 254 single-byte characters. If an error occurs during inputting characters, the input is canceled.

(12) Fixed pattern: Input a fixed pattern registered by specifying the starting point.



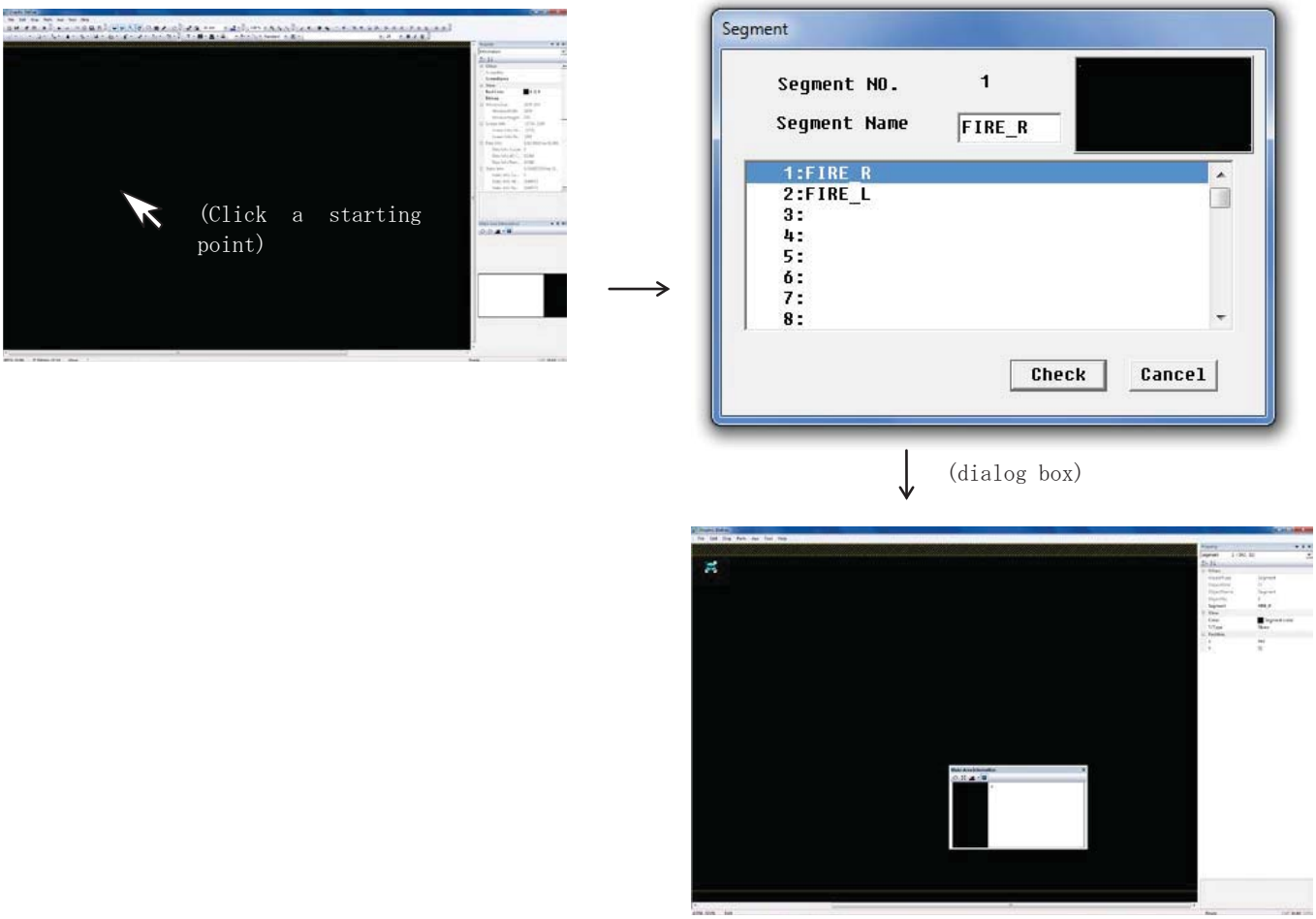
Note 1: The pattern area can be registered even if fixed pattern was not created.

When the pattern is created by using a pattern editor after creating the pattern area, the pattern is shown on the screen.

Note 2: The keyboard becomes available for use only if the fixed pattern window is active.

For more information, refer to Microsoft Windows reference manual.

(13) Segment: Input a segment registered by specifying the starting point.



Note 1) If the segment name is not input, S**** is shown.

If the segment itself is not created even if the segment name was entered, SΔΔΔΔ is shown (ΔΔΔΔ: 1 to 1024).

Note 2) The starting point of the segment is also the starting point of the first command in the segment.

Note 3) It is impossible to call up a part of the graphic registered in the segment. Save and use necessary part of the graphic as a separate segment.

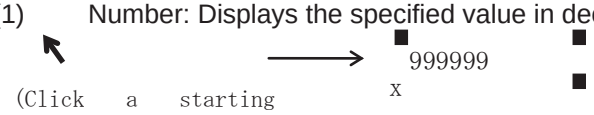
There are two ways to specify a segment:

1. Enter and confirm the name
2. Select and confirm the name from the name list

* To refer to the segment itself, display the segment screen (P.3—2).

2.3.3 How to Operate Variable Object

(1) Number: Displays the specified value in decimal or hexadecimal format.

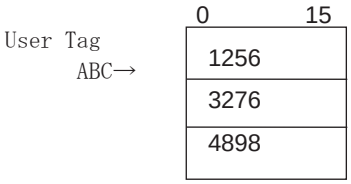


('9' is shown as many as the number of digits)

-	**Other**		**Description**	**Change**
	Object Type	Type of the object	Numeric	No
	Object Kind	Kind of the object	IDDYNAMIC	No
	Object Name	Name of the object	Numeric	No
	Object No	Number of the object	–	No
-	**Position**			
	X	Set the coordinate X of starting point.	0 to 1870	Yes
	Y	Set the coordinate Y of starting point.	0 to 935	Yes
-	**View**			
	Size	Set the character size.	8x8, 16x8, 16x16, 24x12, 24x24, 32x16	Yes
	ForeColor	Set the character color.	Select it from the color dialog.	Yes
	For Type	Set the character color type.	None, Blink, Reverse, Reverse + Blink	Yes
	BackColor	Set the background color.	Select it from the color dialog.	Yes
	Magnifier	Set the zoom ratio.	Standard, two to eight times	Yes
-	**Data**			
	ST No	Set the station No.	Only when specifying the system Tag, specify the physical ST No. (101 to 440)	Yes
	Tag	Set the display Tag.	Number of Tag characters: Up to 16 single-byte characters Selectable from the dialog box	Yes
	Numeric Type	Set the numeric value type.	Decimal two word, hexadecimal two word, decimal one word, and hexadecimal one word	Yes
	Tag Accordance	Set whether to display the tag in the tag definition.	True, False	Yes
	Color Extension	Set the color change extension.	Fixed, Module, Option, Option 2 - Use the option and option 2 when changing the display color and background color. **(See "2.3.3 (17) Color option and option 2")**	Yes
-	**Num Processing**			
	Digit Number	Set the number of display digits.	1 to 16 (decimal point is also counted)	Yes
	Truncate Number	Set the place to be truncated.	-15 to 15 (the data is rounded off)	Yes
	Decimal Place	Set the place of decimal point. Specifiable only for decimal numbers.	-1 to 15 (The Tag definition determines if -1)	Yes
	Zero Suppression	Set the zero suppression.	True, False	Yes
-	**Array**			
	Array Count	Set the number of pieces of continuous data.	1 to 128	Yes
	Array Direction	Set the display direction of continuous data.	Right, Down	Yes
	Array Spacing	Set the display spacing of continuous data.	0 to 256	Yes

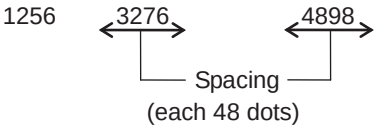
◆ Specifying continuous numeric data

The display data is continuous, as shown in the figure below (three words are continuous from the position indicated in the Tag):

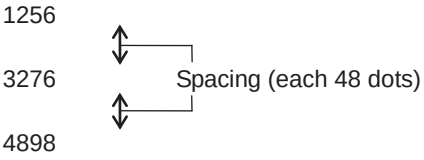


When the numeric data spacing is set to 48 dots, the data is shown as follows:

I) To the right

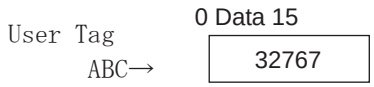


II) Downward



◆ Special usages of Digit Number, Truncate Number, and Decimal Place

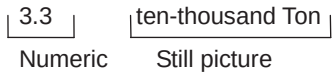
Example 1: To display numeric data to first decimal place in ten thousand yen units on the plant panel at the operator station:



Here, if Digit Number: 3, Truncate Number: 3, and Decimal Place: 4, the data is shown as follows:

* The data is rounded off based on Truncate Number. In this case, the last third digits 767 are to be rounded off.

The last two digits are not supposed to be rounded off.



Example 2: Another example of Digit Number, Truncate Number, and Decimal Place are as follows:

Data: 32754

	Digit Num	Truncate Num	Decimal Pl	Display
①	5	0	0	32754
②	5	1	0	3275
③	5	2	0	328
④	6	0	2	327.54
⑤	6	1	2	327.5
⑥	6	3	2	328
⑦	3	4	0	3
⑧	4	0	0	**** …(*)

* When the data cannot be contained in the display digit, the number is shown as asterisks (*).

◆ Specifying double word data in integrated modules

When specifying double word data in integrated modules, make sure to specify higher-order word data (extension: **H).

The data cannot be shown by specifying lower-order word data (extension: **L).

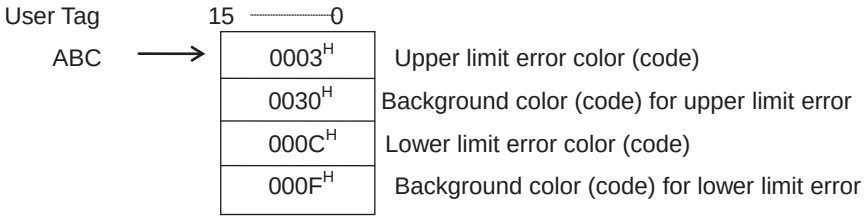
(2) Limiter numeric: Number with upper and lower limits.

('9' is shown as many as the number of digits)

-	**Other**		**Description**	**Change**
	Object Type	Type of the object	LimiterNumeric	No
	Object Kind	Kind of the object	IDDYNAMIC	No
	Object Name	Name of the object	Upper and Lower Limits	No
	Object No	Number of the object	–	No
-	**Position**			
	X	Set the coordinate X of starting point.	0 to 1870	Yes
	Y	Set the coordinate Y of starting point.	0 to 935	Yes
-	**View**			
	Size	Set the character size.	8x8, 16x8, 16x16, 24x12, 24x24, 32x16	Yes
	Color Variation	Set the color change extension.	Fixed, Module, Option, Option 2 - Use the option and option 2 when changing the display color and background color. **(See "2.3.3 (17) Color option and option 2")**	Yes
	ForeColor	Set the character color.	Select it from the color dialog.	Yes
	For Type	Set the character color type.	None, Blink, Reverse, Reverse + Blink	Yes
	Back Color	Set the background color.	Select it from the color dialog.	Yes
	Magnifier	Set the zoom ratio.	Standard, two to eight times	Yes
-	**Data**			
	Tag	Set the display Tag.	Number of Tag characters: Up to 16 single-byte characters Selectable from the dialog box	Yes
	Numeric Type	Set the numeric value type.	Decimal two word, hexadecimal two word, decimal one word, and hexadecimal one word	Yes
	Tag Accordance	Set whether to display the tag in the tag definition.	True, False	Yes
-	**Num Processing**			
	Digit Number	Set the number of display digits.	1 to 16 (decimal point is also counted)	Yes
	Truncate Number	Set the place to be truncated. The data is rounded off.	-15 to 15	Yes
	Decimal Place	Set the place of decimal point. Specifiable only for decimal numbers.	-1 to 15 (The Tag definition determines if -1)	Yes
	Zero Suppression	Set the zero suppression.	True, False	Yes
-	**Limit Condition**			
	Use Limit Color	Set the upper or lower limit error color.	None, Tag setting, Direct	Yes
	Limit Definition	Set whether to define upper and lower limits.	Yes, No	Yes

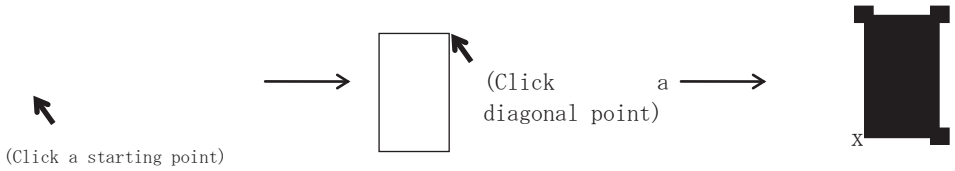
- Specifying tag

When specifying the user Tag "ABC" as upper or lower limit for limit error color definition and when the data is shown in the figure below:



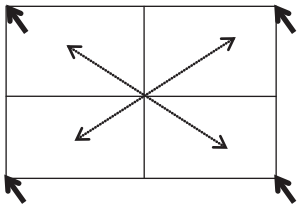
The upper limit error color is red (0003^H) and its background color is blue (0030^H)
The lower limit error color is green ($000C^H$) and its background color is yellow ($000F^H$)
(The Tag always handles raw one word for upper or lower limit error color.)

(3) Bar graph: Displays values with a rectangle graph.



[The mouse can be moved to four directions as shown in the figure below]

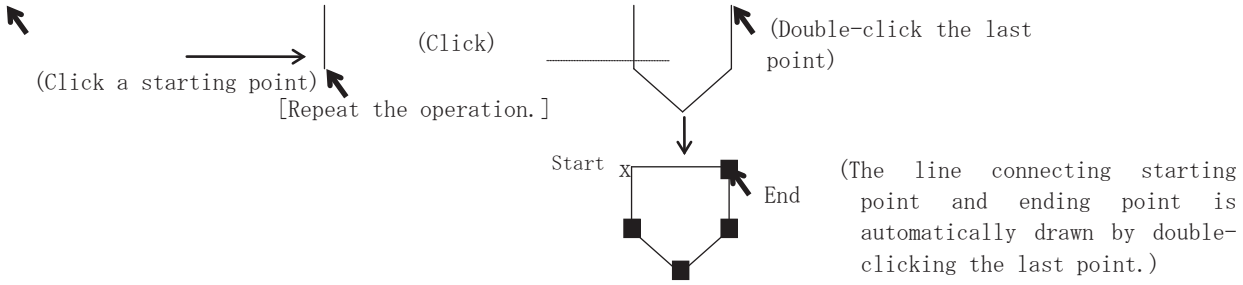
(Maximum range, display width, and display color are shown.)



-	**Other**		**Description**	**Change**
	Object Type	Type of the object	Bar graph	No
	Object Kind	Kind of the object	IDDYNAMIC	No
	Object Name	Name of the object	Bar Graph	No
	Object No	Number of the object	–	No
-	**Position**			
	Start X	Set the coordinate X of starting point.	0 to 1870	Yes
	Start Y	Set the coordinate Y of starting point.	0 to 935	Yes
-	**Data**			
	Cond ST No	Set the condition Tag station No.	Only when specifying the system Tag, specify the physical ST No. (101 to 440)	Yes
	Tag	Set the display Tag.	Number of Tag characters: Up to 16 single-byte characters Selectable from the dialog box	Yes
-	**View**			
	Direction	Set the direction of variable graph.	Up, Down, Left, Right	Yes
	Max Range	Set the maximum variable range.	1 to 935	Yes
	Max Width	Set variable display width.	1 to 1870	Yes
	Fill Pattern	Set a variable fill pattern.	Select it from the pattern dialog.	Yes
	Color Variation	Set the color change extension.	Fixed, Module, Option, Option 2 - Use the option and option 2 when changing the display color and background color. **(See "2.3.3 (17) Color option and option 2")**	Yes
	Erase Color	Set the erasing color.	Select it from the color dialog.	Yes
	Fore Color	Set the pattern color.	Select it from the color dialog.	Yes
	For Type	Set the pattern color type.	None, Blink, Reverse, Reverse + Blink	Yes
	Back Color	Set the background color.	Select it from the color dialog.	Yes

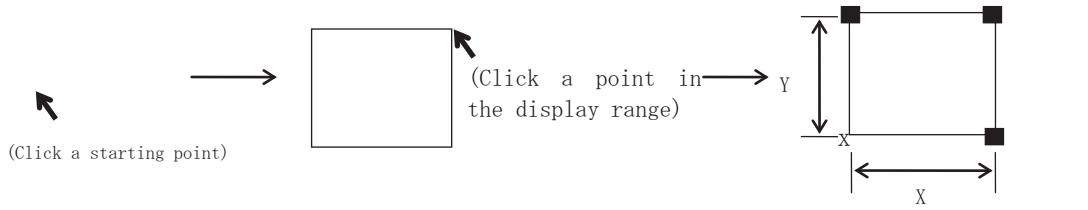
(4) Picture graph: Displays values with graph of other than rectangles (tank pictures, for example).

Create a graphic in the same manner as creating a polygon as a fixed object.

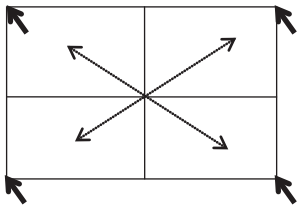


-	**Other**		**Description**	**Change**
	Object Type	Type of the object	Water graph	No
	Object Kind	Kind of the object	IDDYNAMIC	No
	Object Name	Name of the object	Picture Graph	No
	Object No	Number of the object	–	No
-	**View**			
	Line Weight	Set the line width.	1 to 16	Yes
	Line Color	Set the line color.	Select it from the color dialog.	Yes
	Line Type	Set the line width.	1 to 16	Yes
	Direction	Set the direction of graph.	Up, Down, Left, Right	Yes
	Max Range	Set the maximum range.	According to the graphic created	According to the change of graphic
	Fill Pattern	Set a fill pattern.	Select it from the pattern dialog.	Yes
	Color Variation	Set the color change extension.	Fixed, Module, Option, Option 2 - Use the option and option 2 when changing the display color and background color. **(See "2.3.3 (17) Color option and option 2")**	Yes
	Erase Color	Set the erasing color.	Select it from the color dialog.	Yes
	Fore Color	Set the pattern color.	Select it from the color dialog.	Yes
	Fore Type	Set the pattern color type.	None, Blink, Reverse, Reverse + Blink	Yes
	Back Color	Set the background color.	Select it from the color dialog.	Yes
-	**Data**			
	Cond ST No	Set the condition Tag station No.	Only when specifying the system Tag, specify the physical ST No. (101 to 440)	Yes
	Tag	Set the display Tag.	Number of Tag characters: Up to 16 single-byte characters Selectable from the dialog box	Yes
-	**Others**			
	Apexes	Set the number of apexes.	According to the graphic created	Yes
-	**Position**			
	Apex$_n$ ***(n=1 to Apexes)***	Coordinates information of apex$_n$	According to the graphic created	Yes

(5) Trend: Displays specified range of values with historical and real trend graphs.

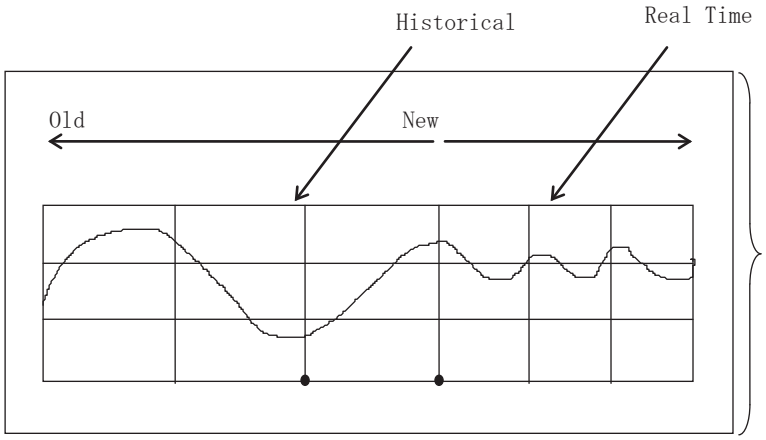


[The mouse can be moved to four directions as shown in the figure below]



(The rectangle with display range X, Y is shown.)

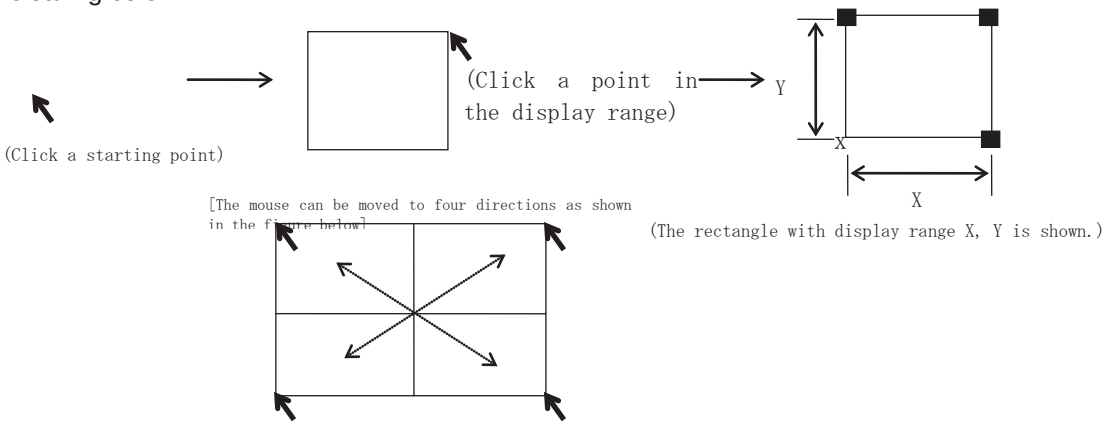
-	**Other**		**Description**	**Change**
	Object Type	Type of the object	Trend Graph	No
	Object Kind	Kind of the object	IDDYNAMIC	No
	Object Name	Name of the object	Trend	No
	Object No	Number of the object	–	No
-	**Position**			
	Start X	Set the coordinate X of starting point.	0 to 1870	Yes
	Start Y	Set the coordinate Y of starting point.	0 to 935	Yes
-	**Data**			
	Tag	Set the display Tag.	Up to 16 single-byte Tag characters Selectable from the dialog box	Yes
	Trend Type	Set the trend type.	Historical, Real Time	Yes
	Trend Unit	Set the unit of the trend cycle.	Hour, Minute, Second	Yes
	Trend Cycle	Set the trend cycle.	2 to 3600	Yes
-	**View**			
	Direction	Set the direction of graph.	Up, Down, Left, Right	Yes
	Range X	Set the display range X.	1 to 1870	Yes
	Range Y	Set the display range Y.	1 to 935	Yes
	Fore Color	Set the line color.	Select it from the color dialog.	Yes



Note: Both historical and real time cannot be specified at the same time.
The figure at the left shows graphs drawn separately from each other.

Online display example

(6) Line: Displays specified range of values with polygonal line and displays previous values with clearing color.



-	**Other**		**Description**	**Change**
	Object Type	Type of the object	Broken Curve	No
	Object Kind	Kind of the object	IDDYNAMIC	No
	Object Name	Name of the object	Trend	No
	Object No	Number of the object	–	No
-	**Position**			
	Start X	Set the coordinate X of starting point.	0 to 1870	Yes
	Start Y	Set the coordinate Y of starting point.	0 to 935	Yes
-	**View**			
	Range X	Set the display range X.	1 to 1870	Yes
	Range Y	Set the display range Y.	1 to 935	Yes
	Full X	Set the full value X.	1 to 32767	Yes
	Full Y	Set the full value Y.	1 to 32767	Yes
	Line Weight	Set the line width.	1 to 16	Yes
	Style	Set the style.	Solid, Dash Dot, Chain Dash2Dot	
	Color Variation	Set the color change extension.	Fixed, Module, Option, Option 2 - Use the option and option 2 when changing the display color and background color. **(See "2.3.3 (17) Color option and option 2")**	Yes
	Fore Color	Set the line color.	Select it from the color dialog.	Yes
	Fore Type	Set the line color type.	None, Blink, Reverse, Reverse + Blink	Yes
	Back Color	Set the background color.	Select it from the color dialog.	Yes
	Erase Color	Set the erasing color.	Select it from the color dialog.	Yes
-	**Data**			
	Coordinate Date Num	Set the number of pieces of coordinate data.	1 to 300	Yes
	ST No	Set the station No.	Only when specifying the system Tag, specify the physical ST No. (101 to 440)	Yes
	Tag	Set the display Tag.	Up to 16 single-byte Tag characters Selectable from the dialog box	Yes
	Controller Type	Set the controller type.	Standard, PCS500	Yes

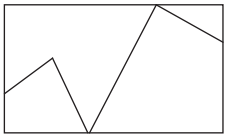
TAGNO.: Specify the start area of apex values, as shown below:

TAGNO. →

X1
Y1
X2
Y2
⋮

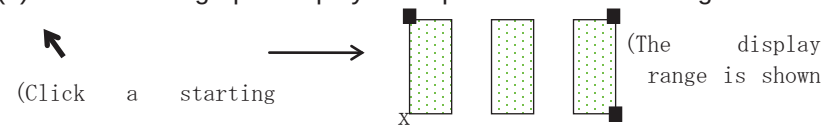
(*) As shown in the figure at the left, Set TAGNO. to the Tag number indicating the X1 area.
If X2 is specified, the apexes X1, Y1 are ignored.

Note: The plant panel editor just indicates that a polygonal line is registered. The display color and line width are displayed and only five apexes including ending point are displayed.



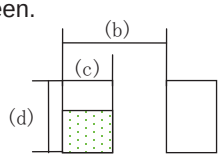
Line

(7) Prot bar graph: Displays multiple values with rectangle bars.

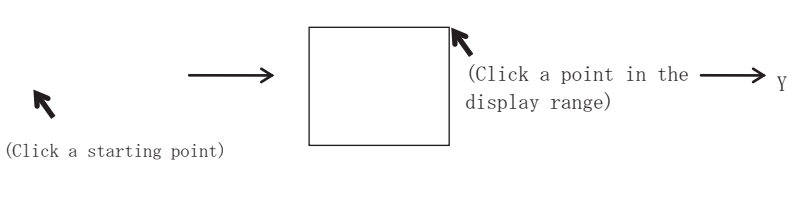


-	**Other**		**Description**	**Change**
	Object Type	Type of the object	Prot Bar Graph	No
	Object Kind	Kind of the object	IDDYNAMIC	No
	Object Name	Name of the object	Prot Bar Graph	No
	Object No	Number of the object	–	No
-	**Position**			
	X	Set the coordinate X of starting point.	0 to 1870	Yes
	Y	Set the coordinate Y of starting point.	0 to 935	Yes
-	**Data**			
	ST No	Set the station No.	Only when specifying the system Tag, specify the physical ST No. (101 to 440)	Yes
	Tag	Set the display Tag.	Up to 16 single-byte Tag characters Selectable from the dialog box	Yes
-	**View**			
	Direction	Set the direction of graph.	Up, Down, Left, Right	Yes
	Fore Color	Set the line color.	Select it from the color dialog.	Yes
	For Type	Set the line color type.	None, Blink, Reverse, Reverse + Blink	Yes
	Back Color	Set the background color.	Select it from the color dialog.	Yes
	Erase Color	Set the erasing color.	Select it from the color dialog.	Yes
	Fill Pattern	Set a fill pattern.	Select it from the pattern dialog.	Yes
	Graph Count	Set the number of bars to display.	1 to 512	Yes
	Graph Spacing	Set the spacing between bars.	1 to 1024 (b in the figure below)	Yes
	Max Range	Set the maximum range.	1 to 935 (d in the figure below)	Yes
	Graph Width	Set the bar width.	1 to 1870 (c in the figure below)	Yes

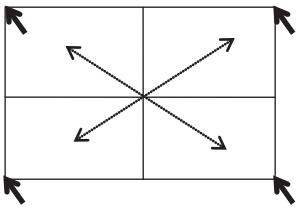
Note 1: Specify the spacing and bar quantity so that all bars can be shown within the screen.



(8) Batch trend: Displays specified range of values with batch trend graph.



[The mouse can be moved to four directions as shown in the figure below]



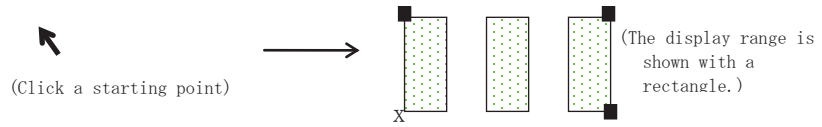
(The rectangle with display range X, Y is shown.)

-	**Other**		**Description**	**Change**
	Object Type	Type of the object	Batch Trend	No
	Object Kind	Kind of the object	IDDYNAMIC	No
	Object Name	Name of the object	Batch Trend	No
	Object No	Number of the object	–	No
-	**Position**			
	X	Set the coordinate X of starting point.	0 to 1870	Yes
	Y	Set the coordinate Y of starting point.	0 to 935	Yes
-	**Data**			
	Group	Set the group number.	1 to 256	Yes
-	**View**			
	Range X	Set the display range X.	1 to 1870	Yes
	Range Y	Set the display range Y.	1 to 935	Yes

Note 1: In MICREX VieW, up to eight groups for the batch trend can be created per panel.
The groups over nine cannot be defined.
(Define the group in the trend record definition.)

Note 2: In the plant panel, up to eight tags can be shown for trend and batch trend graphs.

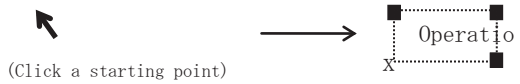
(9) Prot limiter bar graph: This prot bar graph has upper and lower limits.



-	**Other**		**Description**	**Change**
	Object Type	Type of the object	Prot Limiter Bar Graph	No
	Object Kind	Kind of the object	IDDYNAMIC	No
	Object Name	Name of the object	Prot Limiter Bar Graph	No
	Object No	Number of the object	–	No
-	**Position**			
	X	Set the coordinate X of starting point.	0 to 1870	Yes
	Y	Set the coordinate Y of starting point.	0 to 935	Yes
-	**Data**			
	Tag	Set the display Tag.	Up to 16 single-byte Tag characters Selectable from the dialog box	Yes
-	**View**			
	Direction	Set the direction of graph.	Up, Down, Left, Right	Yes
	Fore Color	Set the line color.	Select it from the color dialog.	Yes
	For Type	Set the line color type.	None, Blink, Reverse, Reverse + Blink	Yes
	Back Color	Set the background color.	Select it from the color dialog.	Yes
	Erase Color	Set the erasing color.	Select it from the color dialog.	Yes
	Fill Pattern	Set a fill pattern.	Select it from the pattern dialog.	Yes
	Graph Count	Set the number of bars to display.	1 to 999	Yes
	Graph Spacing	Set the spacing between bars.	1 to 1024	Yes
	Max Range	Set the maximum range.	1 to 935	Yes
	Max Width	Set the display width.	1 to 1870	Yes
-	**Limit Condition**			
	Use Limit Color	Set the upper or lower limit error color.	None, Tag setting, Direct	Yes
	Limit color ST No (Use Limit color for Tag setting)	Set the limit error color Tag station No.	Only when specifying the system Tag, specify the physical ST No. (101 to 440)	Yes
	Limit Color Tag (Use Limit Color for Tag setting)	Set upper or lower limit color Tag.	Up to 16 single-byte Tag characters Selectable from the dialog box	Yes
	Upper Lim Color (Use Limit Color for Direct)	Set the upper limit color.	Select it from the color dialog.	Yes
	Upper Lim Type (Use Limit Color	Set the upper limit color type.	None, Blink, Reverse, Reverse + Blink	Yes

	for Direct)			
	Upper Lim Back Color (Use Limit Color for Direct)	Set the upper limit background color.	Select it from the color dialog.	Yes
	Lower Lim Color (Use Limit Color for Direct)	Set the lower limit color.	Select it from the color dialog.	Yes
	Lower Lim Type (Use Limit Color for Direct)	Set the lower limit color type.	None, Blink, Reverse, Reverse + Blink	Yes
	Lower Lim Back Color (Use Limit Color for Direct)	Set the lower limit background color.	Select it from the color dialog.	Yes
	Limit Definition	Set whether to define upper and lower limits.	Yes, No	Yes
	Upper Limit Type (Limit Definition for Yes)	Set the upper limit data.	Tag setting, Direct	Yes
	Upper Limit ST No (Upper Limit Type for Tag setting)	Set the upper limit Tag station No.	Only when specifying the system Tag, specify the physical ST No. (101 to 440)	Yes
	Upper Limit Tag (Upper Limit Type for Tag setting)	Set upper limit Tag.	Up to 16 single-byte Tag characters Selectable from the dialog box	Yes
	Upper Limit Data (Upper Limit Type for Direct)	Set upper limit data.	Numeric	Yes
	Lower Limit Type (Limit Definition for Yes)	Set the lower limit data.	Tag setting, Direct	Yes
	Lower Limit ST No (Lower Limit Type for Tag setting)	Set the lower limit Tag station No.	Only when specifying the system Tag, specify the physical ST No. (101 to 440)	Yes
	Lower Limit Tag (Lower Limit Type for Tag setting)	Set lower limit Tag.	Up to 16 single-byte Tag characters Selectable from the dialog box	Yes
	Lower Limit Data (Lower Limit Type for Direct)	Set lower limit data.	Numeric	Yes

(10) Exchange text: Used to exchange the text.

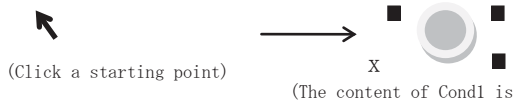


(The content of Cond1 is displayed.)

-	**Other**		**Description**	**Change**
	Object Type	Type of the object	Text Exchange	No
	Object Kind	Kind of the object	IDDYNAMIC	No
	Object Name	Name of the object	Exchange Text	No
	Object No	Number of the object	–	No
-	**Position**			
	X	Set the coordinate X of starting point.	0 to 1870	Yes
	Y	Set the coordinate Y of starting point.	0 to 935	Yes
-	**Data**			
	Cond ST No	Set the exchange condition Tag station No.	Only when specifying the system Tag, specify the physical ST No. (101 to 440)	Yes
	Condition Tag	Set the exchange condition Tag.	Number of Tag characters: Up to 16 single-byte characters Selectable from the dialog box	Yes
-	**View**			
	Direction	Set the direction to display.	Horizontal, Vertical	Yes
	Text Size	Set the character size.	8, 16, 24, 32	Yes
	Cond*n* *(n=1 to 8)*			
	Cond***n*** Text	Set the Cond***n*** text.	Double-byte four characters	Yes
	Cond***n*** Mask	Set the Cond*n* mask.	Hexadecimal four characters **(Refer to section 2.3.3, "(18) Selecting BitPattern")**	Yes
	Cond***n*** Data	Set the Cond*n* data.	Hexadecimal four characters **(Refer to section 2.3.3, "(18) Selecting BitPattern")**	Yes
	Cond***n*** Fore Color	Set the Cond*n* display color.	Select it from the color dialog.	Yes
	Cond***n*** Fore Type	Set the Cond***n*** display color type.	None, Blink, Reverse, Reverse + Blink	Yes
	Cond***n*** Back Color	Set the Cond*n* background color.	Select it from the color dialog.	Yes
-	**Cond Else**			
	Else Text	Set text for data deviating from the exchange condition.	Double-byte four characters	Yes
	Else Fore Color	Set the display color for data deviating from the exchange condition.	Select it from the color dialog.	Yes
	Else Fore Type	Set the display color type for data deviating from the exchange condition.	None, Blink, Reverse, Reverse + Blink	Yes
	Else Back Color	Set the background color for data deviating from the exchange condition.	Select it from the color dialog.	Yes
	Erase Color	Set the erasing color for data deviating from the exchange condition.	Select it from the color dialog.	Yes

To define exchanges of nine patterns or more, define Else Fore Color and Else Back Color with a Transparency option.

(11) Exchange segment: Used to exchange the segment.

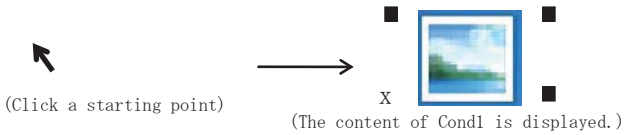


-	**Other**		**Description**	**Change**
	Object Type	Type of the object	Segment Exchange	No
	Object Kind	Kind of the object	IDDYNAMIC	No
	Object Name	Name of the object	Exchange	No
	Object No	Number of the object	–	No
-	**Position**			
	X	Set the coordinate X of starting point.	0 to 1870	Yes
	Y	Set the coordinate Y of starting point.	0 to 935	Yes
-	**Data**			
	Cond ST No	Set the exchange condition Tag station No.	Only when specifying the system Tag, specify the physical ST No. (101 to 440)	Yes
	Condition Tag	Set the exchange condition Tag.	Number of Tag characters: Up to 16 single-byte characters Selectable from the dialog box	Yes
-	**View**			
	Segment Type	Set the segment type.	Segment	No
-	**Cond*n* *(n=1 to 8)***			
	Cond***n*** Segment	Set the Cond*n* segment.	Select it from the segment dialog.	Yes
	Cond***n*** Mask	Set the Cond*n* mask.	Hexadecimal four characters **(Refer to section 2.3.3, "(18) Selecting BitPattern")**	Yes
	Cond***n*** Data	Set the Cond*n* data.	Hexadecimal four characters **(Refer to section 2.3.3, "(18) Selecting BitPattern")**	Yes
	Cond***n*** Fore Color	Set the Cond*n* display color.	Select it from the color dialog.	Yes
	Cond***n*** Fore Type	Set the Cond*n* display color type.	None, Blink, Reverse, Reverse + Blink	Yes
	Cond***n*** Back Color	Set the Cond*n* background color.	Select it from the color dialog.	Yes
-	**Cond Else**			
	Else Segment	Set the segment for data deviating from the exchange condition.	Select it from the segment dialog.	Yes
	Else Fore Color	Set the display color for data deviating from the exchange condition.	Select it from the color dialog.	Yes
	Else Fore Type	Set the display color type for data deviating from the exchange condition.	None, Blink, Reverse, Reverse + Blink	Yes
	Else Back Color	Set the background color for data deviating from the exchange condition.	Select it from the color dialog.	Yes
	Erase Color	Set the erasing color for data	Select it from the color	Yes

		deviating from the exchange condition.	dialog.	

To define exchanges of nine patterns or more, define Else Fore Color and Else Back Color with a Transparency option.

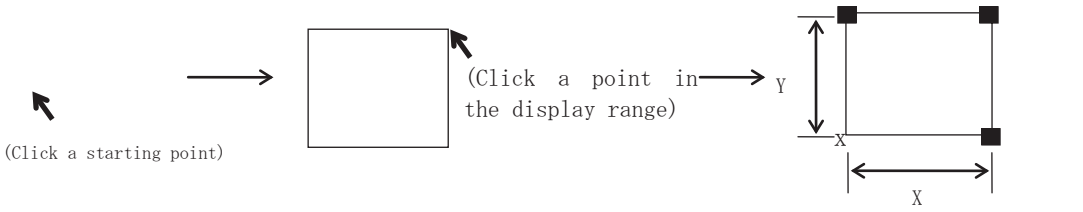
(12) Exchange bitmap: Used to exchange the bitmap.



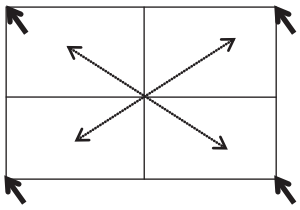
-	**Other**		**Description**	**Change**
	Object Type	Type of the object	Bitmap Exchange	No
	Object Kind	Kind of the object	IDDYNAMIC	No
	Object Name	Name of the object	Exchange	No
	Object No	Number of the object	–	No
-	**Position**			
	X	Set the coordinate X of starting point.	0 to 1870	Yes
	Y	Set the coordinate Y of starting point.	0 to 935	Yes
-	**Data**			
	Cond ST No	Set the exchange condition Tag station No.	Only when specifying the system Tag, specify the physical ST No. (101 to 440)	Yes
	Condition Tag	Set the exchange condition Tag.	Number of Tag characters: Up to 16 single-byte characters Selectable from the dialog box	Yes
-	**BMP*n*** ***(n=1 to 8)***			
	Bitmap***n***	Set the Cond*n* bitmap.	Select it from the bitmap dialog.	Yes
	Bitmap***n*** Mask	Set the Cond*n* mask.	Hexadecimal four characters **(Refer to section 2.3.3, "(18) Selecting BitPattern")**	Yes
	Bitmap***n*** Data	Set the Cond*n* data.	Hexadecimal four characters **(Refer to section 2.3.3, "(18) Selecting BitPattern")**	Yes
-	**Cond Else**			
	Else Bitmap	Set the bitmap for data deviating from the exchange condition.	Select it from the bitmap dialog.	Yes
	Erase Color	Set the erasing color for data deviating from the exchange condition.	Select it from the color dialog.	Yes

To define exchanges of nine patterns or more, define Else Fore Color and Else Back Color with a Transparency option.

(13) Move pattern: Move specified range of patterns by specifying values.



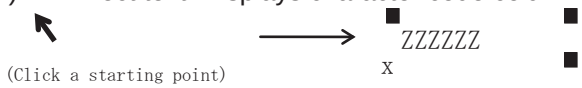
[The mouse can be moved to four directions as shown in the figure below]



(The rectangle with display range X, Y is shown.)

-	**Other**		**Description**	**Change**
	Object Type	Type of the object	Pattern Motion	No
	Object Kind	Kind of the object	IDDYNAMIC	No
	Object Name	Name of the object	Move Pattern	No
	Object No	Number of the object	–	No
-	**Position**			
	StartX	Set the coordinate X of starting point.	0 to 1870	Yes
	StartY	Set the coordinate Y of starting point.	0 to 935	Yes
-	**Data**			
	ST No	Set the station No.	Only when specifying the system Tag, specify the physical ST No. (101 to 440)	Yes
	Tag	Set the display Tag.	Up to 16 single-byte Tag characters Selectable from the dialog box	Yes
-	**View**			
	Range X	Set the display range X.	1 to 1870	Yes
	Range Y	Set the display range Y.	1 to 935	Yes
	Full X	Set the full value X.	1 to 32767	Yes
	Full Y	Set the full value Y.	1 to 32767	Yes
	Pattern Type	Set the pattern type.	Other than pattern size segment: 0 to 79 Pattern size segment: Segment name	Yes
	Pattern Size	Set the pattern size.	8*8, 16*8, 16*16, 24*12, 24*24, 32*16, 32*32, segment	Yes
	Fore Color	Set the display color.	Select it from the color dialog.	Yes
	Fore Type	Set the display color type.	None, Blink, Reverse, Reverse + Blink	Yes
	Back Color	Set the background color.	Select it from the color dialog.	Yes
	Erase Color	Set the erasing color.	Select it from the color dialog.	Yes

(14) Direct text: Displays character code column from the specified place.

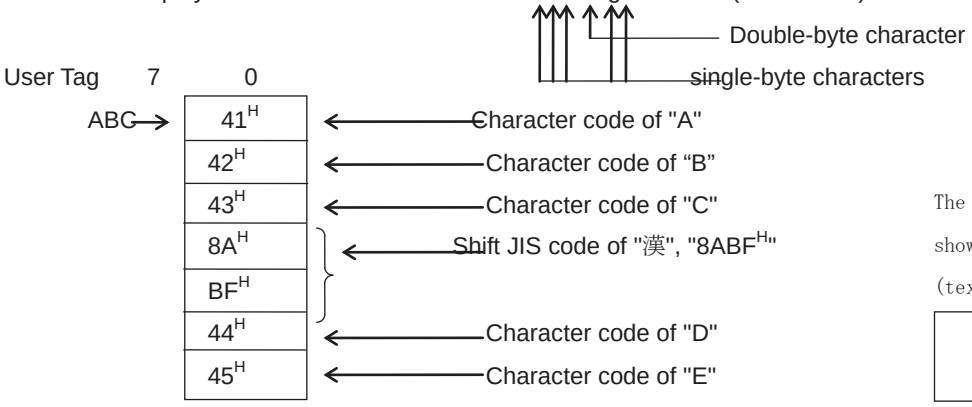


('Z' is shown as many as the number of texts)

-	**Other**		**Description**	**Change**
	Object Type	Type of the object	Direct Text	No
	Object Kind	Kind of the object	IDDYNAMIC	No
	Object Name	Name of the object	Text	No
	Object No	Number of the object	–	No
-	**Position**			
	X	Set the coordinate X of starting point.	0 to 1870	Yes
	Y	Set the coordinate Y of starting point.	0 to 935	Yes
-	**Data**			
	ST No	Set the station No.	Only when specifying the system Tag, specify the physical ST No. (101 to 440)	Yes
	Tag	Set the display Tag.	Up to 16 single-byte Tag characters Selectable from the dialog box	Yes
-	**View**			
	Size	Set the character size.	8*8, 16*8, 16*16, 24*12, 24*24, 32*16	Yes
	Magnifier	Set the zoom ratio.	Standard, 1 to 8 times	Yes
	Direction	Set the text direction.	Horizontal, Vertical	Yes
	Color Variation	Set the color change extension.	Fixed, Module, Option, Option 2 - Use the option and option 2 when changing the display color and background color. **(See "2.3.3 (17) Color option and option 2")**	Yes
	Fore Color	Set the character color.	Select it from the color dialog.	Yes
	Fore Type	Set the character color type.	None, Blink, Reverse, Reverse + Blink	Yes
	Back Color	Set the background color.	Select it from the color dialog.	Yes
-	**Num Processing**			
	Digit Number	Set the number of display characters.	1 to 255	Yes

Specifying text data

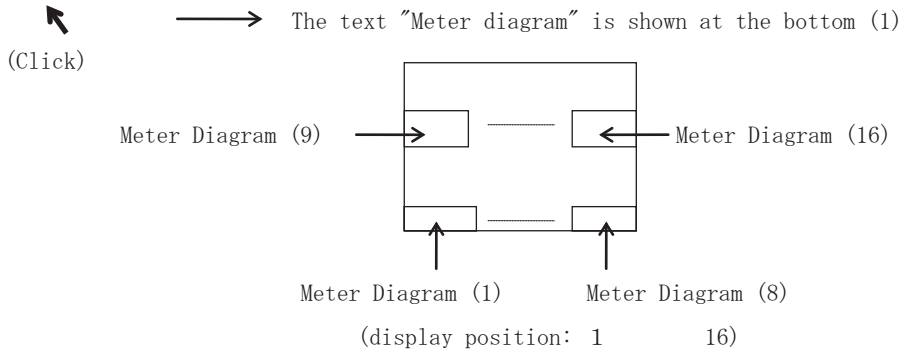
When the display data is consecutive as shown in the figure below (ABC 漢 DE):



The data input is shown as follows:
(text direction: horizontal)

ABC 漢 DE

(15) Meter diagram: Displays a meter diagram at the specified position.



-	**Other**		**Description**	**Change**
	Object Type	Type of the object	Gauge Window	No
	Object Kind	Kind of the object	IDDYNAMIC	No
	Object Name	Name of the object	Meter Diagram	No
	Object No	Number of the object	–	No
-	**Position**			
	X	Set the coordinate X of starting point.	0 to 1870	No
	Y	Set the coordinate Y of starting point.	0 to 935	No
-	**Data**			
	Tag	Set the display Tag.	Up to 16 single-byte Tag characters Selectable from the dialog box	Yes
-	**View**			
	Disp Position	Set the display position.	1 to 8 from the left at lower column 9 to 16 from the left at upper column	Yes
	Loop Link	Set the loop link.	True: Link to loop panel False: Not-treated	Yes

- Display position: 1 to 16

Coordinates of the display position: 1=X0, Y808, 2=X233, Y808, 3=X466, Y808, 4=X699, Y808, 5=X932, Y808, 6=X1165, Y808,

7=X1398, Y808, 8=X1631, Y808, 9=X0, Y392, 10=X233, Y392, 11=X466, Y392, 12=X699, Y392, 13=X932, Y392, 14=X1165, Y392, 15=X1398, Y392, 16=X1631, Y392.

- Loop link: Touch the meter diagram displayed online to specify whether to link the meter diagram to the loop panel.

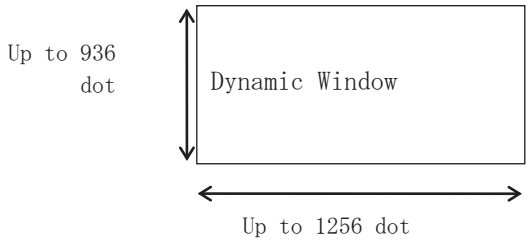
Note 1: In order to prevent duplicate definition, if the number already used is entered into Disp Position, an error occurs.

Note 2: Up to 16 trend, dynamic window, and meter diagram can be shown in a panel.

(16) Dynamic window: Displays the dynamic window at the specified position.

-	**Other**		**Description**	**Change**
	Object Type	Type of the object	Dynamic Window	No
	Object Kind	Kind of the object	IDDYNAMIC	No
	Object Name	Name of the object	Dynamic Wizard	No
	Object No	Number of the object	–	No
-	**View**			
	Disp X	Set initial coordinate X.	0 to 1870	Yes
	Disp Y	Set initial coordinate Y.	0 to 935	Yes
	Link Window	Set the dynamic window name.	Select it from the dialog.	Yes

Note 1: The dynamic window size should be up to 1256x936 dot as shown below:



Note 2: Up to 16 trend, dynamic window, and meter diagram can be shown in a panel.

(17) Color option and option 2

Color Extension: Specify the display color for numeric values.

- Fixed: Specify this to have a fixed display color.
- Module: Specify this to change the display color according to the module alarm when a module Tag is specified.
- Arbitrary: Specify this to change the display color arbitrarily.

[When Arbitrary is selected, a definition item as shown in the table below is added to the property so that a color can be set in the same manner as the Change Color.

-	**Data**		**Description**	**Change**
	Color Ext. ST No	Set the color change Tag station No.	Only when specifying the system Tag, specify the physical ST No. (101 to 440)	Yes
	Color Ext. Tag	Set the color change Tag.	Number of Tag characters: Up to 16 single-byte characters Selectable from the dialog box	Yes
-	**Color*n* *(n=1 to 8)***			
	Color***n*** Mask	Set the Color*n* mask.	Hexadecimal four characters **(Refer to section 2.3.3, "(18) Selecting BitPattern")**	Yes
	Color***n*** Mask	Set the Cond*n* data.	Hexadecimal four characters **(Refer to section 2.3.3, "(18) Selecting BitPattern")**	Yes
	Color***n*** Fore Color	Set the Color*n* display color.	Select it from the color dialog.	Yes
	Color***n*** Fore Type	Set the Color*n* display color type.	None, Blink, Reverse, Reverse + Blink	Yes
-	**Else Color**			
	Else Fore Color	Set the display color for data deviating from the exchange condition.	Select it from the color dialog.	Yes
	Else Fore Type	Set the display color type for data deviating from the exchange condition.	None, Blink, Reverse, Reverse + Blink	Yes

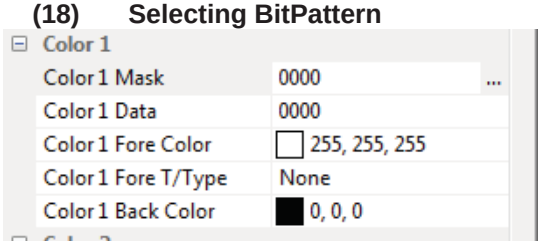
- Arbitrary 2: Specify this to change the display color and background color arbitrarily.

[When Arbitrary is selected, a definition item as shown in the table below is added to the property so that a color can be set in the same manner as the Change Color.

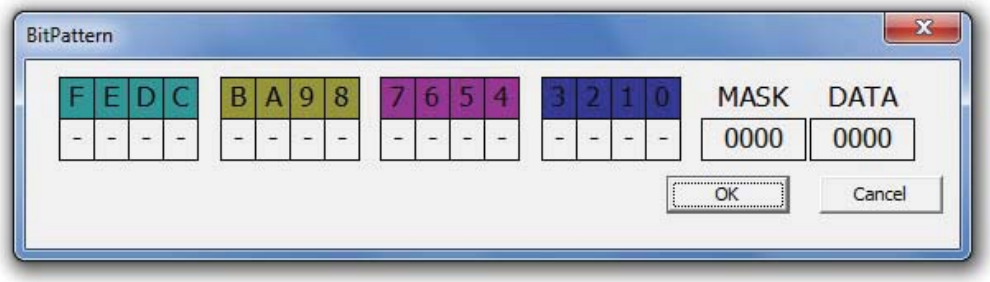
-	**Data**		**Description**	**Change**
	Color Ext. ST No	Set the color change Tag station No.	Only when specifying the system Tag, specify the physical ST No. (101 to 440)	Yes
	Color Ext. Tag	Set the color change Tag.	Number of Tag characters: Up to 16 single-byte characters Selectable from the dialog box	Yes
-	**Color*n* *(n=1 to 8)***			
	Color***n*** Mask	Set the Color*n* mask.	Hexadecimal four characters **(Refer to section 2.3.3, "(18) Selecting BitPattern")**	Yes
	Color***n*** Mask	Set the Color*n* data.	Hexadecimal four characters **(Refer to section 2.3.3, "(18) Selecting BitPattern")**	Yes
	Color***n*** Fore Color	Set the Color*n* display color.	Select it from the color dialog.	Yes
	Color***n*** Fore Type	Set the Color*n* display color type.	None, Blink, Reverse, Reverse + Blink	Yes
	Color***n*** Back Color	Set the Color*n* background color.	Select it from the color dialog.	Yes
-	**Else Color**			
	Else Fore Color	Set the display color for data deviating from the exchange condition.	Select it from the color dialog.	Yes
	Else Fore Type	Set the display color type for data deviating from the exchange condition.	None, Blink, Reverse, Reverse + Blink	Yes

	Else Back Color	Set the background color for data deviating from the exchange condition.	Select it from the color dialog.	Yes

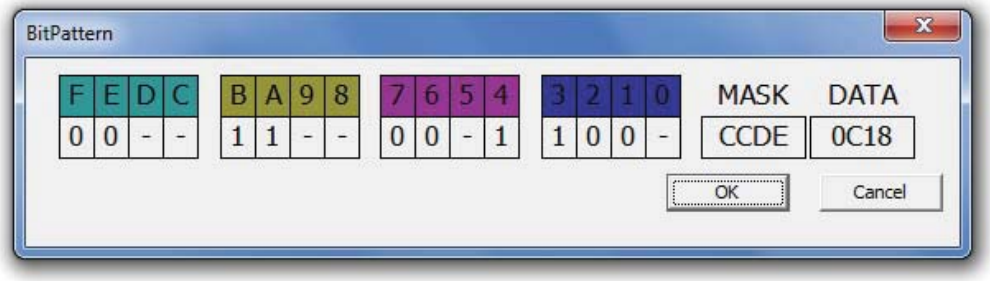
(18) Selecting BitPattern



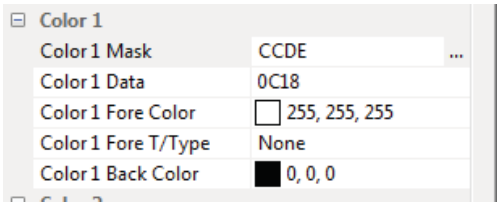
Click the button at the right of Mask, Data,



Click "-" part under each label to create a bit pattern.



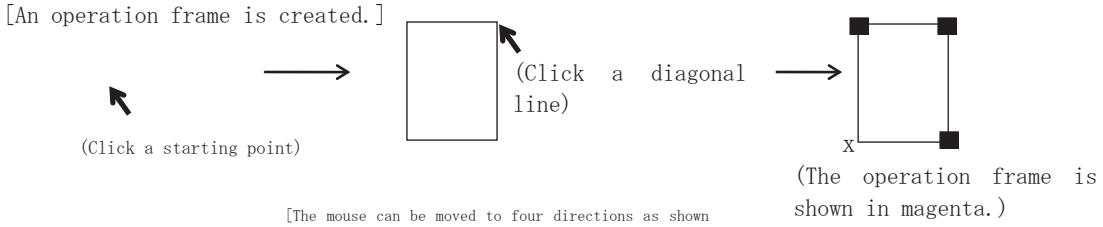
Click the OK button.



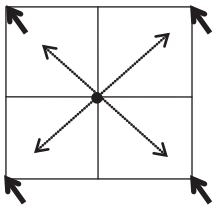
The bit pattern created is put in the property.

2.3.4 How to Operate Operation Object

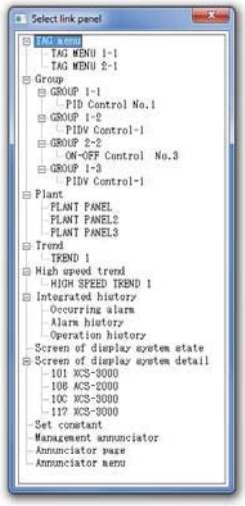
(1) Link screens (single action): Link a screen to another one.



[The mouse can be moved to four directions as shown in the figure below]

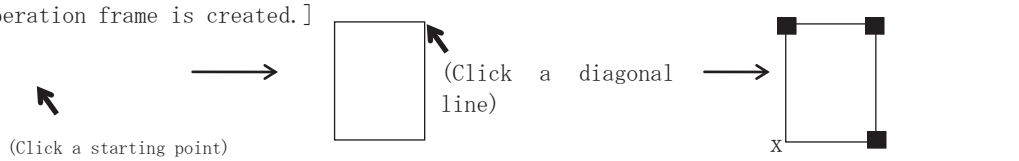


-	**Other**		**Description**	**Change**
	Object Type	Type of the object	Single Action Link	No
	Object Kind	Kind of the object	IDSET	No
	Object Name	Name of the object	Link	No
	Object No	Number of the object	–	No
-	**Position**			
	StartX	Set the coordinate X of starting point of the operation frame.	0 to 1870	Yes
	StartY	Set the coordinate Y of starting point of the operation frame.	0 to 935	Yes
	Width	Set the width of the operation frame.	1 to 1870	Yes
	Height	Set the height of the operation frame.	1 to 935	Yes
-	**Link**			
	LinkID LinkID2	Set the destination screen for the link.	Though there are two places to set, the link screen selection dialog box below is shown by clicking either one. The definition once will complete the setup.	Yes

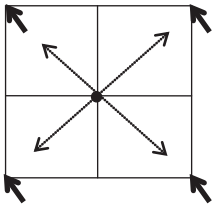
(2) Link screens (two actions): Link a screen to another one.

[An operation frame is created.]

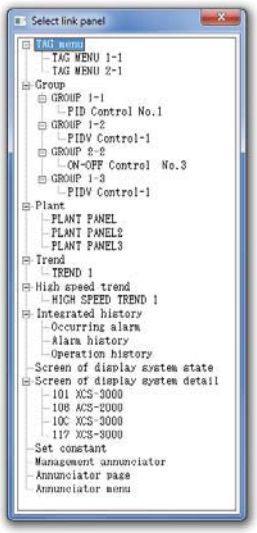


(The operation frame is shown in magenta.)

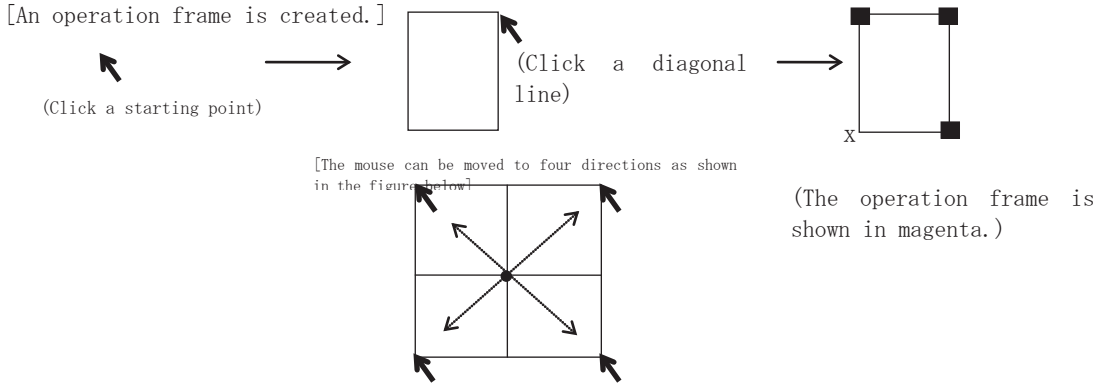
[The mouse can be moved to four directions as shown in the figure below]



-	**Other**		**Description**	**Change**
	Object Type	Type of the object	2 Action Link	No
	Object Kind	Kind of the object	IDSET	No
	Object Name	Name of the object	Link	No
	Object No	Number of the object	–	No
-	**Position**			
	StartX	Set the coordinate X of starting point of the operation frame.	0 to 1870	Yes
	StartY	Set the coordinate Y of starting point of the operation frame.	0 to 935	Yes
	Width	Set the width of the operation frame.	1 to 1870	Yes
	Height	Set the height of the operation frame.	1 to 935	Yes
-	**G*n* Link *(n=1 to 8)***			

	G***n*** LinkID G***n*** LinkID2	Set the destination screen for the link.	Though there are two places to set, the link screen selection dialog box below is shown by clicking either one. The definition once will complete the setup. 	Yes

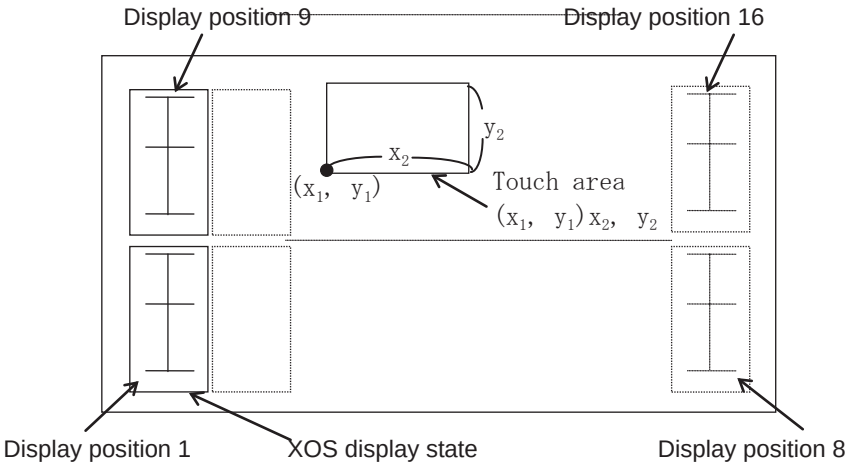
(3) Meter diagram window: Displays the meter diagram window.



-	**Other**		**Description**	**Change**
	Object Type	Type of the object	Gauge Window Link	No
	Object Kind	Kind of the object	IDSET	No
	Object Name	Name of the object	Meter Diagram Window	No
	Object No	Number of the object	–	No
-	**Position**			
	StartX	Set the coordinate X of starting point of the operation frame.	0 to 1870	Yes
	StartY	Set the coordinate Y of starting point of the operation frame.	0 to 935	Yes
	Width	Set the width of the operation frame.	1 to 1870	Yes
	Height	Set the height of the operation frame.	1 to 935	Yes
-	**View**			
	Disp Position	Set the display position.	1 to 8 from the left at lower column 9 to 16 from the left at upper column	Yes
	Loop Link	Set the loop link.	True: Link to loop panel False: Not-treated	Yes
-	**Data**			

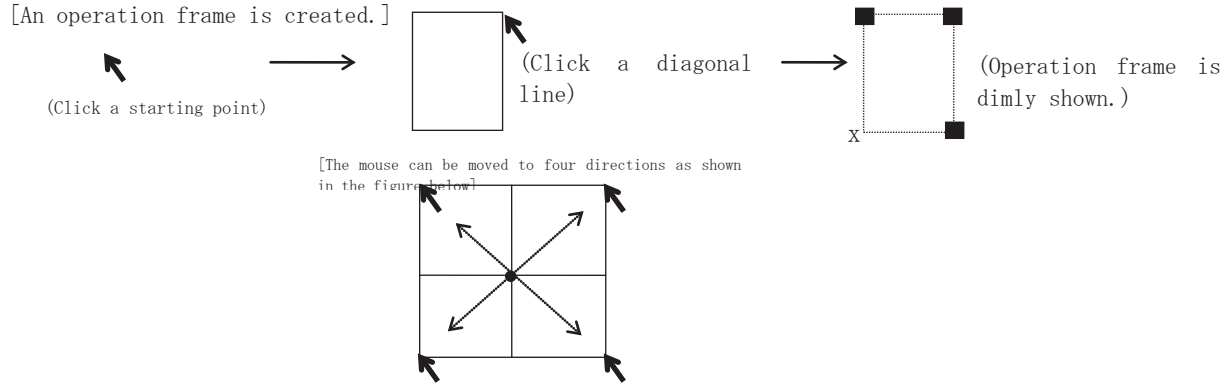
	Tag	Set the display Tag.	Up to 16 single-byte Tag characters Selectable from the dialog box	Yes

- TAG NO. can be defined only for the module Tag. Also, ST NO. (station number) cannot be defined.
- Display position: 1 to 16

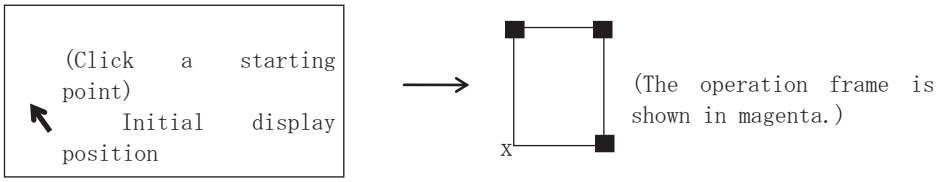


For the display position of the meter diagram, positions 1 to 8 at the lower column and positions 9 to 16 at the upper column can be defined.

(4) Dynamic window: Displays the dynamic window.

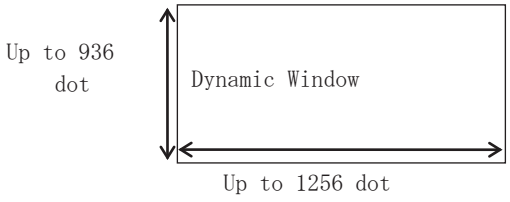


[Click the initial position for dynamic window.]



-	**Other**		**Description**	**Change**
	Object Type	Type of the object	Dynamic Window Link	No
	Object Kind	Kind of the object	IDSET	No
	Object Name	Name of the object	Dynamic Window	No
	Object No	Number of the object	–	No
-	**Position**			
	StartX	Set the coordinate X of starting point of the operation frame.	0 to 1870	Yes
	StartY	Set the coordinate Y of starting point of the operation frame.	0 to 935	Yes
	Width	Set the width of the operation frame.	1 to 1870	Yes
	Height	Set the height of the operation frame.	1 to 935	Yes
-	**View**			
	Disp X	Set initial coordinate X.	0 to 1870	Yes
	Disp Y	Set initial coordinate Y.	0 to 935	Yes
	Link Window	Set the dynamic window name.	Select it from the dialog.	Yes

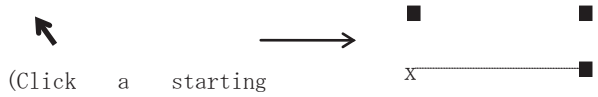
Note1: The dynamic window size should be up to 1256x936 dot as shown below:



Note2: The type "Normal" indicates dynamic window to be linked from the base panel.

"Computer" is the dynamic window to be started by the computer link. (The configuration and specification of each computer need to be confirmed.)

(5) Full screen entry 1: Set numeric value under locked condition in the full screen.



-	**Other**		**Description**	**Change**
	Object Type	Type of the object	Full Screen Input 1	No
	Object Kind	Kind of the object	IDSET	No
	Object Name	Name of the object	Full Screen Entry 1	No
	Object No	Number of the object	–	No
-	**Position**			
	X	Set the coordinate X of starting point.	0 to 1870	Yes
	Y	Set the coordinate Y of starting point.	0 to 935	Yes
-	**Output**			
	Set Tag ST No	Set the entry Tag station No.	Only when specifying the system Tag, specify the physical ST No. (101 to 440)	Yes
	Set Tag	Set the entry Tag.	Specify the Tag to which the data is transmitted. Up to 16 single-byte Tag characters Selectable from the dialog box	Yes
-	**Condition**			
	Action Times	Set the number of actions for the touch operation.	1: "Run" transmits data 2: "Confirm" and then "Run" transmits data ("Confirm" is optional)	Yes
	IL Tag1 ST No	Interlock Set the interlock Tag 1 station No.	Only when specifying the system Tag, specify the physical ST No. (101 to 440)	Yes
	IL Tag1	Set the interlock Tag.	Two interlock Tags are available for use in each entry data. When two Tags are used, set both Tags to ON (or OFF) to set the interlock. ON (masking available): (Interlock Tag AND Interlock Mask) == Interlock Mask OFF (zero setting available): (Interlock Tag AND Interlock Mask) == zero Up to 16 single-byte Tag characters Selectable from the dialog box	Yes
	IL Mask1	Set the interlock Tag1 Mask condition.	Hexadecimal four characters **(Refer to section 2.3.4, "(11) Selecting BitPattern")**	Yes
	IL Tag2 ST No	Interlock Set the interlock Tag 2 station No.	Only when specifying the system Tag, specify the physical ST No. (101 to 440)	Yes
	IL Tag2	Set the interlock Tag.	Up to 16 single-byte Tag characters Selectable from the dialog box	Yes
	IL Mask2	Set the interlock Tag2 mask condition.	Hexadecimal four characters **(Refer to section 2.3.4, "(11) Selecting BitPattern")**	Yes
	IL Mask Behavior	Set the interlock condition.	Operable with the mask value Operable with zero	Yes
-	**Strobe**			
	Strobe	Set whether to use a strobe.	True (use) False (do not use)	Yes
	Strobe Tag ST No (When Strobe is set	Set the strobe Tag station No.	Only when specifying the system Tag,	Yes

	to True)		specify the physical ST No. (101 to 440)	
	Strobe Tag (When Strobe is set to True)	Set the entry Tag for the strobe Tag.	Up to 16 single-byte Tag characters Selectable from the dialog box	Yes
	Strobe Mask (When Strobe is set to True)	Set the mask for the strobe Tag.	Hexadecimal four characters **(Refer to section 2.3.4, "(11) Selecting BitPattern")**	Yes
	Strobe Data (When Strobe is set to True)	Set the data for the strobe Tag.	Hexadecimal four characters **(Refer to section 2.3.4, "(11) Selecting BitPattern")**	Yes
-	**View**			
	FB Tag ST No	Set the source Tag station No.	Only when specifying the system Tag, specify the physical ST No. (101 to 440)	Yes
	FB Tag	Set the source Tag.	The present value is displayed in the entry field based on the source Tag. Up to 16 single-byte Tag characters Selectable from the dialog box	Yes
	Digit Number	Set the number of display digits.	1 to 16 (decimal point is also counted)	Yes
	Magnifier	Set the zoom ratio.	Standard, two to eight times	Yes
	Decimal Place	Set the place of decimal point. Specifiable only for decimal numbers.	-1 to 15	Yes
	Size	Set the character size.	8*8, 16*8, 16*16, 24*12, 24*24, 32*16	Yes
	Last SV	Set whether to display previous value.	True (Yes), False (No)	Yes
-	**Data**			
	Data Type	Set the display type.	Decimal two word, hexadecimal two word, decimal one word, and hexadecimal one word Text	Yes

Note: Difference between full screen entries 1 and 2

The troubleshooting procedures may vary when transmitting data online.

* Full Screen Entry 1

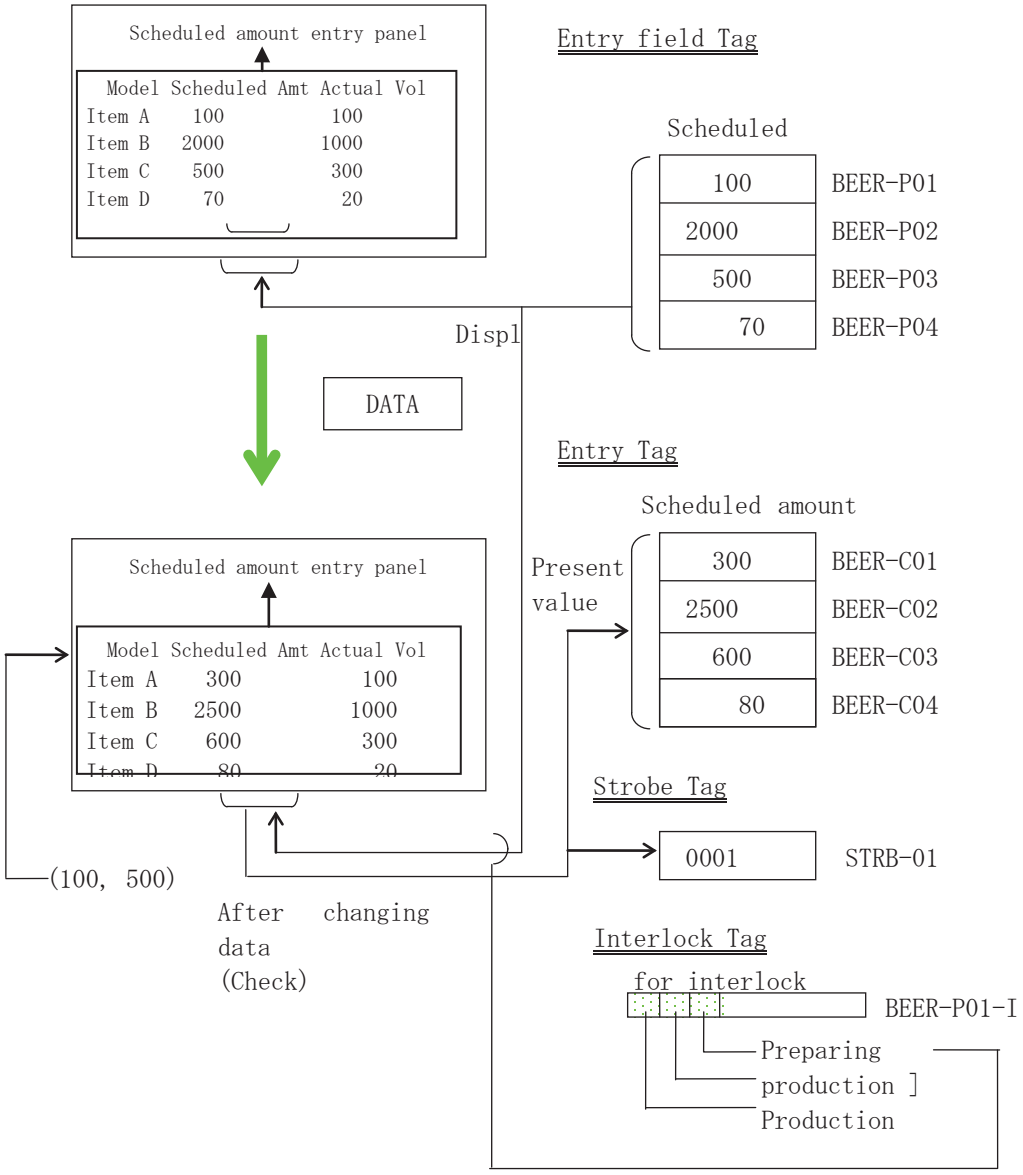
  The entry fields are still shown even for the data sent successfully during data transmission error, thereby waiting for the retransmission.

* Full Screen Entry 2

  The entry fields are cleared for the data sent successfully during data transmission error but the entry fields with data error is still shown.

Full Screen Entry 1 (Example)

Online display & entry example



① Interlock

Two interlock Tags are available for use for each entry data.

When two Tags are used, set both Tags to ON (or OFF) to set the interlock.

ON: (ILOC Tag AND ILOC Mask) == ILOC Mask

OFF: (ILOC Tag AND ILOC Mask) == zero

(6) Full screen entry 2: Set numeric value under locked condition in the full screen.

-	**Other**		**Description**	**Change**
	Object Type	Type of the object	Full Screen Input2	No
	Object Kind	Kind of the object	IDSET	No
	Object Name	Name of the object	Full Screen Entry 2	No
	Object No	Number of the object	–	No
-	**Position**			
	X	Set the coordinate X of starting point.	0 to 1870	Yes
	Y	Set the coordinate Y of starting point.	0 to 935	Yes
-	**Output**			
	Set Tag ST No	Set the entry Tag station No.	Only when specifying the system Tag, specify the physical ST No. (101 to 440)	Yes
	Set Tag	Set the entry Tag.	Specify the Tag to which the data is transmitted. Up to 16 single-byte Tag characters Selectable from the dialog box	Yes
-	**Condition**			
	Action Times	Set the number of actions for the touch operation.	1: "Run" transmits data 2: "Confirm" and then "Run" transmits data ("Confirm" is optional)	Yes
	IL Tag1 ST No	Interlock Set the interlock Tag 1 station No.	Only when specifying the system Tag, specify the physical ST No. (101 to 440)	Yes
	IL Tag1	Set the interlock Tag.	Two interlock Tags are available for use in each entry data. When two Tags are used, set both Tags to ON (or OFF) to set the interlock. ON (masking available): (Interlock Tag AND Interlock Mask) == Interlock Mask OFF (zero setting available): (Interlock Tag AND Interlock Mask) == zero Up to 16 single-byte Tag characters Selectable from the dialog box	Yes
	IL Mask1	Set the interlock Tag1 Mask condition.	Hexadecimal four characters **(Refer to section 2.3.4, "(11) Selecting BitPattern")**	Yes
	IL Tag2 ST No	Interlock Set the interlock Tag 2 station No.	Only when specifying the system Tag, specify the physical ST No. (101 to 440)	Yes
	IL Tag2	Set the interlock Tag.	Up to 16 single-byte Tag characters Selectable from the dialog box	Yes
	IL Mask2	Set the interlock Tag2 mask condition.	Hexadecimal four characters **(Refer to section 2.3.4, "(11) Selecting BitPattern")**	Yes

	IL Mask Behavior	Set the interlock condition.	Operable with the mask value Operable with zero	Yes
-	**Strobe**			
	Strobe	Set whether to use a strobe.	True (use) False (do not use)	Yes
	Strobe Tag ST No (When Strobe is set to True)	Set the strobe Tag station No.	Only when specifying the system Tag, specify the physical ST No. (101 to 440)	Yes
	Strobe Tag (When Strobe is set to True)	Set the entry Tag for the strobe Tag.	Up to 16 single-byte Tag characters Selectable from the dialog box	Yes
	Strobe Mask (When Strobe is set to True)	Set the mask for the strobe Tag.	Hexadecimal four characters **(Refer to section 2.3.4, "(11) Selecting BitPattern")**	Yes
	Strobe Data (When Strobe is set to True)	Set the data for the strobe Tag.	Hexadecimal four characters **(Refer to section 2.3.4, "(11) Selecting BitPattern")**	Yes
-	**View**			
	FB Tag ST No	Set the source Tag station No.	Only when specifying the system Tag, specify the physical ST No. (101 to 440)	Yes
	FB Tag	Set the source Tag.	The present value is displayed in the entry field based on the source Tag. Up to 16 single-byte Tag characters Selectable from the dialog box	Yes
	Digit Number	Set the number of display digits.	1 to 16 (decimal point is also counted)	Yes
	Magnifier	Set the zoom ratio.	Standard, two to eight times	Yes
	Decimal Place	Set the place of decimal point. Specifiable only for decimal numbers.	-1 to 15	Yes
	Size	Set the character size.	8*8, 16*8, 16*16, 24*12, 24*24, 32*16	Yes
	Last SV	Set whether to display previous value.	True (Yes), False (No)	Yes
-	**Data**			
	Data Type	Set the display type.	Decimal two word, hexadecimal two word, decimal one word, and hexadecimal one word Text	Yes

Note: Difference between full screen entries 1 and 2

The troubleshooting procedures may vary when transmitting data online.

* Full screen entry 1

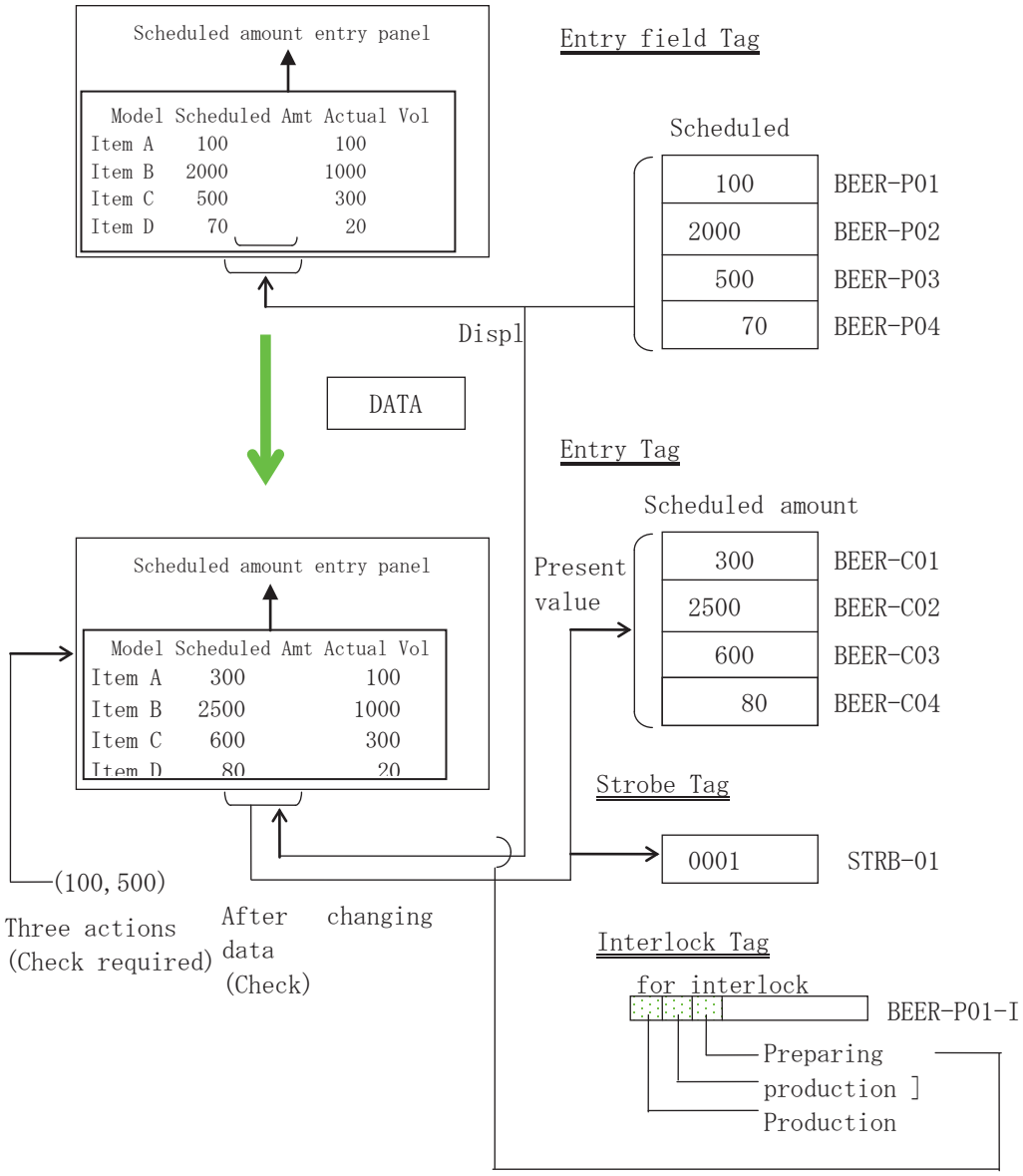
The entry fields are still shown even for the data sent successfully during data entry error, thereby waiting for the retransmission.

* Full screen entry 2

The entry fields are cleared for the data sent successfully during data transmission error but the entry fields with data error is still shown.

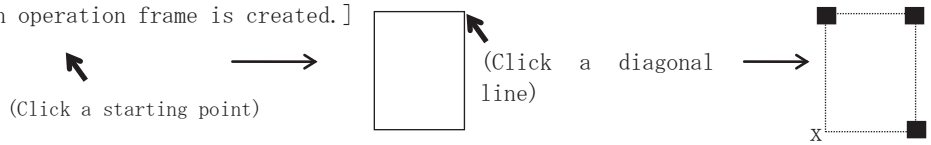
Full Screen Entry 2 (Example)

Online display & entry example

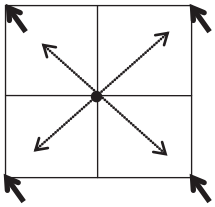


(7) Multiple touch point window: Set numeric value under locked condition in the key-in area.

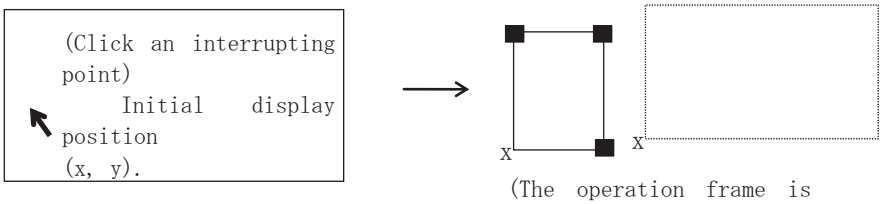
[An operation frame is created.]



[The mouse can be moved to four directions as shown in the figure



[Click the initial position on the data entry window.]



(The operation frame is shown in magenta.)

When the operation frame is selected, the data entry is enclosed with dimmed rectangle. (The size is fixed at 272x100)

The mouse cannot be

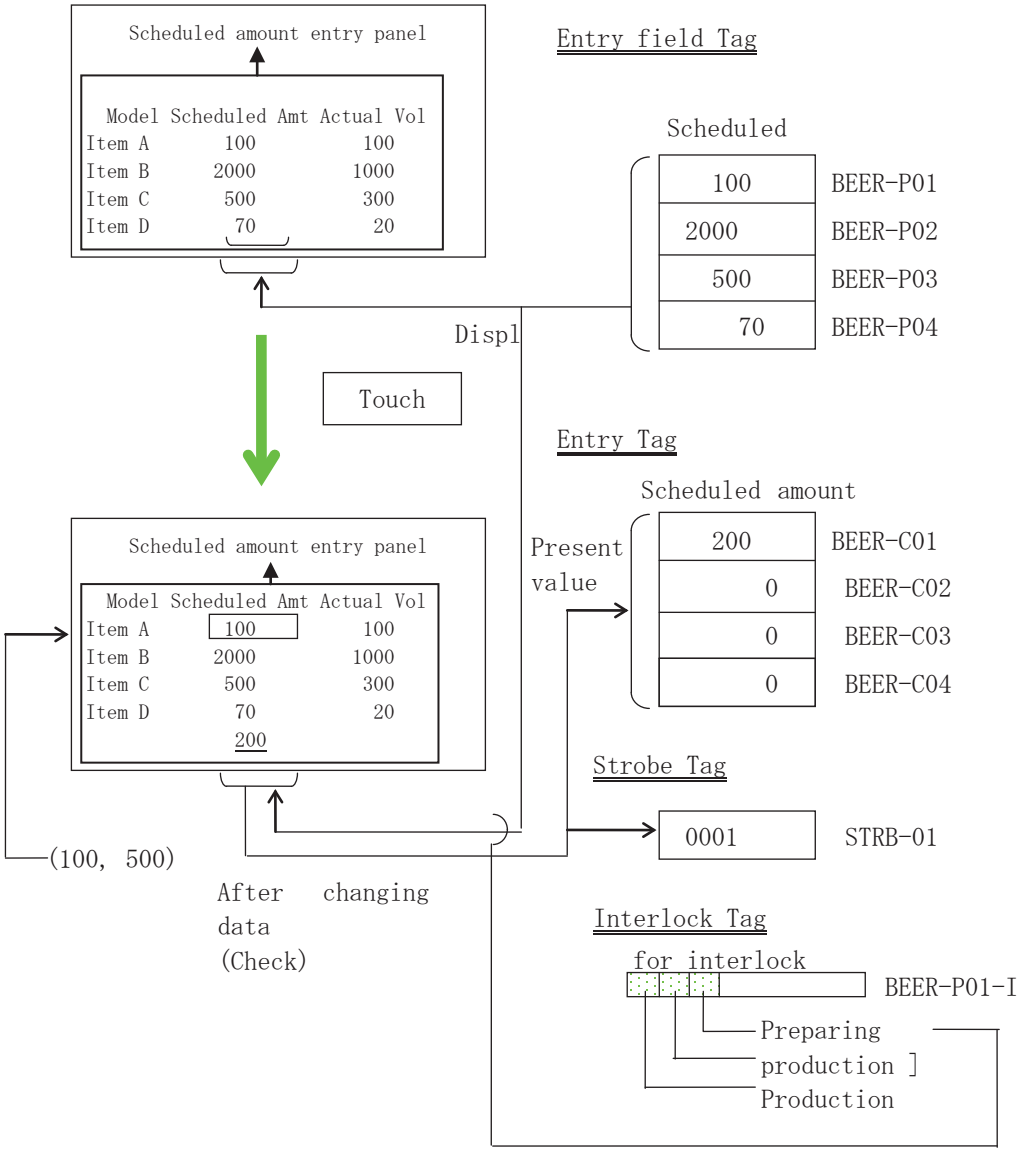
-	**Other**		**Description**	**Change**
	Object Type	Type of the object	Setting Window	No
	Object Kind	Kind of the object	IDSET	No
	Object Name	Name of the object	Multiple Touch Point Window	No
	Object No	Number of the object	–	No
-	**Position**			
	Set StartX	Set the coordinate X of starting point of the operation frame.	0 to 1870	Yes
	Set StartY	Set the coordinate Y of starting point of the operation frame.	0 to 935	Yes
	Width	Set the width of the operation frame.	1 to 1870	Yes
	Height	Set the height of the operation frame.	1 to 935	Yes
	InputX	Set the coordinate X of starting point for the data entry window.	0 to 1870	Yes
	InputY	Set the coordinate Y of starting point for the data entry window.	0 to 935	Yes
-	**Output**			
	Set Tag ST No	Set the entry Tag station No.	Only when specifying the system Tag, specify the physical ST No. (101 to 440)	Yes
	Set Tag	Set the entry Tag.	Specify the Tag to which the data is transmitted. Up to 16 single-byte Tag characters Selectable from the dialog box	Yes
-	**Condition**			
	Action Times	Set the number of actions for the touch operation.	1: "Run" transmits data 2: "Confirm" and then "Run" transmits data ("Confirm" is optional)	Yes

	IL Tag1 ST No	Interlock Set the interlock Tag 1 station No.	Only when specifying the system Tag, specify the physical ST No. (101 to 440)	Yes
	IL Tag1	Set the interlock Tag.	Two interlock Tags are available for use in each entry data. When two Tags are used, set both Tags to ON (or OFF) to set the interlock. ON (masking available): (Interlock Tag AND Interlock Mask) == Interlock Mask OFF (zero setting available): (Interlock Tag AND Interlock Mask) == zero Up to 16 single-byte Tag characters Selectable from the dialog box	Yes
	IL Mask1	Set the interlock Tag1 Mask condition.	Hexadecimal four characters **(Refer to section 2.3.4, "(11) Selecting BitPattern")**	Yes
	IL Tag2 ST No	Interlock Set the interlock Tag 2 station No.	Only when specifying the system Tag, specify the physical ST No. (101 to 440)	Yes
	IL Tag2	Set the interlock Tag.	Up to 16 single-byte Tag characters Selectable from the dialog box	Yes
	IL Mask2	Set the interlock Tag2 mask condition.	Hexadecimal four characters **(Refer to section 2.3.4, "(11) Selecting BitPattern")**	Yes
	IL Mask Behavior	Set the interlock condition.	Operable with the mask value Operable with zero	Yes
-	**Strobe**			
	Strobe	Set whether to use a strobe.	True (use) False (do not use)	Yes
	Strobe Tag ST No (When Strobe is set to True)	Set the strobe Tag station No.	Only when specifying the system Tag, specify the physical ST No. (101 to 440)	Yes
	Strobe Tag (When Strobe is set to True)	Set the entry Tag for the strobe Tag.	Up to 16 single-byte Tag characters Selectable from the dialog box	Yes
	Strobe Mask (When Strobe is set to True)	Set the mask for the strobe Tag.	Hexadecimal four characters **(Refer to section 2.3.4, "(11) Selecting BitPattern")**	Yes
	Strobe Data (When Strobe is set to True)	Set the data for the strobe Tag.	Hexadecimal four characters **(Refer to section 2.3.4, "(11) Selecting BitPattern")**	Yes
-	**View**			
	FB Tag ST No	Set the source Tag station No.	Only when specifying the system Tag, specify the physical ST No. (101 to 440)	Yes
	FB Tag	Set the source Tag.	The present value is displayed in the entry field based on the source Tag. Up to 16 single-byte Tag characters Selectable from the dialog box	Yes

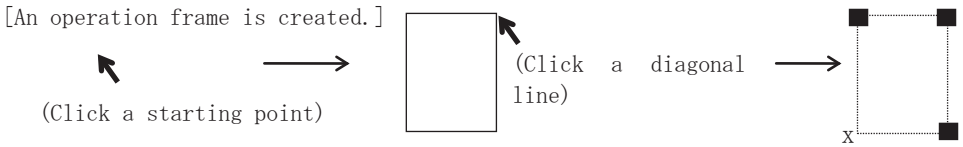
	Digit Number	Set the number of display digits.	1 to 16 (decimal point is also counted)	Yes
	Decimal Place	Set the place of decimal point. Specifiable only for decimal numbers.	-1 to 15	Yes
	Last SV	Set whether to display previous value.	True (Yes), False (No)	Yes
-	**Data**			
	Data Type	Set the display type.	Decimal two word, hexadecimal two word, decimal one word, and hexadecimal one word Text	Yes

Multiple Touch Point Window (Example)

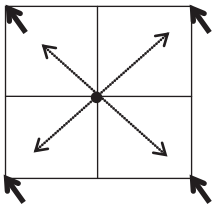
Online display & entry example



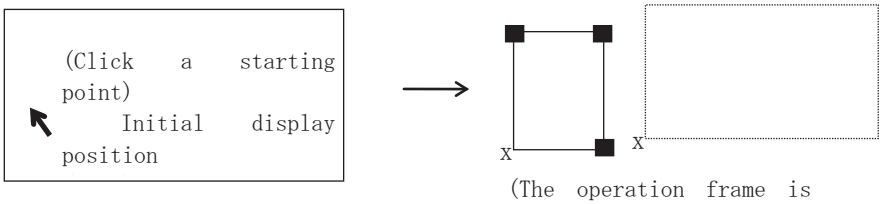
(8) Multiple touch point entry: Set numeric value under locked condition in the key-in area.



[The mouse can be moved to four directions as shown in the figure below]



[Click the initial position on the data entry window.]



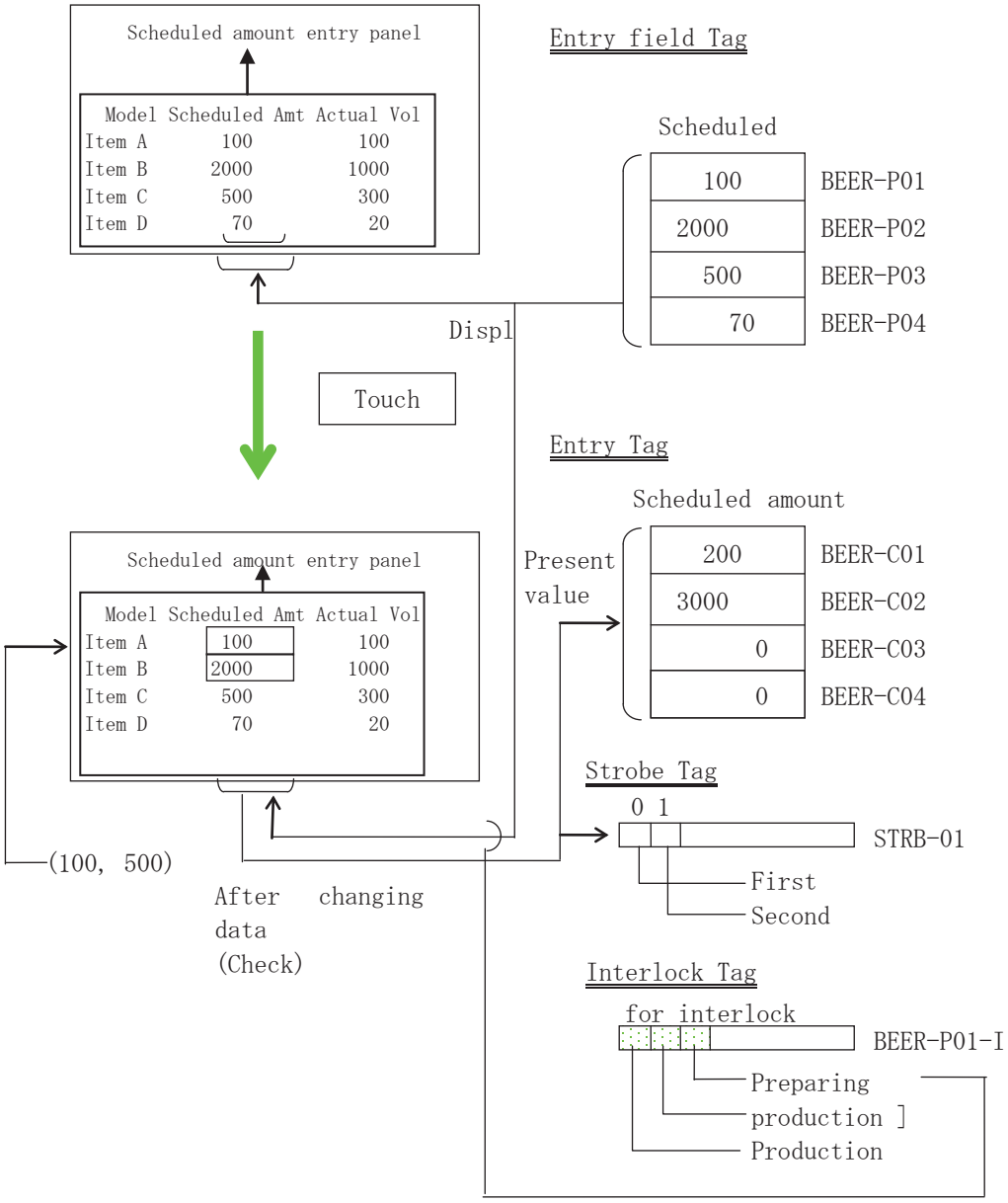
-	**Other**		**Description**	**Change**
	Object Type	Type of the object	Setting Field	No
	Object Kind	Kind of the object	IDSET	No
	Object Name	Name of the object	Multiple Touch Point Entry	No
	Object No	Number of the object	–	No
-	**Position**			
	Set StartX	Set the coordinate X of starting point of the operation frame.	0 to 1870	Yes
	Set StartY	Set the coordinate Y of starting point of the operation frame.	0 to 935	Yes
	Width	Set the width of the operation frame.	1 to 1870	Yes
	Height	Set the height of the operation frame.	1 to 935	Yes
	InputX	Set the coordinate X of starting point for the data entry window.	0 to 1870	Yes
	InputY	Set the coordinate Y of starting point for the data entry window.	0 to 935	Yes
-	**Output**			
	Set Tag ST No	Set the entry Tag station No.	Only when specifying the system Tag, specify the physical ST No. (101 to 440)	Yes
	Set Tag	Set the entry Tag.	Specify the Tag to which the data is transmitted. Up to 16 single-byte Tag characters Selectable from the dialog box	Yes
-	**Condition**			
	Action Times	Set the number of actions for the touch operation.	1: "Run" transmits data 2: "Confirm" and then "Run" transmits data ("Confirm" is optional)	Yes
	IL Tag1 ST No	Interlock Set the interlock Tag 1 station No.	Only when specifying the system Tag, specify the physical ST No. (101 to 440)	Yes

	IL Tag1	Set the interlock Tag.	Two interlock Tags are available for use in each entry data. When two Tags are used, set both Tags to ON (or OFF) to set the interlock. ON (masking available): (Interlock Tag AND Interlock Mask) == Interlock Mask OFF (zero setting available): (Interlock Tag AND Interlock Mask) == zero Up to 16 single-byte Tag characters Selectable from the dialog box	Yes
	IL Mask1	Set the interlock Tag1 Mask condition.	Hexadecimal four characters **(Refer to section 2.3.4, "(11) Selecting BitPattern")**	Yes
	IL Tag2 ST No	Interlock Set the interlock Tag 2 station No.	Only when specifying the system Tag, specify the physical ST No. (101 to 440)	Yes
	IL Tag2	Set the interlock Tag.	Up to 16 single-byte Tag characters Selectable from the dialog box	Yes
	IL Mask2	Set the interlock Tag2 mask condition.	Hexadecimal four characters **(Refer to section 2.3.4, "(11) Selecting BitPattern")**	Yes
	IL Mask Behavior	Set the interlock condition.	Operable with the mask value Operable with zero	Yes
-	**Strobe**			
	Strobe	Set whether to use a strobe.	True (use) False (do not use)	Yes
	Strobe Tag ST No (When Strobe is set to True)	Set the strobe Tag station No.	Only when specifying the system Tag, specify the physical ST No. (101 to 440)	Yes
	Strobe Tag (When Strobe is set to True)	Set the entry Tag for the strobe Tag.	Up to 16 single-byte Tag characters Selectable from the dialog box	Yes
	Strobe Mask (When Strobe is set to True)	Set the mask for the strobe Tag.	Hexadecimal four characters **(Refer to section 2.3.4, "(11) Selecting BitPattern")**	Yes
	Strobe Data (When Strobe is set to True)	Set the data for the strobe Tag.	Hexadecimal four characters **(Refer to section 2.3.4, "(11) Selecting BitPattern")**	Yes
-	**View**			
	FB Tag ST No	Set the source Tag station No.	Only when specifying the system Tag, specify the physical ST No. (101 to 440)	Yes
	FB Tag	Set the source Tag.	The present value is displayed in the entry field based on the source Tag. Up to 16 single-byte Tag characters Selectable from the dialog box	Yes
	Digit Number	Set the number of display digits.	1 to 16 (decimal point is also counted)	Yes

	Magnifier	Set the zoom ratio.	Standard, two to eight times	Yes
	Decimal Place	Set the place of decimal point. Specifiable only for decimal numbers.	-1 to 15	Yes
	Size	Set the character size.	8*8, 16*8, 16*16, 24*12, 24*24, 32*16	Yes
	Last SV	Set whether to display previous value.	True (Yes), False (No)	Yes
-	**Data**			
	Data Type	Set the display type.	Decimal two word, hexadecimal two word, decimal one word, and hexadecimal one word Text	Yes

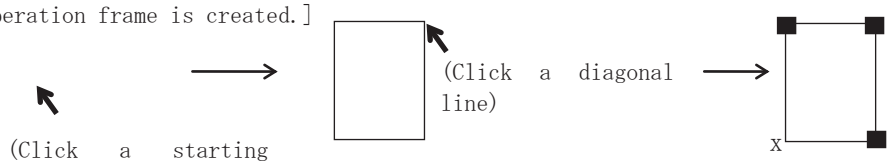
Multiple Touch Point Entry (Example)

Online display & entry example

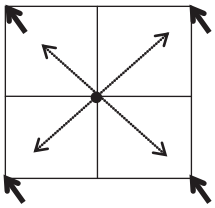


(9) Device operation 1: Set a mode under locked condition in the key-in area.

[An operation frame is created.]



[The mouse can be moved to four



-	**Other**		**Description**	**Change**
	Object Type	Type of the object	Operation1	No
	Object Kind	Kind of the object	IDSET	No
	Object Name	Name of the object	Device Operation 1	No
	Object No	Number of the object	–	No
-	**Position**			
	Operation StartX	Set the coordinate X of starting point of the operation frame.	0 to 1870	Yes
	Operation StartY	Set the coordinate Y of starting point of the operation frame.	0 to 935	Yes
	Width	Set the width of the operation frame.	1 to 1870	Yes
	Height	Set the height of the operation frame.	1 to 935	Yes
-	**Condition**			
	Action Times	Set the number of actions for the touch operation.	2: Select a device and select a command to transmit data. 3: Select a device, select a command, and click Run to transmit data. 4: Select a device, select a command, confirm, and click Run to transmit data. The four actions are not available for use because it is for electronic apparatus field.	Yes
	IL Tag1 ST No	Interlock Set the interlock Tag 1 station No.	Only when specifying the system Tag, specify the physical ST No. (101 to 440)	Yes
	IL Tag1	Set the interlock Tag. (Also used for the display Tag as Disp Tag.)	Two interlock Tags are available for use in each entry data. When two Tags are used, set both Tags to ON (or OFF) to set the interlock. ON (masking available): (Interlock Tag AND Interlock Mask) == Interlock Mask OFF (zero setting available): (Interlock Tag AND Interlock Mask) == zero Up to 16 single-byte Tag characters	Yes

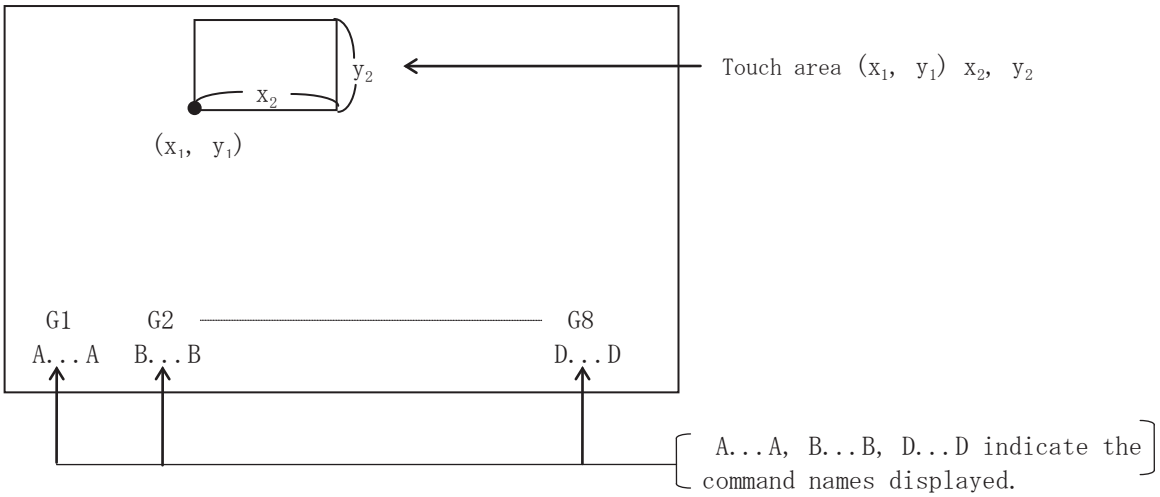
			Selectable from the dialog box	
	IL Tag2 ST No	Interlock Set the interlock Tag 2 station No.	Only when specifying the system Tag, specify the physical ST No. (101 to 440)	Yes
	IL Tag2	Set the interlock Tag. (Also used for the display Tag as Disp Tag.)	Two interlock Tags are available for use in each entry data. When two Tags are used, set both Tags to ON (or OFF) to set the interlock. ON (masking available): (IL Tag AND IL Mask) = IL Mask OFF (zero setting available): (L Tag AND IL Mask) = 0 Up to 16 single-byte Tag characters Selectable from the dialog box	Yes
	IL Mask Behavior	Set the interlock condition.	Operable with the mask value Operable with zero	Yes
-	**Output**			
	Ope Tag1 ST No	Set the operation Tag 1 station No.	Only when specifying the system Tag, specify the physical ST No. (101 to 440)	Yes
	Operation Tag1	Set the operation Tag 1.	Specify the Tag to which the data is transmitted. Up to 16 single-byte Tag characters Selectable from the dialog box	Yes
	Ope Tag2 ST No	Set the operation Tag 2 station No.	Only when specifying the system Tag, specify the physical ST No. (101 to 440)	Yes
	Operation Tag2	Set the operation Tag 2.	Specify the Tag to which the data is transmitted. Up to 16 single-byte Tag characters Selectable from the dialog box	Yes
-	**View**			
	G8 Usage	Set the G8 key usage for operation.	- Run: Click "G8", click "Run", and touch to transmit data * The display on the online panel is "Run". - Release: The "G8" is used for selection similar to "G1" to "G7". Only the "Run" is used for transmission - Status: Only the "Run" is used for transmission. The "G8" is used for displaying the result.	Yes
	Cancel Key	Set whether to use the cancel key and the location.	When the clear option is OFF, the operation area is cleared each time the device is operated. When the clear option is ON, the G key-in area is not cleared so that consecutive operation of the device is enabled. When the clear option is ON, touch the "Clear" G key or click "Cancel" to clear the operation area. The "Clear" G key should be one of the clear display position G1 to G8.	Yes

			Clear display position: When no clear display is selected, the "Clear" G key is not shown. Here, click the "Cancel" key to clear consecutive operation.	
-	**G*n* Behavior *(n=1 to 8)*: When n=8, it is enabled to use G8 for clearing operation.**			
	G***n*** OutTag	Set the output Tag.	Select operation Tag 1 or 2 for the output Tag. The operation Tag selected by the Cancel Key option cannot be selected.	Yes
	G***n*** Command	Select the command name.	Up to single-byte 12 characters are available for use in the comment. The comment input once can be selected from the dialog. (The command name definition that was enabled in existing models is not available now)	Yes
	G***n*** Mask	Set the mask for the operation Tag.	Hexadecimal four characters **(Refer to section 2.3.4, "(11) Selecting BitPattern")**	Yes
	G***n*** Data	Set the data for the operation Tag.	Hexadecimal four characters **(Refer to section 2.3.4, "(11) Selecting BitPattern")**	Yes
-	**G*n* View *(n=1 to 8)*: When n=8, it is enabled to use G8 for clearing operation.**			
	G***n*** Mask1	Set the mask for the interlock Tag1.	When two Tags are used, set both Tags to ON (or OFF) to set the interlock. Hexadecimal four characters **(Refer to section 2.3.4, "(11) Selecting BitPattern")** - Interlock condition ON: (IL Tag AND G***n*** Mask) == G***n*** Mask OFF: (IL Tag AND G***n*** Mask) == zero (For ON, set the interlock condition to mask. For OFF, set the interlock condition to zero.)	Yes
	G***n*** Mask2	Set the mask for the interlock Tag2.		Yes
	G***n*** IL Mask Behaver	Set the interlock condition.		Yes
	G***n*** Disp Tag	Set the Tag for displaying status.	Select IL Tag1 or 2 above.	Yes
	G***n*** Disp Mask	Set the Tag mask for displaying status.	Hexadecimal four characters **(Refer to section 2.3.4, "(11) Selecting BitPattern")**	Yes
	G***n*** Disp Data	Set the Tag data for displaying status.	Hexadecimal four characters **(Refer to section 2.3.4, "(11) Selecting BitPattern")**	Yes

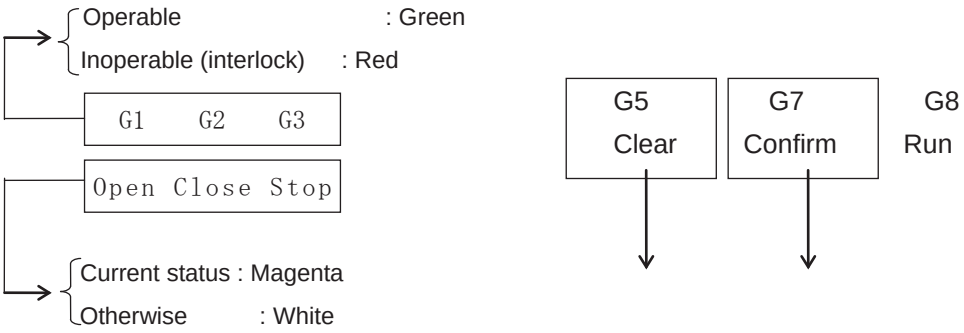
<Behavior of device operation 1 (key-in area)>

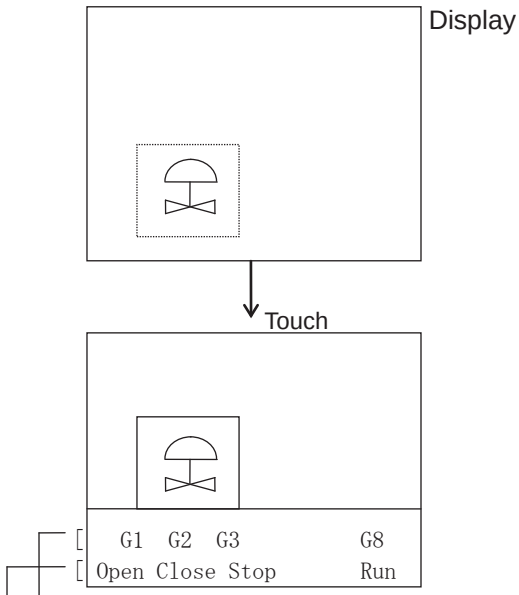
Touch concerned place on the CRT screen while displaying the plant panel to display the key area.

Select an item in the key area to transmit the data.



Displaying device operation key-in area

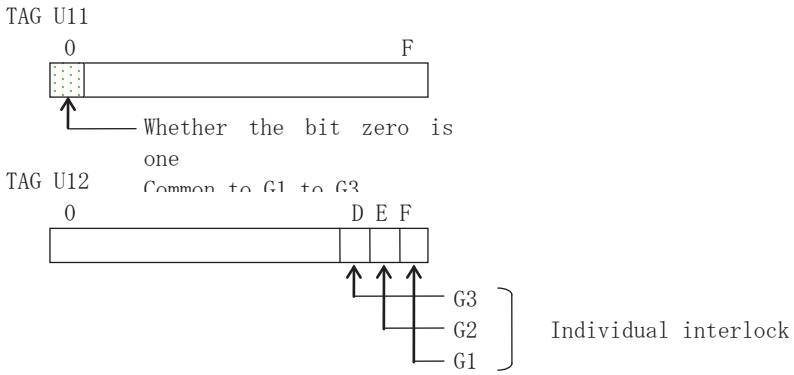




The color is changed according to the interlock information

Operable	The color is according to the system definition
Inoperable	: Red

Interlock information 1



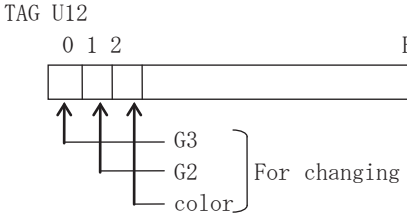
Since the lock condition is mask according to the definition information,

- masked result 1 of interlock TAG U11 and masked result 1 (bit F) of interlock TAG U12 → operability of G1
- masked result 1 of interlock TAG U11 and masked result 1 (bit E) of interlock TAG U12 → operability of G2
- masked result 1 of interlock TAG U11 and masked result 1 (bit D) of interlock TAG U12 → operability of G3

Present status display

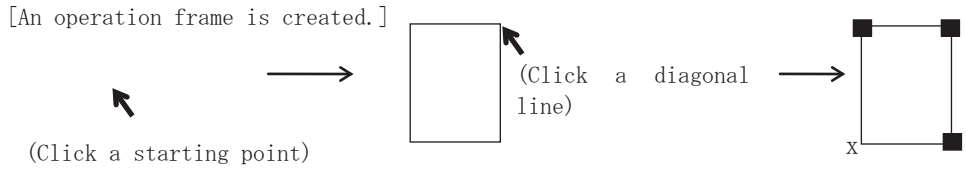
Current status : Magenta	The color is according to the system definition
Otherwise : White	

Since the definition indicates ● Tag2, the Tag 2, that is U12 information, is followed.

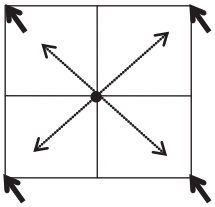


If the bit zero of U12 is one, the G1 Open is shown in magenta
If the bit one of U12 is one, the G2 Close is shown in magenta
If the bit two of U12 is one, the G3 Stop is shown in magenta

(10) Device operation 2: Set a mode under locked condition in the full screen.



[The mouse can be moved to four directions as shown in the figure below]



-	**Other**		**Description**	**Change**
	Object Type	Type of the object	Opration2	No
	Object Kind	Kind of the object	IDSET	No
	Object Name	Name of the object	Device Operation 2	No
	Object No	Number of the object	–	No
-	**Position**			
	Operation StartX	Set the coordinate X of starting point of the operation frame.	0 to 1870	Yes
	Operation StartY	Set the coordinate Y of starting point of the operation frame.	0 to 935	Yes
	Width	Set the width of the operation frame.	1 to 1870	Yes
	Height	Set the height of the operation frame.	1 to 935	Yes
-	**Condition**			
	Action Times	Set the number of actions for the touch operation.	1: Select data to transmit data. 2: Select data and click Run to transmit data.	Yes
	IL Tag1 ST No	Interlock Set the interlock Tag 1 station No.	Only when specifying the system Tag, specify the physical ST No. (101 to 440)	Yes
	IL Tag1	Set the interlock Tag.	Two interlock Tags are available for use in each entry data. When two Tags are used, set both Tags to ON (or OFF) to set the interlock. ON (masking available): (IL Tag AND IL Mask) = IL Mask OFF (zero setting available): (L Tag AND IL Mask) = 0 Up to 16 single-byte Tag characters Selectable from the dialog box	Yes
	IL Mask1	Set the mask for the interlock Tag1.	Hexadecimal four characters **(Refer to section 2.3.4, "(11) Selecting BitPattern")**	Yes

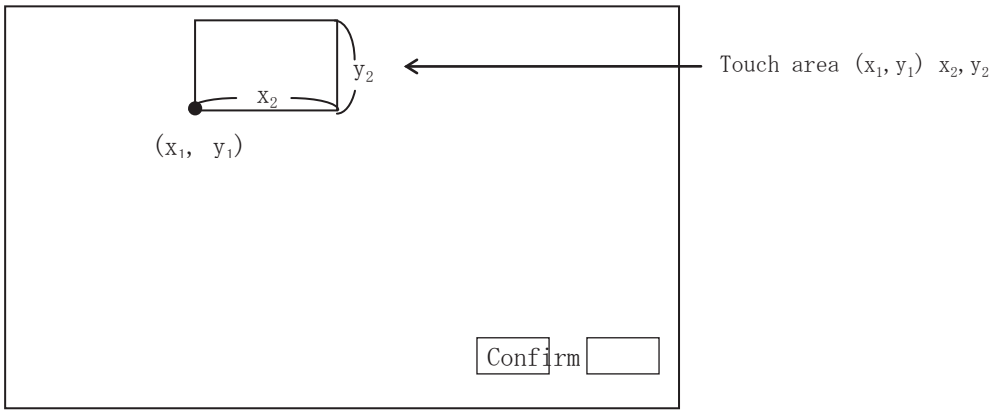
	IL Tag2 ST No	Interlock Set the interlock Tag 2 station No.	Only when specifying the system Tag, specify the physical ST No. (101 to 440)	Yes
	IL Tag2	Set the interlock Tag.	Two interlock Tags are available for use in each entry data. When two Tags are used, set both Tags to ON (or OFF) to set the interlock. ON (masking available): (IL Tag AND IL Mask) = IL Mask OFF (zero setting available): (L Tag AND IL Mask) = 0 Up to 16 single-byte Tag characters Selectable from the dialog box	Yes
	IL Mask2	Set the mask for the interlock Tag2.	Hexadecimal four characters **(Refer to section 2.3.4, "(11) Selecting BitPattern")**	Yes
	IL Mask Behavior	Set the interlock condition.	Operable with the mask value Operable with zero	Yes
-	**Output**			
	Ope Tag1 ST No	Set the operation Tag 1 station No.	Only when specifying the system Tag, specify the physical ST No. (101 to 440)	Yes
	Operation Tag1	Set the operation Tag 1.	Specify the Tag to which the data is transmitted. Up to 16 single-byte Tag characters Selectable from the dialog box	Yes
	Operation Mask	Set the mask for the operation Tag.	Hexadecimal four characters **(Refer to section 2.3.4, "(11) Selecting BitPattern")**	Yes
	Operation Data	Set the data for the operation Tag.	Hexadecimal four characters **(Refer to section 2.3.4, "(11) Selecting BitPattern")**	Yes
	Command	Select the command name.	Up to single-byte 12 characters are available for use in the comment. The comment input once can be selected from the dialog. (The command name definition that was enabled in existing models is not available now)	Yes
	Operation Group	Set the group number.	1 to 64	Yes

<Behavior of device operation 2>

Set a mode under locked condition in the full screen.

Select a group by touching it. Only one can be selected in the same group. Select data for each group to transmit it.

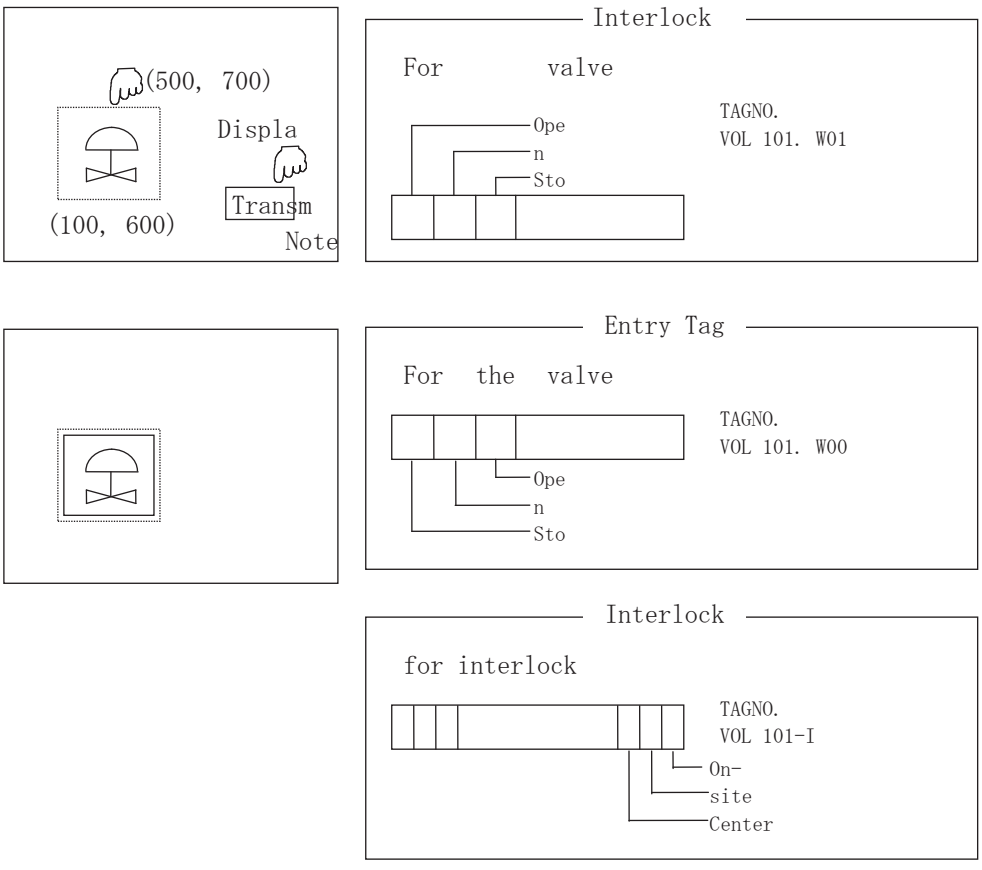
However, the selection is impossible during interlocking.



Online display & operation example

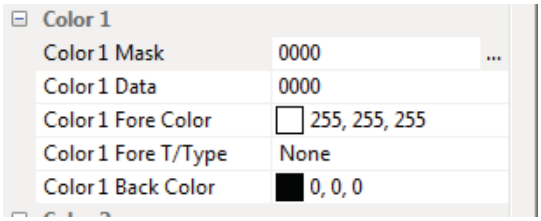
Example of group No. = 1:

Note: Create a frame for transmission by specifying the function code 0031H in the definition of virtual keyboard

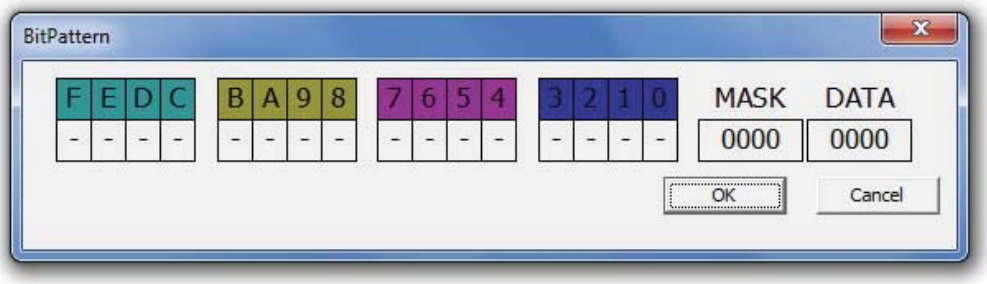


2.

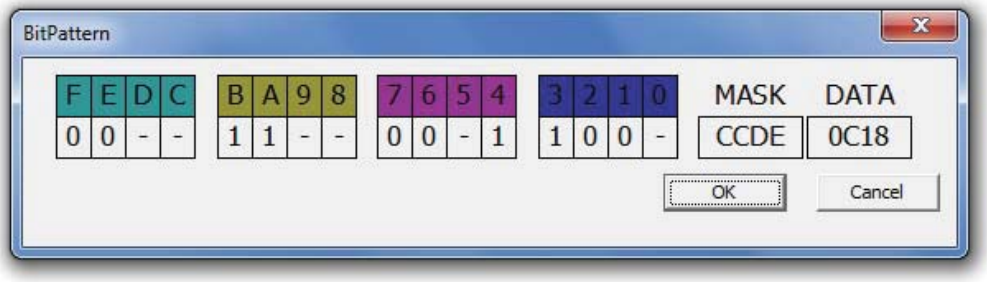
(11) Selecting BitPattern



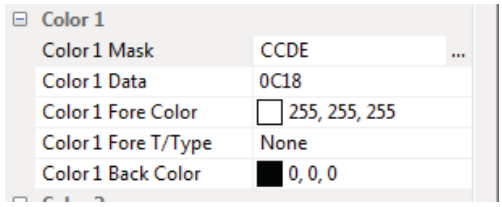
Click the button at the right of Mask, Data,



Click "-" part under each label to create a bit pattern.



Click the OK button.



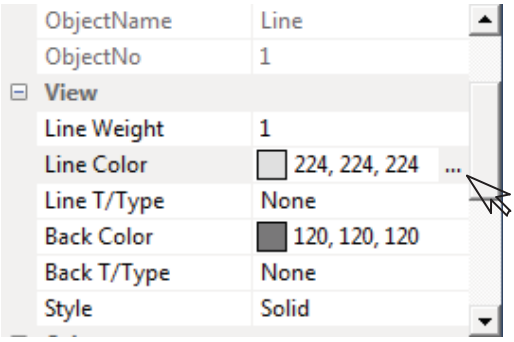
The bit pattern created is put in the property.

2.3.5 Dialog for Property

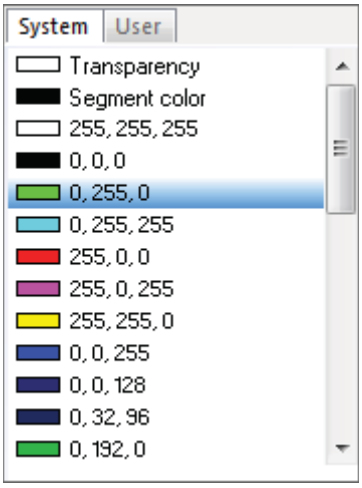
(1) Specifying color

In order to specify a color, click at the right in the definition field to display the color selection dialog.

There are System and User tabs. In each tab, 32 colors, transparency option, and segment colors can be selected.

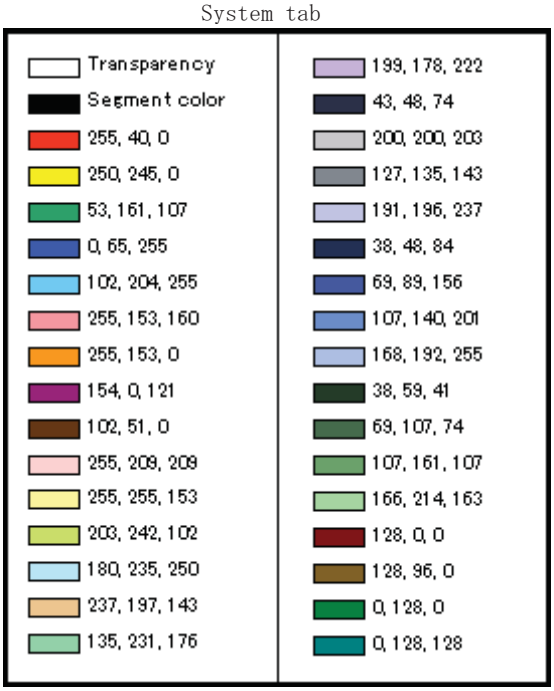


Click the right

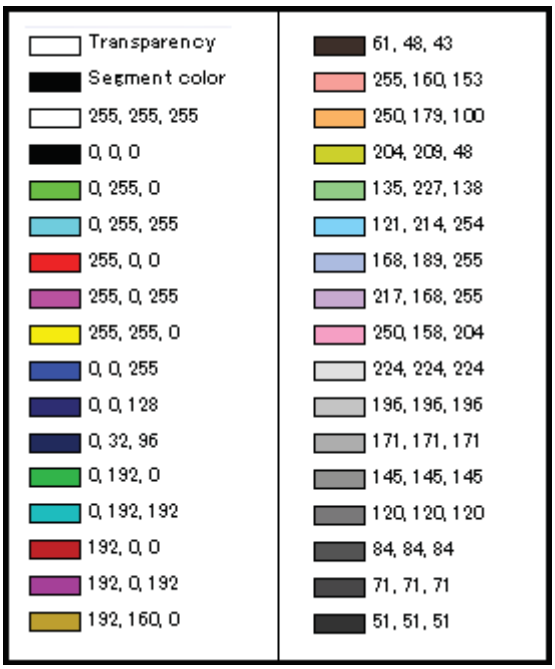


Select a color from the dialog box

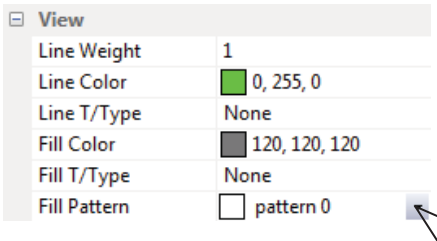
System tab



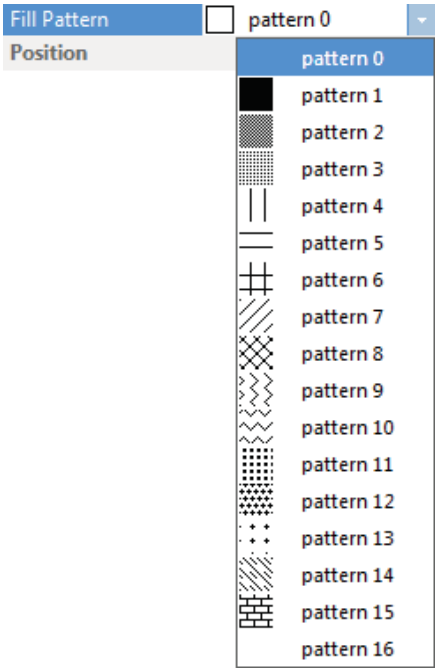
User tab



(2) Pattern

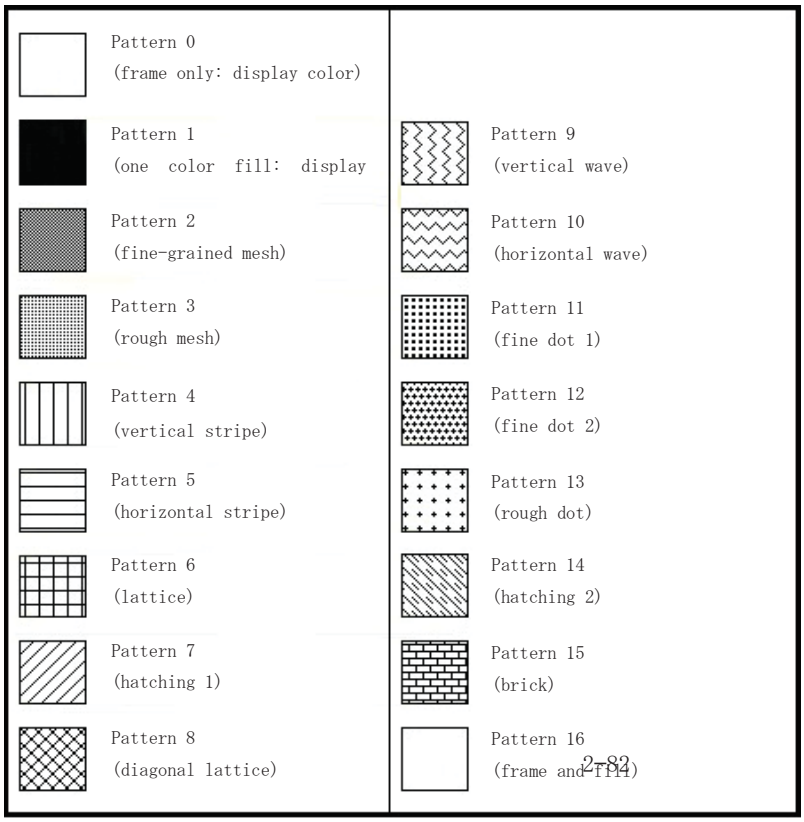


Click the right



Select a pattern from the dialog box

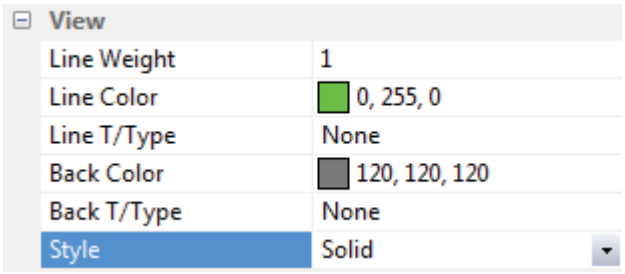
Pattern list

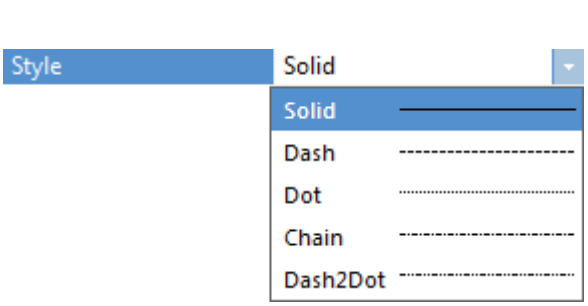


- The pattern 0 is frame line only. Specify its color.
- The pattern 1 is a one color fill. Specify its color.
- The patterns 2 to 15 are device patterns. Specify the display color and background color for black and white parts, respectively.
- For the pattern 16, the frame line is the display color and its inside is filled in background

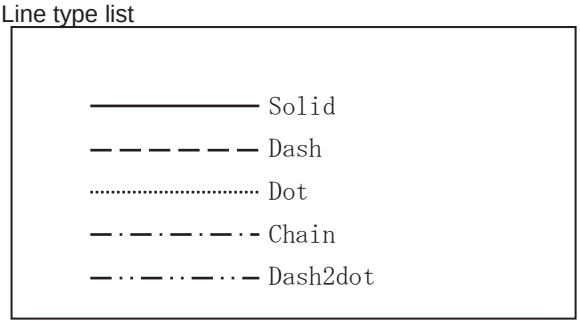
(3) Line type

Only the line attribute can be specified. (line, arc, elliptic arc)





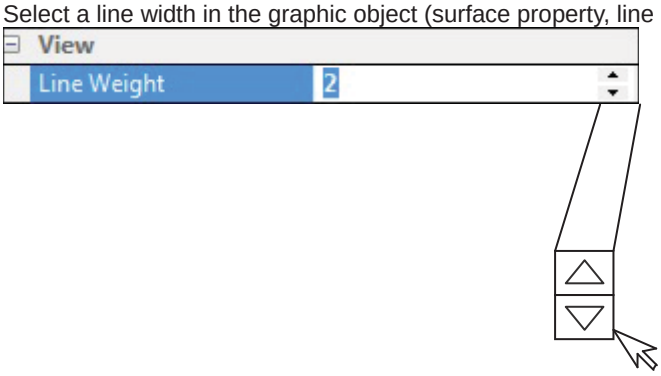
Line type list



Only if the line width is a dot, other than solid line (dash, dot, dashdot, or dash2dot) is selectable.

(4) Line width

Select a line width in the graphic object (surface property, line property).



Click the UP or DOWN button at the right of the item to set the width.

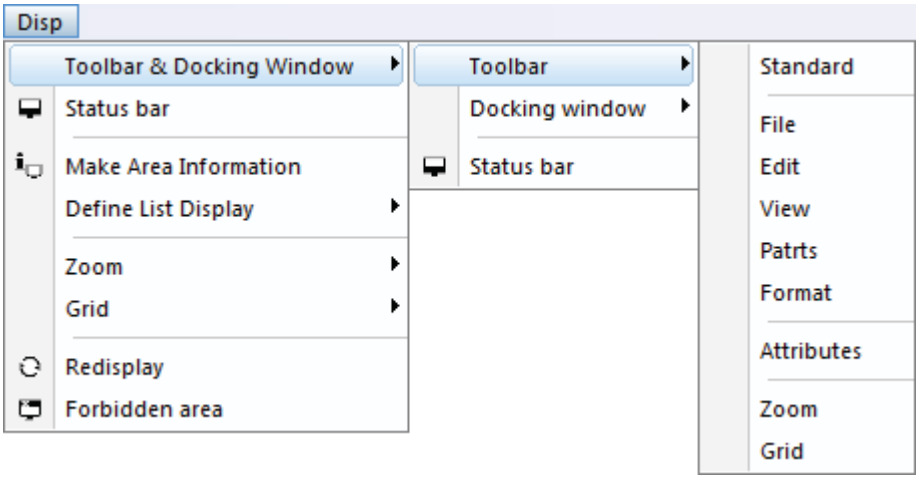
A numeric value can be input directly.

(1 to 16 dots)

Note: Only if the line type is solid, the widths 2 to 16 dot are

2.3.6 How to Operate View

(1) Tool bar and docking window



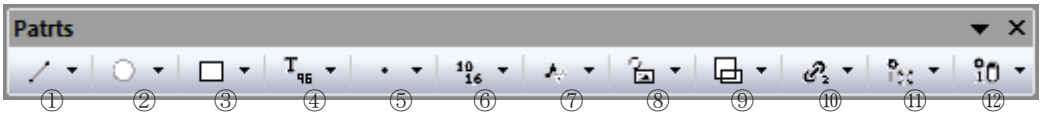
Each tool bar is shown.

- Standard

① Select panel
② Save panel
③ Copy panel
④ Delete panel
⑤ Exit
⑥ Cut
⑦ Copy
⑧ Paste
⑨ Delete
⑩ Undo
⑪ Redo

- Move it vertically or to the right corner to fix it as the menu bar.

- Object



① Line [line/line (horizontal, vertical)]

② Circle/arc [circle/arc/ellipse/elliptic arc/sector]

③ Polygon [rectangle/button type rectangle/polygon]

④ Character [8/16/24/32/48/64/96 dot]

⑤ Other fixed objects [point/fixed pattern/segment/bitmap]

⑥ Numeric [numeric/limiter numeric]

⑦ Graph [bar graph/trend/line/prot bar graph/batch trend/prot limiter bar graph]

⑧ Exchange [exchange text/exchange segment/exchange bitmap]

⑨ Other variable objects [move pattern/text/meter diagram/dynamic window]

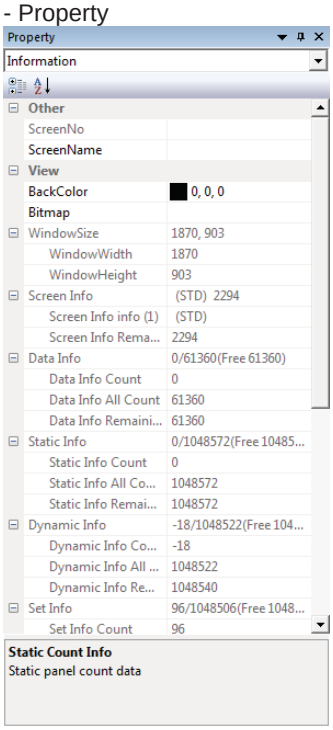
⑩ Link [link screens (single action)/Link screens (two actions)/meter diagram window/dynamic window]

⑪ Entry [full screen entry 1/full screen entry 2/multiple touch point window/multiple touch point entry]

⑫ Operation [device operation 1/device operation 2]

- Move it vertically or to the right corner to fix it as the menu bar.

- Property



Right-click the cursor within the window to display the Property menu.

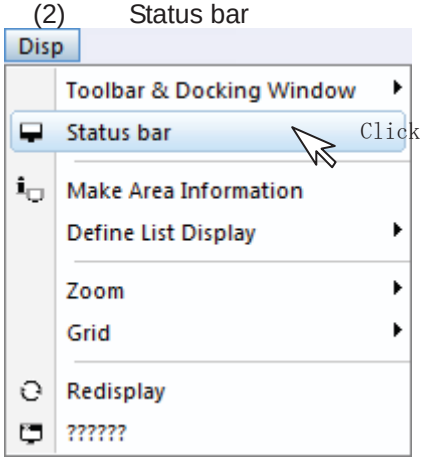
Click "Docking" in the menu to fix the menu at the left corner of the engineering panel.

Description

-	**Other**		**Description**	**Change**
	Screen No	Screen number	–	No
	Screen Name	Screen name	Up to single-byte 32 texts. Single-byte and double-byte texts can be mixed	Yes
-	**View**			
	Back Color	Specify a background color of the screen.	Select it from the color dialog.	Yes
	Bitmap	Specify a background bitmap of the screen.	Select it from the bitmap dialog.	Yes
-	**Window Size**	Screen size	–	No
	Window Width	Specify a width of the screen.	1 to 1870	No
	Window Height	Specify a height of the screen.	1 to 935	No
-	**Screen Info Buffer**	Screen information	–	No
	Screen Info (1)	Screen information 1 (panel type)	(standard) (DW) (horizontal double) (horizontal three times) (horizontal four times) (vertical double) (vertical three times) (vertical four times) (four times)	No
	Screen Remaining	Screen information 2 (remaining information)	Number of remaining records definable in the screen One record = 512 byte	No
-	**Data Info Buffer**	Used data amount	–	No
	Data Info Used	Used data amount, number of pieces of data	–	No
	Data Info Total	Used data amount, total number of pieces of data	–	No
	Data Info Free	Used data amount, remaining data	–	No
-	**Static Info Buffer**	Still picture	–	No
	Static Info Used	Still picture, number of pictures	–	No
	Static Info Total	Still picture, total number of pictures	–	No

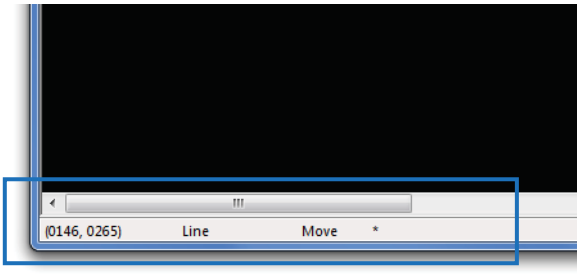
	Static Info Free	Still picture, remaining pictures	–	No
-	**Dynamic Info Buffer**	Operational picture	–	No
	Dynamic Info Used	Operational picture, number of pictures	–	No
	Dynamic Info Total	Operational picture, total number of pictures	–	No
	Dynamic Info Free	Operational picture, remaining pictures	–	No
-	**Set Info Buffer**	Operation picture	–	No
	Set Info Used	Operation picture, number of pictures	–	No
	Set Info Total	Operation picture, total number of pictures	–	No
	Set Info Free	Operation picture, remaining pictures	–	No
-	**Static Parts Count**	Number of still pictures	–	No
	Static Parts Used	Number of still pictures, quantity	–	No
	Static Parts Total	Number of still pictures, total	–	No
	Static Parts Free	Number of still pictures, remaining	–	No
-	**Dynamic Parts Count**	Number of Tags	–	No
	Dynamic Parts Used	Number of Tags, quantity	–	No
	Dynamic Parts Total	Number of Tags, total	–	No
	Dynamic Parts Free	Number of Tags, remaining	–	No
-	**Set Parts Count**	Number of operational pictures	–	No
	Set Parts Used	Number of operational pictures, quantity	–	No
	Set Parts Total	Number of operational pictures, total	–	No
	Set Parts Free	Number of operational pictures, remaining	–	No

(2) Status bar

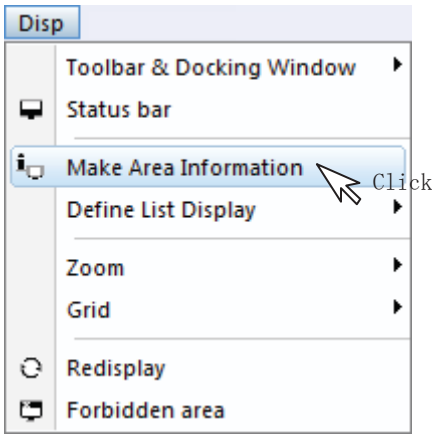


Click the Status Bar item to display the status bar at the bottom of the screen.

An explanation and so on of the item in the menu



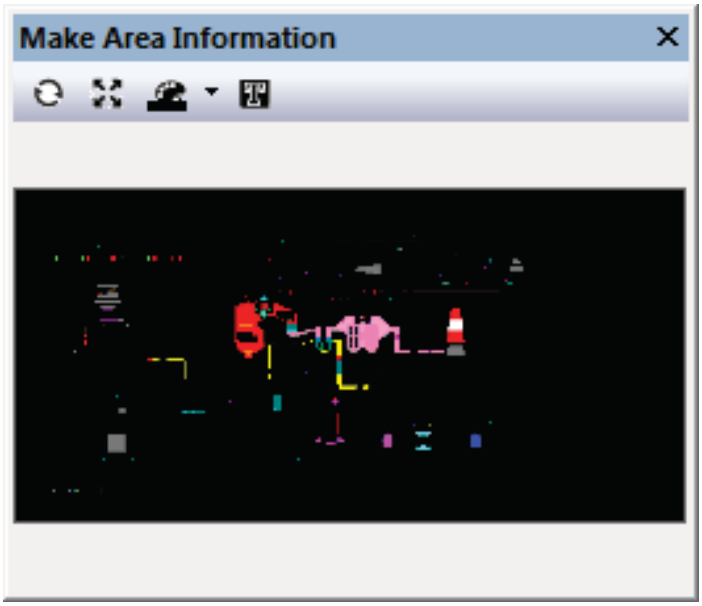
(3) Display area information



Click the Display Area Information to display the display area information window (indicating the position displayed at present on the entire screen).

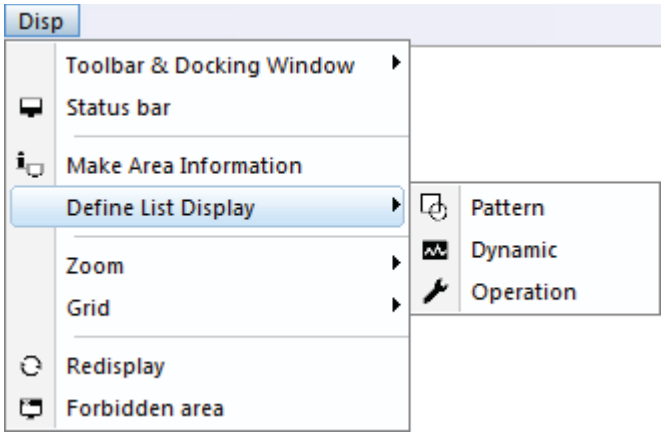
* The window is grayed out and becomes nonselectable while the display area information window is shown.

<Display area information window>

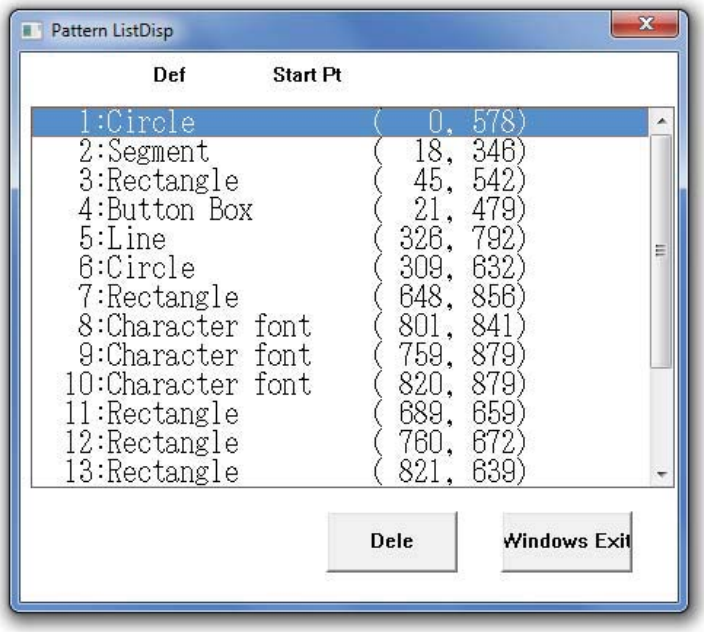


The displayed area is shown inverted. Therefore, the window is displayed in different color from that shown at present.
(Example: screen: red, working area: blue)

(4) Object list



The Object List menu item displays the window similar to the figure shown below for fixed, variable, operation objects.



How to operate

1. Move the mouse cursor to the Object List shown in the View menu to display another submenus (Graphic, Variable, Operation).
2. Click the desired command submenu to display the object list.
3. The dialog box is shown.

Move	Selects the move action.
Delet	Selects the delete action.
Run	Performs the selected action.
Cancel	Cancels the action.
Exit	Closes the object list window.

- How to operate

1. Move, delete

① Click to select an object in the list box. (Multiple selection: Dependent on the selection mode in auxiliary interact drawing menu.)

Click the same object again to deselect the object.

- The object in the list box is displayed inverted.
- The action commands (Move, Delete) are

② Click the Move button (select an

- The action commands (Move, Delete, Run) are grayed out
- The action command (Cancel) is enabled
- The movable range of mouse is restricted (movable only in the dialog box)

⑤ Click the Delete button (select an action)

- The action command (Run, Delete) is grayed out
- The action commands (Cancel, Run) are enabled
- The movable range of mouse is restricted

③ Click the destination for

(The object is moved in front of the destination.)

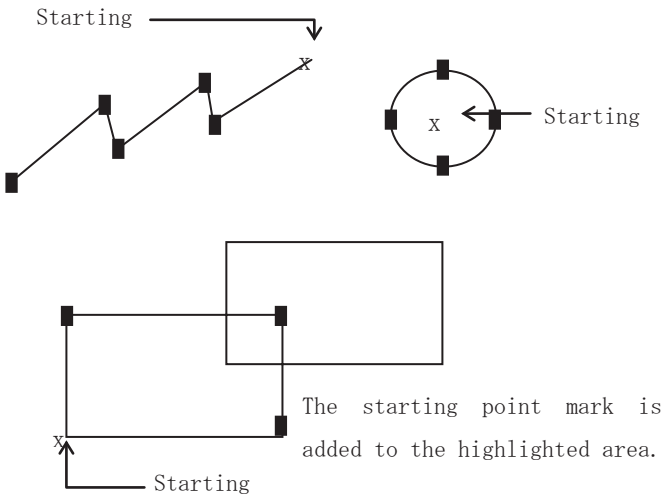
- The action command (Run) is enabled

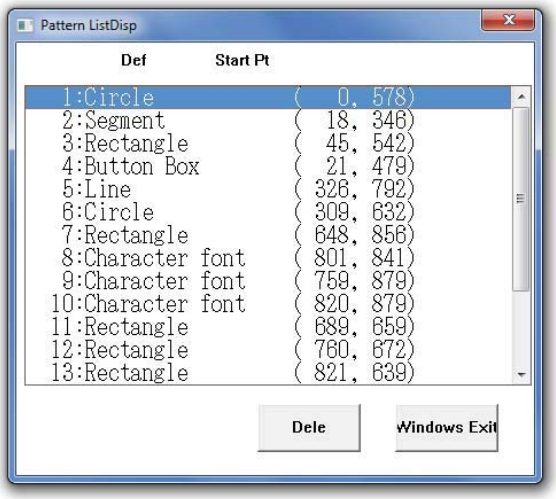
⑥ Click Cancel

- The movable range of mouse is cleared
- The action commands (Run, Cancel) are grayed out

④ Click Run

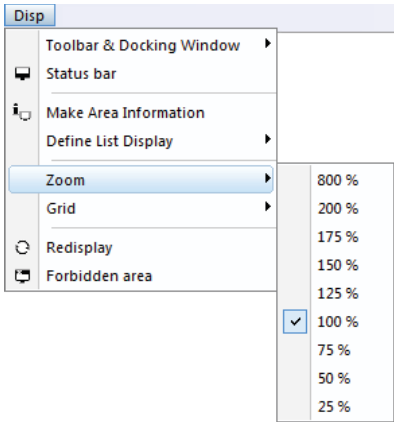
- The object selected is moved (deleted)
  When multiple objects are selected, all of them are moved (in order of objects).
- The movable range of mouse is cleared
- The action commands (Run, Cancel) are grayed out





- Click an object in the list to highlight the object in the drawing screen. Also, click an object in the drawing screen to highlight the object in the list.
- When an object is added, deleted, copied in the drawing screen, the list box is updated.
- When the starting point is changed due to a move, change, etc. in the drawing area, the list box is updated.
- When multiple objects are clicked in the list box, an object displayed inverted is added. When an object is selected, inverted display of other objects is cleared.
- Click a picture object in the drawing screen to display the top of current page. Scroll the list to find the object inverted. Here, the top of the page is 1 to 9 objects of 1 to 128 for page 1 and 129 to 256 for page 2.

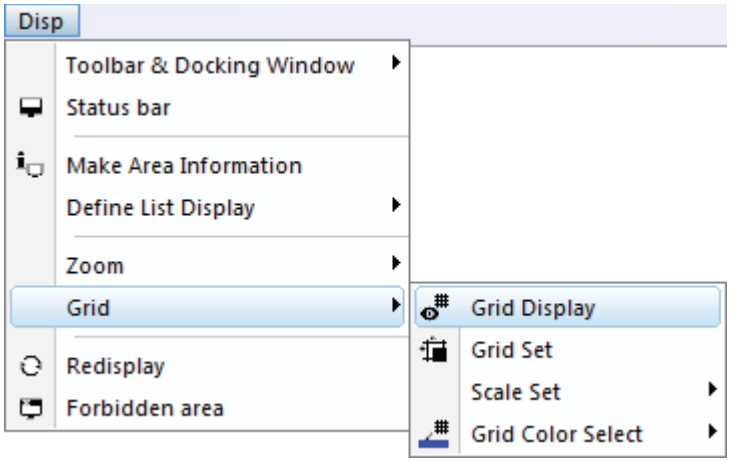
(5) Zoom

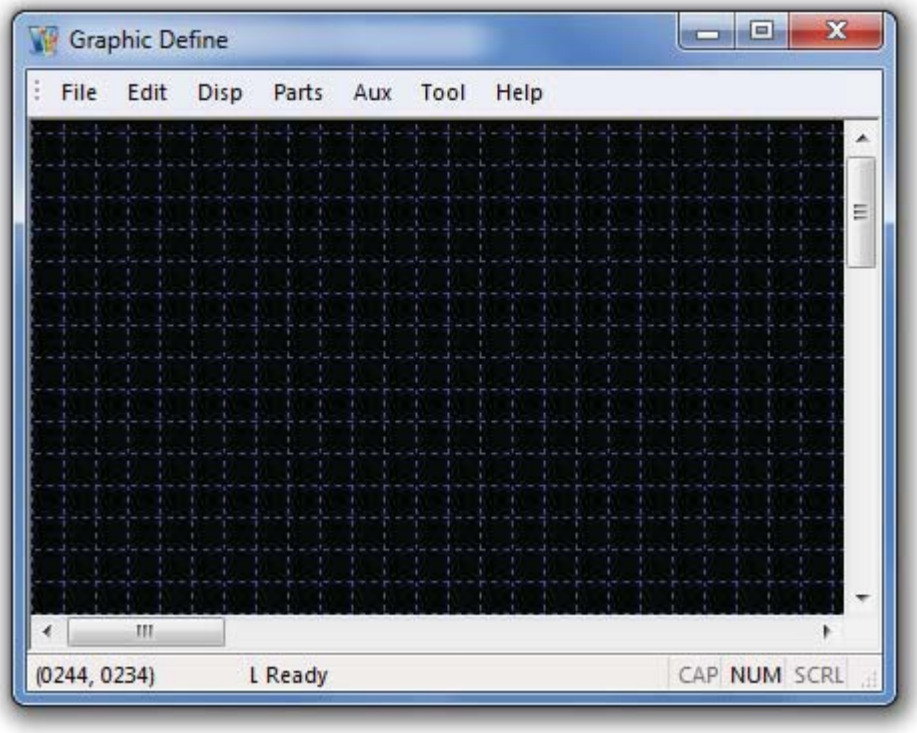


Displays of expansion & reduction ratio (800% - 25%)

Some circles, arcs, sectors, ellipses, or elliptic arcs can not be enlarged. See ″0

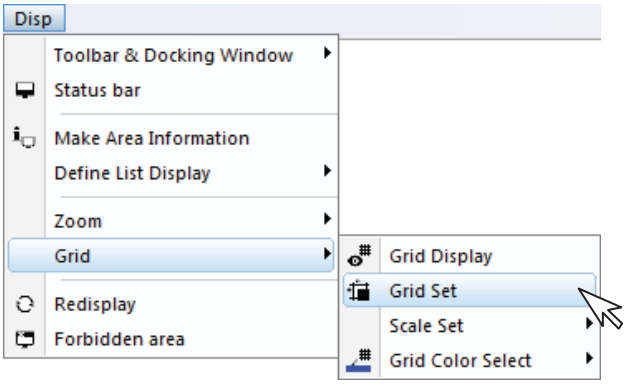
(6) Grid Display





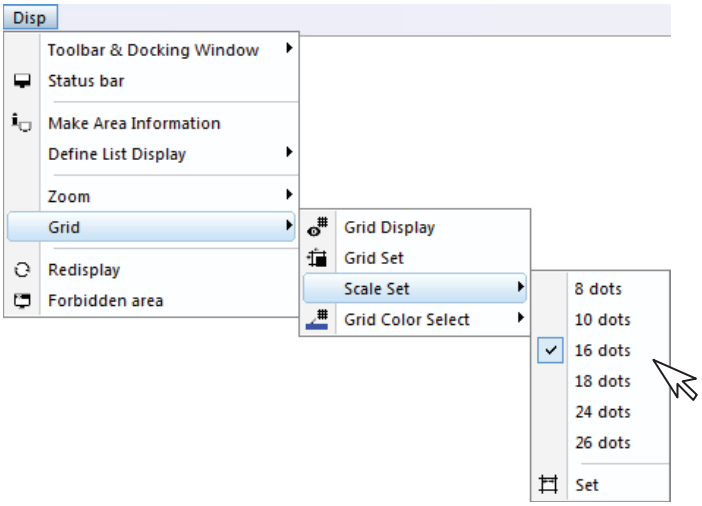
Displays grids on the screen.
The grid size is specified by "(8) Scale Set".

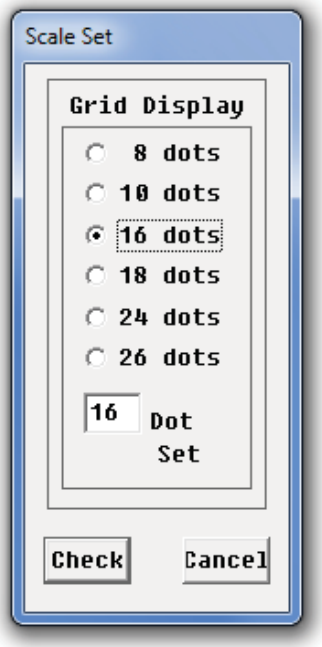
(7) Grid Set



Aligns the display of objects to grids.

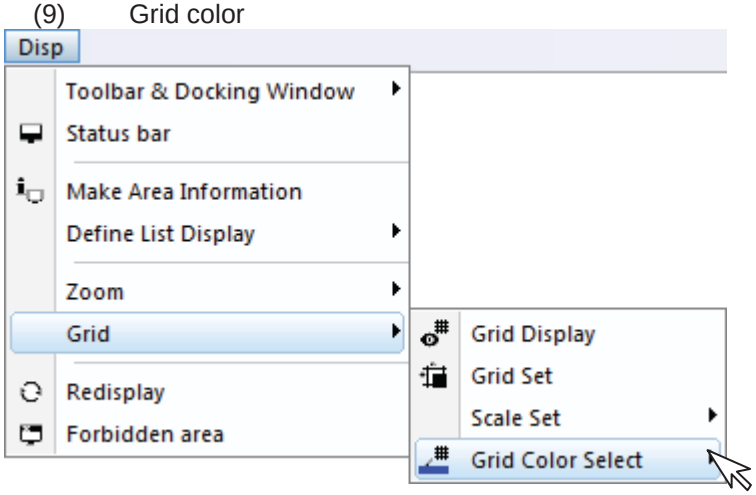
(8) Scale Set

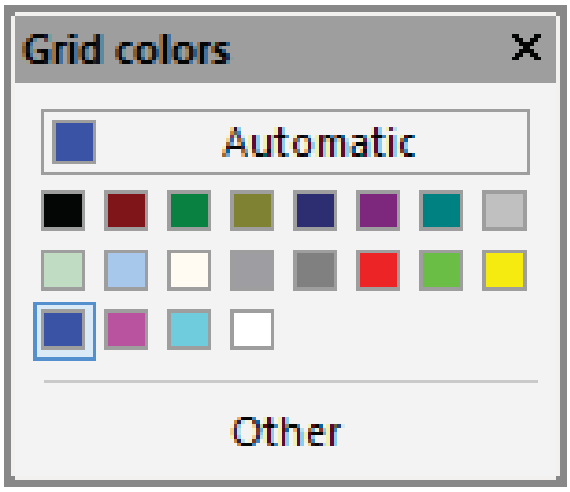




・Any value can be specified for Display Grids and Align to Grid.
The inputtable range is 1 to the (size of both sides)/2.
When an input error occurs, the grids are displayed inverted. Enter a correct value.

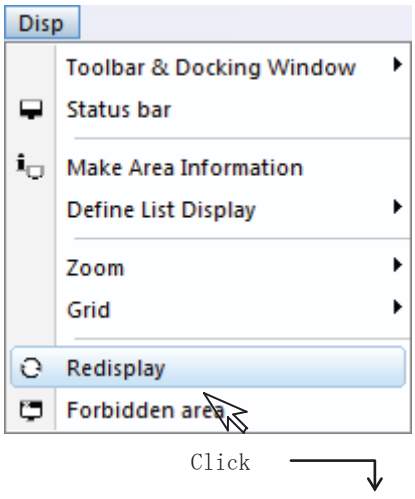
(9) Grid color





・Specify the grid color.
(blue by default)

(10) Redisplay



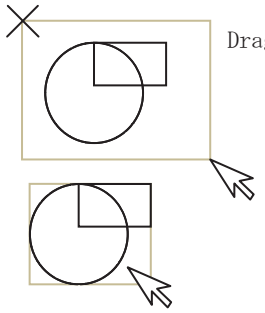
Click

Refreshes the screen.

Note: If you release the left mouse button when the mouse cursor overlaps with an attribute pallet, screen information, object list, display area information, or auxiliary window while moving graphics, the rubber band remains displayed and the mouse is still restricted from moving (move mode).

[ESC] Press any key or click right button to release the move mode and refresh the screen.

(11) Multiple selection with box (pick up)



Drag the mouse to enclose two or more fixed objects.

Right-click in the enclosed area

Select an item in the menu to change attributes of multiple objects, or delete, move, copy, or segment them.

To clear the box selection, select another object(s).

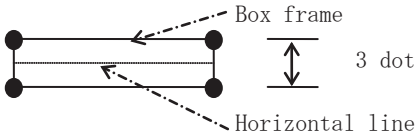
Also, multiple object selected can be deformed (changing graphics).

To clear the box selection, select next box or another object(s).

The larger box frame as shown below is displayed only when the width of graphics enclosed by the box is one dot.

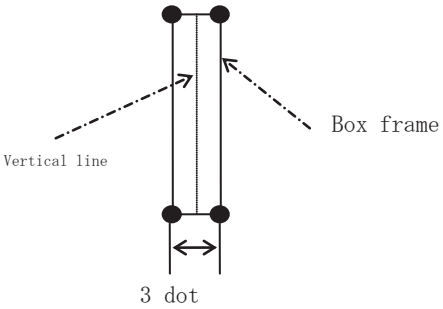
- Horizontal line:

・The box frame is displayed by moving vertically by one dot.



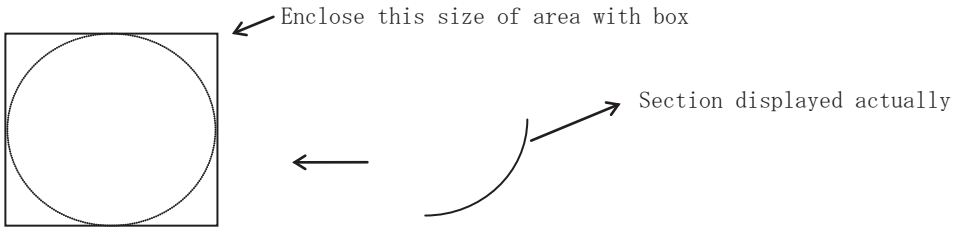
- Vertical line:

・The box frame is displayed by moving horizontally by one dot.



To select graphics including arcs, enclose not only the arc itself but original entire circle with the box.

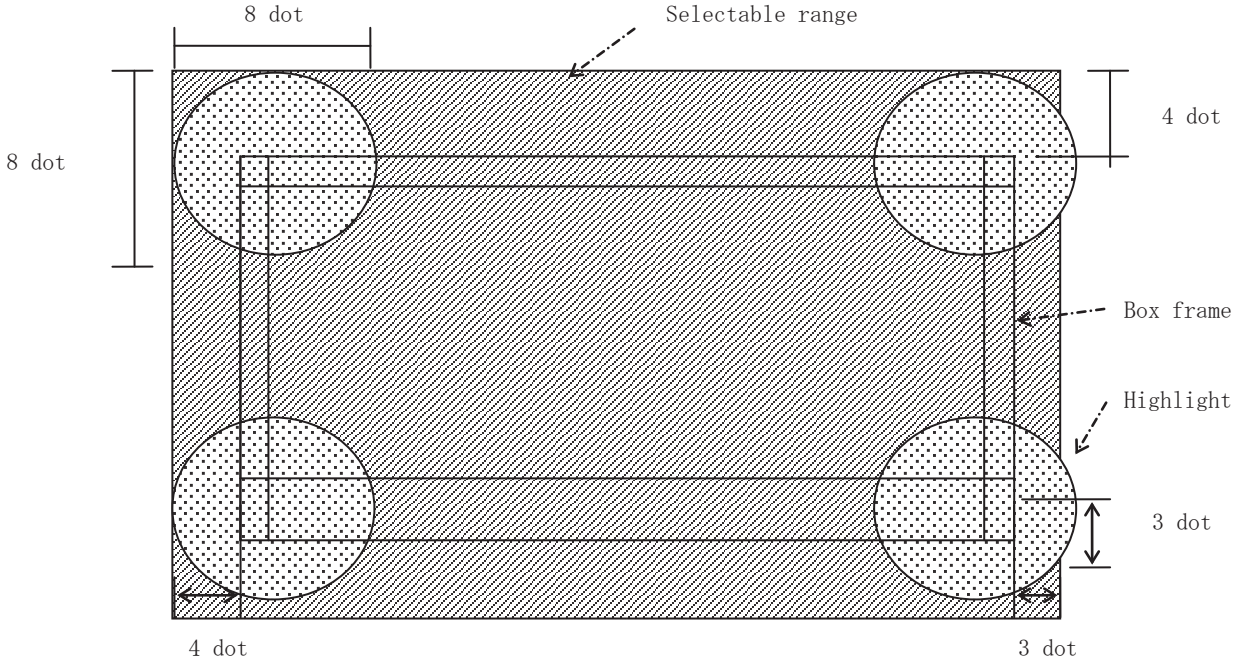
Example: arc



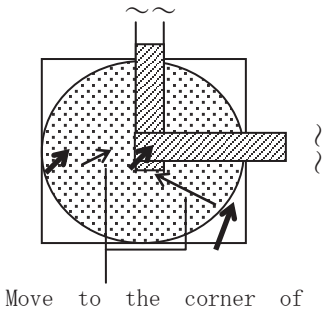
・The same applies to elliptic arc, sector, three point arc, and three point closed arc.

Automatic change of selection area and mouse cursor position during box selection

(1) Selection range is 4 dot wider in upper left direction and 3 dot wider in lower right direction than the box frame.



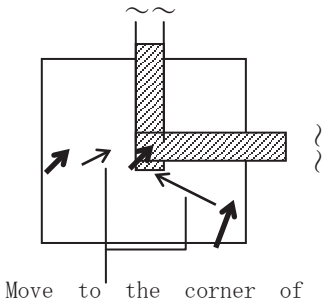
(2) When the highlighted section is selected in standard screen size, the mouse cursor is moved to the corner of the box frame. However, the highlighted rectangle instead of circle is handled.



※Though this is an example at the lower left, the same applies to four corners (upper left, lower left, upper right, lower right).

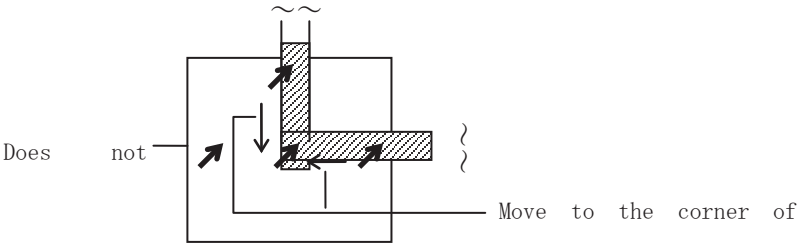
① Standard zoom ratio:

- When the mouse is dragged in the highlighted area, the mouse cursor moves to the corner of the box frame.



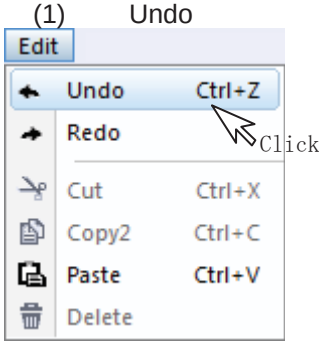
② Magnified ratio:

- When the mouse is dragged on the box line in the highlighted area, the mouse cursor moves to the corner of the box frame.



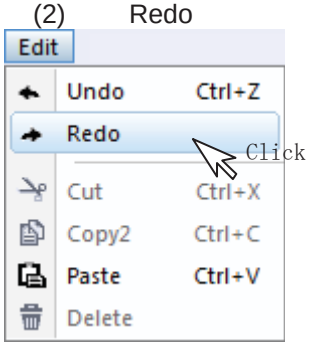
2.3.7 How to Operate Edit

(1) Undo



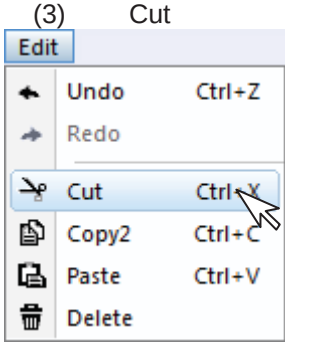
Up to ten actions can be undone.

(2) Redo



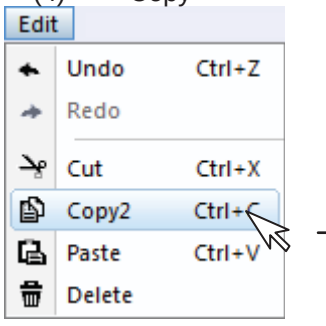
Restores undone action.

(3) Cut



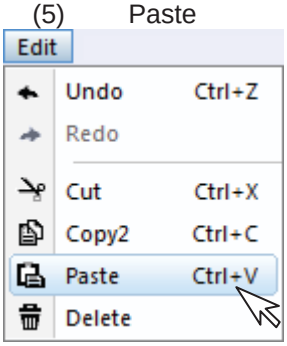
Cuts an object selected.
Cuts an object to clipboard.

(4) Copy



Copies selected object to clipboard.

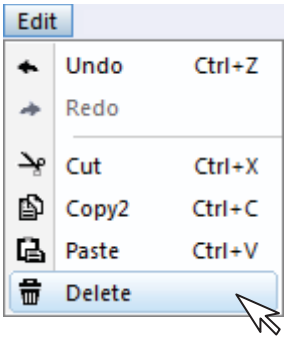
(5) Paste



Click

The object copied or cut to clipboard is pasted.

(6) Delete

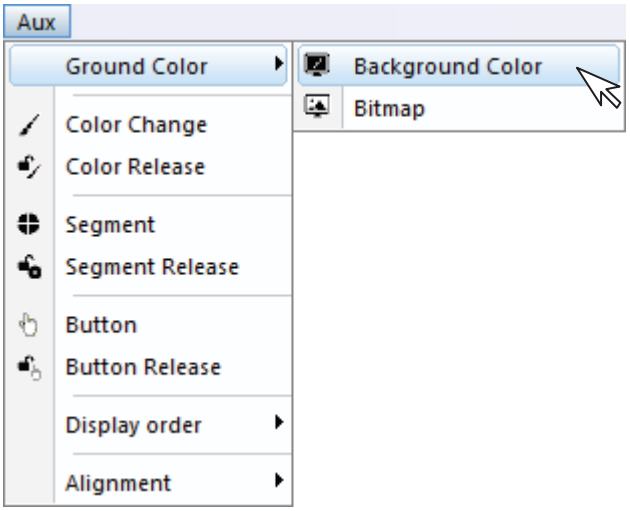


Click

Deletes an object selected.

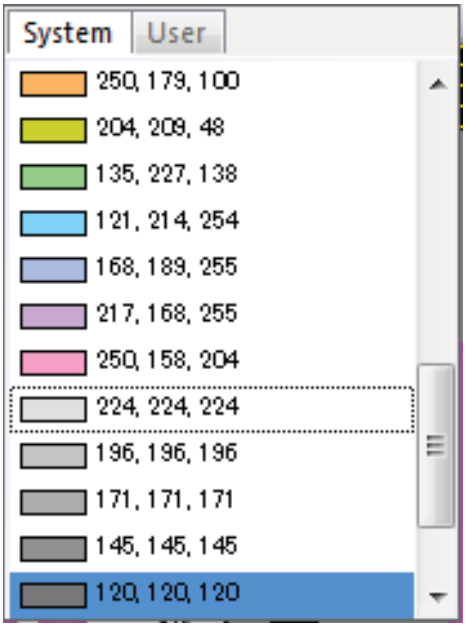
2.3.8 How to Operate Format

(1) Background of screen (background color)

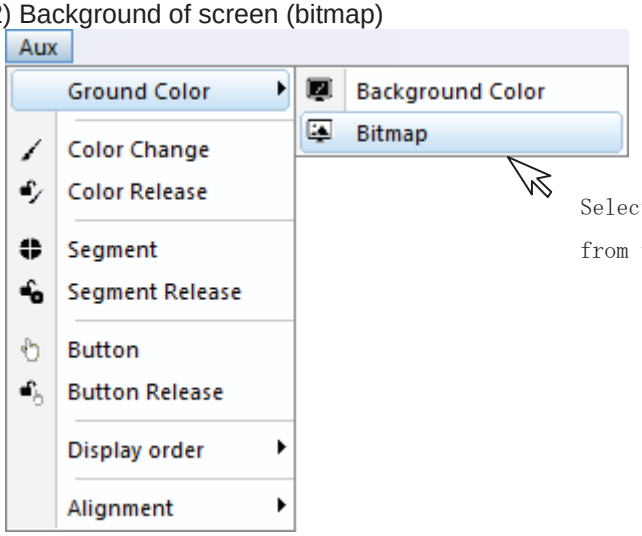


Select a color from the dialog box.

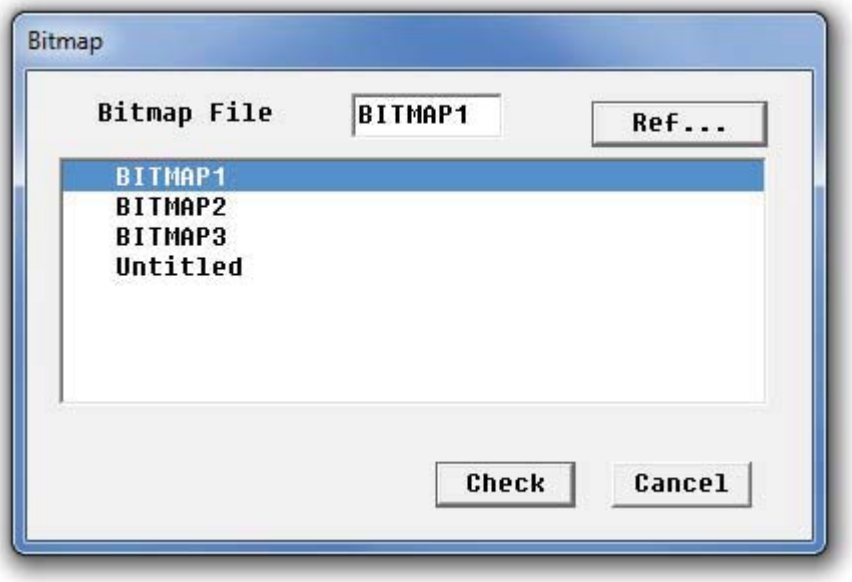
(color dialog box)



(2) Background of screen (bitmap)



Bitmap dialog box



- To register a bitmap in the bitmap dialog box, save the BMP file in the directory below.

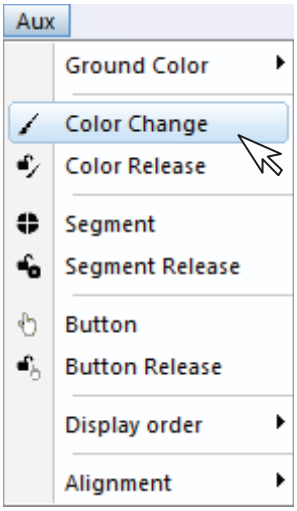
"<Project file name>\PLT\BITMAP"

(3) Color changing

Used to change the color of graphic lines and so on. First, operate a graphic created as follows:

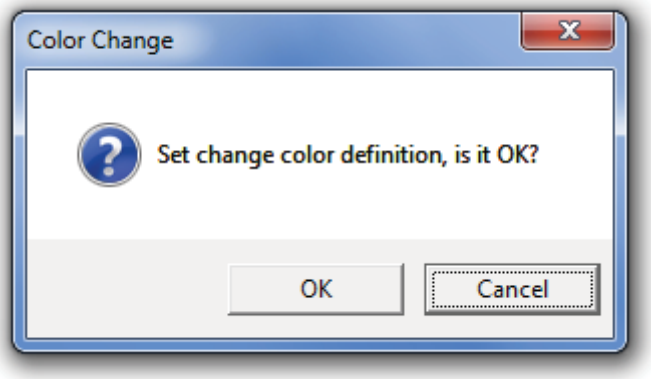
①Create a graphic.

② Select desired graphic.



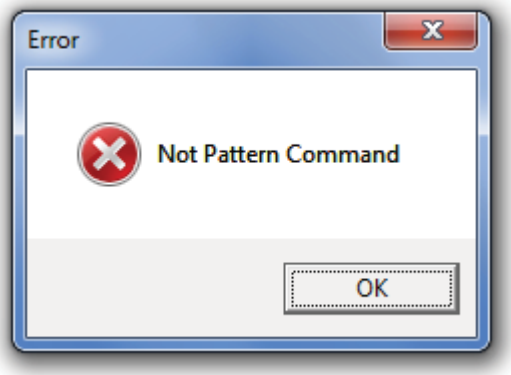
③ Click to select the menu item (click)

④ The confirmation window is shown.



⑤ Click OK to enable to change the color of object by displaying the color property dialog.

Error message when unchangeable object is selected



Color change property

-	**Other**		**Description**	**Change**
	Object Type	Type of the object	Color Change	No
	Object Kind	Kind of the object	IDDYNAMIC	No
	Object Name	Name of the object	Color Change	No
	Object No	Number of the object	–	No
-	**Data**			
	Color Ext ST No	Set the color change Tag station No.	Only when specifying the system Tag, specify the physical ST No. (101 to 440)	Yes
	Color Ext. Tag	Set the color change Tag.	Number of Tag characters: Up to 16 single-byte characters Selectable from the dialog box	Yes
-	**Color*n* *(n=1 to 8)***			
	Color***n*** Mask	Set the Color*n* mask.	Hexadecimal four characters **(Refer to section 2.3.3, "(18) Selecting BitPattern")**	Yes
	Color***n*** Data	Set the Color*n* data.	Hexadecimal four characters **(Refer to section 2.3.3, "(18) Selecting BitPattern")**	Yes
	Color***n*** Fore Color	Set the Color*n* display color.	Select it from the color dialog.	Yes
	Color***n*** Fore Type	Set the Color*n* display color type.	None, Blink, Reverse, Reverse + Blink	Yes
	Color***n*** Back Color	Set the Color*n* background color.	Select it from the color dialog.	Yes
-	**Cond Else**			
	Else Fore Color	Set the display color for data deviating from the exchange condition.	Select it from the color dialog.	Yes
	Else Fore Type	Set the display color type for data deviating from the exchange condition.	None, Blink, Reverse, Reverse + Blink	Yes
	Else Back Color	Set the background color for data deviating from the exchange condition.	Select it from the color dialog.	Yes

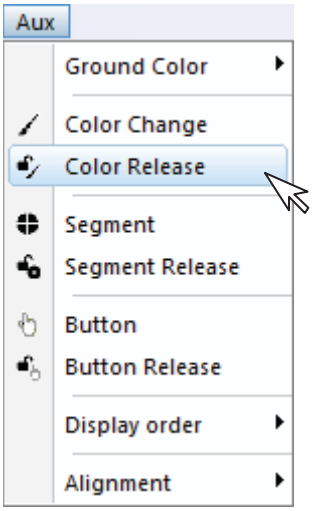
- Changeable graphic list

ITEM TYPE	YES/NO	ITEM TYPE	YES/NO	ITEM TYPE	YES/NO
Line	Yes	Elliptic Arc	Yes	48 Dot Character	Yes
Line (horizontal, vertical)	Yes	Polygon	Yes	32 Dot Character	Yes
Circle	Yes	Sector	Yes	24 Dot Character	Yes
Arc	Yes	Button Type Rectangle	Yes	16 Dot Character	Yes
Rectangle	Yes	Point	Yes	8 Dot Character	Yes
Ruled Line	Yes	96 Dot Character	Yes	Fixed Pattern	Yes
Ellipse	Yes	64 Dot Character	Yes	Segment	Yes

(4) Color Release

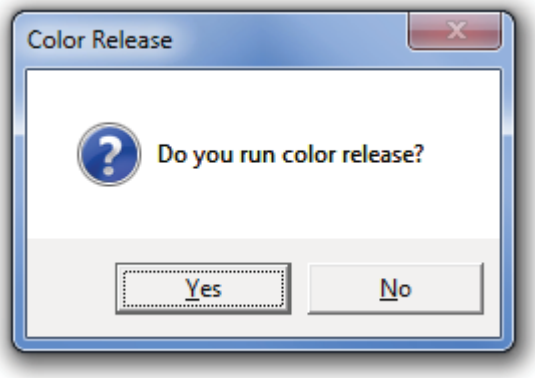
Puts the object changed to variable picture object by changing the color back to original picture object. Here, the graphic colors (display color, background color) are changed to the first ones specified at the color change with alarm flicker.

① Select desired graphic for “Color Release”.



② Click to select the menu item (click)

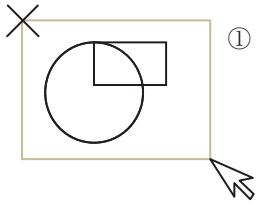
③ The confirmation window is shown.



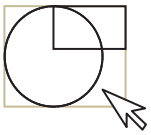
④ Click OK to enable to put the color property back to original.

(5) Segmenting

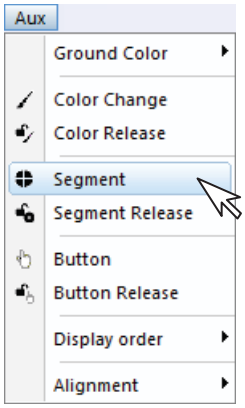
Groups and saves the graphic composed of multiple fixed objects as a segment.



① Drag the mouse to enclose two or more fixed objects.

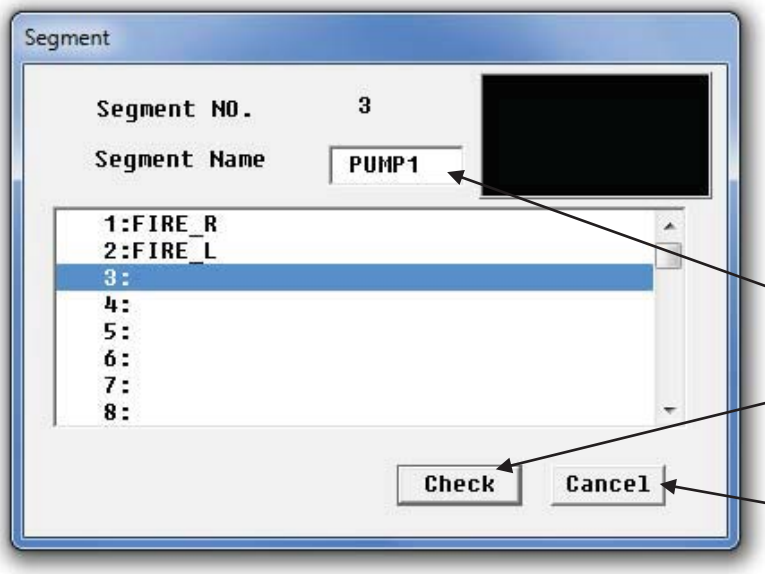


② Right-click in the enclosed area



③ Click to select the menu item

Segment



④ Select a number to register.

⑤ Enter a segment name.

⑥ Confirm to finish the registration

Click the Cancel to cancel the operation.

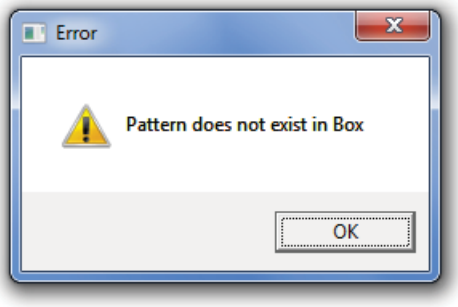
Note 1: Only the fixed objects grouped are segmented.

* The available maximum amount of fixed objects is 1023.

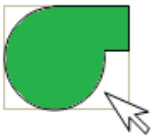
- When the picture objects already segmented are grouped, a message is shown.

Note 2: The object range to be segmented may vary according to the screen types.

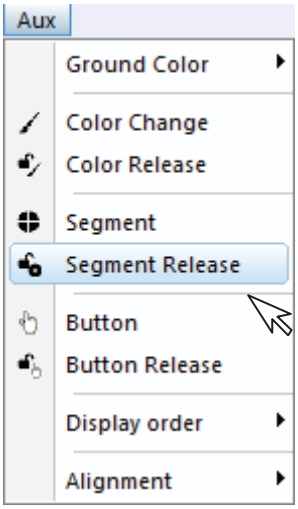
Error message



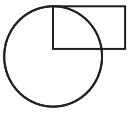
(6) Segment Release



① Select desired graphic for “Segment Release”.

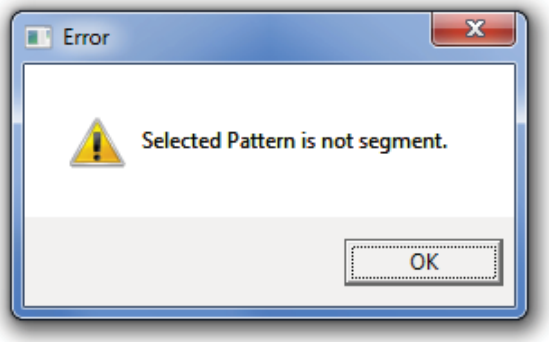


② Click to select the menu item (click)



③ The graphic returns to original individual objects.

Error message



(7) Button

Create buttons from the touch area of the operation picture object selected and rectangle graphic.

Target picture object: operation/link screens

Operation/meter diagram window

Operation/dynamic window

Operation/virtual keyboard 2

Operation/multiple touch point window

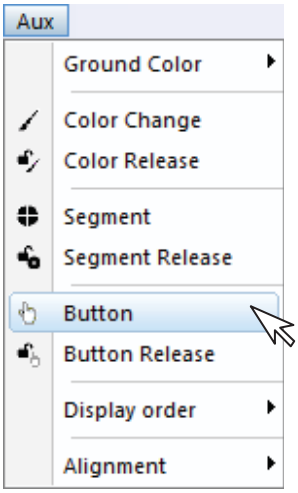
Operation/multiple touch point entry

Operation/device operation 1

Operation/device operation 2

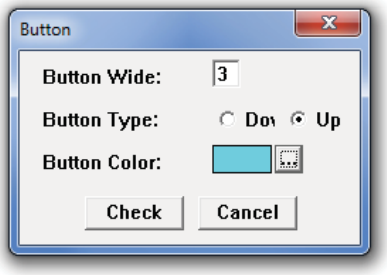
Graphics/rectangle

① Select an operation frame or rectangle created.

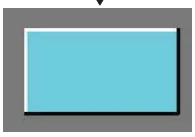


② Click to select the menu item (click)

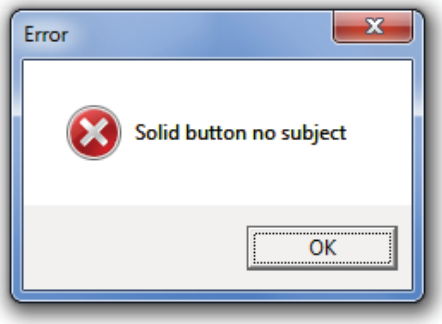
③ The definition window is shown.



Click Confirm after setting



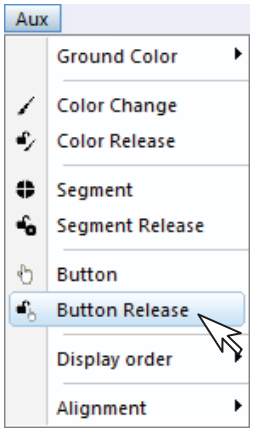
Error message



(8) Button Release

Clear the button made of operation picture object.

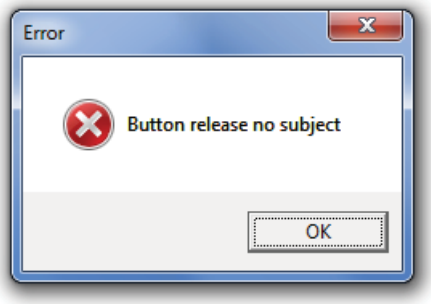
① Select the buttoned object.



② Click to select the menu item (click)

③ The object returns to original operation frame or rectangle.

Error message



Note: The buttoned rectangle cannot be cleared.

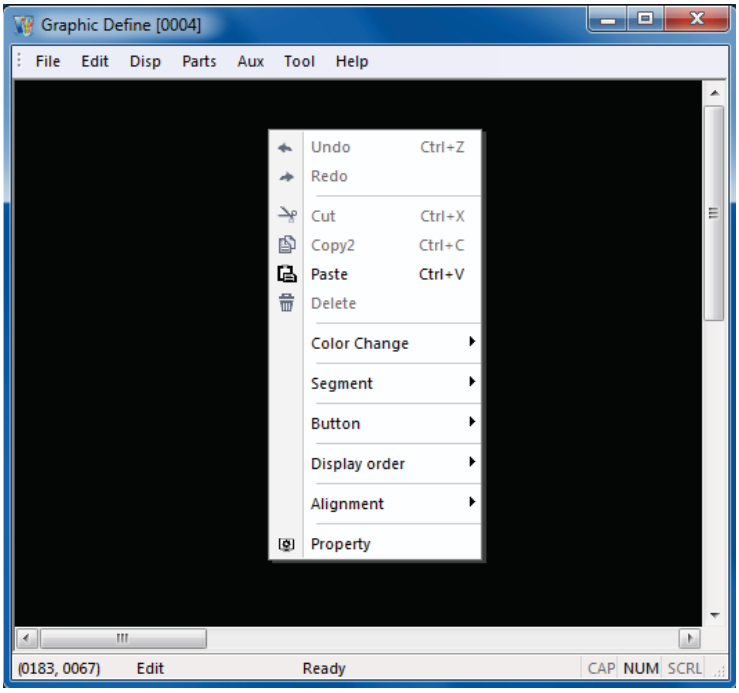
2.4 Right-Button Function

2.4.1 Function Overview

Right-click the mouse on the drawing area in the plant panel to display the menu available for use at the click position. (Popup menu is shown)

2.4.2 How to Operate Popup Menu

① Right-click the mouse on the drawing area in the plant panel.



② Left-click the mouse on desired item in the popup menu.
The item selected is performed.

* For information about the functions, refer to 2.3.7 How to Operate Edit and 2.3.8 How to Operate Format.

2.5 Precautions for Drawing

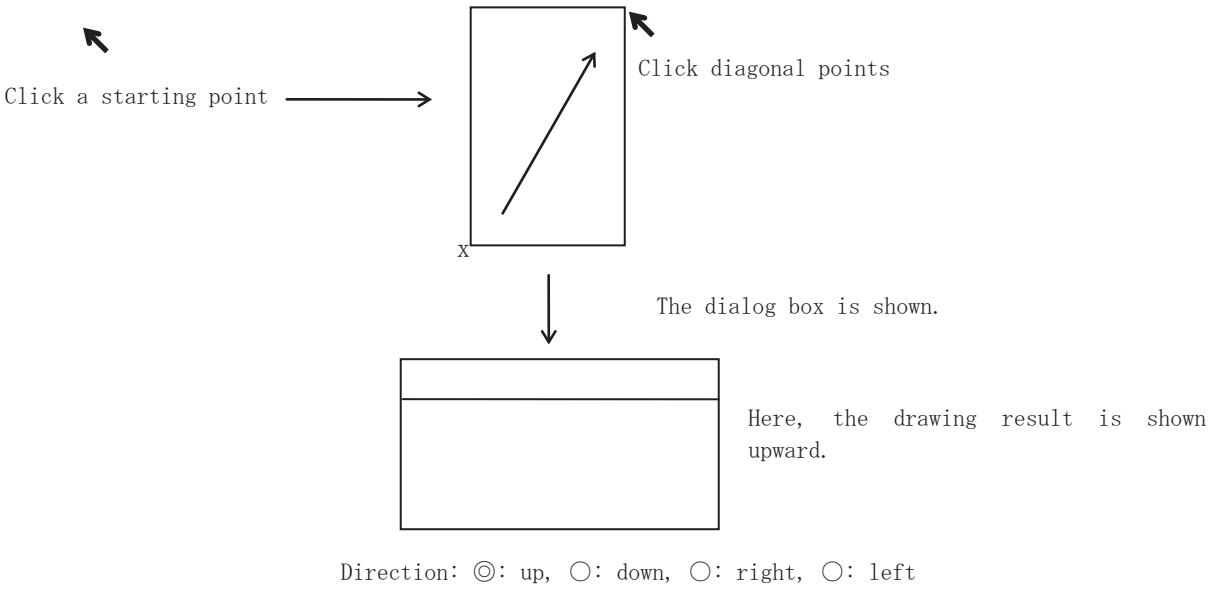
2.5.1 State when the Plant Panel Editor Starts

- Just after the plant panel editor starts, there are no screen number and name and the screen type is standard.
- To create other than standard screen type, select Select Screen in the Screen menu to create a screen by specifying the screen number and type.

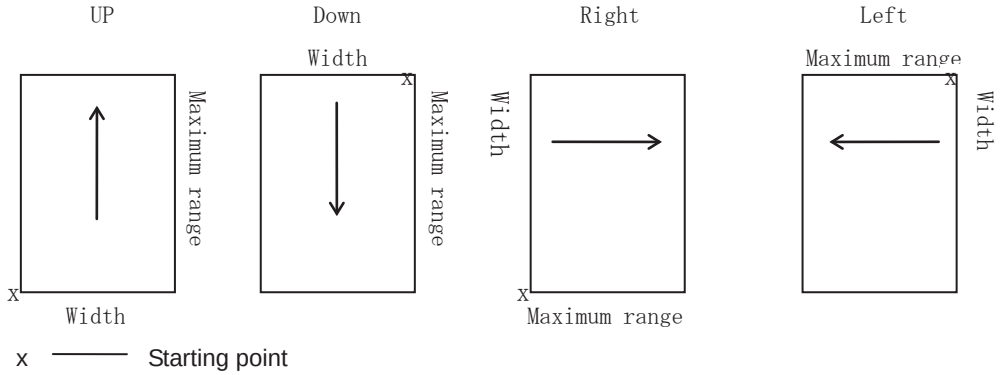
2.5.2 Display Order of Graphic, Variable, Operation

After selecting screen, the display order is graphic commands (drawing order), variable commands (drawing order), and operation commands (drawing order).

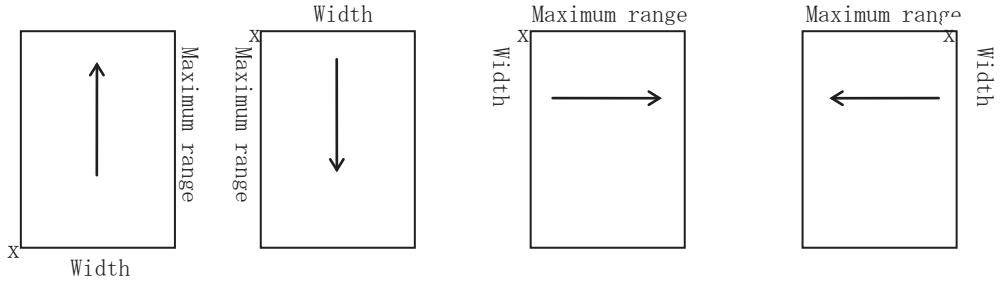
2.5.3 Relationship among Starting Point of each Graph, Display Width of Rectangles, and Maximum Range



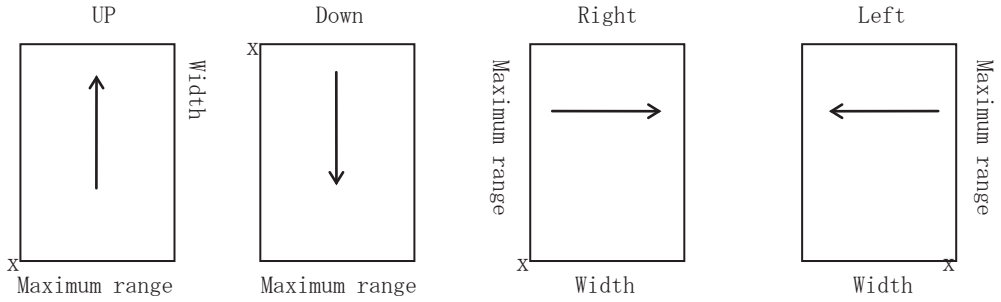
① Bar graph, integrated bar graph, ratio bar graph, limit bar graph:



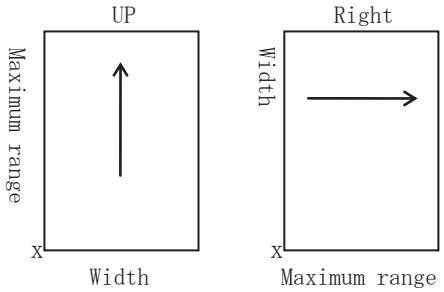
② Prot Bar Graph, Prot Limiter Bar Graph



③ Prot Line Graph, Prot Dot Graph, Prot Step Graph, Prot Step (Bar) Graph, Trend:



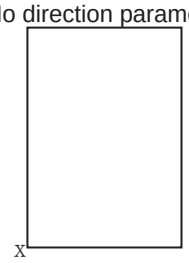
④ Difference bar graph:



No Down and Left

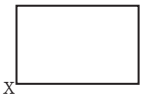
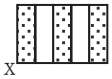
⑤ Batch Trend, Line, XY Step, and others:

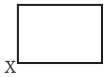
No direction parameter



・The maximum range is to specify the dot quantity and the maximum value is to specify data for the maximum range. There is no magnitude relation between the maximum range and maximum value.

2.5.4 Display of Variable and Operation Objects

Item	Display form	Remarks
Numeric Upper and lower limits	999999	Displays '9' as many as the display digit
Text	ZZZZZZ	Displays 'Z' as many as the display characters (single-byte)
Exchange	*1: 運転	Displays the first pattern specified
Bar Graph	*2: 	Displays all with the specified color
Picture Graph	*2: 	The height and direction are shown with arrows on the created graphic.
Move Pattern Trend Line Batch Trend		The range is displayed with a frame
Meter diagram		The meter diagram is displayed with a frame at the display position
Prot Bar Graph Prot Limiter Bar Graph	*3: 	Displays as many as the specified quantity

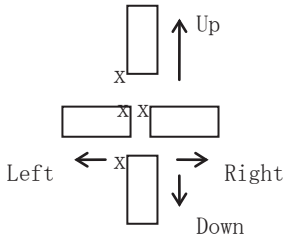
Operation (Common)		Changes the touch frame size
		When the touch frame size is zero, the dot '・' is shown

① Supplemental explanation of display state

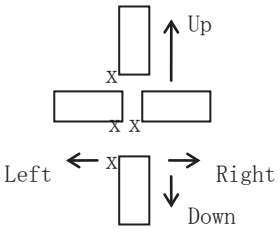
・x: Starting point

*1: Starting point of each pattern

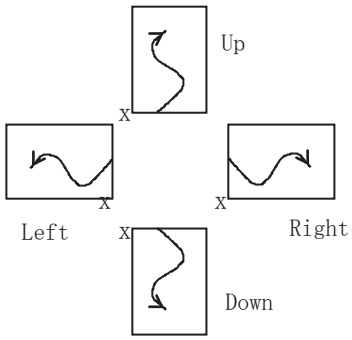
*2: Different according to the display direction.



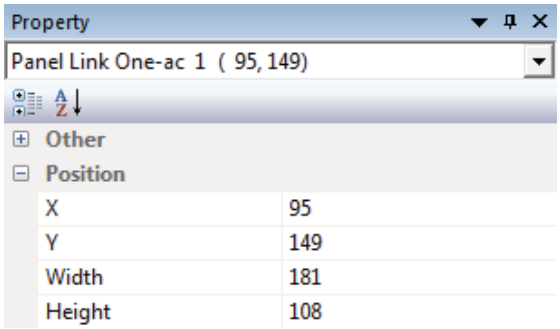
*3: Different according to the display direction.



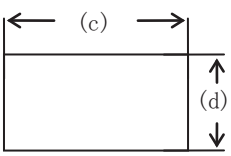
*4: Different according to the display direction.



② Touch area



(a): Start X
(b): Start Y
(c): Width
(d): Height



((a), (b))

※: In the touch area, the operation content of the touch size is automatically displayed.
The items in the dialog box can be changed.

③ How to input STNO. and TAGNO.

Select a number in the dialog.

The number can be input directly.

2.5.5 Vertical Display of Text or Message and Mixed Mode

(1) Precautions for displaying text and message

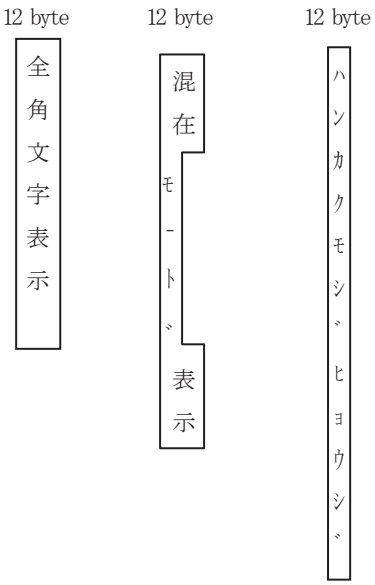
When displaying text vertically or when displaying single-byte text mixed with double-byte text, the following restrictions are applied.

① The characters are aligned at the left side.
The right sides are uneven.

② When displaying same-byte character codes, the text length may vary according to the quantity of double-byte or single-byte characters.

③ When displaying single-byte katakana characters, voiced sound symbol and semivoiced sound symbol are displayed below.

④ The direction of prolonged sound symbol is not changed.



2.5.6 Precautions for Enlarged Display

(1) Phenomenon that arcs cannot be displayed

In the still picture screen, the following graphic may not be enlarged:

① Arc

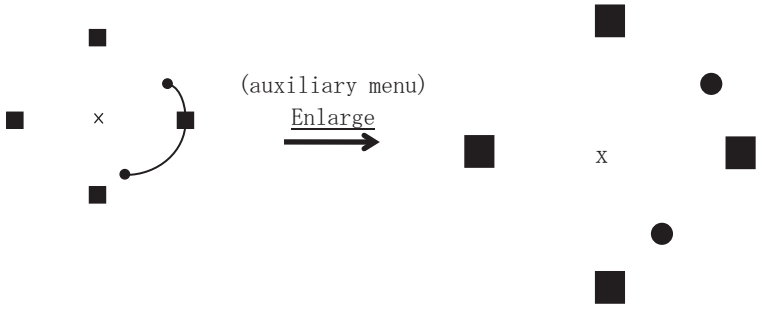
② Elliptic Arc

③ Sector

④ Three point arc

⑤ Three point closed arc

In this case, only the symbols (x, ■, ●) are shown, but the arc is not displayed (see below).



When this phenomenon happens, draw graphics (in standard size).

2.6 Restrictions on Plant Panel Editor

NO.	Restrictions	Drawing Algorithm/hardware	Restrictions on specification
1	Difference of variable/meter diagram drawing		○
2	Difference of variable/meter diagram drawing		○

NO.	1

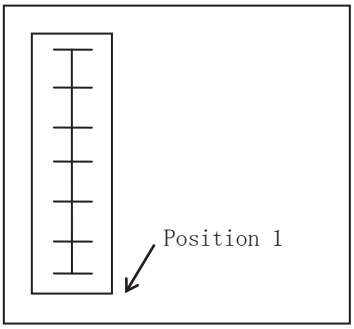
Restrictions

Difference of variable/meter diagram drawing

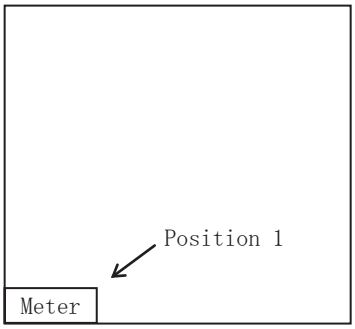
Details

1. At present

① HCI



② Plant panel editor



The online software displays the meter according to specific pattern file.

The plant panel editor displays the type and position only.

2. Countermeasure

This is a restriction based on the specification (plant panel editor displays the definition only).

Remarks

NO.	2

Restrictions

Blinking display

Details

1. Phenomenon

① HCI

It blinks at some interval.

② Plant panel editor

It does not blink.

The online software supports and realizes blinking function.

The plant panel editor does not support blinking function, so it does not blinks.

2. Countermeasure

This is a restriction according to the difference between online software and plant panel editor.

Remarks

3. Segment

3.1 Overview

3.1.1 Overview

XES creates and registers multiple graphic commands as a segment.

The segment can be used in the plant panel as graphic or variable object.

3.1.2 Type and Quantity

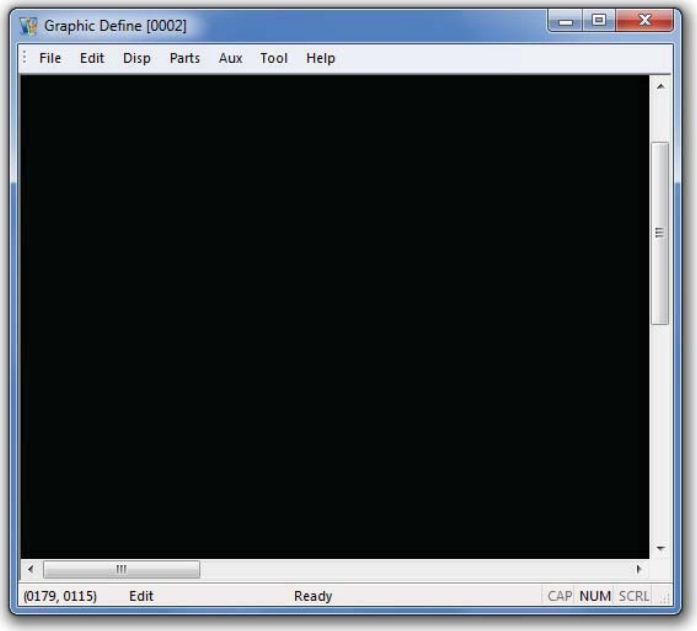
① User segment: 1024, for each user

Note) The segment name can be up to eight single-byte characters (single-byte and double-byte texts can be mixed).

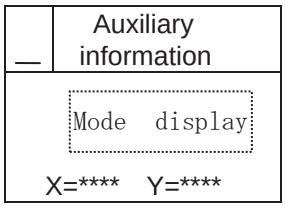
However, only the support side can name the segment. HCI cannot name the segment.

3.2 Display Items and Descriptions

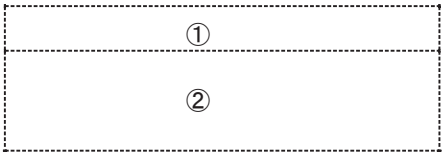
3.2.1 Screen as a Whole



3.2.2 Various Information



＊: Details of the mode display area



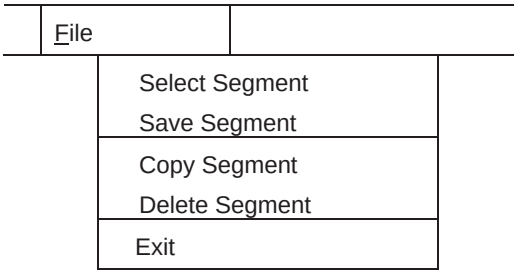
①	②
Graphic	Type (picture)
Move	Type
Copy	Type
Transform	Type
Box	
Select Graphics	Type (last information is displayed)

3.2.3 Cursor

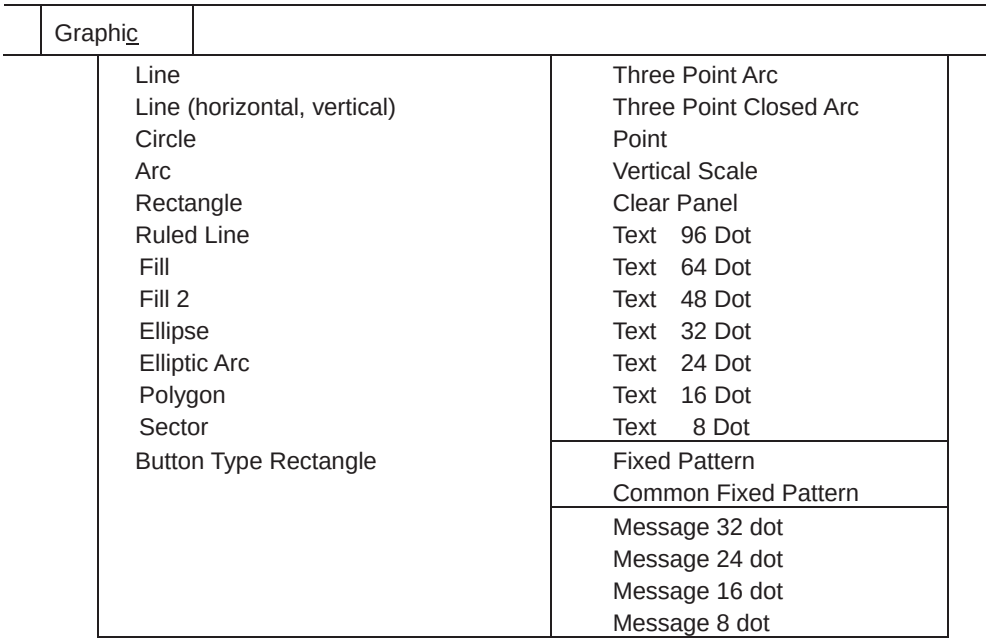
: Cursor for creation
: Cursor for movement
: Cursor for copy
: Cursor for change

3.2.4 Menu Bar

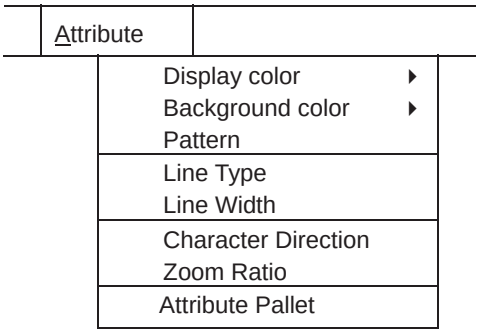
① File



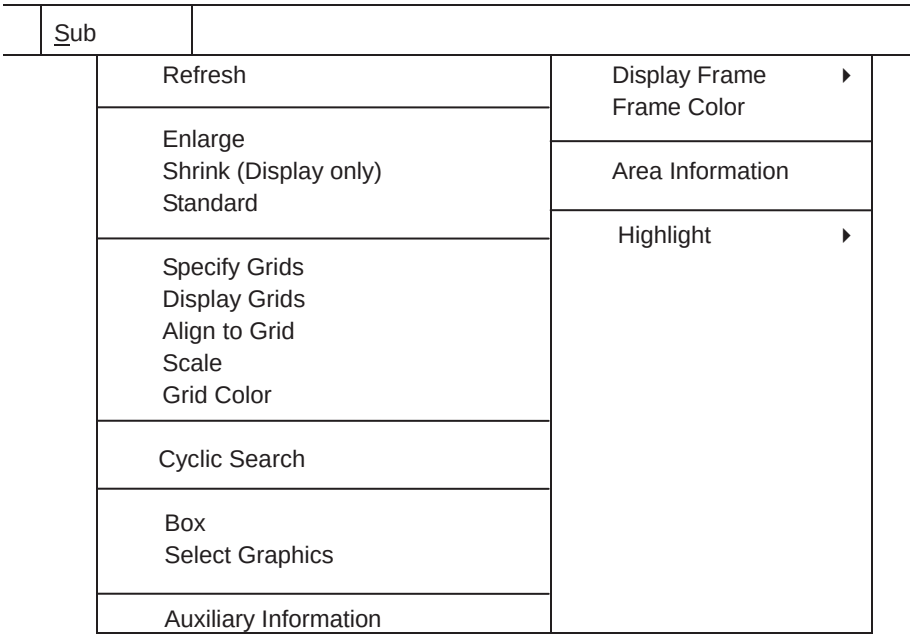
② Graphic



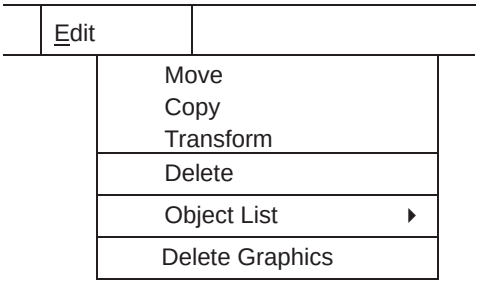
③ Attribute



④ Sub



⑤ Edit



3.2.5 How to Operate

The detailed operation method is the same as the plant panel editor except for the File menu.

For details, see "How to Operate Plant Panel Editor".

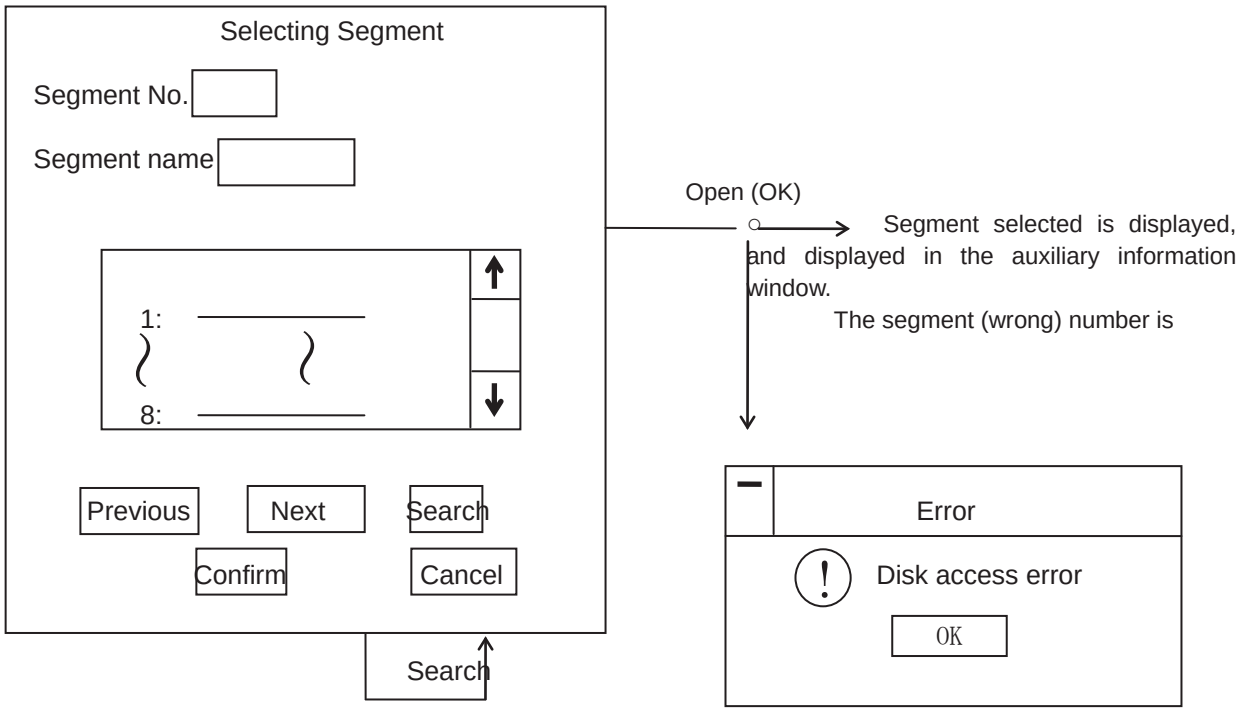
Here, describes how to operate the File menu.

Note 1) The Object List in the Edit menu has Graphic only, unlike with the plant panel editor.

(Since the commands in the segment is only for graphic object.)

3.3 How to Operate Files

3.3.1 Select Segment



The segment name input in segment number field is displayed in the list box.

※How to display the list box

- A page shows 128 segments in the list box.

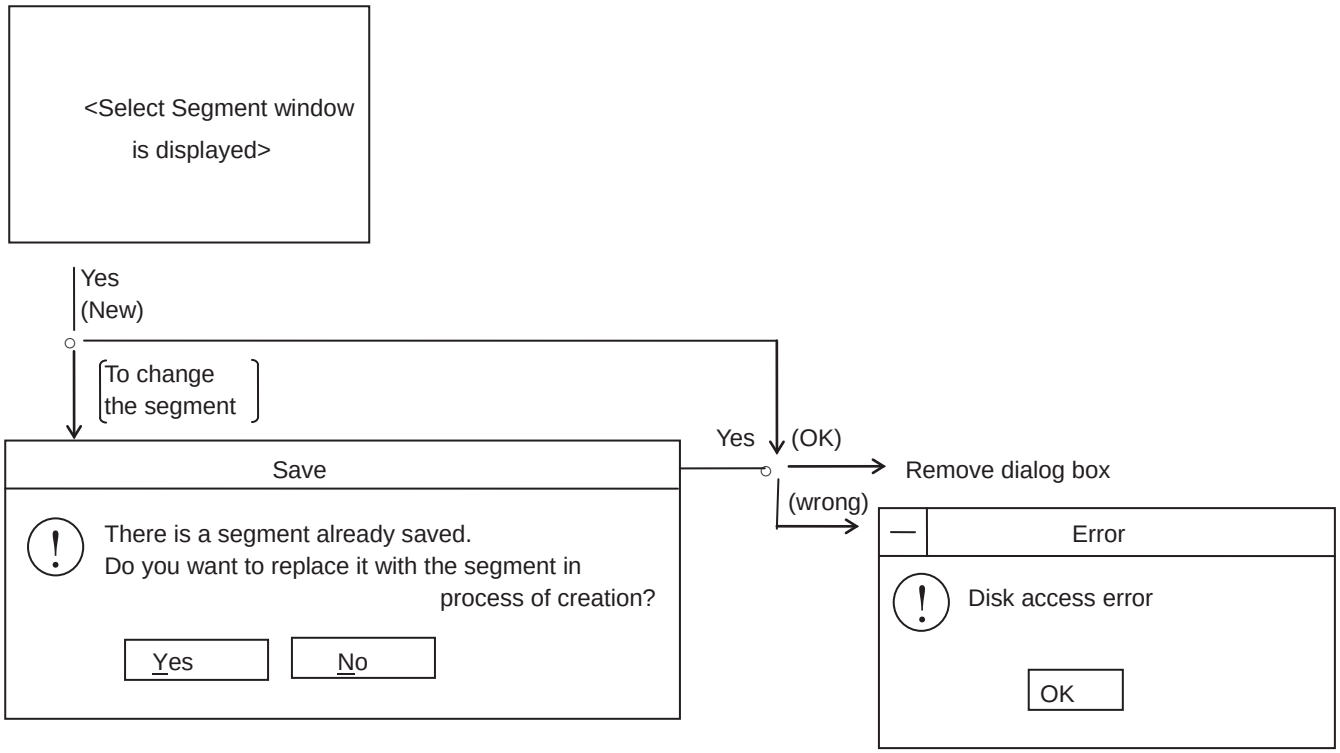
Page 1: 1 to 128

Page 2: 897 to 1024

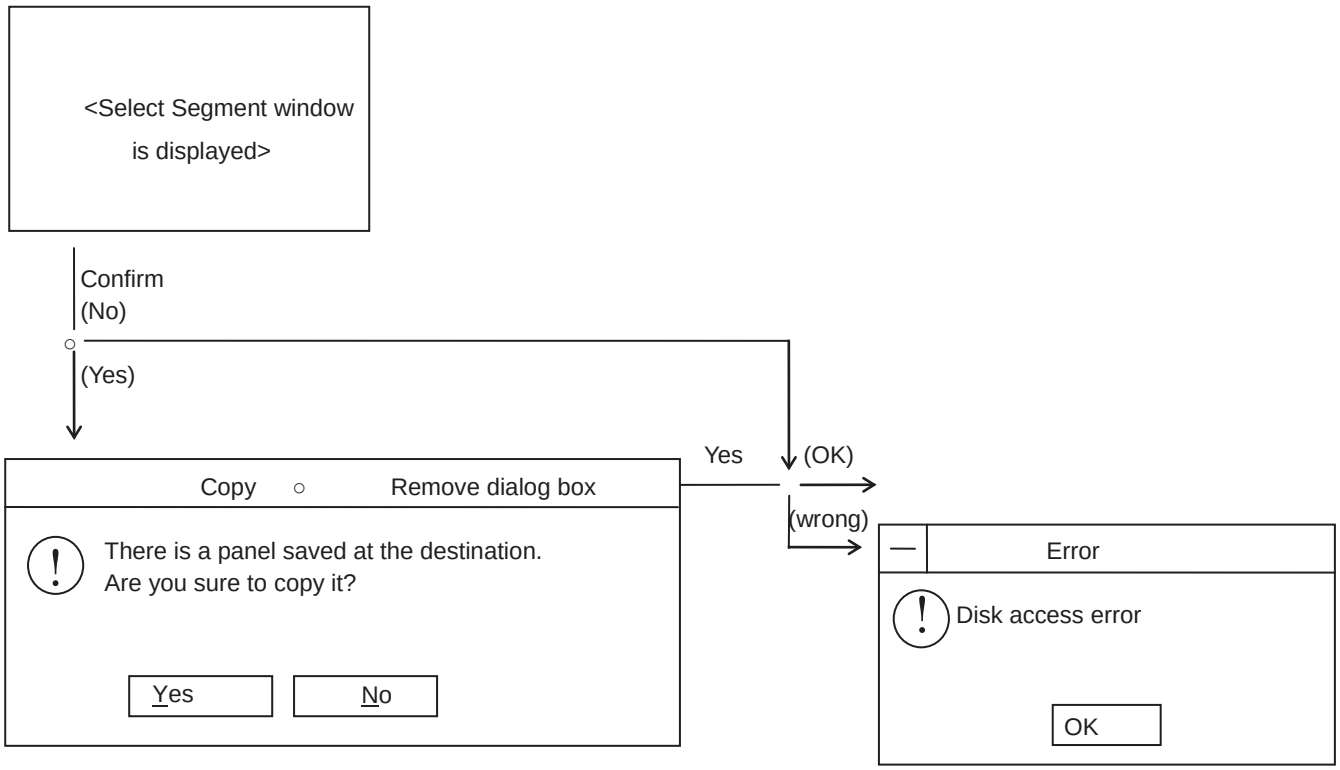
To update the page, click Next/Previous.

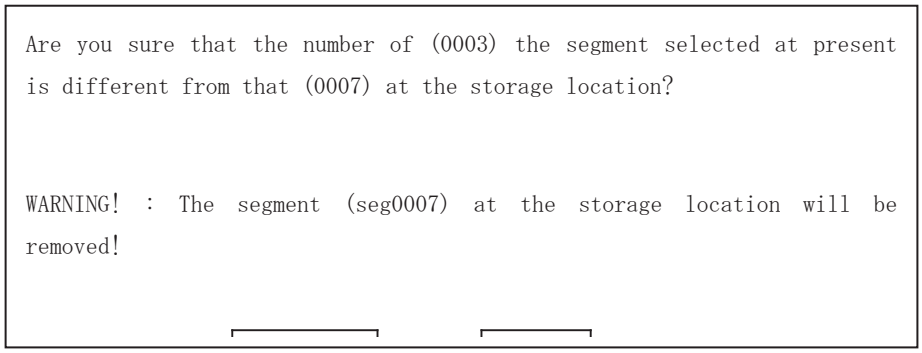
- To update the menu in a page, click the scroll bar at the right side.

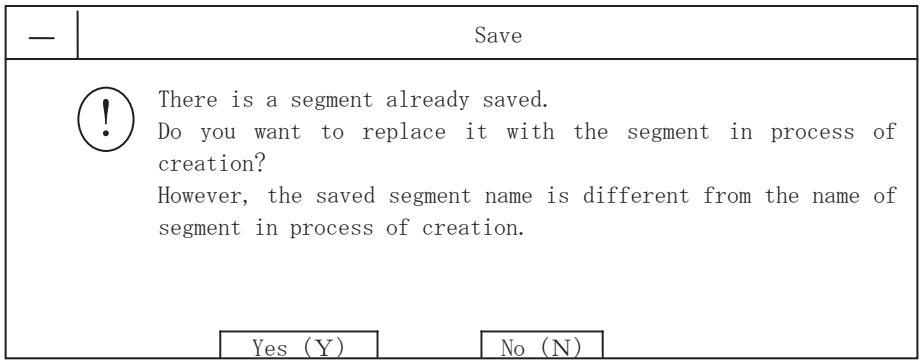
3.3.2 Saving Segment



3.3.3 Copying Segment

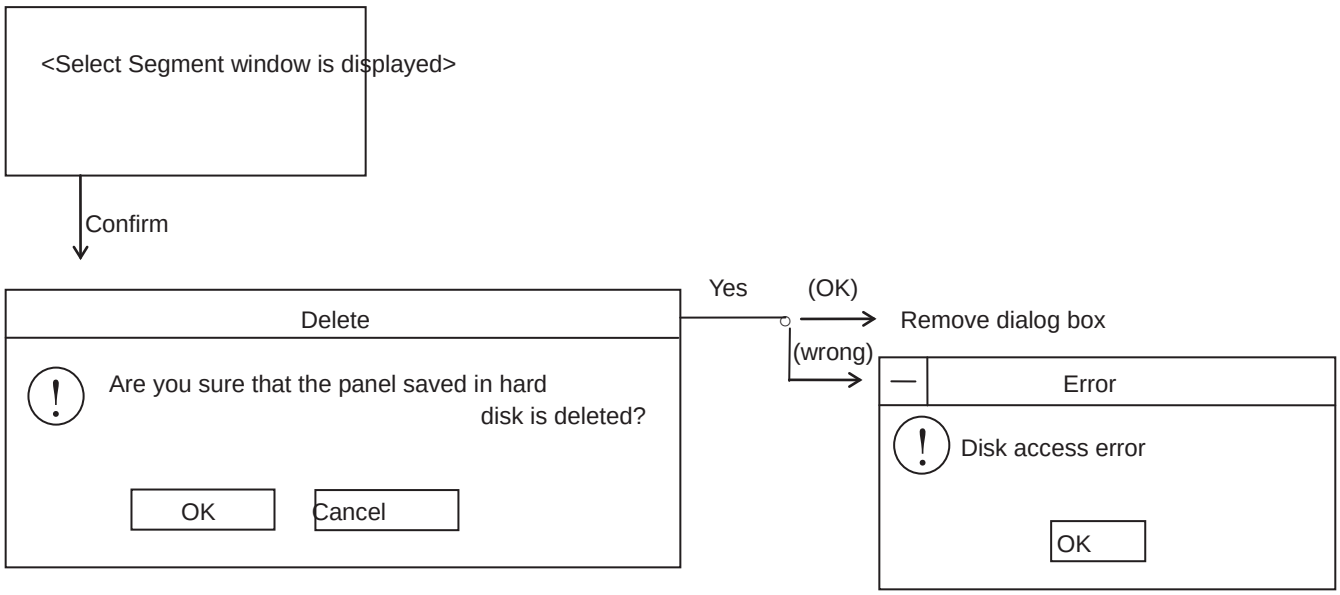






(warning message)

3.3.4 Deleting Segment

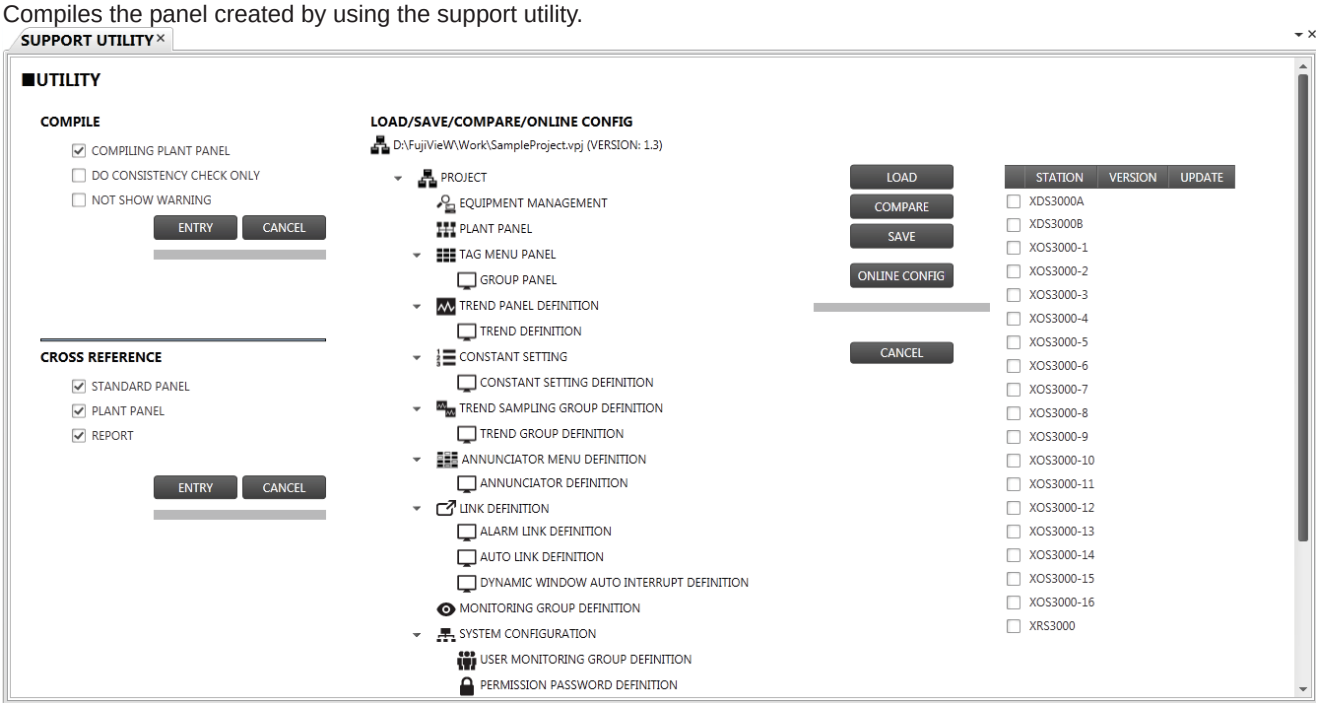


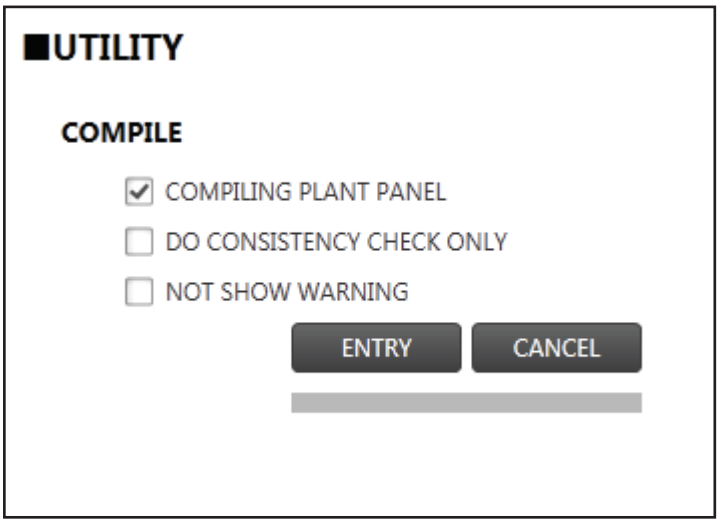
3.3.5 Exit

Performs the process similar to the save, exit the segment, and starts FPROCES-M.

4. Compile

Compiles the panel created by using the support utility.





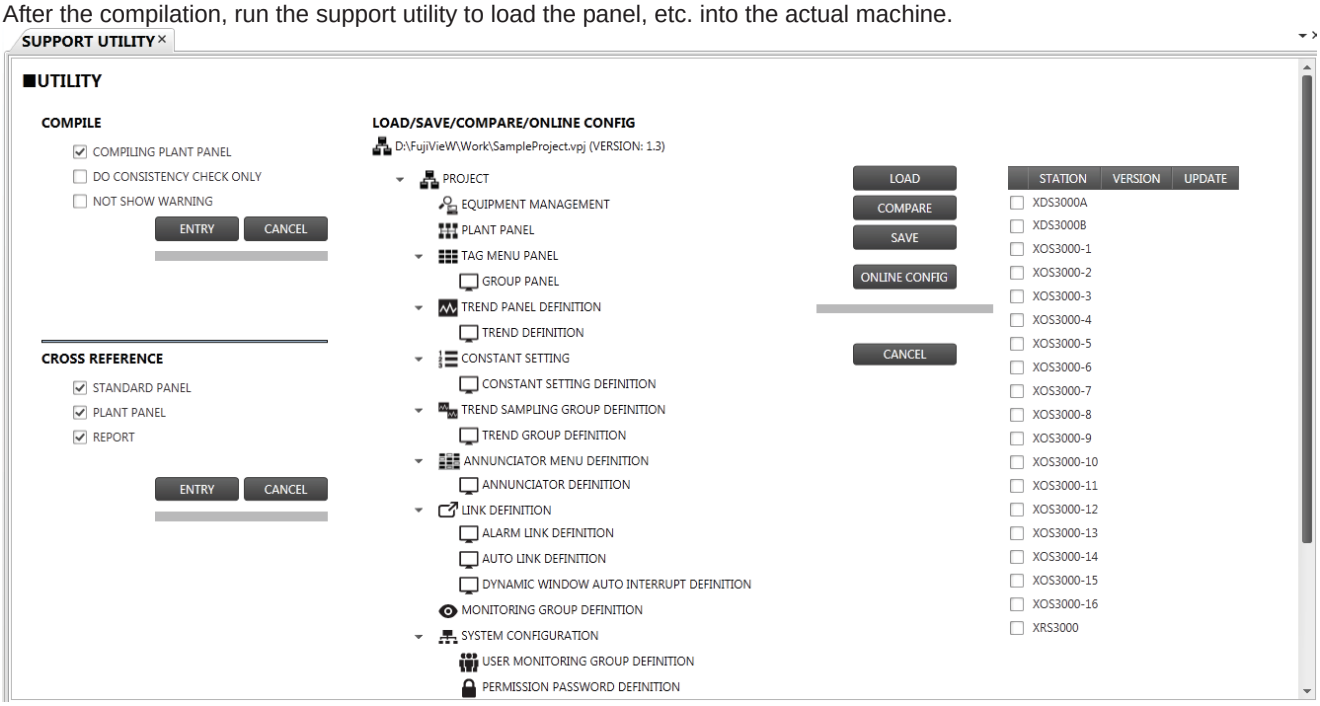
① Click the check box of "Compile Plant Panel" in the Compile area.

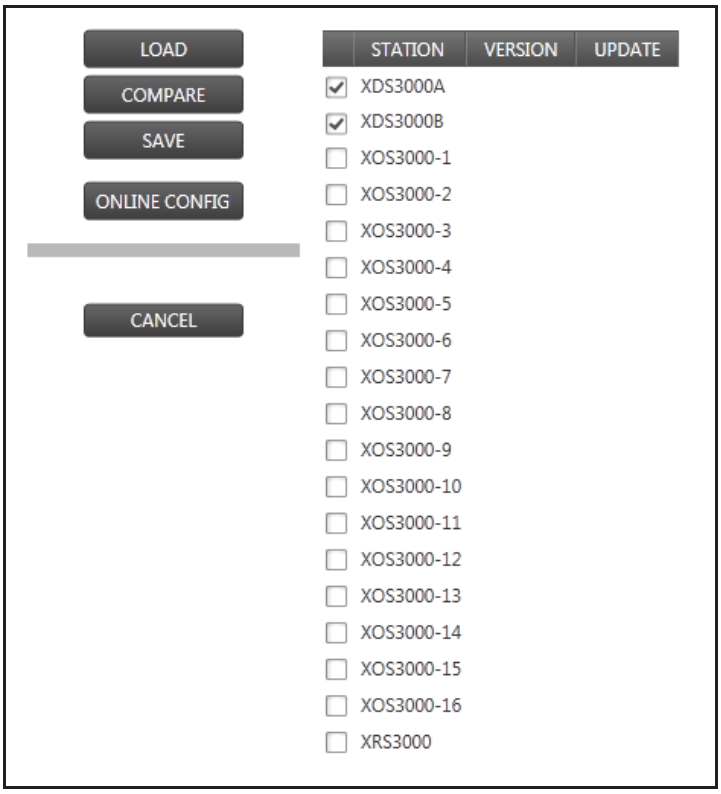
② Click the "Run" button to compile the panel.

③ The compilation result can be confirmed in the Error List and Record areas at the bottom.

5. Load

After the compilation, run the support utility to load the panel, etc. into the actual machine.





① Click the check box of the actual destination machine.

The example above loads the panel, etc. into both systems for XDS duplication

② Click the "Load" button to perform the load.

③ The load result can be confirmed in the Error List and Record areas at the bottom.

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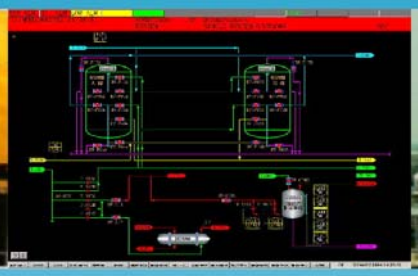

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